Week No. 7

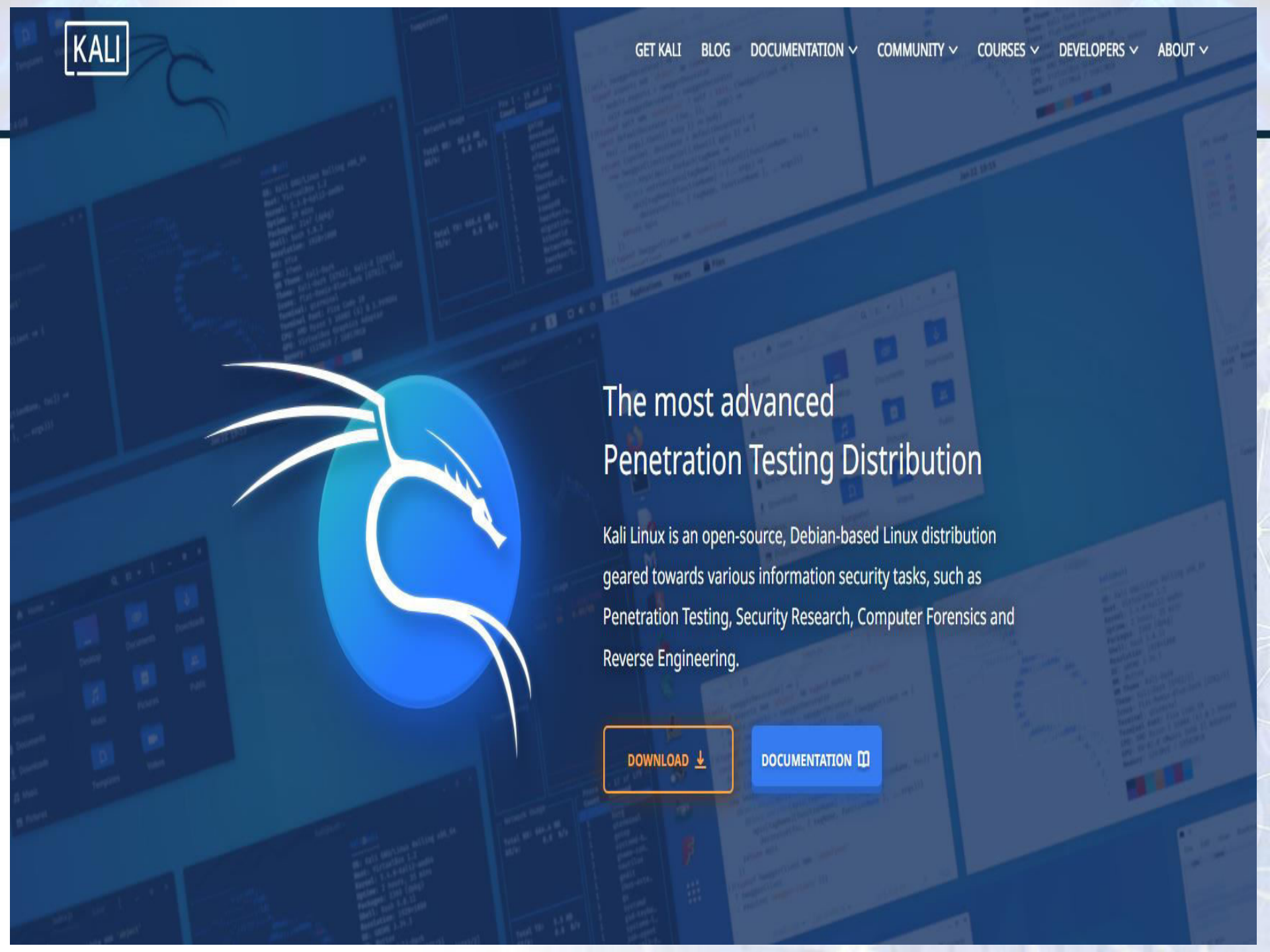
Cyber Security Overview

Week 7 Topic 1

- **Kali Linux**
- **Virtual Box**

- **Virtual Box**
- https://www.virtualbox.org/wiki/Downloads

- **Kali Linux**
- https://www.kali.org/
- https://www.osboxes.org/kali-linux/#kali-linux-2022-3-vbox


KALI
GET KALI
BLOG
DOCUMENTATION
COMMUNITY
COURSES
DEVELOPERS
ABOUT
The most advanced
Penetration Testing Distribution
Kali Linux is an open-source, Debian-based Linux distribution geared towards various information security tasks, such as Penetration Testing, Security Research, Computer Forensics and Reverse Engineering.
DOWNLOAD
DOCUMENTATION

VirtualBox

search...
Login Preferen

Download VirtualBox

Here you will find links to VirtualBox binaries and its source code.

About
Screenshots
Downloads
Documentation
End-user docs
Technical docs
Contribute
Community

VirtualBox binaries

By downloading, you agree to the terms and conditions of the respective license.

If you're looking for the latest VirtualBox 6.1 packages, see VirtualBox 6.1 builds. Version 6.1 will remain supported until December 2023.

VirtualBox 7.0.6 platform packages

- Windows hosts
- macOS / Intel hosts
- Developer preview for macOS / Arm64 (M1/M2) hosts
- Linux distributions
- Solaris hosts
- Solaris 11 IPS hosts

The binaries are released under the terms of the GPL version 3.

See the changelog for what has changed.

You might want to compare the checksums to verify the integrity of downloaded packages. *The SHA256 checksums should be favored as the MD5 algorithm must be treated as insecure!*

- SHA256 checksums, MD5 checksums

Note: After upgrading VirtualBox it is recommended to upgrade the guest additions as well.

VirtualBox 7.0.6 Oracle VM VirtualBox Extension Pack

- All supported platforms

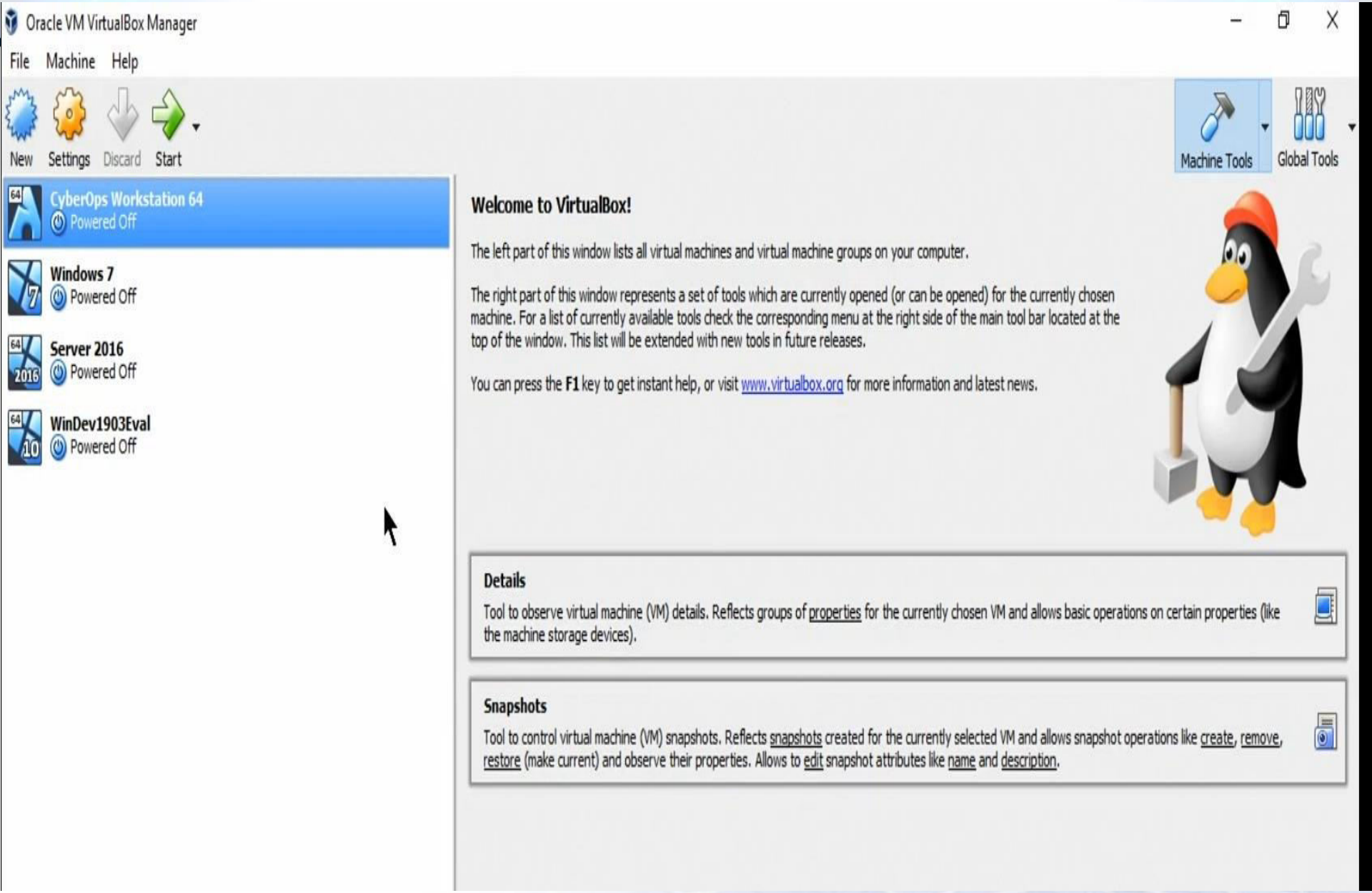
Support VirtualBox RDP, disk encryption, NVMe and PXE boot for Intel cards. See this chapter from the User Manual for an introduction to this Extension Pack. The Extension Pack binaries are released under the VirtualBox Personal Use and Evaluation License (PUEL). *Please install the same version extension pack as your installed version of VirtualBox.*

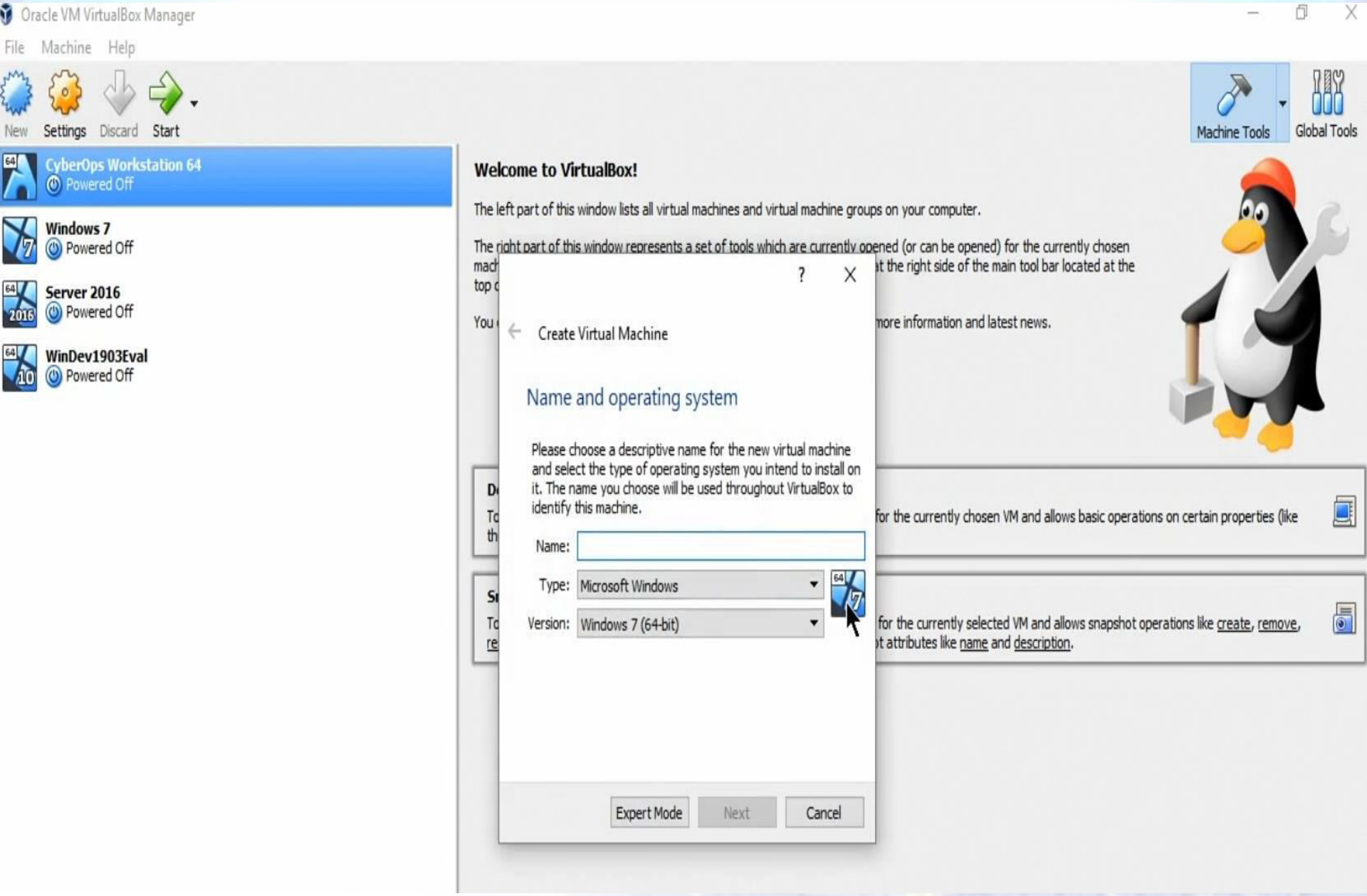
VirtualBox 7.0.6 Software Developer Kit (SDK)

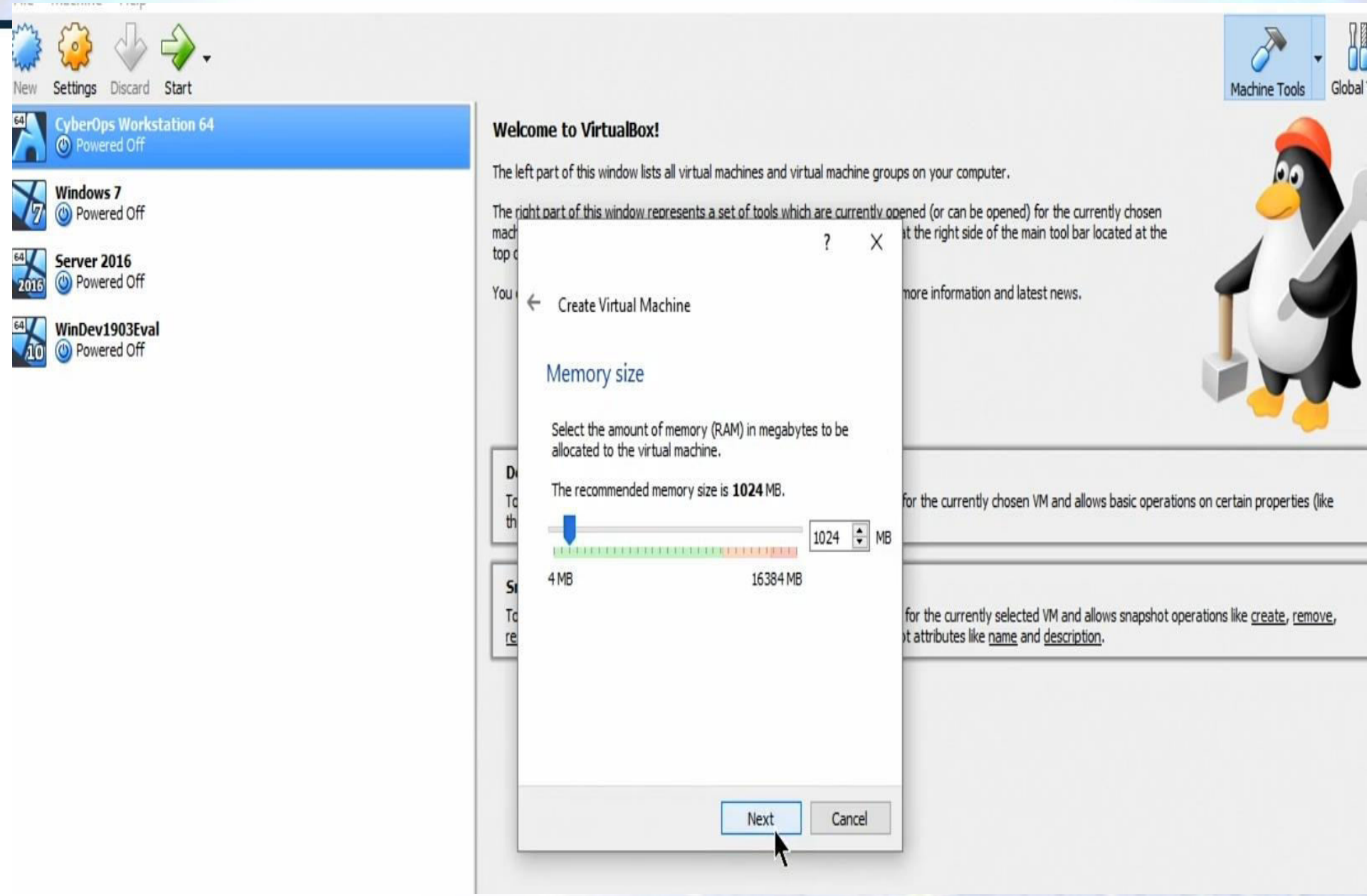
- All platforms

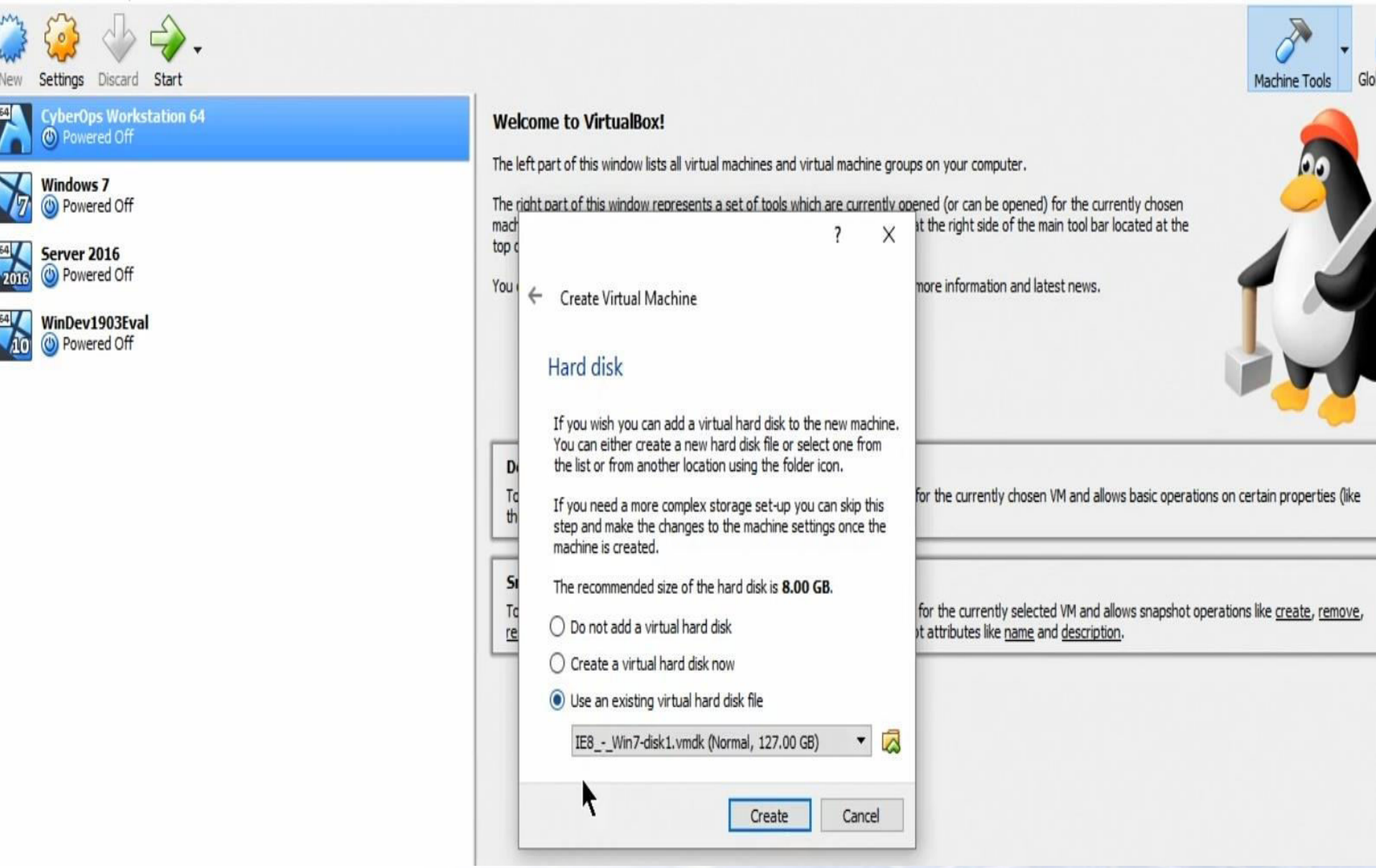
User Manual

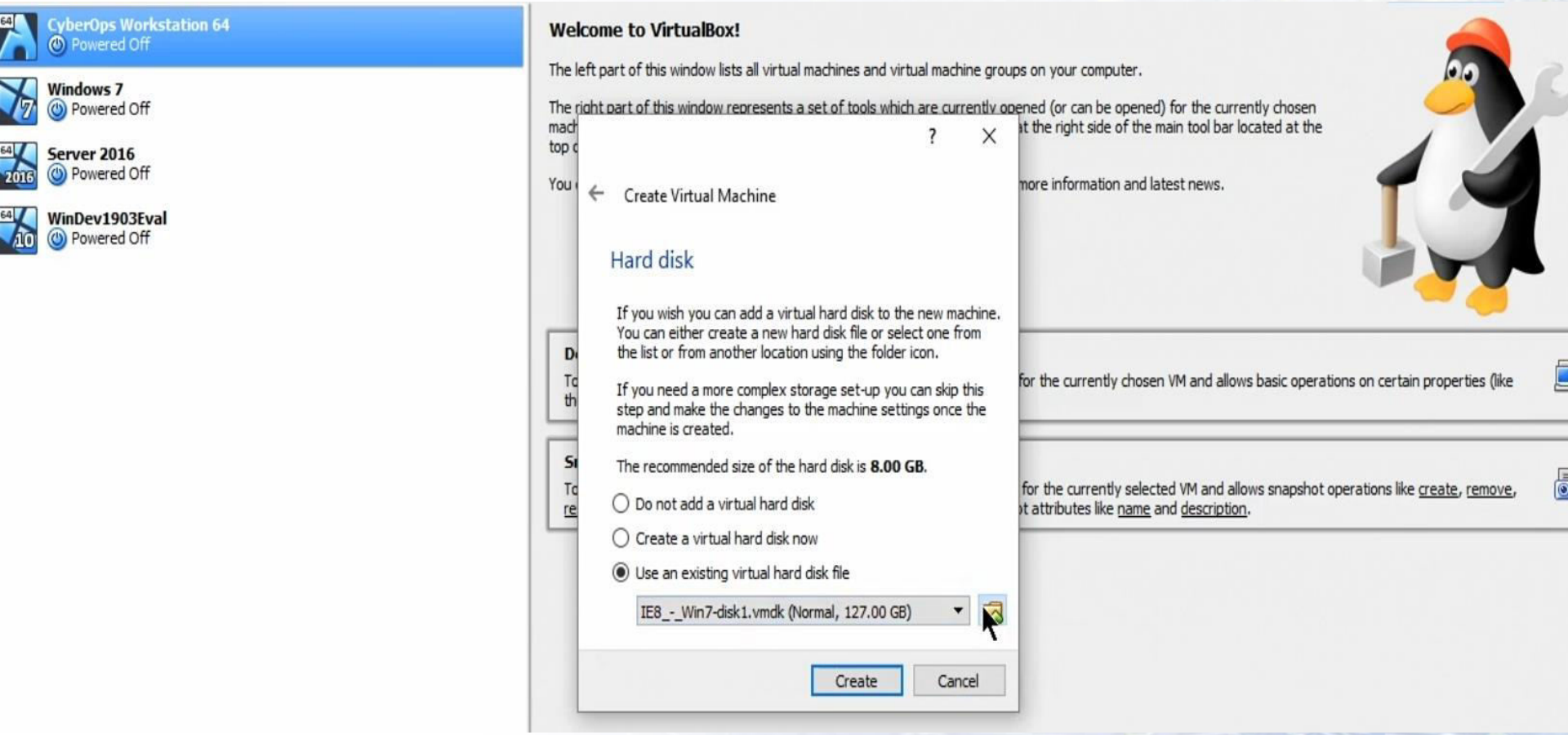
CentOS
archlinux
fedora
openSUSE
& Many More
debian
x86
ORACLE
VirtualBox
copyright: osboxes.org

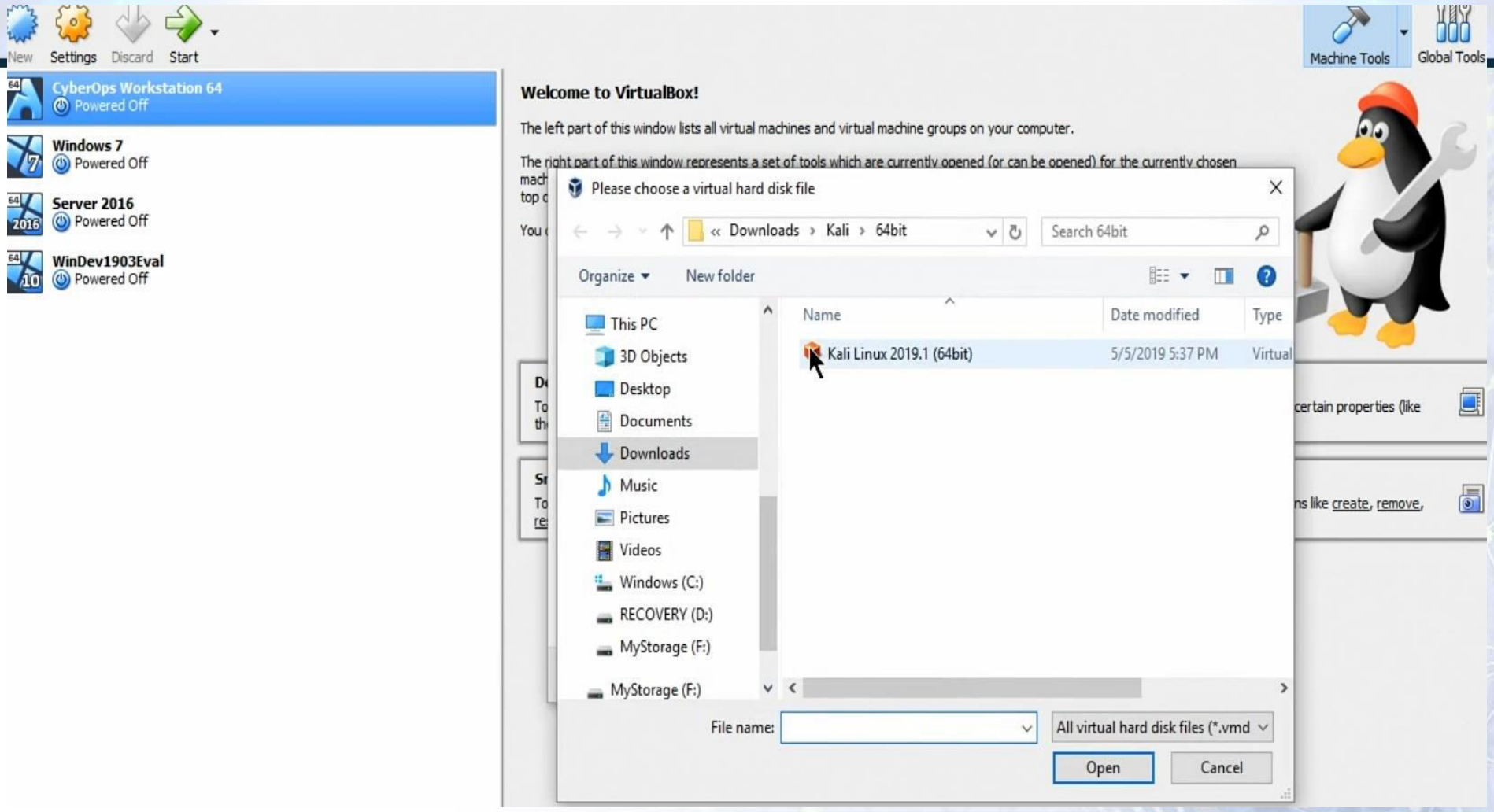

Oracle VM VirtualBox Manager
File
Machine
Help
New
Settings
Discard
Start
Machine Tools
Global Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
The right part of this window represents a set of tools which are currently opened (or can be opened) for the currently chosen machine. For a list of currently available tools check the corresponding menu at the right side of the main tool bar located at the top of the window. This list will be extended with new tools in future releases.
You can press the F1 key to get instant help, or visit www.virtualbox.org for more information and latest news.
Details
Tool to observe virtual machine (VM) details. Reflects groups of properties for the currently chosen VM and allows basic operations on certain properties (like the machine storage devices).
Snapshots
Tool to control virtual machine (VM) snapshots. Reflects snapshots created for the currently selected VM and allows snapshot operations like create, remove, restore (make current) and observe their properties. Allows to edit snapshot attributes like name and description.

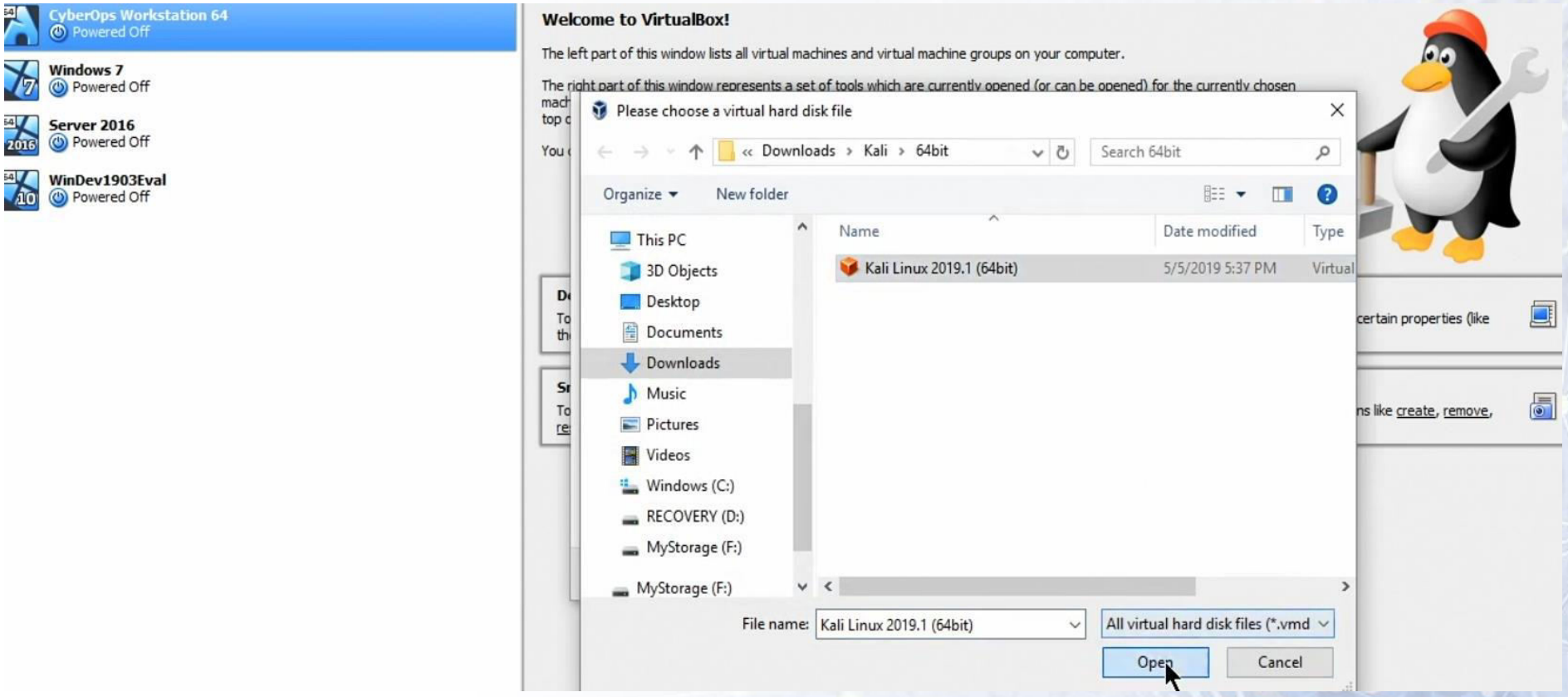

Oracle VM VirtualBox Manager
File Machine Help
New Settings Discard Start
Machine Tools
Global Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
The right part of this window represents a set of tools which are currently opened (or can be opened) for the currently chosen
Create Virtual Machine
Name and operating system
Please choose a descriptive name for the new virtual machine and select the type of operating system you intend to install on it. The name you choose will be used throughout VirtualBox to identify this machine.
Name:
Type: Microsoft Windows
Version: Windows 7 (64-bit)
Expert Mode
Next
Cancel

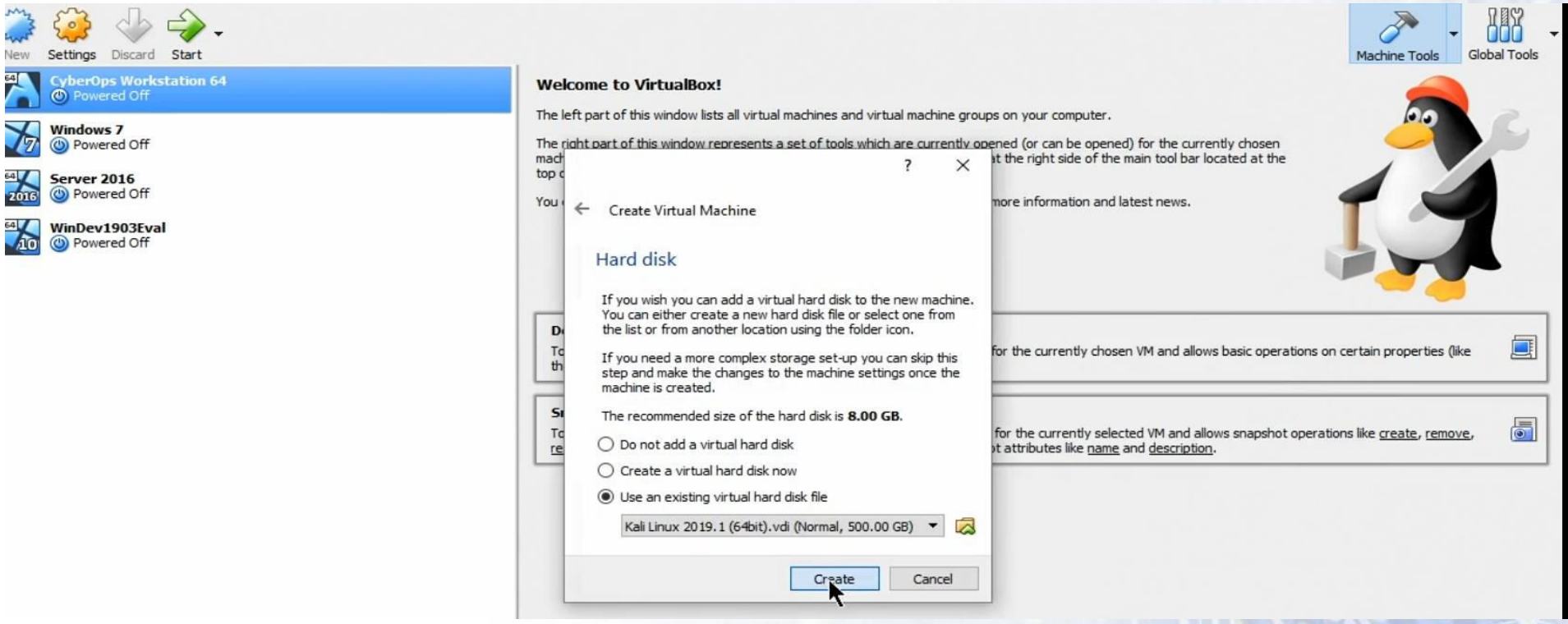

New
Settings
Discard
Start
Machine Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
The right part of this window represents a set of tools which are currently opened (or can be opened) for the currently chosen
Create Virtual Machine
Memory size
Select the amount of memory (RAM) in megabytes to be allocated to the virtual machine.
The recommended memory size is 1024 MB.
1024
MB
4 MB
16384 MB
Next
Cancel

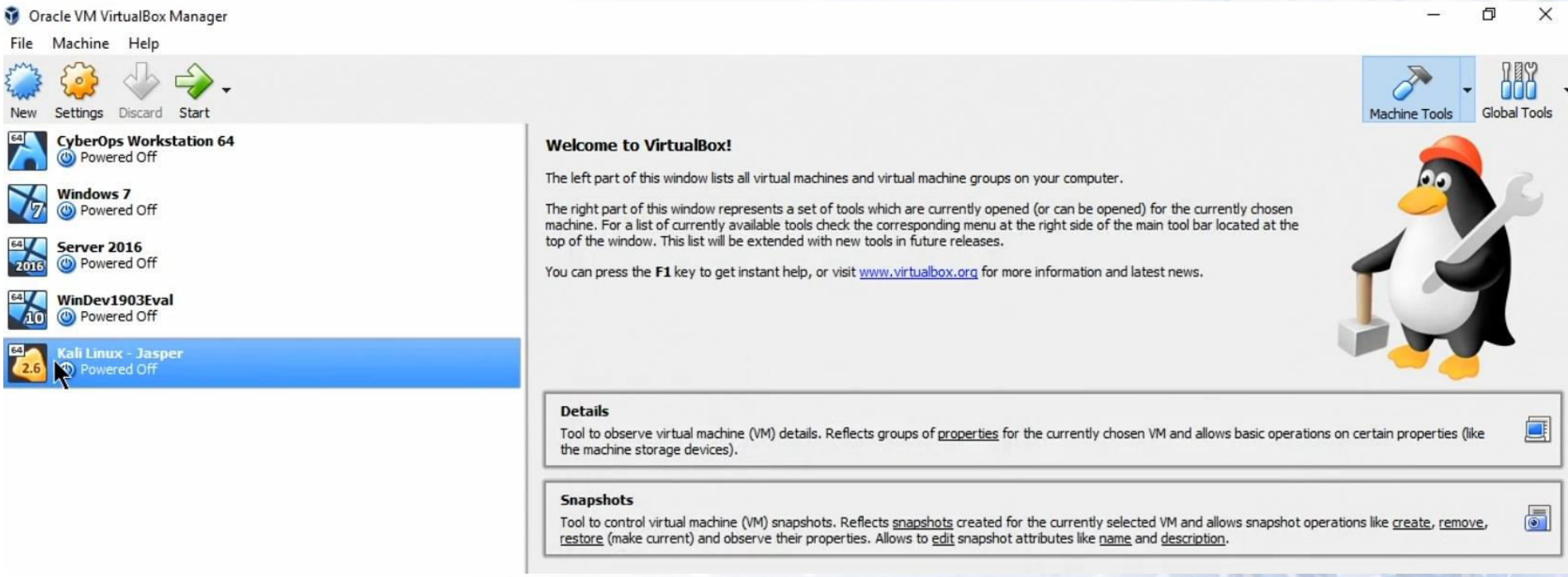

New
Settings
Discard
Start
Machine Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
Create Virtual Machine
Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select one from the list or from another location using the folder icon.
If you need a more complex storage set-up you can skip this step and make the changes to the machine settings once the machine is created.
The recommended size of the hard disk is 8.00 GB.
Do not add a virtual hard disk
Create a virtual hard disk now
Use an existing virtual hard disk file
IE8_-_Win7-disk1.vmdk (Normal, 127.00 GB)
Create
Cancel

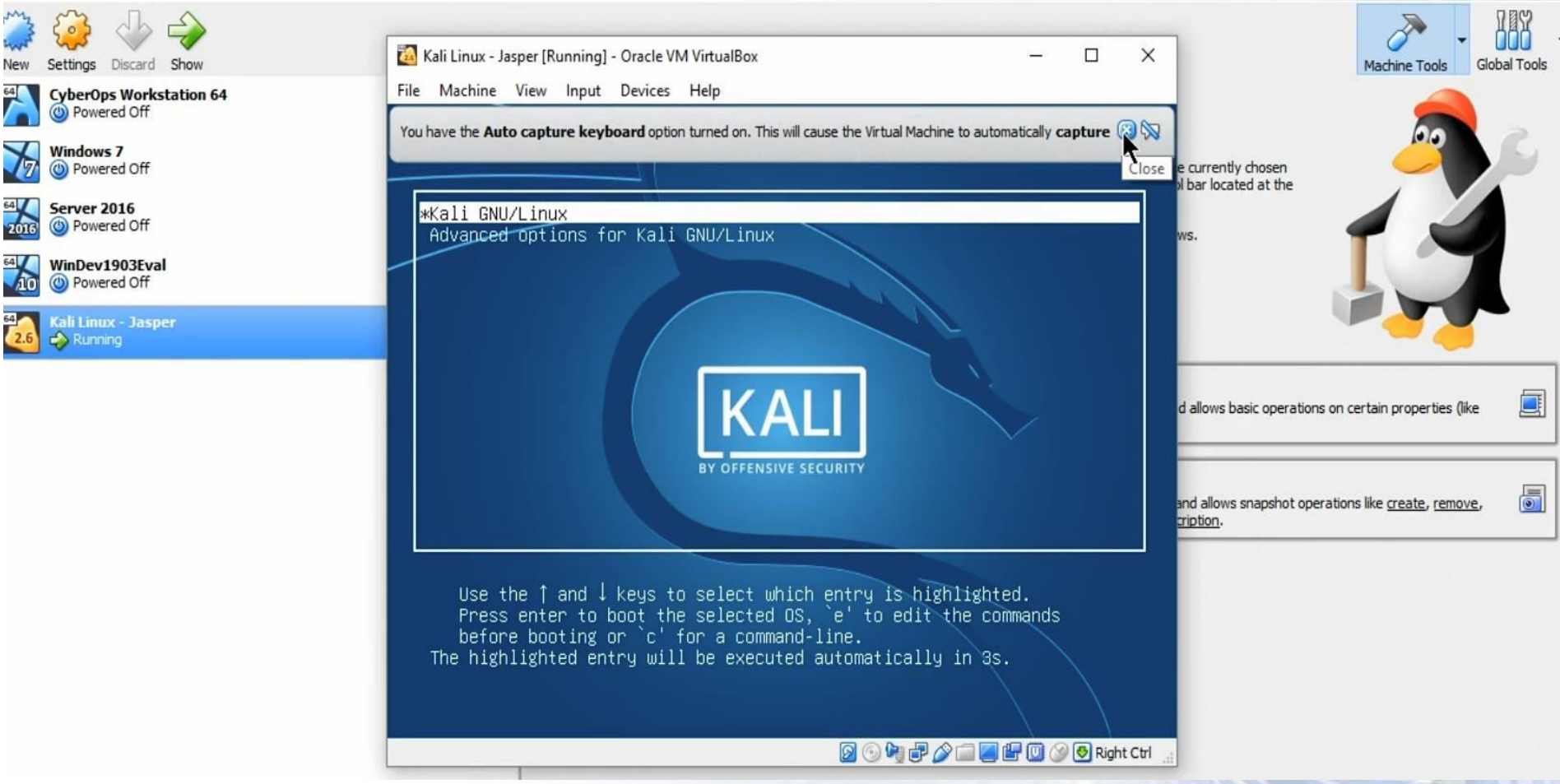

CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
Create Virtual Machine
Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select one from the list or from another location using the folder icon.
If you need a more complex storage set-up you can skip this step and make the changes to the machine settings once the machine is created.
The recommended size of the hard disk is 8.00 GB.
Do not add a virtual hard disk
Create a virtual hard disk now
Use an existing virtual hard disk file
IE8_-_Win7-disk1.vmdk (Normal, 127.00 GB)
Create
Cancel

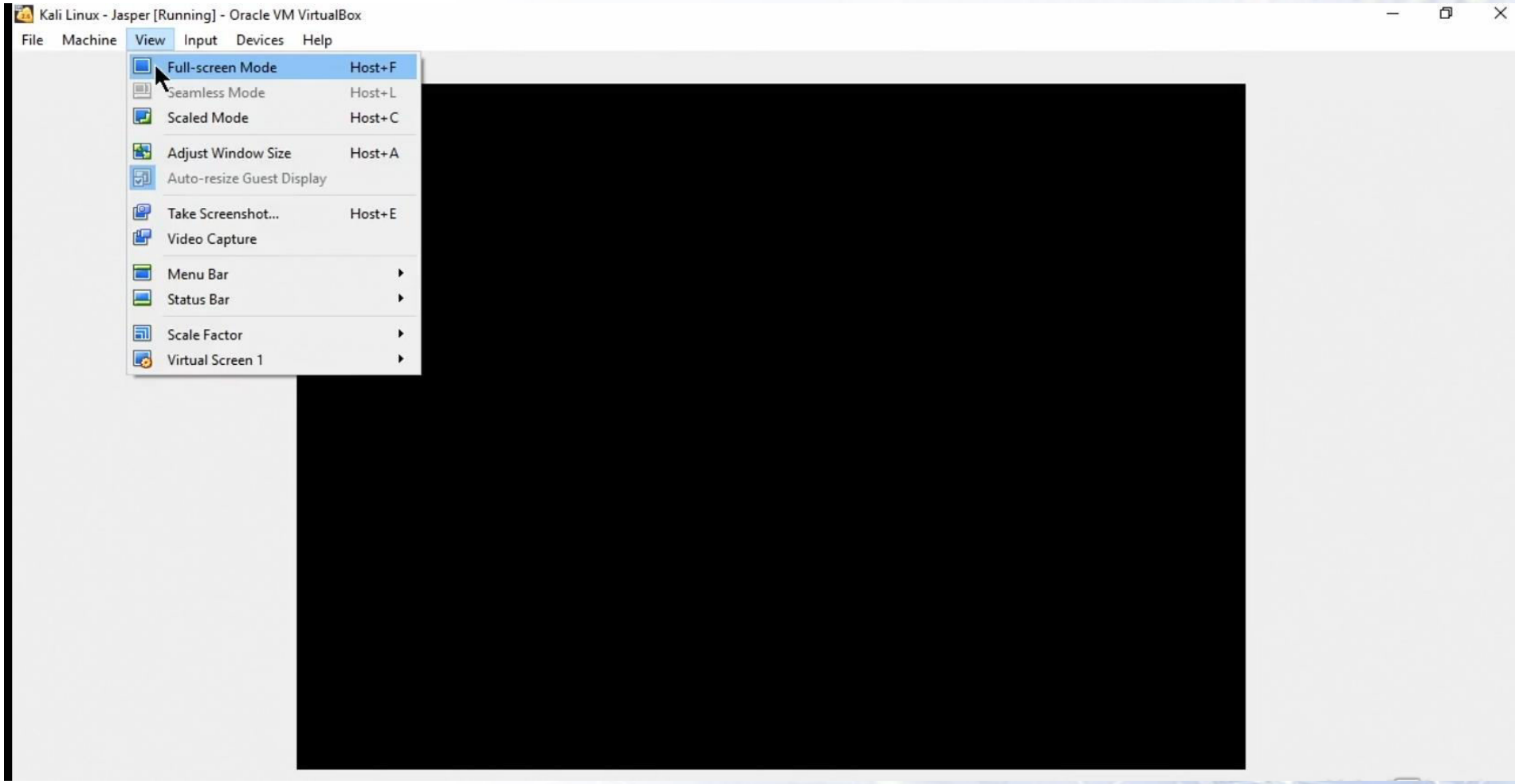

New
Settings
Discard
Start
Machine Tools
Global Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
Please choose a virtual hard disk file
Downloads › Kali › 64bit
Search 64bit
Organize
New folder
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
RECOVERY (D:)
MyStorage (F:)
MyStorage (F:)
Name
Date modified
Type
Kali Linux 2019.1 (64bit)
5/5/2019 5:37 PM
Virtual
File name:
All virtual hard disk files (*.vmd
Open
Cancel
certain properties (like
ns like create, remove,

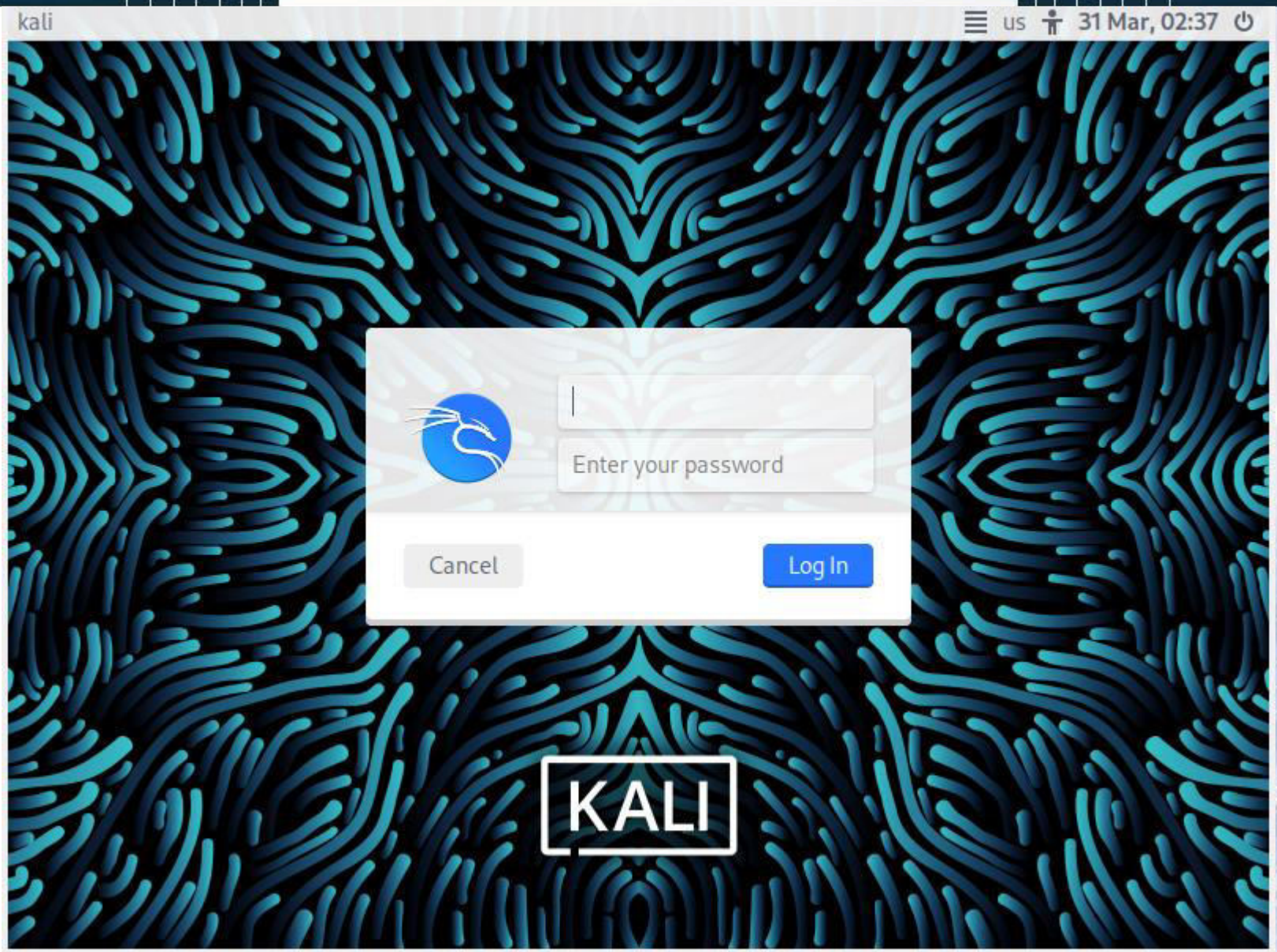

CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
The right part of this window represents a set of tools which are currently opened (or can be opened) for the currently chosen
Please choose a virtual hard disk file
« Downloads › Kali › 64bit
Search 64bit
Organize
New folder
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
RECOVERY (D:)
MyStorage (F:)
MyStorage (F:)
Name
Date modified
Type
Kali Linux 2019.1 (64bit)
5/5/2019 5:37 PM
Virtual
certain properties (like
ns like create, remove,
File name:
Kali Linux 2019.1 (64bit)
All virtual hard disk files (*.vmd
Open
Cancel

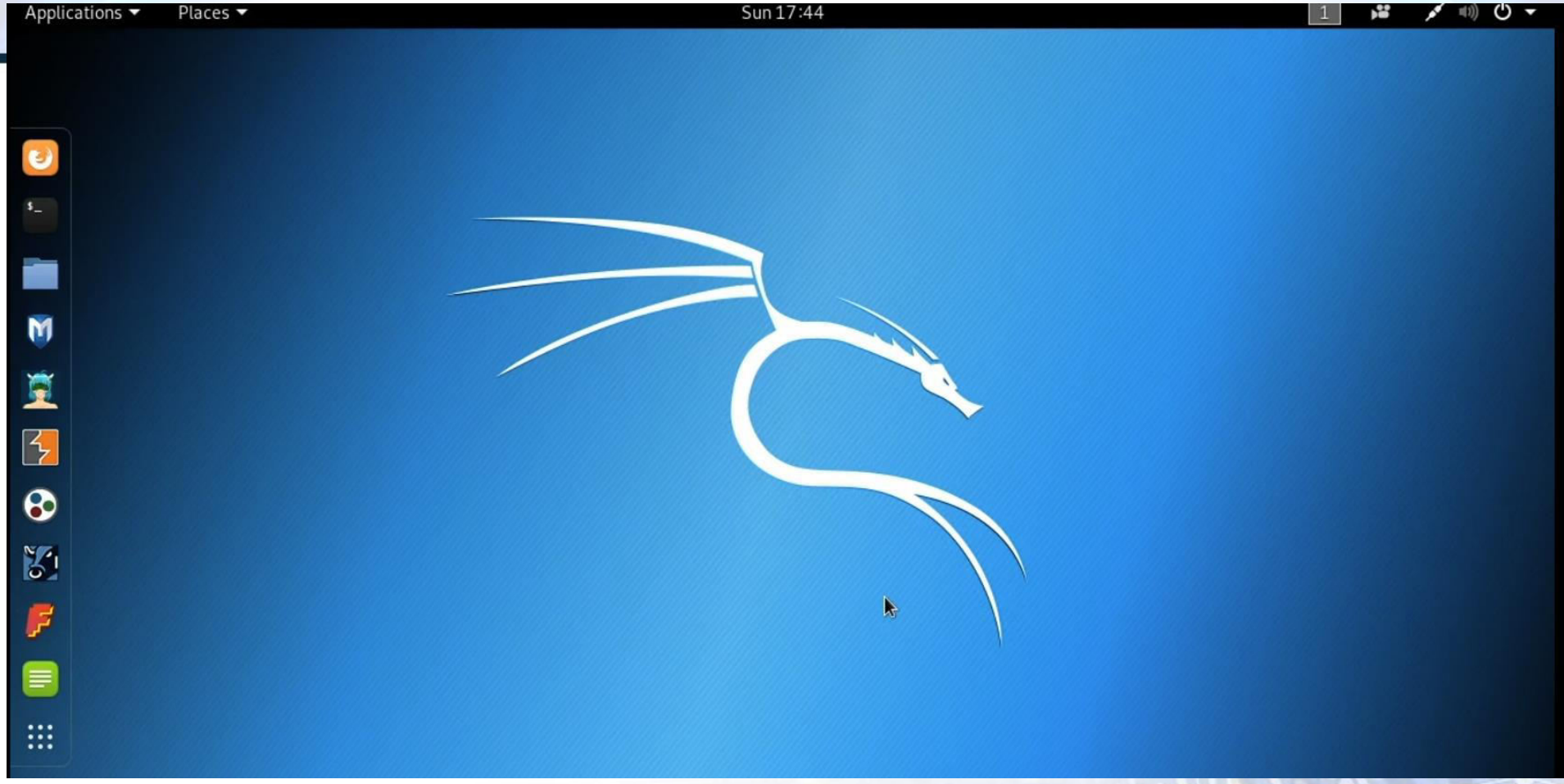

New
Settings
Discard
Start
Machine Tools
Global Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
Create Virtual Machine
Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select one from the list or from another location using the folder icon.
If you need a more complex storage set-up you can skip this step and make the changes to the machine settings once the machine is created.
The recommended size of the hard disk is 8.00 GB.
Do not add a virtual hard disk
Create a virtual hard disk now
Use an existing virtual hard disk file
Kali Linux 2019.1 (64bit).vdi (Normal, 500.00 GB)
Create
Cancel

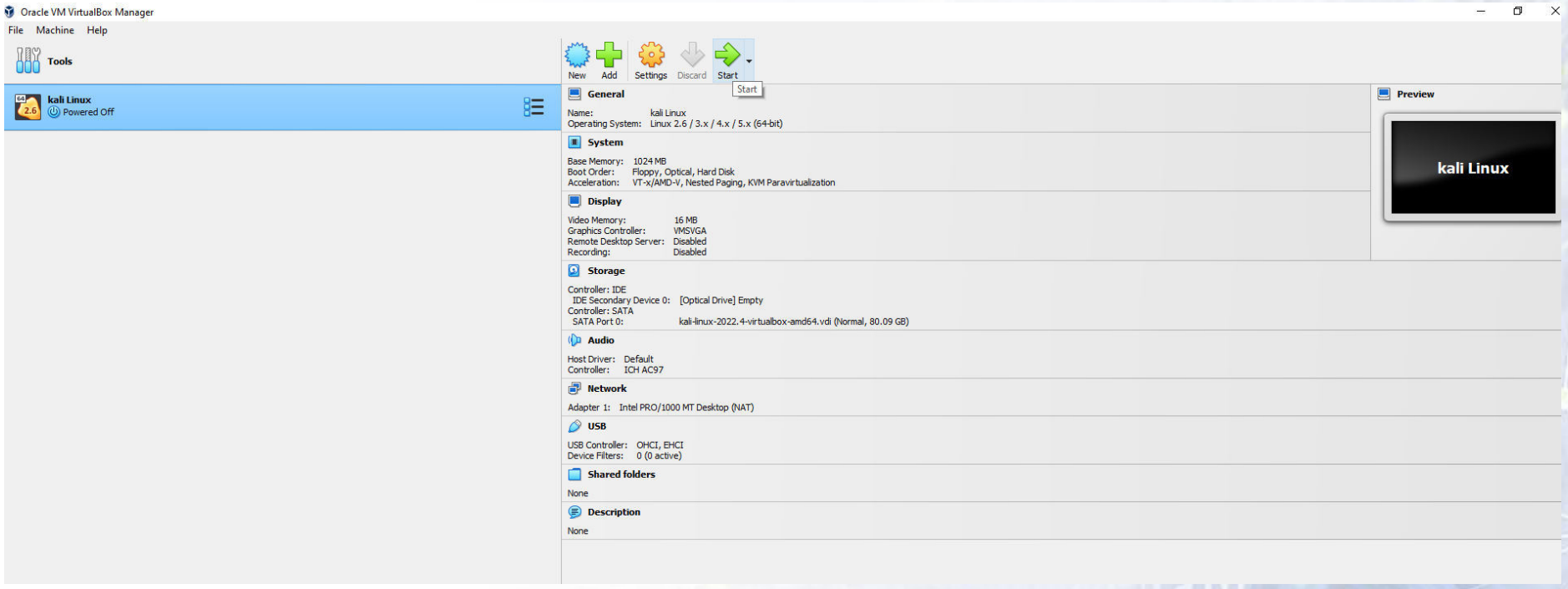

Oracle VM VirtualBox Manager
File
Machine
Help
New
Settings
Discard
Start
Machine Tools
Global Tools
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Kali Linux - Jasper
Powered Off
Welcome to VirtualBox!
The left part of this window lists all virtual machines and virtual machine groups on your computer.
The right part of this window represents a set of tools which are currently opened (or can be opened) for the currently chosen machine. For a list of currently available tools check the corresponding menu at the right side of the main tool bar located at the top of the window. This list will be extended with new tools in future releases.
You can press the F1 key to get instant help, or visit www.virtualbox.org for more information and latest news.
Details
Tool to observe virtual machine (VM) details. Reflects groups of properties for the currently chosen VM and allows basic operations on certain properties (like the machine storage devices).
Snapshots
Tool to control virtual machine (VM) snapshots. Reflects snapshots created for the currently selected VM and allows snapshot operations like create, remove, restore (make current) and observe their properties. Allows to edit snapshot attributes like name and description.


New
Settings
Discard
Show
CyberOps Workstation 64
Powered Off
Windows 7
Powered Off
Server 2016
Powered Off
WinDev1903Eval
Powered Off
Kali Linux - Jasper
Running
Machine Tools
Global Tools
Kali Linux - Jasper [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
You have the Auto capture keyboard option turned on. This will cause the Virtual Machine to automatically capture
Close
*Kali GNU/Linux
Advanced options for Kali GNU/Linux
KALI
BY OFFENSIVE SECURITY
Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, `e' to edit the commands
before booting or `c' for a command-line.
The highlighted entry will be executed automatically in 3s.
Right Ctrl
currently chosen
bar located at the
ws.
allows basic operations on certain properties (like
and allows snapshot operations like create, remove,
cription.


Kali Linux - Jasper [Running] - Oracle VM VirtualBox
File
Machine
View
Input
Devices
Help
Full-screen Mode Host+F
Seamless Mode Host+L
Scaled Mode Host+C
Adjust Window Size Host+A
Auto-resize Guest Display
Take Screenshot... Host+E
Video Capture
Menu Bar
Status Bar
Scale Factor
Virtual Screen 1


kali
us
31 Mar, 02:37
Enter your password
Cancel
Log In
KALI


Applications
Places
Sun 17:44
1


Oracle VM VirtualBox Manager
File
Machine
Help
Tools
kali Linux
Powered Off
New
Add
Settings
Discard
Start
Start
General
Name: kali Linux
Operating System: Linux 2.6 / 3.x / 4.x / 5.x (64-bit)
System
Base Memory: 1024 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: VT-x/AMD-V, Nested Paging, KVM Paravirtualization
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: kali-linux-2022.4-virtualbox-amd64.vdi (Normal, 80.09 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None
Preview
kali Linux

Week 7 Topic 2

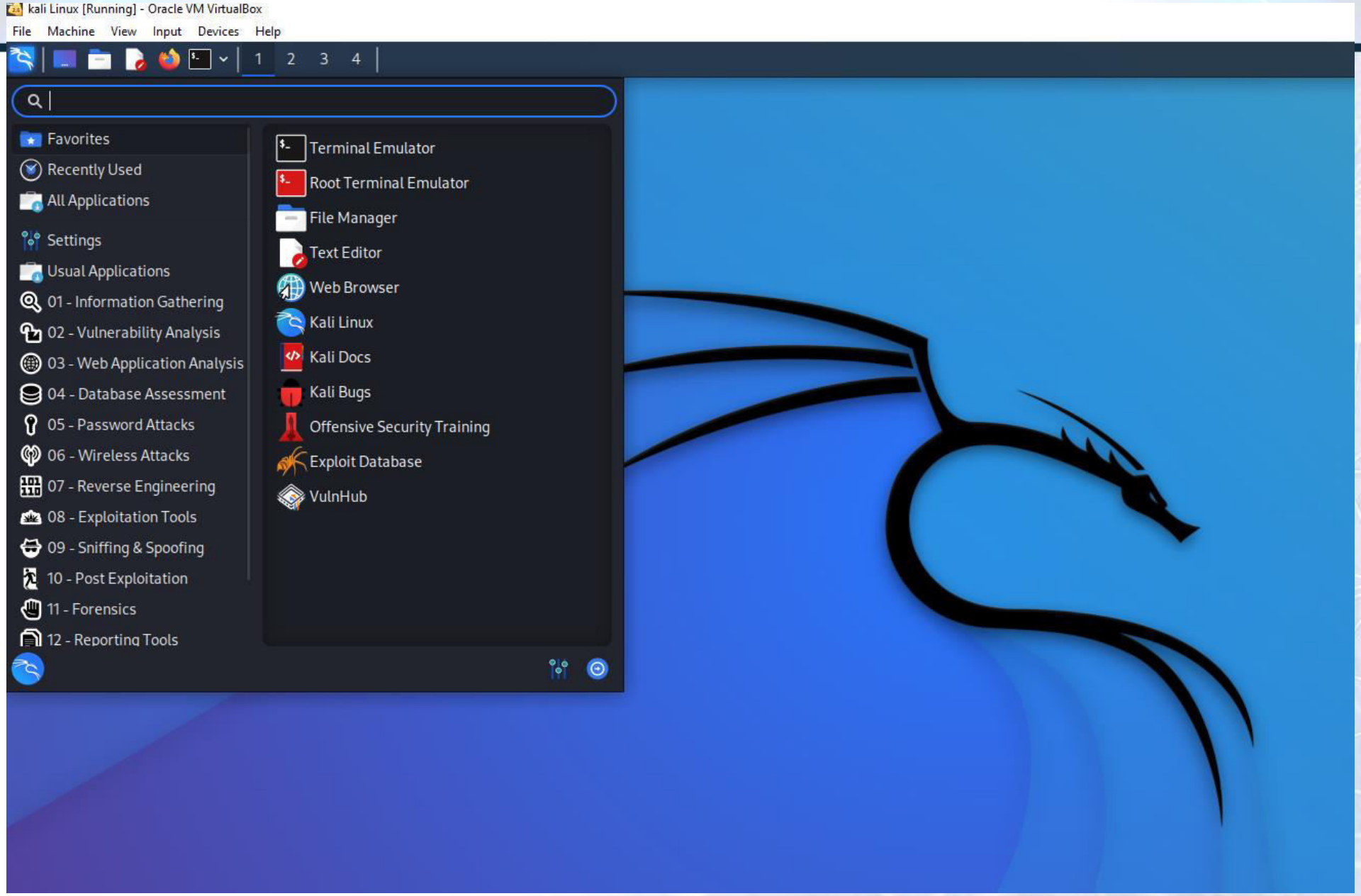
- **Nikto**

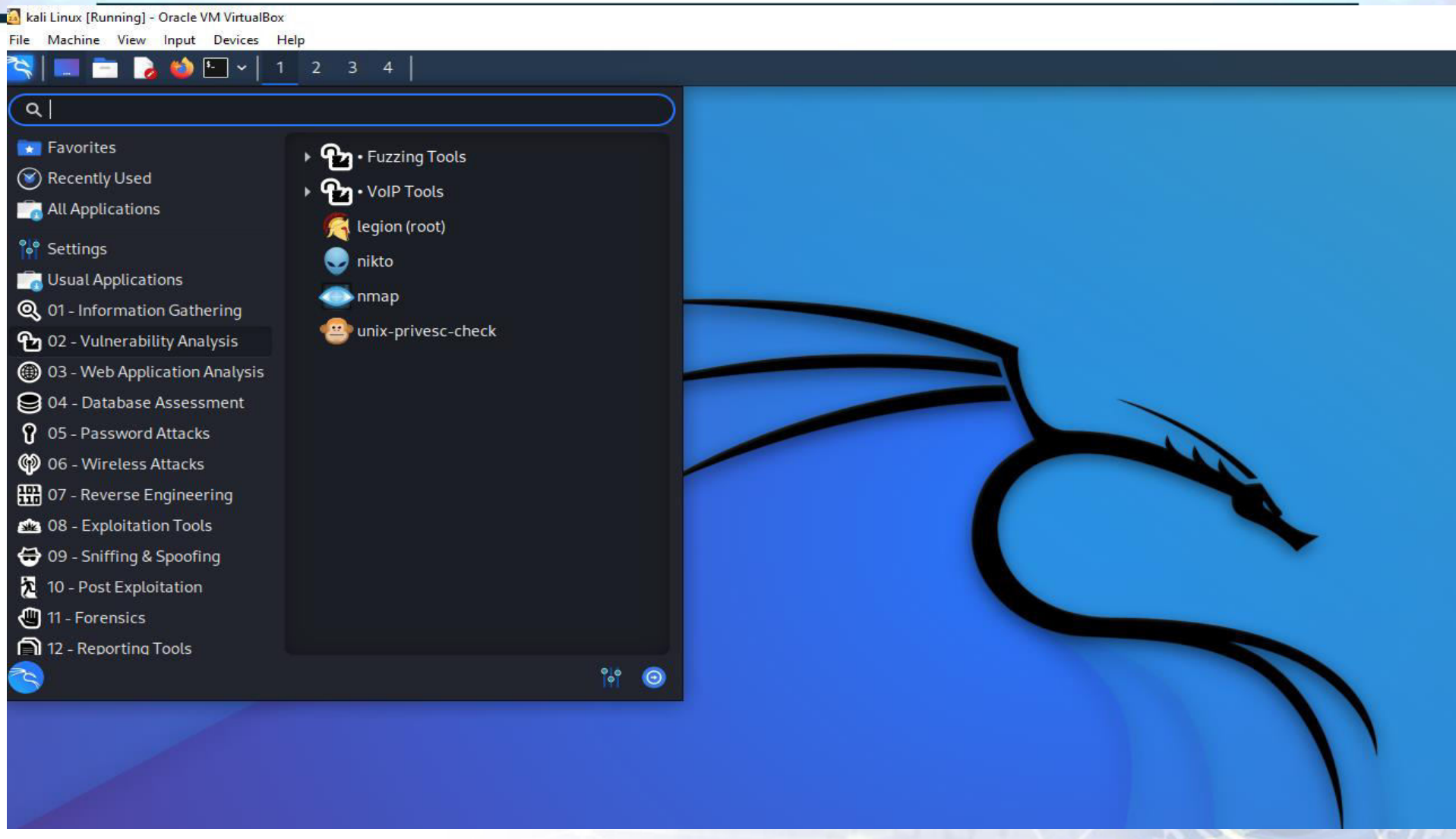
- Web server scanning is a black box test that identifies security vulnerabilities.

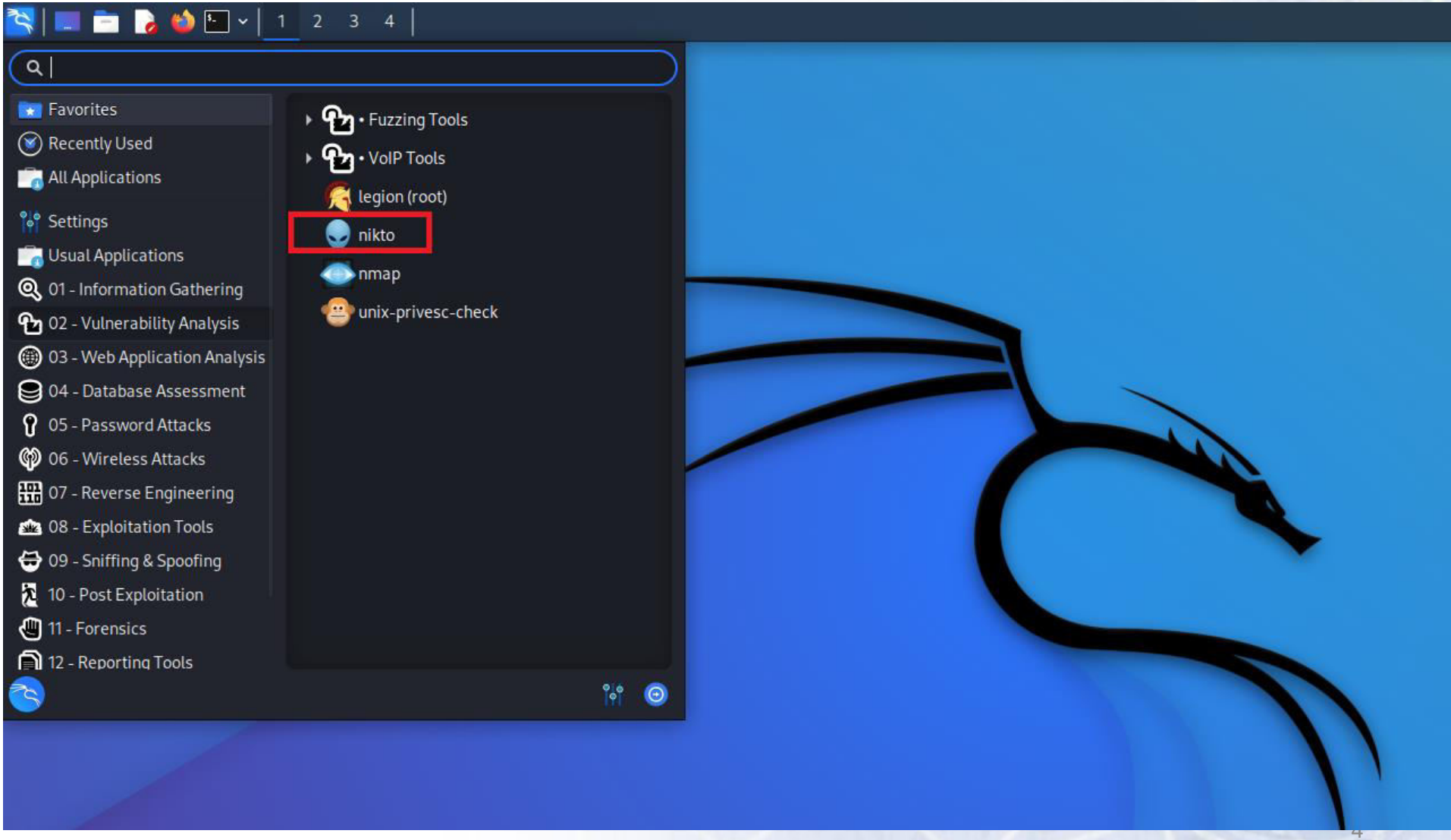
Built into Kali Linux and is an open-source web server scanner

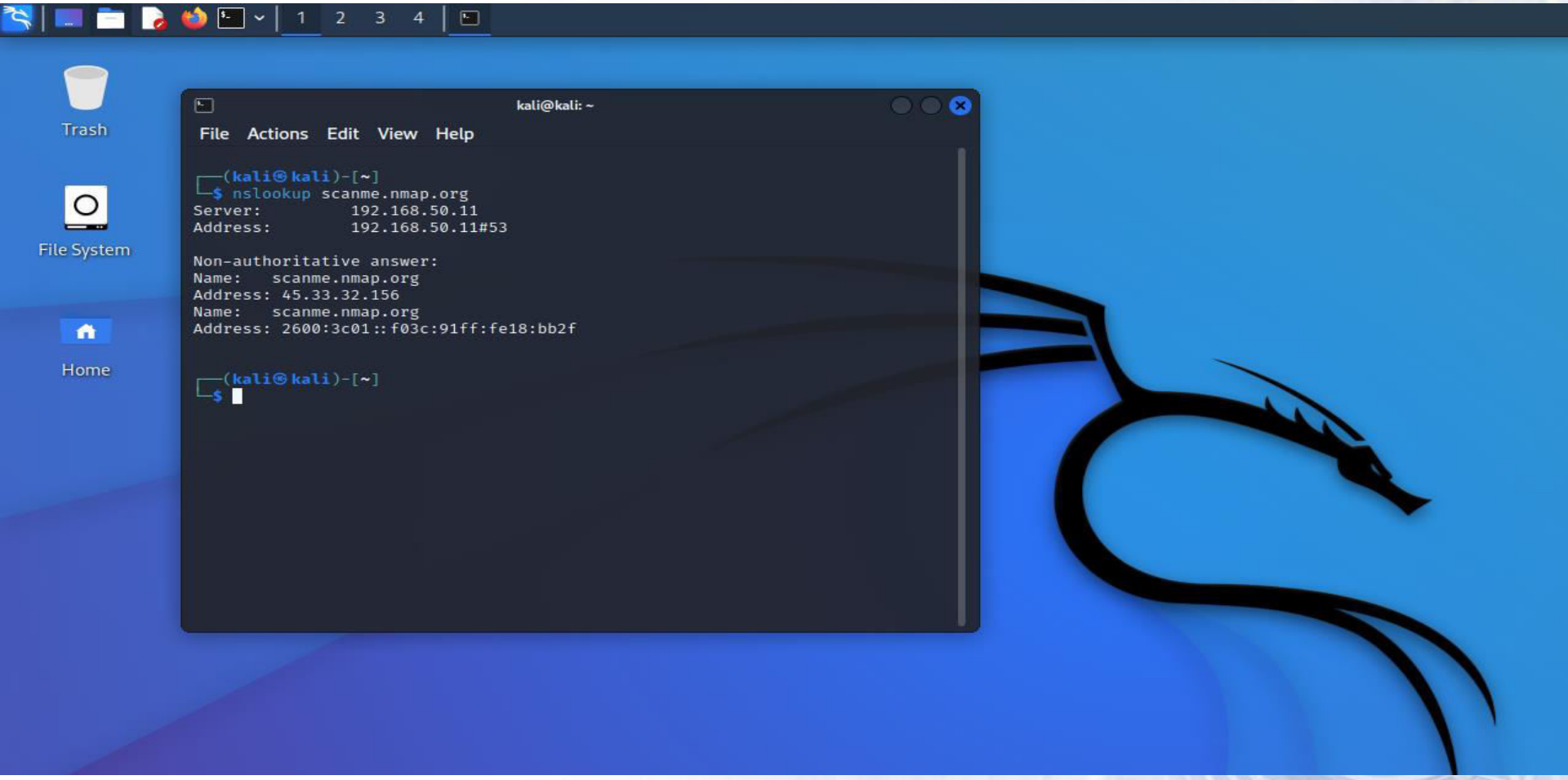
Performs complete testing on web servers for vulnerabilities

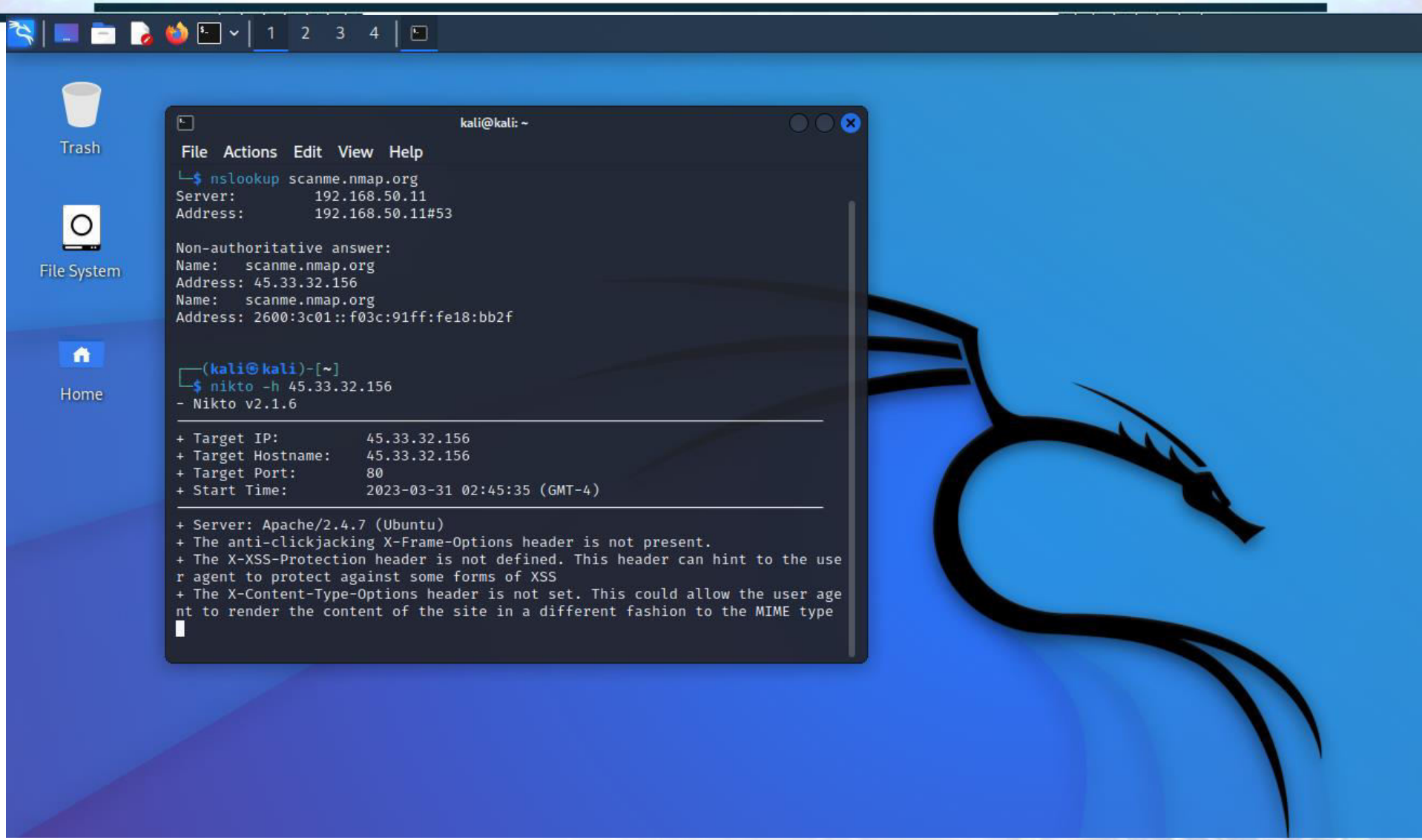
Scans for outdated versions of servers, server-specific problems, missing headers and dangerous files and CGIs


kali Linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
Favorites
Recently Used
All Applications
Settings
Usual Applications
01 - Information Gathering
02 - Vulnerability Analysis
03 - Web Application Analysis
04 - Database Assessment
05 - Password Attacks
06 - Wireless Attacks
07 - Reverse Engineering
08 - Exploitation Tools
09 - Sniffing & Spoofing
10 - Post Exploitation
11 - Forensics
12 - Reporting Tools
Terminal Emulator
Root Terminal Emulator
File Manager
Text Editor
Web Browser
Kali Linux
Kali Docs
Kali Bugs
Offensive Security Training
Exploit Database
VulnHub


kali Linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
Favorites
Recently Used
All Applications
Settings
Usual Applications
01 - Information Gathering
02 - Vulnerability Analysis
03 - Web Application Analysis
04 - Database Assessment
05 - Password Attacks
06 - Wireless Attacks
07 - Reverse Engineering
08 - Exploitation Tools
09 - Sniffing & Spoofing
10 - Post Exploitation
11 - Forensics
12 - Reporting Tools
• Fuzzing Tools
• VoIP Tools
legion (root)
nikto
nmap
unix-privesc-check


1
2
3
4
Favorites
Recently Used
All Applications
Settings
Usual Applications
01 - Information Gathering
02 - Vulnerability Analysis
03 - Web Application Analysis
04 - Database Assessment
05 - Password Attacks
06 - Wireless Attacks
07 - Reverse Engineering
08 - Exploitation Tools
09 - Sniffing & Spoofing
10 - Post Exploitation
11 - Forensics
12 - Reporting Tools
• Fuzzing Tools
• VoIP Tools
legion (root)
nikto
nmap
unix-privesc-check


1 2 3 4
Trash
File System
Home
kali@kali: ~
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nslookup scanme.nmap.org
Server: 192.168.50.11
Address: 192.168.50.11#53
Non-authoritative answer:
Name: scanme.nmap.org
Address: 45.33.32.156
Name: scanme.nmap.org
Address: 2600:3c01::f03c:91ff:fe18:bb2f
┌──(kali㉿kali)-[~]
└─$


1 2 3 4
Trash
File System
Home
kali@kali: ~
File Actions Edit View Help
└─$ nslookup scanme.nmap.org
Server: 192.168.50.11
Address: 192.168.50.11#53
Non-authoritative answer:
Name: scanme.nmap.org
Address: 45.33.32.156
Name: scanme.nmap.org
Address: 2600:3c01::f03c:91ff:fe18:bb2f
┌──(kali㉿kali)-[~]
└─$ nikto -h 45.33.32.156
- Nikto v2.1.6
+ Target IP: 45.33.32.156
+ Target Hostname: 45.33.32.156
+ Target Port: 80
+ Start Time: 2023-03-31 02:45:35 (GMT-4)
+ Server: Apache/2.4.7 (Ubuntu)
+ The anti-clickjacking X-Frame-Options header is not present.
+ The X-XSS-Protection header is not defined. This header can hint to the use
r agent to protect against some forms of XSS
+ The X-Content-Type-Options header is not set. This could allow the user age
nt to render the content of the site in a different fashion to the MIME type

Week 7 Topic 3

Nmap(Network Mapper)

Nmap, short for Network Mapper, is a free and open source utility network discovery and security auditing tool.

It is the world's leading port scanner,known for its simple and easy to remember flags that provide powerful scanning options.

- Guess the operating system running on a target machine
- Get accurate packet routes till the target machine
- Discovering connected hosts on a network
- Analyzing free ports on a target host
- Detecting running services on a system along with version number
- Uncovering network vulnerabilities and system loopholes

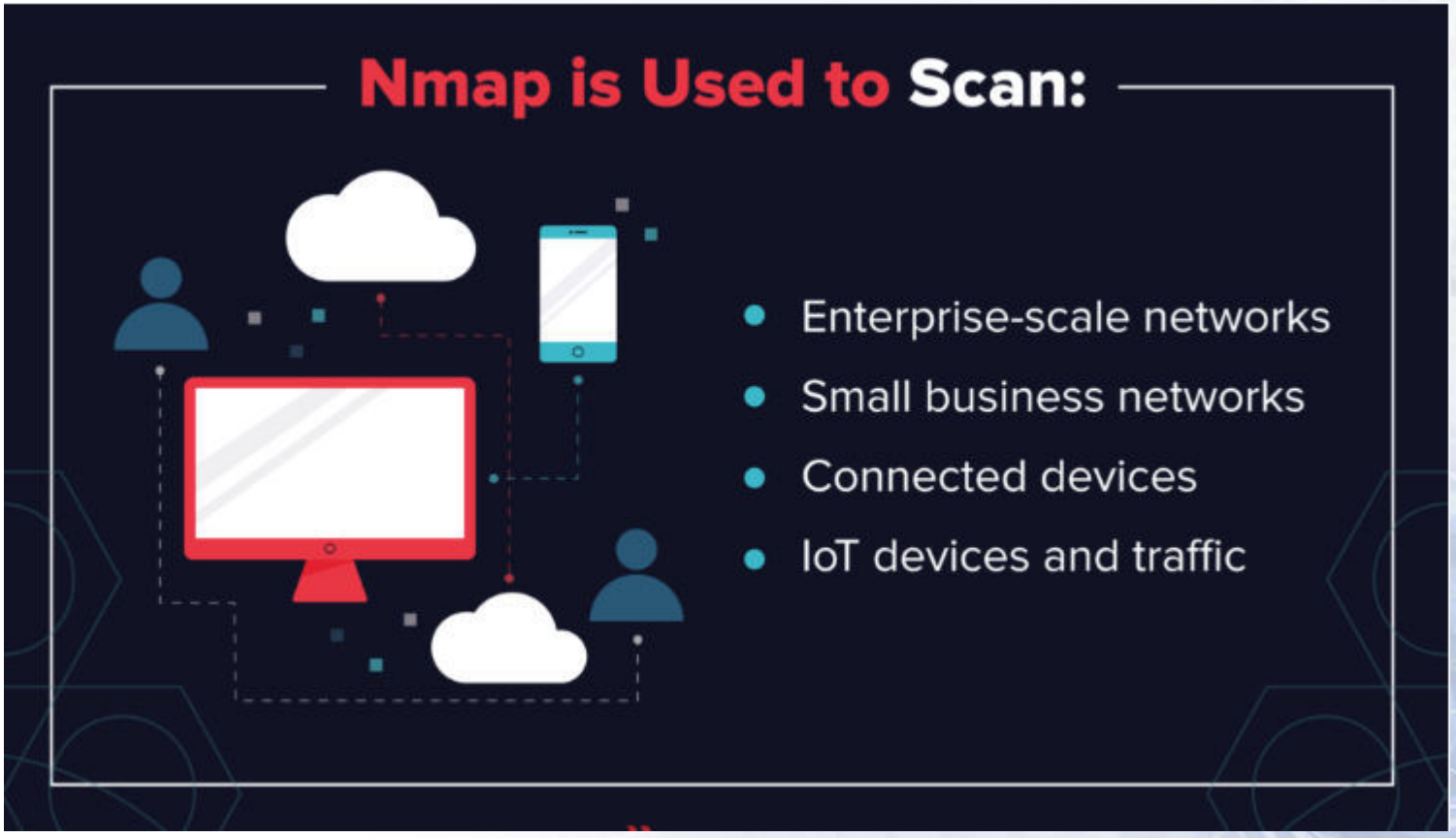
Nmap

Network administrators often use

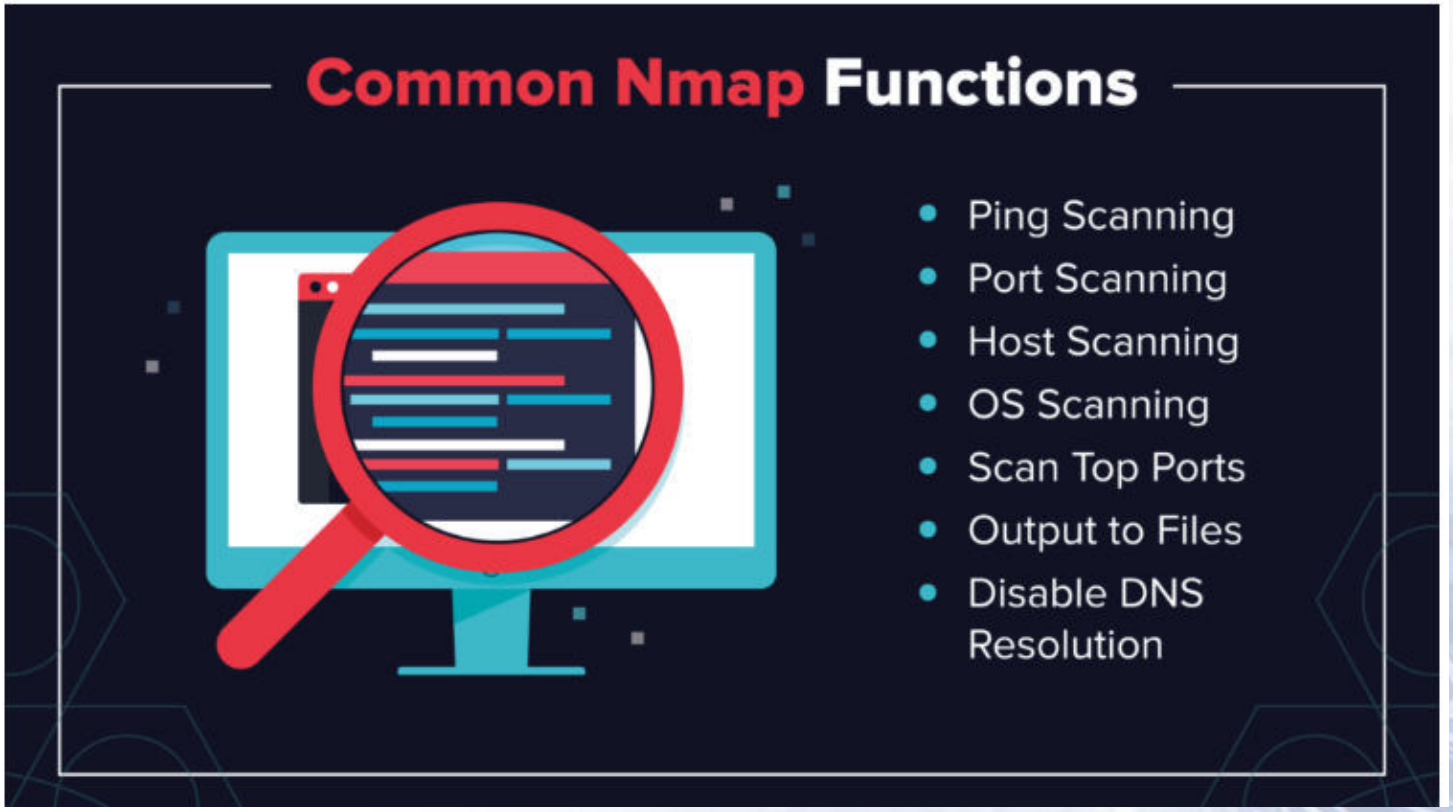
- to detect the devices that are presently working on the system
- the port number to which the devices are attached
- the free ports that may be used.
- what services (application name and version) those hosts are offering
- what operating systems (and OS versions) they are running

Nmap

- helpful while performing network penetration testing.
- not only gives network information but also assists in discovering security issues in the system.
- Nmap is platform-agnostic and may be run on various standard operating systems, including Linux, Windows, macOS.
- It is straightforward to use and includes both a command-line interface(CLI) and a graphical user interface(GUI).

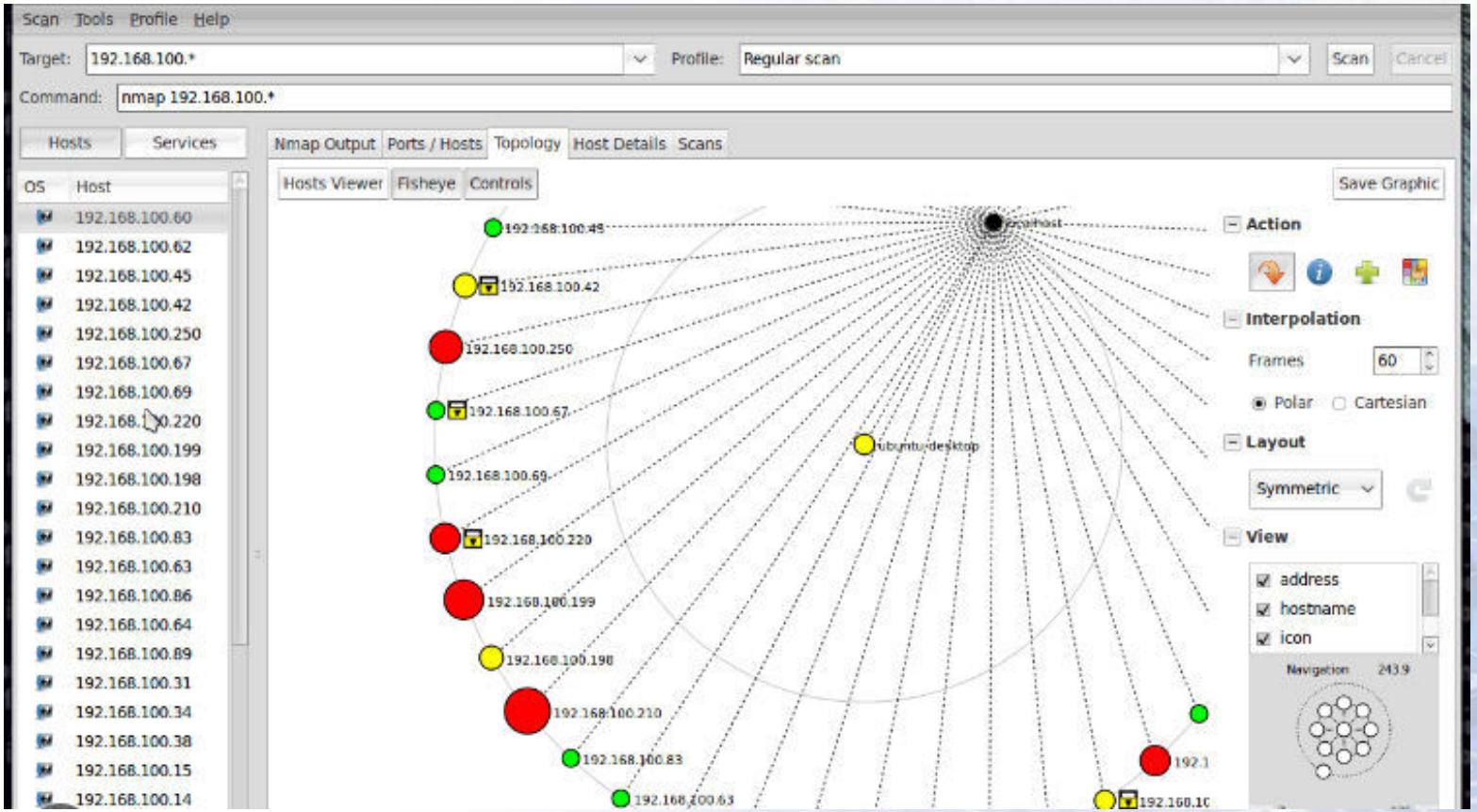

Nmap is Used to Scan:
Enterprise-scale networks
Small business networks
Connected devices
IoT devices and traffic

Nmap Commands



Zenmap GUI

Nmap has a graphical user interface called Zenmap. It helps you develop visual mappings of a network for better usability and reporting.



Nmap Commands

- various commands you can use in Nmap along with their flag and usage description with an example on how to use it.

Basic Nmap Scan against IP or host

nmap 1.1.1.1

- Now, if you want to scan a hostname, simply replace the IP for the host, as you see below:

nmap scanme.com

Nmap Commands with flags

Flag	Use	Example
-Pn	only port scan	nmap -Pn192.168.1.1
-sn	only host discover	nmap -sn192.168.1.1
-PR	arp discovery on a local network	nmap -PR192.168.1.1
-n	disable DNS resolution	nmap -n 192.168.1.1

Flag	Use	Example
-p	specify a port or port range	nmap -p 1-30 192.168.1.1
-p-	scan all ports	nmap -p- 192.168.1.1
-F	fast port scan	nmap -F 192.168.1.1

Flag	Use	Example
-sV	detect the version of services running	nmap -sV 192.168.1.1
-A	aggressive scan	nmap -A 192.168.1.1
-O	detect operating system of the target	nmap -O 192.168.1.1

Week 7 Topic 4

Nmap **Benefits** and **Installation**

- Time tested security and networking tool
- Open source package with large community
- Available for most operating systems
- Free!

- **Download Link**

  https://nmap.org/download

Npcap.com Seclists.org Sectools.org Insecure.org

Site Search

Download | Reference Guide | Book | Docs | Zenmap GUI | In the Movies

Downloading Nmap

Get the latest Nmap for your system:

- Windows
- macOS
- Linux (RPM)
- Any other OS (source code)

Older versions (and sometimes newer test releases) are available from the Nmap release archive (and really old ones are in dist-old). For the more security-paranoid (smart) users, GPG detached signatures and SHA-1 hashes for each release are available in the sigs directory (verification instructions). Before downloading, be sure to read the relevant sections for your platform from the Nmap Install Guide. The most important changes (features, bugfixes, etc) in each Nmap version are described in the Changelog. Using Nmap is covered in the Reference Guide, and don't forget to read the other available documentation, particularly the official book Nmap Network Scanning!

Nmap users are encouraged to subscribe to the *Nmap-hackers* mailing list. It is a low volume (7 posts in 2015), moderated list for the most important announcements about Nmap, Insecure.org, and related projects. You can join the 128,953 current subscribers (as of September 2017) by submitting your email address here:

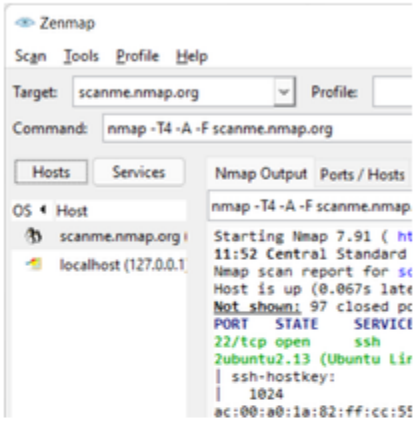
Subscribe to Nmap-hackers

(or subscribe with custom options from the Nmap-hackers list info page)

...so get updates by liking Nmap on Facebook or following us @nmap on Twitter.

html#windows

Microsoft Windows binaries



Please read the Windows section of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap's dependencies and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 R2 and newer. We also maintain a guide for users who must run Nmap on earlier Windows releases..

Note: The version of Npcap included in our installers may not always be the latest version. If you experience problems or just want the latest and greatest version, download and install the latest Npcap release.

Latest stable release self-installer: nmap-7.93-setup.exe
Latest Npcap release self-installer: npcap-1.73.exe

We have written post-install usage instructions. Please notify us if you encounter any problems or have suggestions for the installer.

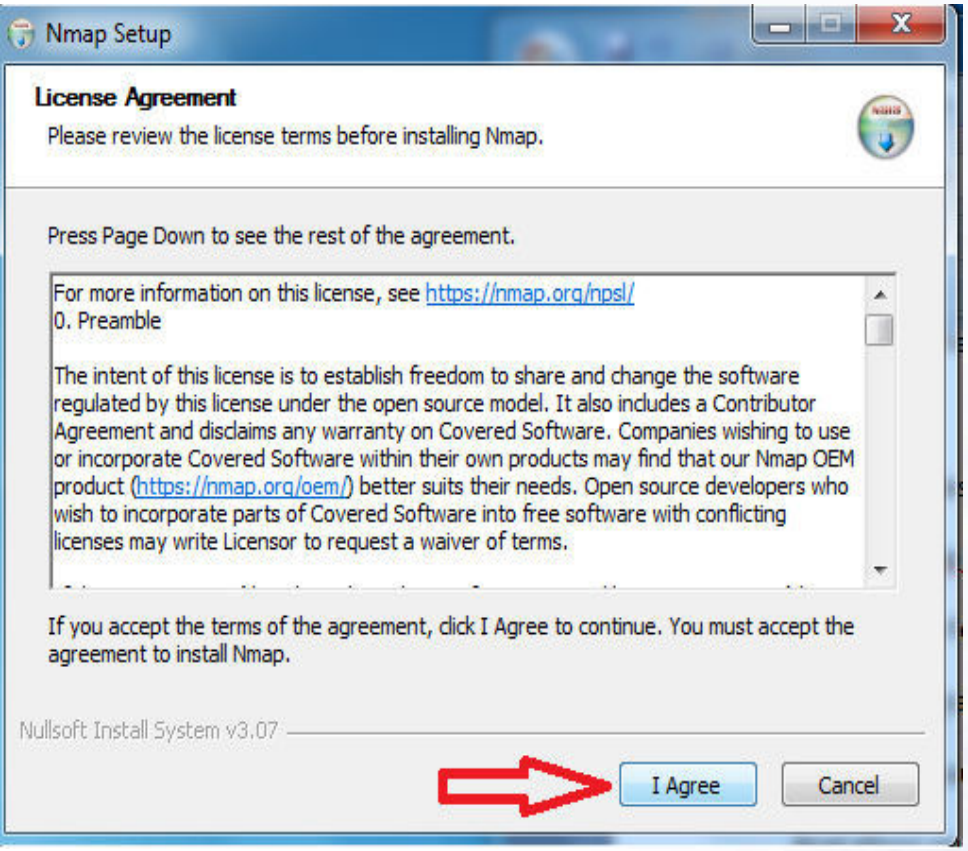
Linux RPM Source and Binaries

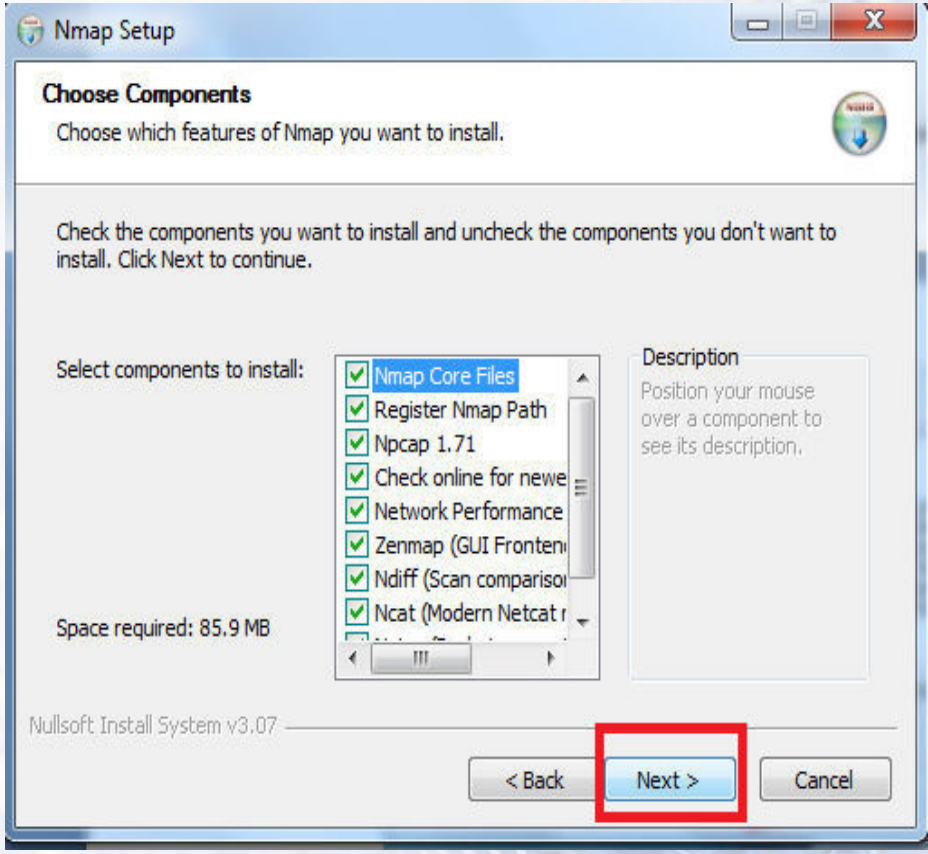
Many popular Linux distributions (Redhat, Mandrake, Suse, etc) use the RPM package management system for quick and easy binary package installation. We have written a detailed guide to installing our RPM packages, though these simple commands usually do the trick:

```
rpm -vhU https://nmap.org/dist/nmap-7.93-1.x86_64.rpm
rpm -vhU https://nmap.org/dist/zenmap-7.93-1.noarch.rpm
rpm -vhU https://nmap.org/dist/ncat-7.93-1.x86_64.rpm
rpm -vhU https://nmap.org/dist/nping-0.7.93-1.x86_64.rpm
```

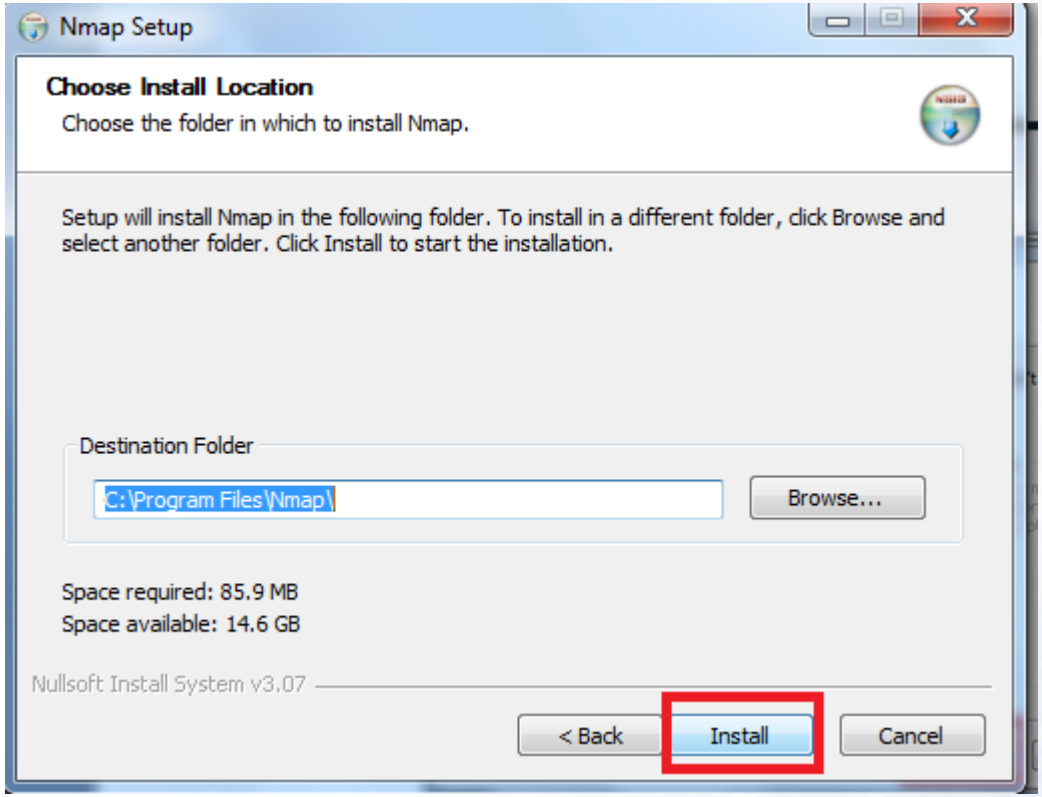
You can also download and install the RPMs yourself:

Latest stable release:
x86-64 (64-bit Linux) Nmap RPM: nmap-7.93-1.x86_64.rpm
x86-64 (64-bit Linux) Ncat RPM: ncat-7.93-1.x86_64.rpm
x86-64 (64-bit Linux) Nping RPM: nping-0.7.93-1.x86_64.rpm
Optional Zenmap GUI (all platforms): zenmap-7.93-1.noarch.rpm
Source RPM (includes Nmap, Zenmap, Ncat, and Nping): nmap-7.93-1.src.rpm

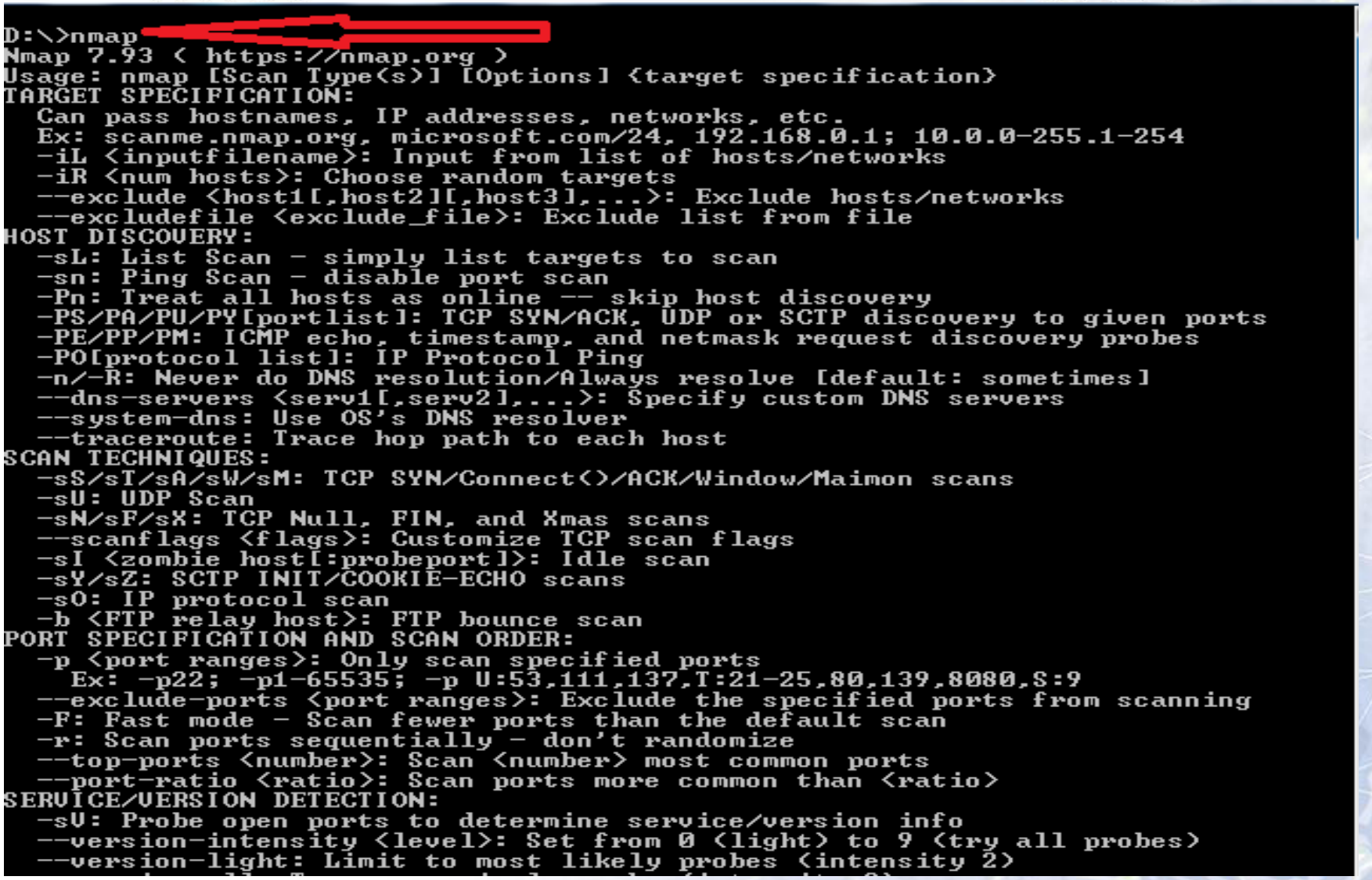

Nmap Setup
License Agreement
Please review the license terms before installing Nmap.
Press Page Down to see the rest of the agreement.
For more information on this license, see https://nmap.org/npsl/
0. Preamble
The intent of this license is to establish freedom to share and change the software regulated by this license under the open source model. It also includes a Contributor Agreement and disclaims any warranty on Covered Software. Companies wishing to use or incorporate Covered Software within their own products may find that our Nmap OEM product (https://nmap.org/oem/) better suits their needs. Open source developers who wish to incorporate parts of Covered Software into free software with conflicting licenses may write Licensor to request a waiver of terms.
If you accept the terms of the agreement, click I Agree to continue. You must accept the agreement to install Nmap.
Nullsoft Install System v3.07
I Agree
Cancel


Nmap Setup
Choose Components
Choose which features of Nmap you want to install.
Check the components you want to install and uncheck the components you don't want to install. Click Next to continue.
Select components to install:
Nmap Core Files
Register Nmap Path
Npcap 1.71
Check online for newe
Network Performance
Zenmap (GUI Fronten
Ndiff (Scan comparisor
Ncat (Modern Netcat
Description
Position your mouse over a component to see its description.
Space required: 85.9 MB
Nullsoft Install System v3.07
< Back
Next >
Cancel

- **Nmap core files includes the Nmap scanning engine**
- **Register Nmap path allows you to execute Nmap from the command line in any directory**
- **Network performance improvements optimization your system for running scans.**
- **Zenmap is a graphical front end for Nmap**
- **Ndiff allows you to compare the results of two Nmap scans.**
- **Ncat allows you to read and write data to network connections from the command line**
- **Nping is a packet generation utility that provides enhancements over the basic ping command.**


Nmap Setup
Choose Install Location
Choose the folder in which to install Nmap.
Setup will install Nmap in the following folder. To install in a different folder, click Browse and select another folder. Click Install to start the installation.
Destination Folder
C:\Program Files\Nmap\
Browse...
Space required: 85.9 MB
Space available: 14.6 GB
Nullsoft Install System v3.07
< Back
Install
Cancel

Nmap -
Zenmap GUI


D:\>nmap
Nmap 7.93 (https://nmap.org)
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
Can pass hostnames, IP addresses, networks, etc.
Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
-iL <inputfilename>: Input from list of hosts/networks
-iR <num hosts>: Choose random targets
--exclude <host1[,host2][,host3],...>: Exclude hosts/networks
--excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
-sL: List Scan - simply list targets to scan
-sn: Ping Scan - disable port scan
-Pn: Treat all hosts as online -- skip host discovery
-PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
-PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
-PO[protocol list]: IP Protocol Ping
-n/-R: Never do DNS resolution/Always resolve [default: sometimes]
--dns-servers <serv1[,serv2],...>: Specify custom DNS servers
--system-dns: Use OS's DNS resolver
--traceroute: Trace hop path to each host
SCAN TECHNIQUES:
-sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
-sU: UDP Scan
-sN/sF/sX: TCP Null, FIN, and Xmas scans
--scanflags <flags>: Customize TCP scan flags
-sI <zombie host[:probeport]>: Idle scan
-sY/sZ: SCTP INIT/COOKIE-ECHO scans
-sO: IP protocol scan
-b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
-p <port ranges>: Only scan specified ports
Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
--exclude-ports <port ranges>: Exclude the specified ports from scanning
-F: Fast mode - Scan fewer ports than the default scan
-r: Scan ports sequentially - don't randomize
--top-ports <number>: Scan <number> most common ports
--port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
-sV: Probe open ports to determine service/version info
--version-intensity <level>: Set from 0 (light) to 9 (try all probes)
--version-light: Limit to most likely probes (intensity 2)

Nmap syntax is the same regardless of the operating system.

```
D:\>nmap -v
Starting Nmap 7.93 ( https://nmap.org ) at 2023-03-31 22:44 Pakistan Standard Ti
me
Read data files from: C:\Program Files\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.08 seconds
           Raw packets sent: 0 (0B) | Rcvd: 0 (0B)

D:\>_
```

```
D:\>nmap scanme.nmap.org
Starting Nmap 7.93 ( https://nmap.org ) at 2023-03-31 22:49 Pakistan Standard T
me
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.34s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 33.13 seconds

D:\>_
```

Use the Snap container manager to install Nmap on Ubuntu and Debian systems.

```
ubuntu@ip-172-31-93-59:~$ sudo snap install nmap
nmap 7.91 from Maximiliano Bertacchini (maxiberta) installed
ubuntu@ip-172-31-93-59:~$
```

```
ubuntu@ip-172-31-93-59:~$ sudo snap install nmap
nmap 7.91 from Maximiliano Bertacchini (maxiberta) installed
ubuntu@ip-172-31-93-59:~$ nmap -V
Nmap version 7.91 ( https://nmap.org )
Platform: x86_64-unknown-linux-gnu
Compiled with: liblua-5.3.3 openssl-1.1.1 libssh2-1.8.0 libz-1.2.11 libpcre-8.39 libpcap-1.8.1 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
ubuntu@ip-172-31-93-59:~$
```

Week 7 Topic 5

Port Stats &
Running Simple Namp command

Port Stats in Nmap

State	Description
open	Accepting connection requests
closed	No service responding to requests
filtered	Blocked by a firewall
unfiltered	Accessible, but scanner was unable to determine whether open or closed

Special cases of port states

- Nmap might be unable to make a definitive statement about the state of the port
- If the Namp is unsure of the state, it may reports two options.

- Open | filtered means that the port is either open or filtered
- Closed | filtered means that the port is either closed or filtered

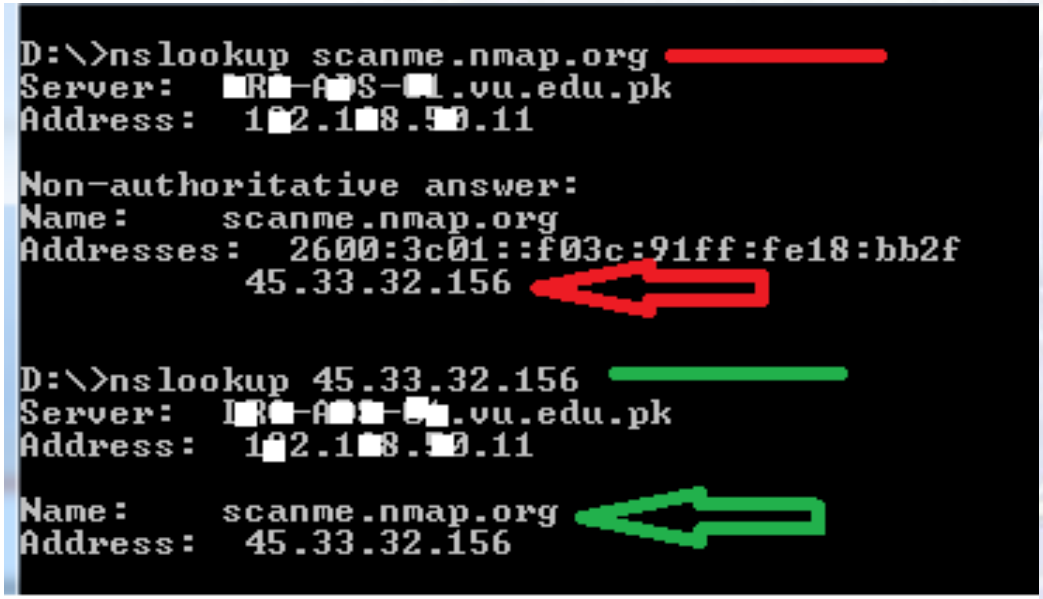
Scan with DNS

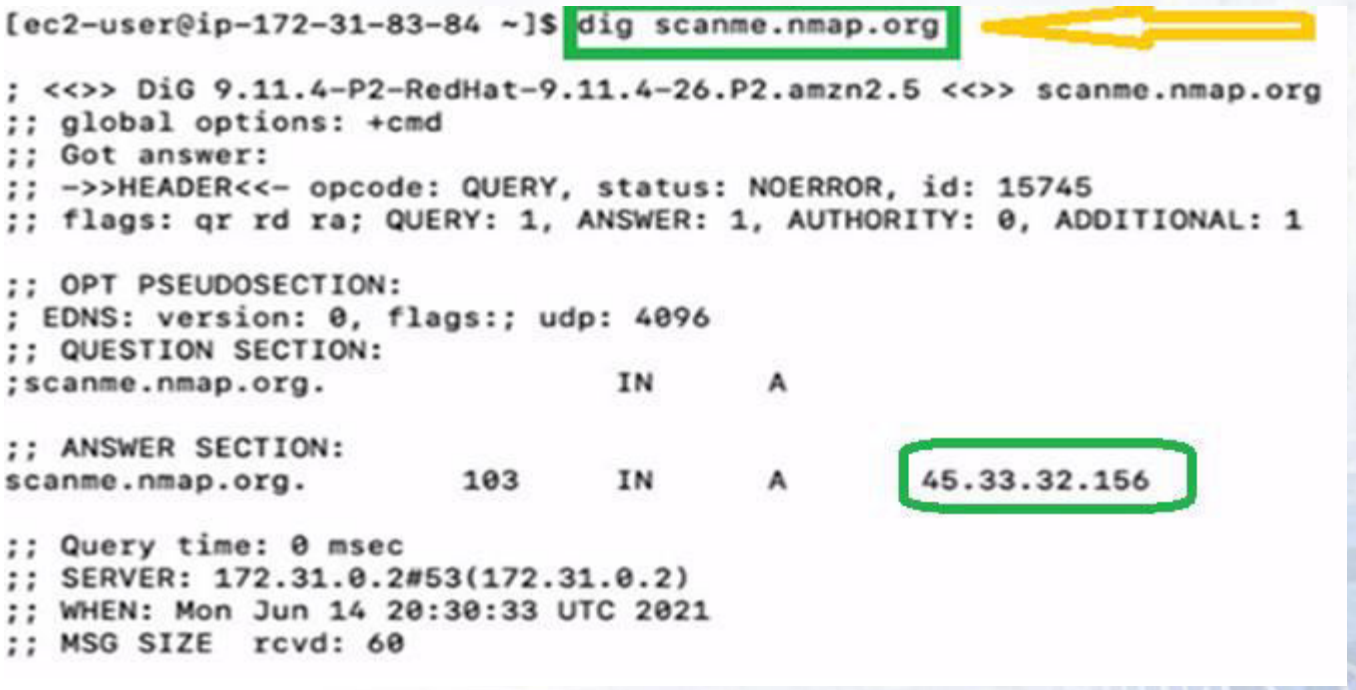
```
[ec2-user@ip-172-31-83-84 ~]$ nmap scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-14 20:30 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.072s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 1.18 seconds
[ec2-user@ip-172-31-83-84 ~]$
```


```
[ec2-user@ip-172-31-83-84 ~]$ nmap scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-14 20:30 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.072s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 1.18 seconds
```

- (Microsoft Windows uses nslookup, while Mac OS X and Linux use dig.)

- Both perform DNS queries, and to test out your DNS configuration.
- determine the IP address associated with a domain name




[ec2-user@ip-172-31-83-84 ~]$ dig scanme.nmap.org
; <<>> DiG 9.11.4-P2-RedHat-9.11.4-26.P2.amzn2.5 <<>> scanme.nmap.org
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 15745
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;scanme.nmap.org. IN A
;; ANSWER SECTION:
scanme.nmap.org. 103 IN A 45.33.32.156
;; Query time: 0 msec
;; SERVER: 172.31.0.2#53(172.31.0.2)
;; WHEN: Mon Jun 14 20:30:33 UTC 2021
;; MSG SIZE rcvd: 60

Scan with server IP

```
[ec2-user@ip-172-31-83-84 ~]$ nmap 45.33.32.156
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-14 20:31 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.071s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 1.19 seconds
[ec2-user@ip-172-31-83-84 ~]$
```

```
D:\>nmap scanme.nmap.org
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-01 14:53 Pakistan Standard Ti
me
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 34.77 seconds

D:\>nmap 45.33.32.156
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-01 15:05 Pakistan Standard Ti
me
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.34s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 55.42 seconds
```

Week 7 Topic 6

OS and service Version Detection

Nmap is able to probe deeper than simply open reports Guesses the target OS.

Activate operating system detection with the -O flag.

```
Mikes-Mac-mini:~ root# nmap -O scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:23 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.080s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE
22/tcp    open     ssh
25/tcp    filtered smtp
80/tcp    open     http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo
31337/tcp open     Elite
Aggressive OS guesses: Linux 2.6.32 (93%), Linux 2.6.32 or 3.10 (93%), Linux 4.4 (92%), Linux 2.6.32 - 2.6.35 (91%), Linux 2.6.32 - 2.6.39 (91%), Linux 4.0 (90
%), Linux 5.4 (90%), Linux 3.11 - 4.1 (89%), Linux 3.2 - 3.8 (89%), Linux 2.6.18 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 16 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.54 seconds
```

```
Mikes-Mac-mini:~ root# nmap -O 192.168.1.54
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:27 EDT
Nmap scan report for 192.168.1.54
Host is up (0.012s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
28201/tcp open  unknown
MAC Address: F0:18:98:5A:B1:A0 (Apple)
Device type: general purpose|media device
Running: Apple OS X 10.11.X, Apple iOS 12.X
OS CPE: cpe:/o:apple:mac_os_x:10.11 cpe:/o:apple:iphone_os:12
OS details: Apple Mac OS X 10.11 (El Capitan) or iOS 12.4 - 13.0 (Darwin 15.6.0 - 19.0.0)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.98 seconds
```

• Service version detection

- Guesses the service versions running on an open port of target system

Activate service version detection with the **-sV** flag.

```
Mikes-Mac-mini:~ root# nmap -sV scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:30 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.089s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.80 seconds
```

```
Mikes-Mac-mini:~ root# nmap -sV scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:30 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.089s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.80 seconds
```

Week 7 Topic 7

Port selection Controls scan behavior

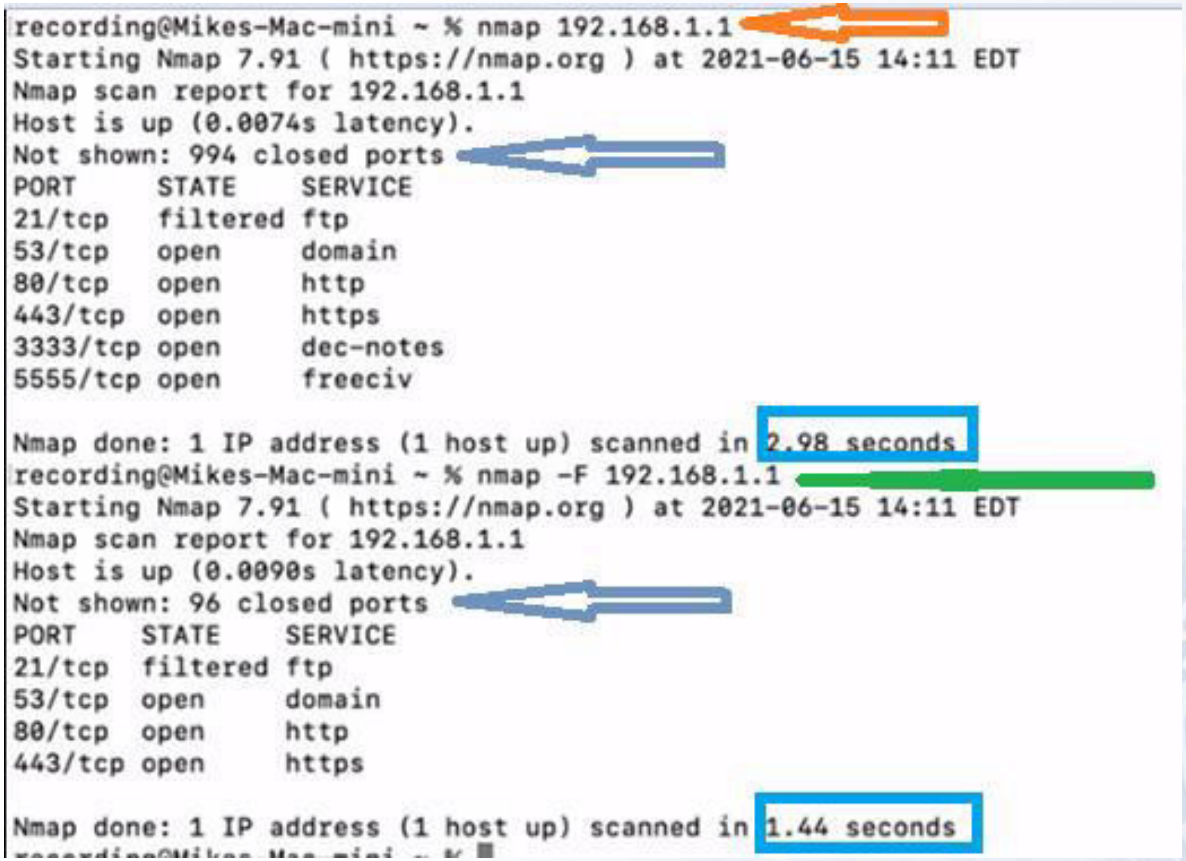
- By default, Nmap scans the 1000 most commonly used TCP ports
- Many of those probably are not used on our network
- Scanning each unused port takes some time
- There are 65535 TCP ports on a system

- If a service is running on any of those ports other than the 1000 most commonly used ports then default Nmap scan will miss it
- There are some options that allow us to customize Nmap scan

The -F flag requests a fast scan of the top 100 TCP ports.

```
recording@Mikes-Mac-mini ~ % nmap 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:11 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0074s latency).
Not shown: 994 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv
```

```
recording@Mikes-Mac-mini ~ % nmap 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:11 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0074s latency).
Not shown: 994 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv

Nmap done: 1 IP address (1 host up) scanned in 2.98 seconds
recording@Mikes-Mac-mini ~ % nmap -F 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:11 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0090s latency).
Not shown: 96 closed ports
PORT    STATE    SERVICE
21/tcp  filtered ftp
53/tcp  open     domain
80/tcp  open     http
443/tcp open     https

Nmap done: 1 IP address (1 host up) scanned in 1.44 seconds
```


recording@Mikes-Mac-mini ~ % nmap 192.168.1.1
Starting Nmap 7.91 (https://nmap.org) at 2021-06-15 14:11 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0074s latency).
Not shown: 994 closed ports
PORT STATE SERVICE
21/tcp filtered ftp
53/tcp open domain
80/tcp open http
443/tcp open https
3333/tcp open dec-notes
5555/tcp open freeciv
Nmap done: 1 IP address (1 host up) scanned in 2.98 seconds
recording@Mikes-Mac-mini ~ % nmap -F 192.168.1.1
Starting Nmap 7.91 (https://nmap.org) at 2021-06-15 14:11 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0090s latency).
Not shown: 96 closed ports
PORT STATE SERVICE
21/tcp filtered ftp
53/tcp open domain
80/tcp open http
443/tcp open https
Nmap done: 1 IP address (1 host up) scanned in 1.44 seconds

The -p flag allows us to specify the ports we'd like to scan.

```
recording@Mikes-Mac-mini ~ % nmap -p 80 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:12 EDT
Nmap scan report for 192.168.1.1
Host is up (0.015s latency).

PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

```
recording@Mikes-Mac-mini ~ % nmap -p 80,443 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:13 EDT
Nmap scan report for 192.168.1.1
Host is up (0.012s latency).

PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

Scan Service

```
recording@Mikes-Mac-mini ~ % nmap -p http,https 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:14 EDT
Nmap scan report for 192.168.1.1
Host is up (0.013s latency).

PORT     STATE  SERVICE
80/tcp   open   http
443/tcp  open   https
8008/tcp closed http

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

```
recording@Mikes-Mac-mini ~ % nmap -p 80 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:12 EDT
Nmap scan report for 192.168.1.1
Host is up (0.015s latency).

PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
recording@Mikes-Mac-mini ~ % nmap -p 80,443 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:13 EDT
Nmap scan report for 192.168.1.1
Host is up (0.012s latency).

PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
recording@Mikes-Mac-mini ~ % nmap -p http,https 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:14 EDT
Nmap scan report for 192.168.1.1
Host is up (0.013s latency).

PORT     STATE  SERVICE
80/tcp   open   http
443/tcp  open   https
8008/tcp closed http

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
```

The `-p-` flag requests a full TCP scan of the target system(s).

```
recording@Mikes-Mac-mini ~ % nmap -p- 192.168.1.1
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 14:15 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0078s latency).
Not shown: 65528 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv
7272/tcp open     watchme-7272

Nmap done: 1 IP address (1 host up) scanned in 121.38 seconds
```

Week 7 Topic 8

Specifying Scan Targets

- Single IP address or DNS name
- List of addresses and names, separated by spaces
- Range of IP addresses
- Subnet mask
- File containing addresses and names
- Scan IPv6

```
recording@Mikes-Mac-mini ~ % nmap 192.168.1.1 192.168.1.3 192.168.1.6
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:38 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0081s latency).
Not shown: 994 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv

Nmap scan report for 192.168.1.3
Host is up (0.012s latency).
Not shown: 996 closed ports
PORT     STATE SERVICE
23/tcp   open  telnet
80/tcp   open  http
111/tcp  open  rpcbind
8080/tcp open  http-proxy

Nmap scan report for 192.168.1.6
Host is up (0.0090s latency).
Not shown: 998 closed ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 3 IP addresses (3 hosts up) scanned in 7.16 seconds
recording@Mikes-Mac-mini ~ %
```

```
recording@Mikes-Mac-mini ~ % nmap 192.168.1.1,3,6
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:39 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0087s latency).
Not shown: 994 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv

Nmap scan report for 192.168.1.3
Host is up (0.0087s latency).
Not shown: 996 closed ports
PORT     STATE SERVICE
23/tcp   open  telnet
80/tcp   open  http
111/tcp  open  rpcbind
8080/tcp open  http-proxy

Nmap scan report for 192.168.1.6
Host is up (0.014s latency).
Not shown: 998 closed ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 3 IP addresses (3 hosts up) scanned in 7.75 seconds
```

```
recording@Mikes-Mac-mini ~ % nmap 192.168.1.1-6
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:40 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0083s latency).
Not shown: 994 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv

Nmap scan report for 192.168.1.3
Host is up (0.011s latency).
Not shown: 996 closed ports
PORT     STATE SERVICE
23/tcp   open  telnet
80/tcp   open  http
111/tcp  open  rpcbind
8080/tcp open  http-proxy

Nmap scan report for 192.168.1.4
Host is up (0.011s latency).
All 1000 scanned ports on 192.168.1.4 are closed

Nmap scan report for 192.168.1.5
Host is up (0.013s latency).
All 1000 scanned ports on 192.168.1.5 are closed

Nmap scan report for 192.168.1.6
Host is up (0.019s latency).
Not shown: 998 closed ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 6 IP addresses (5 hosts up) scanned in 16.02 seconds
```

Scan subnet

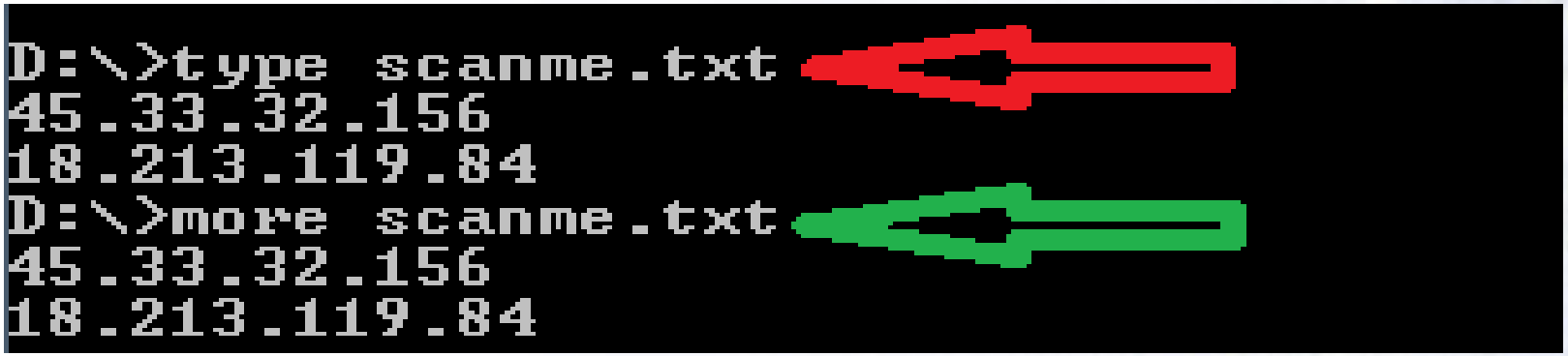
```
recording@Mikes-Mac-mini ~ % nmap 192.168.1.0/29
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:41 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0080s latency).
Not shown: 994 closed ports
PORT     STATE    SERVICE
21/tcp   filtered ftp
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3333/tcp open     dec-notes
5555/tcp open     freeciv

Nmap scan report for 192.168.1.3
Host is up (0.010s latency).
Not shown: 996 closed ports
PORT     STATE SERVICE
23/tcp   open  telnet
80/tcp   open  http
111/tcp  open  rpcbind
8080/tcp open  http-proxy

Nmap scan report for 192.168.1.4
Host is up (0.014s latency).
All 1000 scanned ports on 192.168.1.4 are closed

Nmap scan report for 192.168.1.5
Host is up (0.014s latency).
All 1000 scanned ports on 192.168.1.5 are closed

Nmap scan report for 192.168.1.6
Host is up (0.015s latency).
Not shown: 998 closed ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap scan report for 192.168.1.7
Host is up (0.014s latency).
All 1000 scanned ports on 192.168.1.7 are closed

Nmap done: 8 IP addresses (6 hosts up) scanned in 14.75 seconds
```

Scan from file

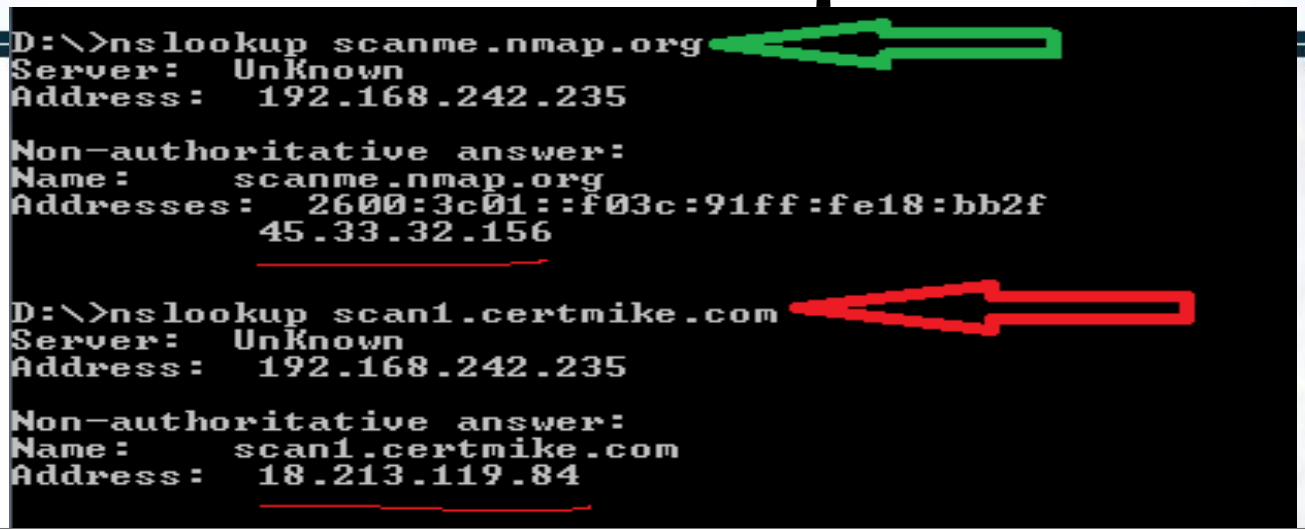
```
recording@Mikes-Mac-mini ~ % cat scan_targets.txt
192.168.1.1
192.168.1.3
192.168.1.6
192.168.1.8
192.168.1.38
192.168.1.40
192.168.1.42
recording@Mikes-Mac-mini ~ % nmap -iL scan_targets.txt
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:42 EDT
```

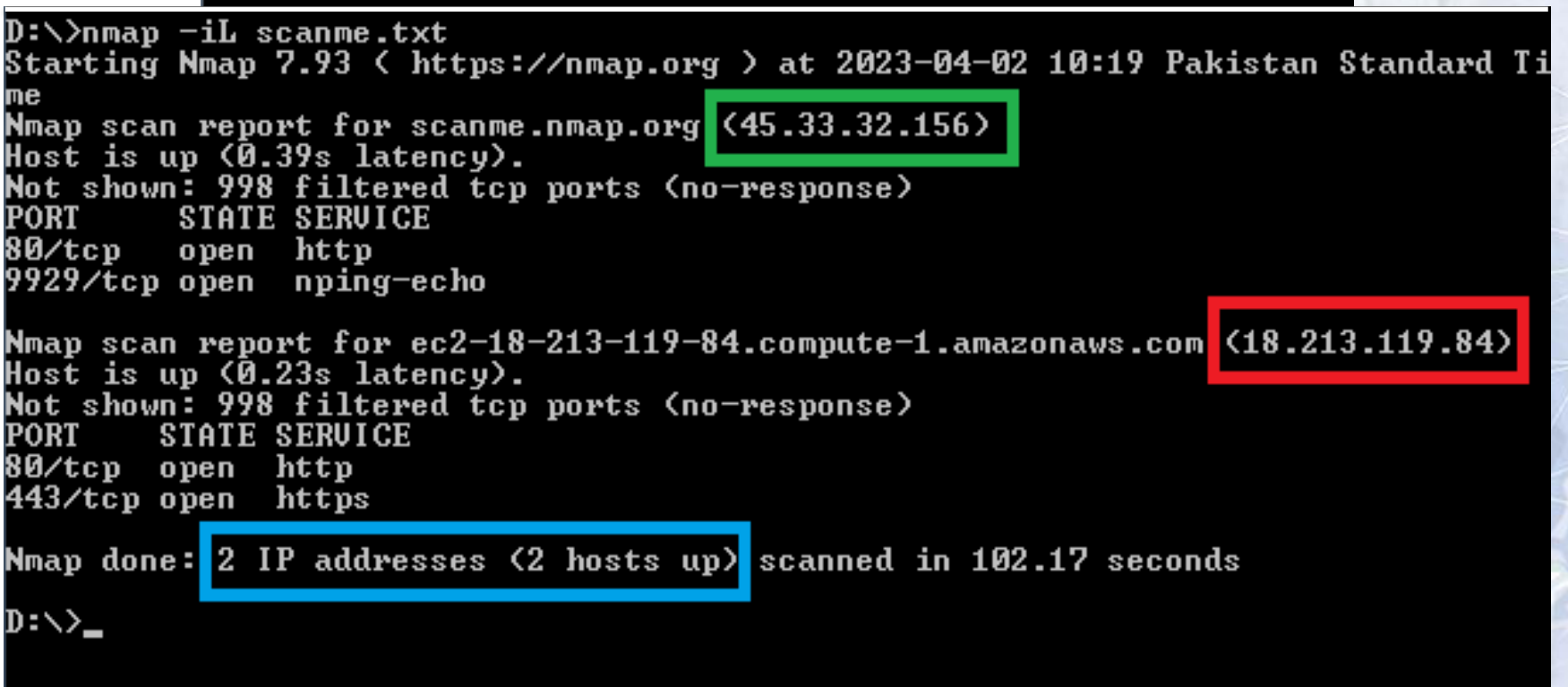
```
Host is up (0.012s latency).
Not shown: 998 closed ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap scan report for 192.168.1.8
Host is up (0.0098s latency).
Not shown: 998 closed ports
PORT     STATE SERVICE
5555/tcp open  freeciv
8009/tcp open  ajp13

Nmap scan report for 192.168.1.38
Host is up (0.011s latency).
Not shown: 993 closed ports
PORT     STATE SERVICE
21/tcp   open  ftp
23/tcp   open  telnet
80/tcp   open  http
443/tcp  open  https
515/tcp  open  printer
631/tcp  open  ipp
9100/tcp open  jetdirect

Nmap scan report for 192.168.1.40
Host is up (0.013s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
62078/tcp open  iphone-sync

Nmap scan report for 192.168.1.42
Host is up (0.010s latency).
Not shown: 994 closed ports
PORT      STATE SERVICE
5000/tcp  open  upnp
7000/tcp  open  afs3-fileserver
7100/tcp  open  font-service
49152/tcp open  unknown
49153/tcp open  unknown
62078/tcp open  iphone-sync

Nmap done: 7 IP addresses (7 hosts up) scanned in 27.92 seconds
```

Scan from file on windows



Nslookup

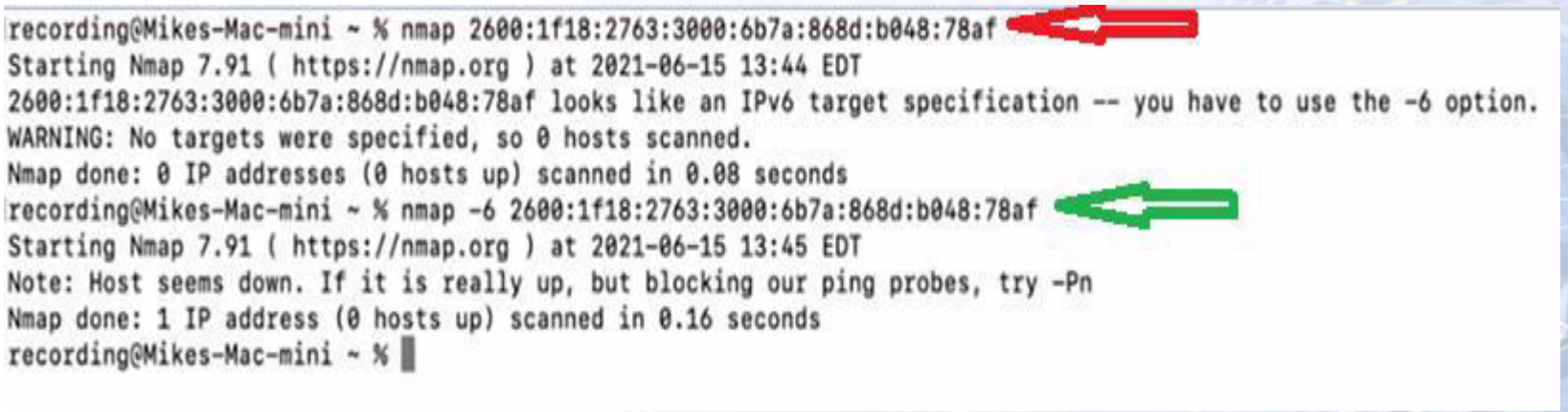

D:\>nslookup scanme.nmap.org
Server: UnKnown
Address: 192.168.242.235
Non-authoritative answer:
Name: scanme.nmap.org
Addresses: 2600:3c01::f03c:91ff:fe18:bb2f
45.33.32.156
D:\>nslookup scan1.certmike.com
Server: UnKnown
Address: 192.168.242.235
Non-authoritative answer:
Name: scan1.certmike.com
Address: 18.213.119.84


D:\>nmap -iL scanme.txt
Starting Nmap 7.93 (https://nmap.org) at 2023-04-02 10:19 Pakistan Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.39s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
9929/tcp open nping-echo
Nmap scan report for ec2-18-213-119-84.compute-1.amazonaws.com (18.213.119.84)
Host is up (0.23s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
443/tcp open https
Nmap done: 2 IP addresses (2 hosts up) scanned in 102.17 seconds
D:\>

Scanning IPv6

Nmap is able to scan IPv6 addresses with the -6 flag.

```
recording@Mikes-Mac-mini ~ % nmap 2600:1f18:2763:3000:6b7a:868d:b048:78af
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:44 EDT
2600:1f18:2763:3000:6b7a:868d:b048:78af looks like an IPv6 target specification -- you have to use the -6 option.
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.08 seconds
recording@Mikes-Mac-mini ~ %
```



Week 7 Topic 9

Scan Shortcut and Disable DNS option

- The **–A** flag offers a scanning shortcut that activate several popular options

-A Options

- Remote operating system detection
- Service and version detection
- Traceroute
- Nmap Scripting engine (NSE)

```
Mikes-Mac-mini:~ root# nmap -A -T4 scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:33 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.079s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Aggressive OS guesses: Linux 2.6.32 (93%), Linux 2.6.32 or 3.10 (93%), Linux 4.4 (93%), Linux 2.6.32 - 2.6.35 (91
%), Linux 2.6.32 - 3.0 (90%), Linux 3.11 - 4.1 (89%), Linux 3.2 - 3.8 (89%), Linux 5.0 - 5.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 16 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 5900/tcp)
HOP RTT      ADDRESS
1   8.82 ms  192.168.1.1
2   12.34 ms 10.0.0.1
3   30.27 ms 96.120.26.197
4   30.27 ms 96.110.154.133
5   21.36 ms 68.87.231.137
6   30.29 ms be-167-ar01.area4.il.chicago.comcast.net (162.151.144.101)
7   31.34 ms 24.153.89.45
8   31.33 ms be-32241-cs04.350ecermak.il.ibone.comcast.net (96.110.40.61)
```

```
Mikes-Mac-mini:~ root# nmap -A -T4 scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:33 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.079s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Aggressive OS guesses: Linux 2.6.32 (93%), Linux 2.6.32 or 3.10 (93%), Linux 4.4 (93%), Linux 2.6.32 - 2.6.35 (91%), Linux 4.0
%), Linux 2.6.32 - 3.0 (90%), Linux 3.11 - 4.1 (89%), Linux 3.2 - 3.8 (89%), Linux 5.0 - 5.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 16 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 5900/tcp)
HOP RTT      ADDRESS
1   8.82 ms  192.168.1.1
2   12.34 ms 10.0.0.1
3   30.27 ms 96.120.26.197
```

```
Mikes-Mac-mini:~ root# nmap -A -T4 scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:33 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.079s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Aggressive OS guesses: Linux 2.6.32 (93%), Linux 2.6.32 or 3.10 (93%), Linux 4.4 (93%), Linux 2.6.32 - 2.6.35 (91%), Lin
%), Linux 2.6.32 - 3.0 (90%), Linux 3.11 - 4.1 (89%), Linux 3.2 - 3.8 (89%), Linux 5.0 - 5.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 16 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 5900/tcp)
HOP RTT      ADDRESS
1   8.82 ms  192.168.1.1
2   12.34 ms 10.0.0.1
```

Namp Scrpting engine that retrieves the SSH key as presented by the host

```
Mikes-Mac-mini:~ root# nmap -A -T4 scanme.nmap.org
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 15:33 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.079s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Aggressive OS guesses: Linux 2.6.32 (93%), Linux 2.6.32 or 3.10 (93%), Linux 4.4 (93%), Linux 2.6.3
%), Linux 2.6.32 - 3.0 (90%), Linux 3.11 - 4.1 (89%), Linux 3.2 - 3.8 (89%), Linux 5.0 - 5.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 16 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 5900/tcp)
HOP RTT      ADDRESS
1   8.82 ms  192.168.1.1
2   12.34 ms 10.0.0.1
3   30.27 ms 96.120.26.197
```

```
25/tcp    filtered smtp
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
Aggressive OS guesses: Linux 2.6.32 (93%), Linux 2.6.32 or 3.10 (93%), Linux 4.4 (93%), Linux 2.6.32 - 
%), Linux 2.6.32 - 3.0 (90%), Linux 3.11 - 4.1 (89%), Linux 3.2 - 3.8 (89%), Linux 5.0 - 5.4 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 16 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 5900/tcp)
HOP RTT      ADDRESS
1   8.82 ms  192.168.1.1
2   12.34 ms 10.0.0.1
3   30.27 ms 96.120.26.197
4   30.27 ms 96.110.154.133
5   21.36 ms 68.87.231.137
6   30.29 ms be-167-ar01.area4.il.chicago.comcast.net (162.151.144.101)
7   31.34 ms 24.153.89.45
8   31.33 ms be-32241-cs04.350ecermak.il.ibone.comcast.net (96.110.40.61)
9   33.63 ms be-2211-pe11.350ecermak.il.ibone.comcast.net (96.110.33.198)
10  27.18 ms ae-3.r08.chcgil09.us.bb.gin.ntt.net (129.250.8.65)
11  31.36 ms ae-4.r23.chcgil09.us.bb.gin.ntt.net (129.250.2.205)
12  96.78 ms ae-1.r24.snjsca04.us.bb.gin.ntt.net (129.250.5.17)
13  96.79 ms ae-40.r02.snjsca04.us.bb.gin.ntt.net (129.250.3.121)
14  96.79 ms ce-0-18-0-1.r02.snjsca04.us.ce.gin.ntt.net (192.80.17.170)
15  96.79 ms if-1-4.csw6-fnc1.linode.com (173.230.159.83)
16  96.81 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.98 seconds
```

DNS resolution

- Adds information, but consumes time
- By default, Nmap perform DNS resolution to identify the names of any systems that are scanned by the IP addresses

The -n flag disables reverse DNS resolution.

```
recording@Mikes-Mac-mini ~ % nmap 45.33.32.156
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:56 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.077s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE
22/tcp    open     ssh
25/tcp    filtered smtp
80/tcp    open     http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo
31337/tcp open     Elite

Nmap done: 1 IP address (1 host up) scanned in 17.09 seconds
```

```
recording@Mikes-Mac-mini ~ % nmap 45.33.32.156
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:56 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.077s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE
22/tcp    open     ssh
25/tcp    filtered smtp
80/tcp    open     http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo
31337/tcp open     Elite

Nmap done: 1 IP address (1 host up) scanned in 17.09 seconds
recording@Mikes-Mac-mini ~ % nmap -n 45.33.32.156
Starting Nmap 7.91 ( https://nmap.org ) at 2021-06-15 13:57 EDT
Nmap scan report for 45.33.32.156
Host is up (0.079s latency).
Not shown: 992 closed ports
PORT      STATE    SERVICE
22/tcp    open     ssh
25/tcp    filtered smtp
80/tcp    open     http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open     nping-echo
31337/tcp open     Elite
```

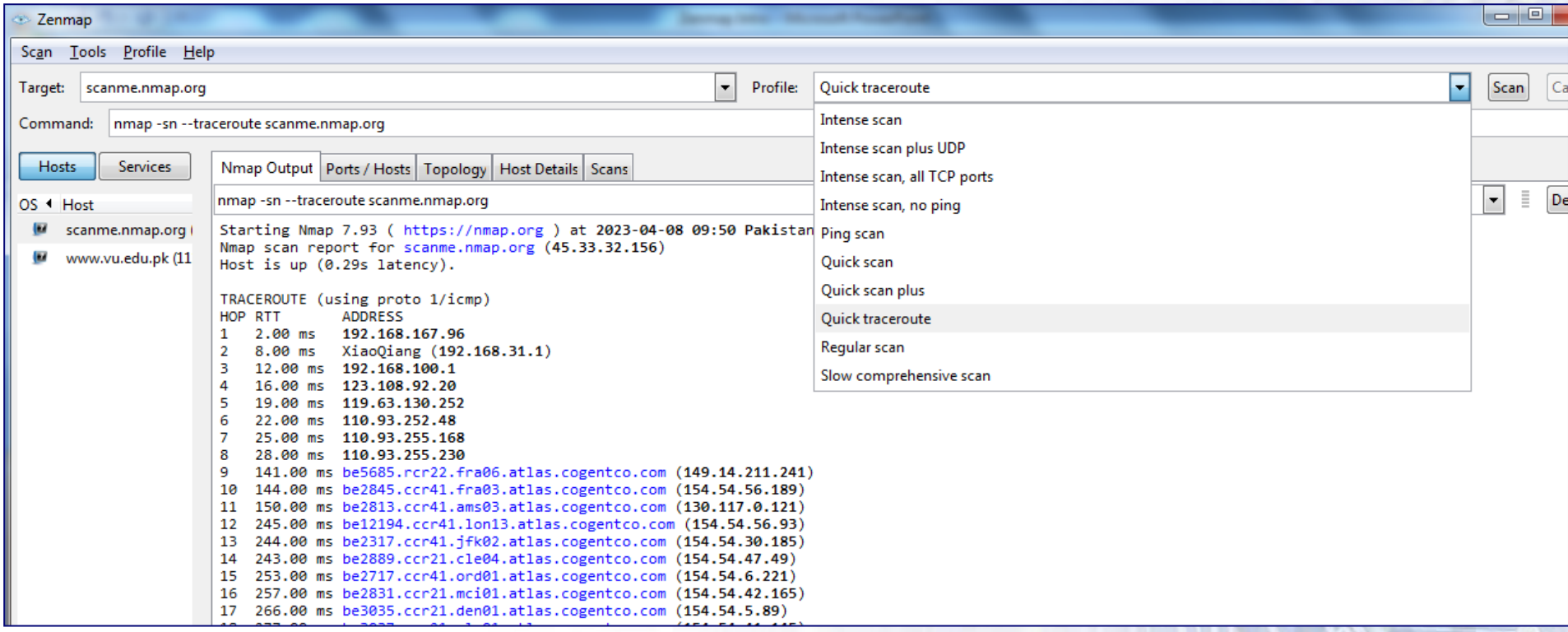
Week 7 Topic 10

Zenmap

- Zenmap is the official graphical user interface (GUI) for the Nmap Security Scanner.

- Easy for beginners to use while providing advanced features for experienced Nmap users.

https://nmap.org/book/zenmap-results.html


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Quick traceroute
Scan
Command: nmap -sn --traceroute scanme.nmap.org
Hosts
Services
OS Host
scanme.nmap.org
www.vu.edu.pk (11
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -sn --traceroute scanme.nmap.org
Starting Nmap 7.93 (https://nmap.org) at 2023-04-08 09:50 Pakistan
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
TRACEROUTE (using proto 1/icmp)
HOP RTT ADDRESS
1 2.00 ms 192.168.167.96
2 8.00 ms XiaoQiang (192.168.31.1)
3 12.00 ms 192.168.100.1
4 16.00 ms 123.108.92.20
5 19.00 ms 119.63.130.252
6 22.00 ms 110.93.252.48
7 25.00 ms 110.93.255.168
8 28.00 ms 110.93.255.230
9 141.00 ms be5685.rcr22.fra06.atlas.cogentco.com (149.14.211.241)
10 144.00 ms be2845.ccr41.fra03.atlas.cogentco.com (154.54.56.189)
11 150.00 ms be2813.ccr41.ams03.atlas.cogentco.com (130.117.0.121)
12 245.00 ms be12194.ccr41.lon13.atlas.cogentco.com (154.54.56.93)
13 244.00 ms be2317.ccr41.jfk02.atlas.cogentco.com (154.54.30.185)
14 243.00 ms be2889.ccr21.cle04.atlas.cogentco.com (154.54.47.49)
15 253.00 ms be2717.ccr41.ord01.atlas.cogentco.com (154.54.6.221)
16 257.00 ms be2831.ccr21.mci01.atlas.cogentco.com (154.54.42.165)
17 266.00 ms be3035.ccr21.den01.atlas.cogentco.com (154.54.5.89)
Intense scan
Intense scan plus UDP
Intense scan, all TCP ports
Intense scan, no ping
Ping scan
Quick scan
Quick scan plus
Quick traceroute
Regular scan
Slow comprehensive scan

Scan Tools Profile Help

Target: scanme.nmap.org Profile: Regular scan Scan Canc

Command: nmap scanme.nmap.org

Hosts Services

Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

scanme.nmap.org

www.vu.edu.pk (11

nmap scanme.nmap.org Detai

```
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-08 09:54 Pakistan Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.30s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 20.49 seconds
```

Zenmap

Scan Tools Profile Help

Target: scanme.nmap.org Profile: Quick traceroute Scan

Command: nmap -sn --traceroute scanme.nmap.org

Hosts Services

Nmap Output Ports / Hosts Topology Host Details Scans

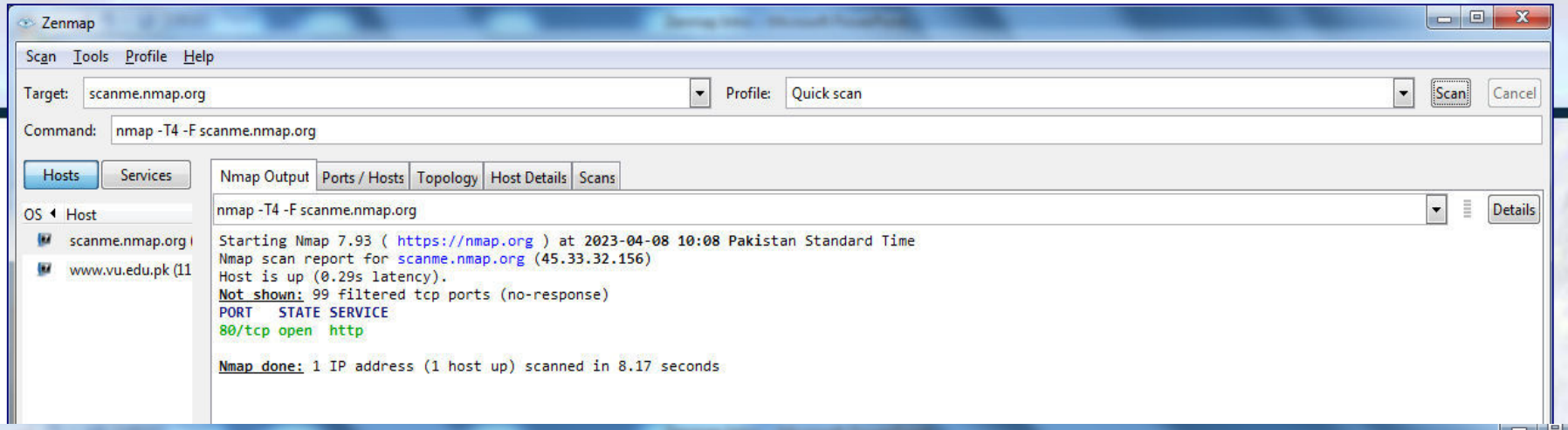
OS Host

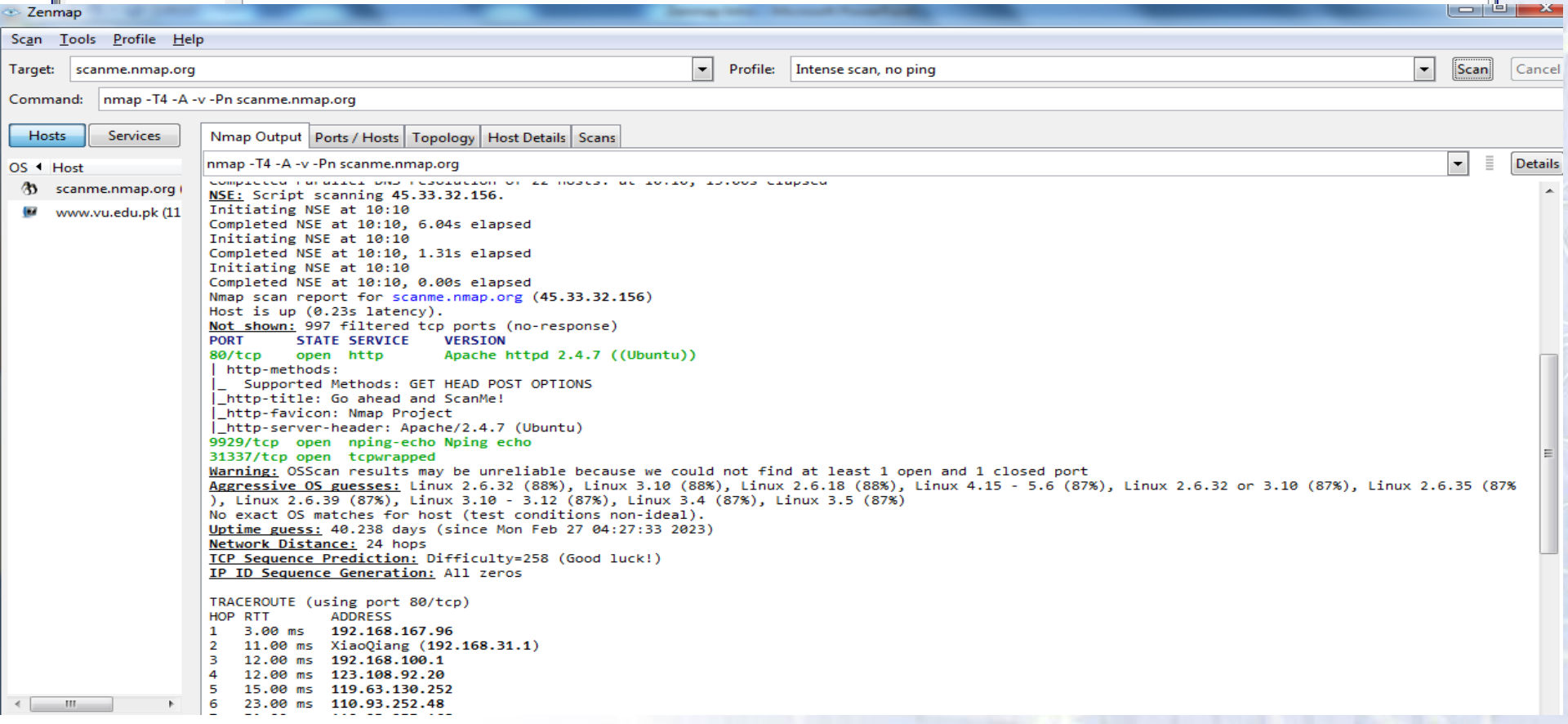
scanme.nmap.org

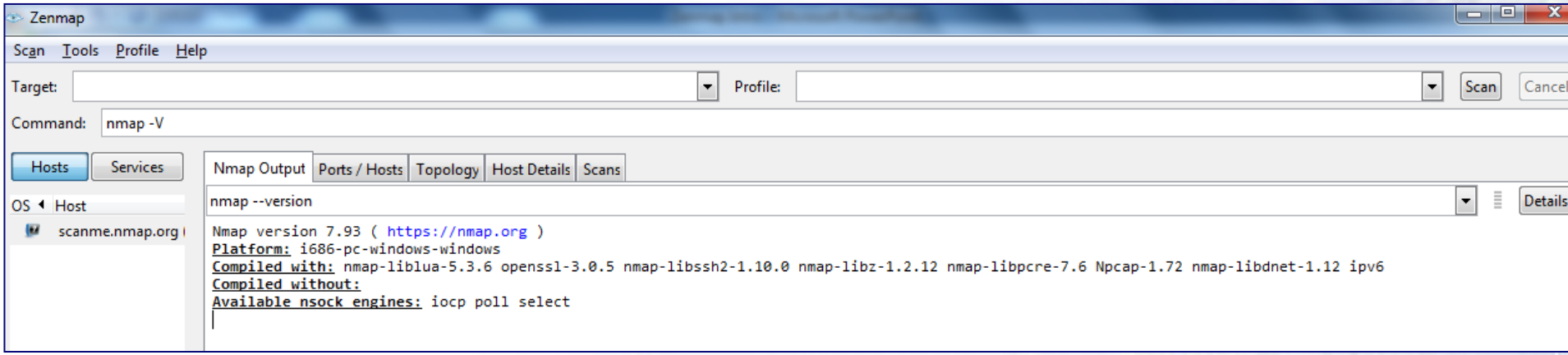
www.vu.edu.pk (11

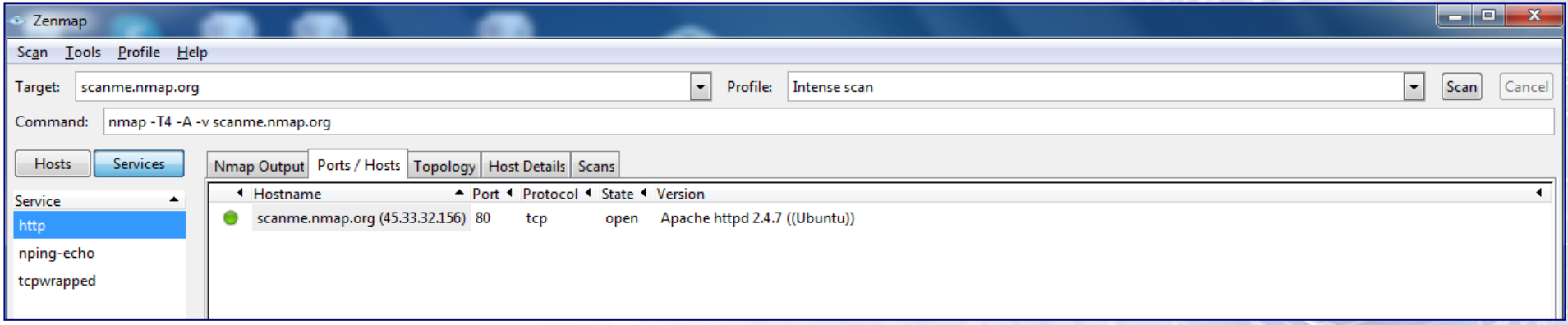
nmap -sn --traceroute scanme.nmap.org

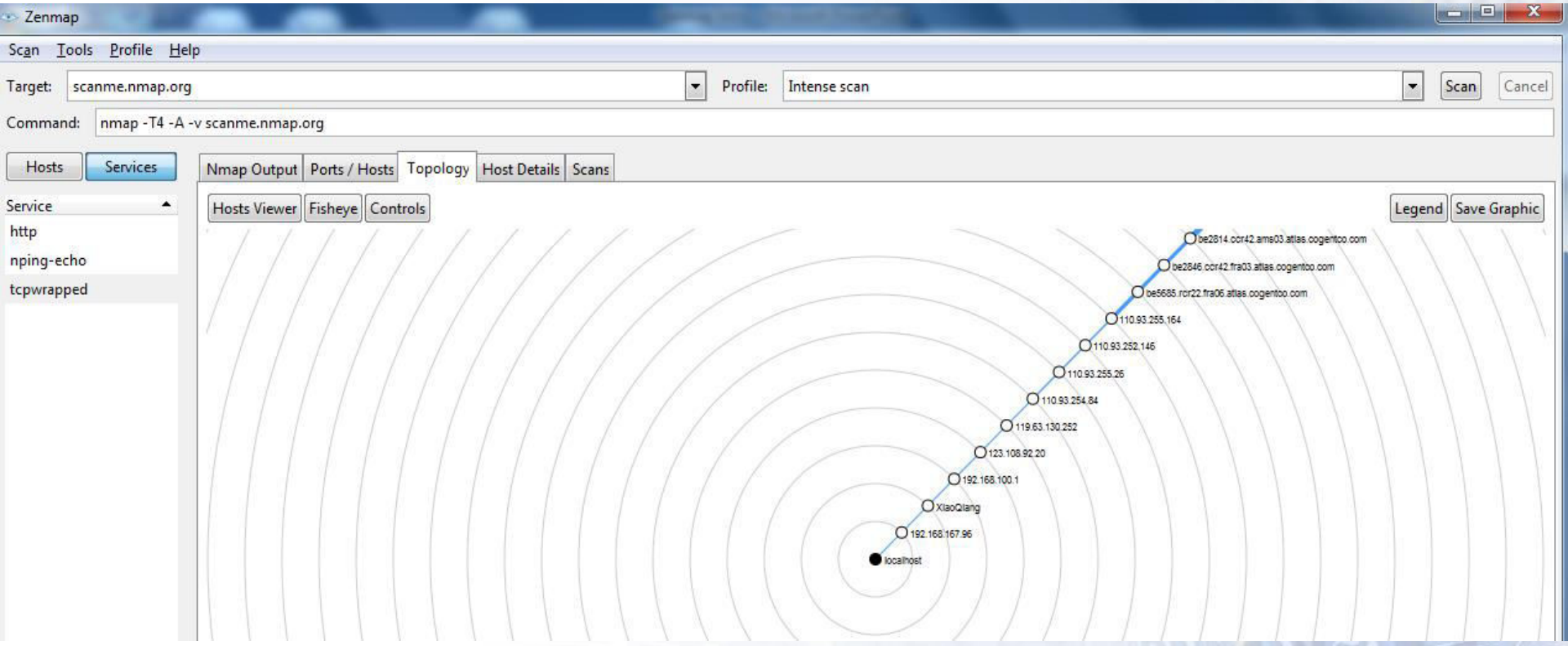
```
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-08 09:50 Pakistan Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).

TRACEROUTE (using proto 1/icmp)
HOP RTT       ADDRESS
1   2.00 ms   192.168.167.96
2   8.00 ms   XiaoQiang (192.168.31.1)
3   12.00 ms  192.168.100.1
4   16.00 ms  123.108.92.20
5   19.00 ms  119.63.130.252
6   22.00 ms  110.93.252.48
7   25.00 ms  110.93.255.168
8   28.00 ms  110.93.255.230
9   141.00 ms be5685.rcr22.fra06.atlas.cogentco.com (149.14.211.241)
10  144.00 ms be2845.ccr41.fra03.atlas.cogentco.com (154.54.56.189)
11  150.00 ms be2813.ccr41.ams03.atlas.cogentco.com (130.117.0.121)
12  245.00 ms be12194.ccr41.lon13.atlas.cogentco.com (154.54.56.93)
13  244.00 ms be2317.ccr41.jfk02.atlas.cogentco.com (154.54.30.185)
14  243.00 ms be2889.ccr21.cle04.atlas.cogentco.com (154.54.47.49)
15  253.00 ms be2717.ccr41.ord01.atlas.cogentco.com (154.54.6.221)
16  257.00 ms be2831.ccr21.mci01.atlas.cogentco.com (154.54.42.165)
17  266.00 ms be3035.ccr21.den01.atlas.cogentco.com (154.54.5.89)
18  277.00 ms be3037.ccr21.slc01.atlas.cogentco.com (154.54.41.145)
19  294.00 ms be3109.ccr21.sfo01.atlas.cogentco.com (154.54.44.137)
20  293.00 ms be3178.ccr21.sjc01.atlas.cogentco.com (154.54.43.70)
21  295.00 ms be2063.rcr21.b001848-1.sjc01.atlas.cogentco.com (154.54.1.162)
22  291.00 ms 38.142.11.154
23  307.00 ms if-1-4.csw5-fnc1.linode.com (173.230.159.81)
24  287.00 ms scanme.nmap.org (45.33.32.156)

Nmap done: 1 IP address (1 host up) scanned in 15.82 seconds
```


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Quick scan
Scan
Cancel
Command: nmap -T4 -F scanme.nmap.org
Hosts
Services
OS Host
scanme.nmap.org
www.vu.edu.pk (11
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -F scanme.nmap.org
Details
Starting Nmap 7.93 (https://nmap.org) at 2023-04-08 10:08 Pakistan Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Not shown: 99 filtered tcp ports (no-response)
PORT STATE SERVICE
80/tcp open http
Nmap done: 1 IP address (1 host up) scanned in 8.17 seconds


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Intense scan, no ping
Scan
Cancel
Command: nmap -T4 -A -v -Pn scanme.nmap.org
Hosts
Services
OS Host
scanme.nmap.org
www.vu.edu.pk (11
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -A -v -Pn scanme.nmap.org
Details
NSE: Script scanning 45.33.32.156.
Initiating NSE at 10:10
Completed NSE at 10:10, 6.04s elapsed
Initiating NSE at 10:10
Completed NSE at 10:10, 1.31s elapsed
Initiating NSE at 10:10
Completed NSE at 10:10, 0.00s elapsed
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT STATE SERVICE VERSION
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
| http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_http-title: Go ahead and ScanMe!
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp open nping-echo Nping echo
31337/tcp open tcpwrapped
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.32 (88%), Linux 3.10 (88%), Linux 2.6.18 (88%), Linux 4.15 - 5.6 (87%), Linux 2.6.32 or 3.10 (87%), Linux 2.6.35 (87%), Linux 2.6.39 (87%), Linux 3.10 - 3.12 (87%), Linux 3.4 (87%), Linux 3.5 (87%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 40.238 days (since Mon Feb 27 04:27:33 2023)
Network Distance: 24 hops
TCP Sequence Prediction: Difficulty=258 (Good luck!)
IP ID Sequence Generation: All zeros
TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 3.00 ms 192.168.167.96
2 11.00 ms XiaoQiang (192.168.31.1)
3 12.00 ms 192.168.100.1
4 12.00 ms 123.108.92.20
5 15.00 ms 119.63.130.252
6 23.00 ms 110.93.252.48


Zenmap
Scan Tools Profile Help
Target:
Profile:
Scan
Cancel
Command: nmap -V
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS Host
scanme.nmap.org
nmap --version
Details
Nmap version 7.93 (https://nmap.org)
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.3.6 openssl-3.0.5 nmap-libssh2-1.10.0 nmap-libz-1.2.12 nmap-libpcre-7.6 Npcap-1.72 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: iocp poll select


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v scanme.nmap.org
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Service
http
nping-echo
tcpwrapped
Hostname
Port
Protocol
State
Version
scanme.nmap.org (45.33.32.156)
80
tcp
open
Apache httpd 2.4.7 ((Ubuntu))


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Intense scan
Scan
Cancel
Command: nmap -T4 -A -v scanme.nmap.org
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
Service
http
nping-echo
tcpwrapped
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
be2814.ccr42.ams03.atlas.cogentco.com
be2846.ccr42.fra03.atlas.cogentco.com
be5685.rcr22.fra06.atlas.cogentco.com
110.93.255.164
110.93.252.146
110.93.255.26
110.93.254.84
119.63.130.252
123.108.92.20
192.168.100.1
XiaoQiang
192.168.167.96
localhost

Zenmap

Scan Tools Profile Help

Target: scanme.nmap.org Profile: Intense scan Scan Cancel

Command: nmap -T4 -A -v scanme.nmap.org

Hosts Services

OS Host

scanme.nmap.org

Filter Hosts

Nmap Output Ports / Hosts Topology Host Details Scans

scanme.nmap.org (45.33.32.156)

Host Status

State: up

Open ports: 3

Filtered ports: 997

Closed ports: 0

Scanned ports: 1000

Up time: 3476823

Last boot: Mon Feb 27 04:27:33 2023

Addresses

IPv4: 45.33.32.156

IPv6: Not available

MAC: Not available

Hostnames

Name - Type: scanme.nmap.org - user

Name - Type: scanme.nmap.org - PTR

Operating System

Name: Linux 2.6.18

Accuracy: 88%

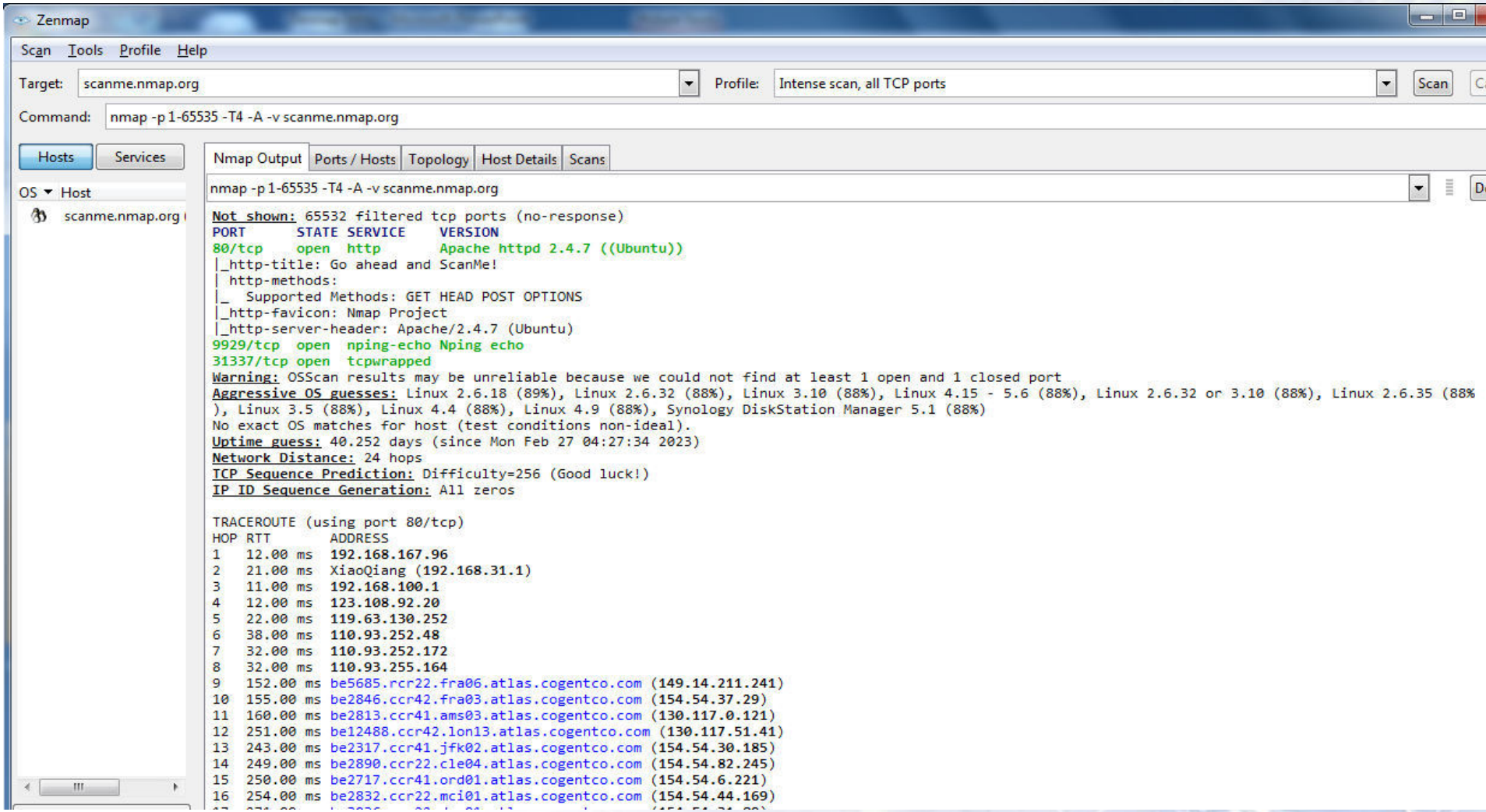
Ports used

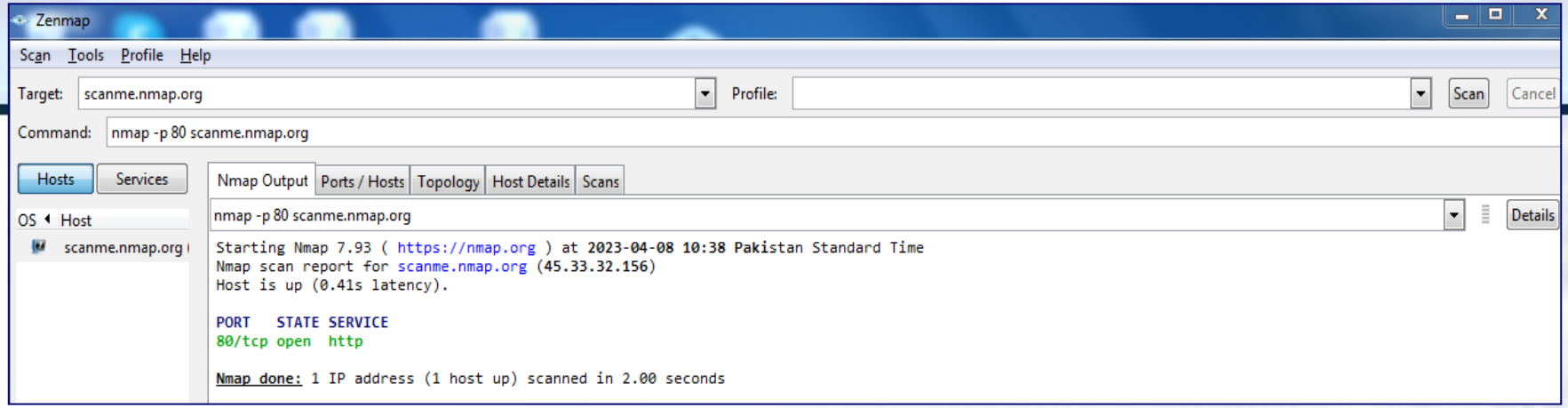
Port-Protocol-State: 80 - tcp - open

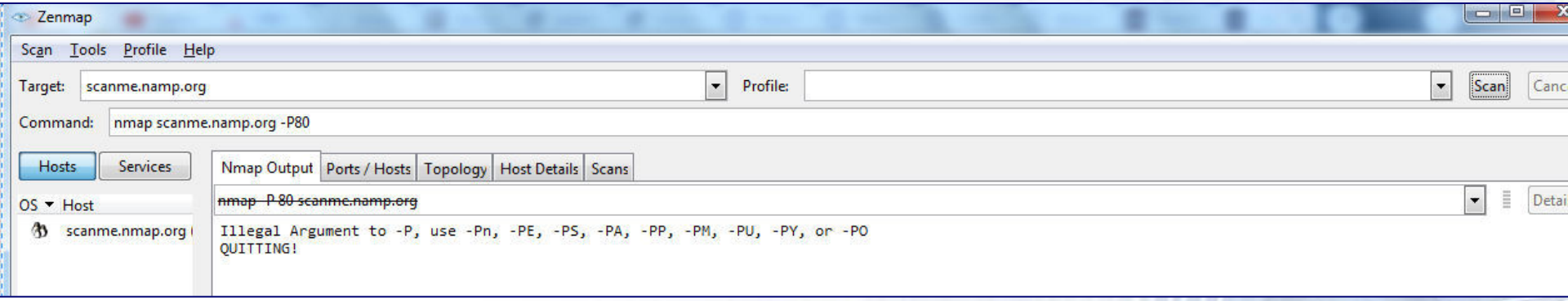
Port-Protocol-State: 30700 - udp - closed

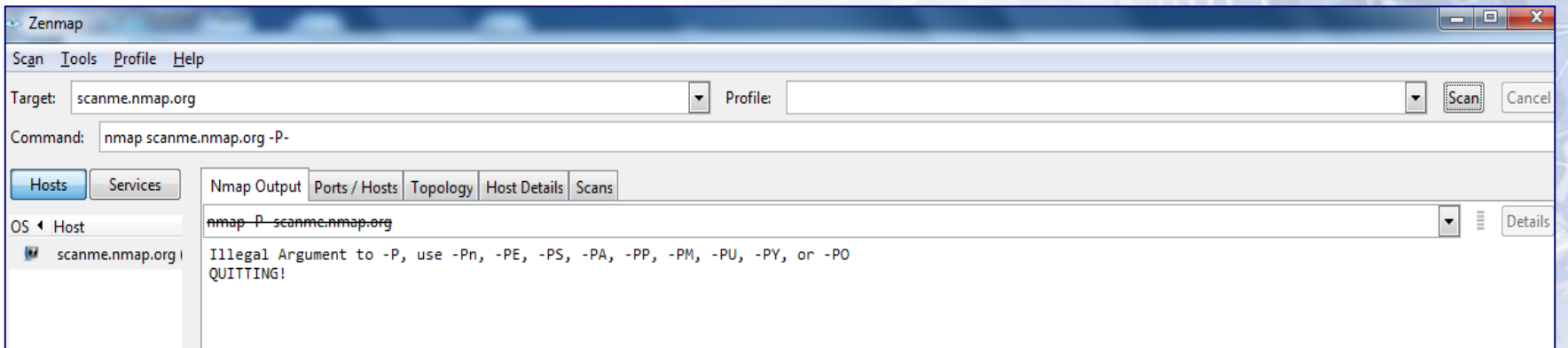
OS Classes

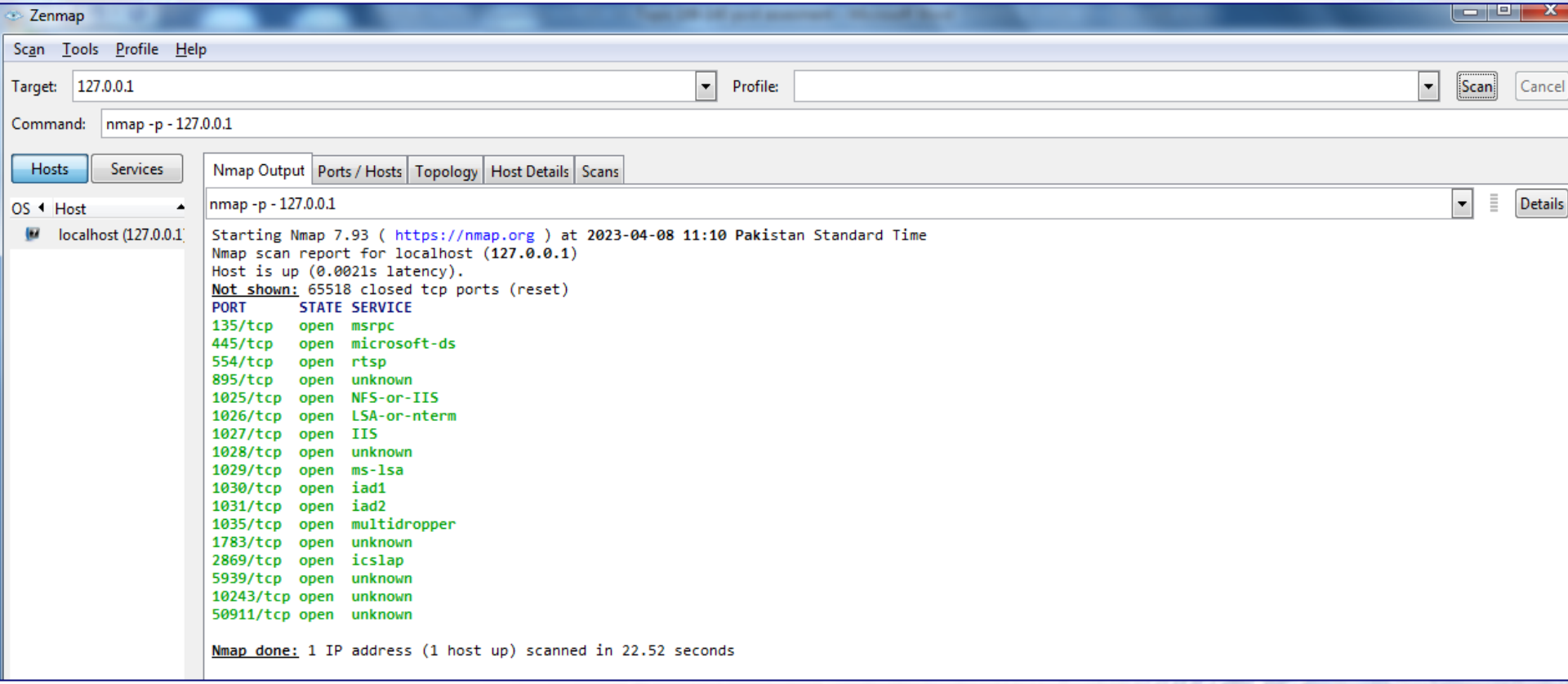
TCP Sequence

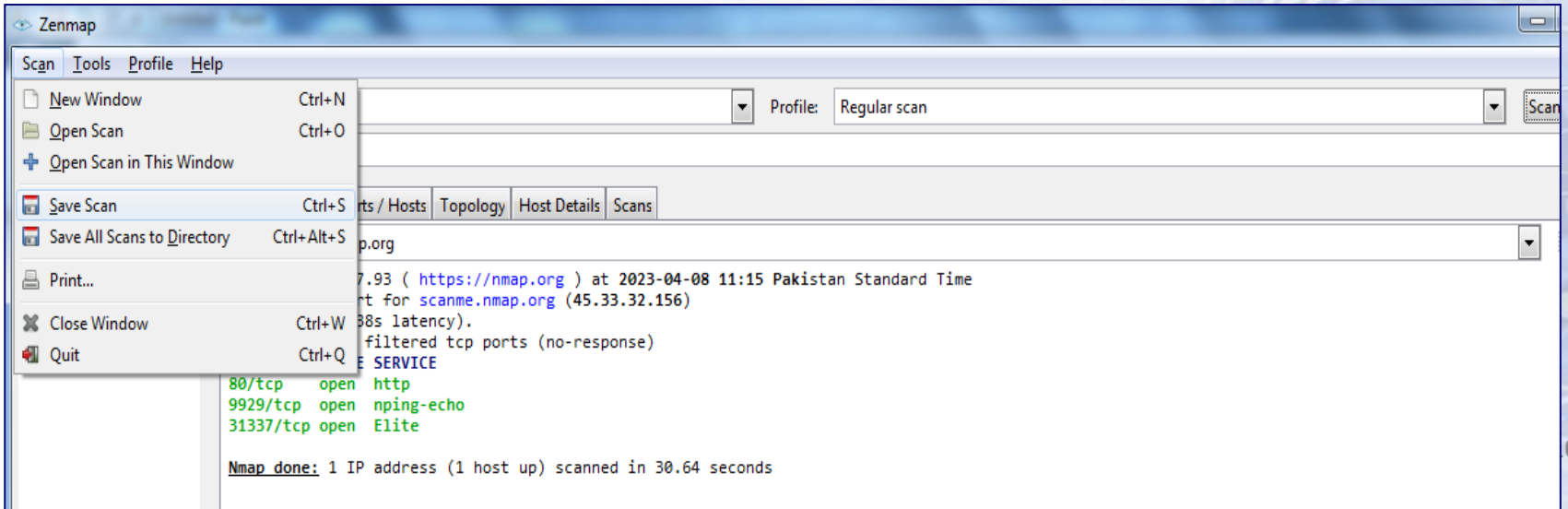

Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Intense scan, all TCP ports
Scan
Command: nmap -p 1-65535 -T4 -A -v scanme.nmap.org
Hosts Services
OS Host
scanme.nmap.org
Nmap Output Ports / Hosts Topology Host Details Scans
nmap -p 1-65535 -T4 -A -v scanme.nmap.org
Not shown: 65532 filtered tcp ports (no-response)
PORT STATE SERVICE VERSION
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
|_http-title: Go ahead and ScanMe!
| http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp open nping-echo Nping echo
31337/tcp open tcpwrapped
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.18 (89%), Linux 2.6.32 (88%), Linux 3.10 (88%), Linux 4.15 - 5.6 (88%), Linux 2.6.32 or 3.10 (88%), Linux 2.6.35 (88%), Linux 3.5 (88%), Linux 4.4 (88%), Linux 4.9 (88%), Synology DiskStation Manager 5.1 (88%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 40.252 days (since Mon Feb 27 04:27:34 2023)
Network Distance: 24 hops
TCP Sequence Prediction: Difficulty=256 (Good luck!)
IP ID Sequence Generation: All zeros
TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 12.00 ms 192.168.167.96
2 21.00 ms XiaoQiang (192.168.31.1)
3 11.00 ms 192.168.100.1
4 12.00 ms 123.108.92.20
5 22.00 ms 119.63.130.252
6 38.00 ms 110.93.252.48
7 32.00 ms 110.93.252.172
8 32.00 ms 110.93.255.164
9 152.00 ms be5685.rcr22.fra06.atlas.cogentco.com (149.14.211.241)
10 155.00 ms be2846.ccr42.fra03.atlas.cogentco.com (154.54.37.29)
11 160.00 ms be2813.ccr41.ams03.atlas.cogentco.com (130.117.0.121)
12 251.00 ms be12488.ccr42.lon13.atlas.cogentco.com (130.117.51.41)
13 243.00 ms be2317.ccr41.jfk02.atlas.cogentco.com (154.54.30.185)
14 249.00 ms be2890.ccr22.cle04.atlas.cogentco.com (154.54.82.245)
15 250.00 ms be2717.ccr41.ord01.atlas.cogentco.com (154.54.6.221)
16 254.00 ms be2832.ccr22.mci01.atlas.cogentco.com (154.54.44.169)


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile:
Scan
Cancel
Command: nmap -p 80 scanme.nmap.org
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS Host
scanme.nmap.org
nmap -p 80 scanme.nmap.org
Details
Starting Nmap 7.93 (https://nmap.org) at 2023-04-08 10:38 Pakistan Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.41s latency).
PORT STATE SERVICE
80/tcp open http
Nmap done: 1 IP address (1 host up) scanned in 2.00 seconds

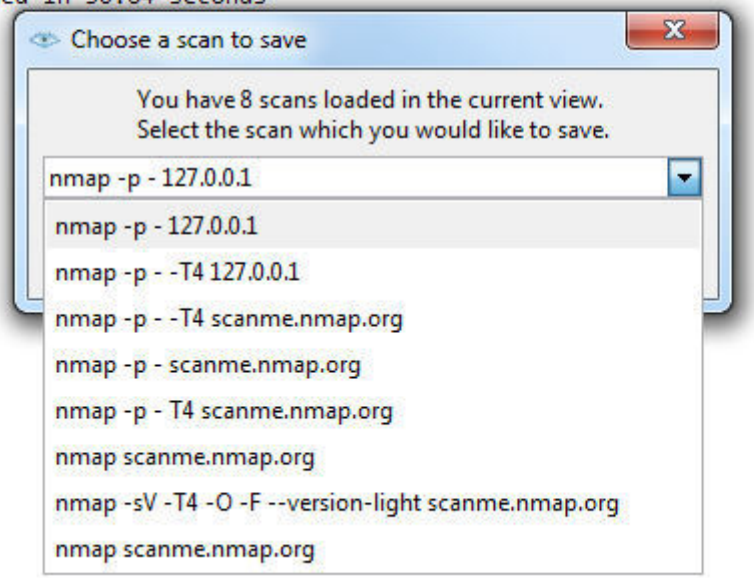

Zenmap
Scan Tools Profile Help
Target: scanme.namp.org
Profile:
Scan
Command: nmap scanme.namp.org -P80
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS Host
scanme.nmap.org
nmap -P 80 scanme.namp.org
Illegal Argument to -P, use -Pn, -PE, -PS, -PA, -PP, -PM, -PU, -PY, or -PO
QUITTING!


Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile:
Scan
Cancel
Command: nmap scanme.nmap.org -P-
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS Host
scanme.nmap.org
nmap -P- scanme.nmap.org
Details
Illegal Argument to -P, use -Pn, -PE, -PS, -PA, -PP, -PM, -PU, -PY, or -PO
QUITTING!


Zenmap
Scan Tools Profile Help
Target: 127.0.0.1
Profile:
Scan
Cancel
Command: nmap -p - 127.0.0.1
Hosts
Services
OS Host
localhost (127.0.0.1)
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -p - 127.0.0.1
Details
Starting Nmap 7.93 (https://nmap.org) at 2023-04-08 11:10 Pakistan Standard Time
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0021s latency).
Not shown: 65518 closed tcp ports (reset)
PORT STATE SERVICE
135/tcp open msrpc
445/tcp open microsoft-ds
554/tcp open rtsp
895/tcp open unknown
1025/tcp open NFS-or-IIS
1026/tcp open LSA-or-nterm
1027/tcp open IIS
1028/tcp open unknown
1029/tcp open ms-lsa
1030/tcp open iad1
1031/tcp open iad2
1035/tcp open multidropper
1783/tcp open unknown
2869/tcp open icslap
5939/tcp open unknown
10243/tcp open unknown
50911/tcp open unknown
Nmap done: 1 IP address (1 host up) scanned in 22.52 seconds

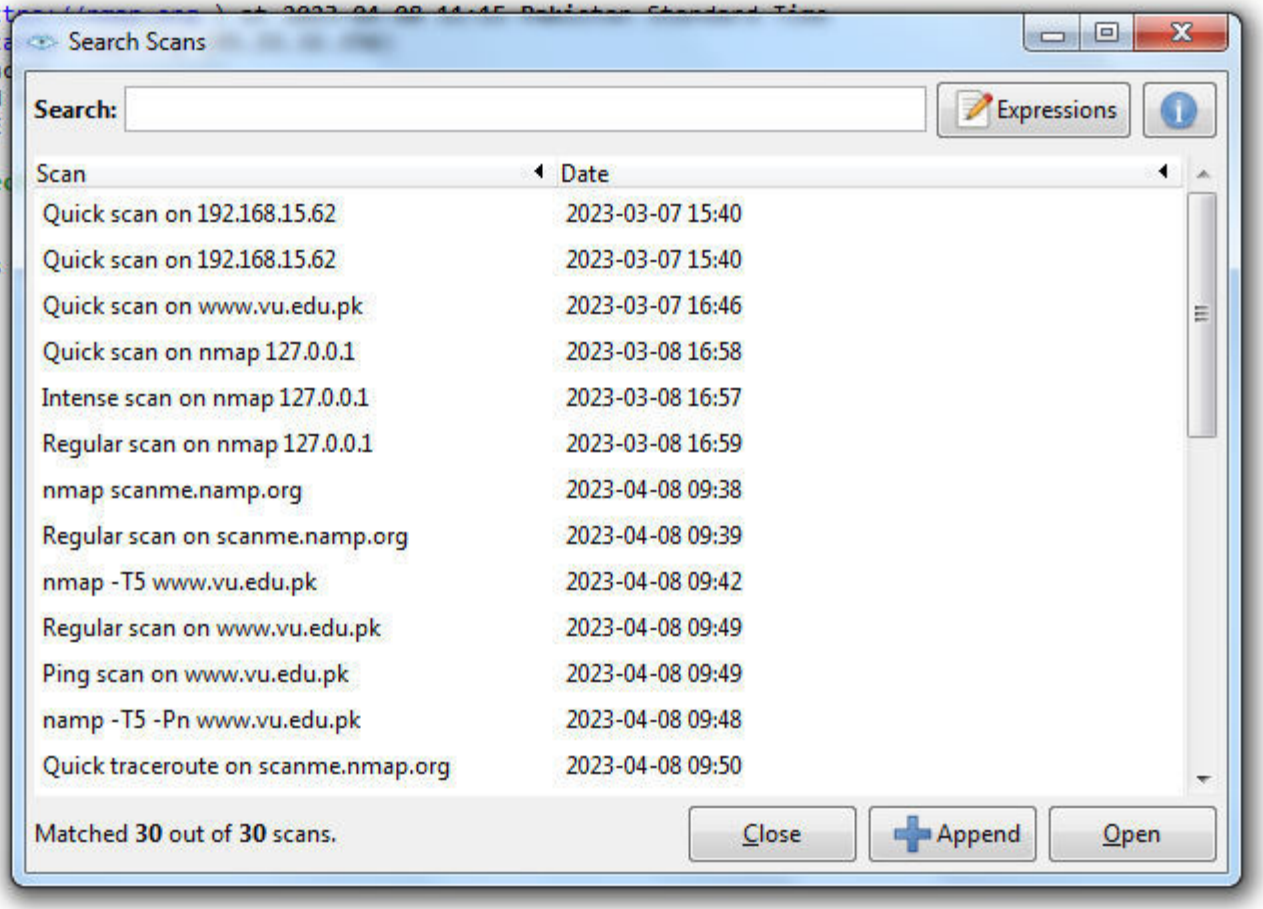

Zenmap
Scan Tools Profile Help
New Window Ctrl+N
Open Scan Ctrl+O
Open Scan in This Window
Save Scan Ctrl+S
Save All Scans to Directory Ctrl+Alt+S
Print...
Close Window Ctrl+W
Quit Ctrl+Q
Profile: Regular scan
Topology
Host Details
Scans
7.93 (https://nmap.org) at 2023-04-08 11:15 Pakistan Standard Time
for scanme.nmap.org (45.33.32.156)
filtered tcp ports (no-response)
SERVICE
80/tcp open http
9929/tcp open nping-echo
31337/tcp open Elite
Nmap done: 1 IP address (1 host up) scanned in 30.64 seconds

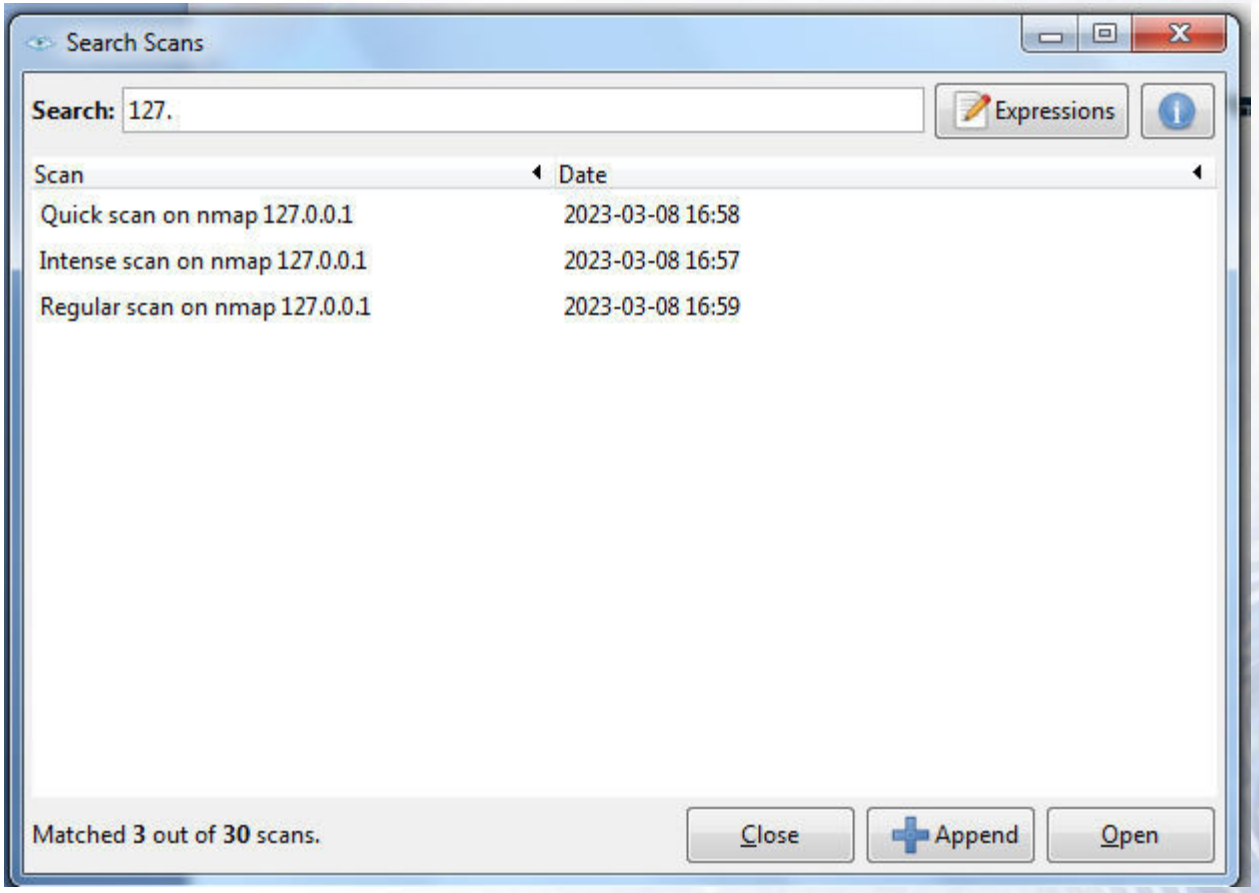
```
nmap scanme.nmap.org

Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-08 11:15 Pakistan Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.38s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 30.64 seconds
```



nmap scanme.nmap.org

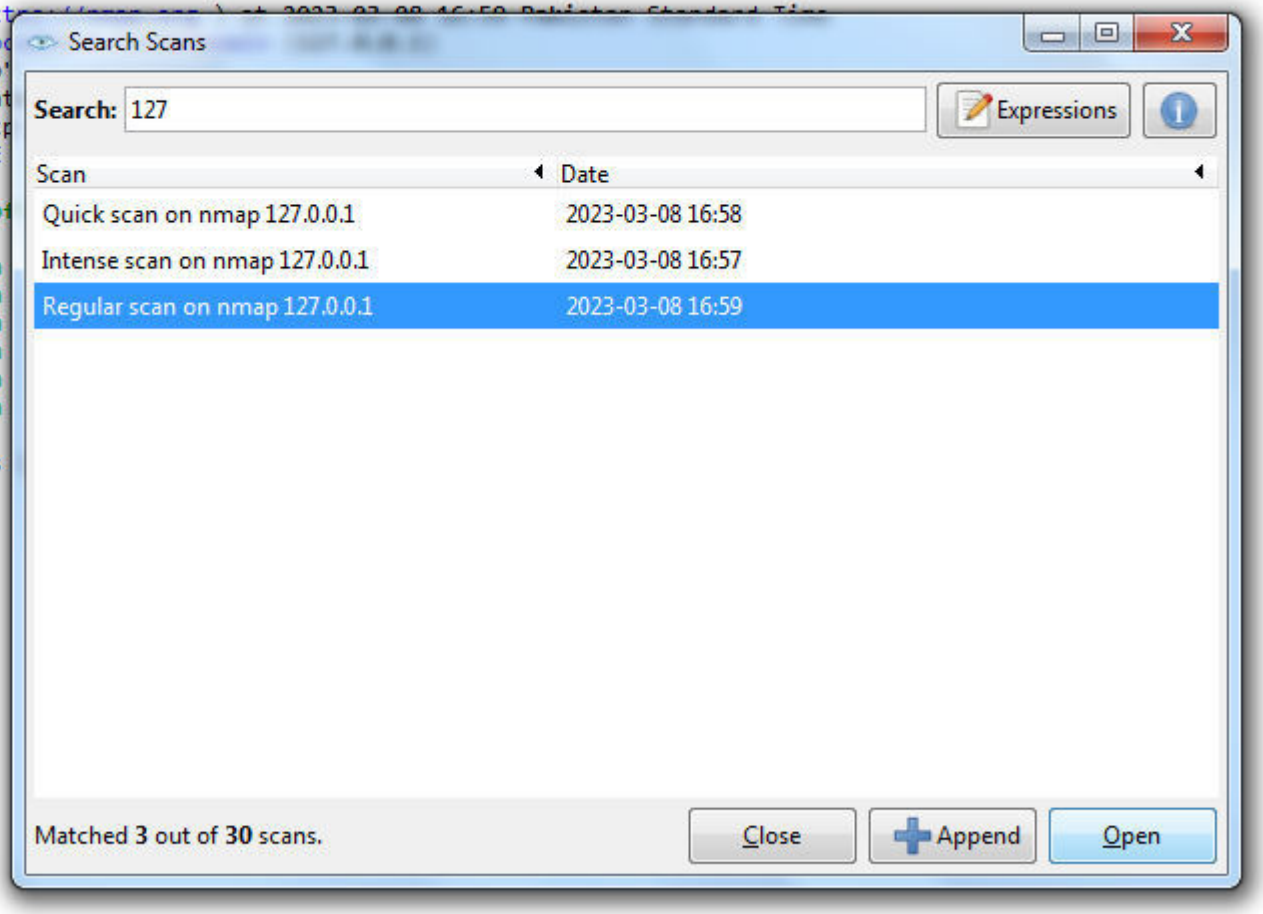
```
Starting Nmap 7.93 ( http
Nmap scan report for sca
Host is up (0.38s laten
Not shown: 997 filtered
PORT      STATE SERVICE
80/tcp    open  http
9929/tcp  open  nping-e
31337/tcp open  Elite

Nmap done: 1 IP address
```

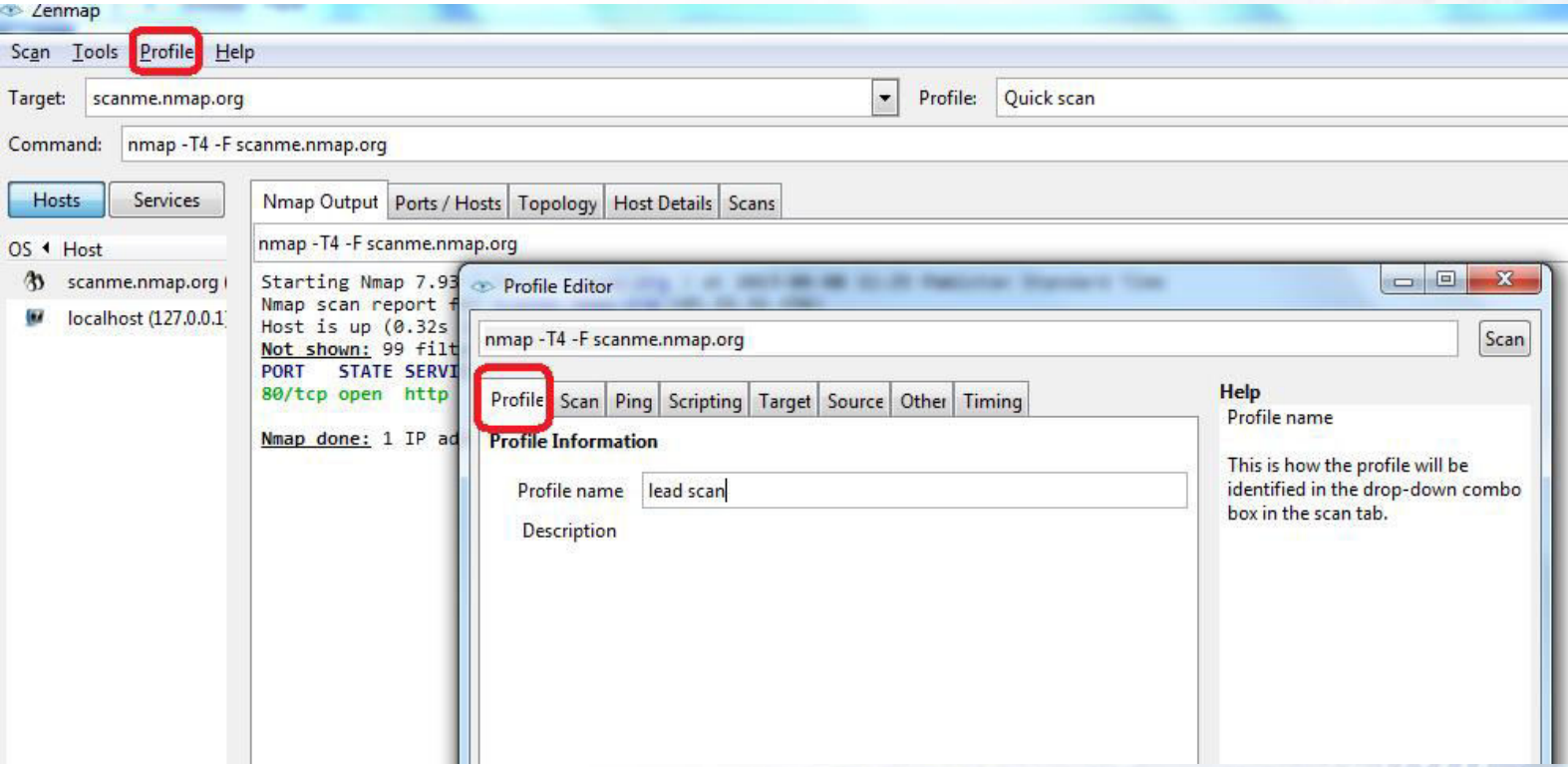


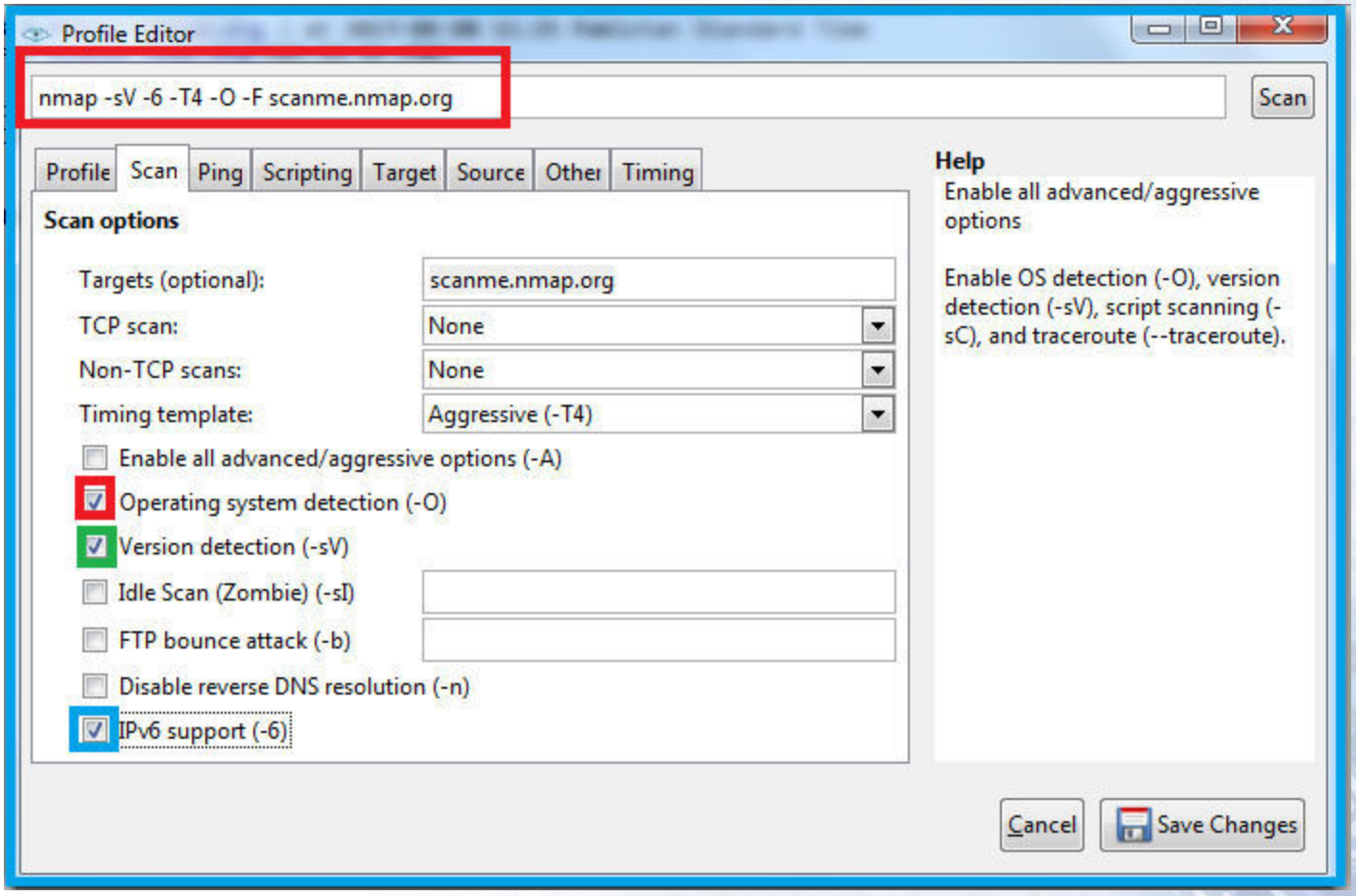

Search Scans
Search: 127.
Expressions
Scan
Date
Quick scan on nmap 127.0.0.1
2023-03-08 16:58
Intense scan on nmap 127.0.0.1
2023-03-08 16:57
Regular scan on nmap 127.0.0.1
2023-03-08 16:59
Matched 3 out of 30 scans.
Close
Append
Open

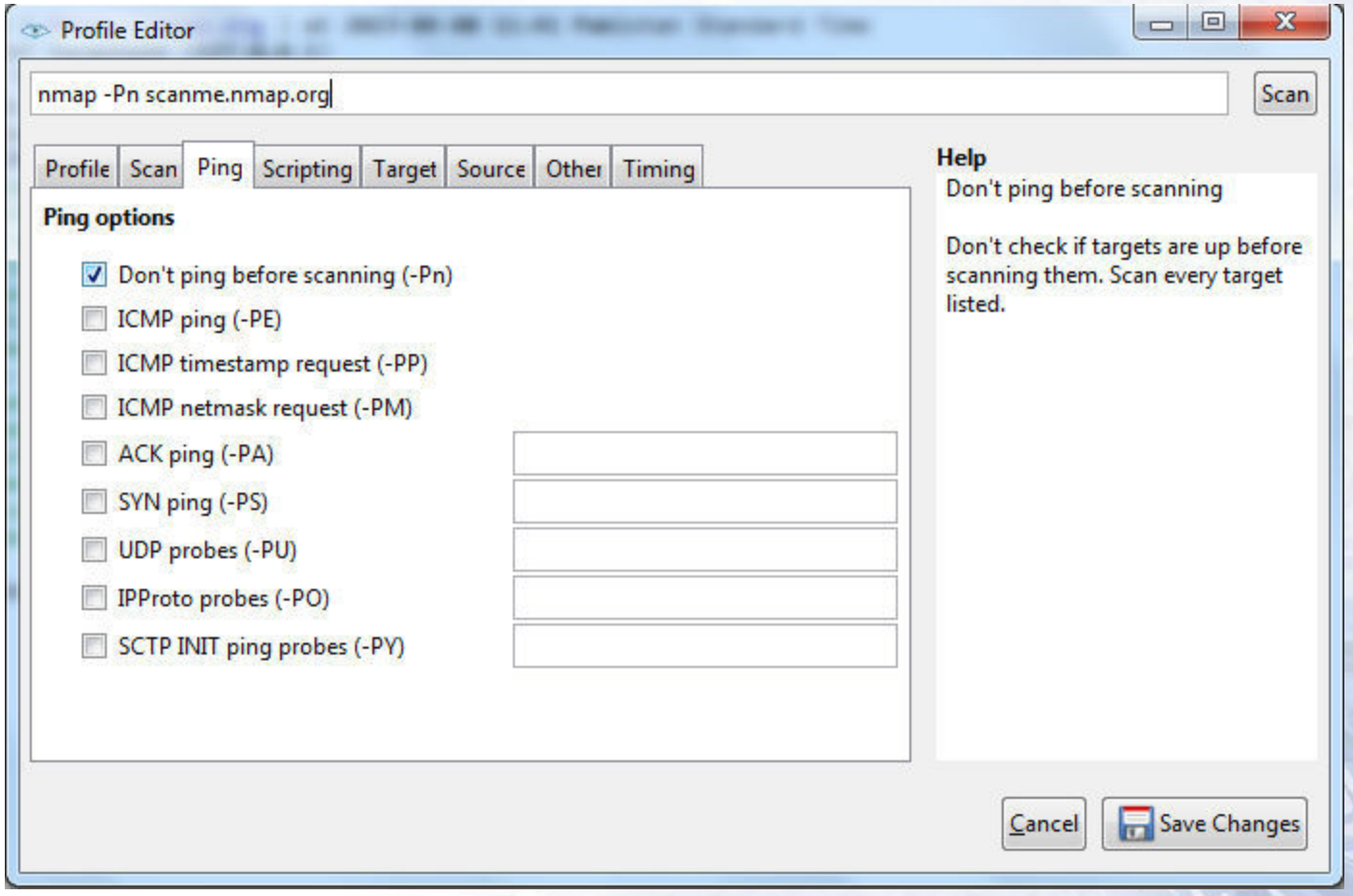
nmap -F nmap 127.0.0.1

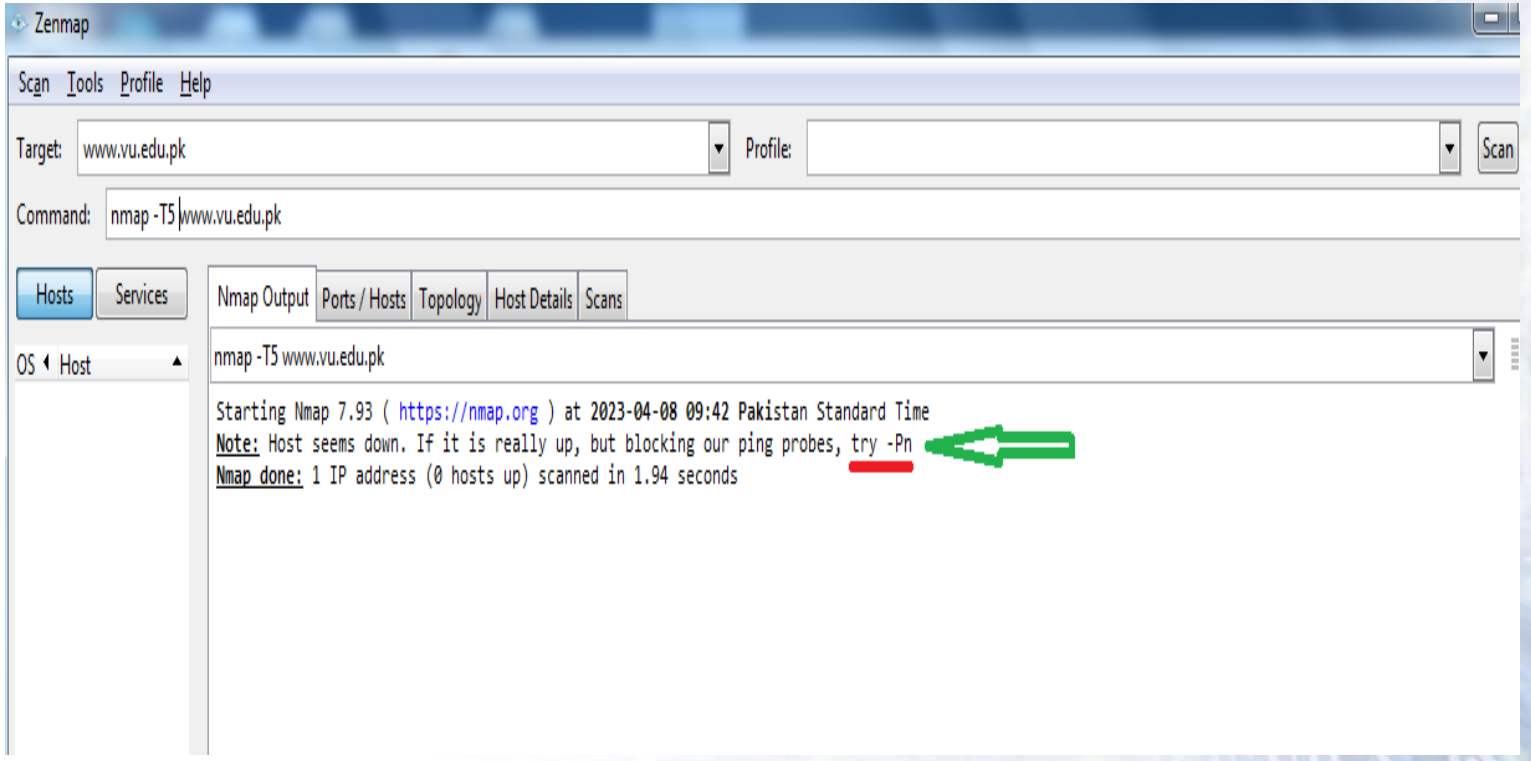
```
Starting Nmap 7.93 ( http
Nmap scan report for loc
Failed to resolve "nmap"
Host is up (0.00015s lat
Not shown: 91 closed tcp
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsof
554/tcp   open  rtsp
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown

Nmap done: 1 IP address
```




Zenmap
Scan Tools Profile Help
Target: scanme.nmap.org
Profile: Quick scan
Scan
Command: nmap -T4 -F scanme.nmap.org
Hosts Services
Nmap Output Ports / Hosts Topology Host Details Scans
OS Host
scanme.nmap.org
localhost (127.0.0.1)
nmap -T4 -F scanme.nmap.org
Starting Nmap 7.93
Nmap scan report f
Host is up (0.32s
Not shown: 99 filt
PORT STATE SERVI
80/tcp open http
Nmap done: 1 IP ad
Profile Editor
nmap -T4 -F scanme.nmap.org
Scan
Profile Scan Ping Scripting Target Source Other Timing
Profile Information
Profile name lead scan
Description
Help
Profile name
This is how the profile will be identified in the drop-down combo box in the scan tab.

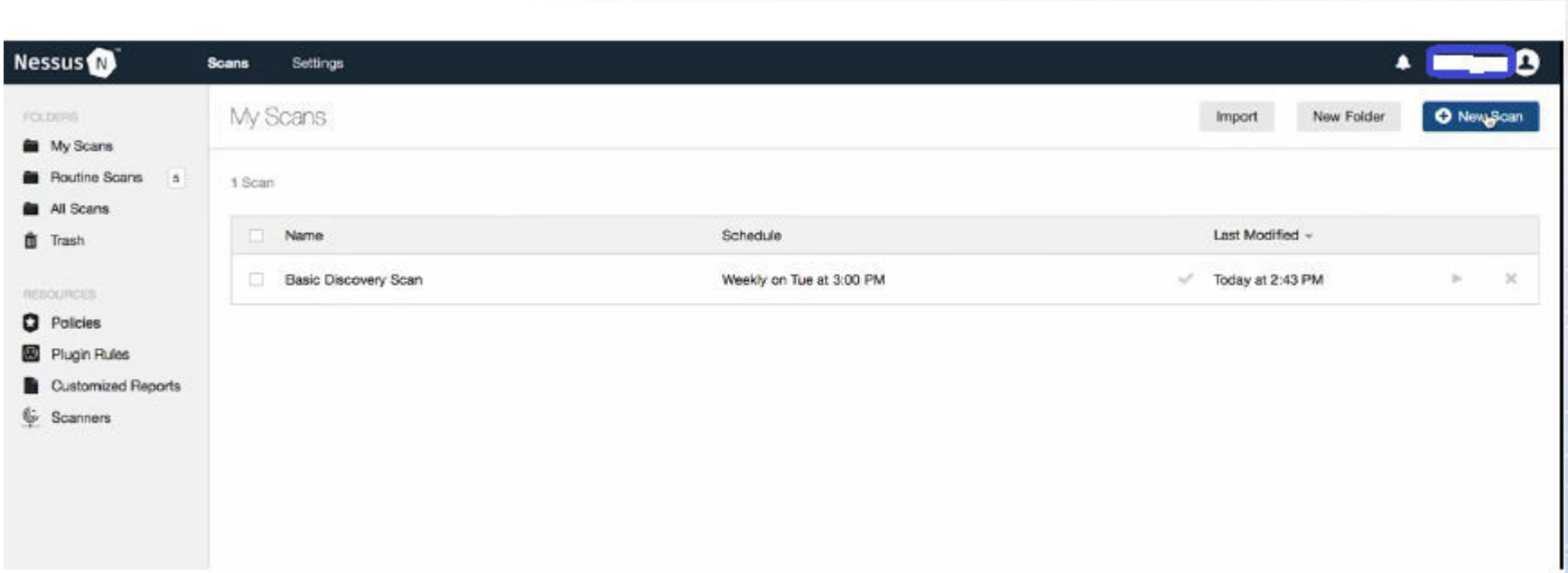

Profile Editor
nmap -sV -6 -T4 -O -F scanme.nmap.org
Scan
Profile
Scan
Ping
Scripting
Target
Source
Other
Timing
Scan options
Targets (optional):
scanme.nmap.org
TCP scan:
None
Non-TCP scans:
None
Timing template:
Aggressive (-T4)
Enable all advanced/aggressive options (-A)
Operating system detection (-O)
Version detection (-sV)
Idle Scan (Zombie) (-sI)
FTP bounce attack (-b)
Disable reverse DNS resolution (-n)
IPv6 support (-6)
Help
Enable all advanced/aggressive options
Enable OS detection (-O), version detection (-sV), script scanning (-sC), and traceroute (--traceroute).
Cancel
Save Changes

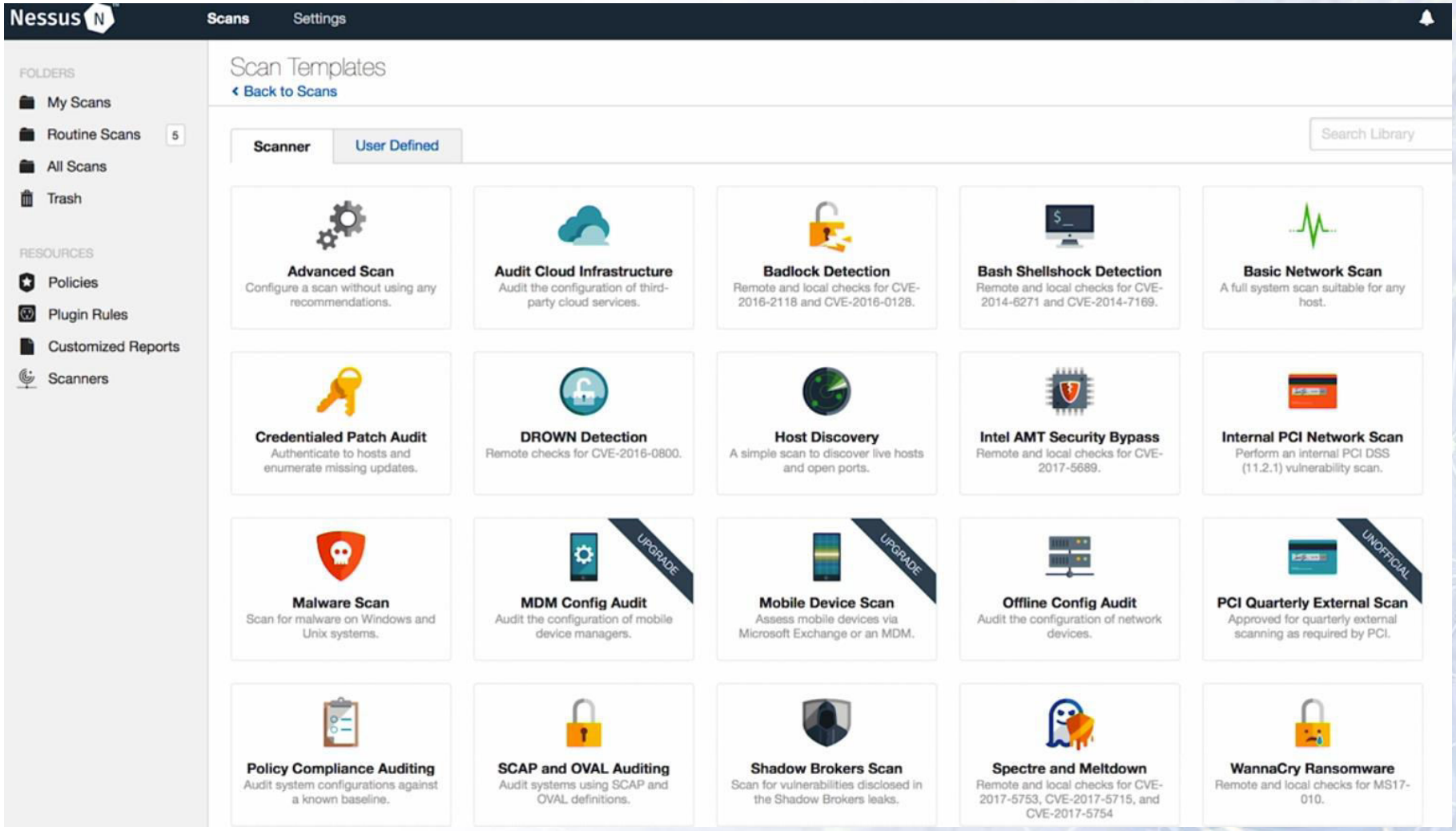

Profile Editor
nmap -Pn scanme.nmap.org
Scan
Profile
Scan
Ping
Scripting
Target
Source
Other
Timing
Ping options
Don't ping before scanning (-Pn)
ICMP ping (-PE)
ICMP timestamp request (-PP)
ICMP netmask request (-PM)
ACK ping (-PA)
SYN ping (-PS)
UDP probes (-PU)
IPProto probes (-PO)
SCTP INIT ping probes (-PY)
Help
Don't ping before scanning
Don't check if targets are up before scanning them. Scan every target listed.
Cancel
Save Changes

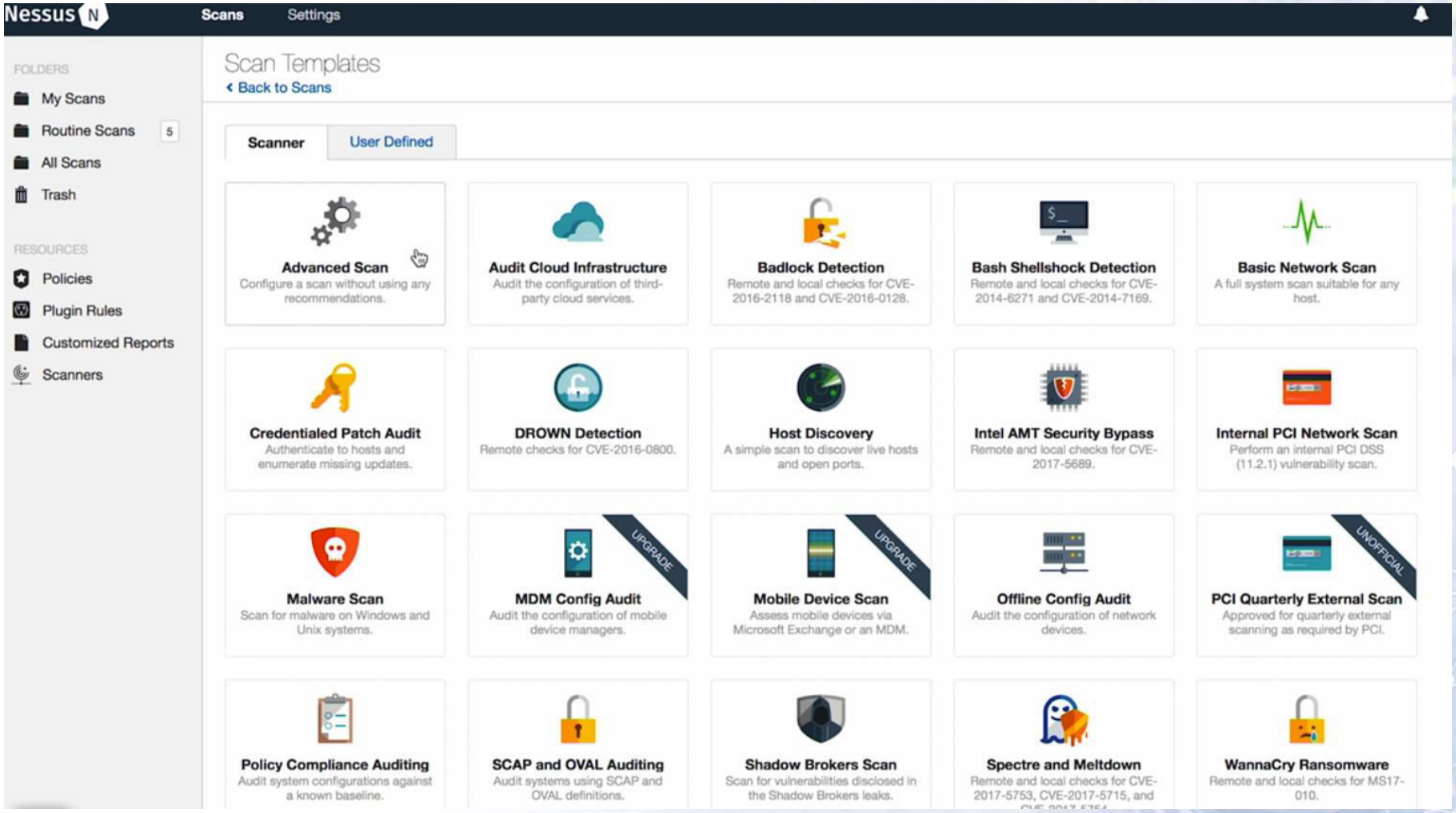

Zenmap
Scan Tools Profile Help
Target: www.vu.edu.pk
Profile:
Scan
Command: nmap -T5 www.vu.edu.pk
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS
Host
nmap -T5 www.vu.edu.pk
Starting Nmap 7.93 (https://nmap.org) at 2023-04-08 09:42 Pakistan Standard Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.94 seconds

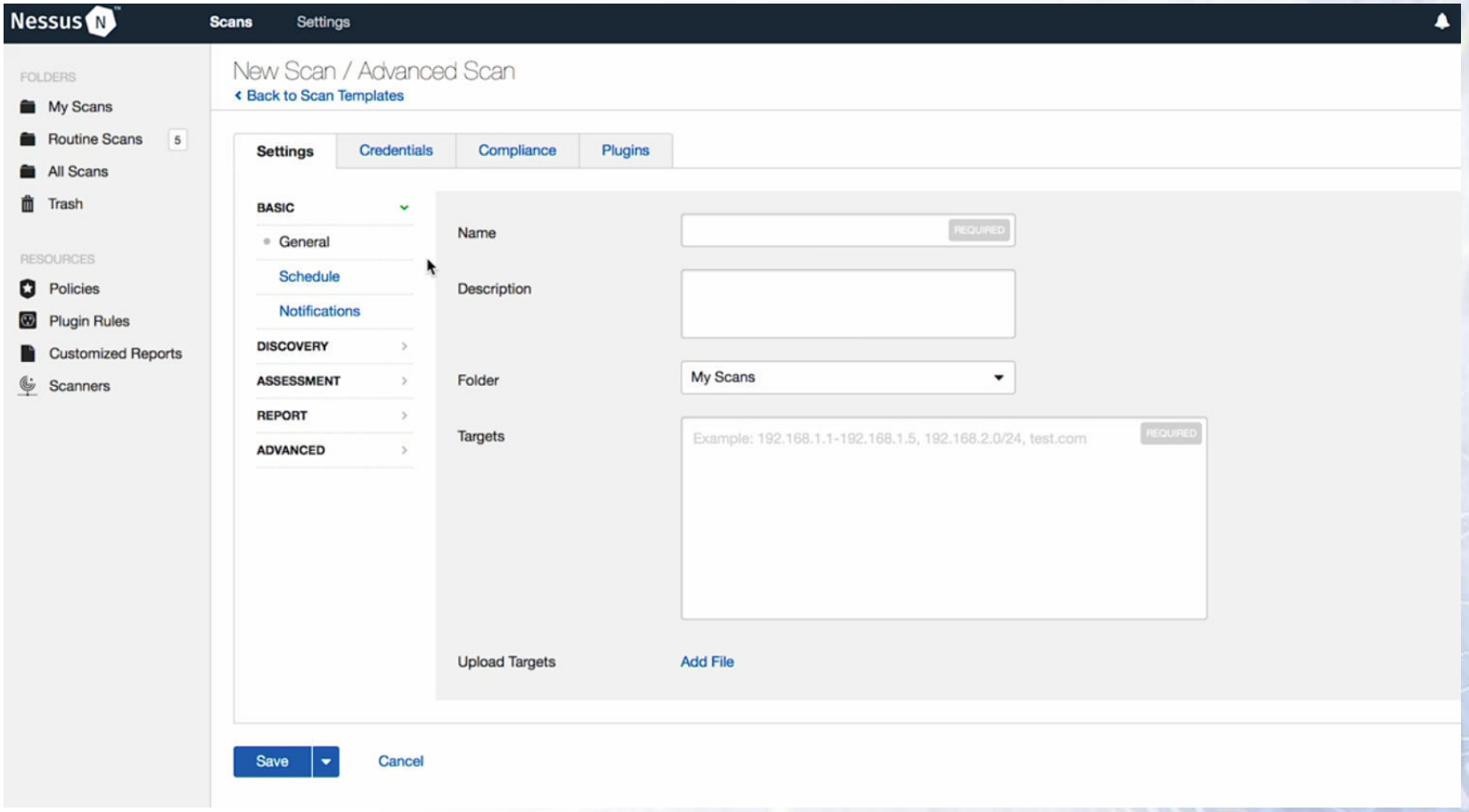
Week 7 Topic 11

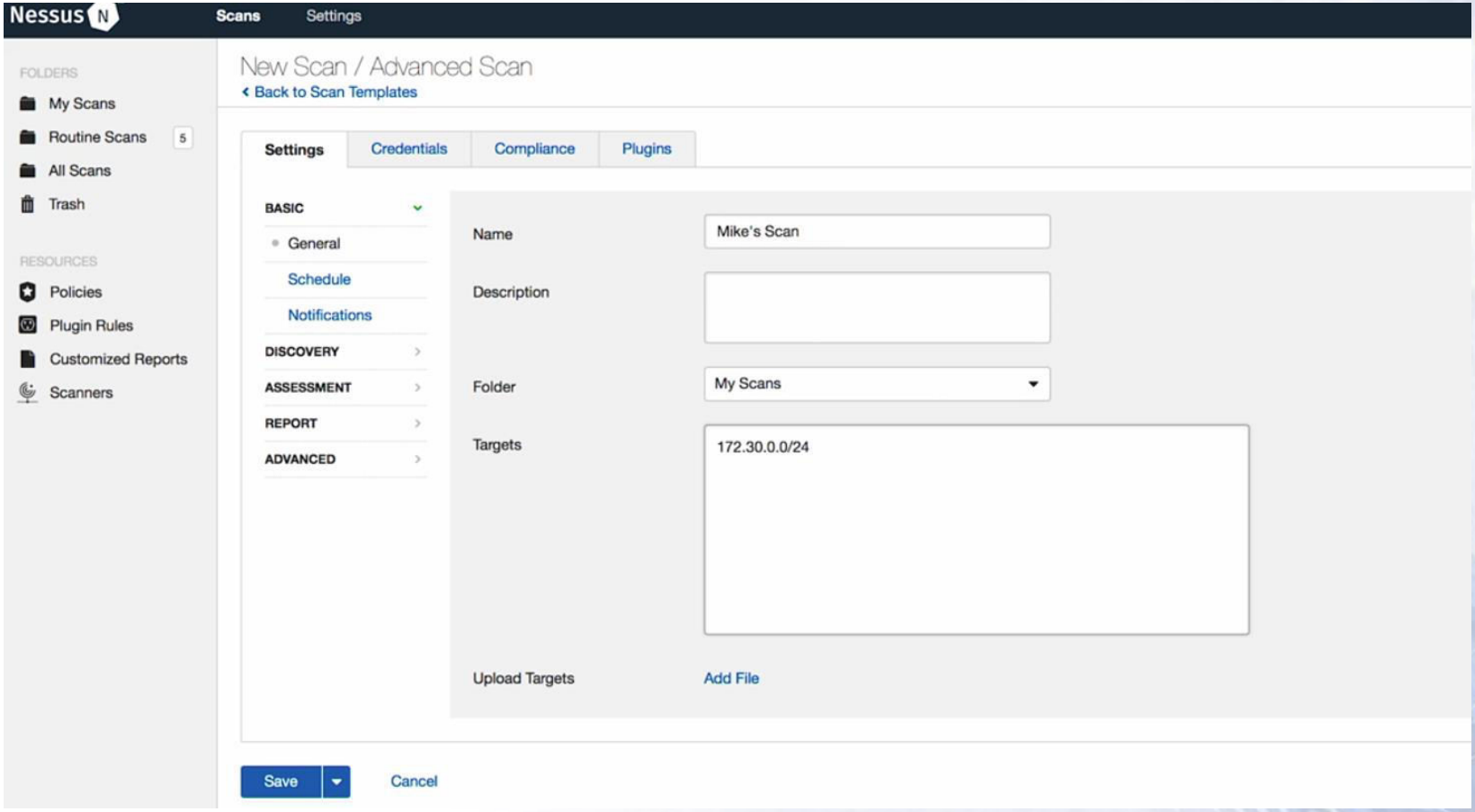
Nessus New scan

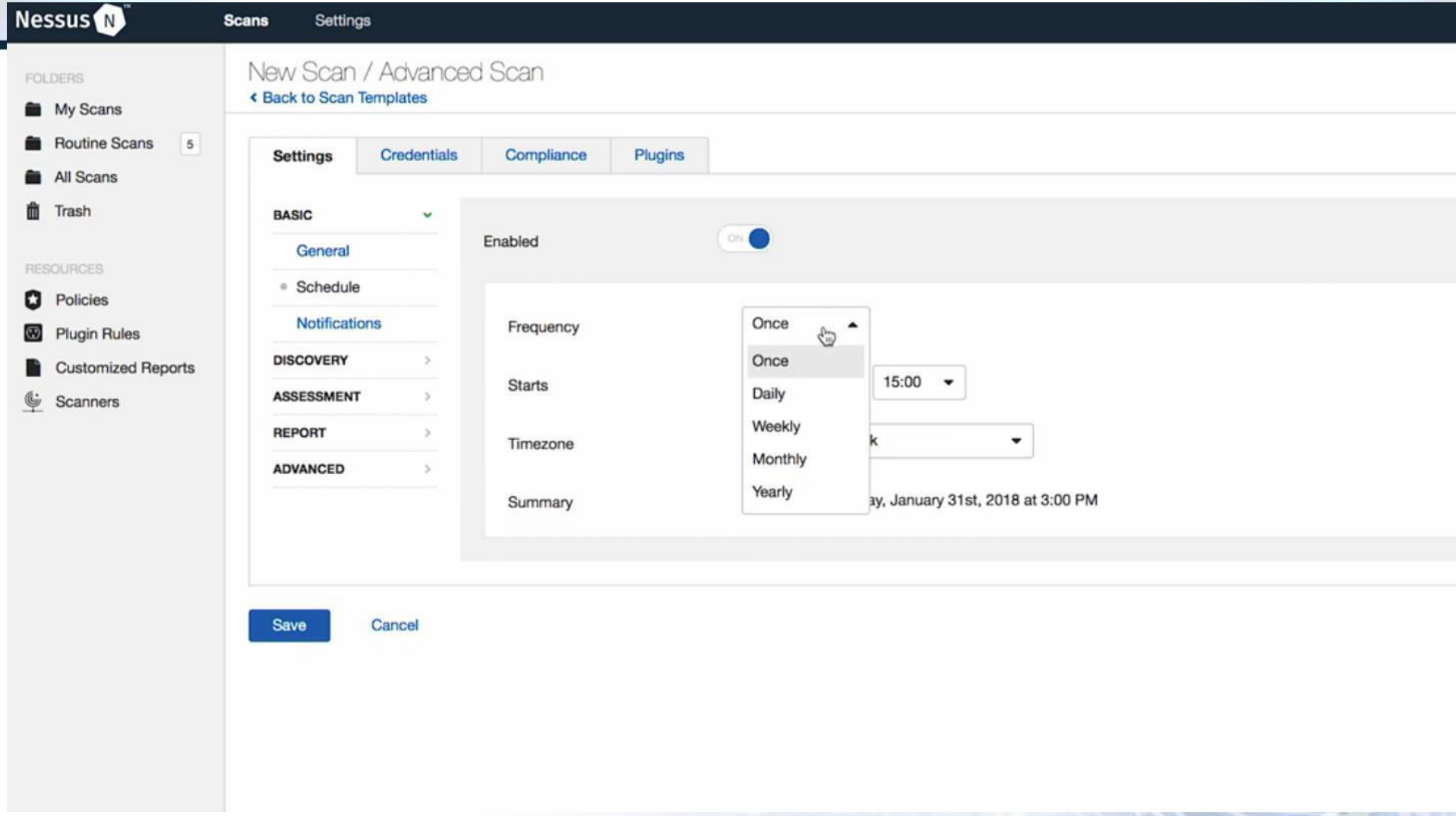


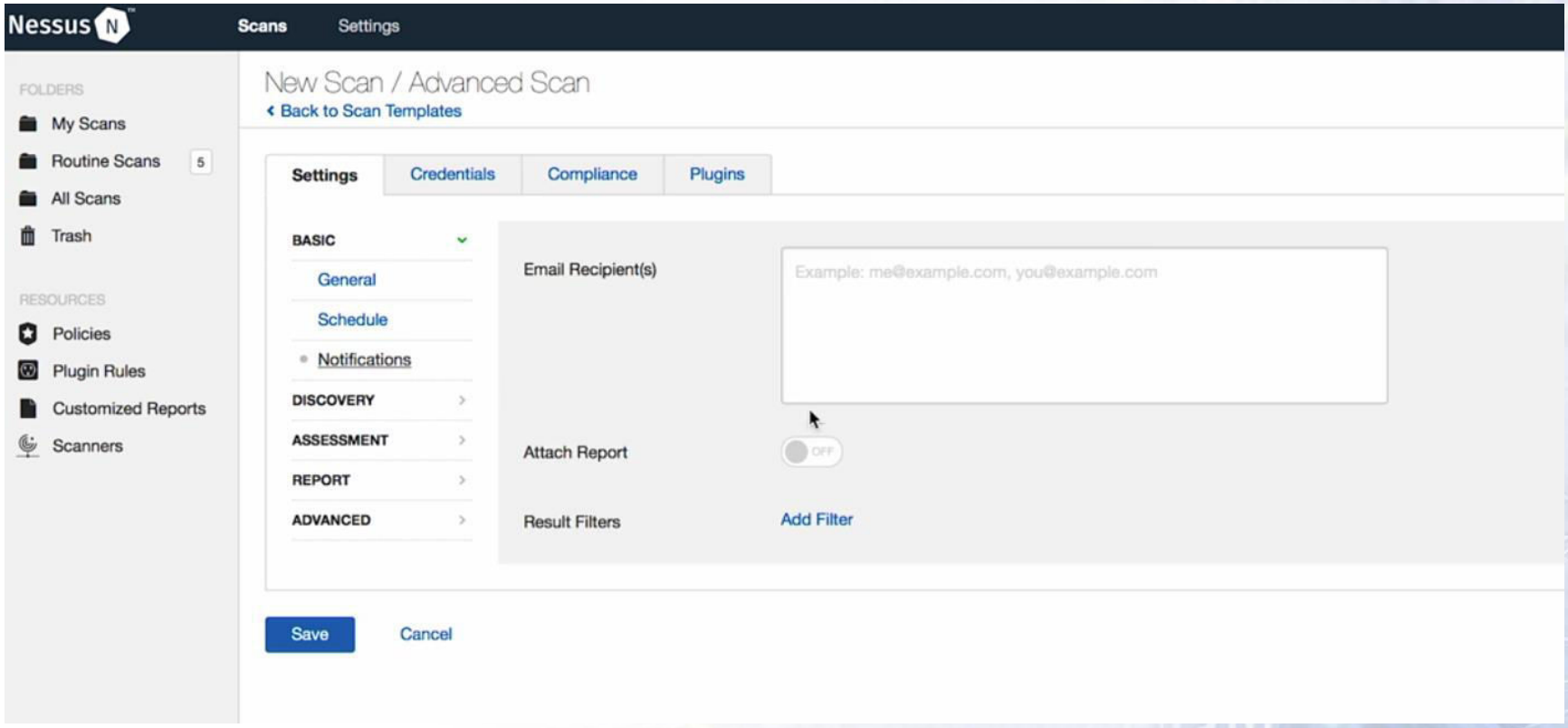

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Scan Templates
Back to Scans
Scanner
User Defined
Search Library
Advanced Scan
Configure a scan without using any recommendations.
Audit Cloud Infrastructure
Audit the configuration of third-party cloud services.
Badlock Detection
Remote and local checks for CVE-2016-2118 and CVE-2016-0128.
Bash Shellshock Detection
Remote and local checks for CVE-2014-6271 and CVE-2014-7169.
Basic Network Scan
A full system scan suitable for any host.
Credentialed Patch Audit
Authenticate to hosts and enumerate missing updates.
DROWN Detection
Remote checks for CVE-2016-0800.
Host Discovery
A simple scan to discover live hosts and open ports.
Intel AMT Security Bypass
Remote and local checks for CVE-2017-5689.
Internal PCI Network Scan
Perform an internal PCI DSS (11.2.1) vulnerability scan.
Malware Scan
Scan for malware on Windows and Unix systems.
UPGRADE
MDM Config Audit
Audit the configuration of mobile device managers.
UPGRADE
Mobile Device Scan
Assess mobile devices via Microsoft Exchange or an MDM.
Offline Config Audit
Audit the configuration of network devices.
UNOFFICIAL
PCI Quarterly External Scan
Approved for quarterly external scanning as required by PCI.
Policy Compliance Auditing
Audit system configurations against a known baseline.
SCAP and OVAL Auditing
Audit systems using SCAP and OVAL definitions.
Shadow Brokers Scan
Scan for vulnerabilities disclosed in the Shadow Brokers leaks.
Spectre and Meltdown
Remote and local checks for CVE-2017-5753, CVE-2017-5715, and CVE-2017-5754
WannaCry Ransomware
Remote and local checks for MS17-010.

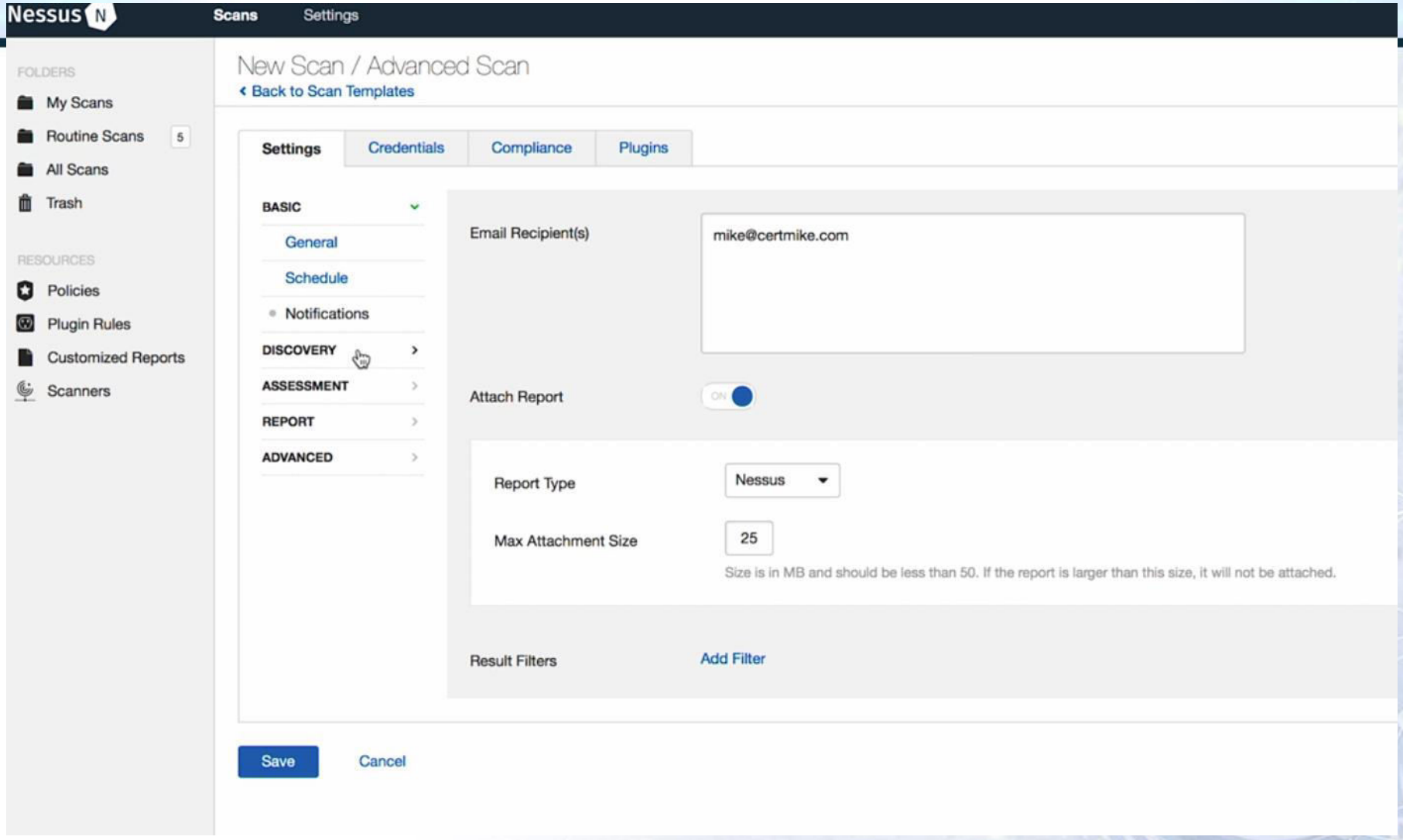

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Scan Templates
Back to Scans
Scanner
User Defined
Advanced Scan
Configure a scan without using any recommendations.
Audit Cloud Infrastructure
Audit the configuration of third-party cloud services.
Badlock Detection
Remote and local checks for CVE-2016-2118 and CVE-2016-0128.
Bash Shellshock Detection
Remote and local checks for CVE-2014-6271 and CVE-2014-7169.
Basic Network Scan
A full system scan suitable for any host.
Credentialed Patch Audit
Authenticate to hosts and enumerate missing updates.
DROWN Detection
Remote checks for CVE-2016-0800.
Host Discovery
A simple scan to discover live hosts and open ports.
Intel AMT Security Bypass
Remote and local checks for CVE-2017-5689.
Internal PCI Network Scan
Perform an internal PCI DSS (11.2.1) vulnerability scan.
Malware Scan
Scan for malware on Windows and Unix systems.
MDM Config Audit
Audit the configuration of mobile device managers.
UPGRADE
Mobile Device Scan
Assess mobile devices via Microsoft Exchange or an MDM.
UPGRADE
Offline Config Audit
Audit the configuration of network devices.
PCI Quarterly External Scan
Approved for quarterly external scanning as required by PCI.
UNOFFICIAL
Policy Compliance Auditing
Audit system configurations against a known baseline.
SCAP and OVAL Auditing
Audit systems using SCAP and OVAL definitions.
Shadow Brokers Scan
Scan for vulnerabilities disclosed in the Shadow Brokers leaks.
Spectre and Meltdown
Remote and local checks for CVE-2017-5753, CVE-2017-5715, and
WannaCry Ransomware
Remote and local checks for MS17-010.

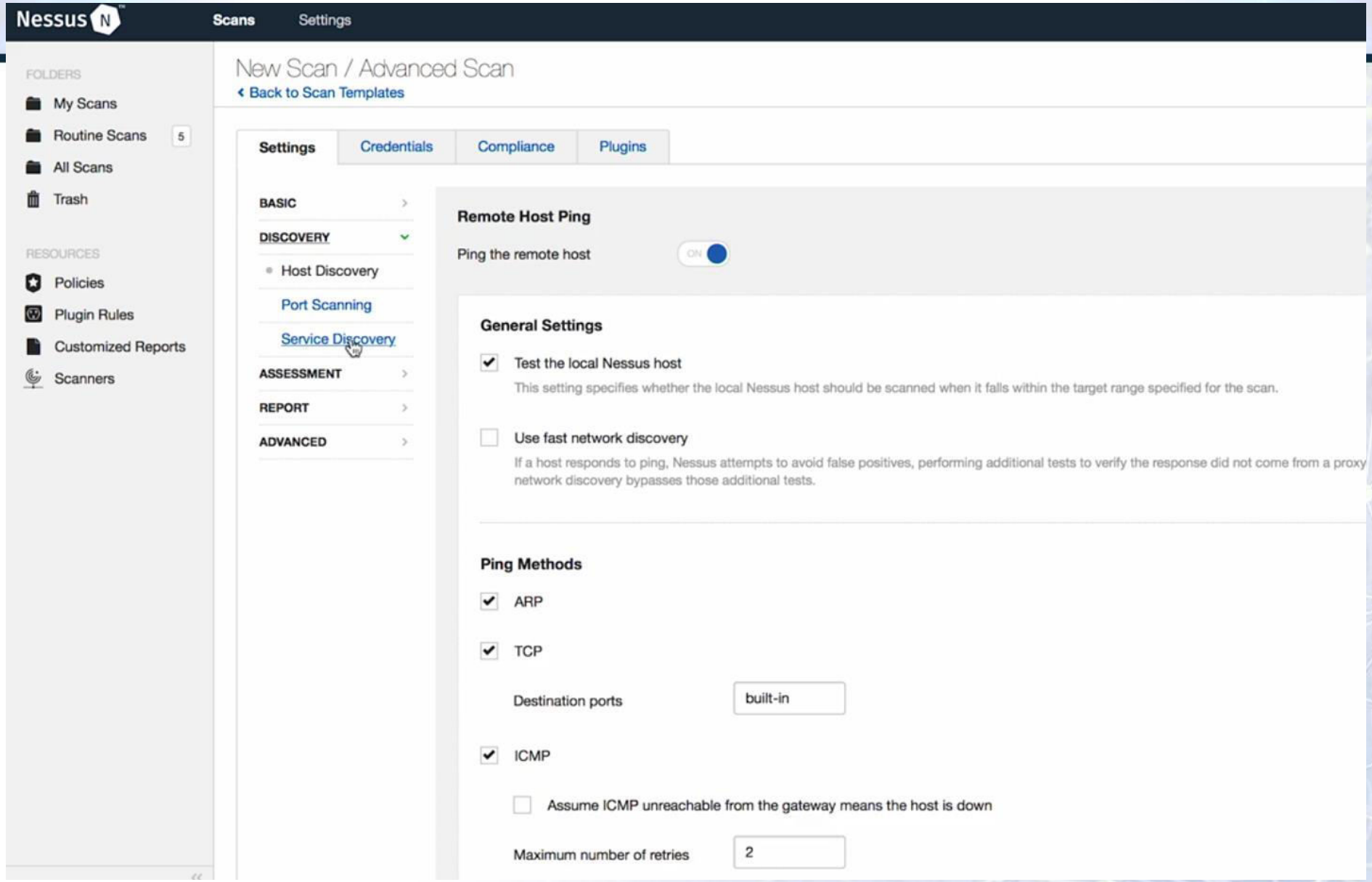

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
General
Schedule
Notifications
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
Name
REQUIRED
Description
Folder
My Scans
Targets
Example: 192.168.1.1-192.168.1.5, 192.168.2.0/24, test.com
REQUIRED
Upload Targets
Add File
Save
Cancel

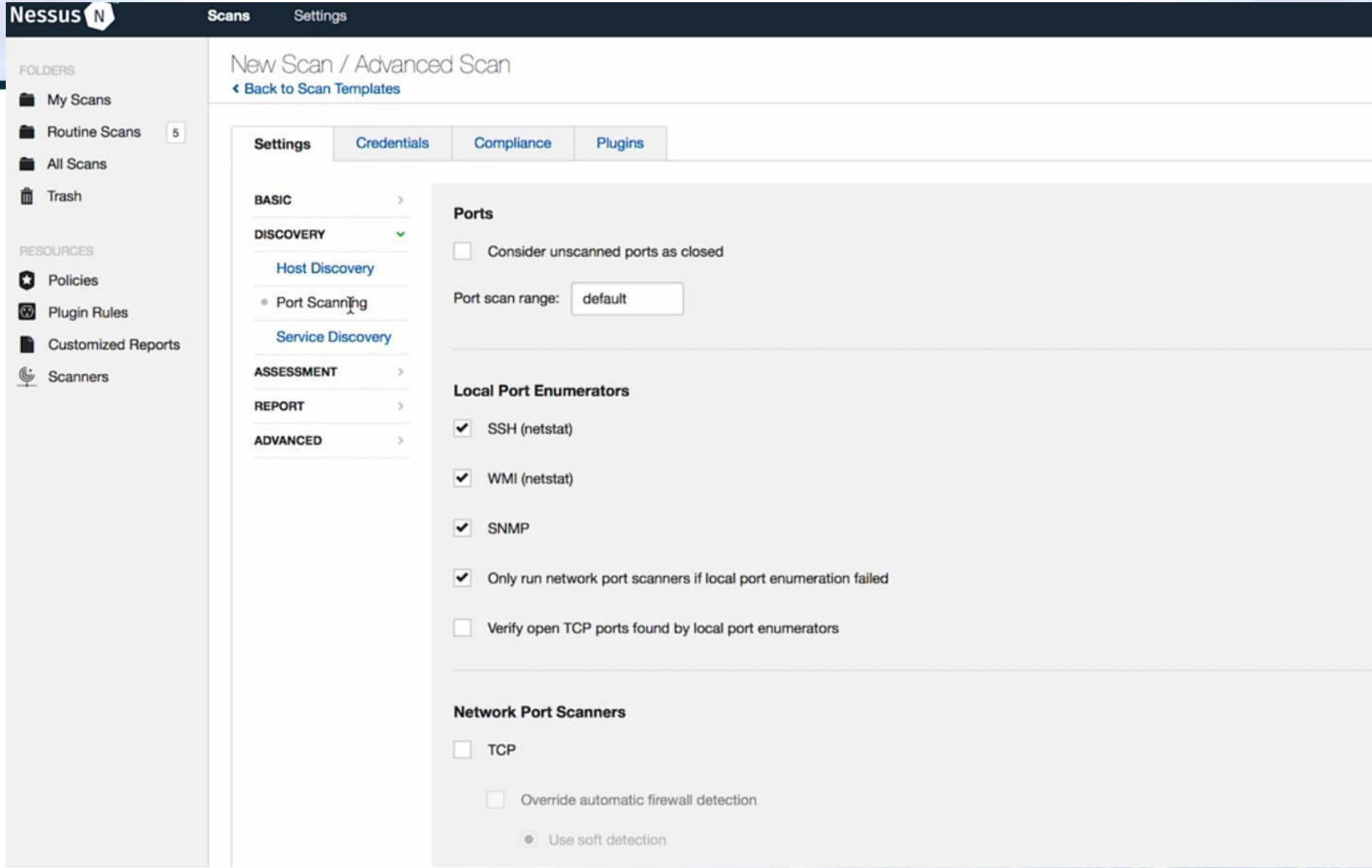

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
General
Schedule
Notifications
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
Name
Mike's Scan
Description
Folder
My Scans
Targets
172.30.0.0/24
Upload Targets
Add File
Save
Cancel

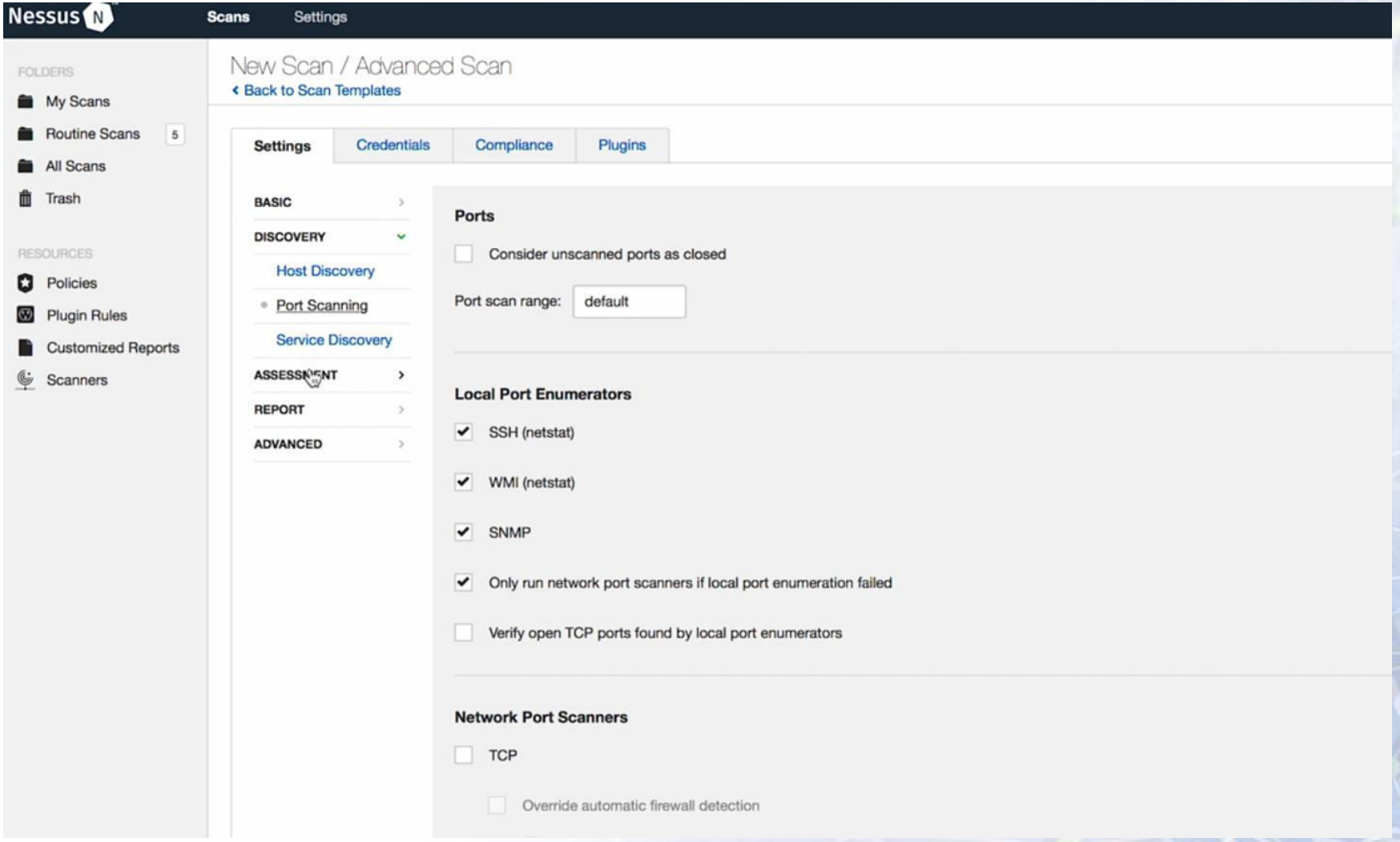

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
General
Schedule
Notifications
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
Enabled
ON
Frequency
Once
Once
Daily
Weekly
Monthly
Yearly
Starts
15:00
Timezone
Summary
ay, January 31st, 2018 at 3:00 PM
Save
Cancel

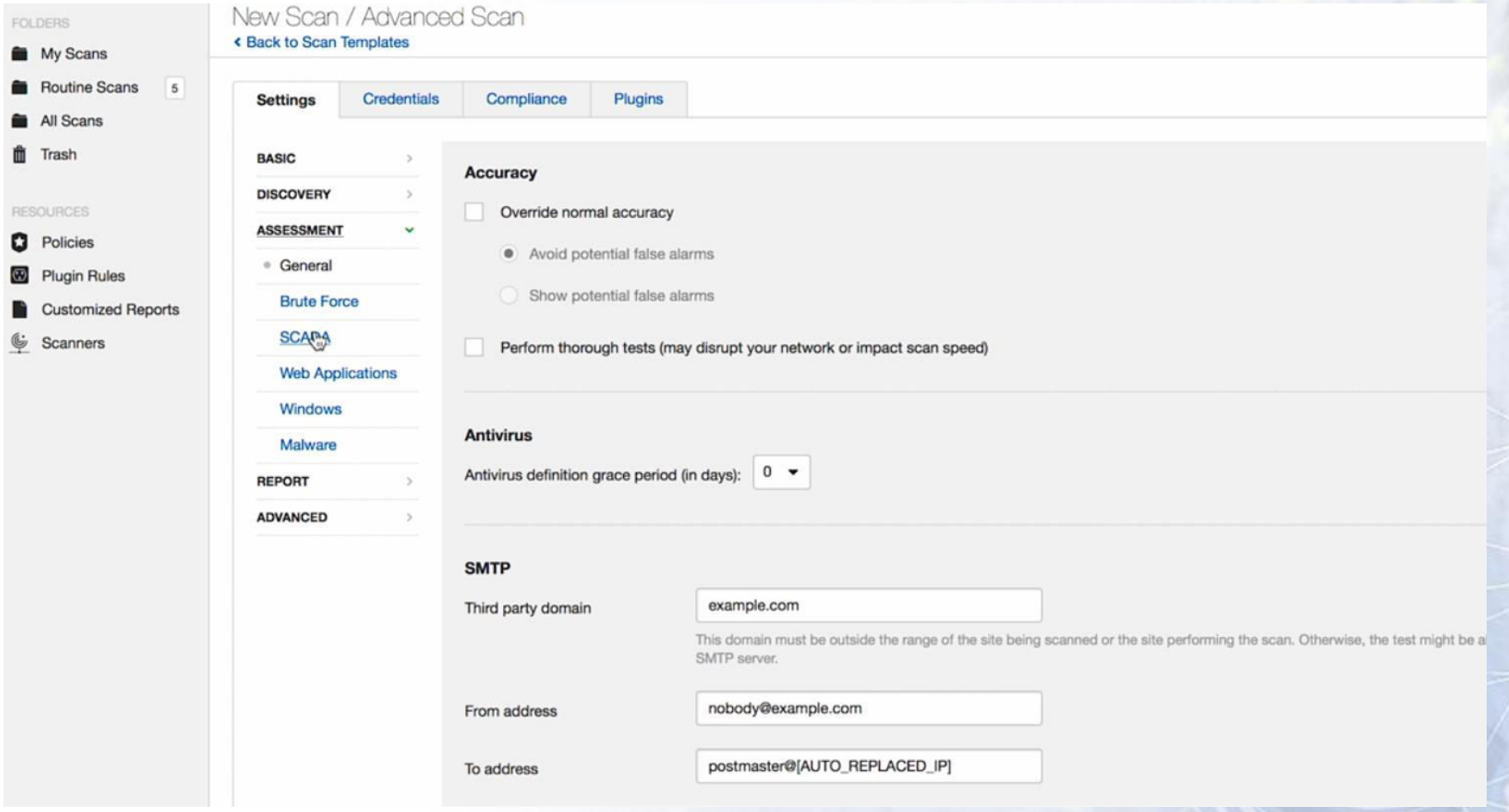

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
General
Schedule
Notifications
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
Email Recipient(s)
Example: me@example.com, you@example.com
Attach Report
OFF
Result Filters
Add Filter
Save
Cancel

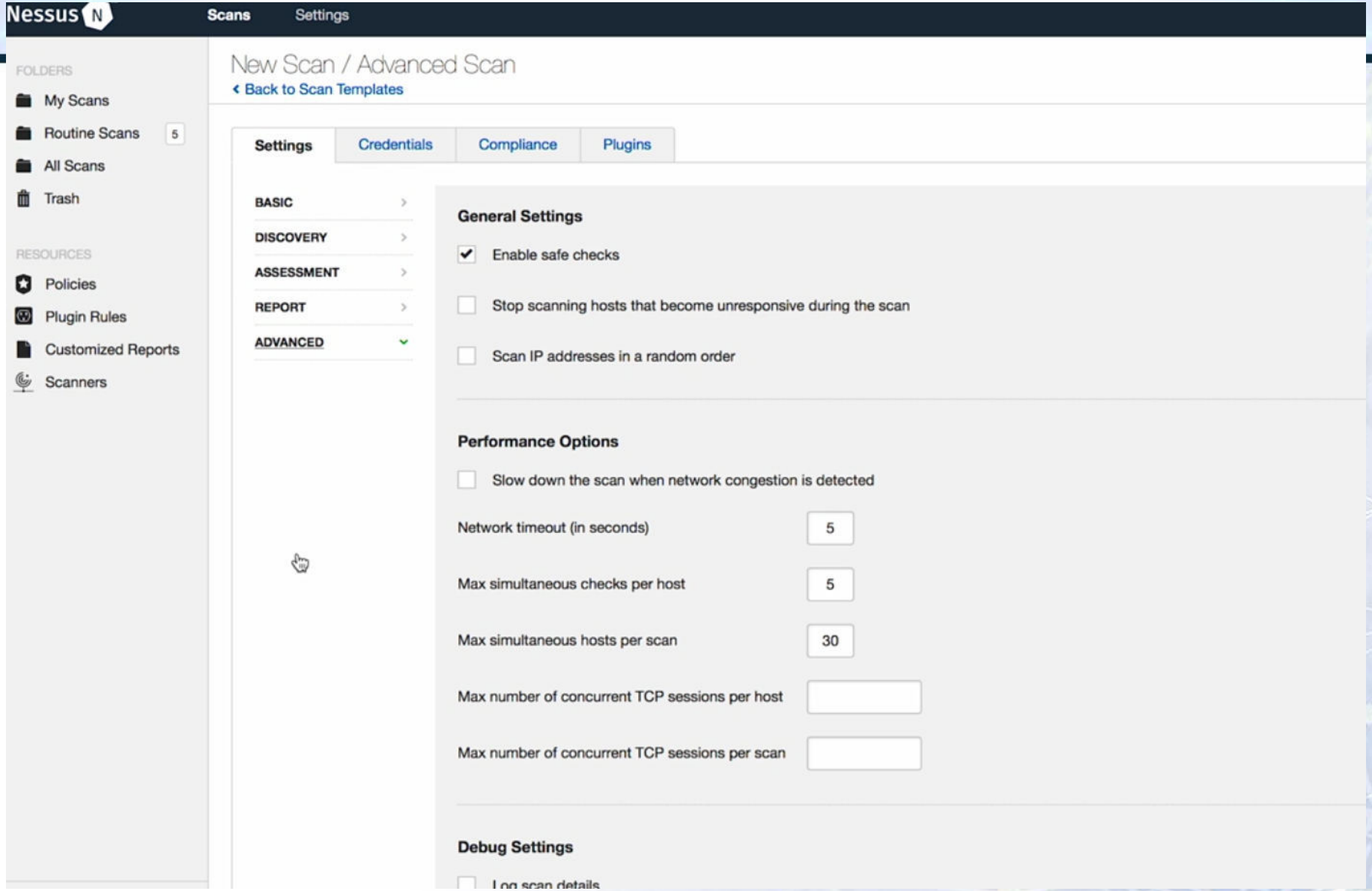

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
General
Schedule
Notifications
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
Email Recipient(s)
mike@certmike.com
Attach Report
ON
Report Type
Nessus
Max Attachment Size
25
Size is in MB and should be less than 50. If the report is larger than this size, it will not be attached.
Result Filters
Add Filter
Save
Cancel

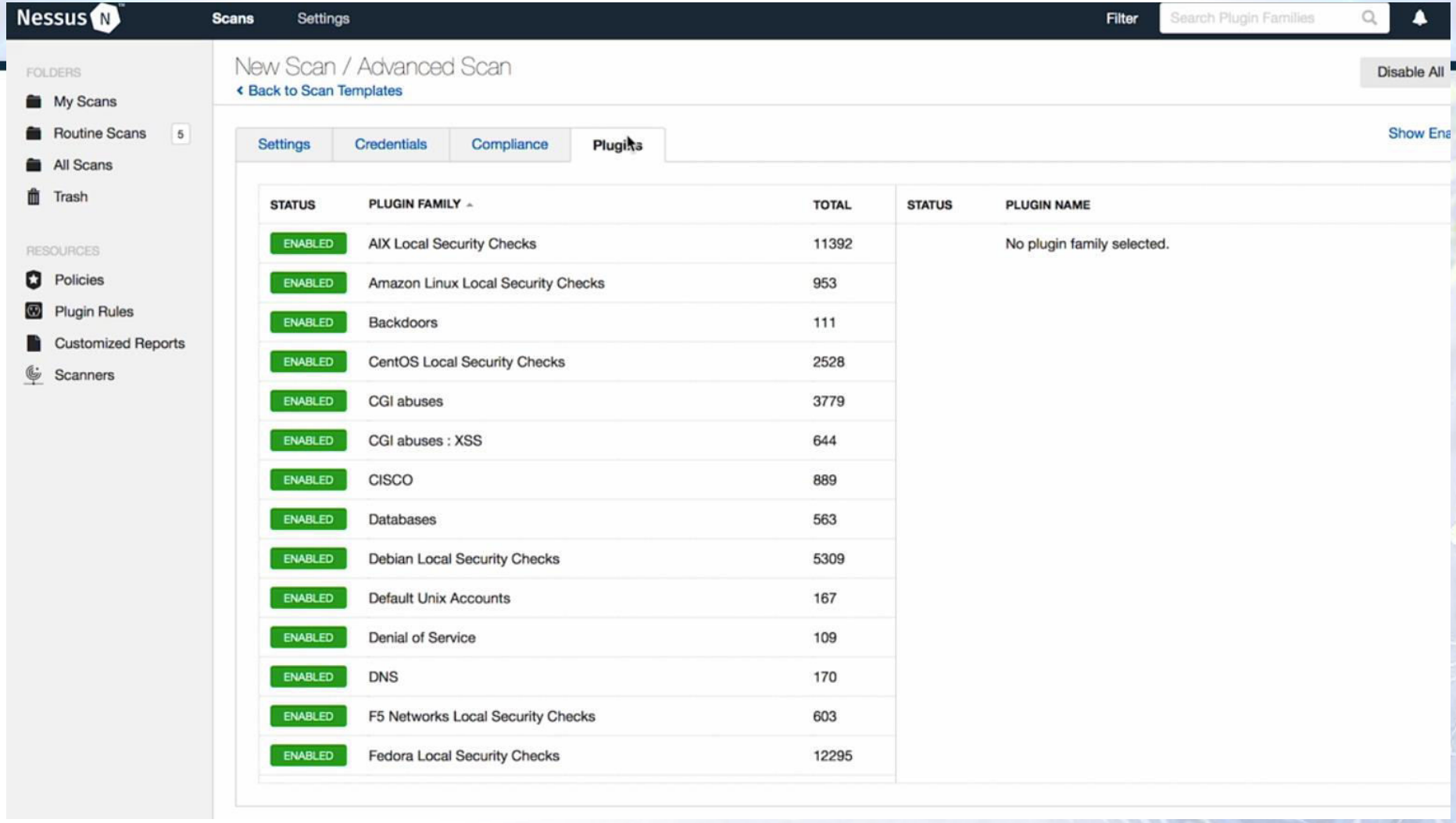

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
DISCOVERY
Host Discovery
Port Scanning
Service Discovery
ASSESSMENT
REPORT
ADVANCED
Remote Host Ping
Ping the remote host
ON
General Settings
Test the local Nessus host
This setting specifies whether the local Nessus host should be scanned when it falls within the target range specified for the scan.
Use fast network discovery
If a host responds to ping, Nessus attempts to avoid false positives, performing additional tests to verify the response did not come from a proxy
network discovery bypasses those additional tests.
Ping Methods
ARP
TCP
Destination ports
built-in
ICMP
Assume ICMP unreachable from the gateway means the host is down
Maximum number of retries
2

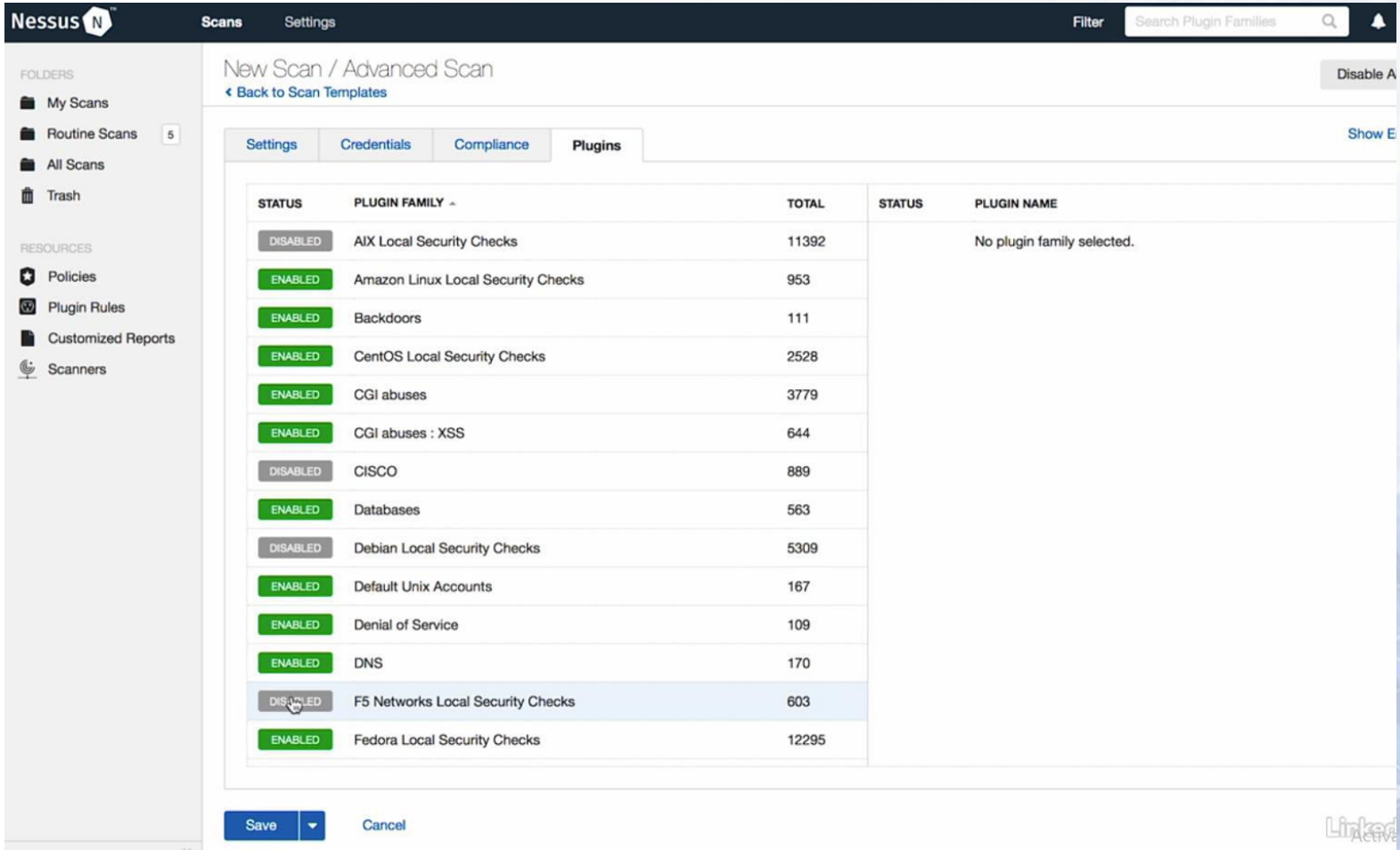

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
DISCOVERY
Host Discovery
Port Scanning
Service Discovery
ASSESSMENT
REPORT
ADVANCED
Ports
Consider unscanned ports as closed
Port scan range:
default
Local Port Enumerators
SSH (netstat)
WMI (netstat)
SNMP
Only run network port scanners if local port enumeration failed
Verify open TCP ports found by local port enumerators
Network Port Scanners
TCP
Override automatic firewall detection
Use soft detection

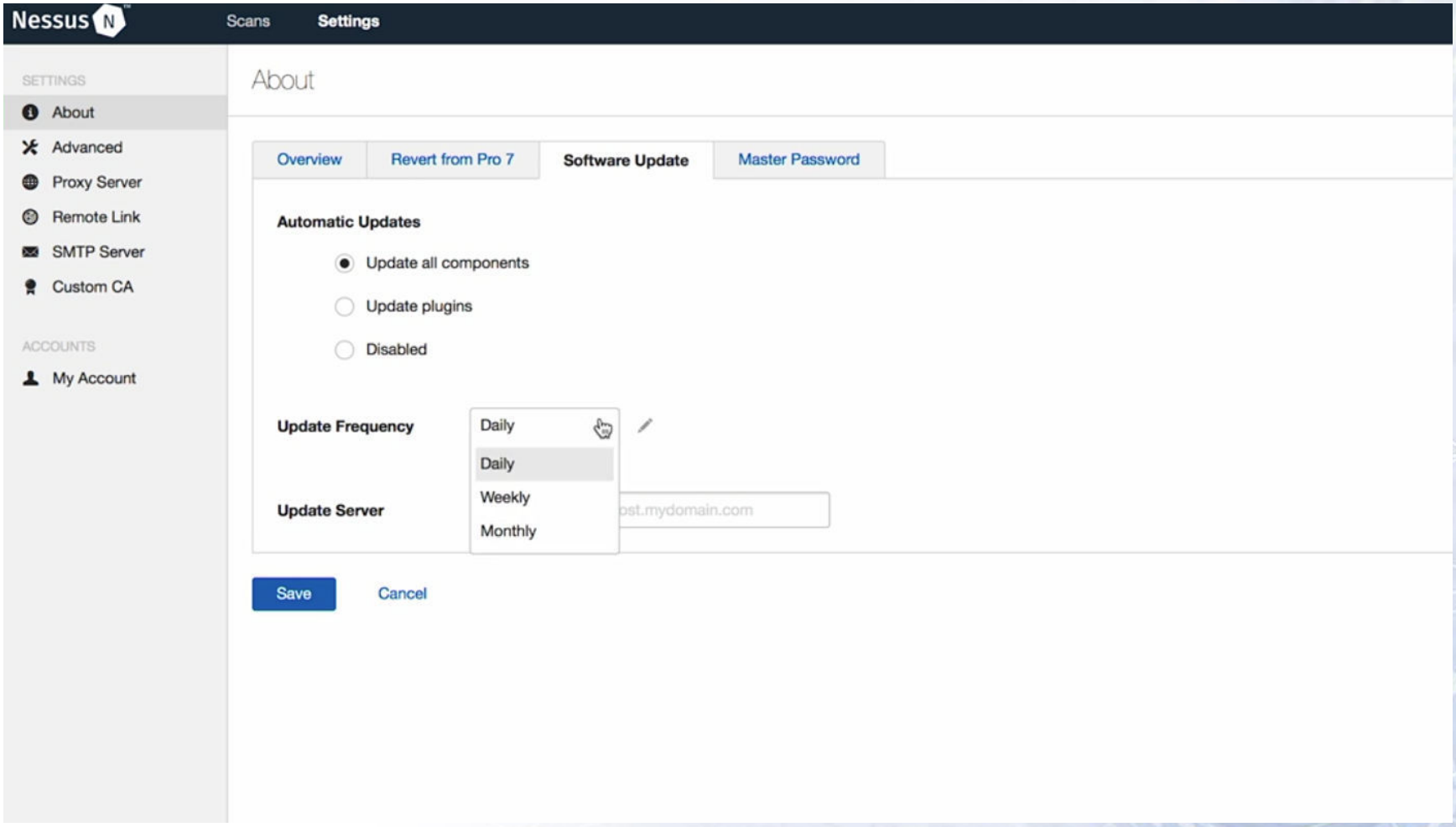

Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
DISCOVERY
Host Discovery
Port Scanning
Service Discovery
ASSESSMENT
REPORT
ADVANCED
Ports
Consider unscanned ports as closed
Port scan range:
default
Local Port Enumerators
SSH (netstat)
WMI (netstat)
SNMP
Only run network port scanners if local port enumeration failed
Verify open TCP ports found by local port enumerators
Network Port Scanners
TCP
Override automatic firewall detection


FOLDERS
My Scans
Routine Scans 5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
DISCOVERY
ASSESSMENT
General
Brute Force
SCADA
Web Applications
Windows
Malware
REPORT
ADVANCED
Accuracy
Override normal accuracy
Avoid potential false alarms
Show potential false alarms
Perform thorough tests (may disrupt your network or impact scan speed)
Antivirus
Antivirus definition grace period (in days): 0
SMTP
Third party domain
example.com
This domain must be outside the range of the site being scanned or the site performing the scan. Otherwise, the test might be a
SMTP server.
From address
nobody@example.com
To address
postmaster@[AUTO_REPLACED_IP]


Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Settings
Credentials
Compliance
Plugins
BASIC
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
General Settings
Enable safe checks
Stop scanning hosts that become unresponsive during the scan
Scan IP addresses in a random order
Performance Options
Slow down the scan when network congestion is detected
Network timeout (in seconds)
5
Max simultaneous checks per host
5
Max simultaneous hosts per scan
30
Max number of concurrent TCP sessions per host
Max number of concurrent TCP sessions per scan
Debug Settings
Log scan details


Nessus
Scans
Settings
Filter
Search Plugin Families
FOLDERS
My Scans
Routine Scans 5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Disable All
Settings
Credentials
Compliance
Plugins
Show Ena
STATUS
PLUGIN FAMILY
TOTAL
STATUS
PLUGIN NAME
ENABLED AIX Local Security Checks 11392
ENABLED Amazon Linux Local Security Checks 953
ENABLED Backdoors 111
ENABLED CentOS Local Security Checks 2528
ENABLED CGI abuses 3779
ENABLED CGI abuses : XSS 644
ENABLED CISCO 889
ENABLED Databases 563
ENABLED Debian Local Security Checks 5309
ENABLED Default Unix Accounts 167
ENABLED Denial of Service 109
ENABLED DNS 170
ENABLED F5 Networks Local Security Checks 603
ENABLED Fedora Local Security Checks 12295
No plugin family selected.


Nessus
Scans
Settings
Filter
Search Plugin Families
FOLDERS
My Scans
Routine Scans 5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
New Scan / Advanced Scan
Back to Scan Templates
Disable A
Settings
Credentials
Compliance
Plugins
Show E
STATUS
PLUGIN FAMILY
TOTAL
STATUS
PLUGIN NAME
DISABLED AIX Local Security Checks 11392
ENABLED Amazon Linux Local Security Checks 953
ENABLED Backdoors 111
ENABLED CentOS Local Security Checks 2528
ENABLED CGI abuses 3779
ENABLED CGI abuses : XSS 644
DISABLED CISCO 889
ENABLED Databases 563
DISABLED Debian Local Security Checks 5309
ENABLED Default Unix Accounts 167
ENABLED Denial of Service 109
ENABLED DNS 170
DISABLED F5 Networks Local Security Checks 603
ENABLED Fedora Local Security Checks 12295
No plugin family selected.
Save
Cancel

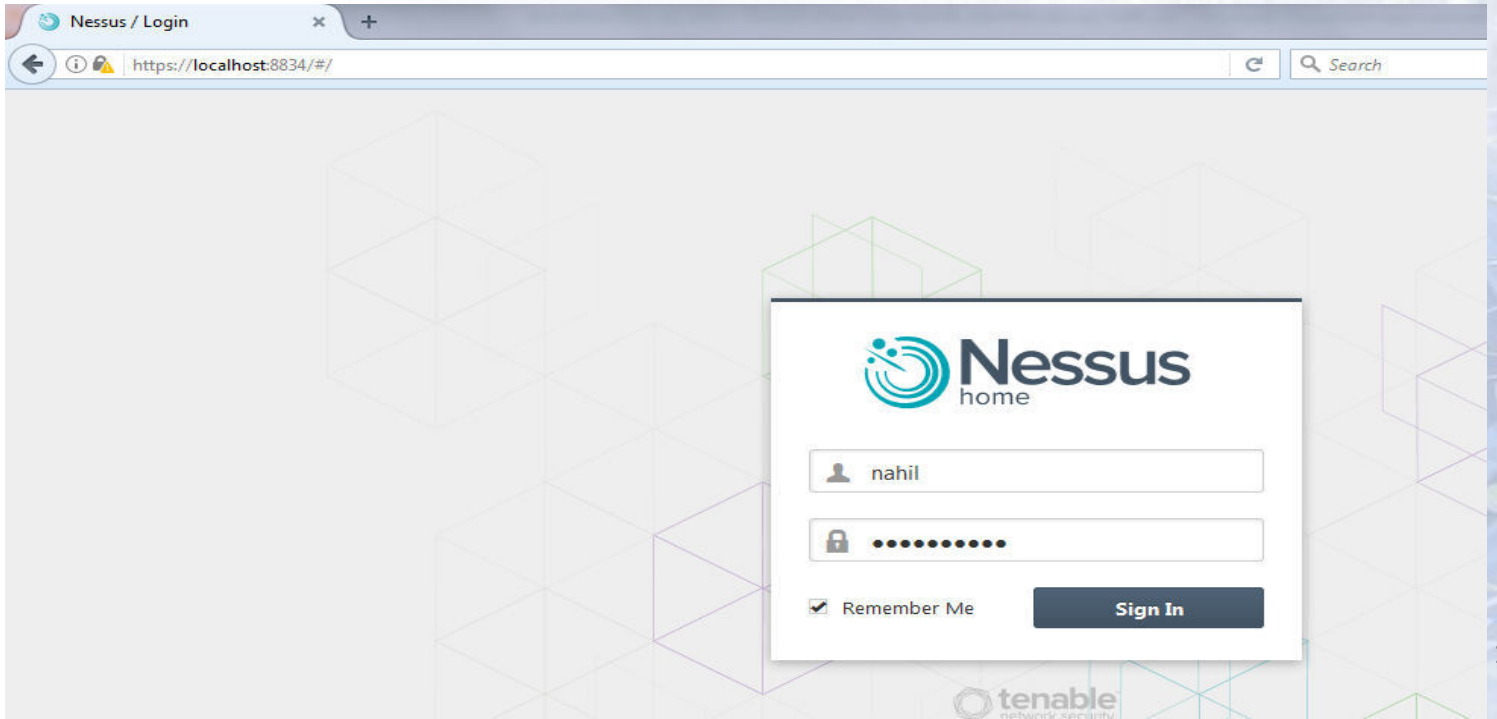

Nessus
Scans
Settings
SETTINGS
About
Advanced
Proxy Server
Remote Link
SMTP Server
Custom CA
ACCOUNTS
My Account
About
Overview
Revert from Pro 7
Software Update
Master Password
Automatic Updates
Update all components
Update plugins
Disabled
Update Frequency
Daily
Daily
Weekly
Monthly
Update Server
st.mydomain.com
Save
Cancel

Week 7 Topic 12

Nessus Demo - 1

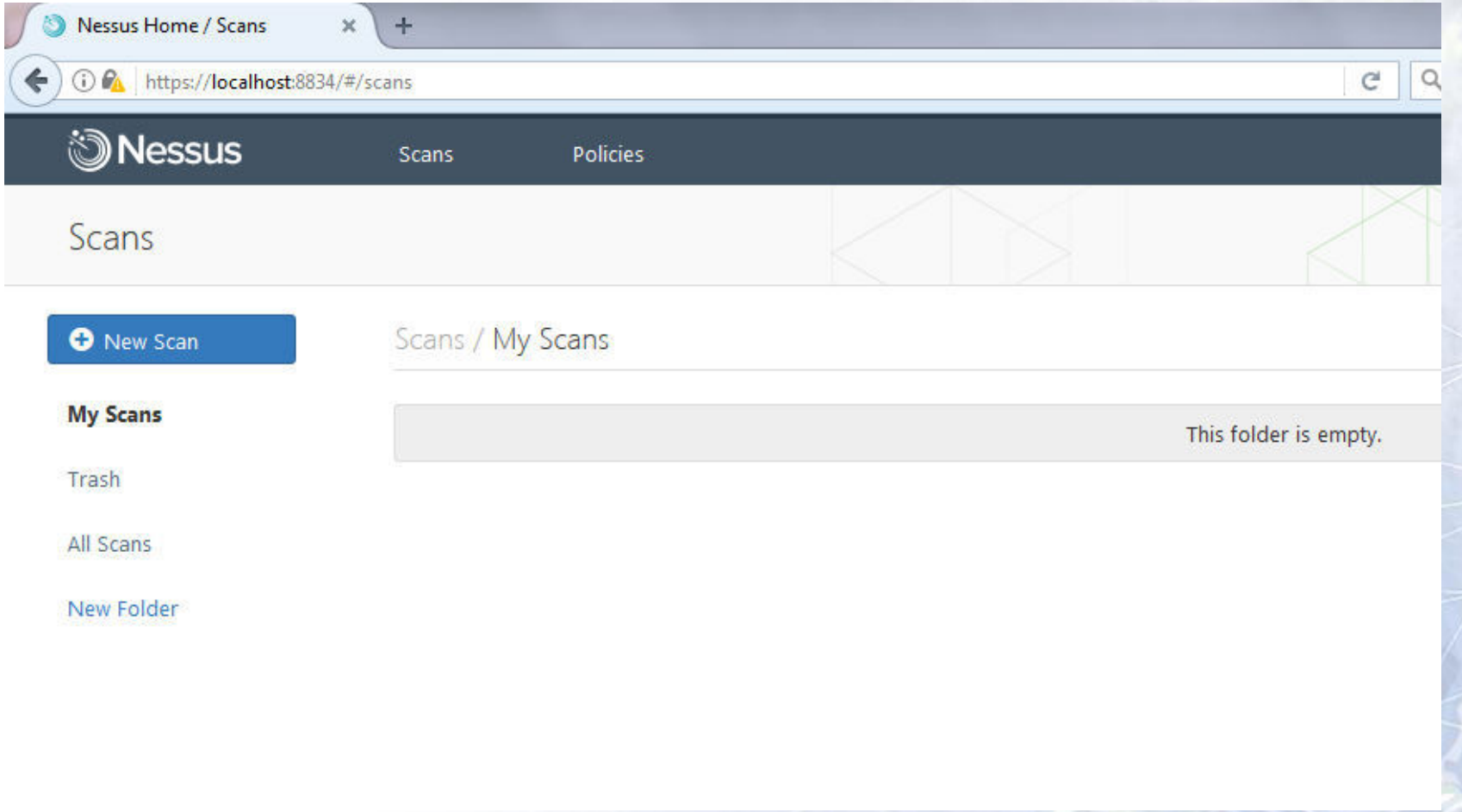
- Lets take a look at Nessus Demo
- https://www.tenable.com/products/nessus/nessus-professional/evaluate
- Download free 7 day trial
- Get activation key from website

LOGIN SCREEN



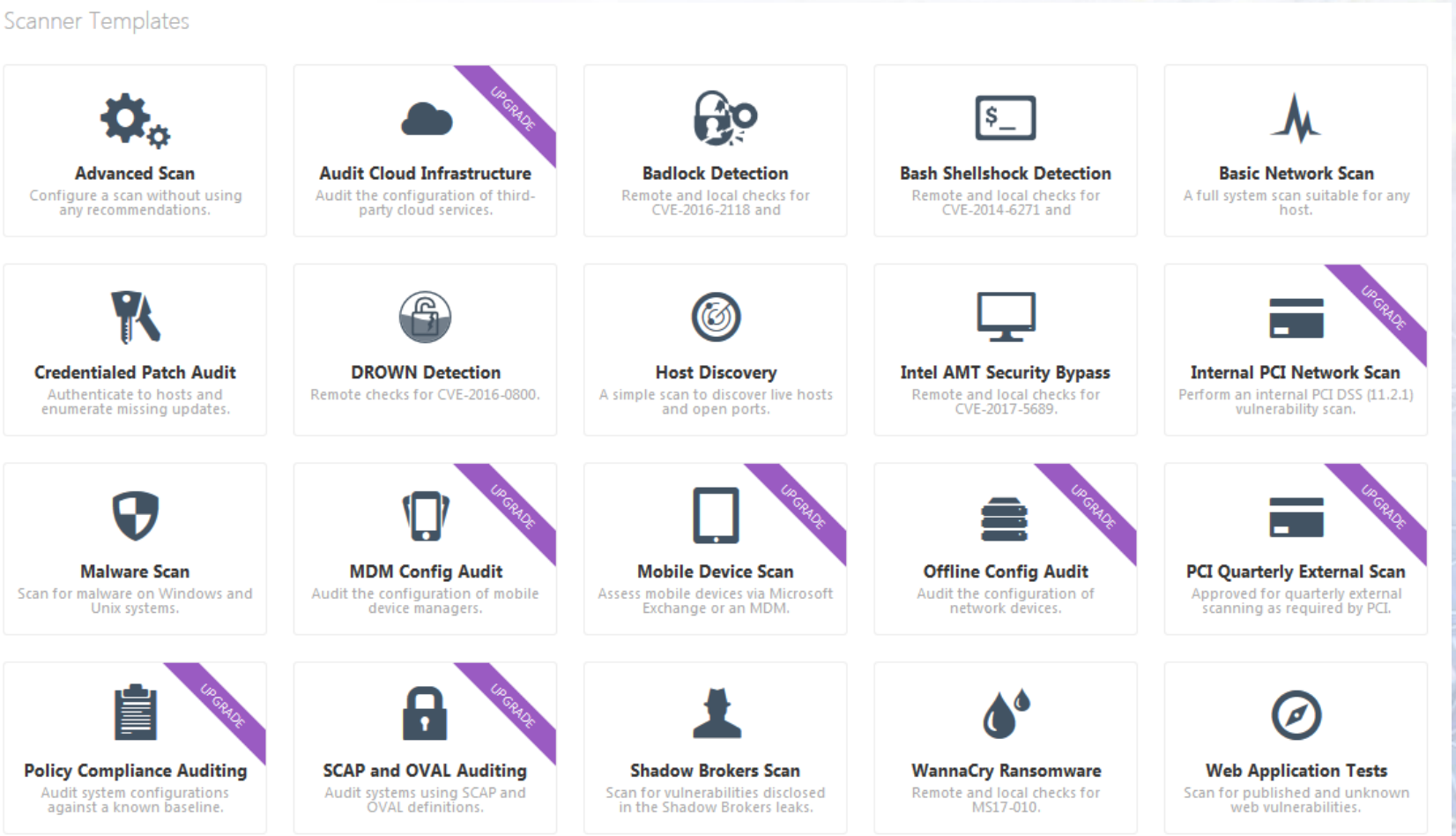
Nessus Demo - 1

DASHBOARD



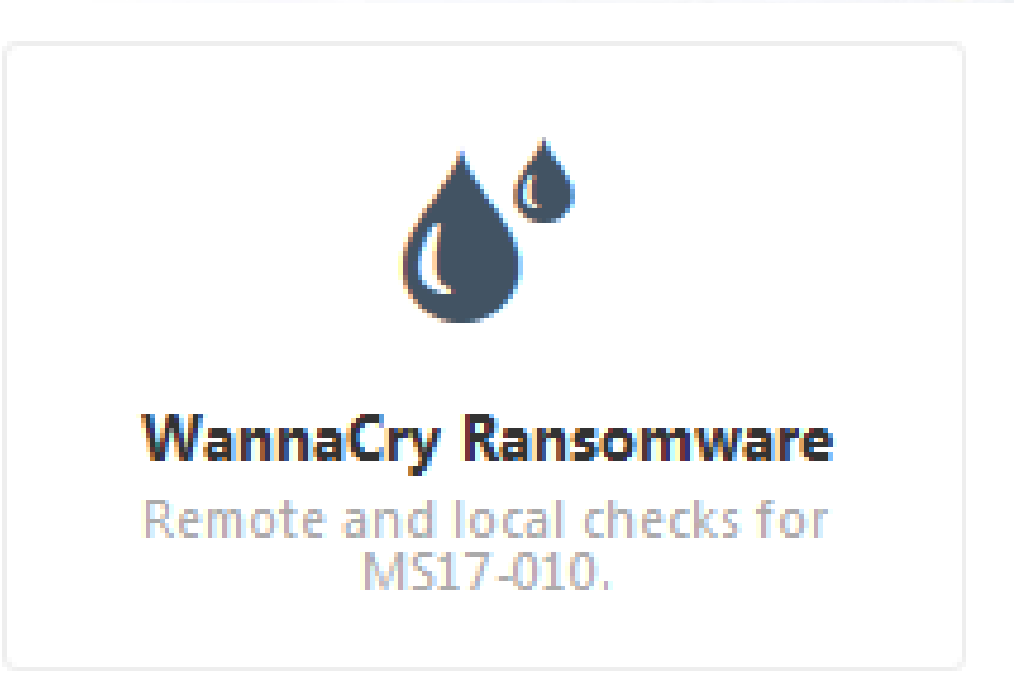
Nessus Demo - 1

NEW SCAN



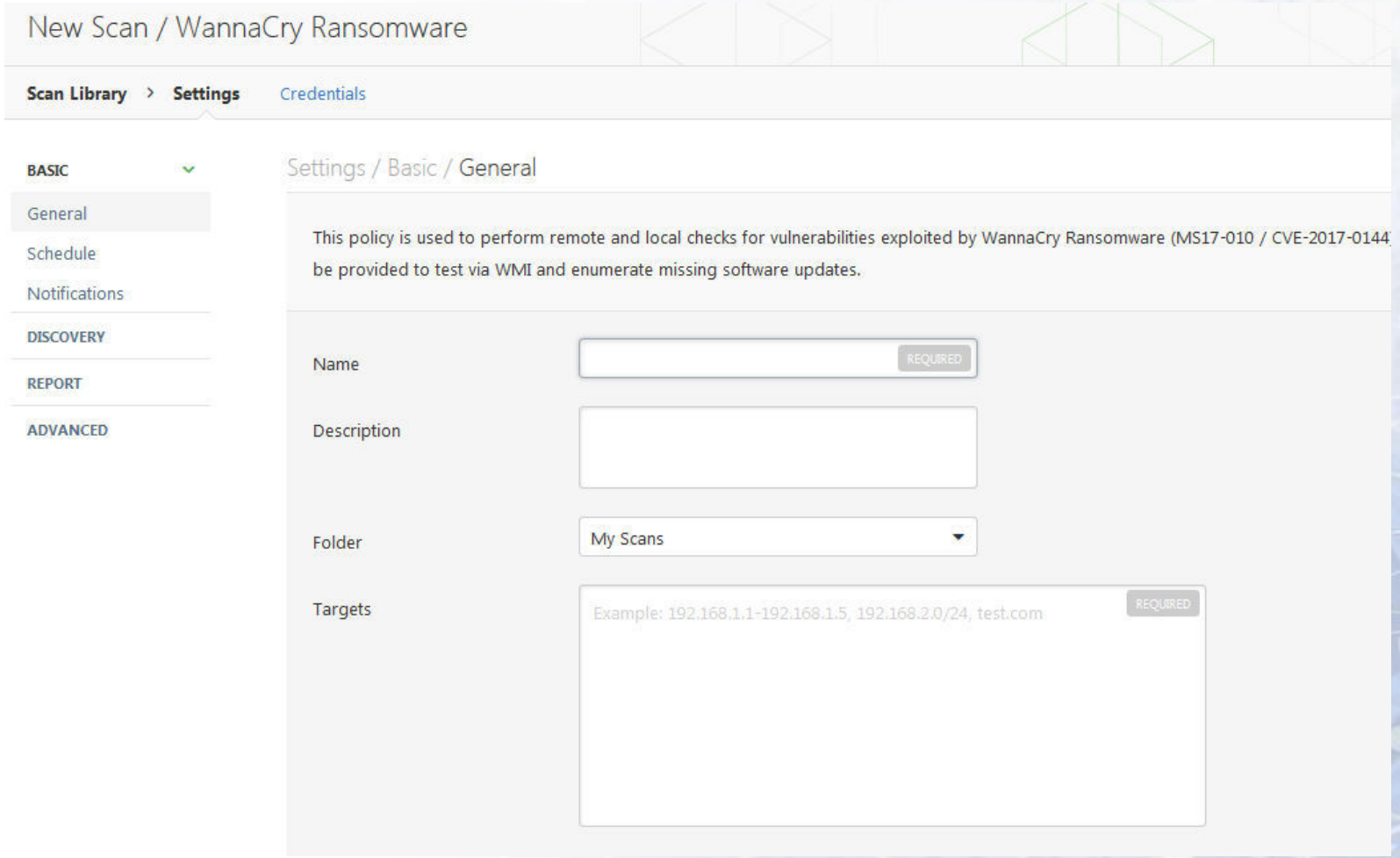
Nessus Demo - 1

WANNACRY RANSOMWARE SCAN



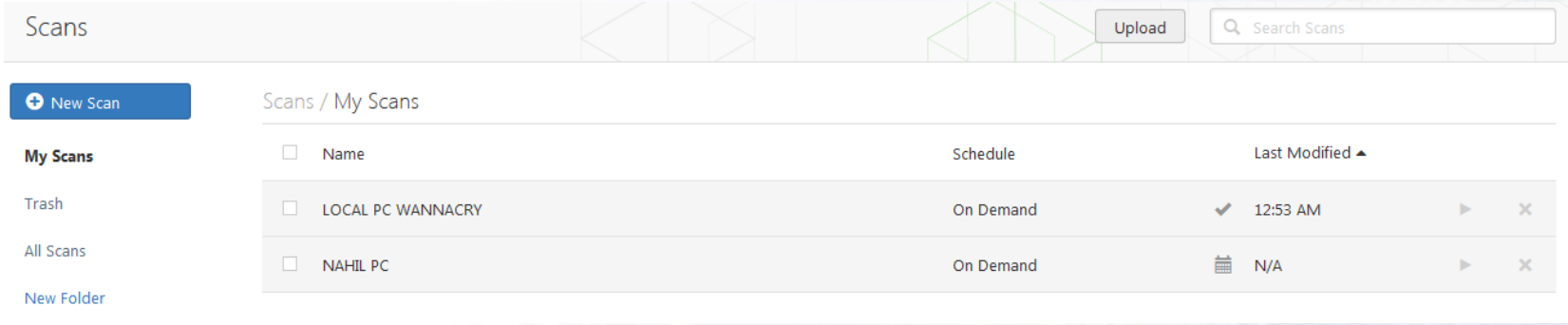
Nessus Demo - 1

NEW SCAN WINDOW



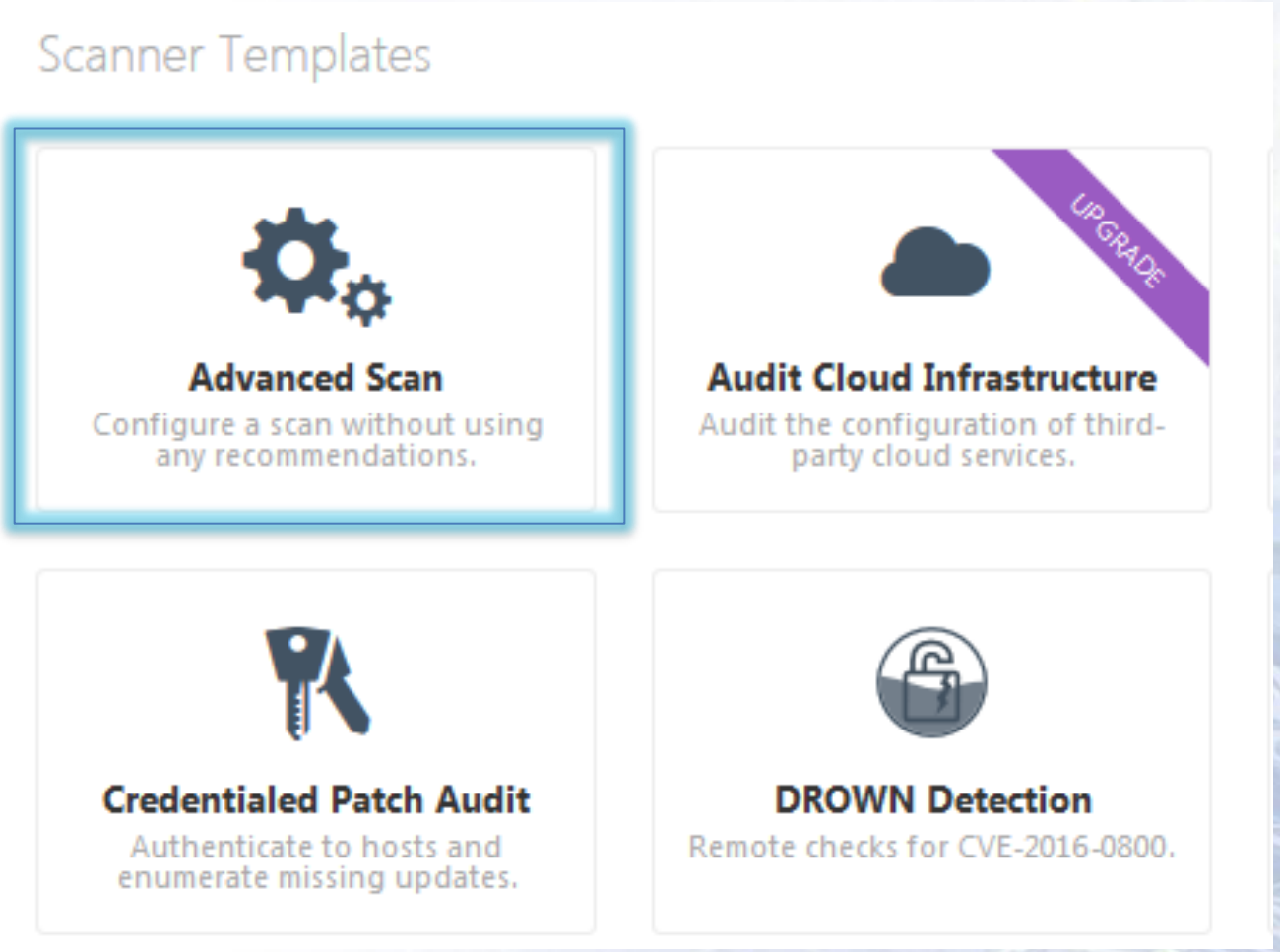
Nessus Demo - 1

DASHBOARD VIEW WITH SCANS



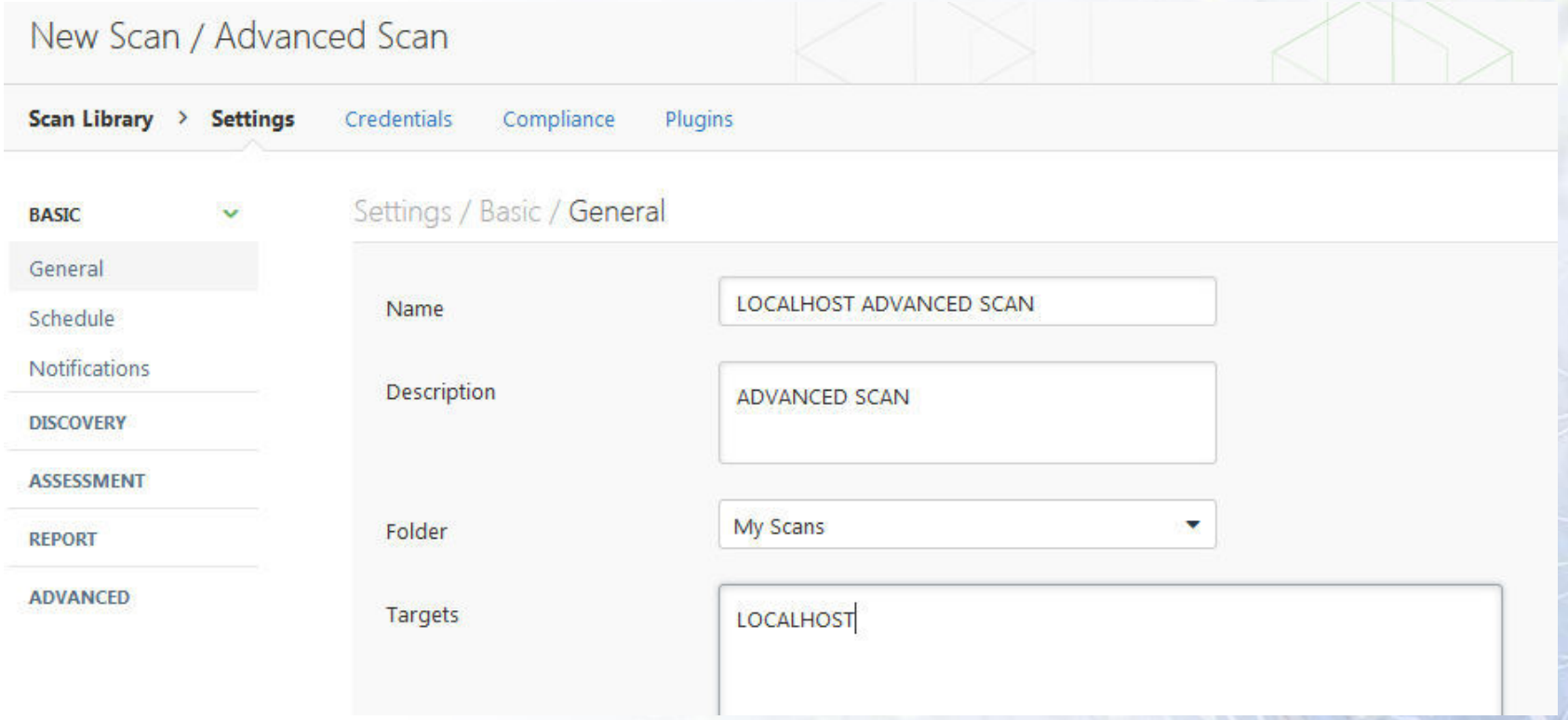
Nessus Demo - 1

NEW SCAN…



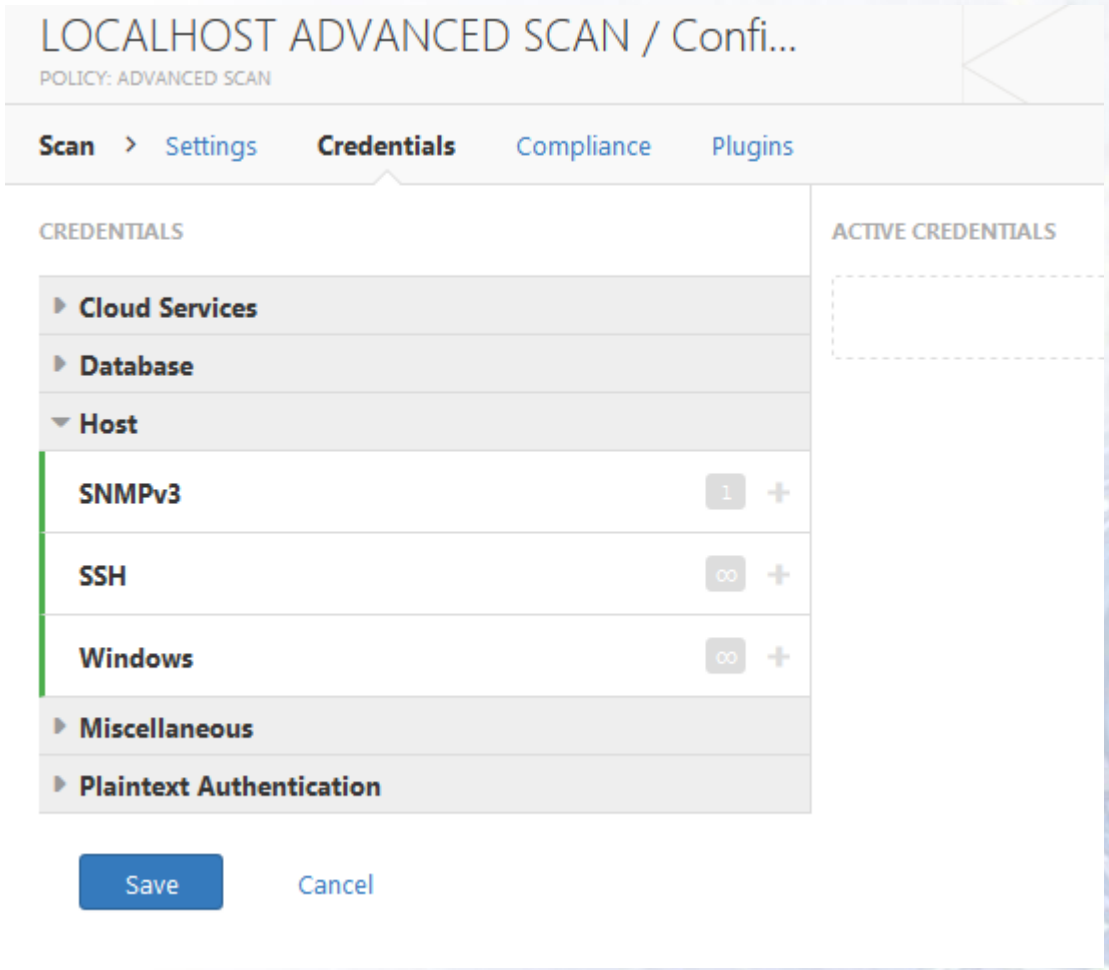
Nessus Demo - 1

ENTER SCAN DETAILS



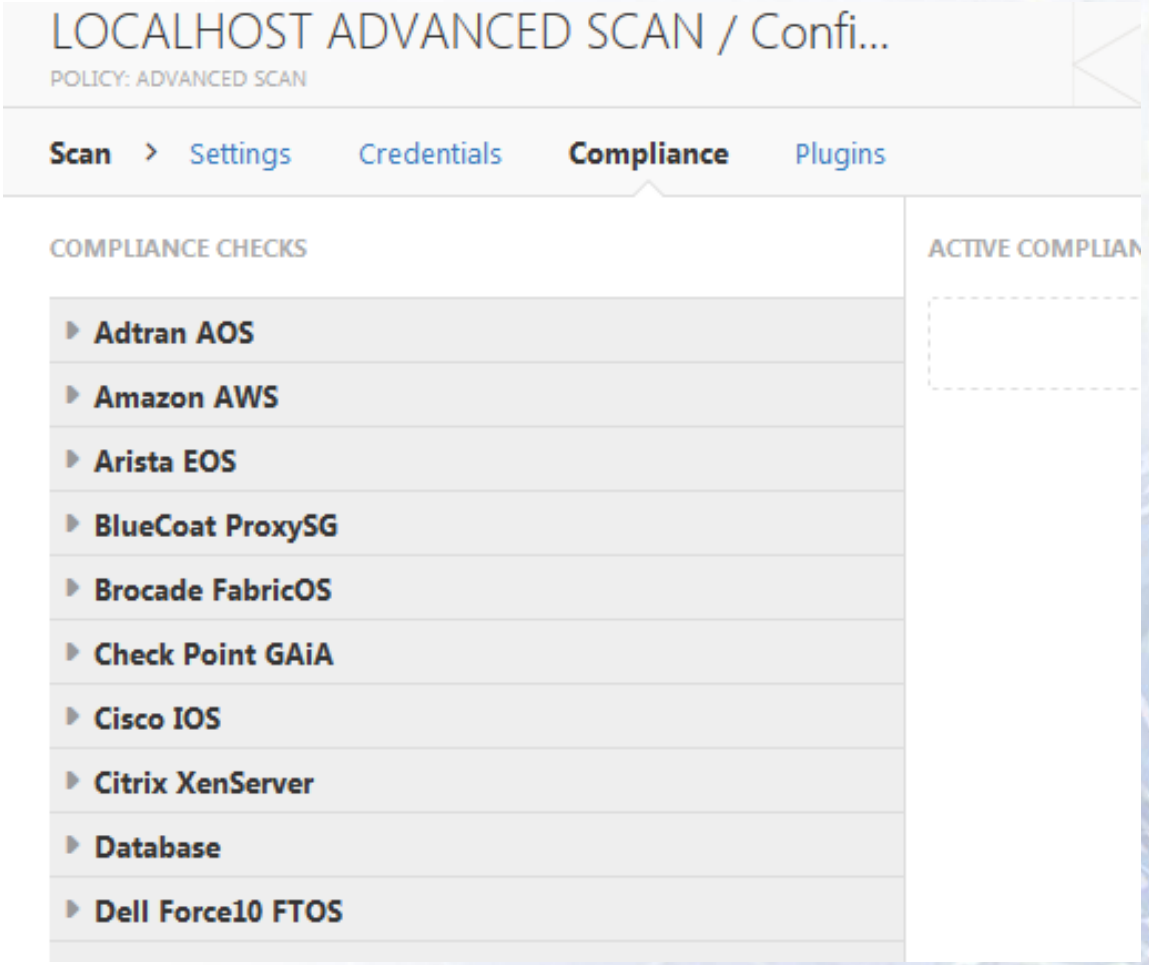
Nessus Demo - 1

CREDENTIAL SCAN



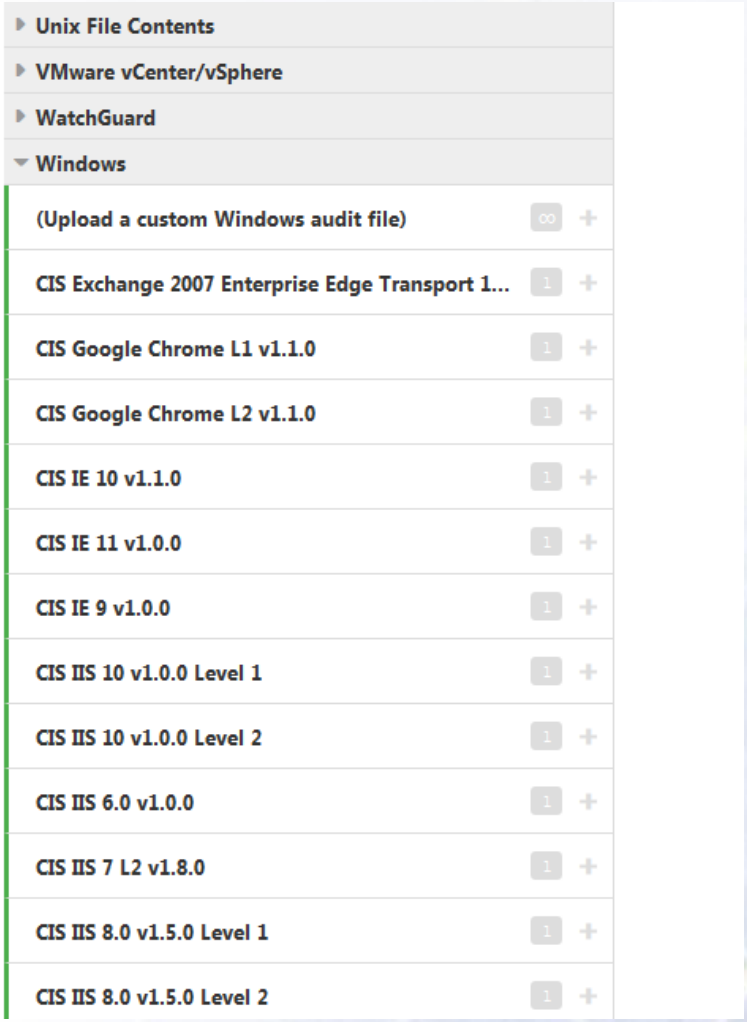
Nessus Demo - 1

COMPLIANCE SCAN



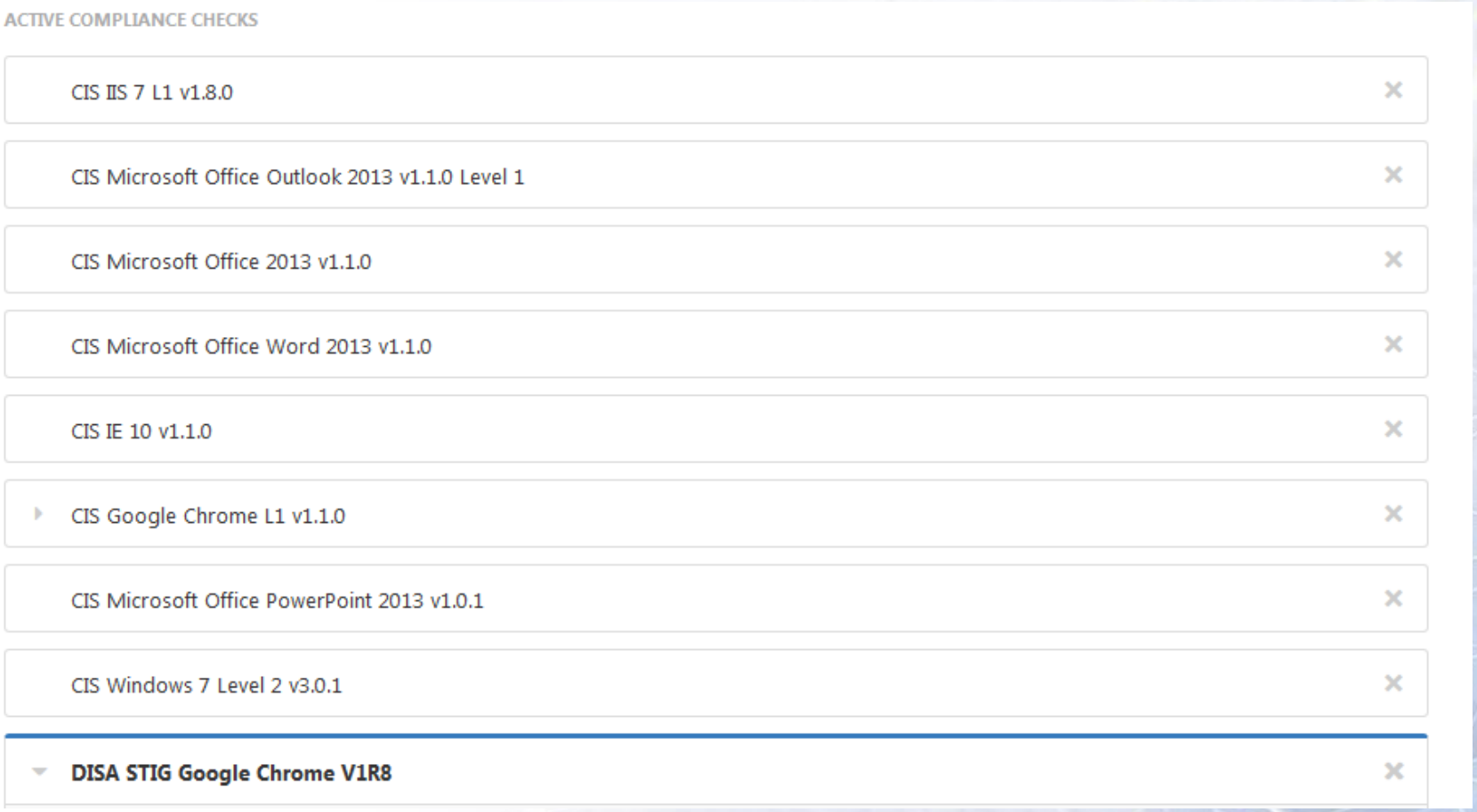
Nessus Demo - 1

WINDOWS COMPLIANCE MENU (CIS)



Nessus Demo - 1

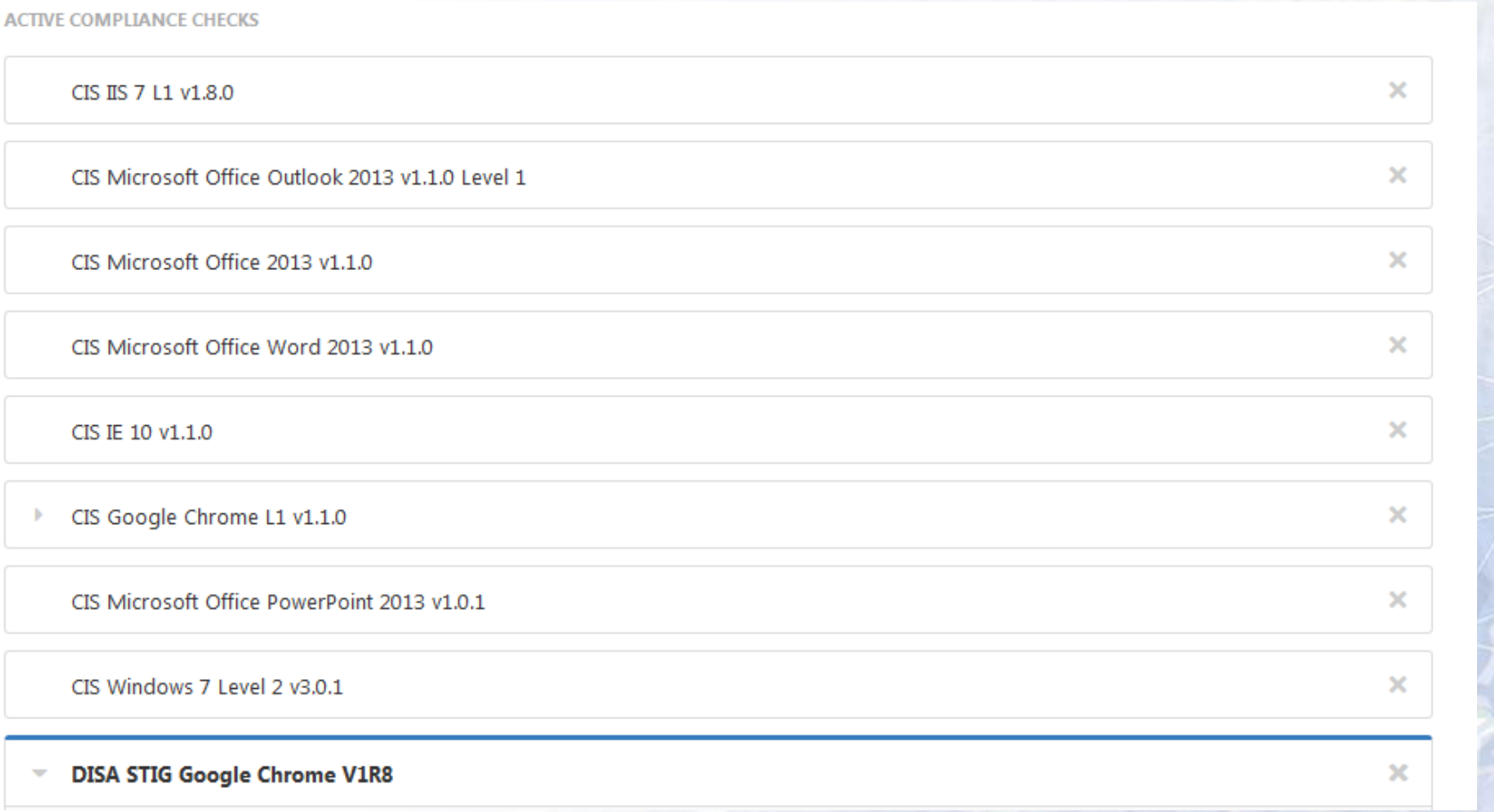
WINDOWS COMPLIANCE MENU (CIS)…



Week 7 Topic 13

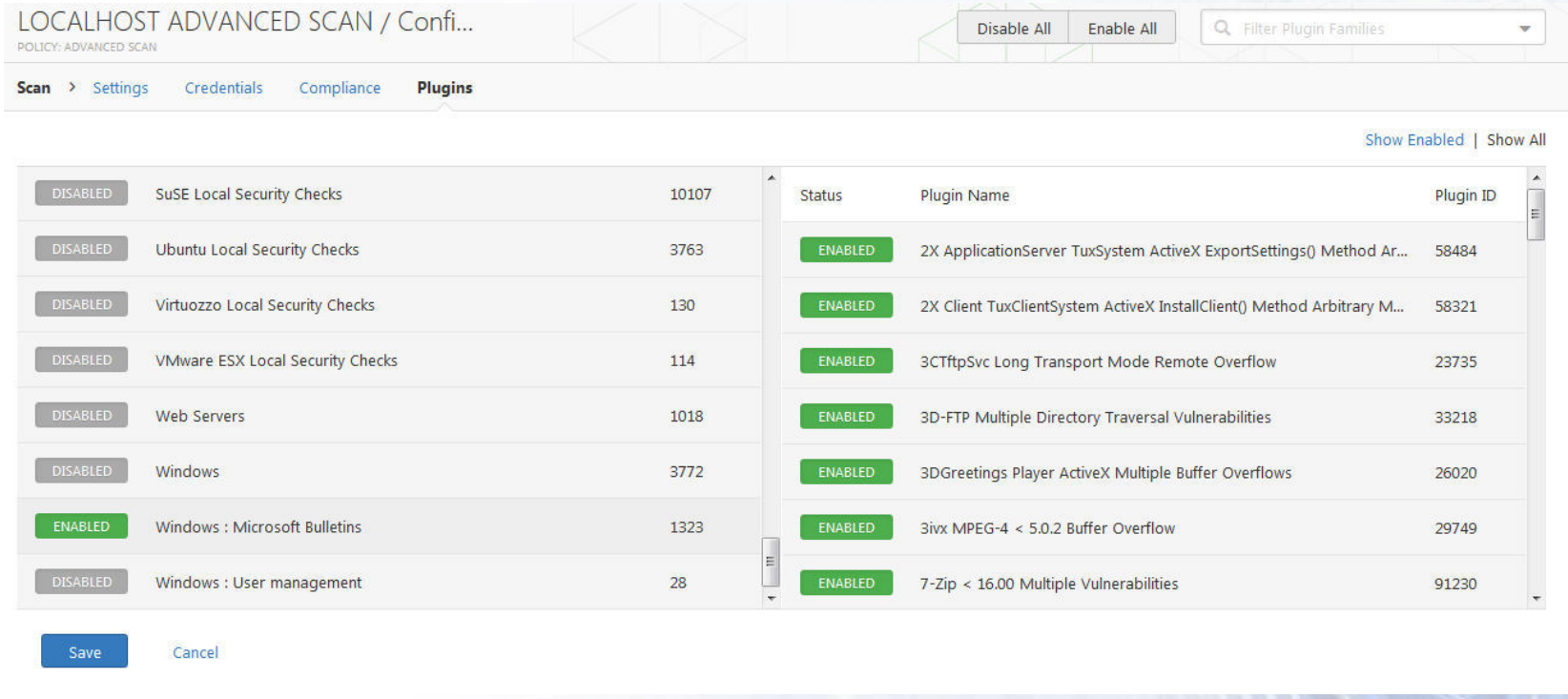
Nessus Demo - 2

ADVANCED SCAN / COMPLIANCE



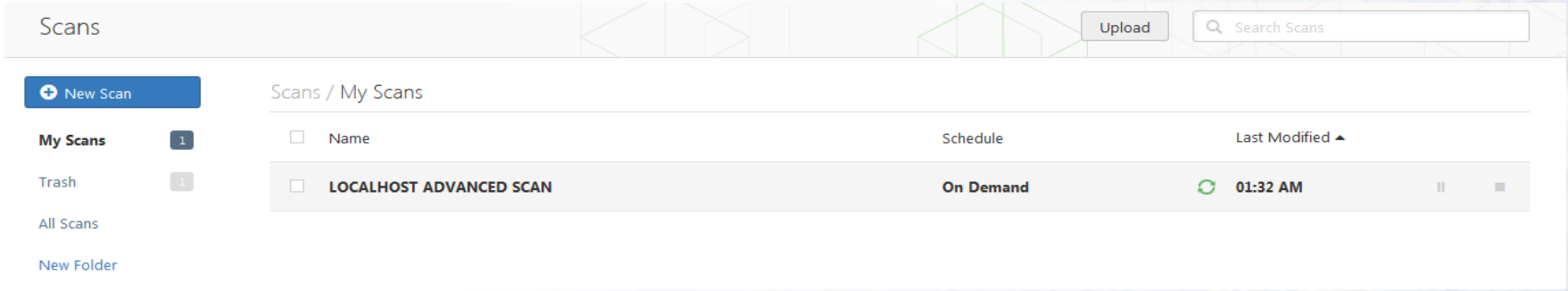
Nessus Demo - 2

ADVANCED SCAN / PLUG-INS

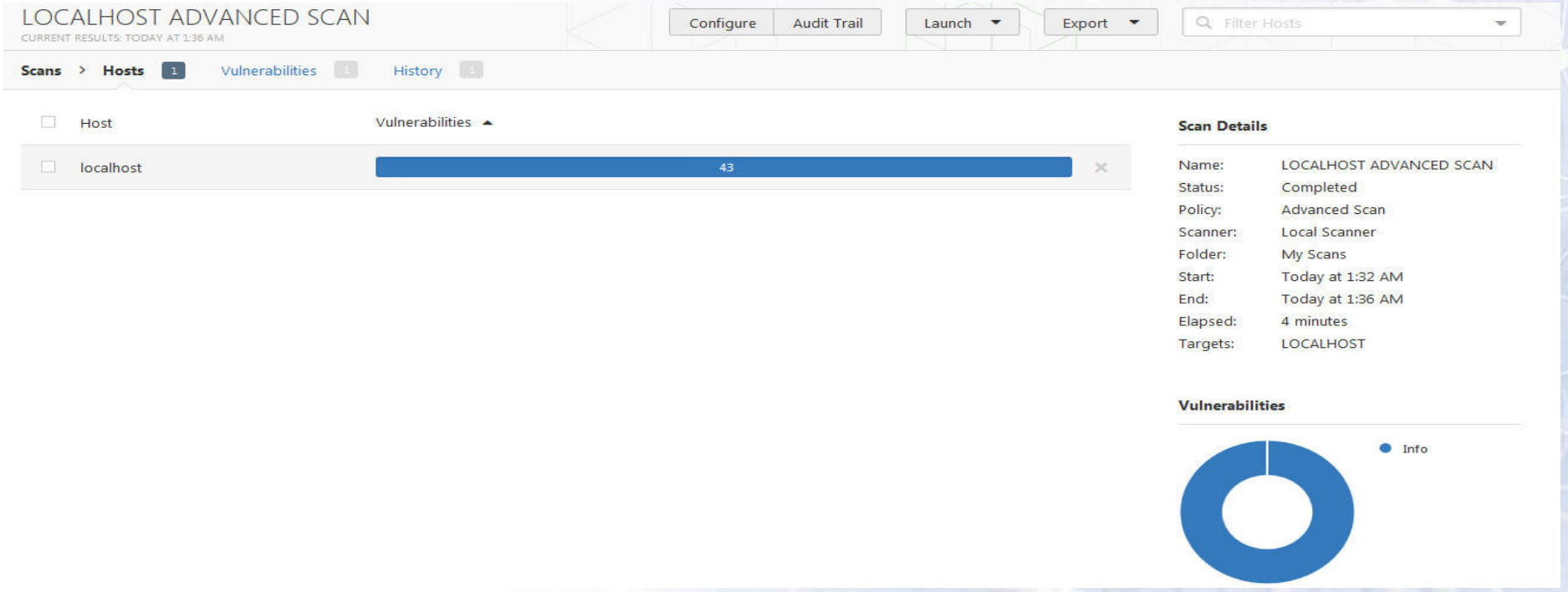


Nessus Demo - 2

SCAN…IN PROGRESS

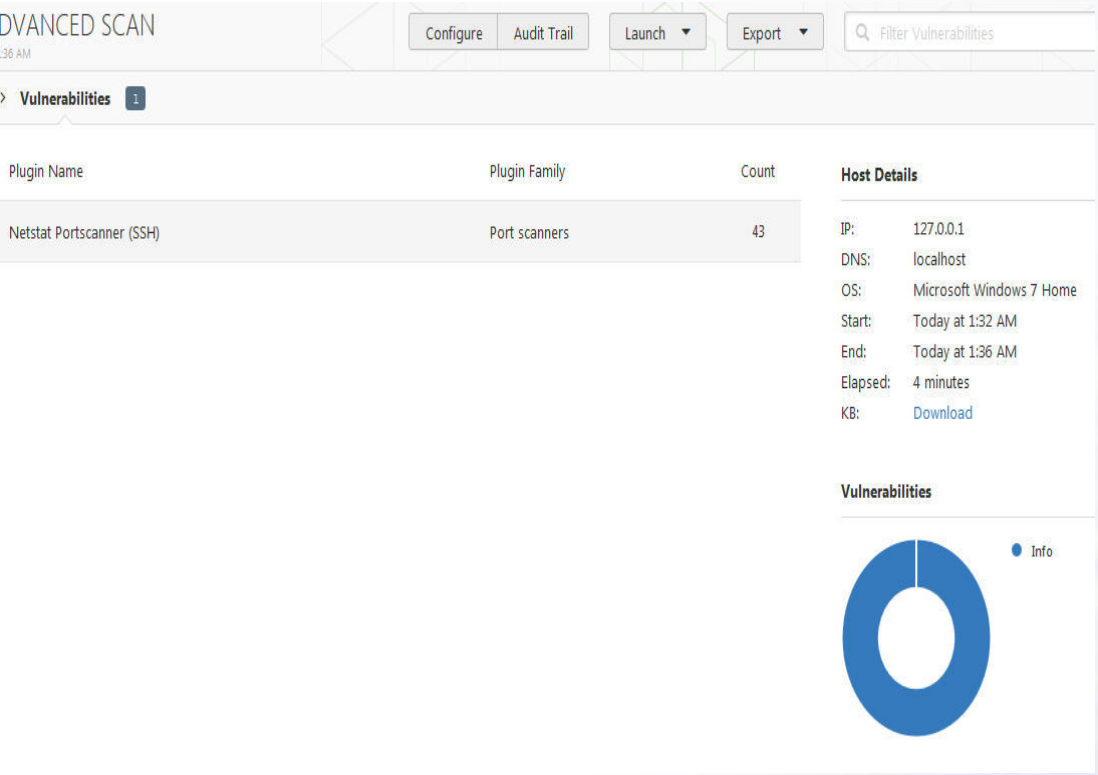


SCAN REPORT [43 INFO]



Nessus Demo - 2

SCAN REPORT [DETAILS]



https://en.wikipedia.org/wiki/Netstat

Output

```
Port 68/udp was found to be open
```

Port	Hosts
68 / udp	localhost

```
Port 123/udp was found to be open
```

Port	Hosts
123 / udp	localhost

```
Port 135/tcp was found to be open
```

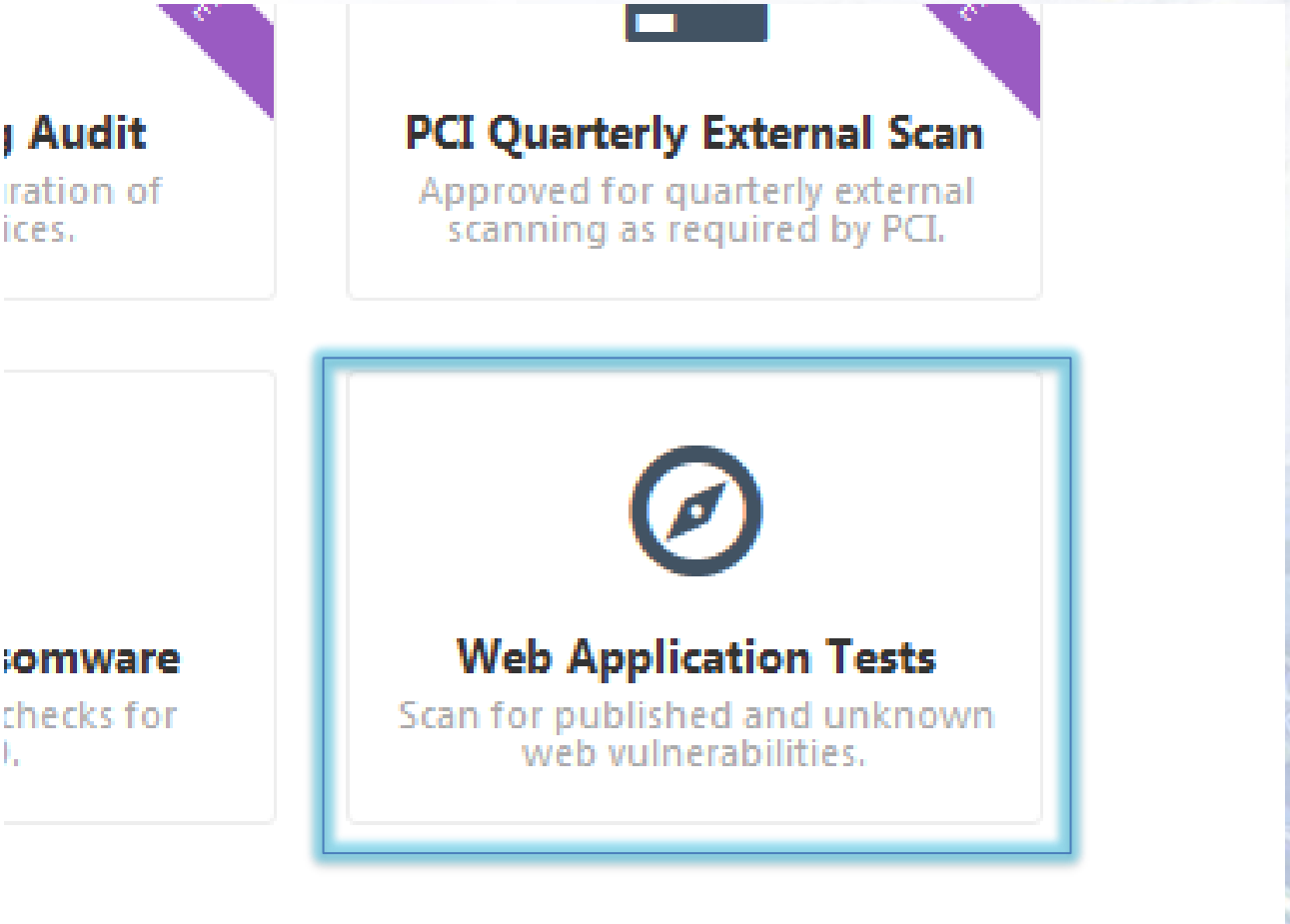
Port	Hosts
135 / tcp / epmap	localhost

```
Port 137/udp was found to be open
```

Port	Hosts
137 / udp	localhost

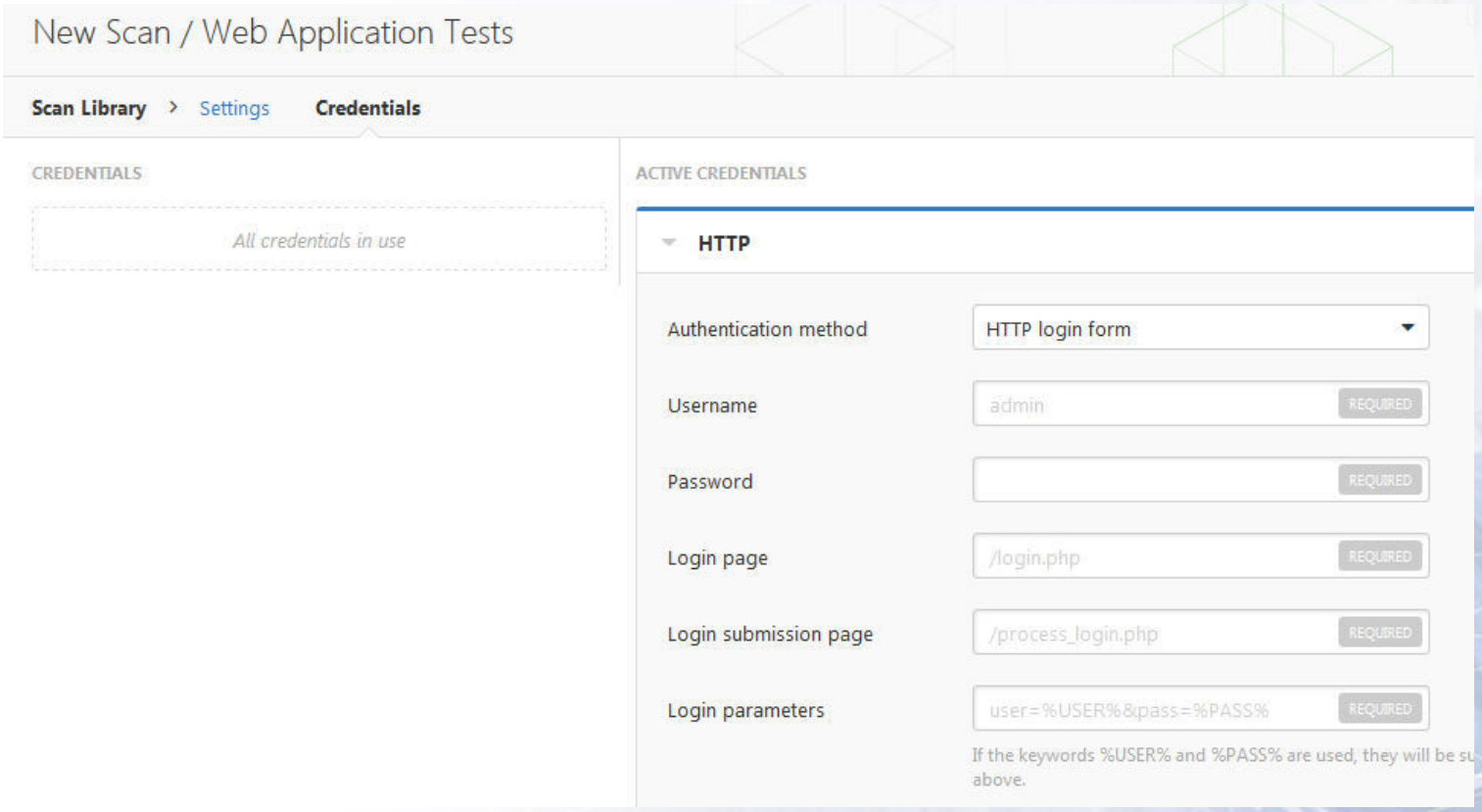
Nessus Demo - 2

WEB APPLICATION TEST



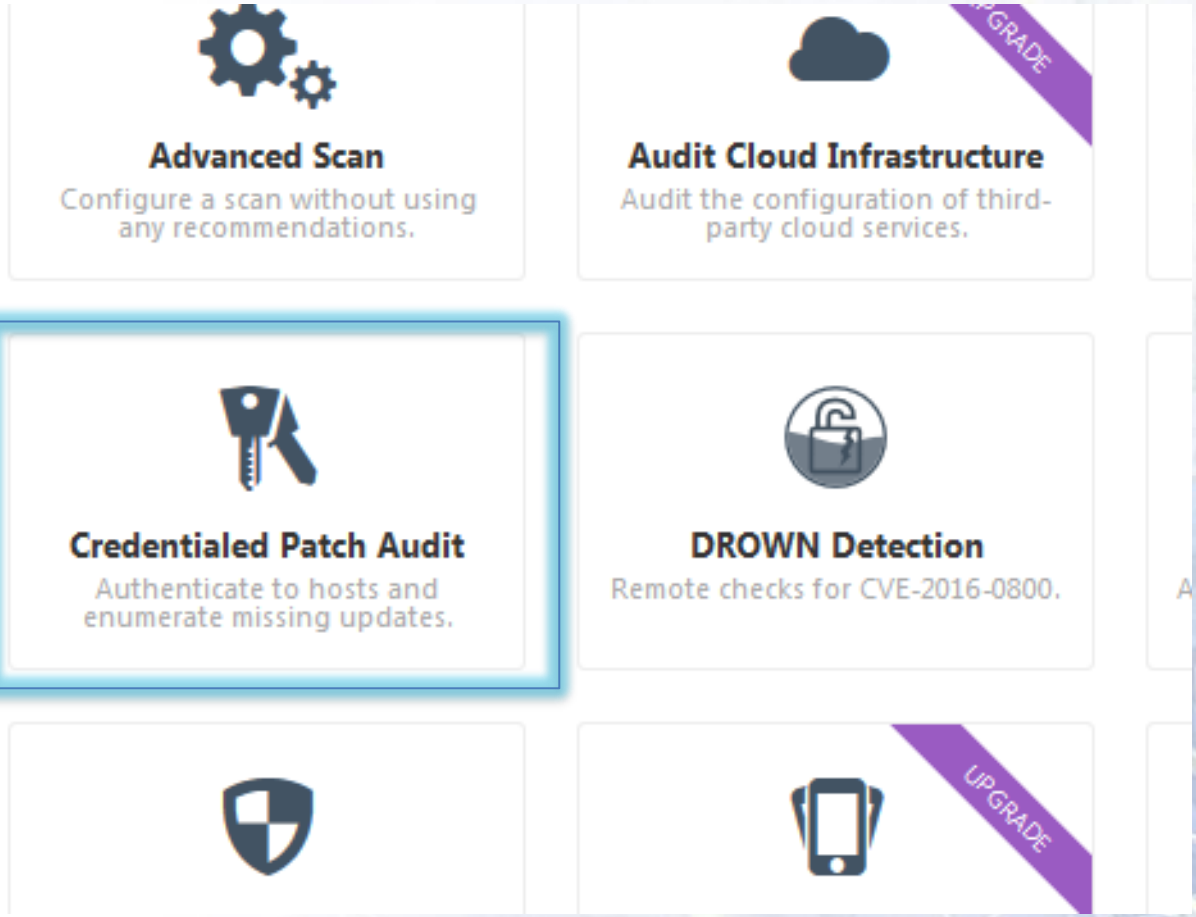
Nessus Demo - 2

WEB APPLICATION TEST - CREDENTIALS



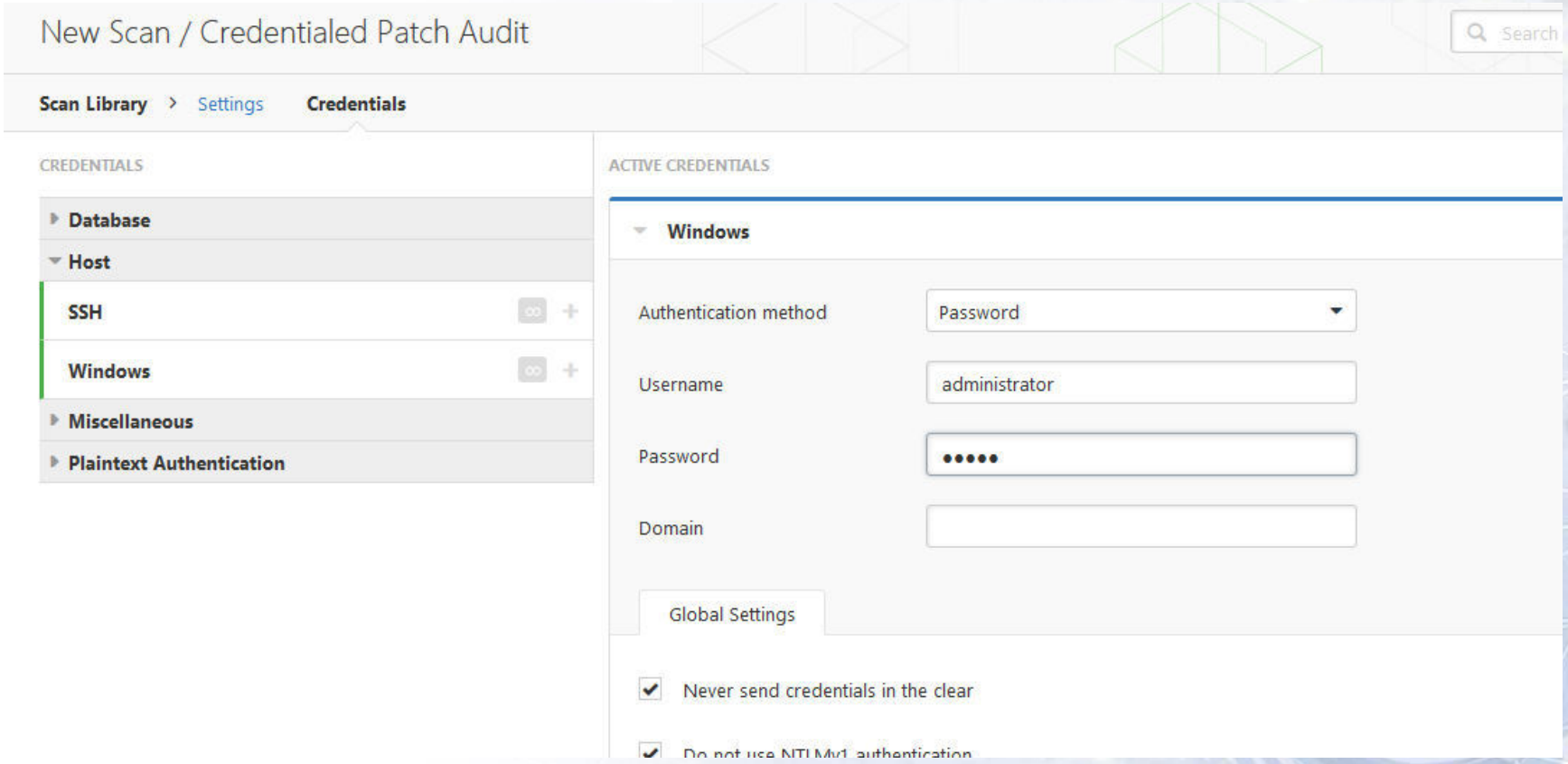
Nessus Demo - 2

CREDENTIALED PATCH AUDIT



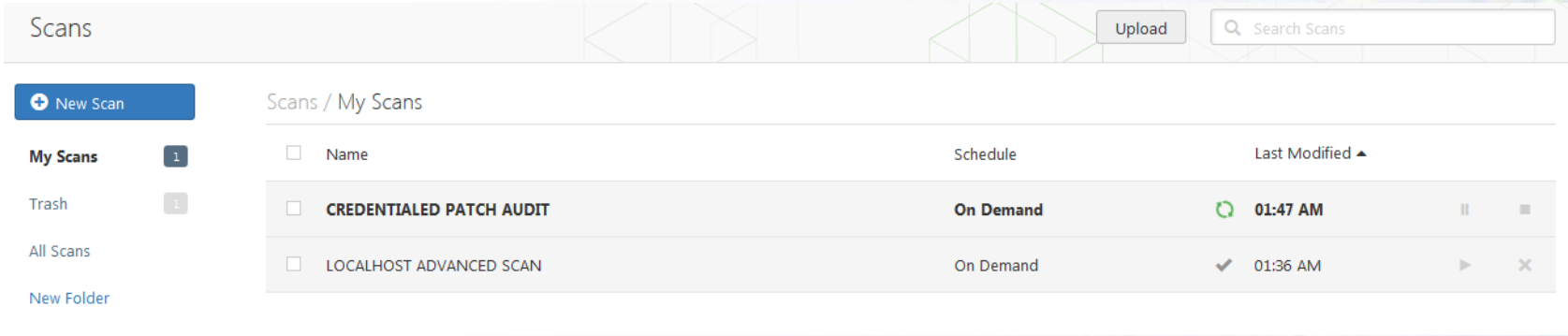
Nessus Demo - 2

CREDENTIALED PATCH AUDIT

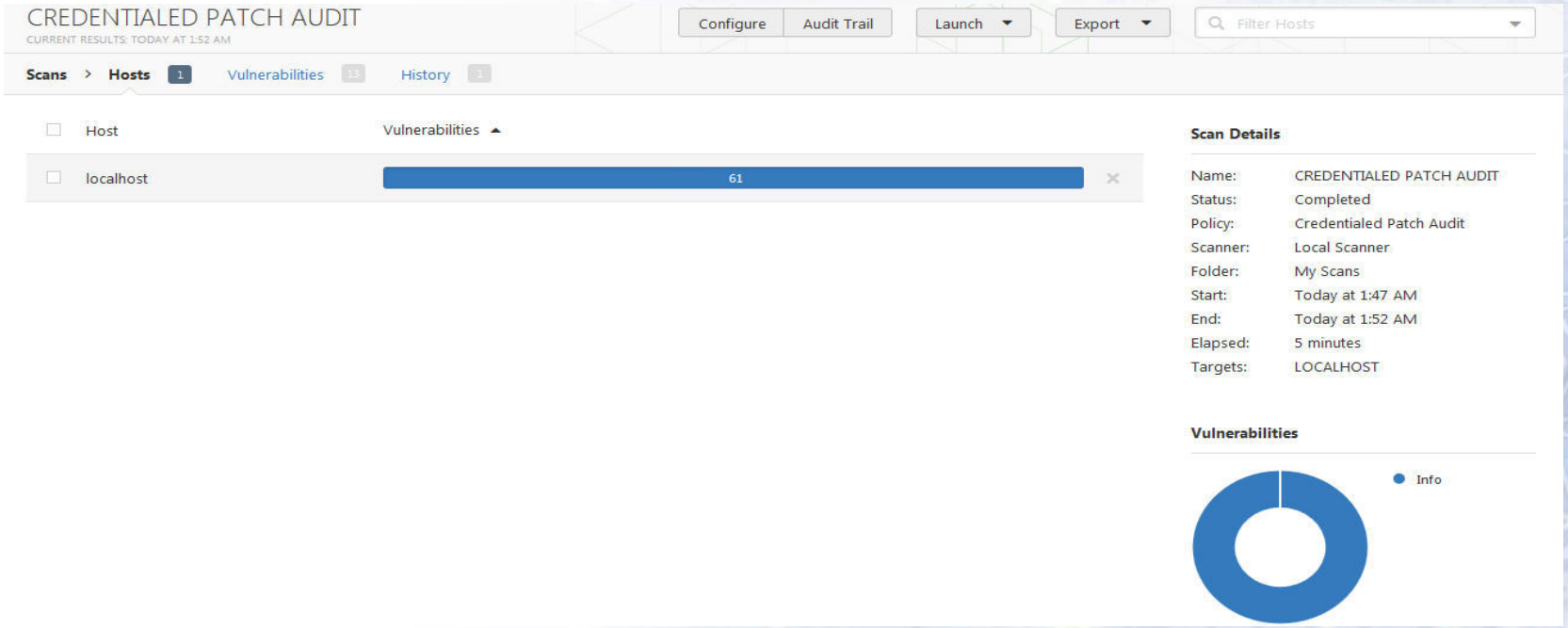


Nessus Demo - 2

SCANS DASHBOARD

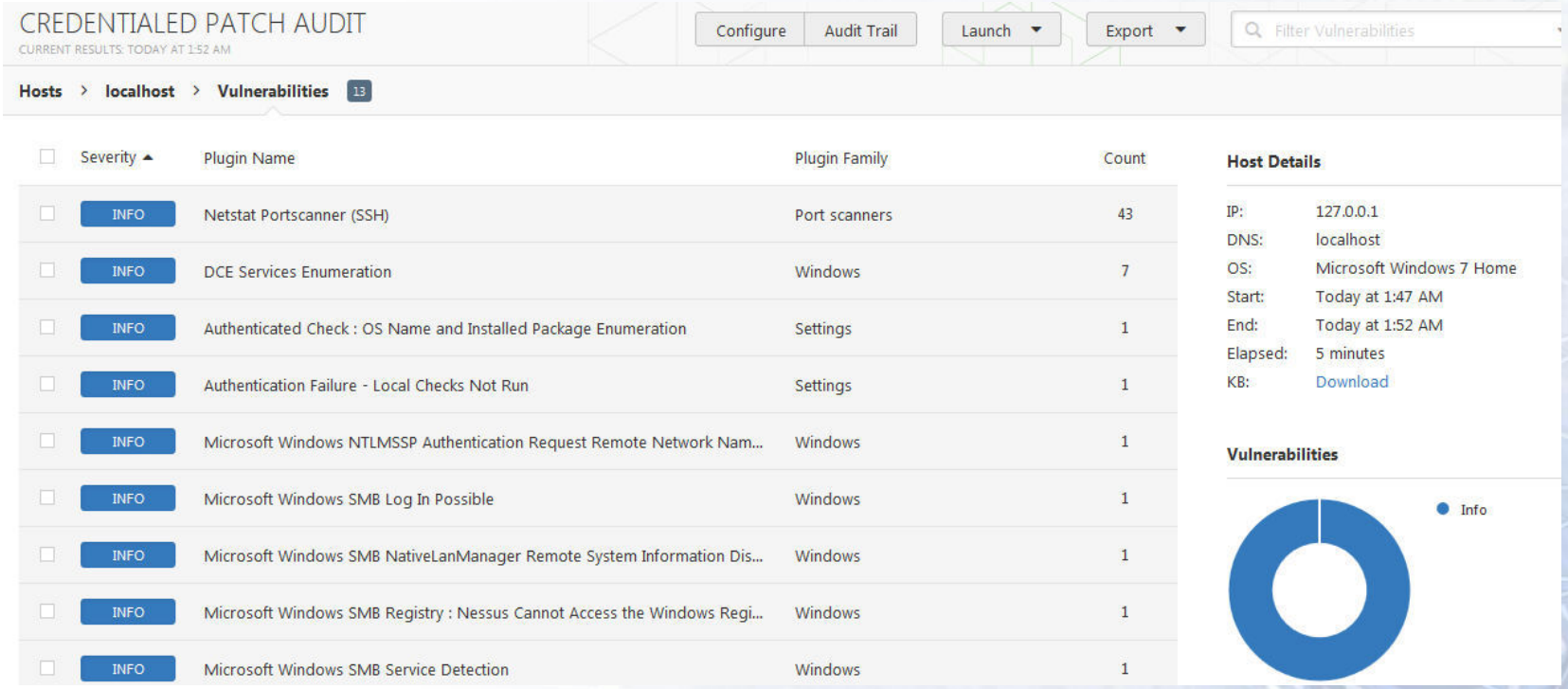


CREDENTIALED AUDIT SCAN RESULTS [61 INFO]



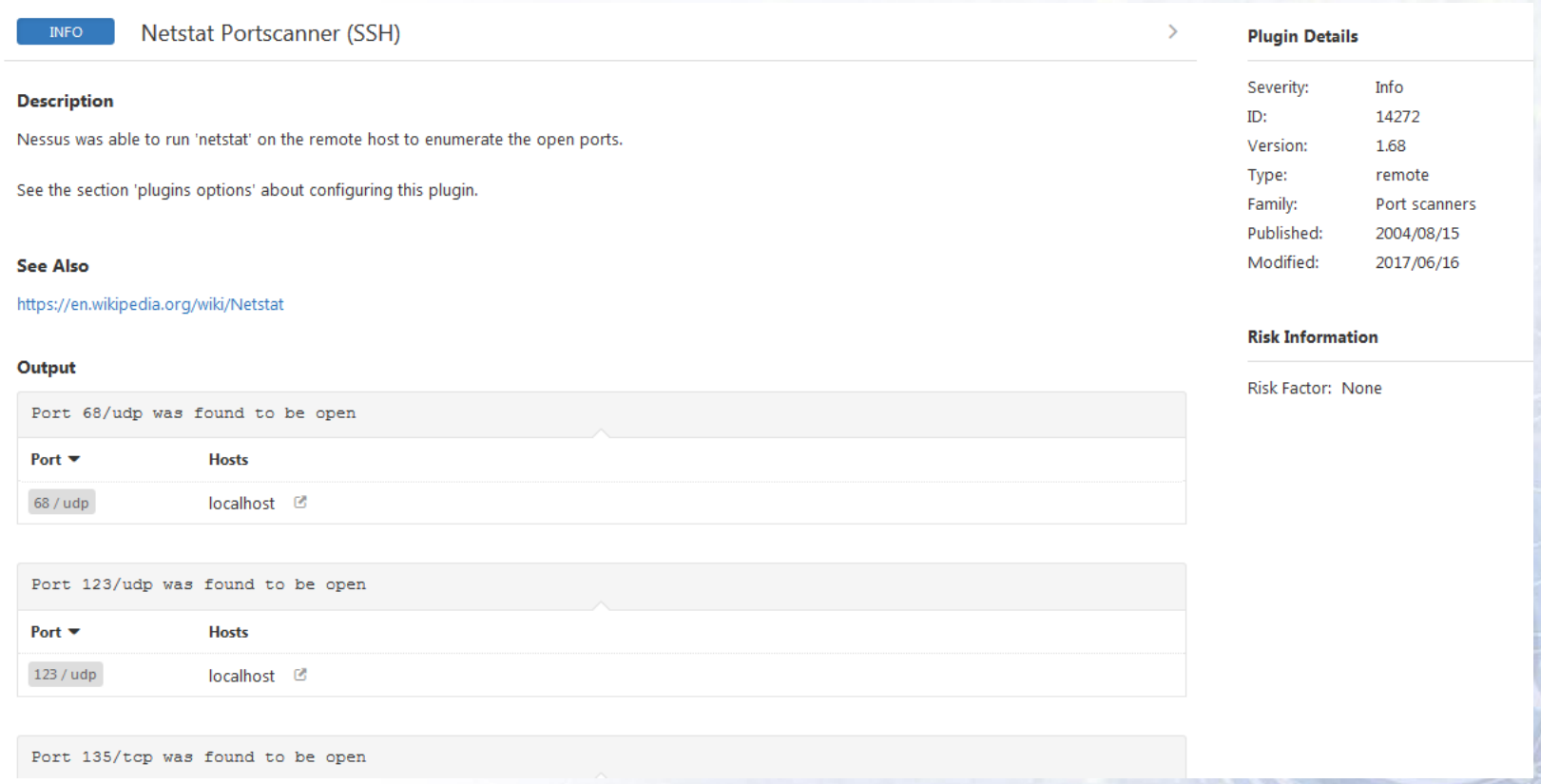
Nessus Demo - 2

CREDENTIALED AUDIT SCAN RESULTS [61 INFO]



Nessus Demo - 2

CREDENTIALED AUDIT SCAN RESULTS [DETAILS]



INFO Netstat Portscanner (SSH)

Description

Nessus was able to run 'netstat' on the remote host to enumerate the open ports.

See the section 'plugins options' about configuring this plugin.

See Also

https://en.wikipedia.org/wiki/Netstat

Output

```
Port 68/udp was found to be open
```

Port	Hosts
68 / udp	localhost

```
Port 123/udp was found to be open
```

Port	Hosts
123 / udp	localhost

```
Port 135/tcp was found to be open
```

Plugin Details

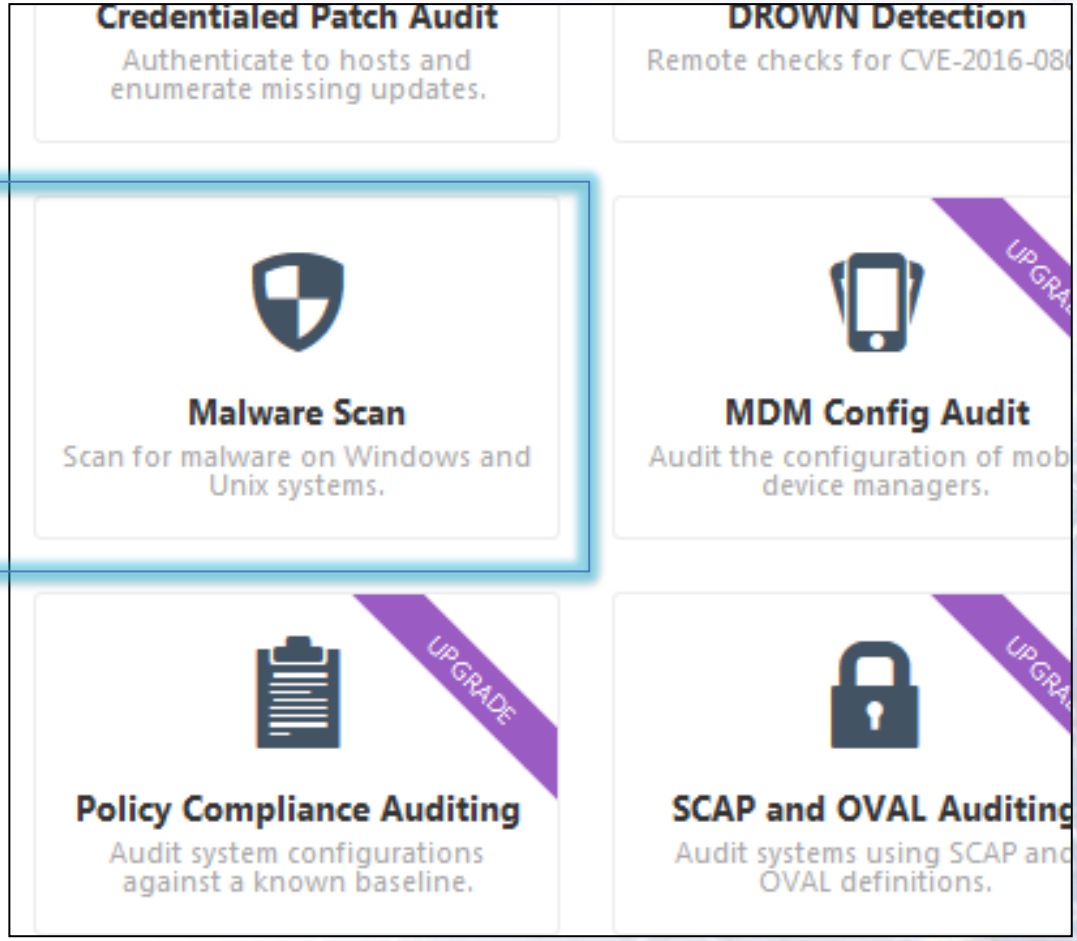
Severity:	Info
ID:	14272
Version:	1.68
Type:	remote
Family:	Port scanners
Published:	2004/08/15
Modified:	2017/06/16

Risk Information

Risk Factor: None

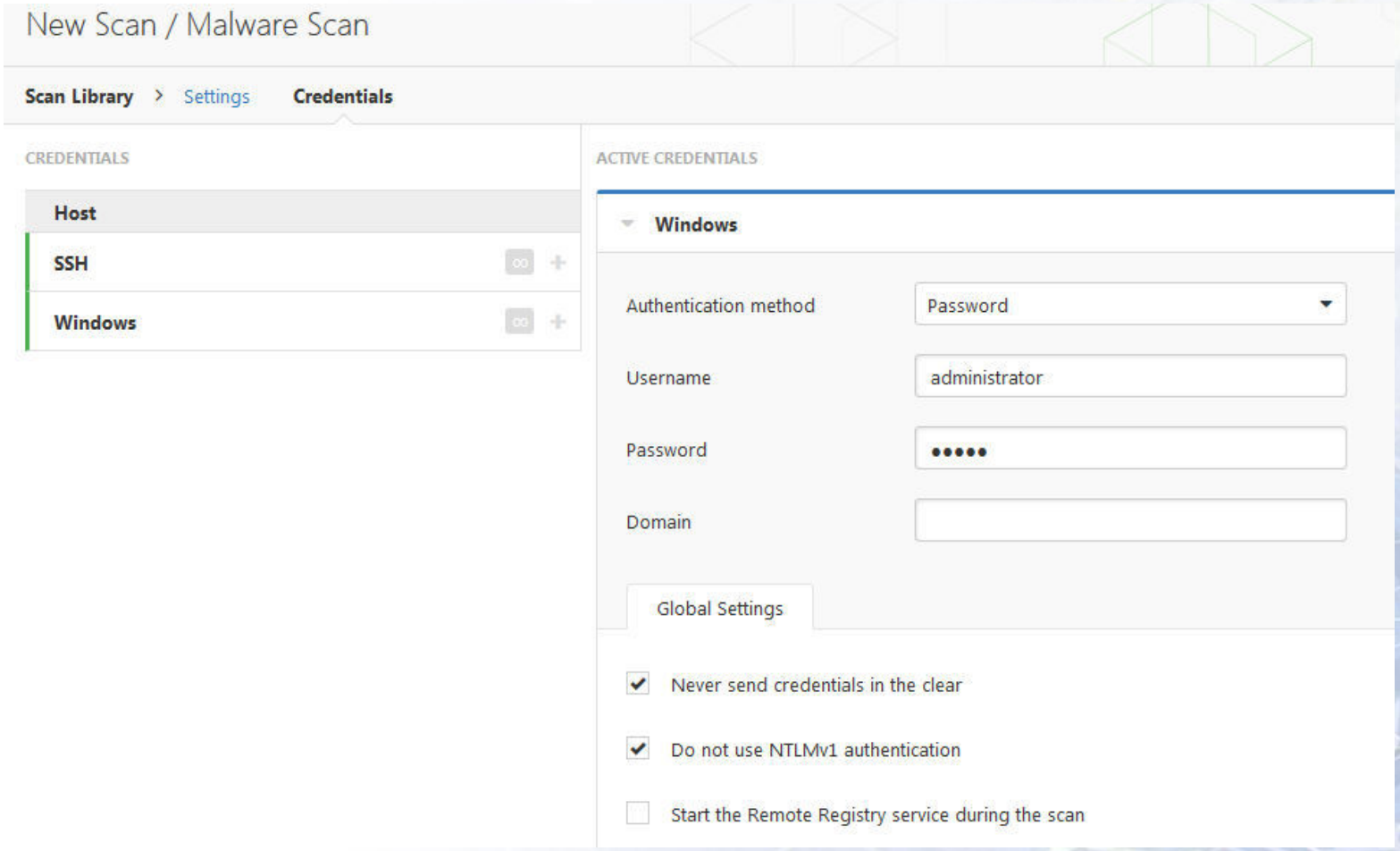
Nessus Demo - 2

SYSTEM MALWARE SCAN



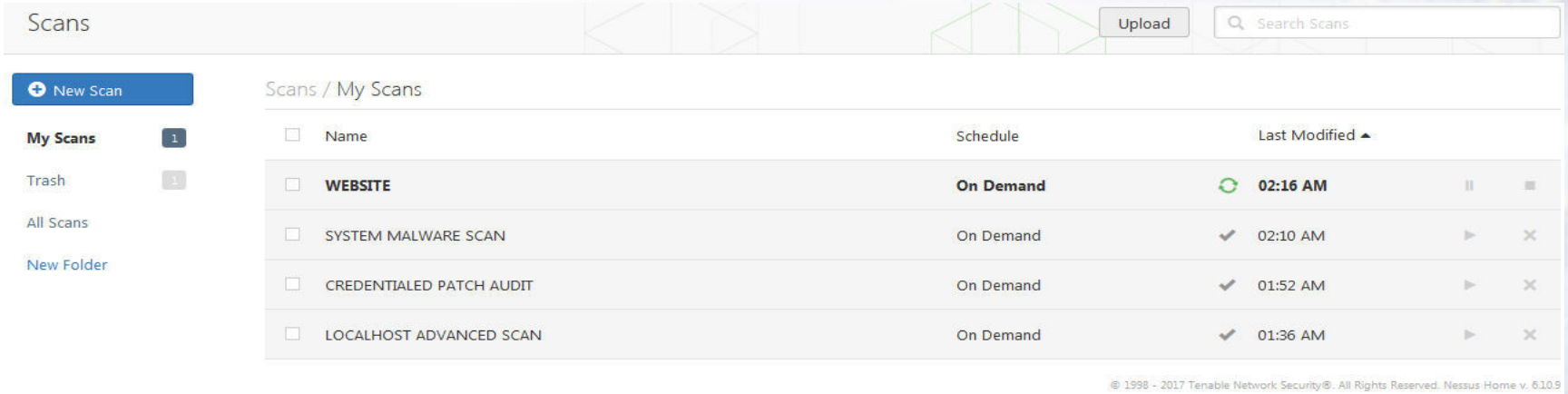
Nessus Demo - 2

SYSTEM MALWARE SCAN

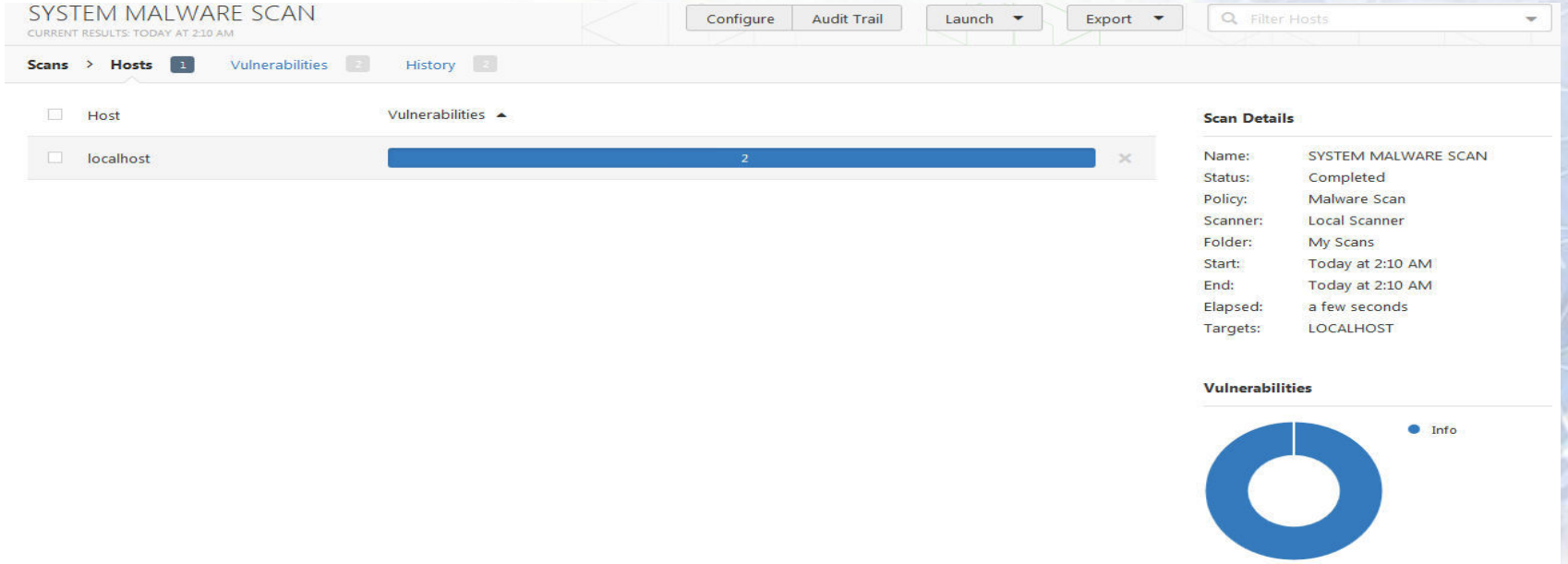


Nessus Demo - 2

SCAN DASHBOARD

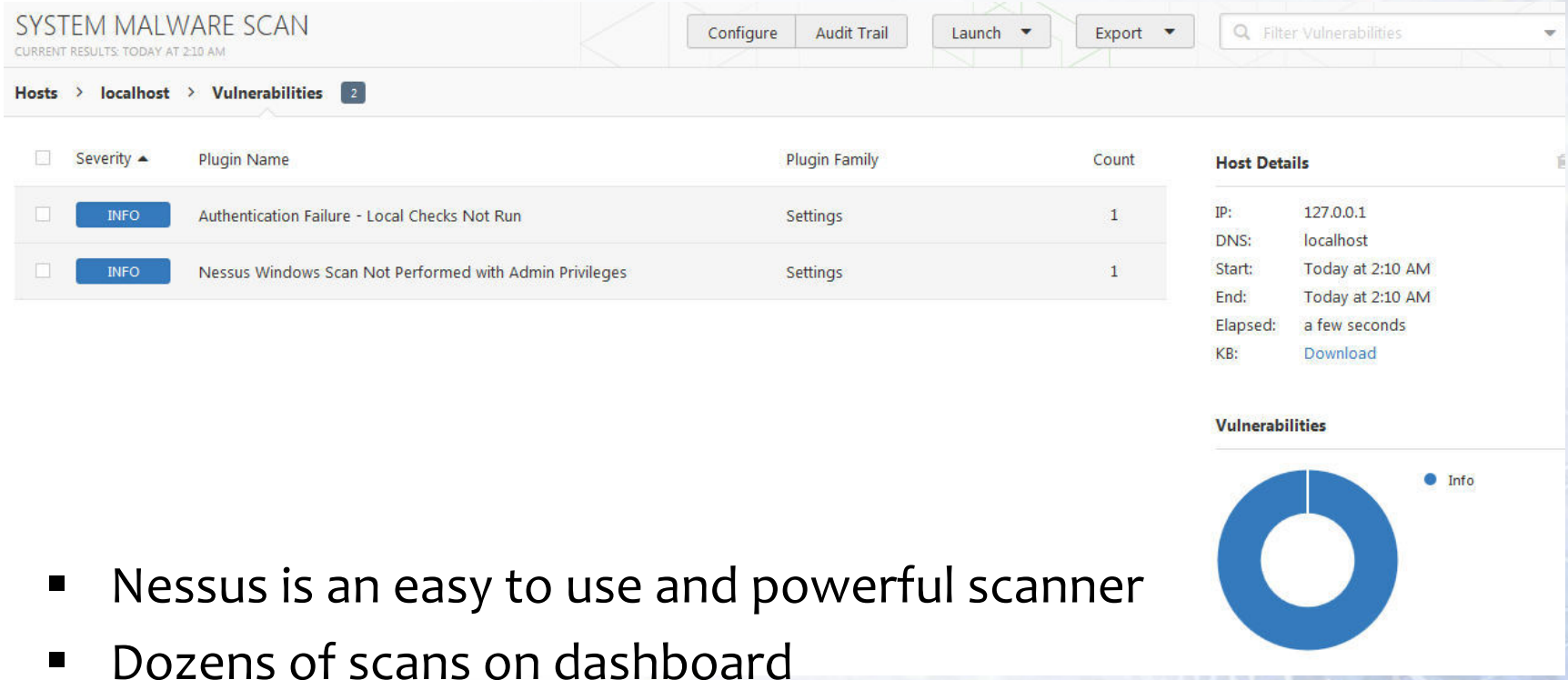


SYSTEM MALWARE SCAN [2 INFO]



Nessus Demo - 2

SYSTEM MALWARE SCAN



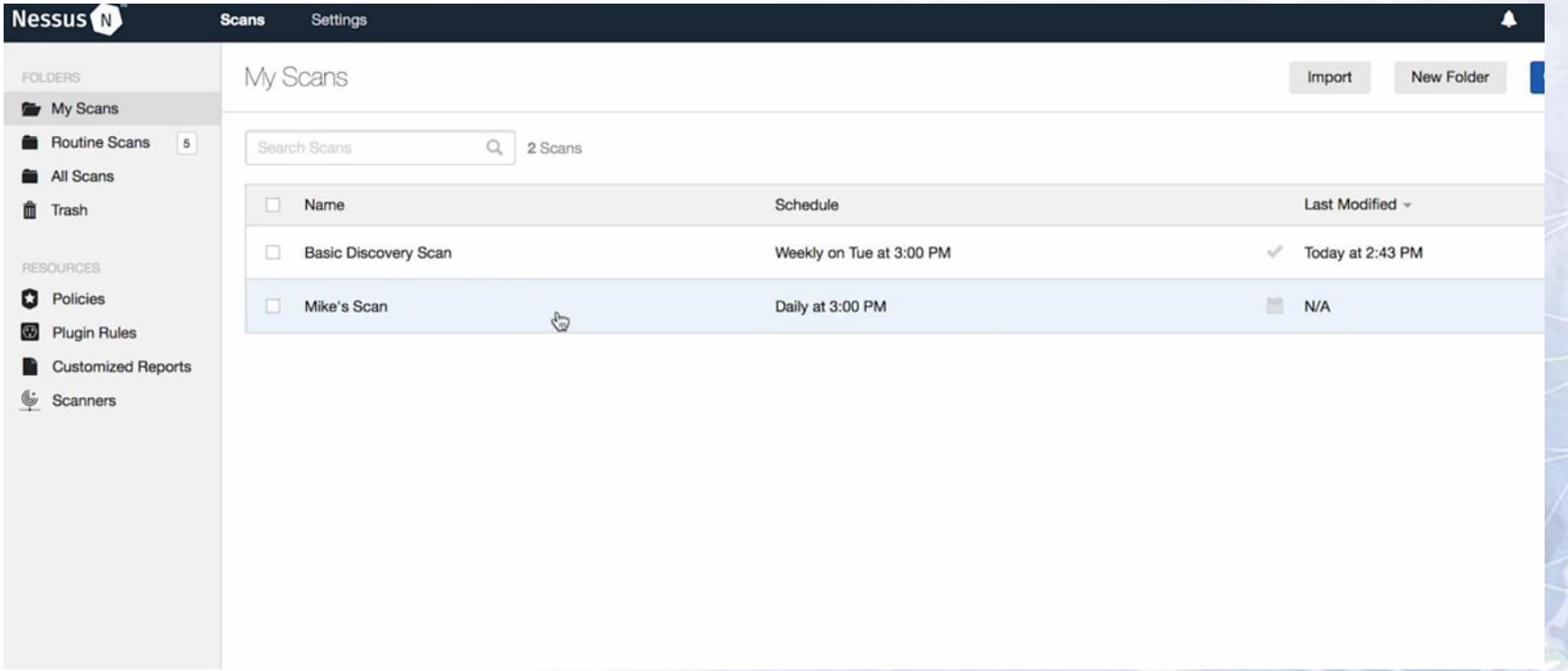
- Nessus is an easy to use and powerful scanner
- Dozens of scans on dashboard
- Setup a new scan
- Setup a profile as scan template
- Credentialed scan
- Thousands of plugins from CIS/DISA etc

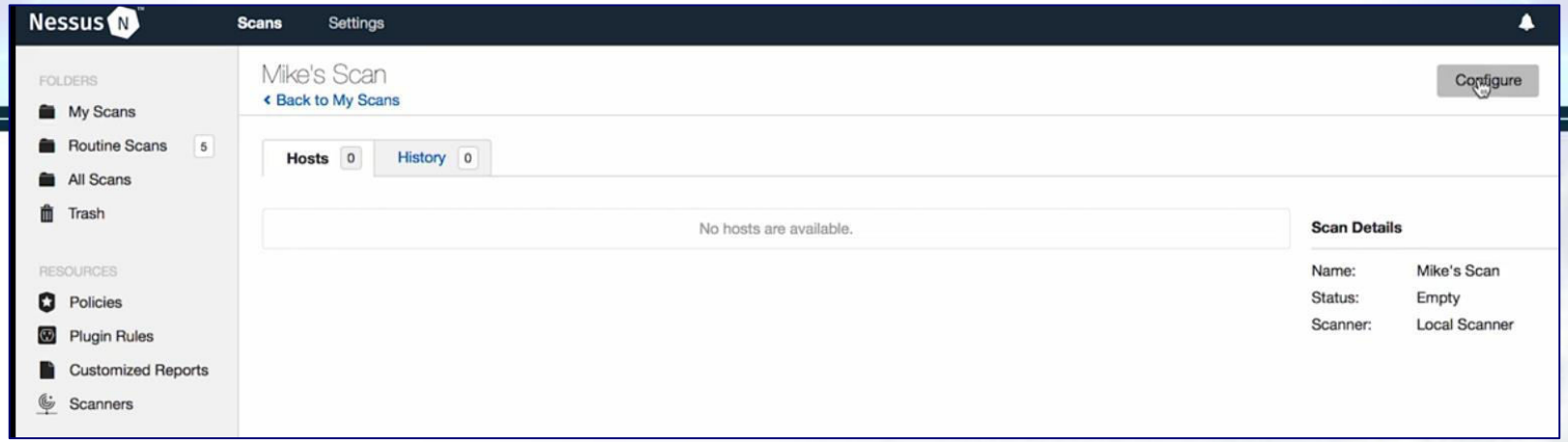
Week 7 Topic 14

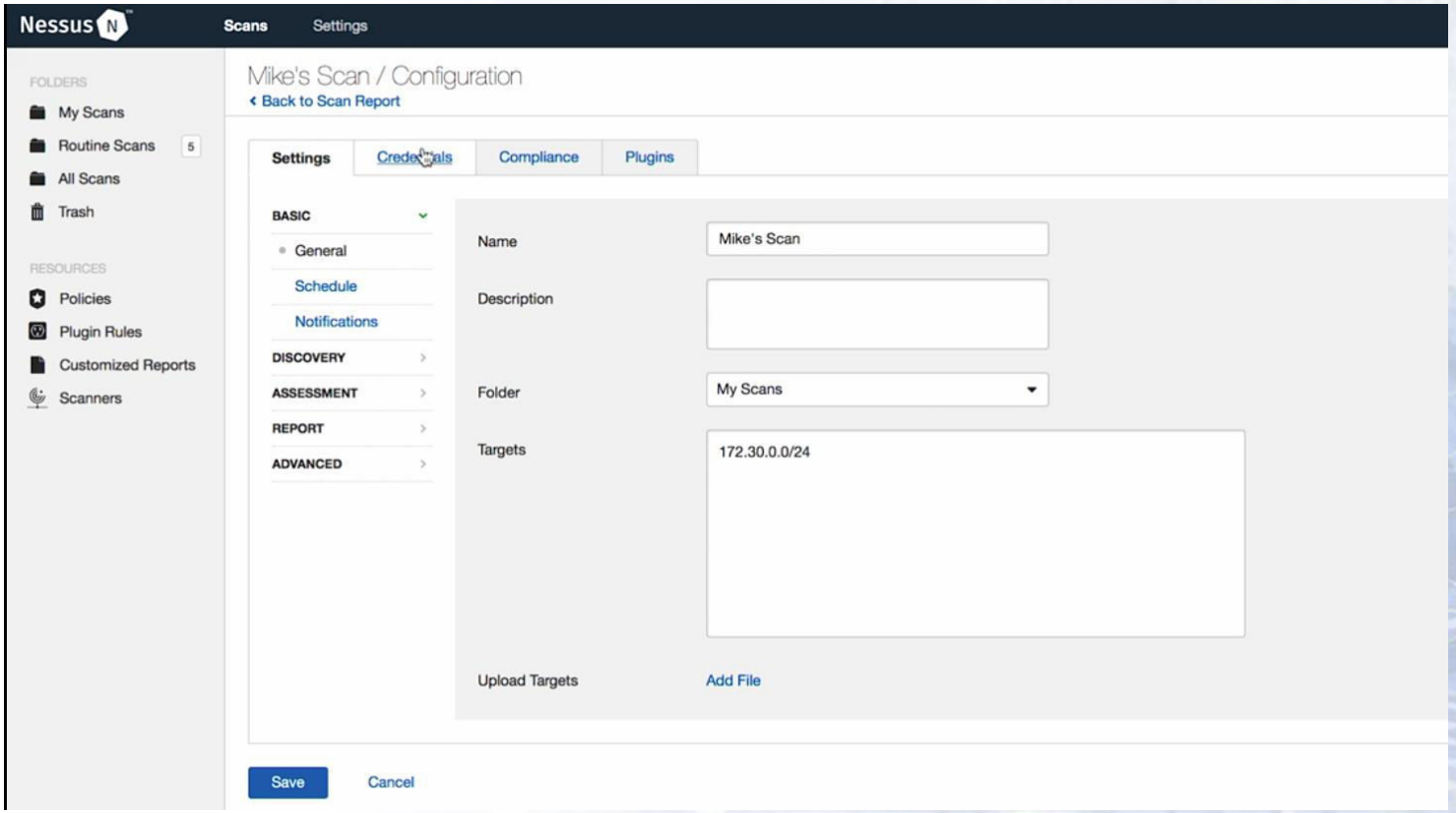
Nessus scan Types

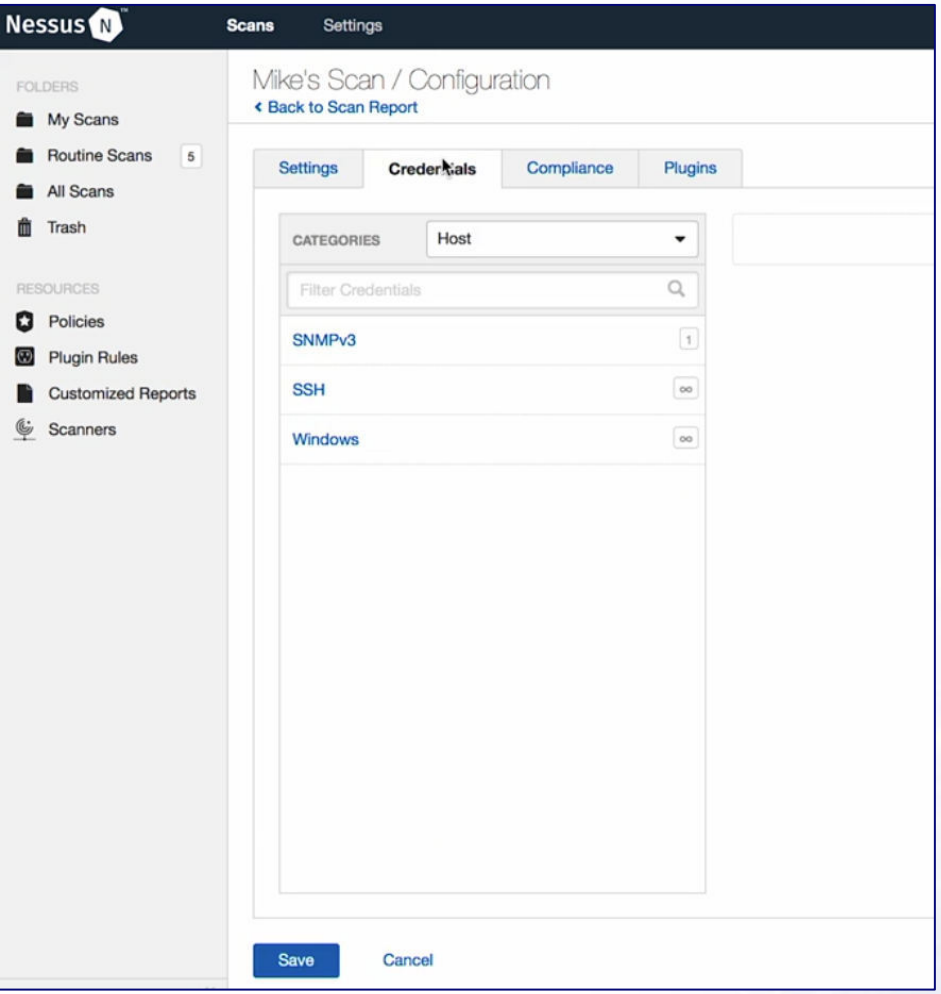
Credentialed Scanning

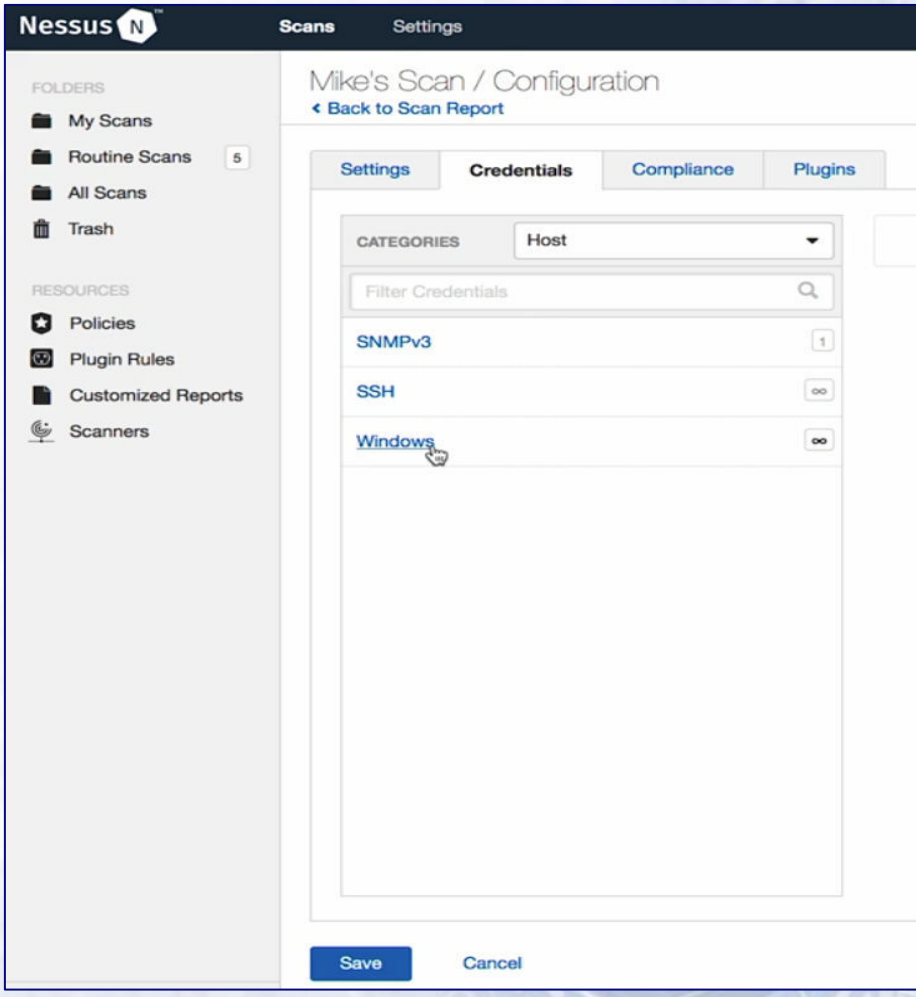
Uses passwords to log into systems

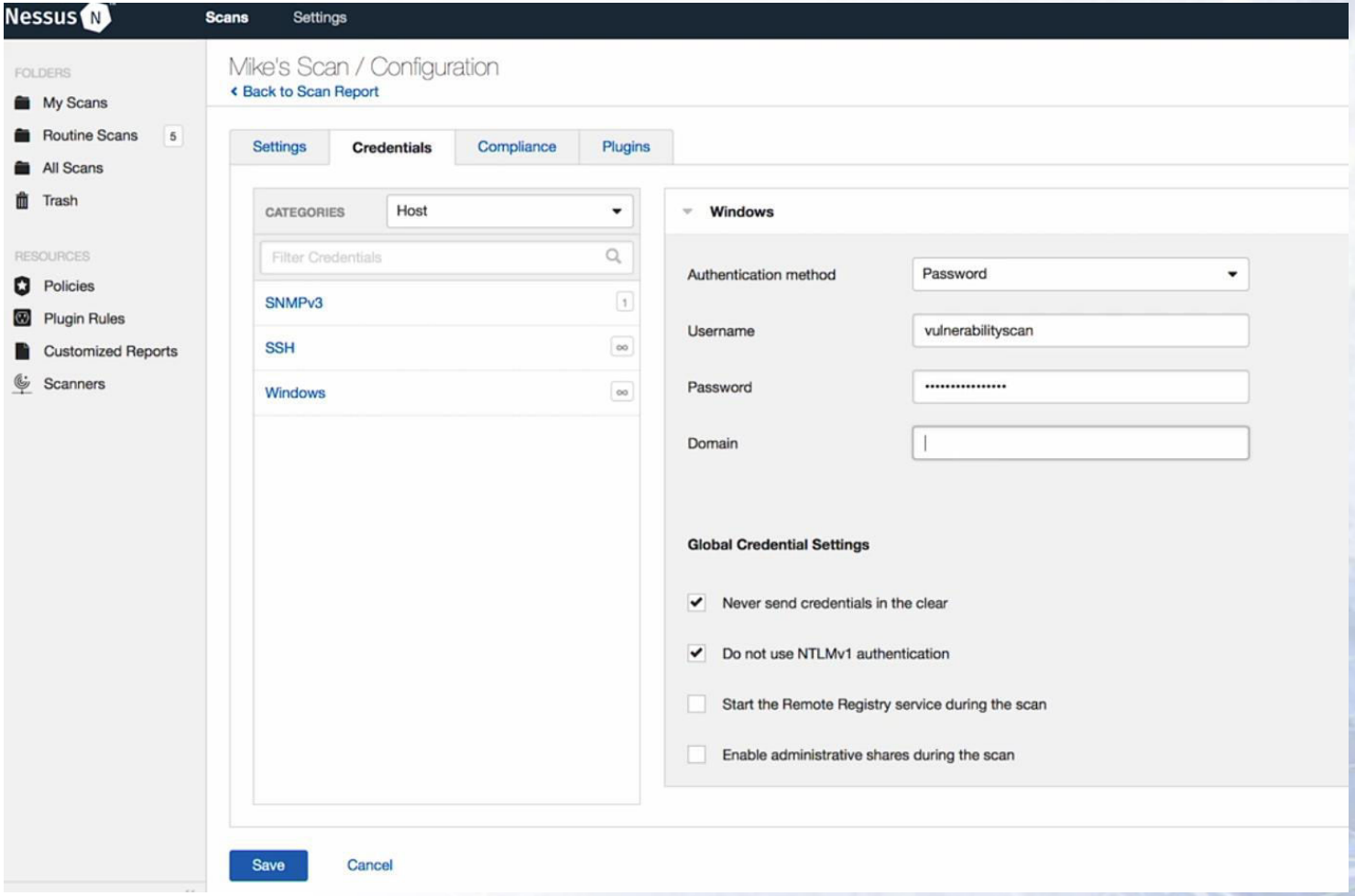



Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Mike's Scan
Back to My Scans
Configure
Hosts 0
History 0
No hosts are available.
Scan Details
Name: Mike's Scan
Status: Empty
Scanner: Local Scanner


Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Mike's Scan / Configuration
Back to Scan Report
Settings
Credentials
Compliance
Plugins
BASIC
General
Schedule
Notifications
DISCOVERY
ASSESSMENT
REPORT
ADVANCED
Name
Mike's Scan
Description
Folder
My Scans
Targets
172.30.0.0/24
Upload Targets
Add File
Save
Cancel


Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans 5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Mike's Scan / Configuration
Back to Scan Report
Settings
Credentials
Compliance
Plugins
CATEGORIES
Host
Filter Credentials
SNMPv3 1
SSH ∞
Windows ∞
Save
Cancel


Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans 5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Mike's Scan / Configuration
Back to Scan Report
Settings
Credentials
Compliance
Plugins
CATEGORIES
Host
Filter Credentials
SNMPv3 1
SSH ∞
Windows ∞
Save
Cancel


Nessus
Scans
Settings
FOLDERS
My Scans
Routine Scans
5
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Customized Reports
Scanners
Mike's Scan / Configuration
Back to Scan Report
Settings
Credentials
Compliance
Plugins
CATEGORIES
Host
Filter Credentials
SNMPv3
1
SSH
Windows
Windows
Authentication method
Password
Username
vulnerabilityscan
Password
Domain
Global Credential Settings
Never send credentials in the clear
Do not use NTLMv1 authentication
Start the Remote Registry service during the scan
Enable administrative shares during the scan
Save
Cancel

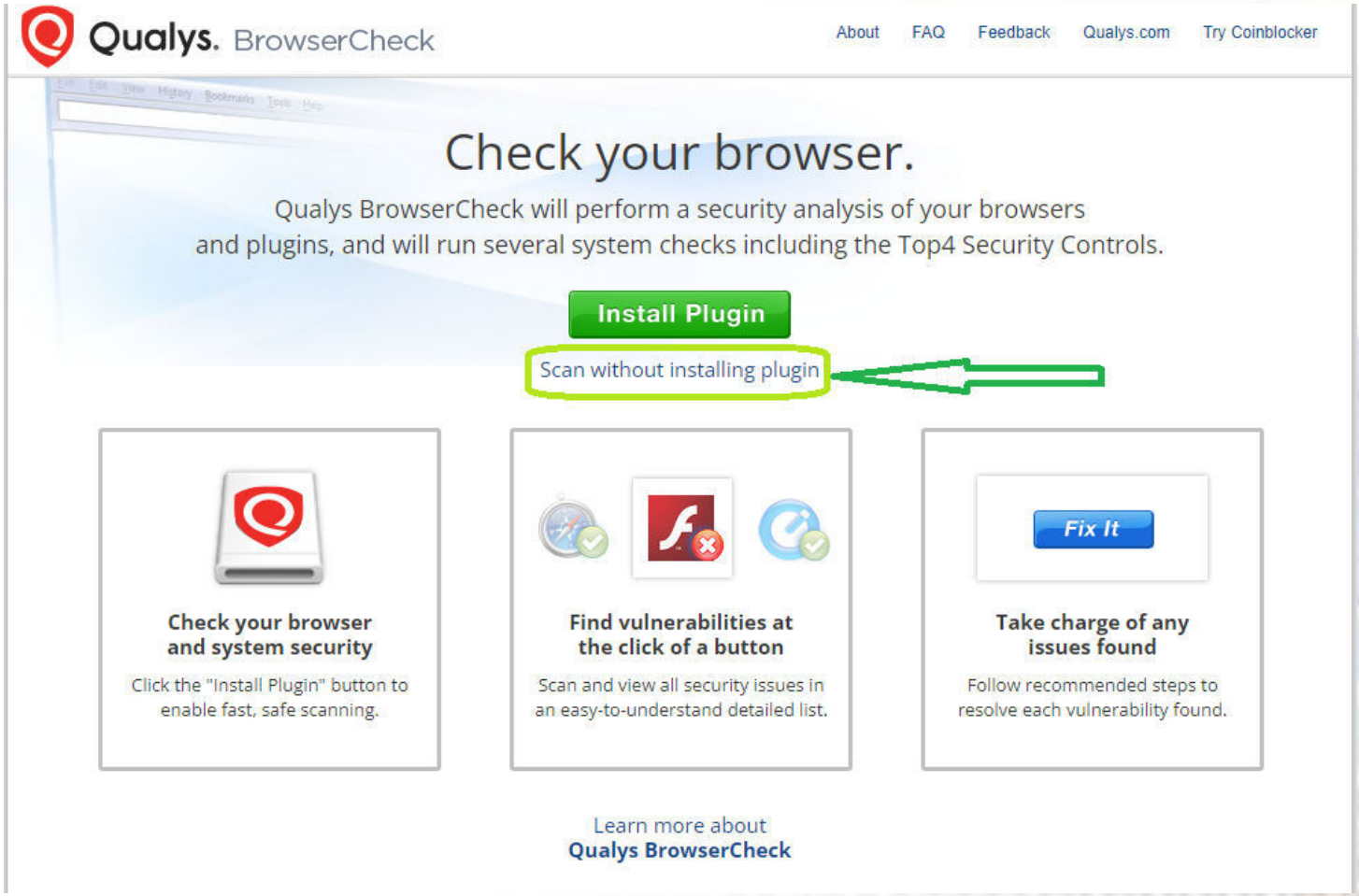
Week 7 Topic 15

Qualys Demo - 1

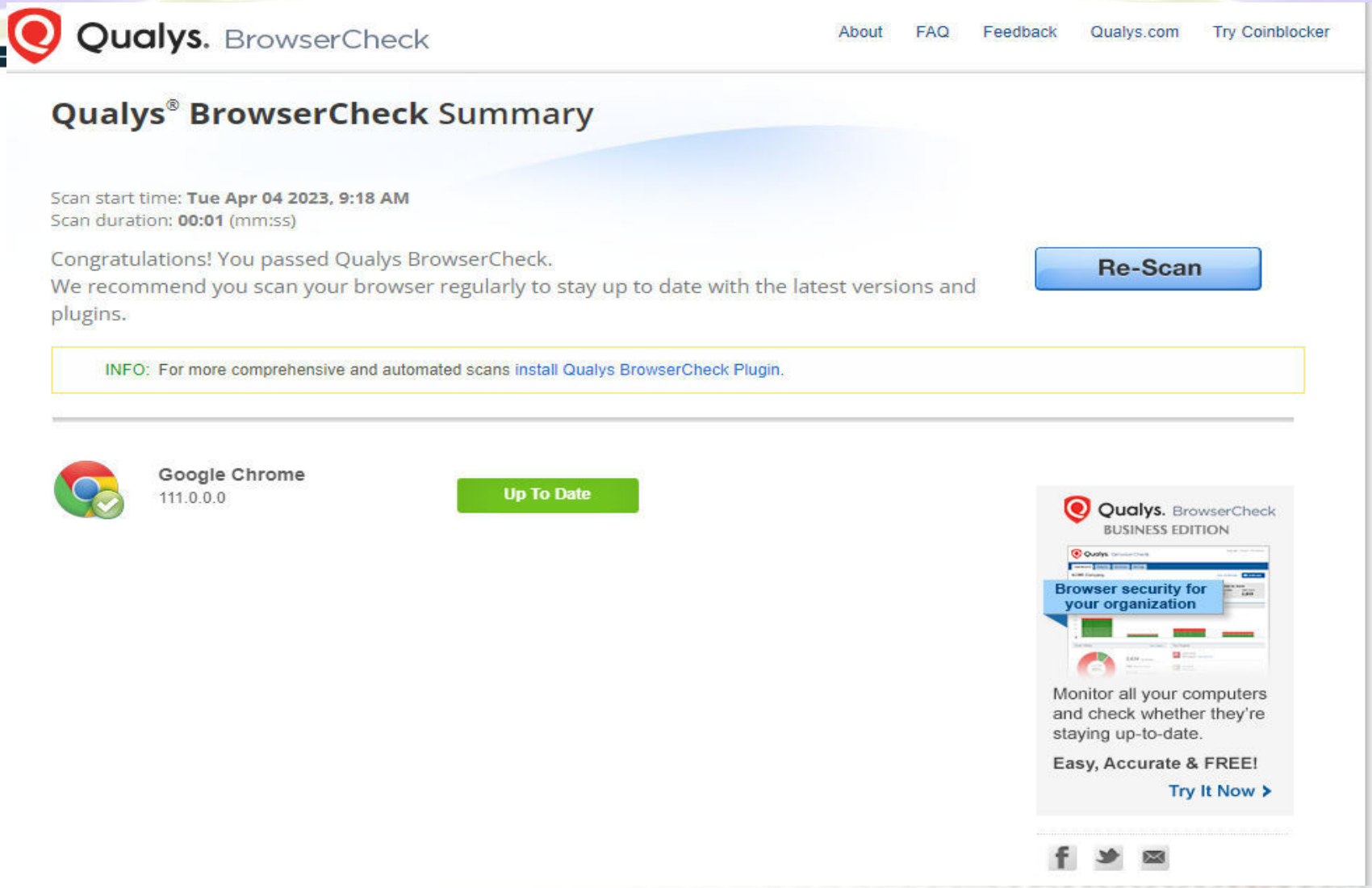
Qualys is a powerful cloud-based vulnerability management tool
Several online free tools
Advanced web application security testing

- Lets take a look at Qualys Demo
- **Free Qualys tools:**
  - BrowserCheck
  - SSL
- **Qualys FreeScan**
  - Vulnerability
  - OWASP
  - Patch Tuesday
  - SCAP

Qualys Demo - 1



https://www.qualys.com/free-services/



https://www.qualys.com/free-services/

Qualys. SSL Labs

Home Projects Qualys Free Trial Contact

HOW WELL DO YOU KNOW SSL?

If you want to learn more about the technology that protects the Internet, you've come to the right place.

Test your server »
Test your site's certificate and configuration

Test your browser »
Test your browser's SSL implementation

SSL Pulse »
See how other web sites are doing

Documentation »
Learn how to deploy SSL/TLS correctly

Books

Bulletproof SSL and TLS is a complete guide to deploying secure servers and web applications. This book, which provides comprehensive coverage of the ever-changing field of SSL/TLS and Web PKI, is intended for IT security professionals, system administrators, and developers, with the main focus on getting things done. MORE »

News

LinkedIn Faux Pas Shines Light on Certificate Management
May 24, 2019

Zombie POODLE and GOLDENDOODLE Vulnerabilities
April 22, 2019

PCI & SSL/Early TLS QIDs 38601, 42366
March 14, 2019

About SSL Labs

SSL Labs is a collection of documents, tools and thoughts related to SSL. It's an attempt to better understand how SSL is deployed, and an attempt to make it better. I hope that, in time, SSL Labs will grow into a forum where SSL will be discussed and improved.

SSL Labs is a non-commercial research effort, and we welcome participation from any individual and organization interested in SSL.

-- *Ivan Ristić, Qualys*

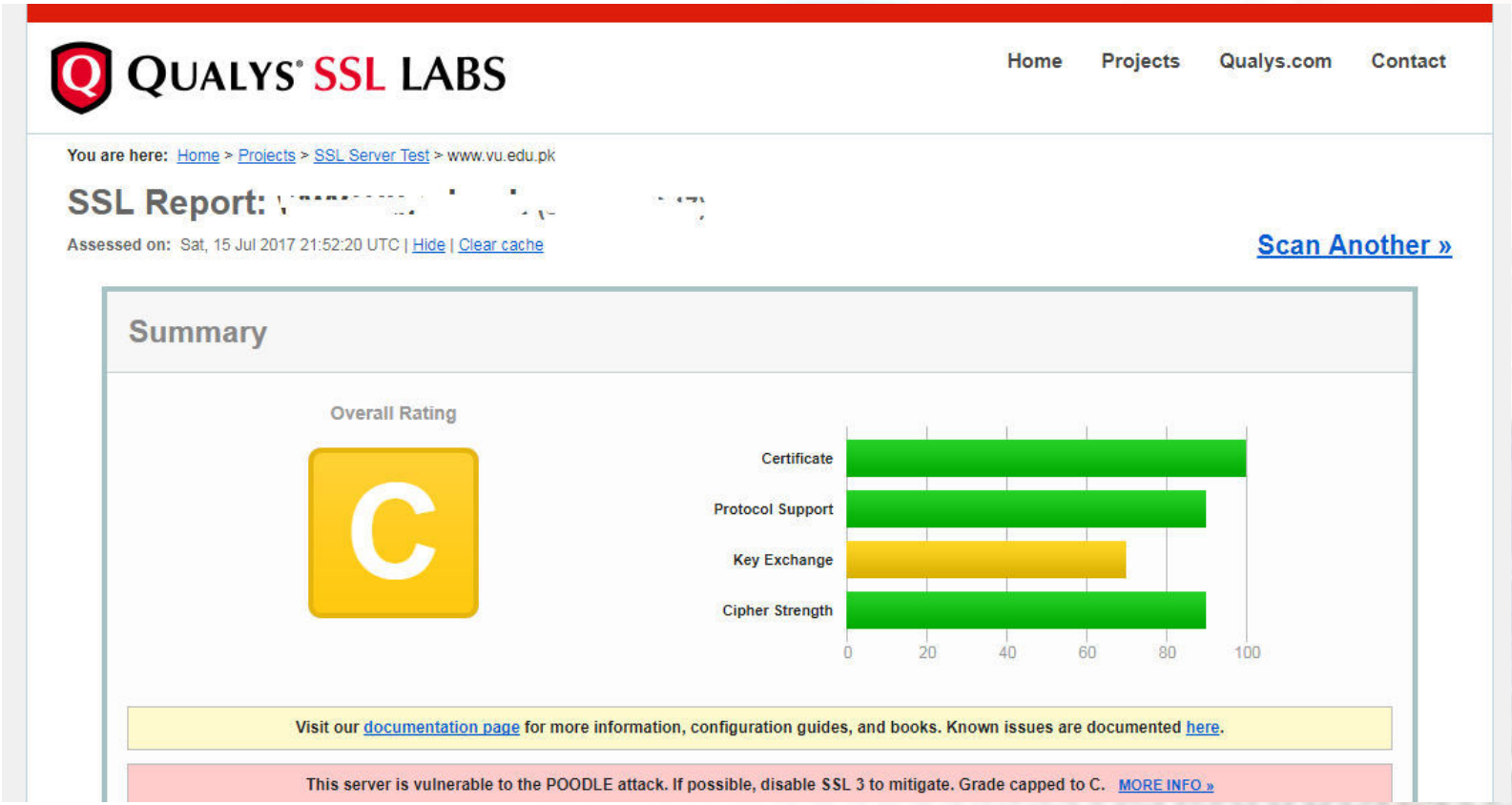
Copyright © 2009-2023 Qualys, Inc. All Rights Reserved.

Terms and Conditions

Try Qualys for free! Experience the award-winning Qualys Cloud Platform and the entire collection of Qualys Cloud Apps, including certificate security solutions.

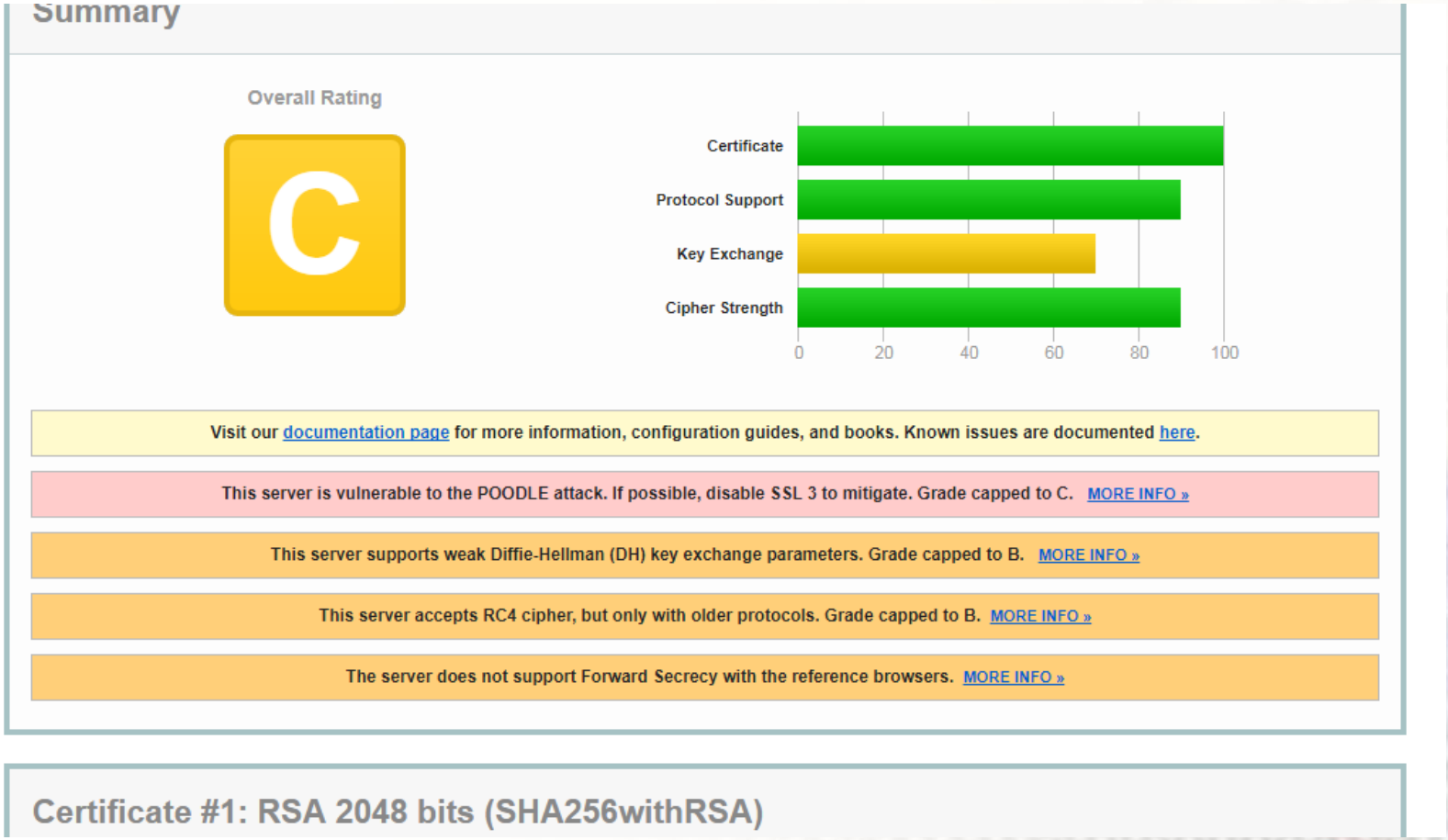
Qualys Demo - 1

SSL LABS REPORT



Qualys Demo - 1

SSL LABS REPORT



https://www.qualys.com/free-services/
https://www.ssllabs.com/

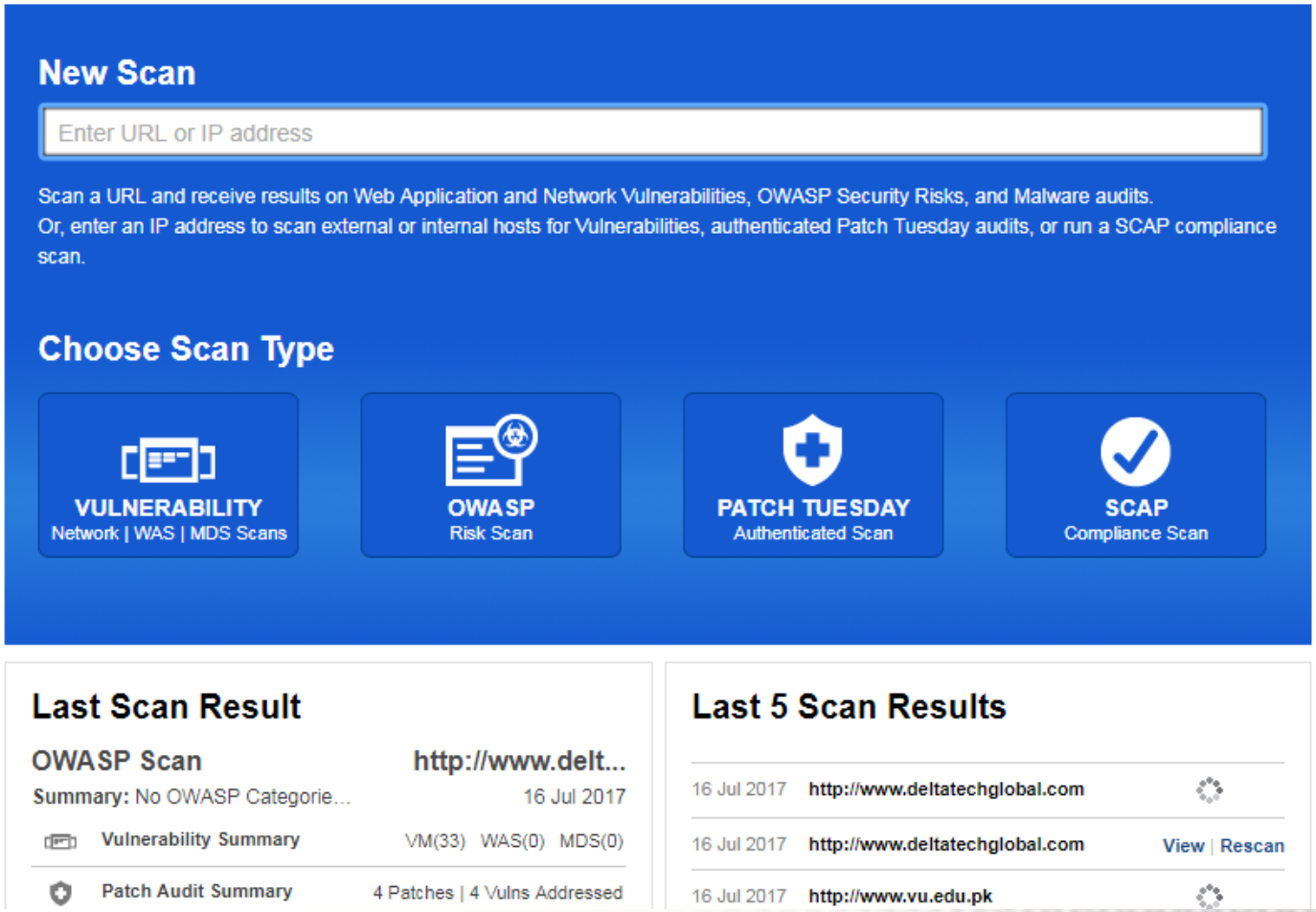
Qualys Demo - 1

PROTOCOL DETAILS

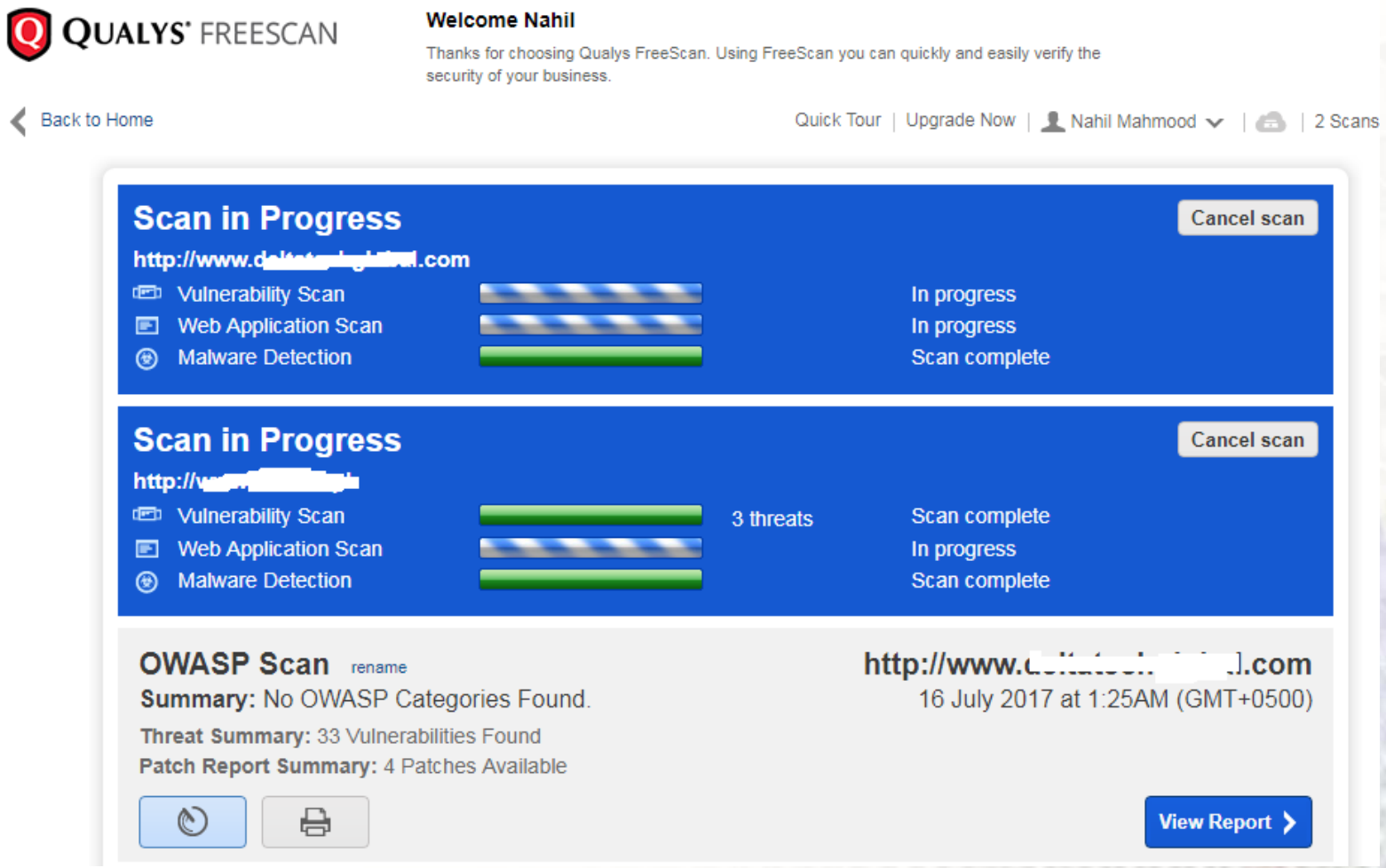
BEAST attack	Not mitigated server-side (more info) SSL 3: 0xa, TLS 1.0: 0xc014
POODLE (SSLv3)	Vulnerable INSECURE (more info) SSL 3: 0xa
POODLE (TLS)	No (more info)
Downgrade attack prevention	No, TLS_FALLBACK_SCSV not supported (more info)
SSL/TLS compression	No
RC4	Yes INSECURE (more info)
Heartbeat (extension)	No
Heartbleed (vulnerability)	No (more info)
Ticketbleed (vulnerability)	No (more info)
OpenSSL CCS vuln. (CVE-2014-0224)	No (more info)
OpenSSL Padding Oracle vuln. (CVE-2016-2107)	No (more info)
Forward Secrecy	Weak key exchange WEAK
ALPN	No

Qualys Demo - 1

QUALYS FREESCAN

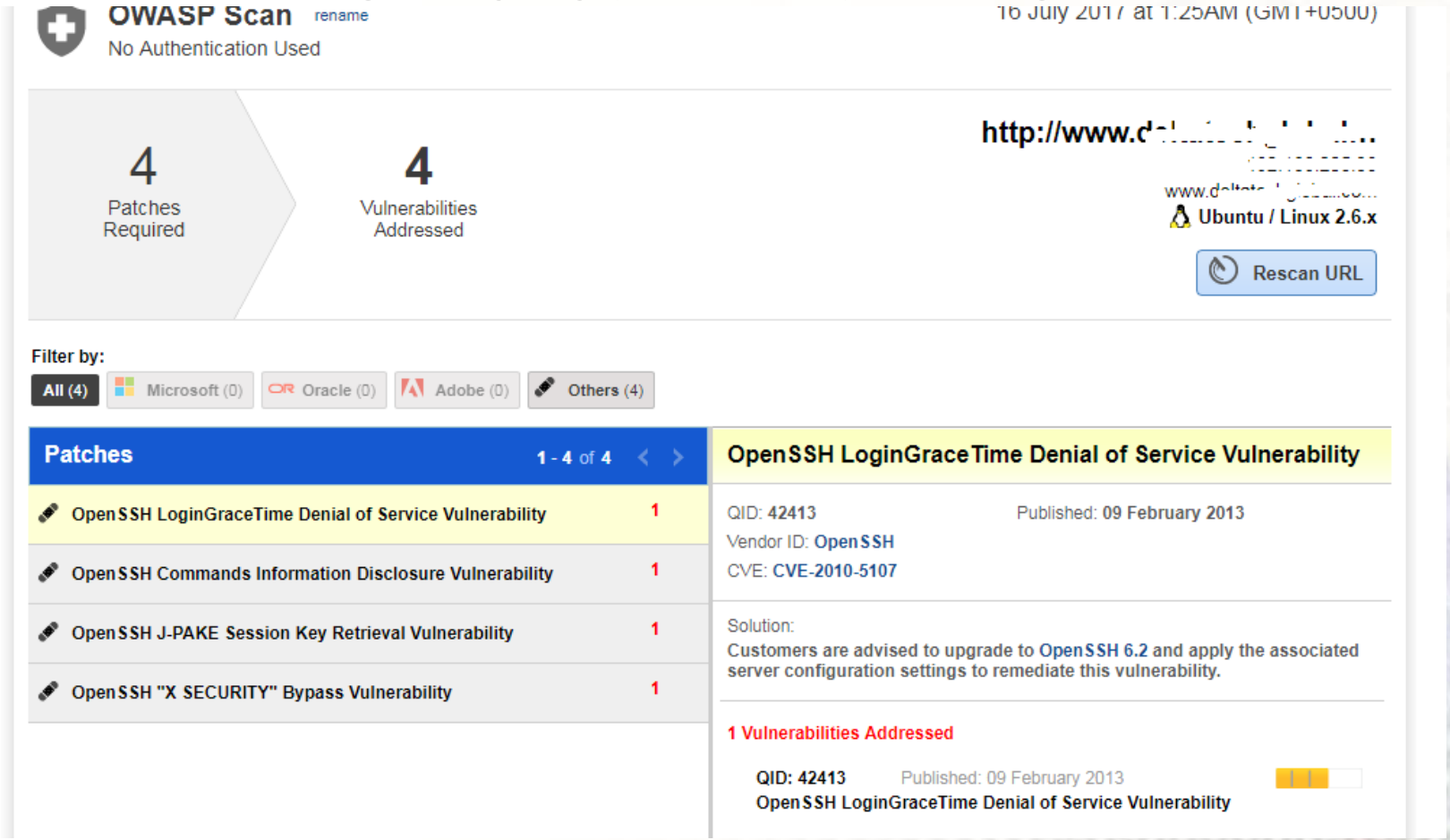


Qualys Demo - 1



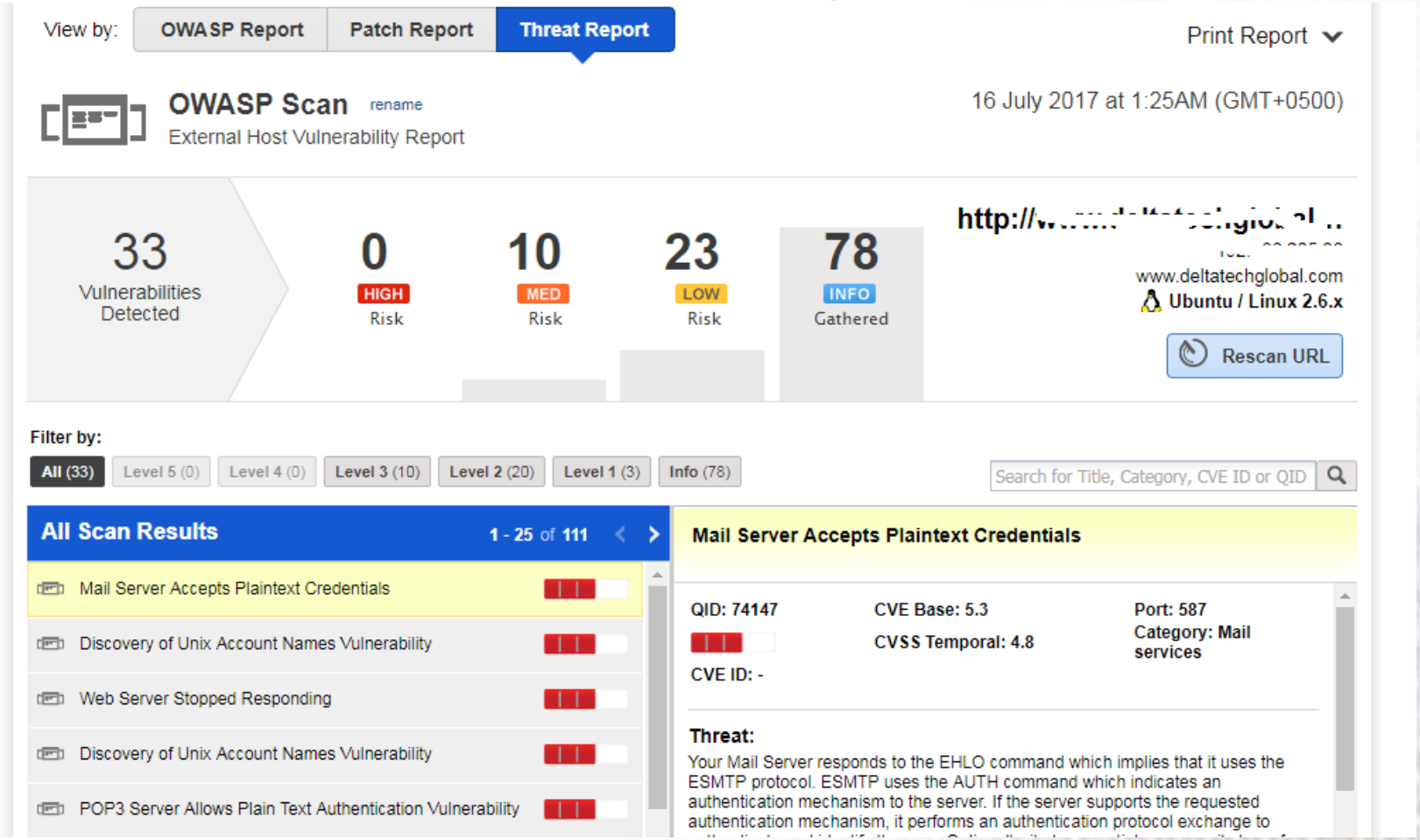
Qualys Demo - 1

OWASP SCAN PATCH REPORT



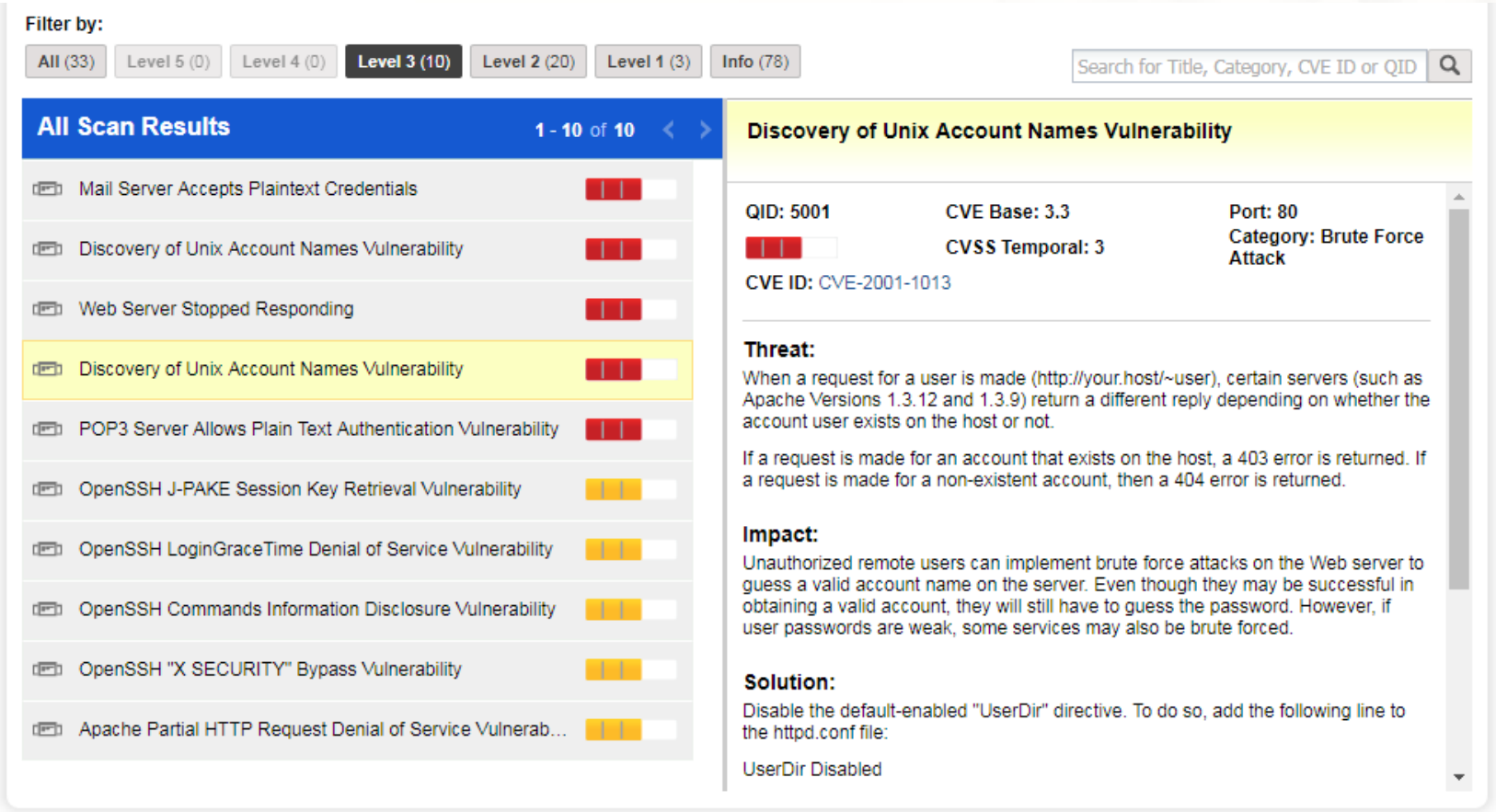
Qualys Demo - 1

THREAT REPORT



Qualys Demo - 1

THREAT REPORT



Qualys Demo - 1

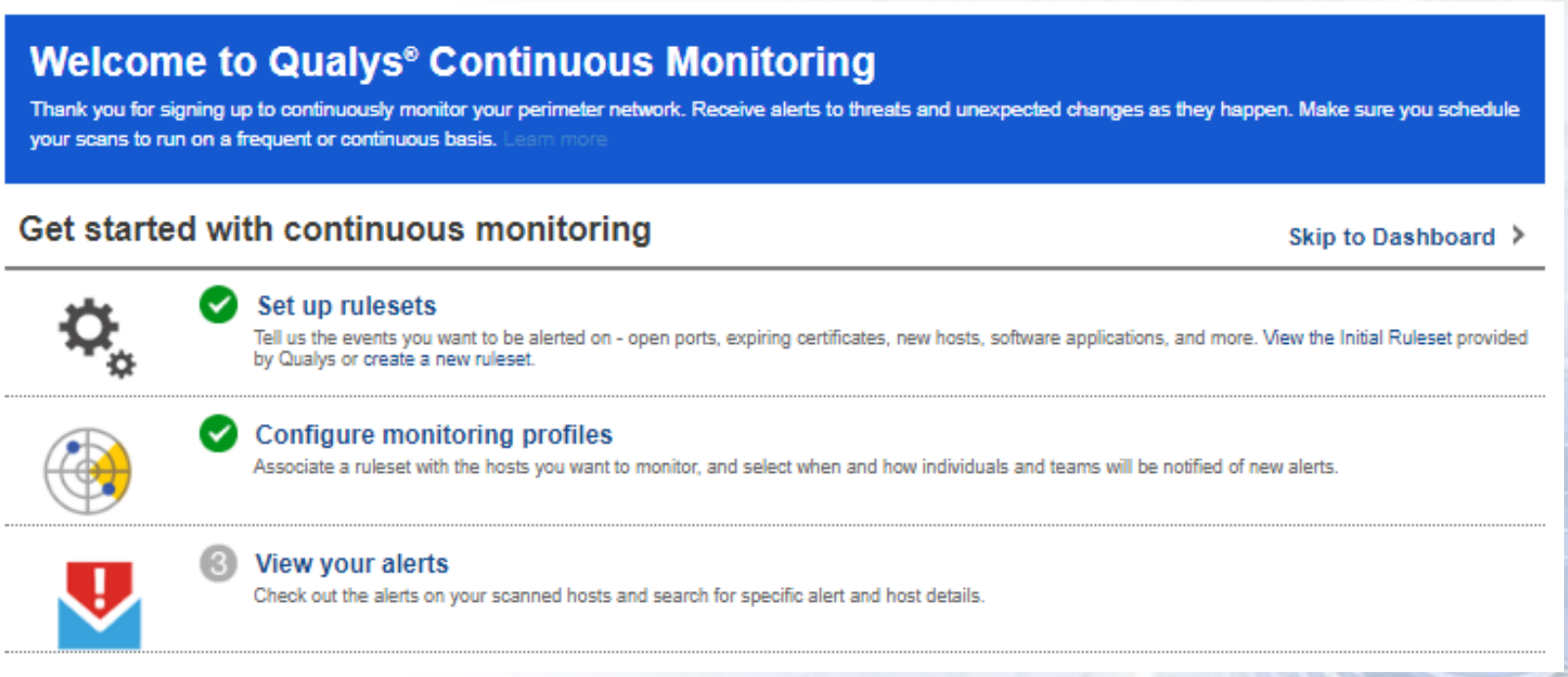
- Qualys is a powerful cloud-based vulnerability management tool
- Several online free tools
- Advanced web application security testing

END

Week 7 Topic 16

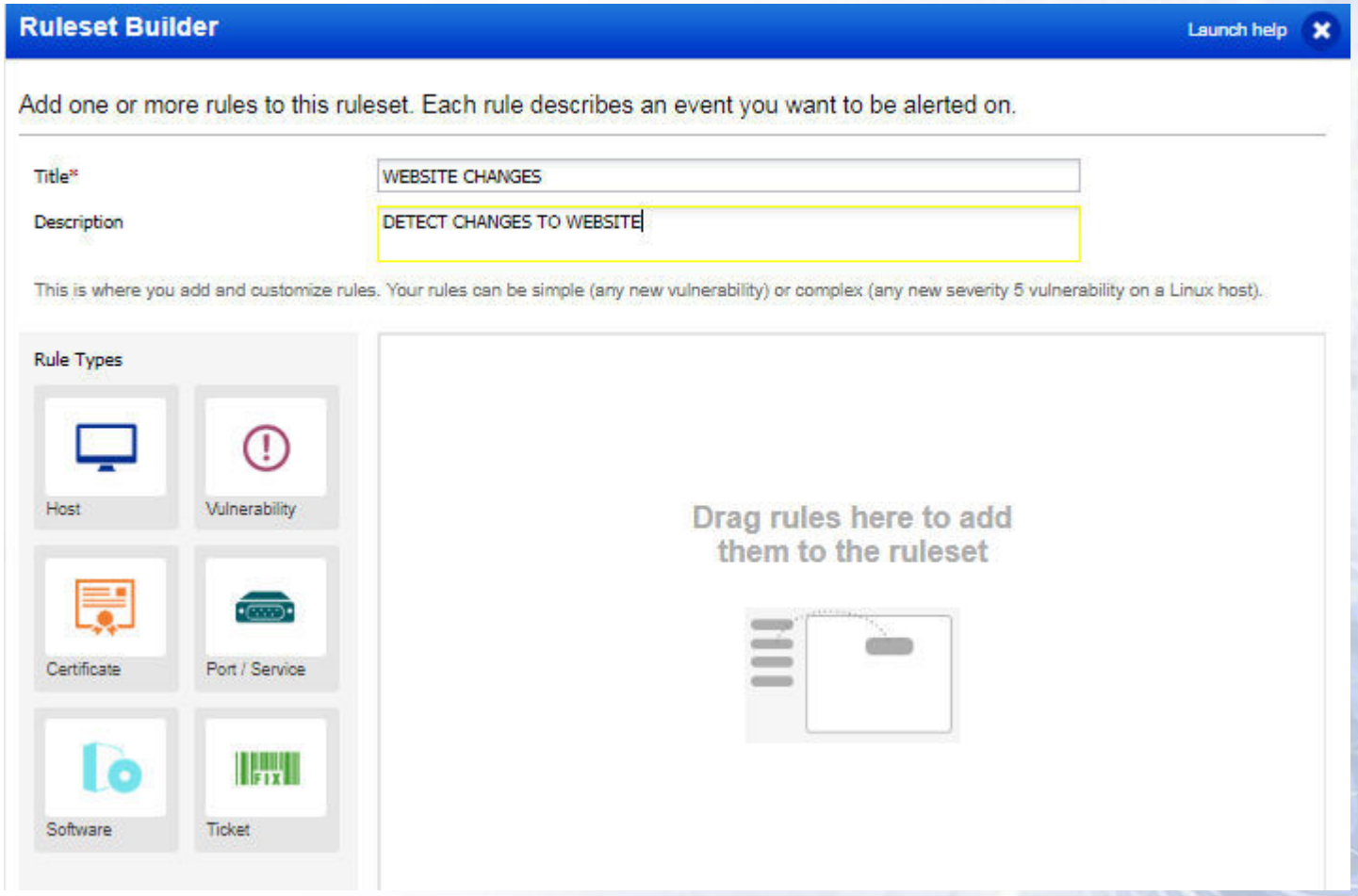
QUALYS DEMO - PART 3

- Lets have a look at some further features of Qualys:
- Qualys Continuous Monitoring



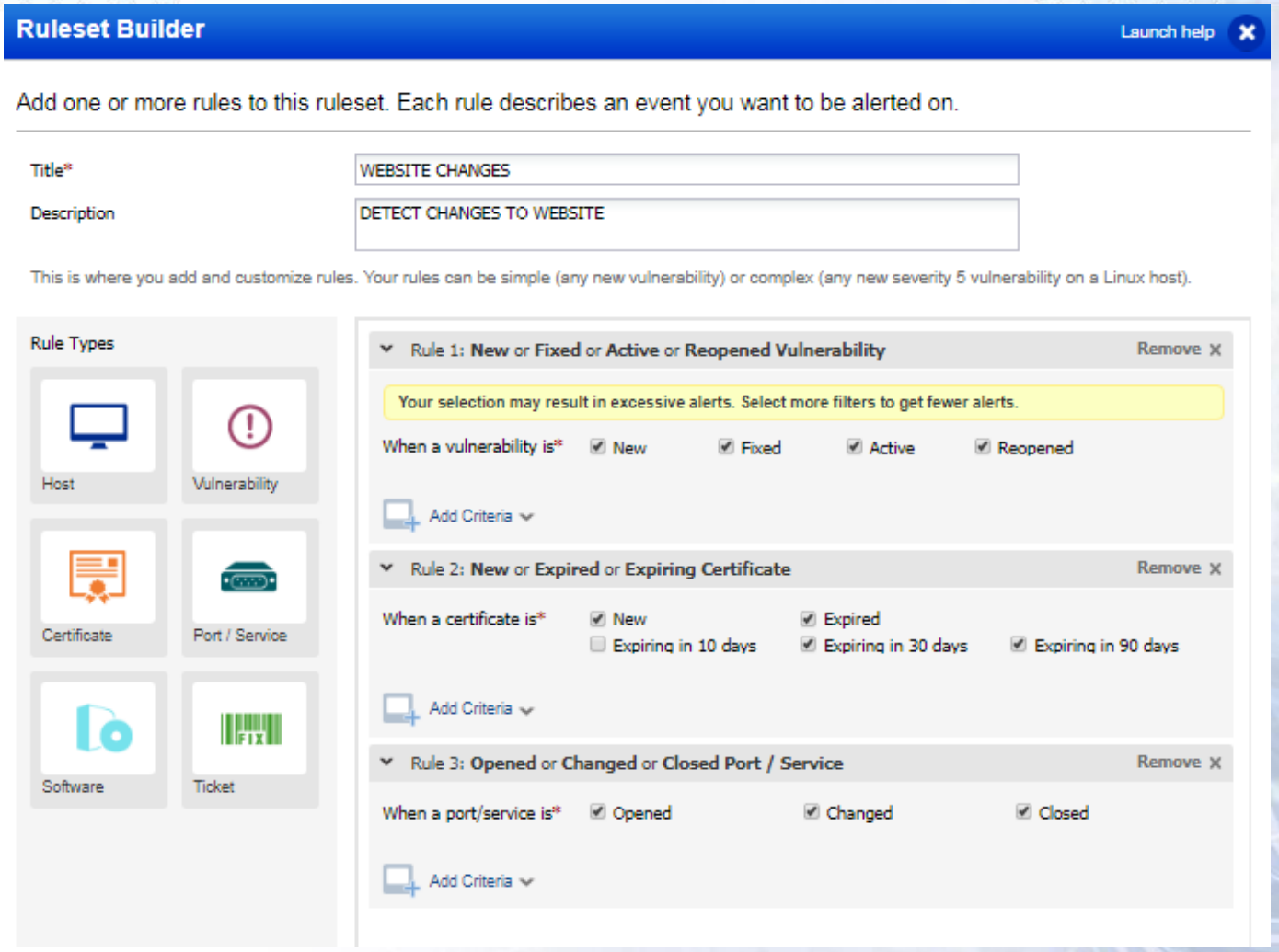
QUALYS CONTINUOUS MONITORING INTERFACE

QUALYS DEMO - PART 3



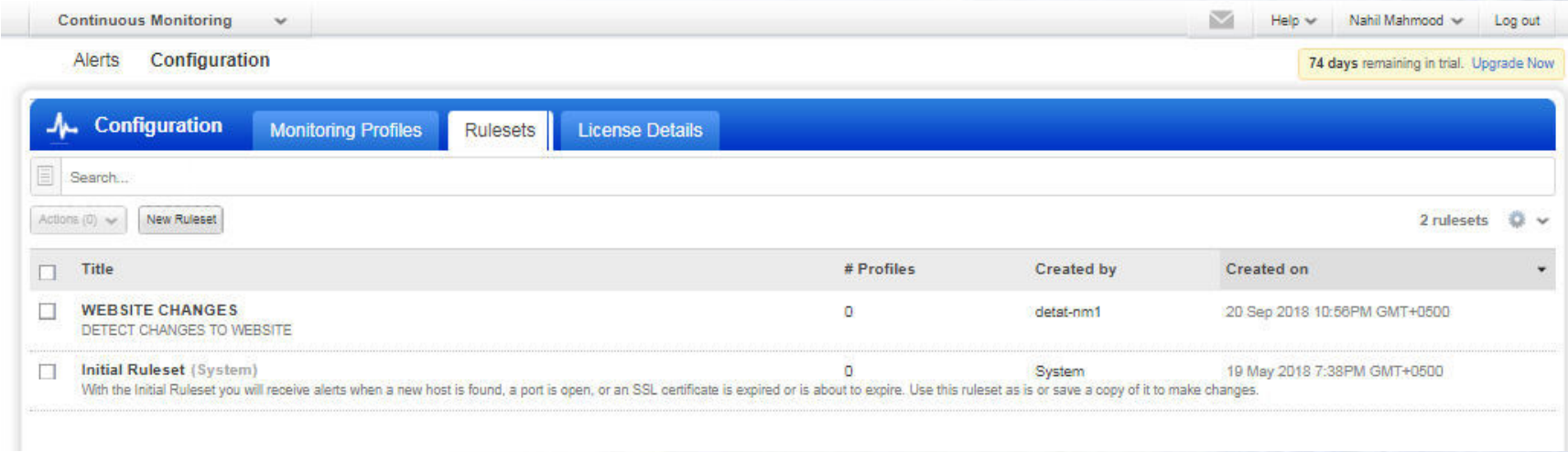
RULESET BUILDER

QUALYS DEMO - PART 3



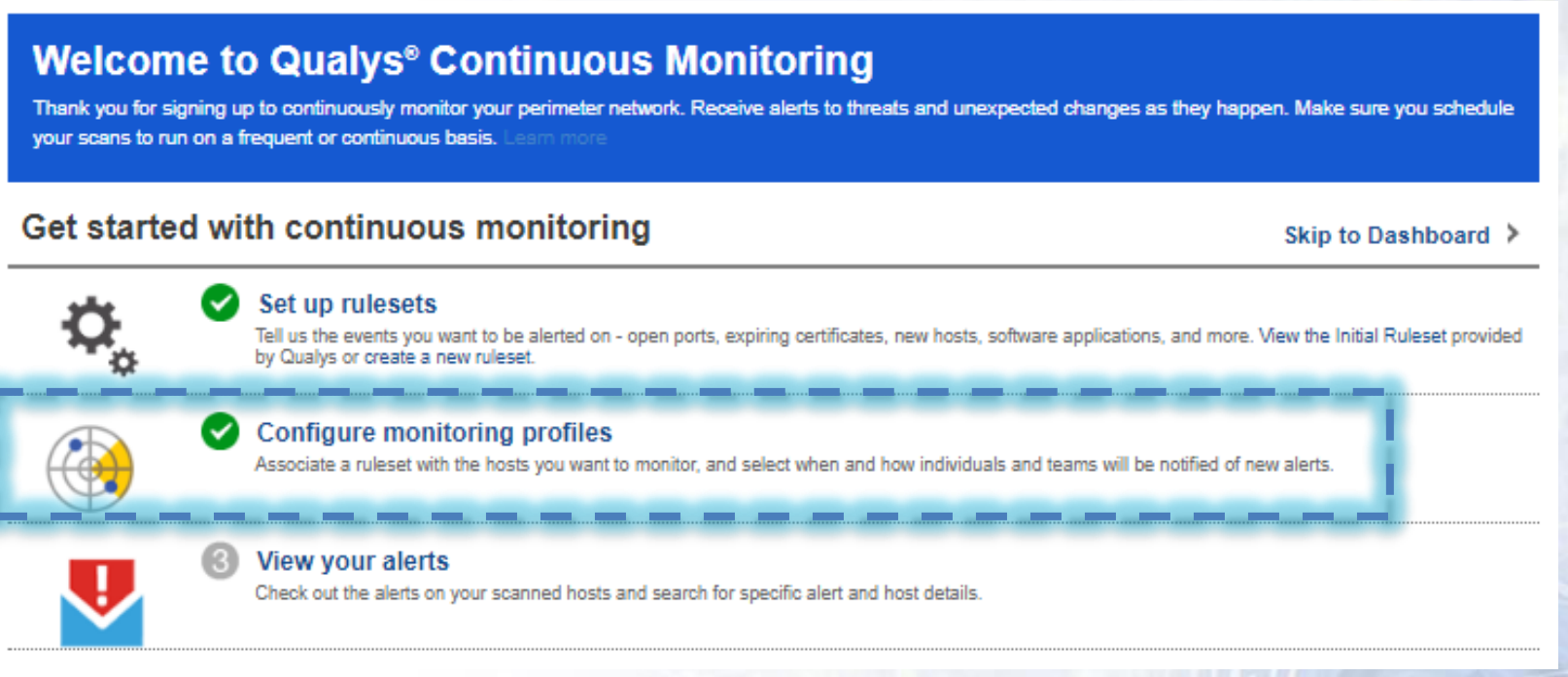
RULETYPES & CONFIGURATION

QUALYS DEMO - PART 3



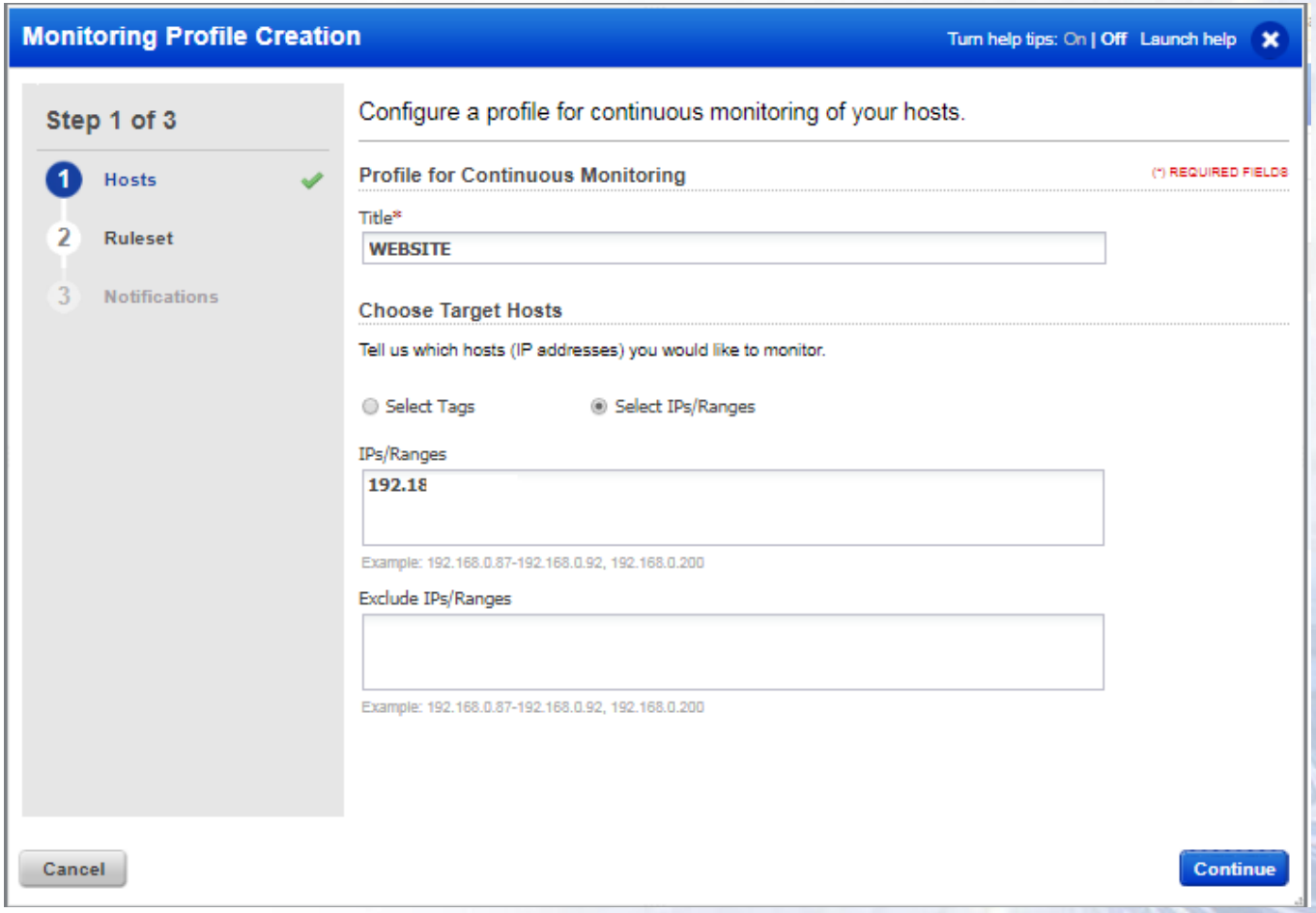
RULESETS INTERFACE

QUALYS DEMO - PART 3



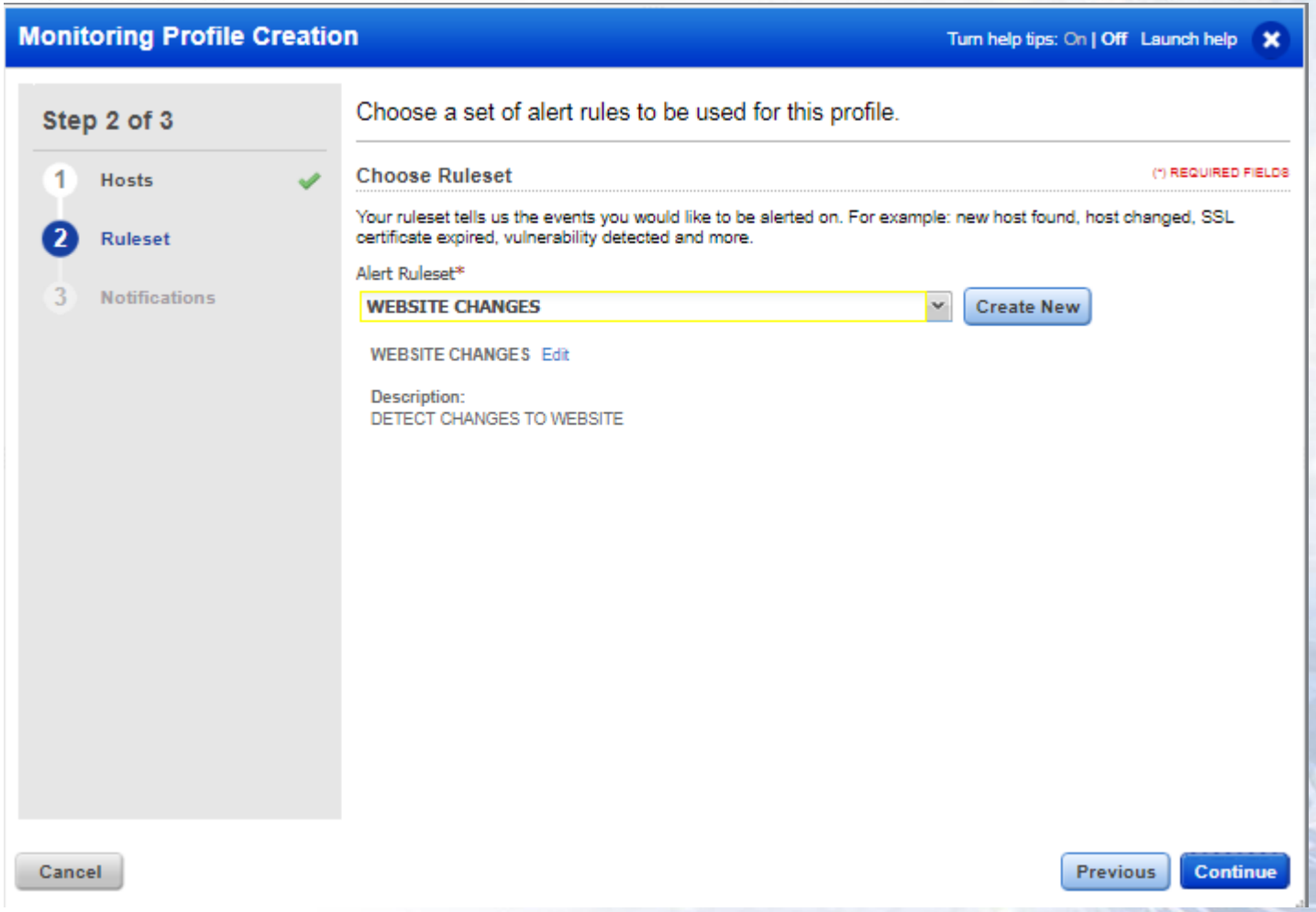
QUALYS CONTINUOUS MONITORING INTERFACE

QUALYS DEMO - PART 3



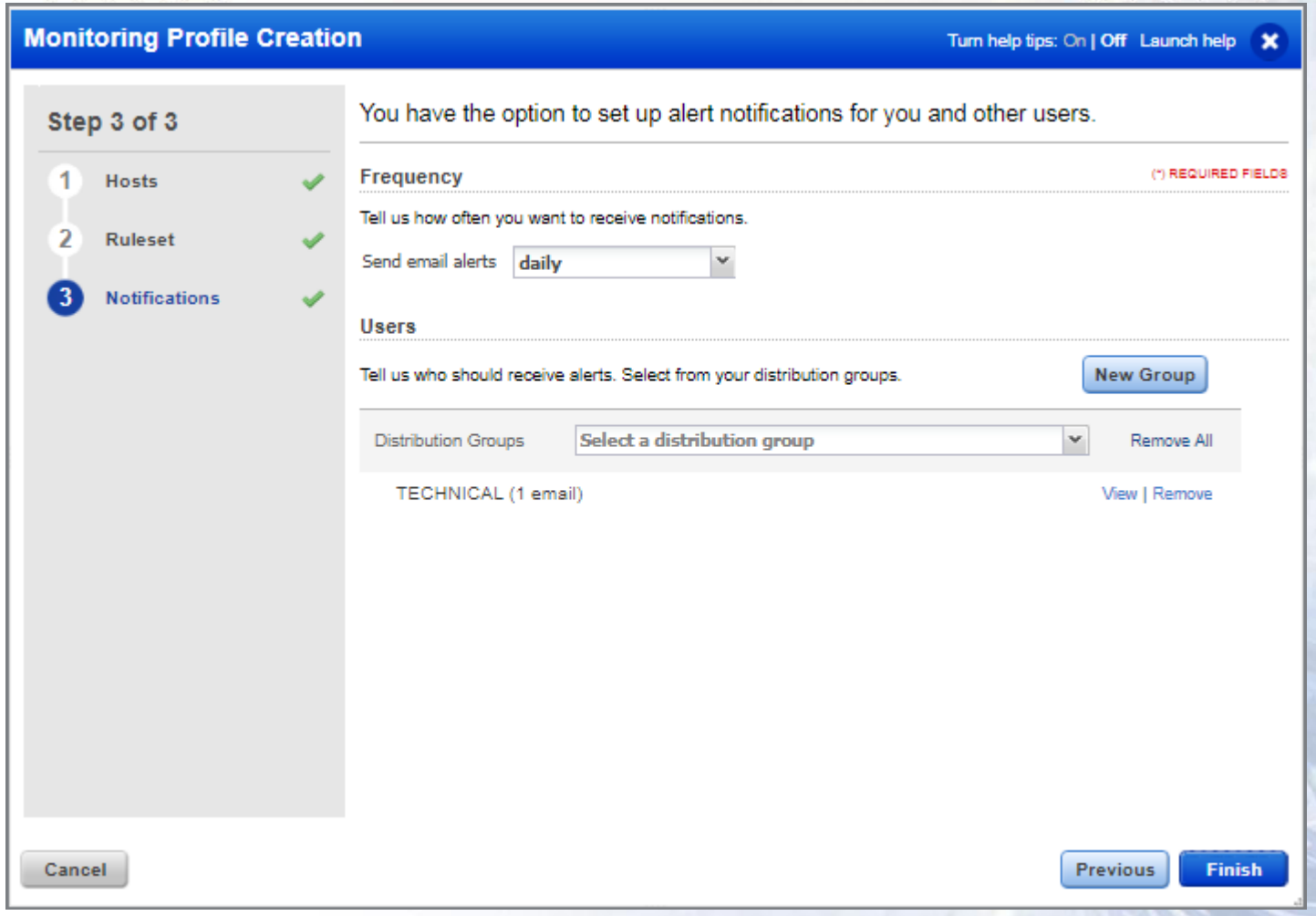
PROFILE CREATION

QUALYS DEMO - PART 3



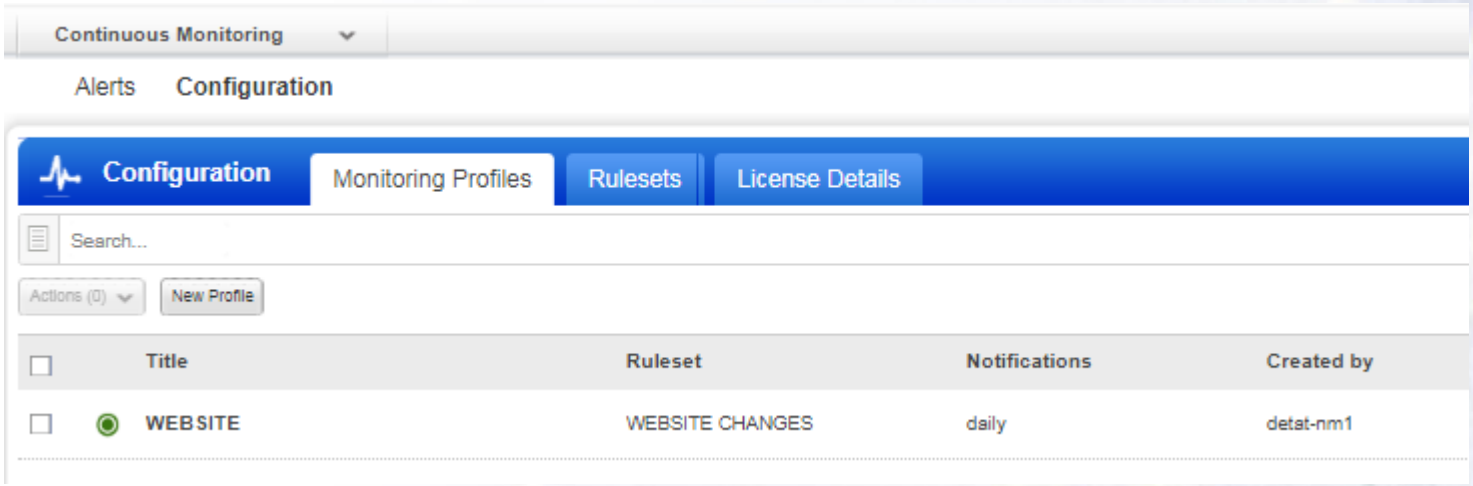
CHOOSE RULESET

QUALYS DEMO - PART 3



NOTIFICATIONS

QUALYS DEMO - PART 3



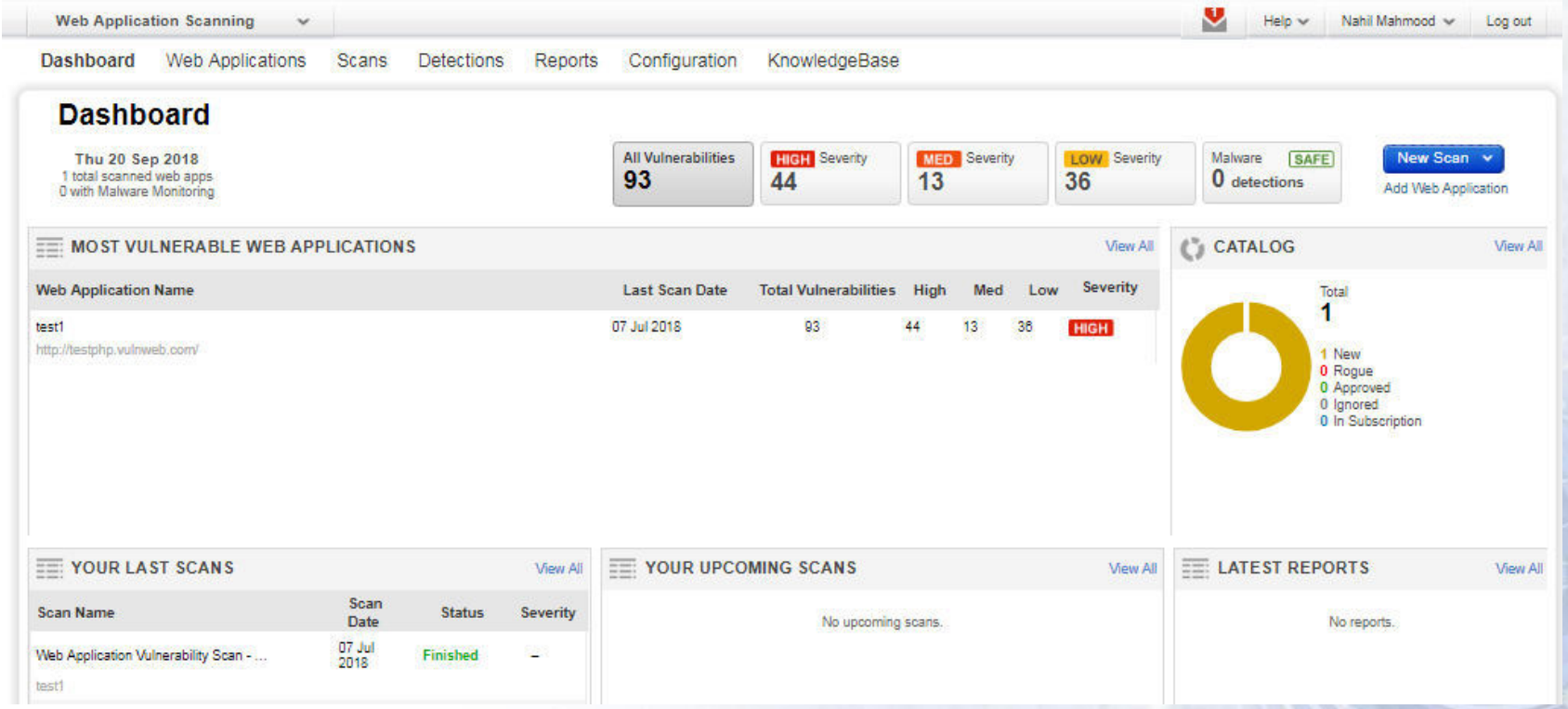
MONITORING PROFILE ACTIVE

- Qualys Continuous Monitoring is very useful for watching critical changes that may impact security

Week 7 Topic 17

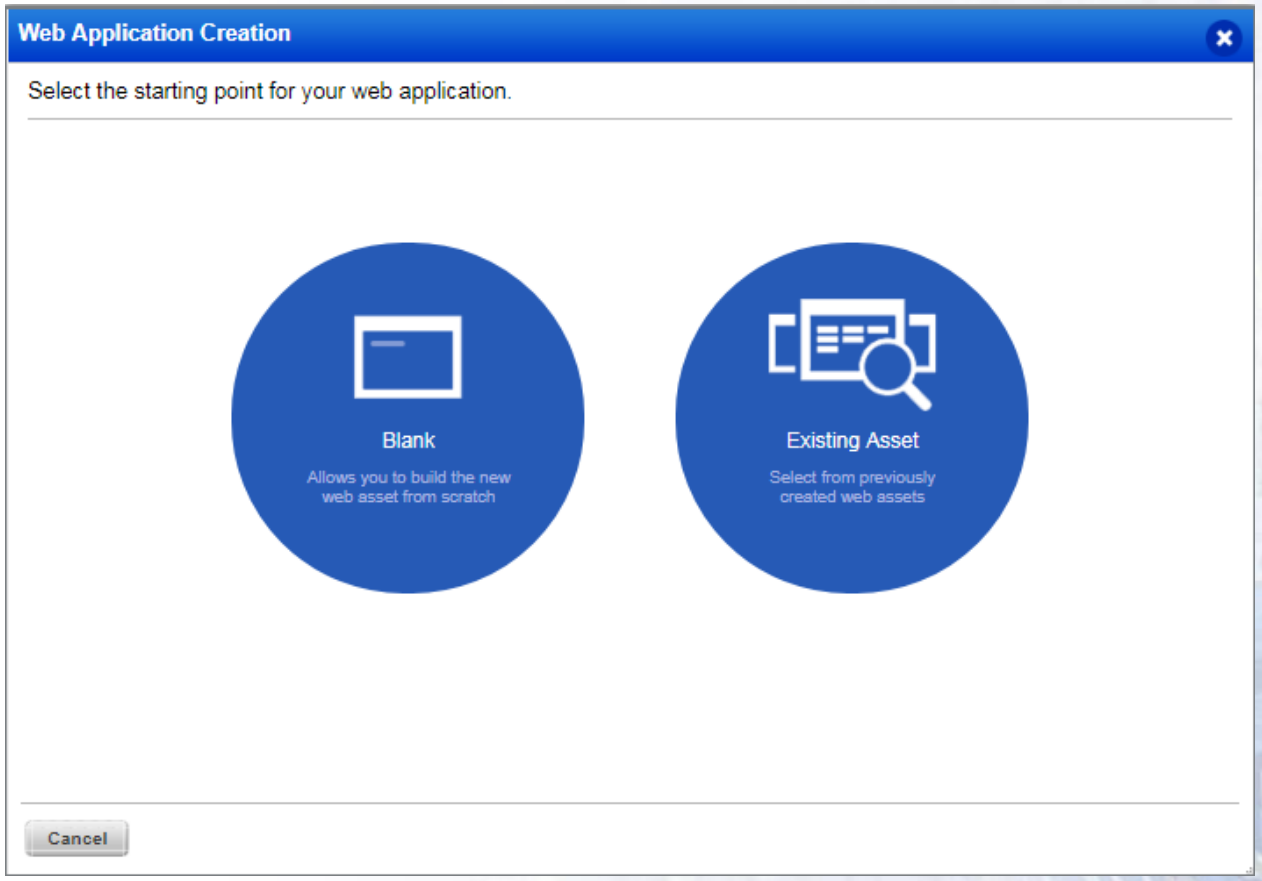
QUALYS WEB APPLICATION SCANNING

- Lets have a look at Qualys Web Application Scanning



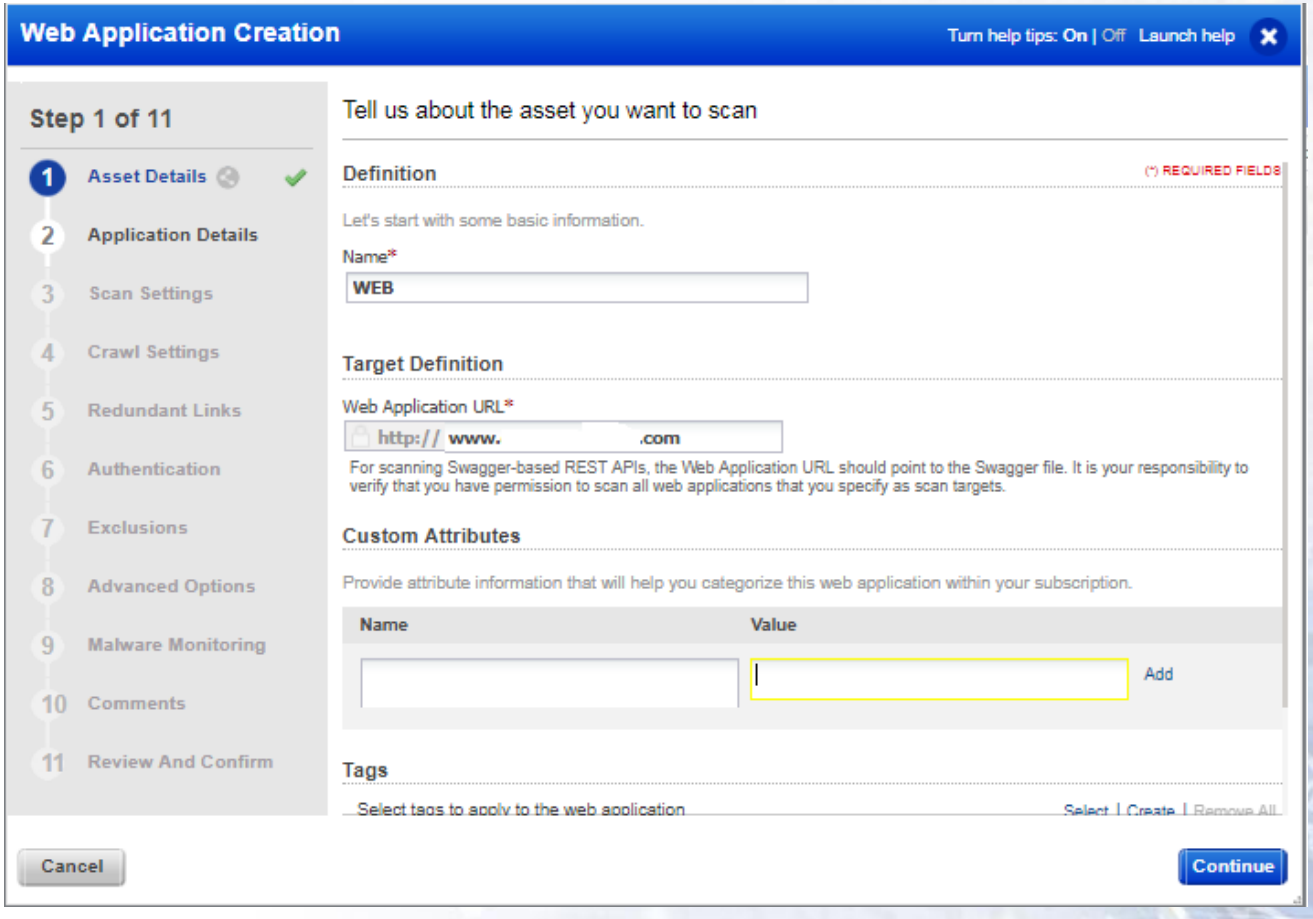
DASHBOARD

QUALYS WEB APPLICATION SCANNING



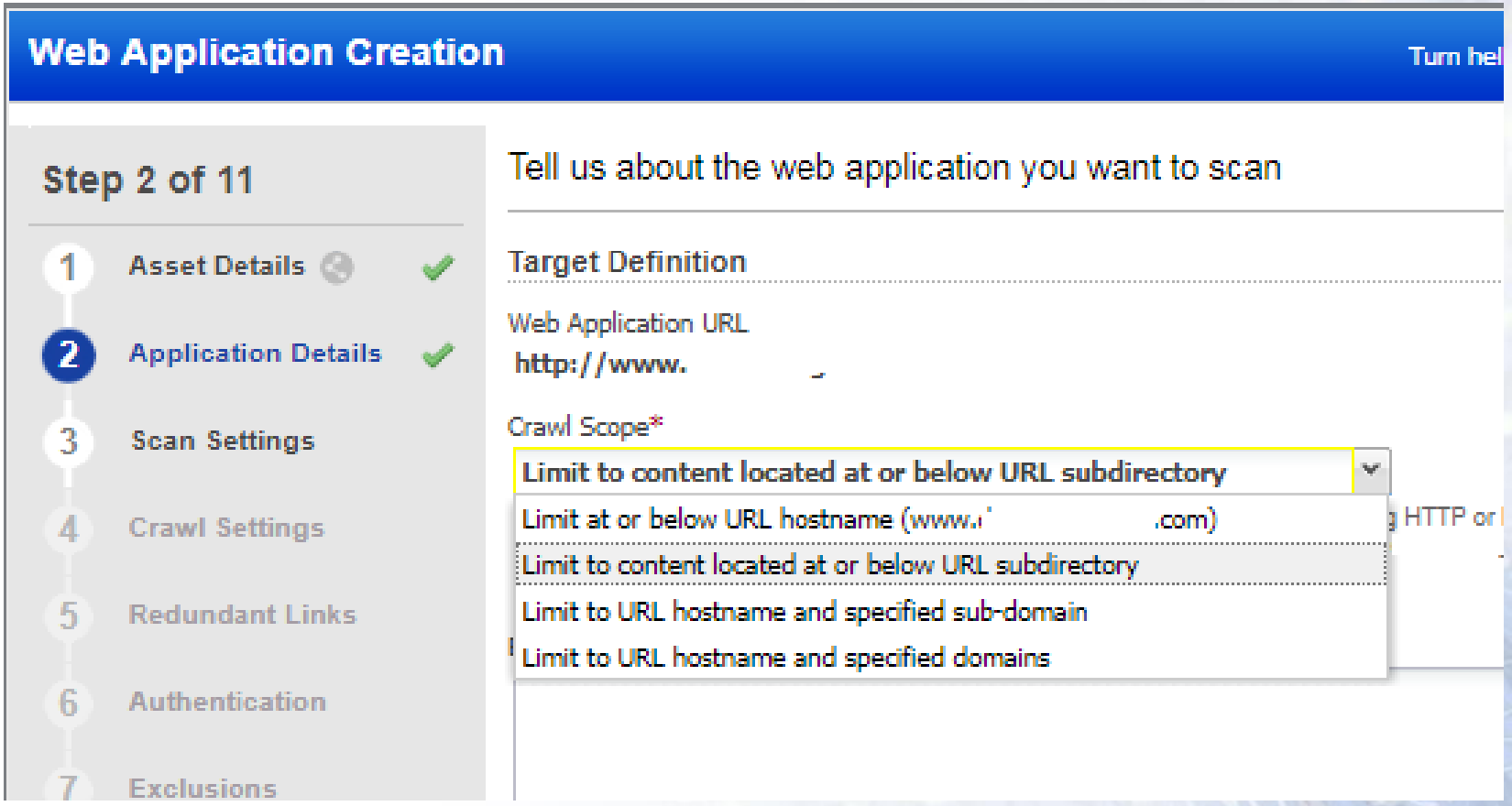
CREATION – BLANK OR EXISTING ASSET

QUALYS WEB APPLICATION SCANNING



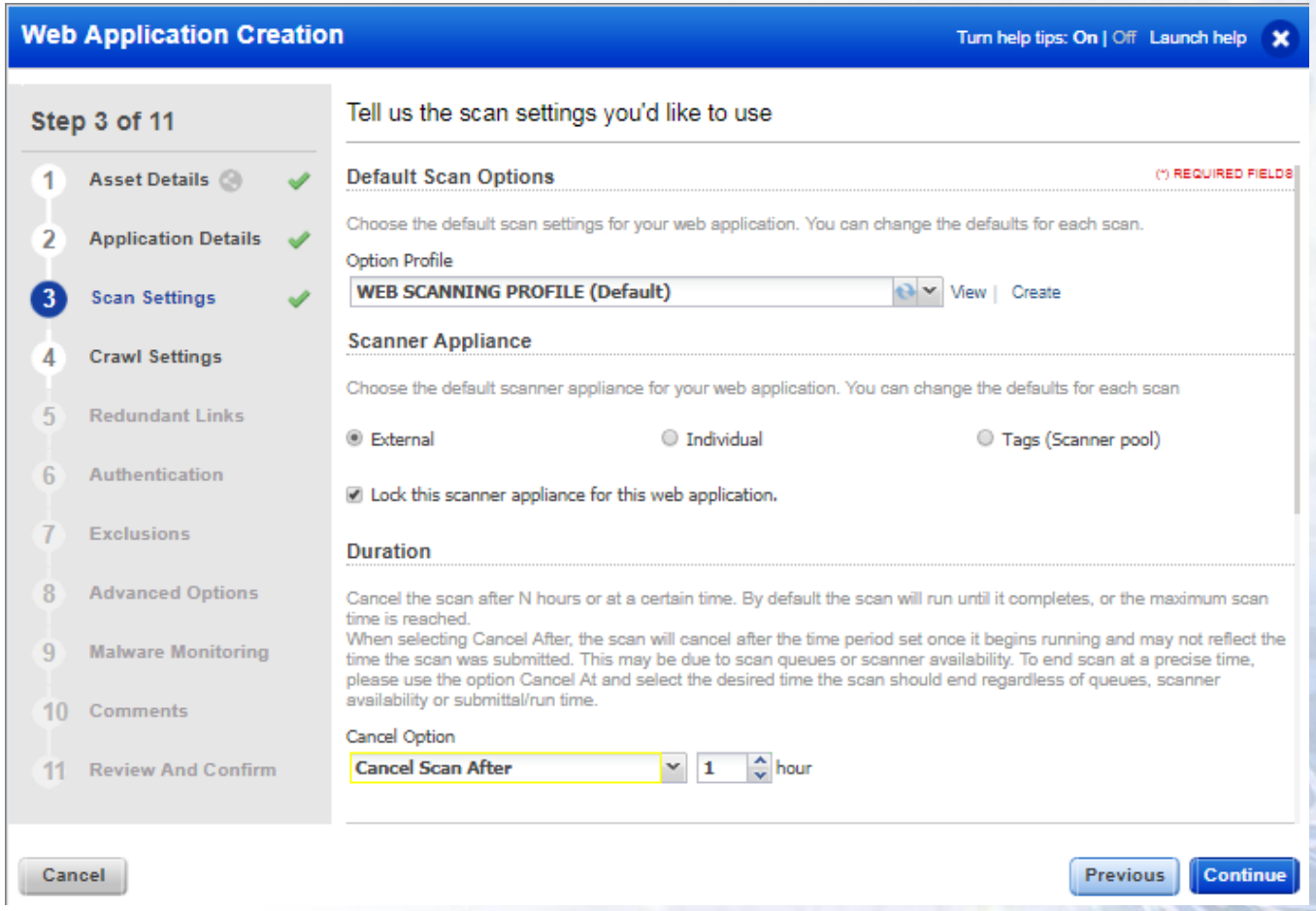
ASSET DETAILS

QUALYS WEB APPLICATION SCANNING



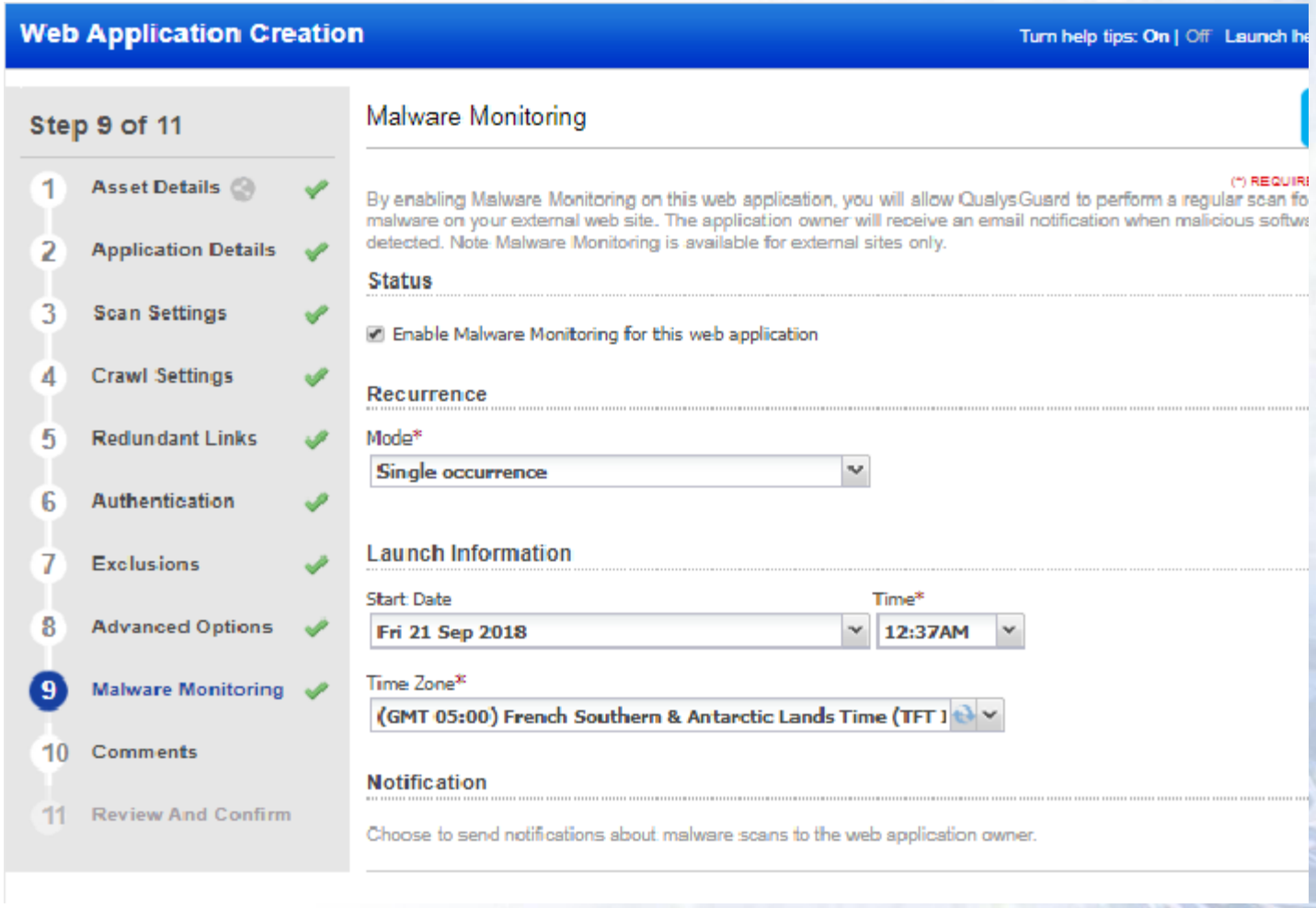
CRAWL SCOPE

QUALYS WEB APPLICATION SCANNING



SCAN SETTINGS

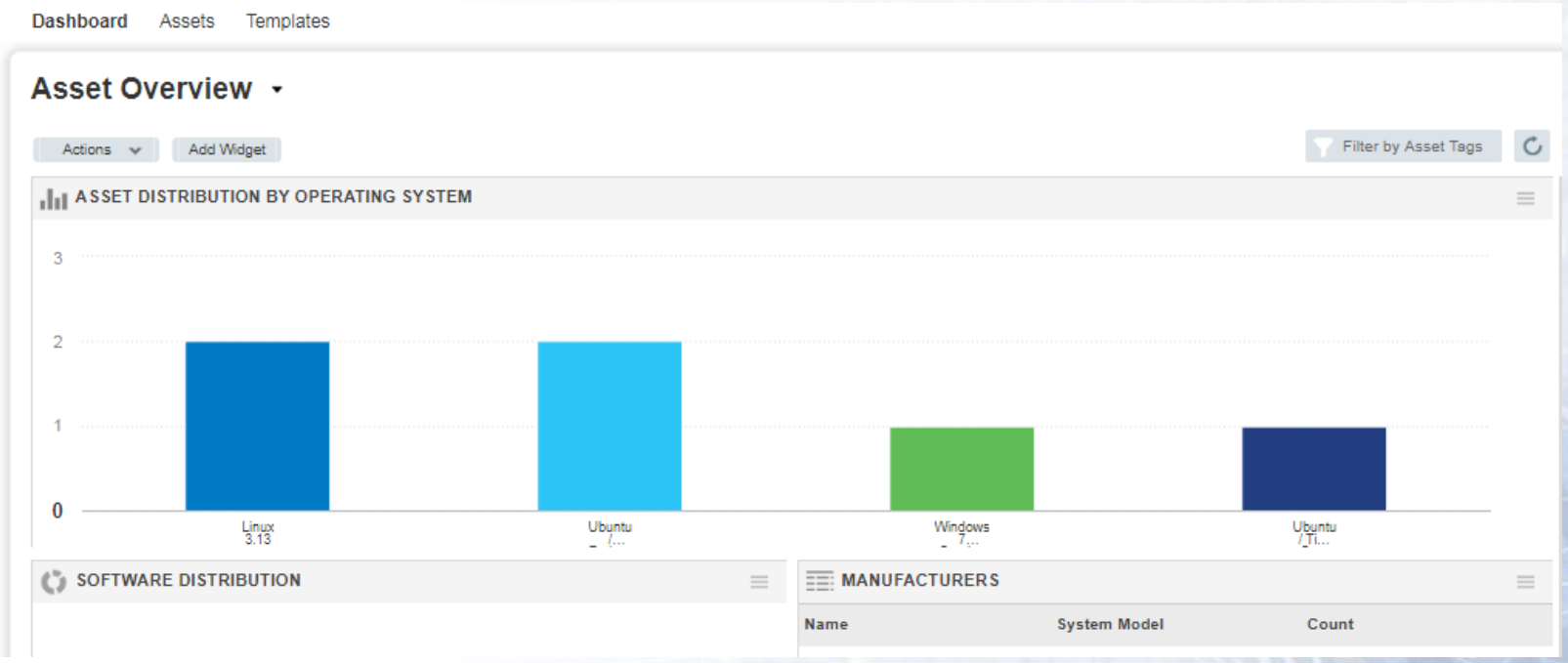
QUALYS WEB APPLICATION SCANNING



MALWARE MONITORING

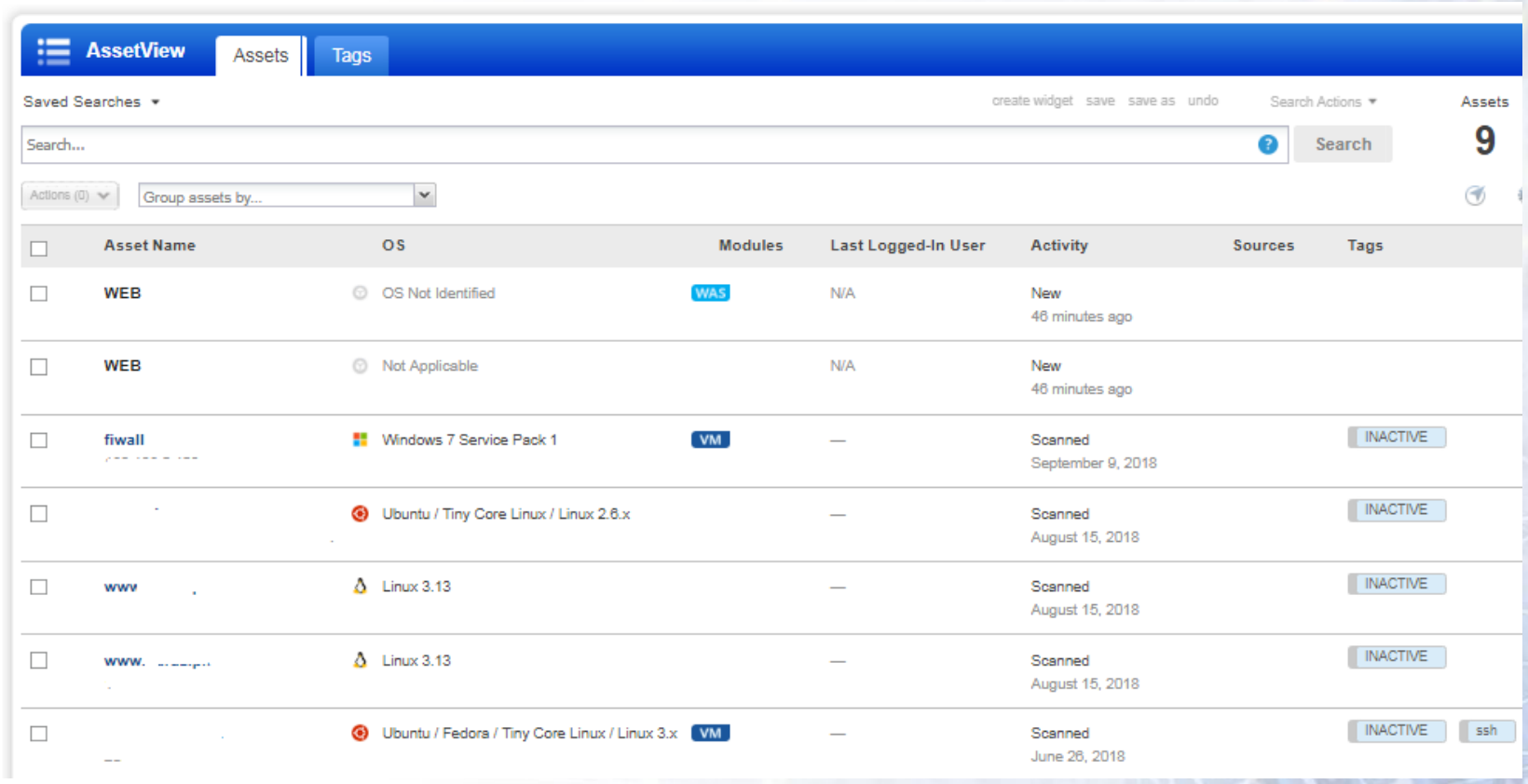
Week 7 Topic 18

ASSET MANAGEMENT THROUGH QUALYS



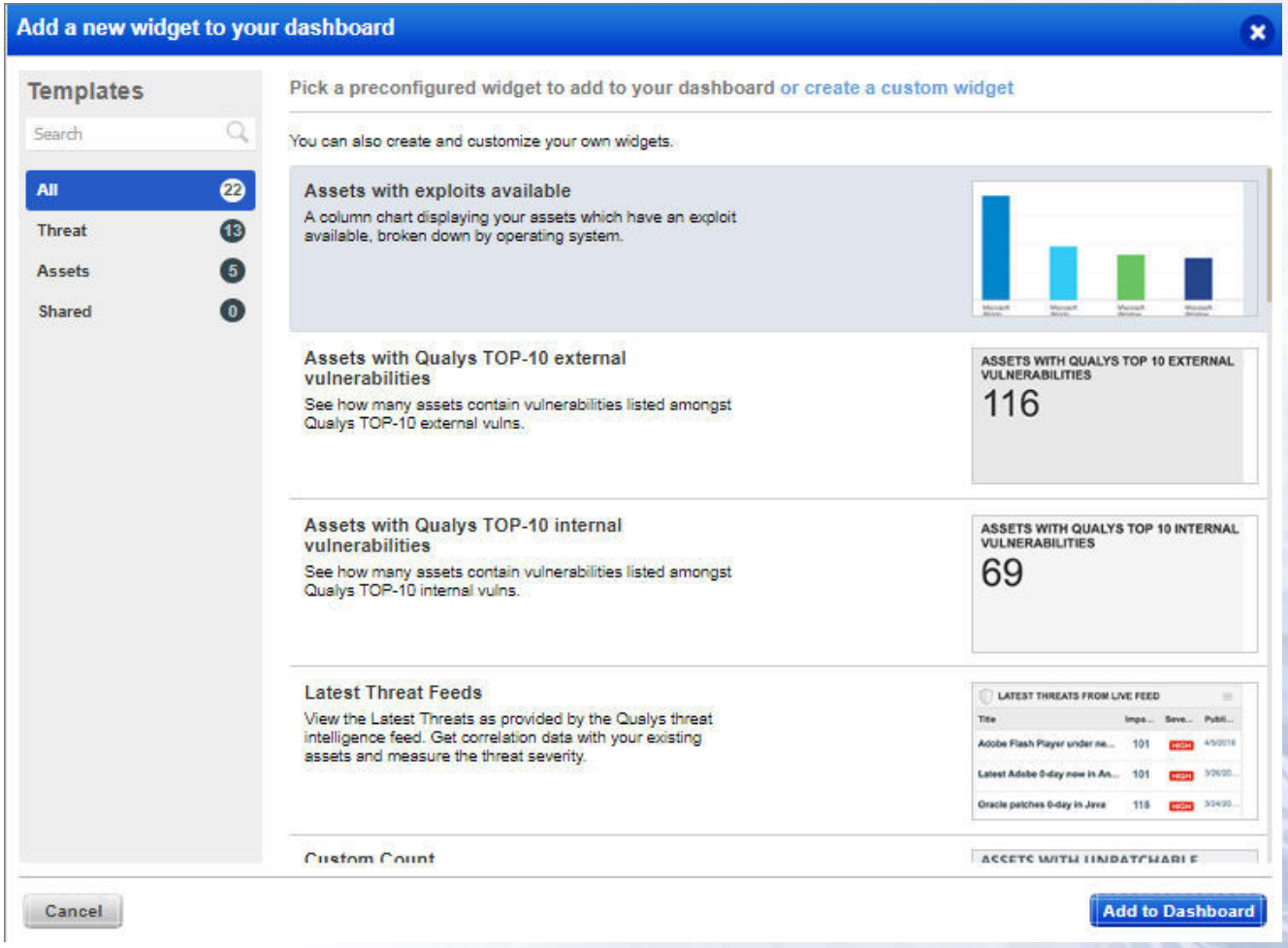
QUALYS ASSET OVERVIEW DASHBOARD

ASSET MANAGEMENT THROUGH QUALYS



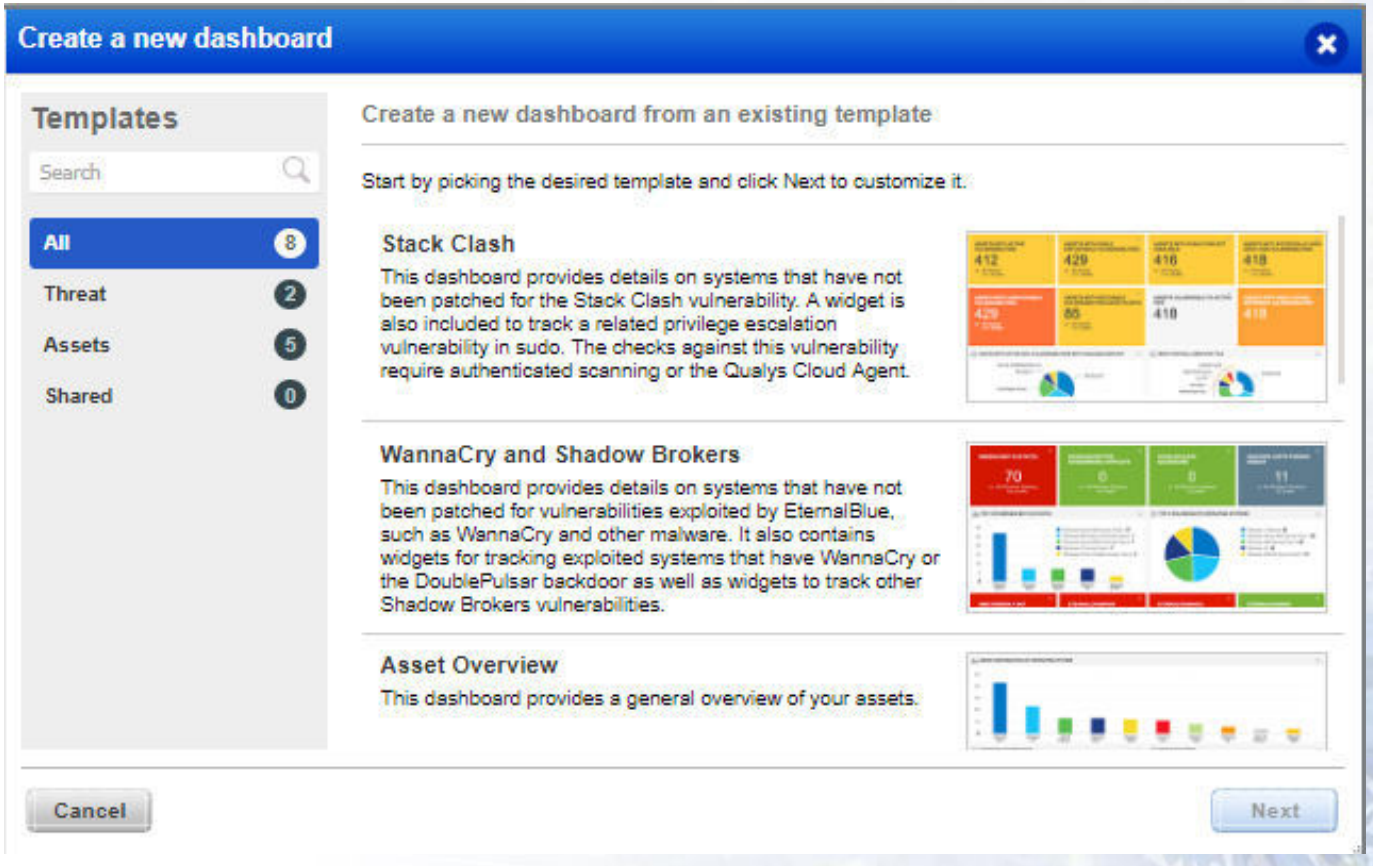
ASSETS LIST

ASSET MANAGEMENT THROUGH QUALYS



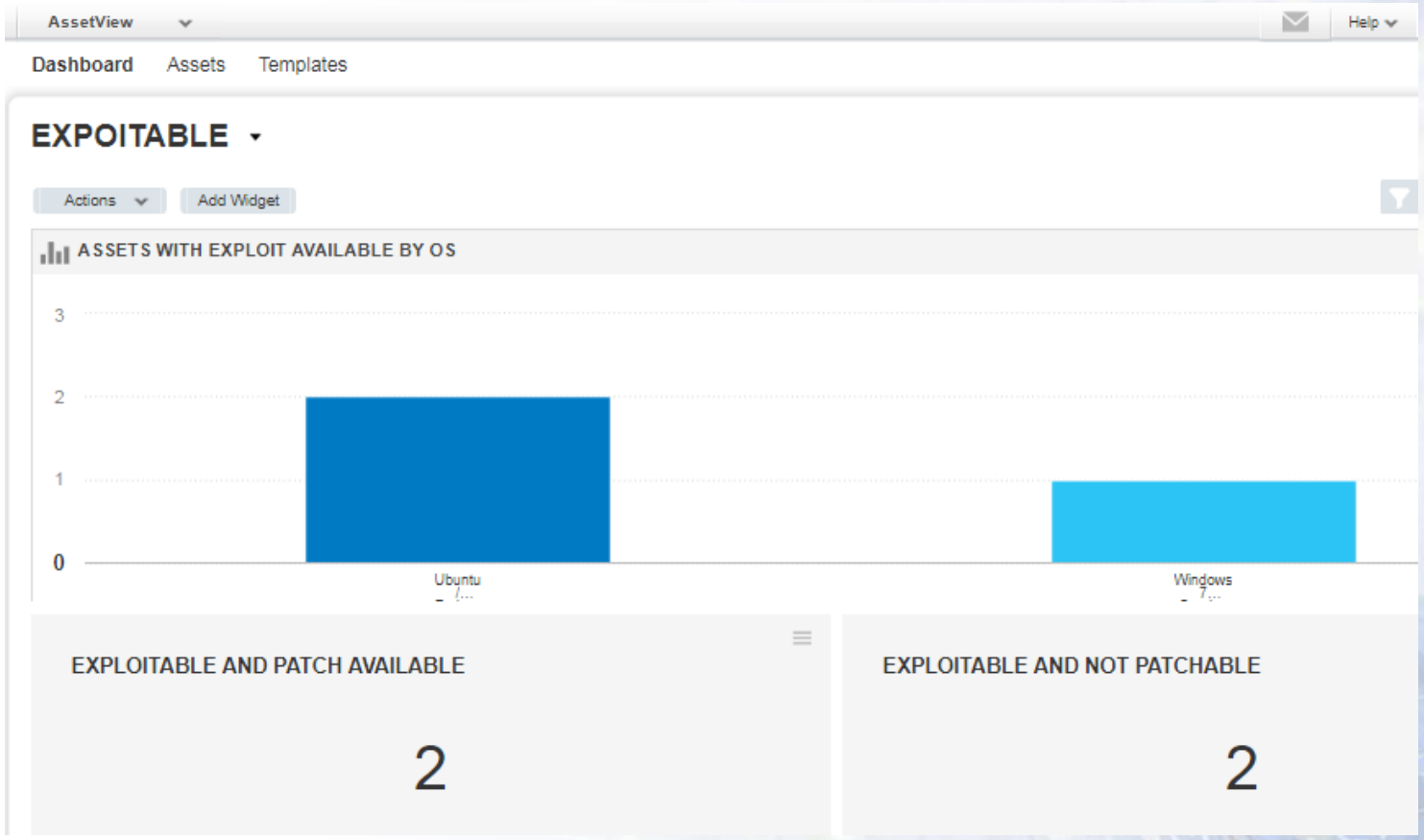
ADD NEW ASSET WIDGET TO DASHBOARD

ASSET MANAGEMENT THROUGH QUALYS



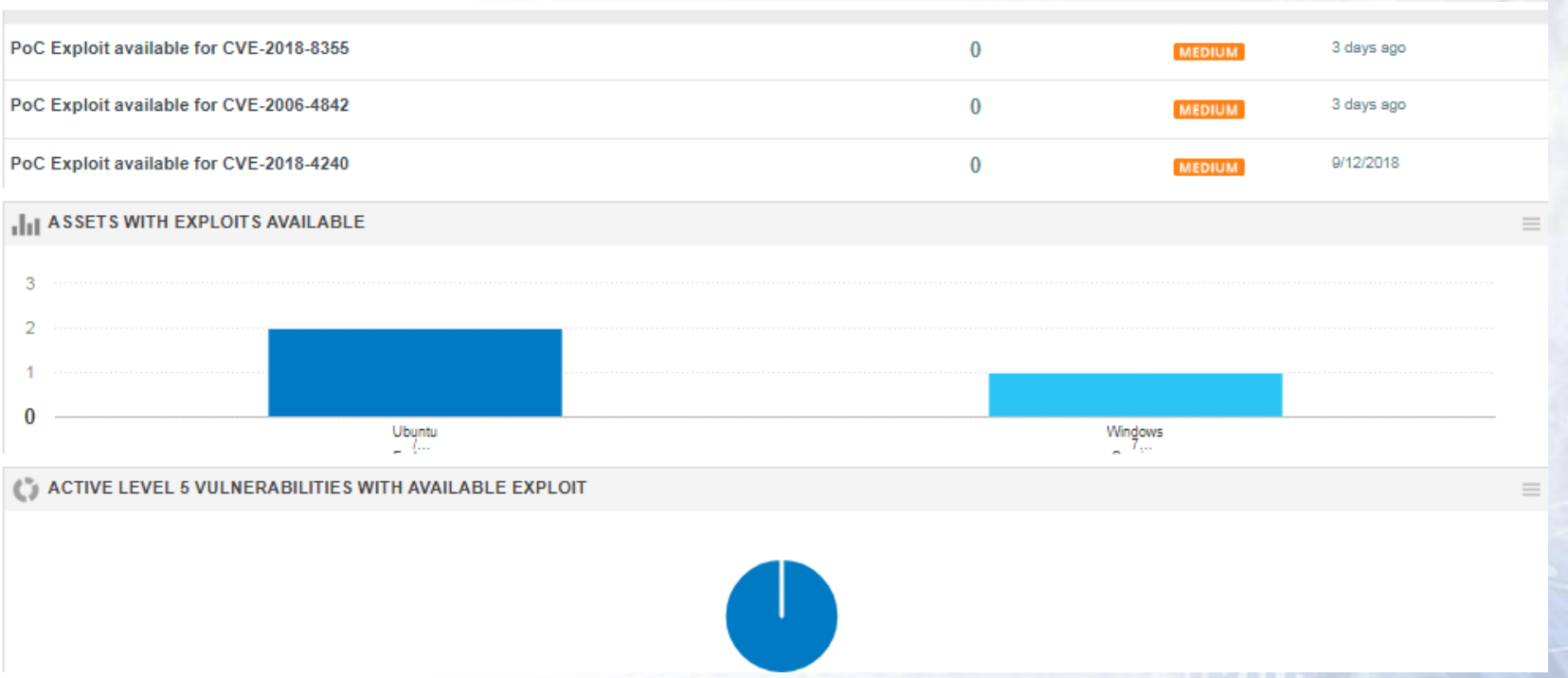
CREATE A NEW DASHBOARD

ASSET MANAGEMENT THROUGH QUALYS



NEW DASHBOARD CREATED

ASSET MANAGEMENT THROUGH QUALYS



ASSETS WITH EXPLOITS AVAILABLE

Qualys allows various reporting and views on assets and their characteristics

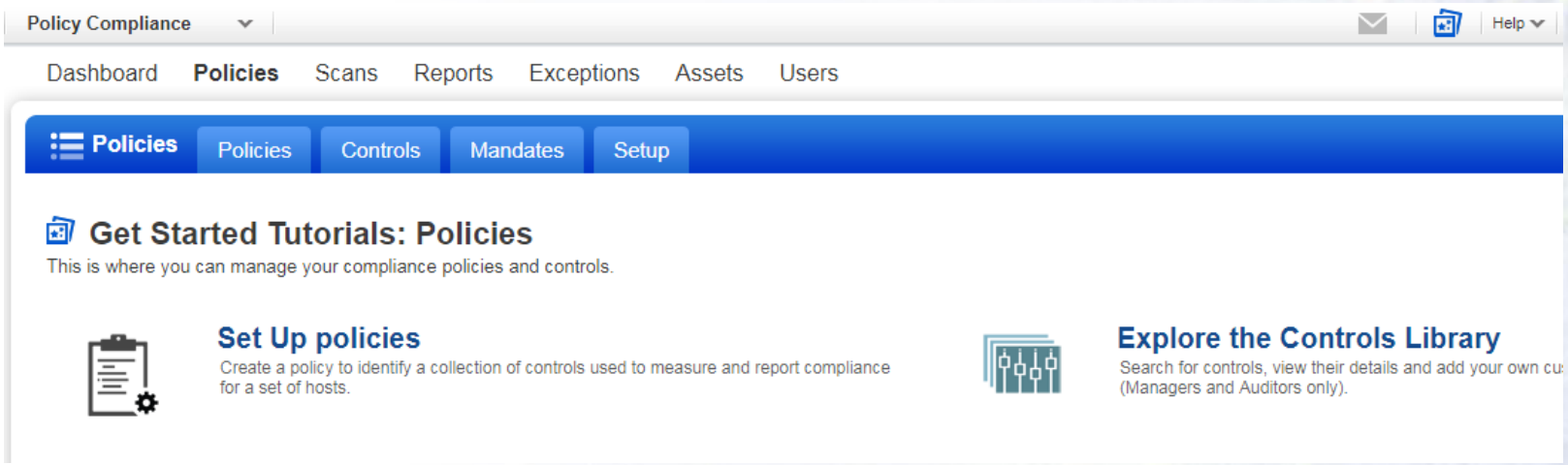
Week 7 Topic 19

QUALYS POLICY LIBRARIES

- Lets have a detailed look at Qualys built-in libraries for creating scanning policies:
  - CIS
  - QUALYS
  - MANDATE
  - DISA
  - VENDOR

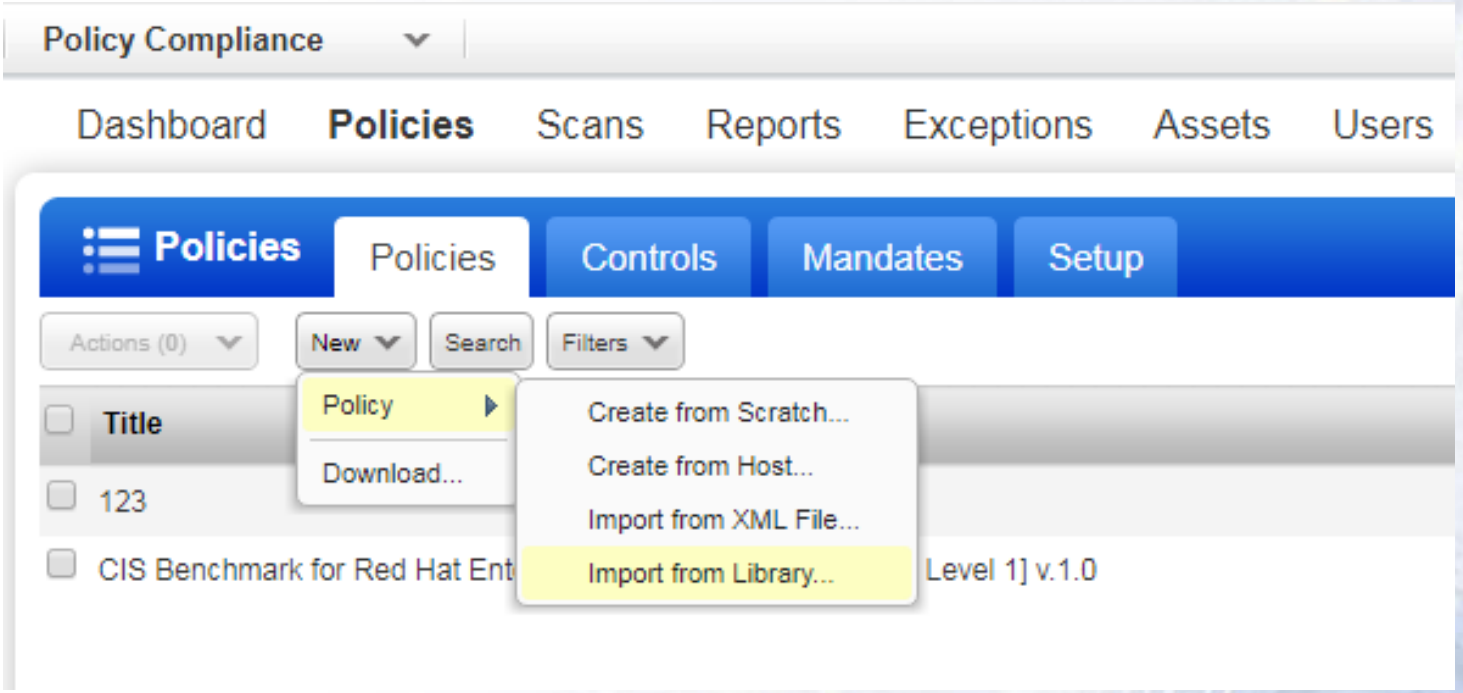
Qualys has a vast number of options for Compliance Scans, and these should be fully explored through the Qualys trial

QUALYS POLICY LIBRARIES



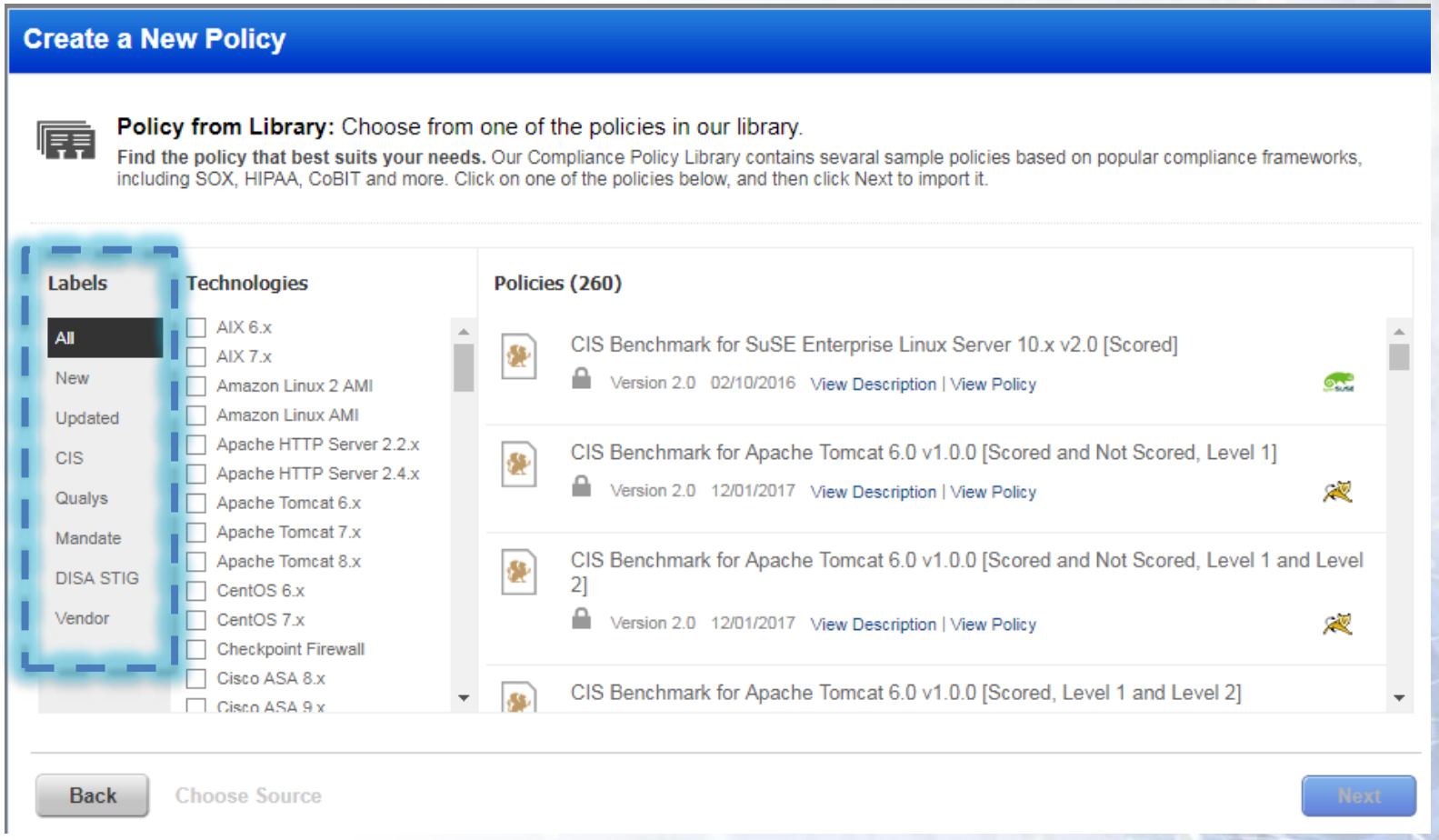
EXPLORE THE CONTROLS LIBRARY

QUALYS POLICY LIBRARIES



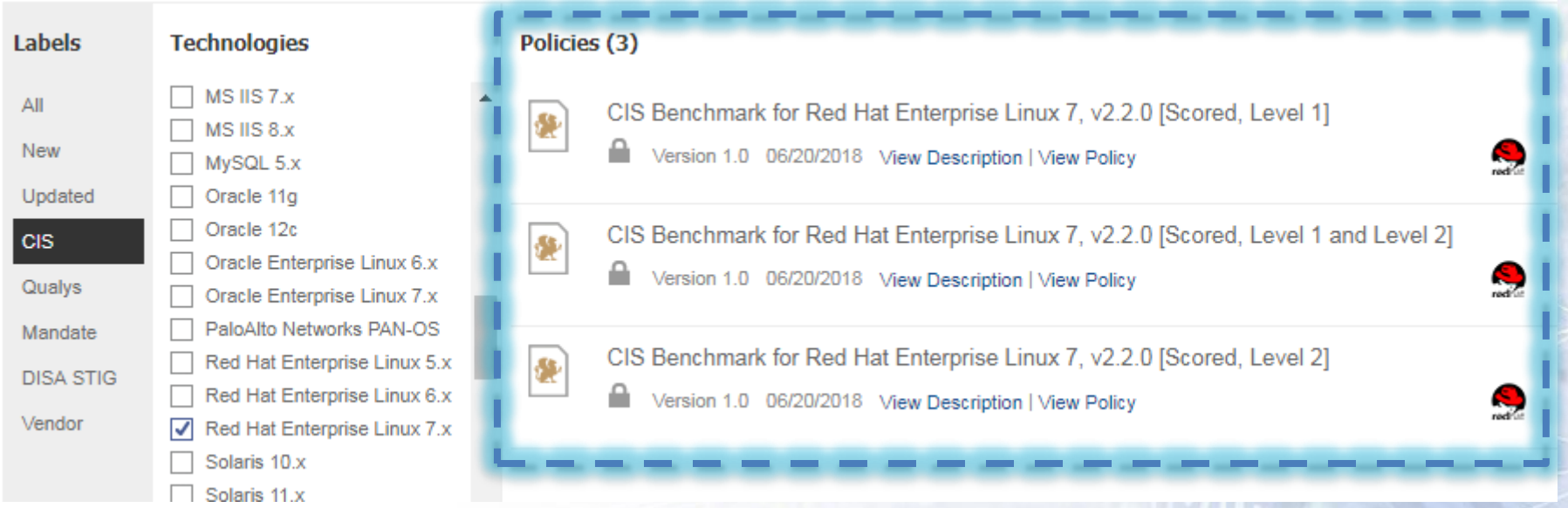
CREATE NEW POLICY > IMPORT FROM LIBRARY

QUALYS POLICY LIBRARIES



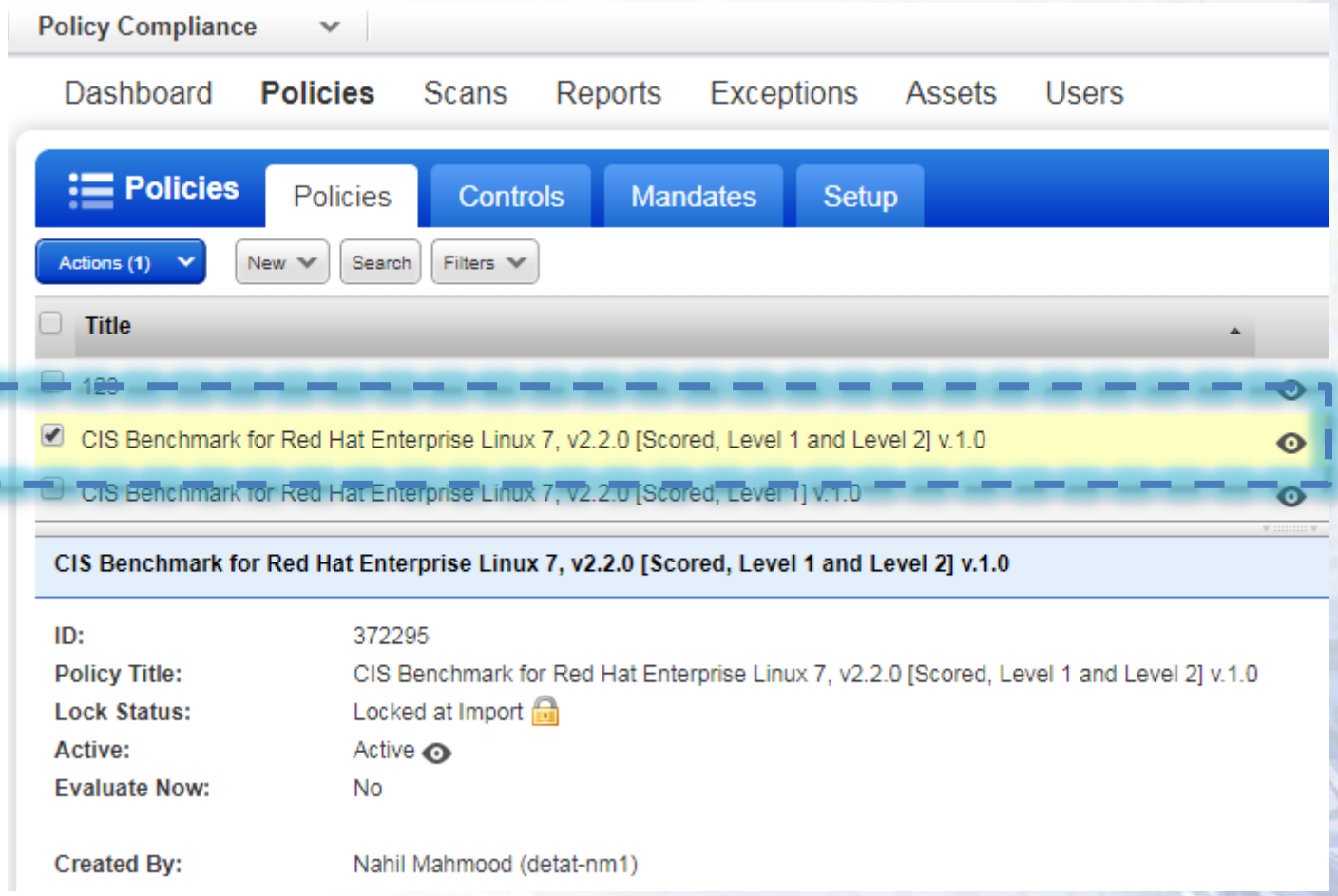
CREATE A NEW POLICY

QUALYS POLICY LIBRARIES



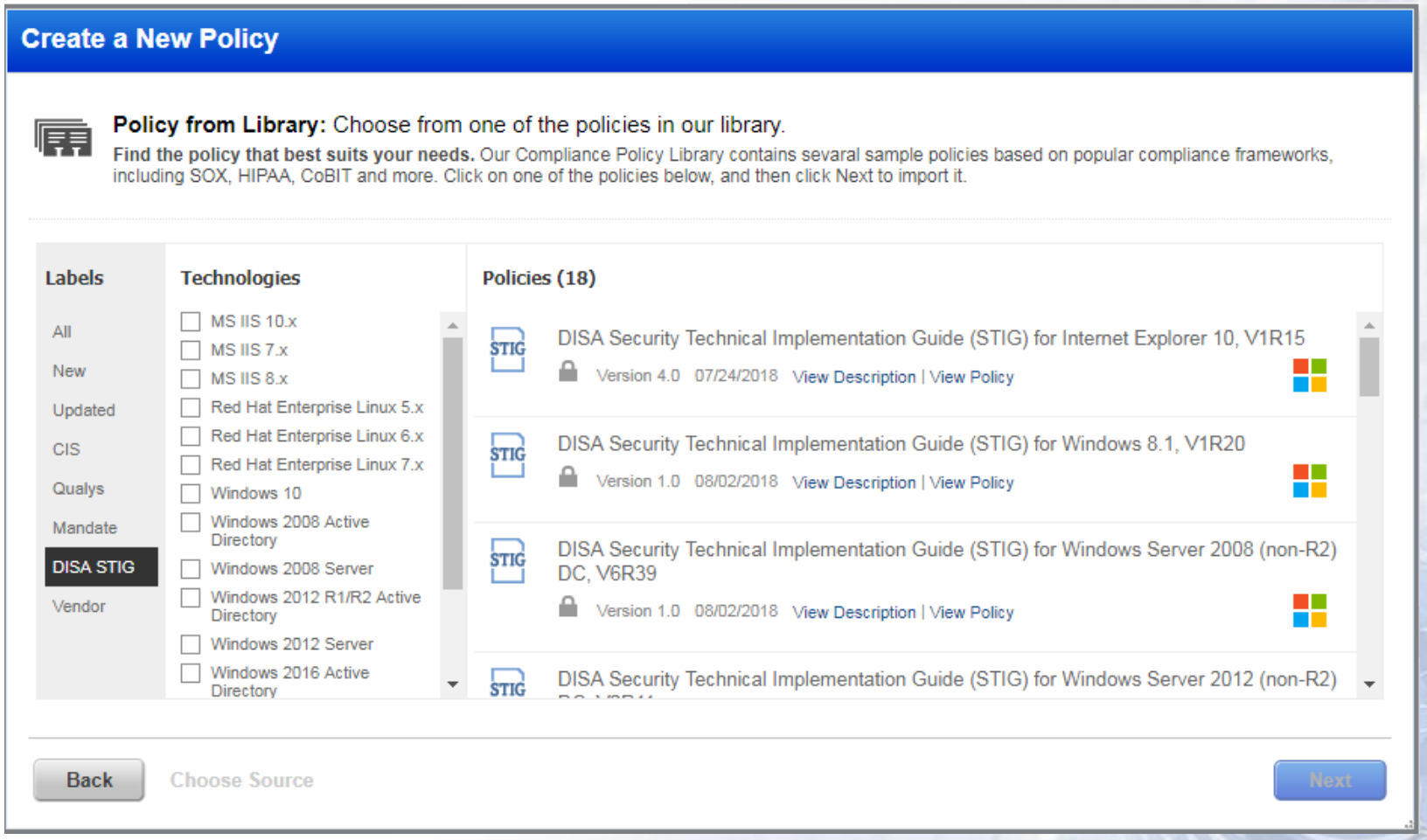
CIS > RED HAT ENT. LINUX 7.X

QUALYS POLICY LIBRARIES



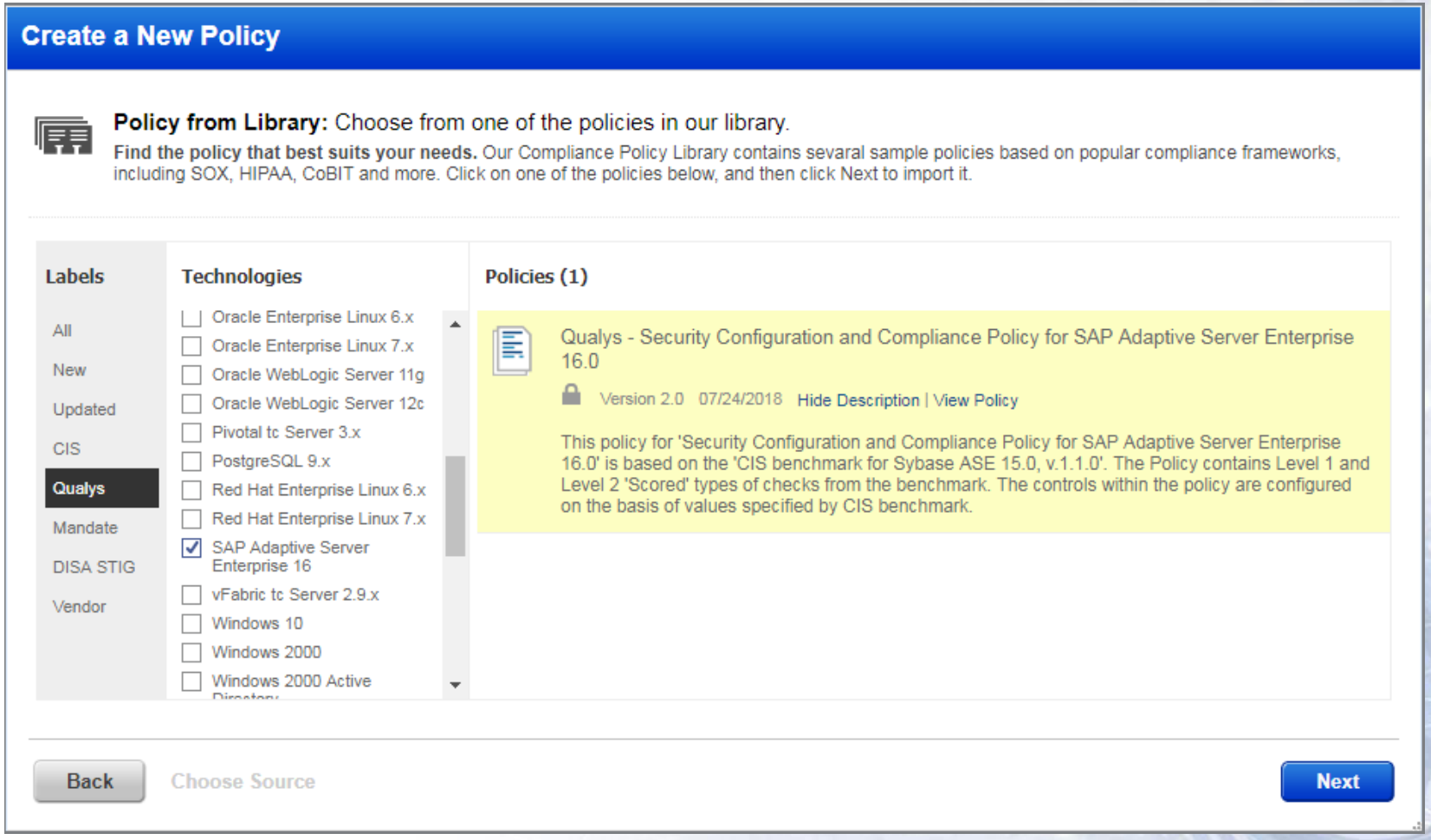
POLICIES DASHBOARD

QUALYS POLICY LIBRARIES



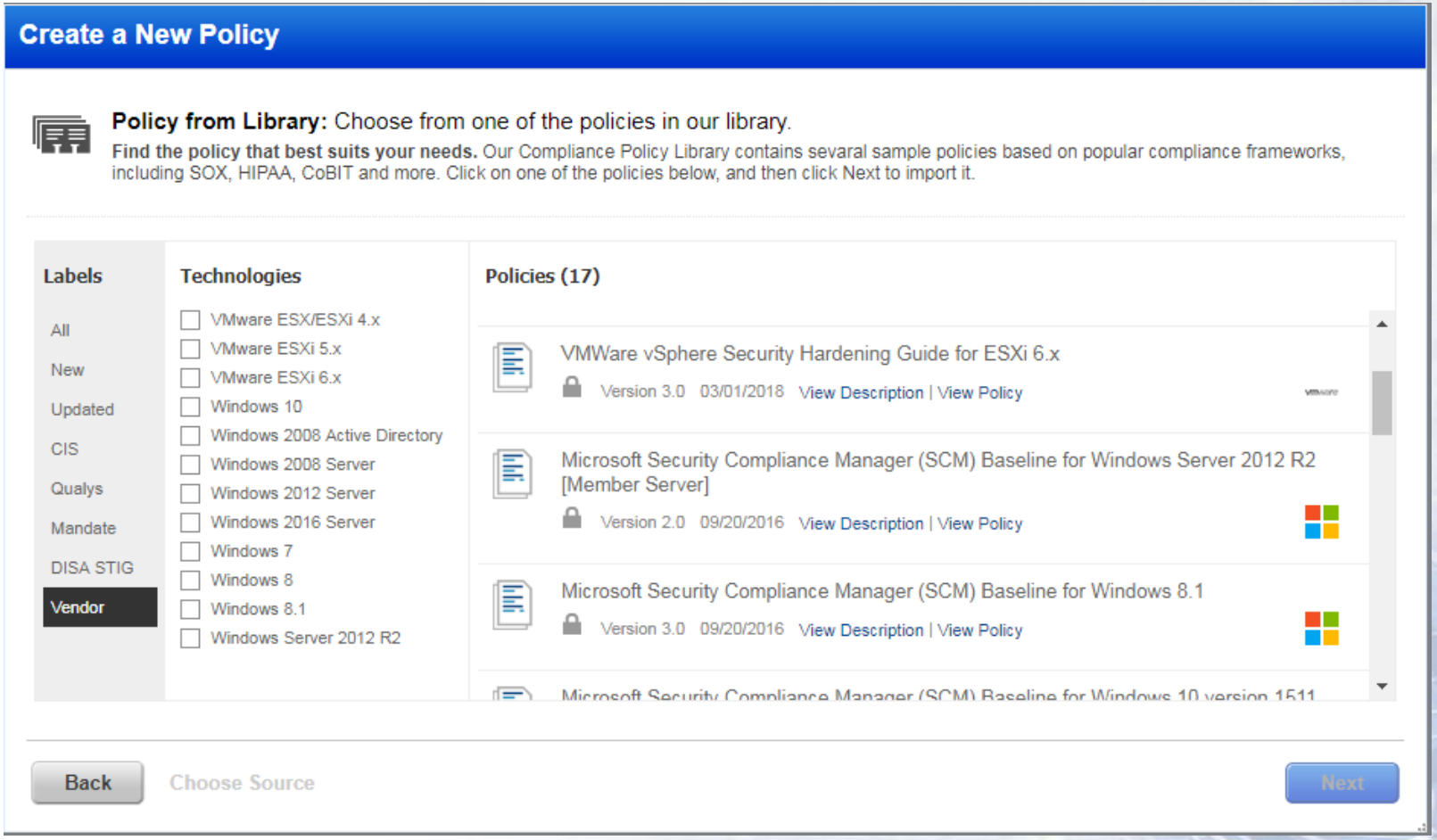
DISA STIG

QUALYS POLICY LIBRARIES



QUALYS SAP ADAPTIVE SERVER ENT 16

QUALYS POLICY LIBRARIES



VENDOR POLICIES

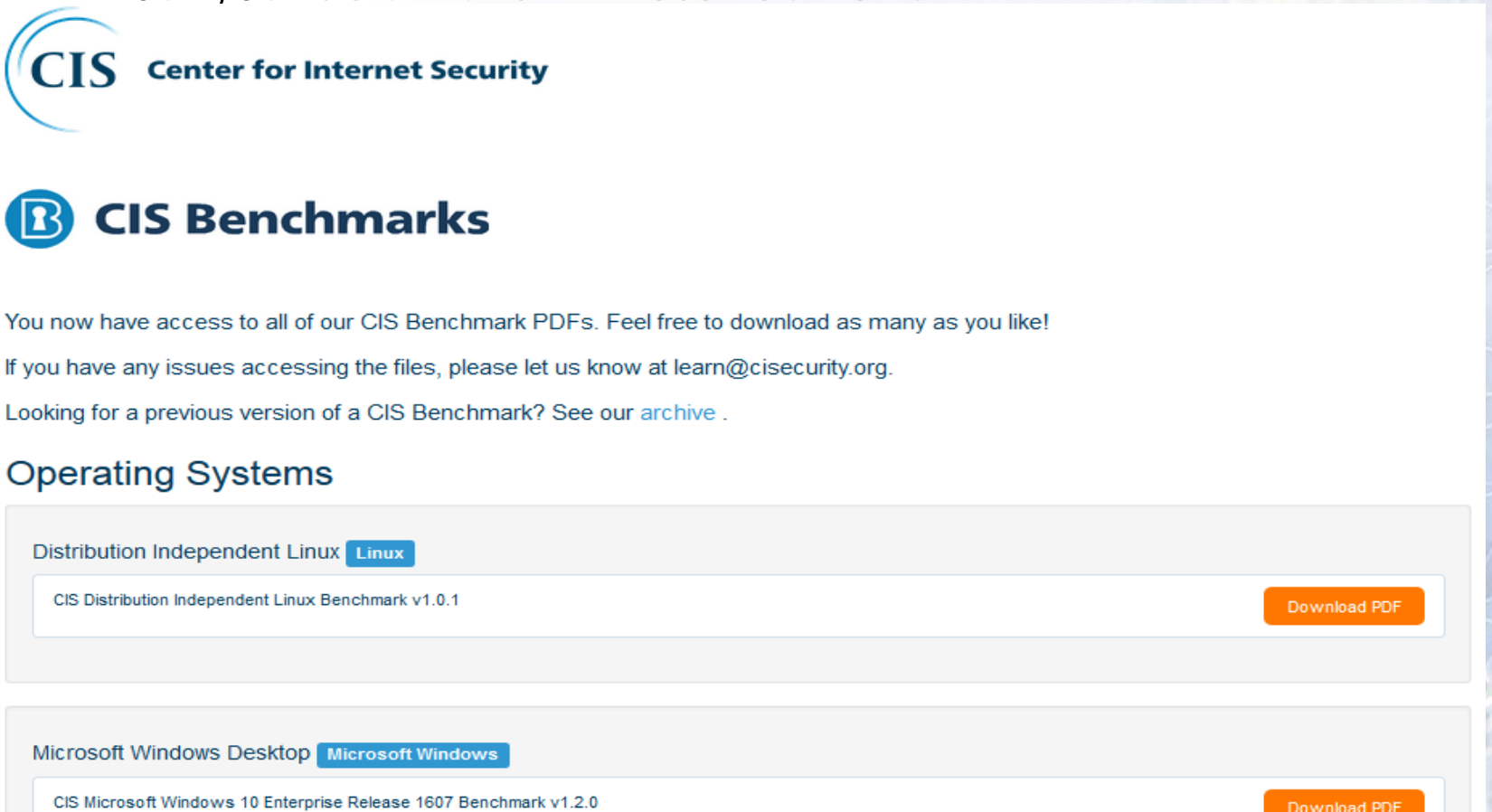
Week No. 8

Security Benchmarks and Framework

Week 8 Topic 1

A Look At CIS Security Benchmarks (1)

- Center for Internet Security (CIS)
  - https://www.cisecurity.org/cis-benchmarks/
  - Fill out your details and will receive an email with link



A Look At CIS Security Benchmarks (2)

#	OVERALL CIS BENCHMARK CATEGORIES	TOTAL
1	OPERATING SYSTEMS	36
2	SERVER SOFTWARE	33
3	CLOUD PROVIDERS	2
4	MOBILE DEVICES	8
5	NETWORK DEVICES	6
6	DESKTOP SOFTWARE	21
7	MULTIFUNCTION PRINT DEVICES	1
	GRAND TOTAL CIS BENCHMARKS	**107**

A Look At CIS Security Benchmarks (1)

#	OPERATING SYSTEMS	TOTAL
1	DISTRIBUTION INDEPENDENT LINUX	1
2	MICROSOFT WINDOWS DESKTOP	5
3	DEBIAN LINUX	2
4	UBUNTU LINUX	3
5	AMAZON LINUX	1
6	CENTOS LINUX	2
7	ORACLE LINUX	2

A Look At CIS Security Benchmarks (1)

#	OPERATING SYSTEMS (CONTD)…	TOTAL
8	REDHAT LINUX	3
9	SUSE LINUX	2
10	APPLE OS (UNIX)	5
11	IBM AIX (UNIX)	1
12	ORACLE SOLARIS (UNIX)	3
13	MS WINDOWS SERVER	6
	TOTAL BENCH MARKS OPERATING SYSTEMS	**36**

A Look At CIS Security Benchmarks (1)

Microsoft Windows Desktop

Microsoft Windows

- CIS Microsoft Windows 10 Enterprise Release 1607 Benchmark v1.2.0
- CIS Microsoft Windows 8.1 Workstation Benchmark v2.2.1
- CIS Microsoft Windows 7 Workstation Benchmark v3.0.1
- CIS Microsoft Windows XP Benchmark v3.1.0
- CIS Microsoft Windows 8 Benchmark v1.0.0

A Look At CIS Security Benchmarks (1)

Microsoft Windows Server **Microsoft Windows**

- CIS Microsoft Windows Server 2016 RTM (Release 1607) Benchmark v1.0.0
- CIS Microsoft Windows Server 2008 (non-R2) Benchmark v3.0.1
- CIS Microsoft Windows Server 2008 R2 Benchmark v3.0.1
- CIS Microsoft Windows Server 2012 R2 Benchmark v2.2.1
- CIS Microsoft Windows Server 2012 (non-R2) Benchmark v2.0.1
- CIS Microsoft Windows Server 2003 Benchmark v3.1.0

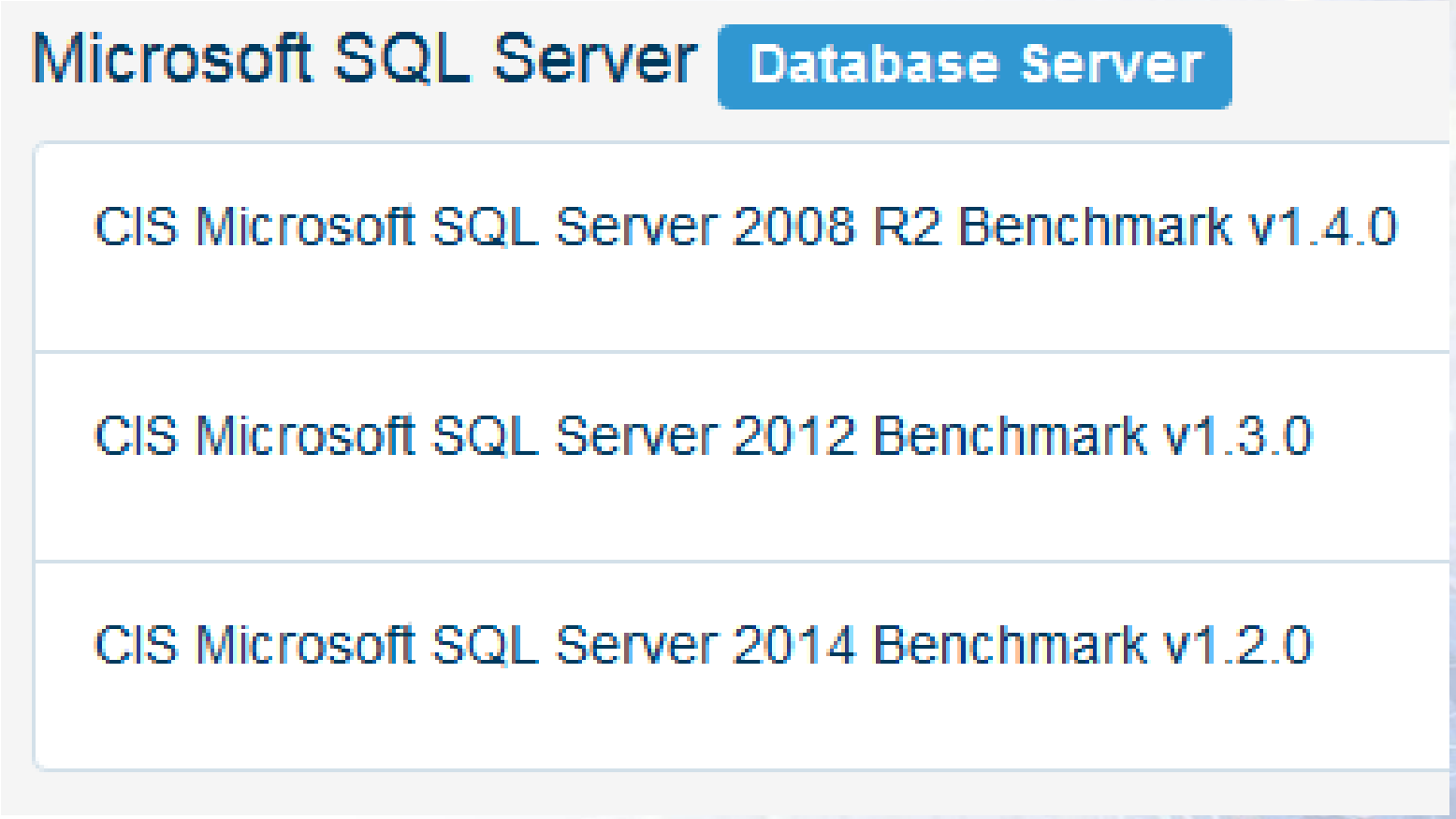
A Look At CIS Security Benchmarks (1)

#	SERVER SOFTWARE	TOTAL
1	MICROSOFT IIS (WEB SERVER)	3
2	VMWARE (VIRTUALIZATION)	2
3	MONGODB (DATABASE SERVER)	3
4	IBM DB2 (DATABASE SERVER)	3
5	BIND (DNS SERVER)	1
6	APACHE TOMCAT (WEB SERVER)	2
7	MICROSOFT SQL SERVER (DB SERVER)	3

A Look At CIS Security Benchmarks (1)

#	SERVER SOFTWARE (CONTD)…	TOTAL
8	APACHE (HTTP SERVER)	2
9	DOCKER (VIRTUALIZATION)	5
10	ORACLE (DATABASE SERVER)	3
11	KUBERNETES (VIRTUALIZATION)	1
12	MIT KERBEROS (AUTHENTICATION)	1
13	ORACLE MySQL (DB SERVER)	4
	TOTAL BENCH MARKS SERVER SOFTWARE	**33**

A Look At CIS Security Benchmarks (1)



A Look At CIS Security Benchmarks (1)

#	CLOUD PROVIDERS	TOTAL
1	AMAZON WEB SERVICES	2
	TOTAL CLOUD PROVIDERS	**2**

Week 8 Topic 2

A Look At CIS Security Benchmarks (2)

- Mobile devices, network devices, desktop software, multifunction print devices

#	MOBILE DEVICES	TOTAL
1	APPLE IOS	5
2	GOOGLE ANDROID	3
	TOTAL BENCH MARKS MOBILE DEVICES	**8**

#	NETWORK DEVICES	TOTAL
1	CISCO	4
2	PALO ALTO NETWORKS	2
	TOTAL BENCH MARKS NETWORK DEVICES	**6**

A Look At CIS Security Benchmarks (2)

#	DESKTOP SOFTWARE	TOTAL
1	MICROSOFT OFFICE	13
2	GOOGLE CHROME (WEB BROWSER)	1
3	MS EXCHANGE SERVER	3
4	MS INTERNET EXPLORER	2
5	MOZILLA FIREFOX	2
	TOTAL BENCH MARKS DESKTOP SOFTWARE	**21**

A Look At CIS Security Benchmarks (2)

#	MULTIFUNCTION PRINT DEVICES	TOTAL
1	MULTIFUNCTION DEVICE	1
	TOTAL BENCH MARKS MULTIFUNCTION PRINT DEVICES	**1**

A Look At CIS Security Benchmarks (2)

Apple iOS

- CIS Apple iOS 10 Benchmark v2.0.0
- CIS Apple iOS 9 Benchmark v1.0.0
- CIS Apple iOS 8 Benchmark v1.0.0
- CIS Apple iOS 7 Benchmark v1.1.0
- CIS Apple iOS 6 Benchmark v1.0.0

Google Android

- CIS Google Android 7 Benchmark v1.0.0
- CIS Google Android 4 Benchmark v1.0.0
- CIS Google Android 2.3 Benchmark v1.1.0

A Look At CIS Security Benchmarks (2)

Cisco

- CIS Cisco Firewall Benchmark v4.0.0
- CIS Cisco IOS 12 Benchmark v4.0.0
- CIS Cisco IOS 15 Benchmark v4.0.0
- CIS Cisco Wireless LAN Controller 7 Benchmark v1.1.0

A Look At CIS Security Benchmarks (2)

Microsoft Office **Productivity Software**

- CIS Microsoft Office 2016 Benchmark v1.1.0
- CIS Microsoft Office PowerPoint 2016 Benchmark v1.0.1
- CIS Microsoft Office PowerPoint 2013 Benchmark v1.0.1
- CIS Microsoft Office Excel 2016 Benchmark v1.0.1
- CIS Microsoft Office Excel 2013 Benchmark v1.0.1
- CIS Microsoft Office Access 2016 Benchmark v1.0.1
- CIS Microsoft Office Access 2013 Benchmark v1.0.1
- CIS Microsoft Office 2013 Benchmark v1.1.0
- CIS Microsoft Office Word 2016 Benchmark v1.1.0
- CIS Microsoft Office Word 2013 Benchmark v1.1.0
- CIS Microsoft Office Outlook 2016 Benchmark v1.1.0
- CIS Microsoft Office Outlook 2013 Benchmark v1.1.0
- CIS Microsoft Outlook 2010 Benchmark v1.0.0

A Look At CIS Security Benchmarks (3)

CIS Benchmarks example (Network Devices

#	OVERALL CIS BENCHMARK CATEGORIES	TOTAL
1	OPERATING SYSTEMS	36
2	SERVER SOFTWARE	33
3	CLOUD PROVIDERS	2
4	MOBILE DEVICES	8
5	NETWORK DEVICES	6
6	DESKTOP SOFTWARE	21
7	MULTIFUNCTION PRINT DEVICES	1
	GRAND TOTAL CIS BENCHMARKS	**107**

Week 8 Topic 3

CIS Benchmarks

- The CIS Benchmarks are prescriptive configuration recommendations for more than 25+ vendor product families. They represent the consensus-based effort of cybersecurity experts globally to help you protect your systems against threats more confidently.

CIS Benchmarks

CIS Benchmarks from the Center of Internet Security (CIS):

- a set of globally recognized and consensus-driven best practices to help security practitioners implement and manage their cybersecurity defenses.
- Developed with a global community of security experts, the guidelines help organizations proactively safeguard against emerging risks.
- Companies implement the CIS Benchmark guidelines to limit configuration-based security vulnerabilities in their digital assets

When you implement CIS Benchmarks, you can better secure your legacy systems against common and emerging risks by taking steps such as these:

- Disabling unused ports
- Removing unnecessary app permissions
- Limiting administrative privileges

CIS levels

- Level 1: Basic easily implementable configurations designed to lower the attack surface without impacting performance.

- Level 2: Configuration recommendations that may create system conflicts and are intended to provide "defense in depth" for environments that need enhanced security

Domain Controller& Memeber Server

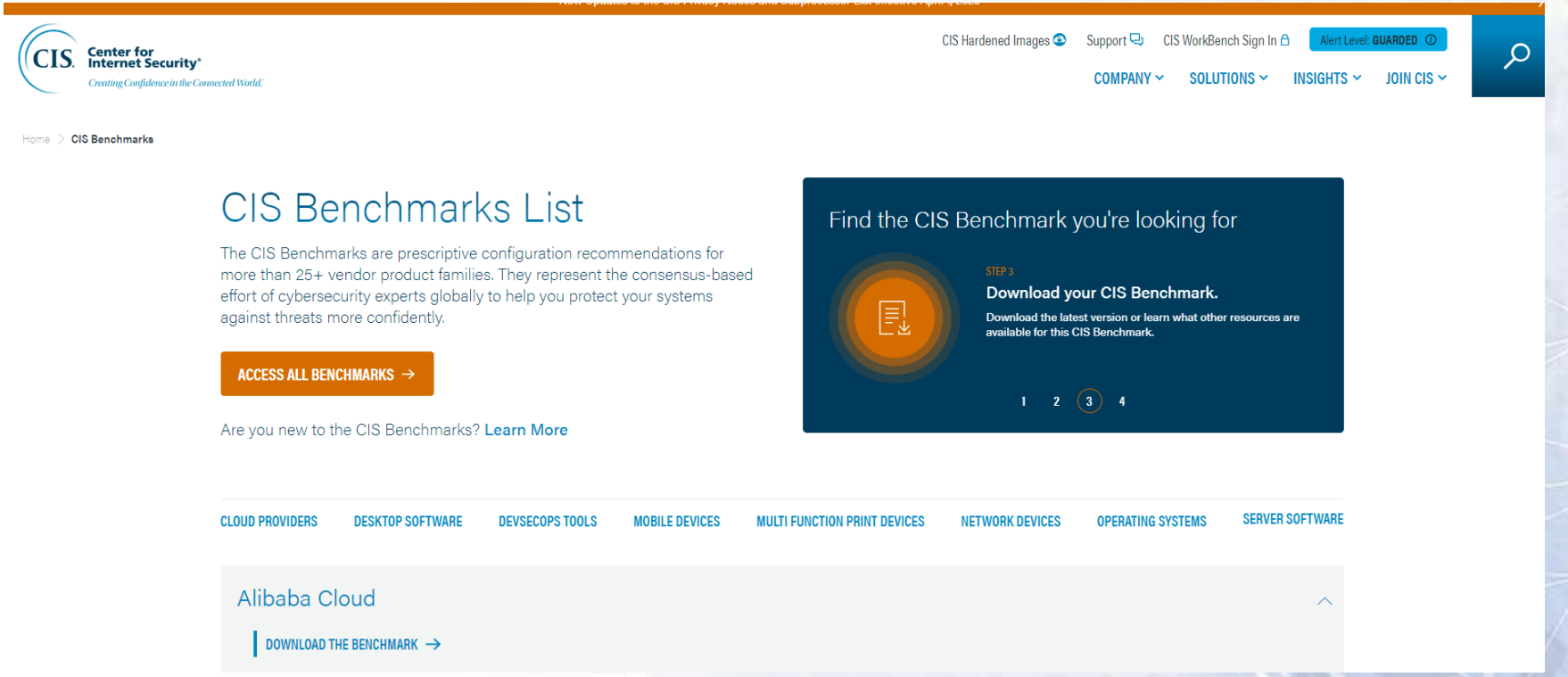
- A computer with a server OS and Active Directory installed is a Domain Controller.
- Any other computer with a server OS or for example with SQL Server installed, which works as a server and is joined to the domain can be called a member server. A member server does not have Active Directory installed.

- If a computer has DHCP or DNS, but not Active Directory, it's a member server.

Scored and Unscored

- Each recommendation is either 'scored' or 'not scored.
- 'Scored' recommendations are mandatory to achieve CIS compliance, and if not met, they lower the total benchmark score.

- Recommendations that are 'not scored' have no impact on the overall score of the benchmark.



https://www.cisecurity.org/cis-benchmarks

CIS Critical Security Controls

- Formerly the SANS Critical Security Controls (SANS Top 20) these are now officially called the CIS Critical Security Controls (CIS Controls).

- CIS Top 20 Critical Security Controls are a prioritized set of cybersecurity best practices that can help protect enterprises from the most pervasive and dangerous attacks.

- The CIS Critical Security Controls (CIS Controls) are a prescriptive, prioritized, and simplified set of best practices that you can use to strengthen your cybersecurity posture.

- Thousands of cybersecurity practitioners from around the world use the CIS Controls and/or contribute to their development via a community consensus process.

- https://www.cisecurity.org/controls

CIS-CAT

- CIS-CAT Pro Assessor evaluates the cybersecurity posture of a system against recommended policy settings. The tool helps organizations save time and resources by employing automated content with policy setting recommendations based on 100+ globally recognized CIS Benchmarks.

- The CIS Controls Self Assessment Tool (CSAT) is a companion tool that helps IT security teams track their implementation of every CIS Control and Sub-Control

There are two versions of CIS CSAT.

- The CIS-hosted CSAT is free to every organization for use in a non-commercial capacity.
- CIS CSAT Pro is an on-premises version that offers additional features and benefits and is only available to CIS SecureSuite Members

https://www.cisecurity.org/insights/blog/cis-csat-free-tool-assessing-implementation-of-cis-controls

Week 8 Topic 4

A Look At CIS Security Benchmarks (4)

- CIS Benchmarks example (Operating Systems)
  - MS Windows Server 2012-R2

A Look At CIS Security Benchmarks (4)

#	OVERALL CIS BENCHMARK CATEGORIES	TOTAL
1	OPERATING SYSTEMS	36
2	SERVER SOFTWARE	33
3	CLOUD PROVIDERS	2
4	MOBILE DEVICES	8
5	NETWORK DEVICES	6
6	DESKTOP SOFTWARE	21
7	MULTIFUNCTION PRINT DEVICES	1
	GRAND TOTAL CIS BENCHMARKS	**107**

A Look At CIS Security Benchmarks (4)

- January 31, 2017
- 760 pages PDF doc

A Look At CIS Security Benchmarks (4)

- Profile applicability:
  - Level 1 domain controller
  - Level 1 member server
  - **Level 2 domain controller**
  - **Level 2 member server**

- **Level 1:** Items in this profile intend to:
  - be practical and prudent;
  - provide a clear security benefit; and
  - not inhibit the utility of the technology beyond acceptable means

A Look At CIS Security Benchmarks (4)

- **Level 2: extends the Level 1 - profile**
  - intended for environments or use cases where security is paramount
  - acts as defense in depth measure
  - may negatively inhibit the utility or performance of the technology
- Control content:
  - Profile applicability (ASA 8.X, ASA 9.X)
  - Description
  - Rationale
  - Audit
  - Remediation
  - **Impact**
  - Default value
  - References

A Look At CIS Security Benchmarks (4)

- **1.1.2 [L1]:** *Ensure 'Maximum password age' is set to '60 or fewer days, but not 0' (Scored)*
  - *Profile applicability: Level 1 Domain Controller, Level 1 Member Server*
- **1.1.2 [L1] Description:**
  - This policy setting defines how long a user can use their password before it expires.
  - Values for this policy setting range from 0 to 999 days. If you set the value to 0, the password will never expire.
- **1.1.2 [L1] Audit:**
  - Navigate to the UI Path articulated in the Remediation section and confirm it is set as prescribed.

A Look At CIS Security Benchmarks (4)

Remediation:

To establish the recommended configuration via GP, set the following UI path to `60 or fewer days, but not 0`:

```
Computer Configuration\Policies\Windows Settings\Security Settings\Account Policies\Password Policy\Maximum password age
```

- 1.1.2 [L1] Default Value: 42 days
- 1.1.2 [L1] Reference: CCE-37167-4
  - Common Configuration Enumeration (Unique identifiers for common system config issues)

Week 8 Topic 5

A Look At CIS Security Benchmarks (3)

- June 29, 2016
- 174 pages PDF doc

A Look At CIS Security Benchmarks (3)

- Control content:
  - Profile applicability (ASA 8.X, ASA 9.X)
  - Description
  - Rationale
  - Audit
  - Remediation
  - Default value
  - References

A Look At CIS Security Benchmarks (3)

- 1.8 (page 88); Session Timeout
  - Profile applicability: Level 1, Cisco ASA9.X
  - Description: Sets the idle timeout for a console session before the security appliance terminates it.

- 1.8 (page 88); Session Timeout
  - Rationale: Limiting session timeout prevents unauthorized users from using abandoned sessions to perform malicious activities.

A Look At CIS Security Benchmarks (3)

Audit:

- Step 1: Run the following command to show what the console timeout is set to

```
hostname#sh run console | in timeout.5
```

The output should look like

```
console timeout 5
```

Example:

```
Asa-fw#sh run console | in timeout.5
console timeout 5
```

Here the session timeout is 5 minutes

- Step 2: If an output is displayed, the system is compliant. If not, there is a finding.

Remediation:

- Step 1: Run the following command to set the console timeout to less than or equal to 5 minutes

```
hostname(config)# console timeout 5
```

A Look At CIS Security Benchmarks (3)

- 1.8 (page 88); Session Timeout
  - Default Value: The default timeout is 0, which means the console session will not time out

- 1.8 (page 88); Session Timeout
  - Reference: CLI Book 1: Cisco ASA Series General Operations CLI Configuration Guide, 9.1

Week 8 Topic 7

A Look At DISA STIGs (1)

- Defense Information Systems Agency (DISA) provides technical guides referred to as Security Technical Implementation Guides (STIGs).

- USA DoD
- Security Technical Implementation Guides (STIGs)
- Most expansive security benchmarks available
- Most regularly updated
- 425 STIGs available

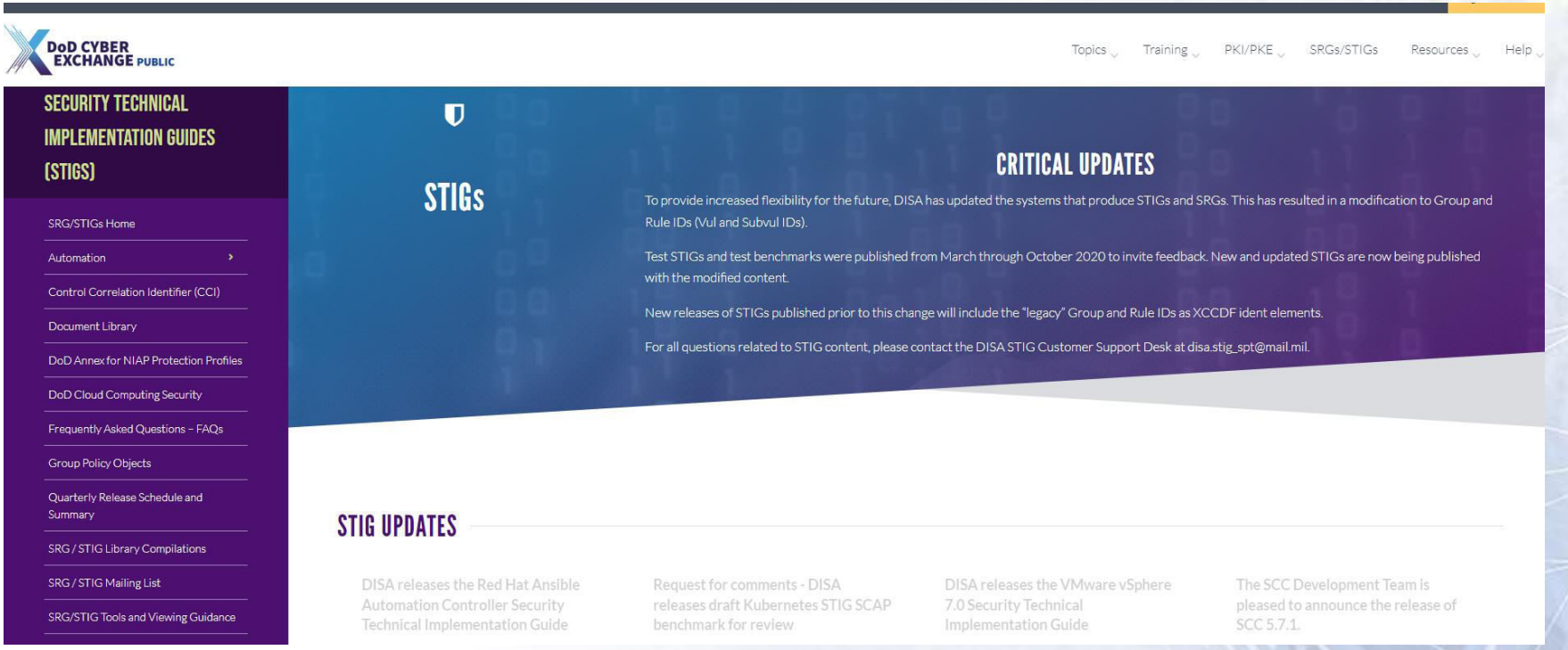
A Look At DISA STIGs (1)

- **STIGs master list (A-Z):**
  - https://public.cyber.mil/stigs/downloads/

  - **STIG viewer:**
  - https://public.cyber.mil/stigs/srg-stig-tools/

A Look At DISA STIGs (1)

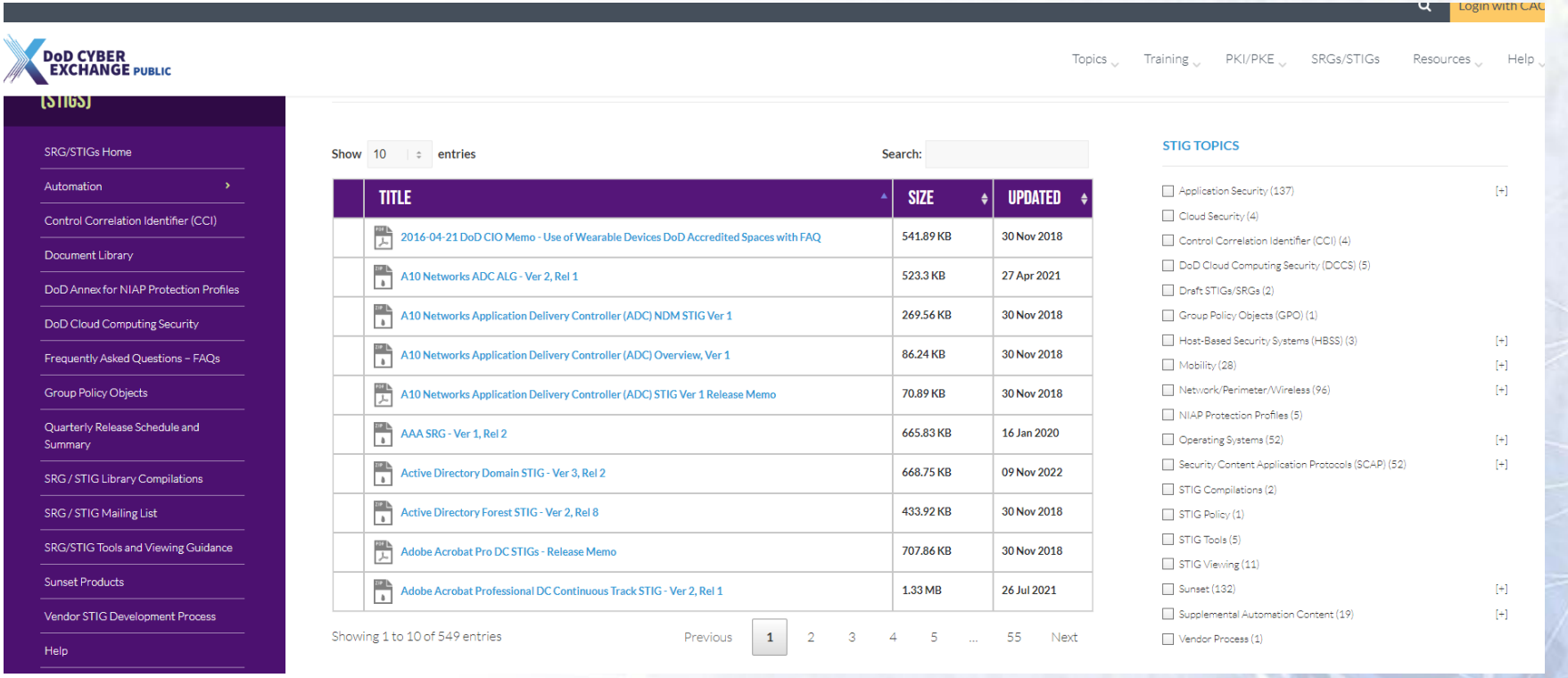
STIGs HOME



https://public.cyber.mil/stigs/

A Look At DISA STIGs (1)

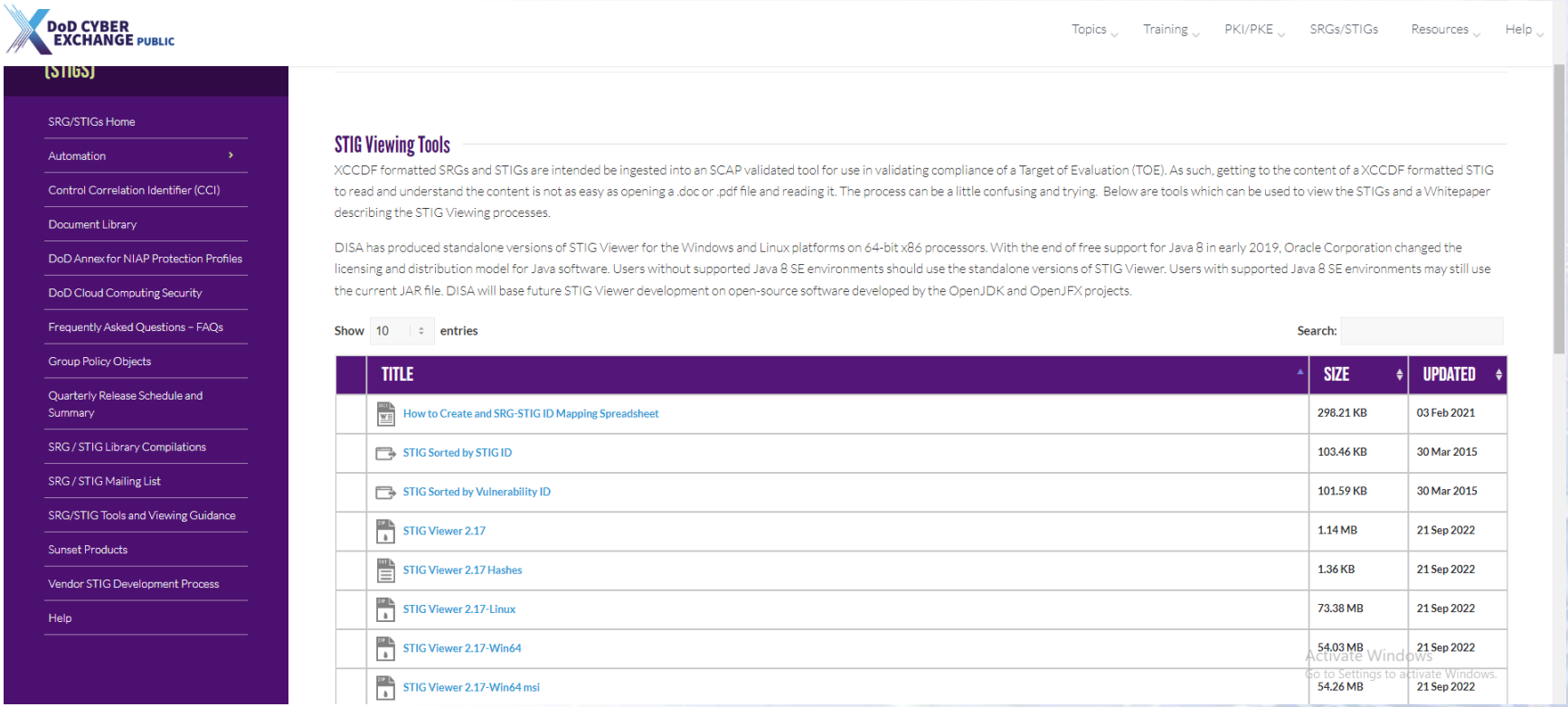
STIGs Master List



https://public.cyber.mil/stigs/downloads/

A Look At DISA STIGs (1)

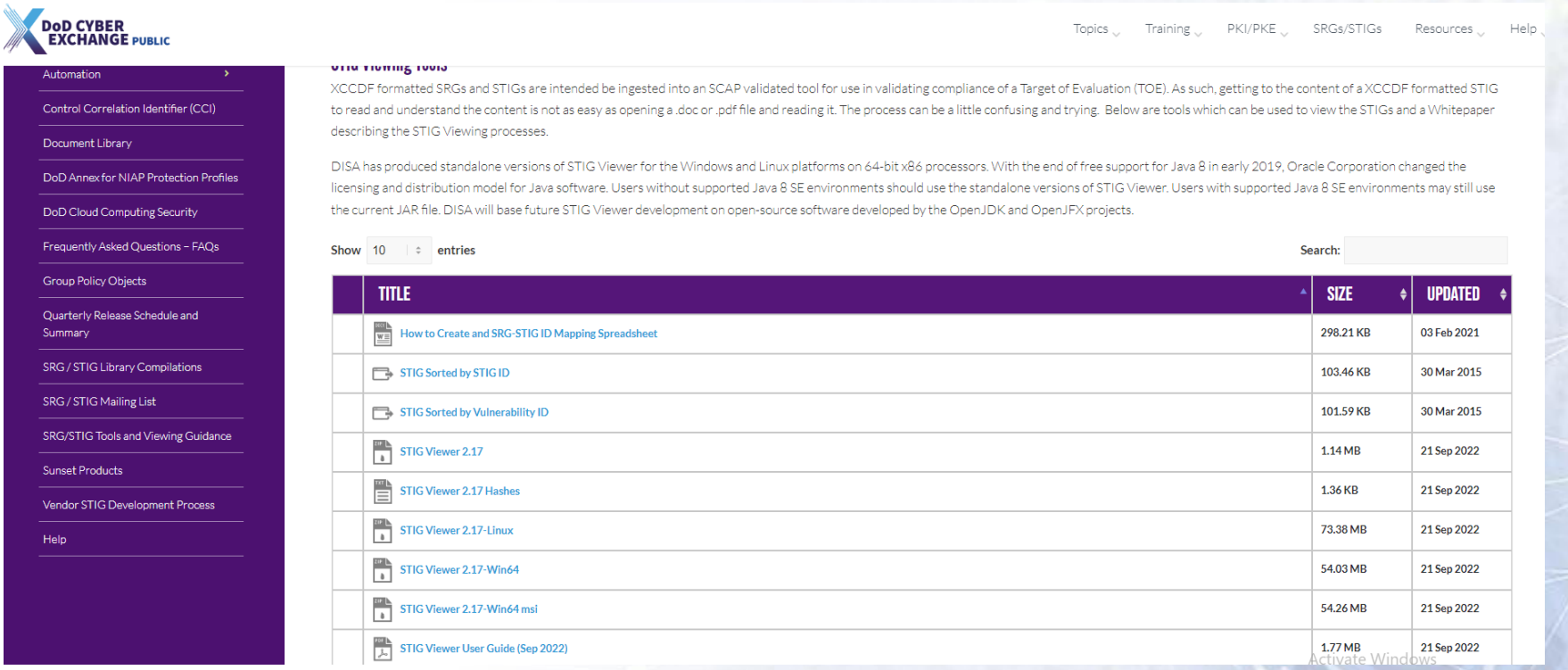
STIGs Viewer



https://public.cyber.mil/stigs/srg-stig-tools/

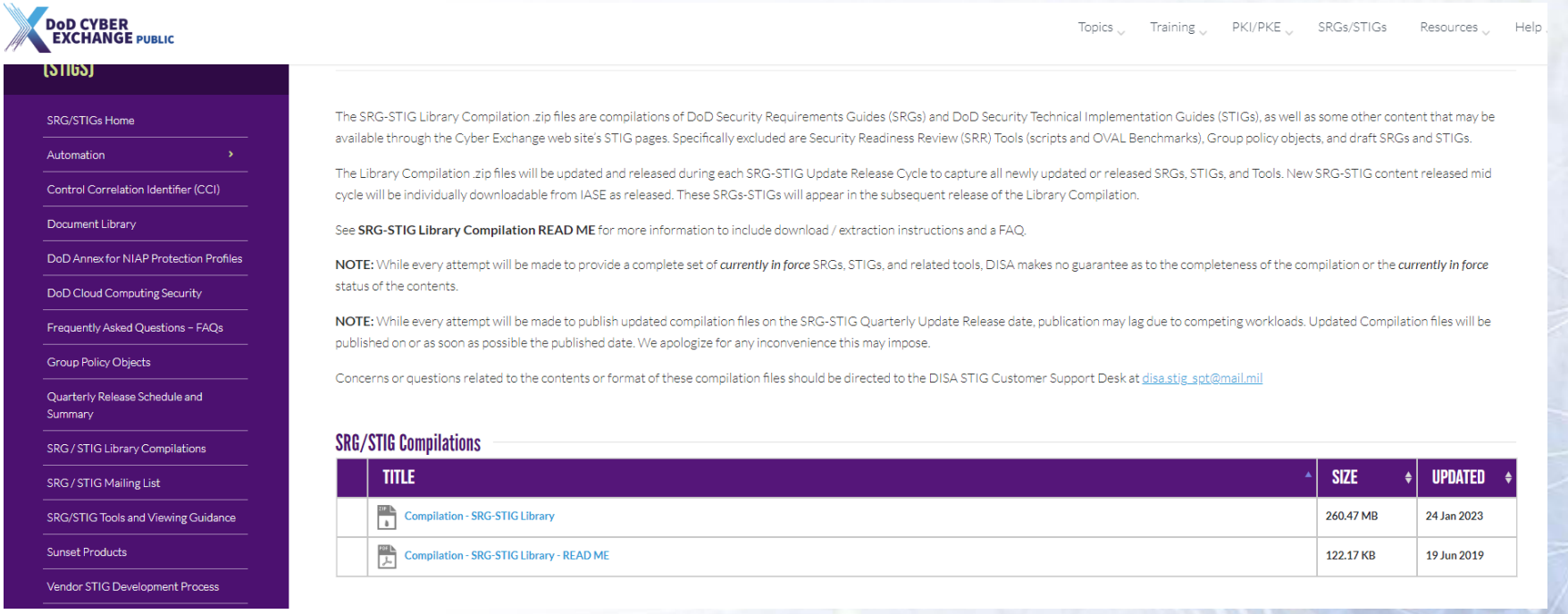
A Look At DISA STIGs (1)

STIG Viewer Download



A Look At DISA STIGs (1)

STIG Library Compilation

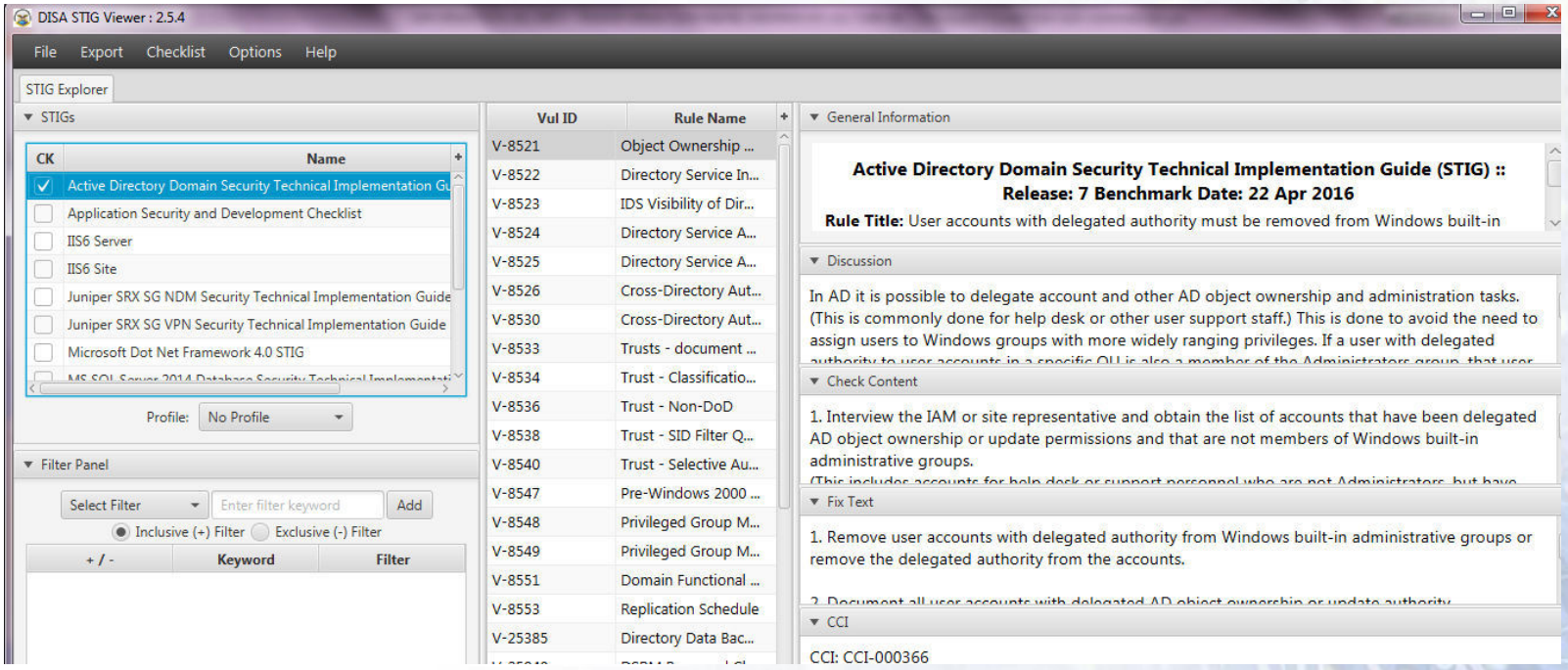


- There are three categories of severity, ranked on level of risk or vulnerability.
- These are known as Severity Category Codes (CAT), with CAT 1, CAT 2 and CAT 3 levels of risk.

- CAT 1 controls cover the most severe vulnerabilities and risks.
- CAT I weaknesses should receive the highest priority for correction or mitigation.

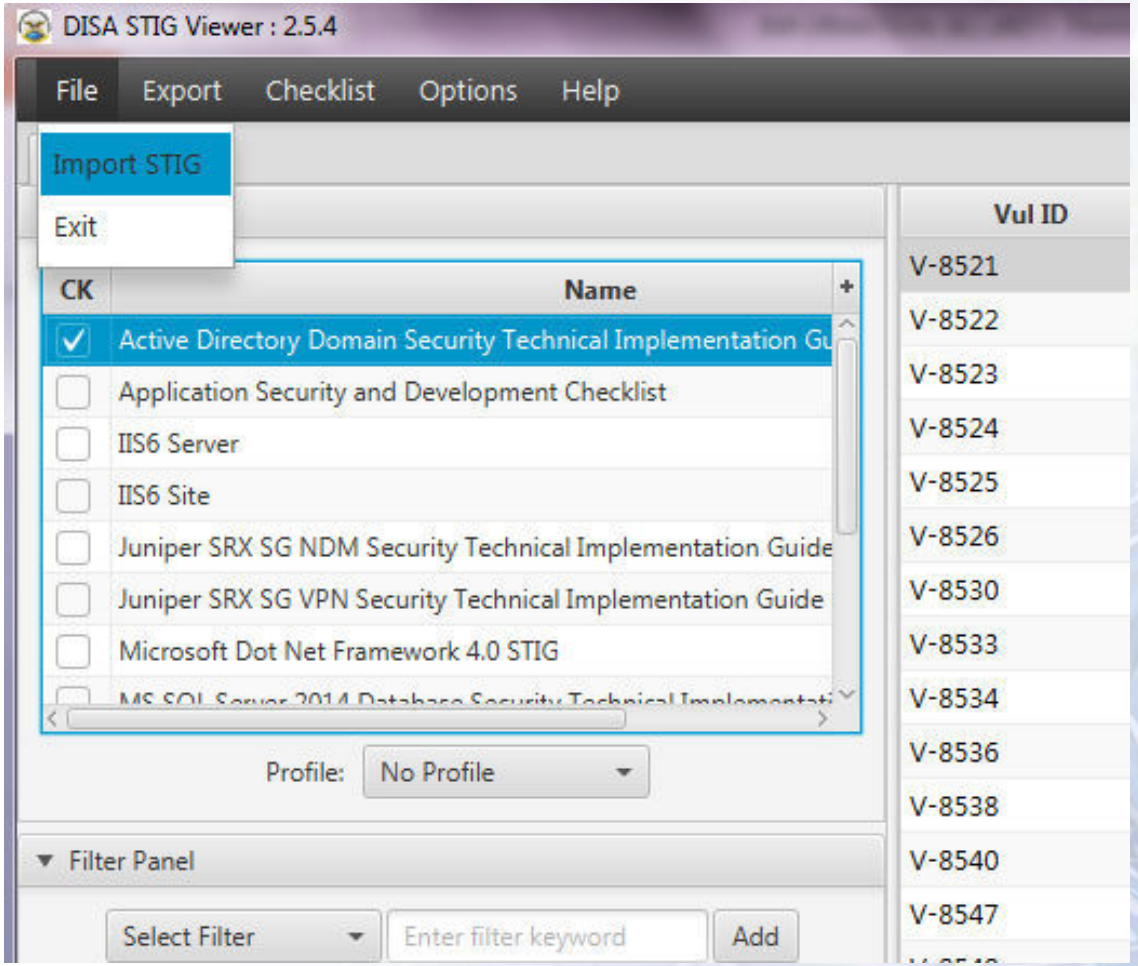
A Look At DISA STIGs (1)

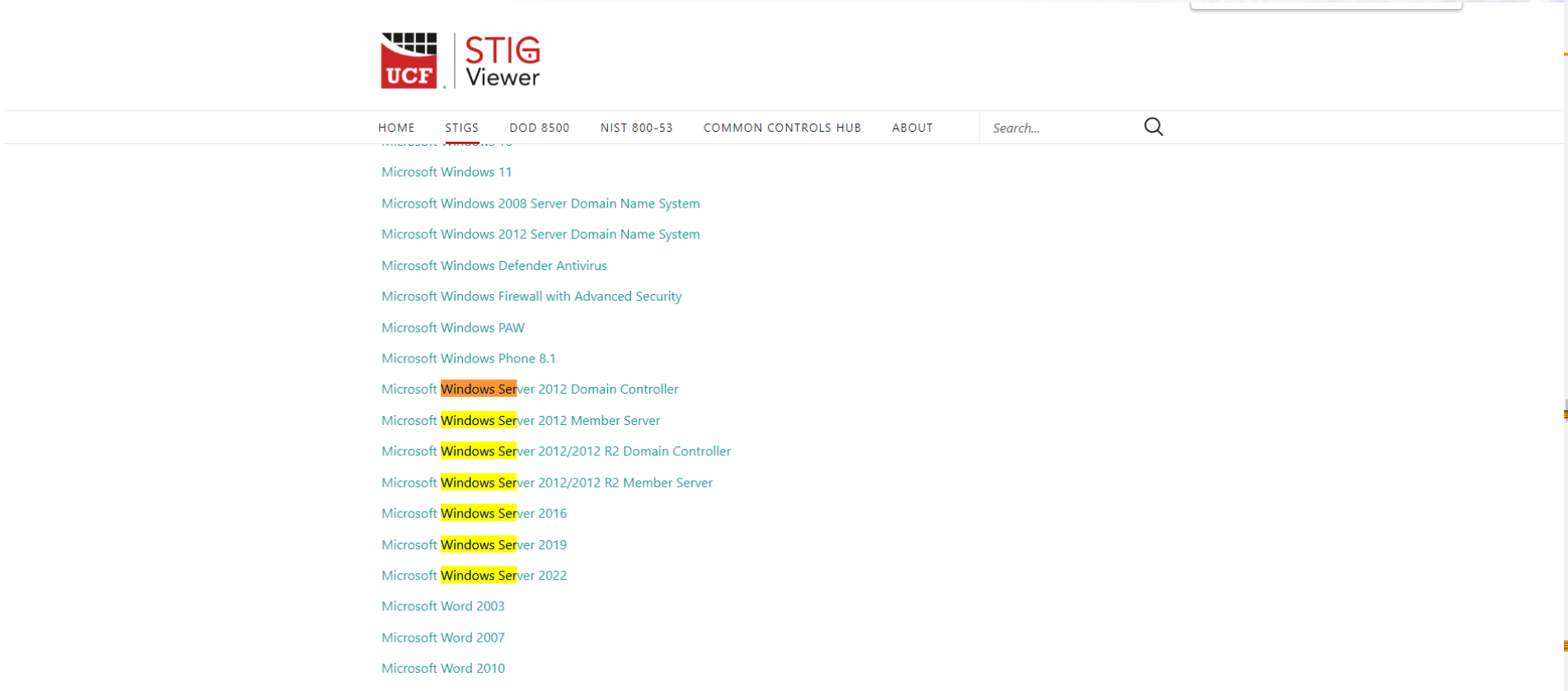
STIG Viewer Window



A Look At DISA STIGs (1)

Import STIG





https://www.stigviewer.com/stigs

UCF
STIG Viewer
HOME
STIGS
DOD 8500
NIST 800-53
COMMON CONTROLS HUB
ABOUT
Search...
Microsoft Windows Server 2012/2012 R2 Member Server Security Technical Implementation Guide
Overview
Version
Date
Finding Count (330)
Downloads
3
2022-09-06
CAT I (High): 33
CAT II (Med): 238
CAT III (Low): 59
Excel
JSON
XML
STIG Description
This Security Technical Implementation Guide is published as a tool to improve the security of Department of Defense (DoD) information systems. The requirements are derived from the National Institute of Standards and Technology (NIST) 800-53 and related documents. Comments or proposed revisions to this document should be sent via email to the following address: disa.stig_spt@mail.mil.
Available Profiles

- Completely different mechanism for DISA STIGs

Week 8 Topic 8

A Look At DISA STIGs (2)

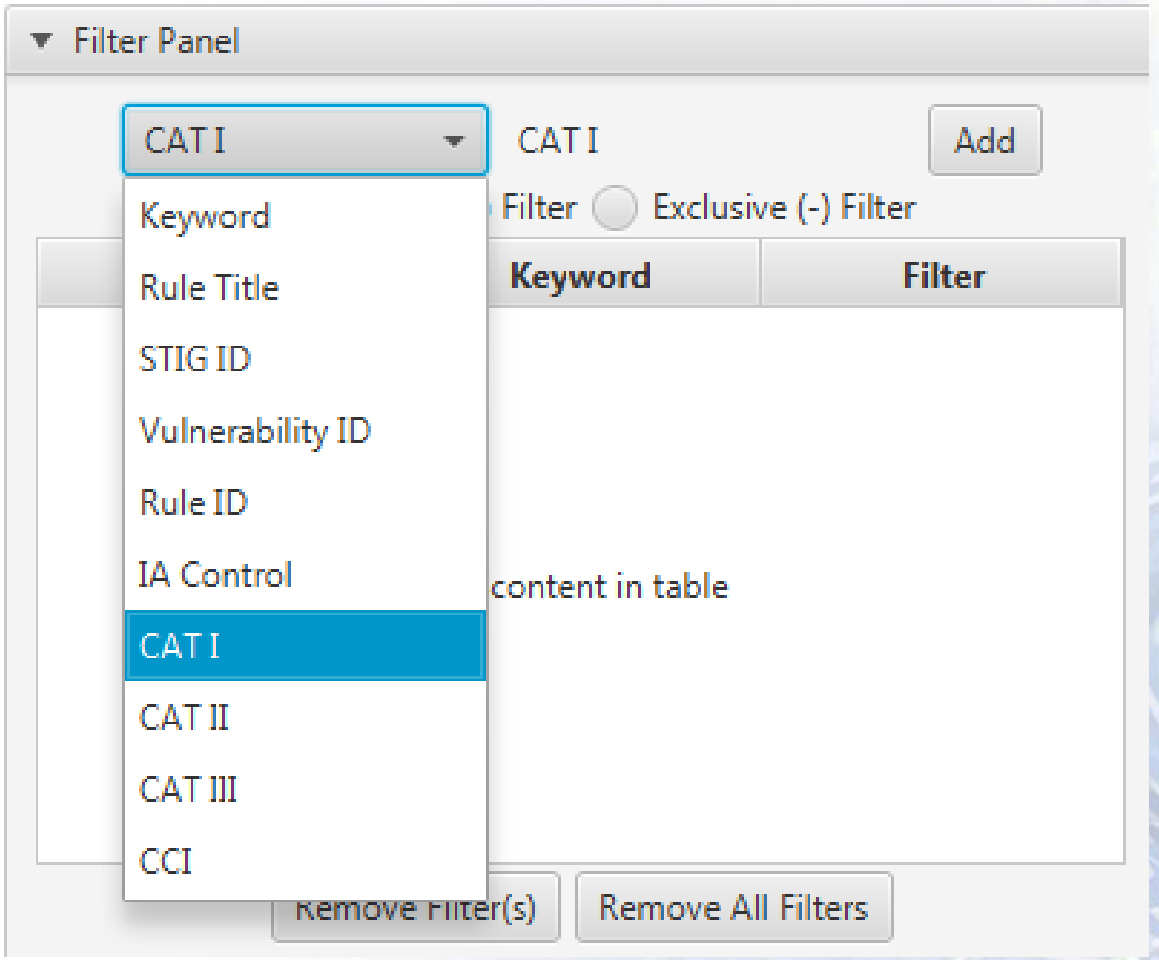
- STIG content:
  - General information (title)
  - Discussion
  - Check content
  - Fix text
  - CCI (References)

A Look At DISA STIGs (2)

SEVERITY	DISA CATEGORY CODE GUIDELINES
CAT 1	Any vulnerability, the exploitation of which will directly and immediately result in loss of Confidentiality, Availability, or Integrity.
CAT 2	Any vulnerability, the exploitation of which has a potential to result in loss of Confidentiality, Availability, or Integrity.
CAT 3	Any vulnerability, the existence of which degrades measures to protect against loss of Confidentiality, Availability, or Integrity

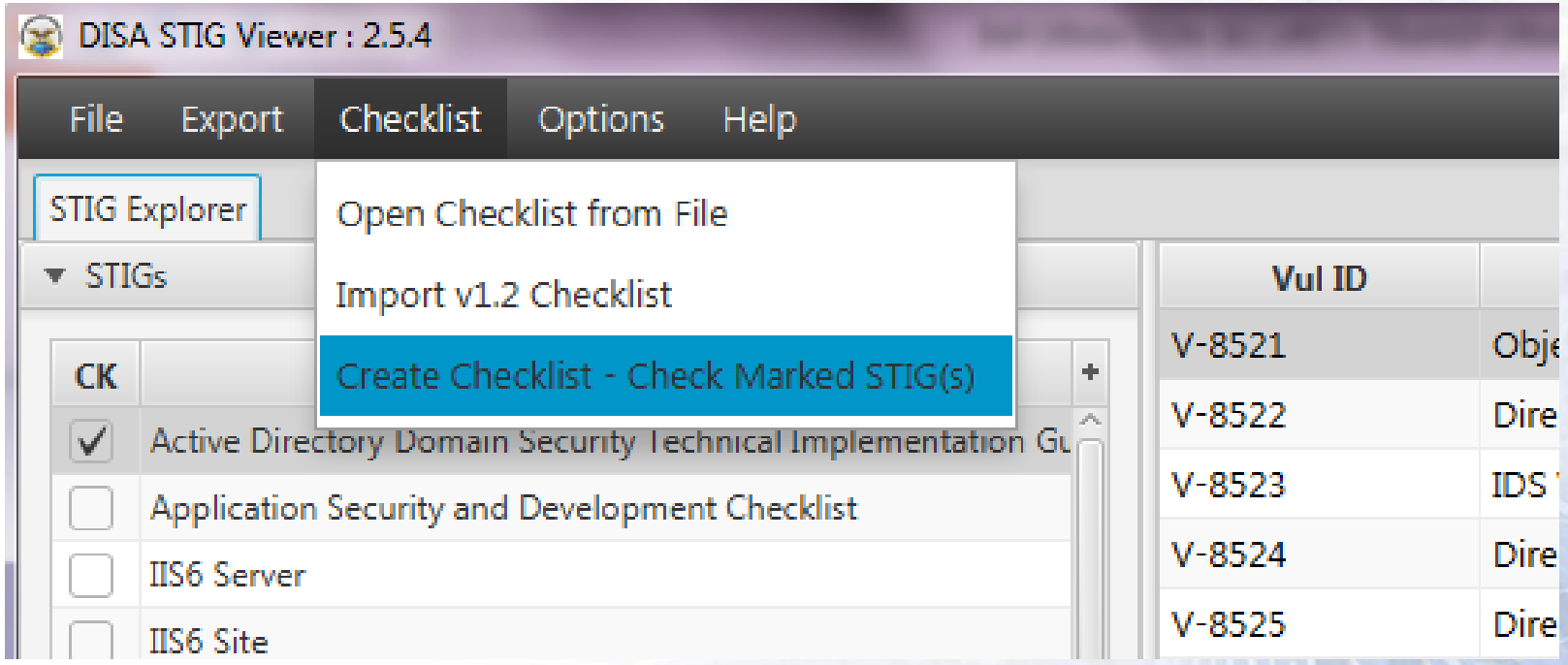
A Look At DISA STIGs (2)

FILTER PANEL



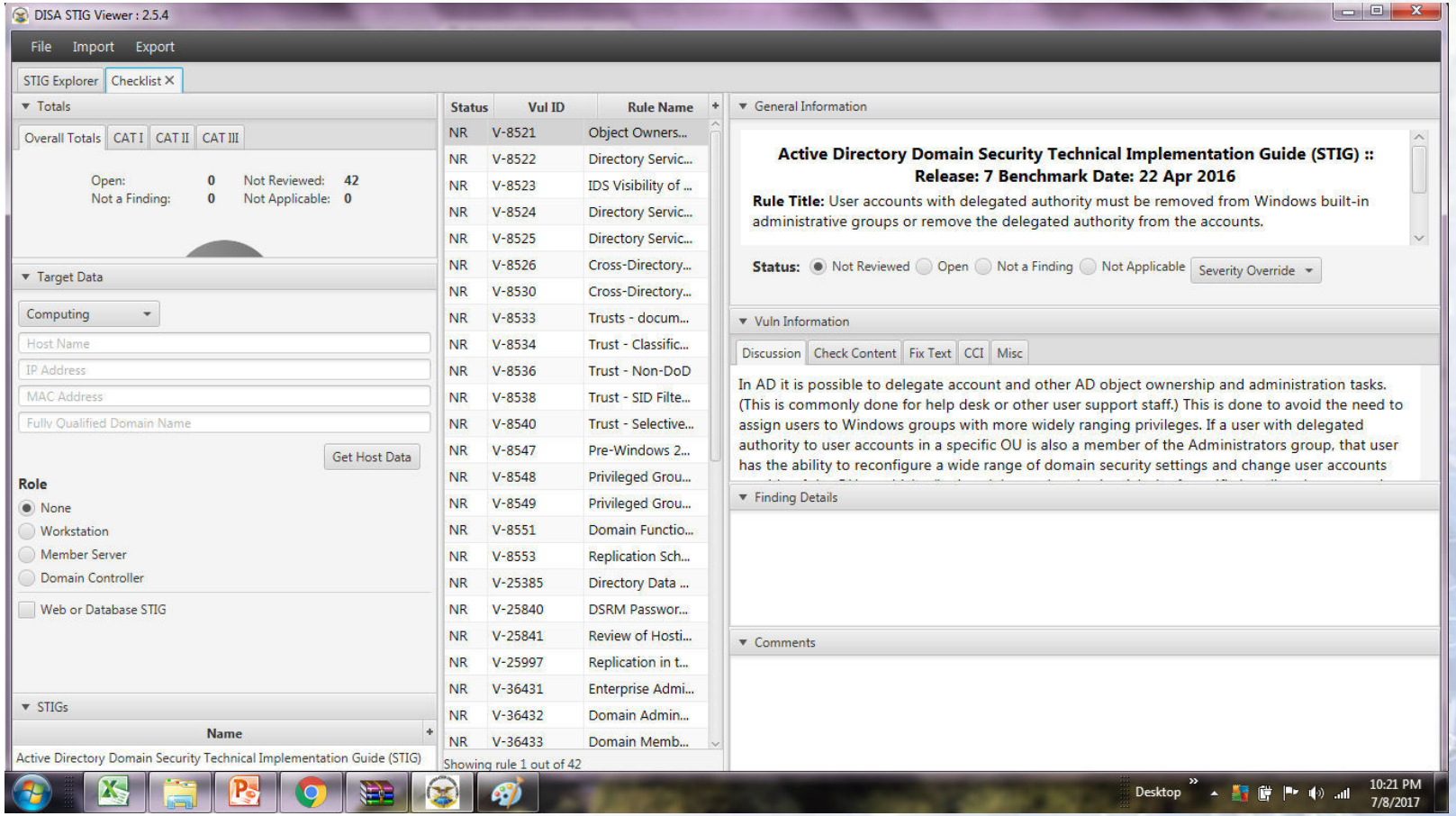
A Look At DISA STIGs (2)

CREATE CHECKLIST



A Look At DISA STIGs (2)

CHECKLIST

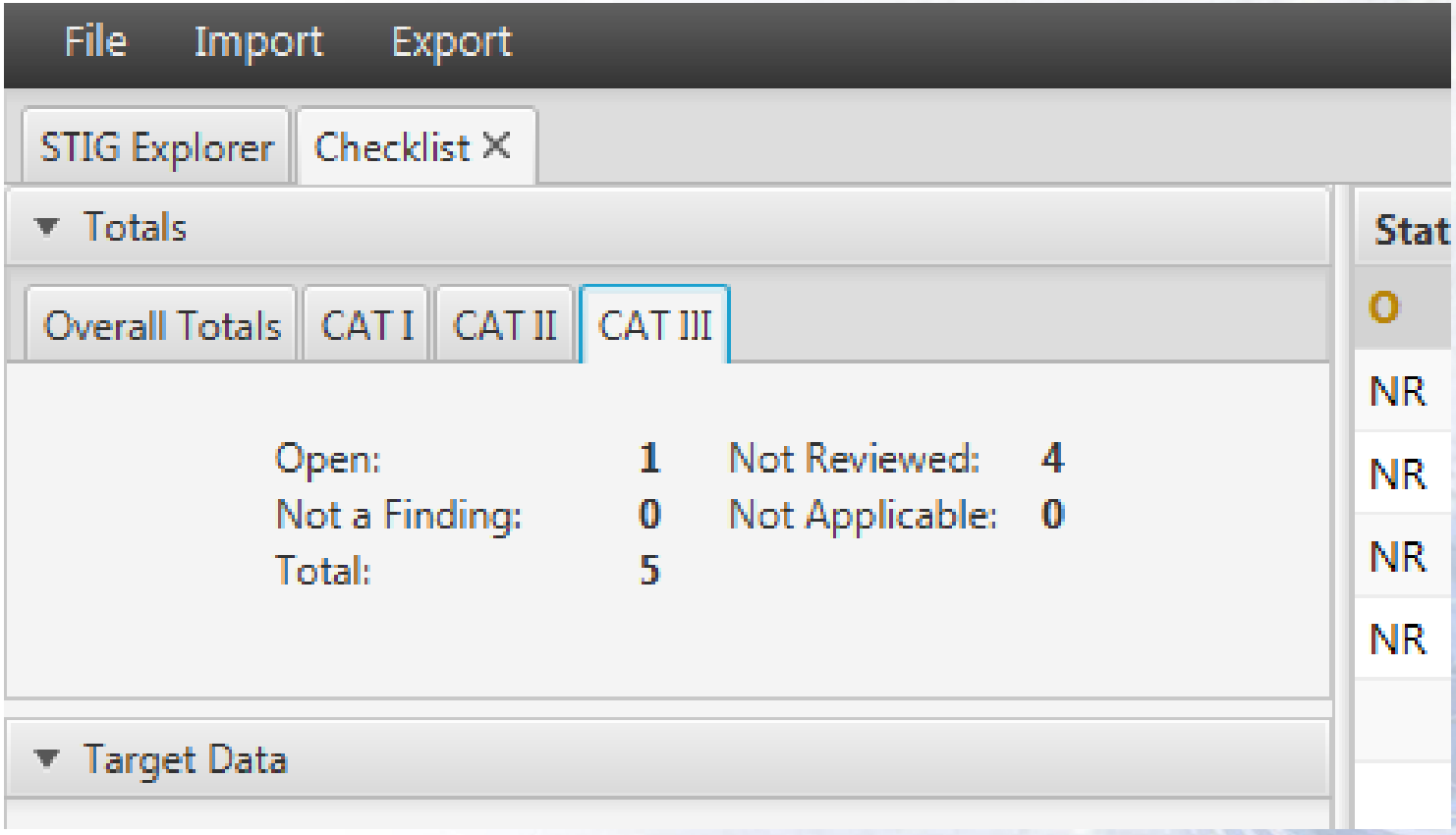


A Look At DISA STIGs (2)

- **Checklist screens:**
  - Overall totals
  - Target data
  - Role
  - Finding details
  - Comments
- **Checklist screens (STATUS):**
  - Not reviewed
  - Open
  - Not a finding
  - Not applicable

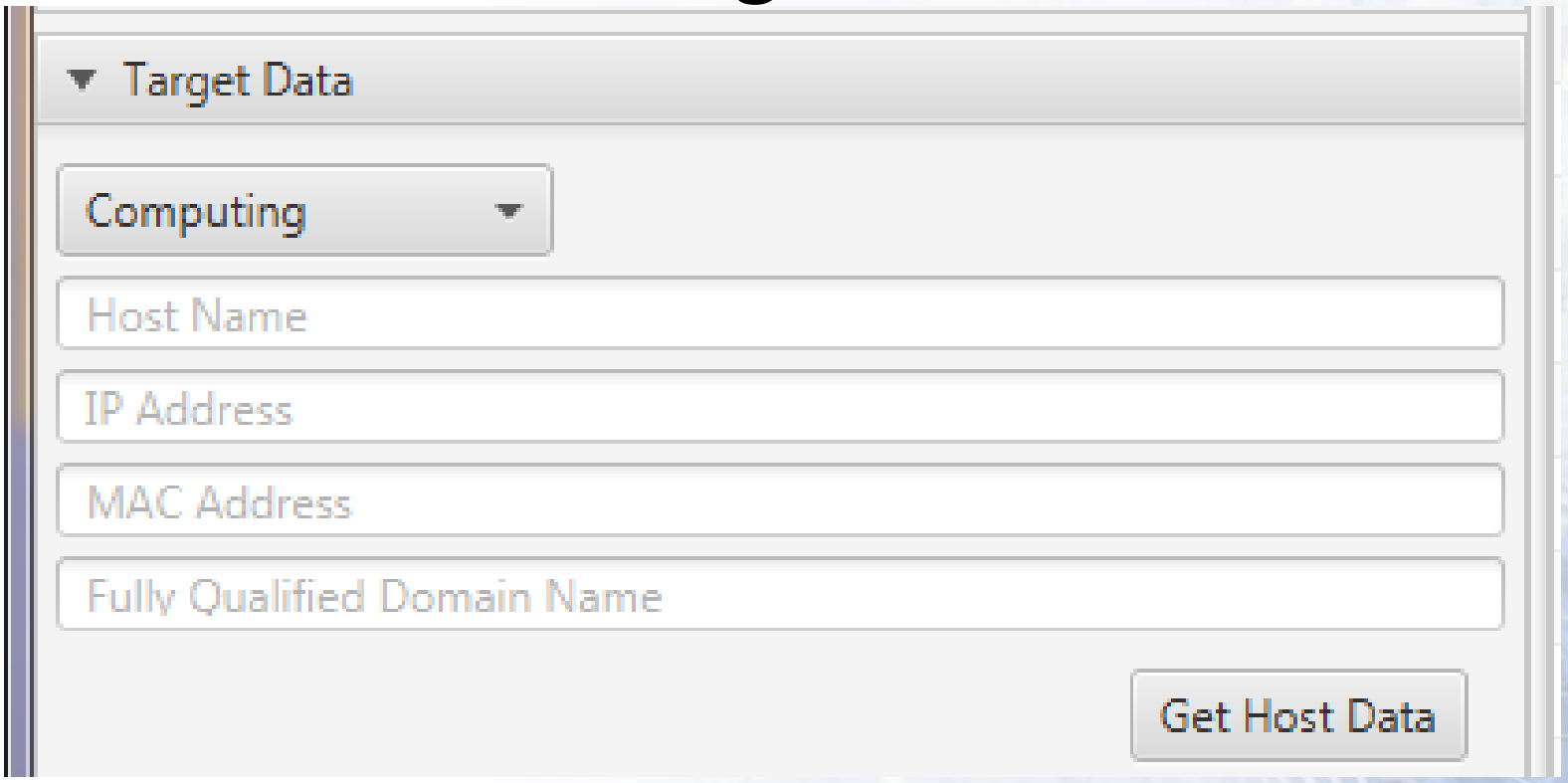
A Look At DISA STIGs (2)

Totals



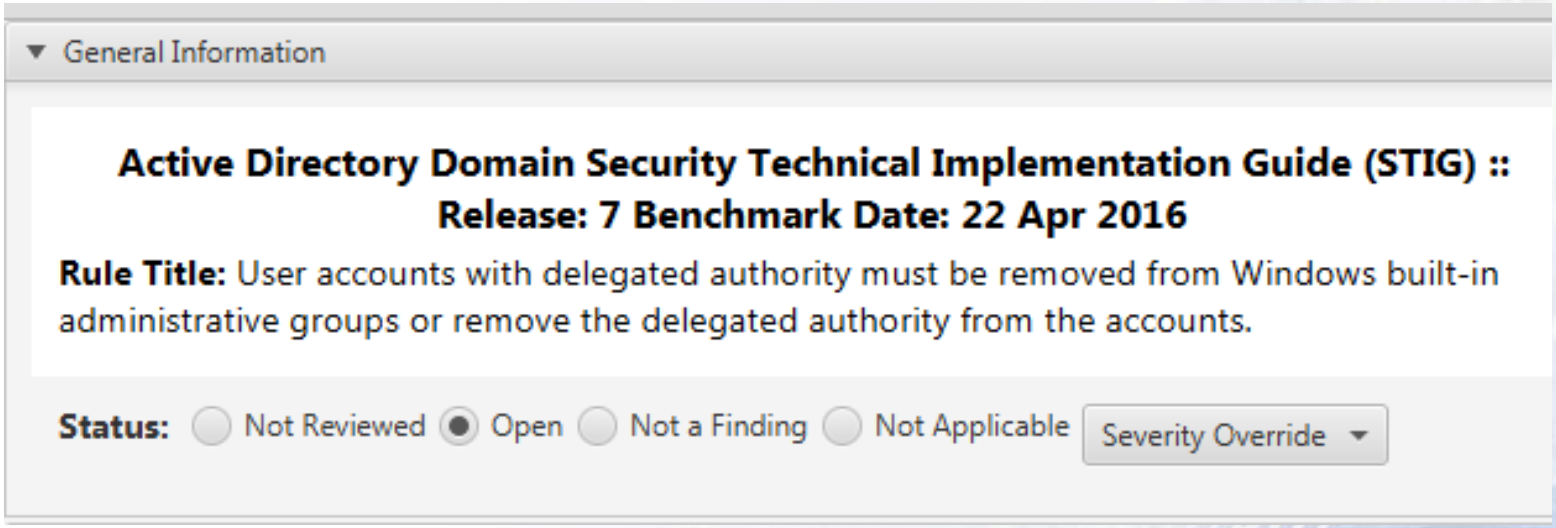
A Look At DISA STIGs (2)

Target Data



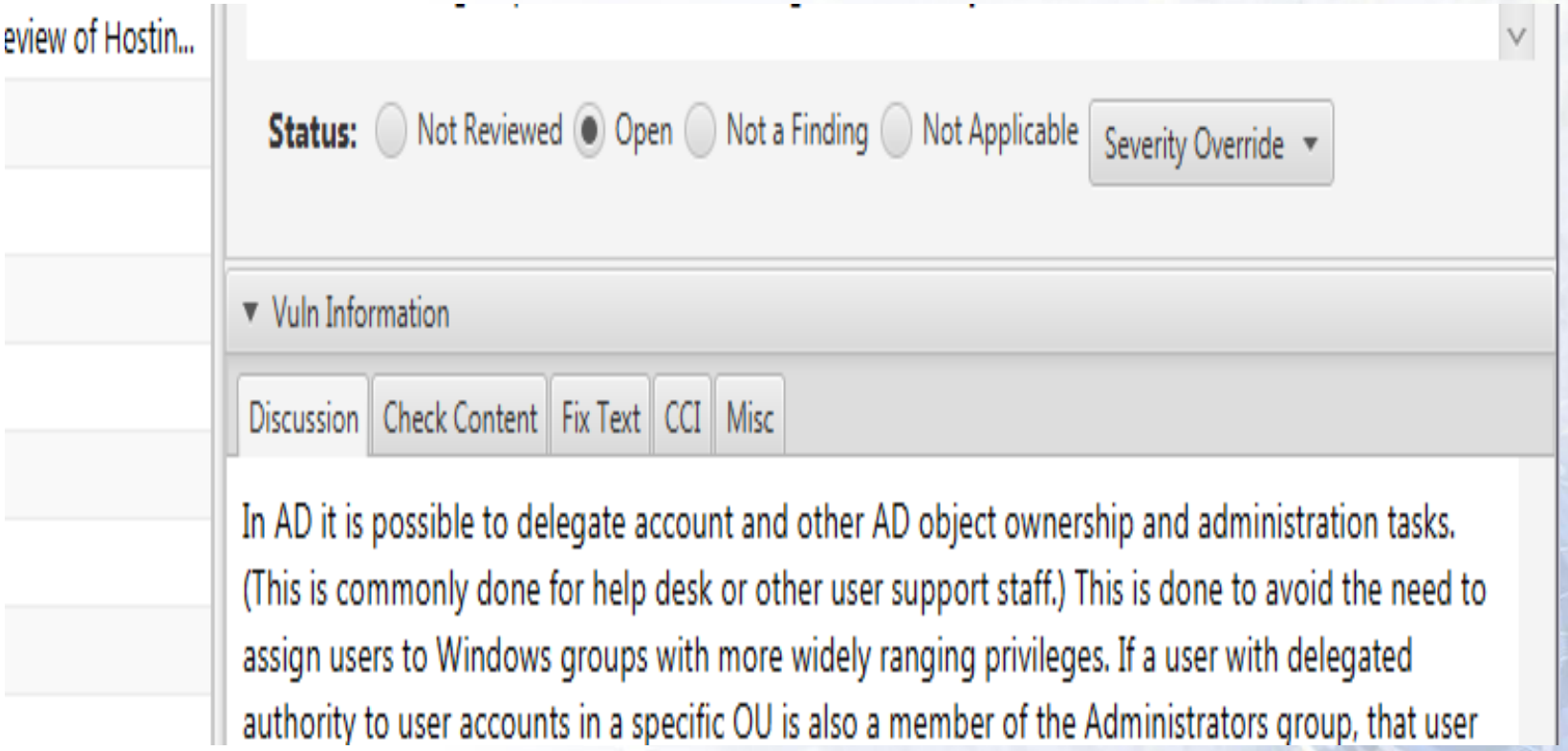
A Look At DISA STIGs (2)

Status



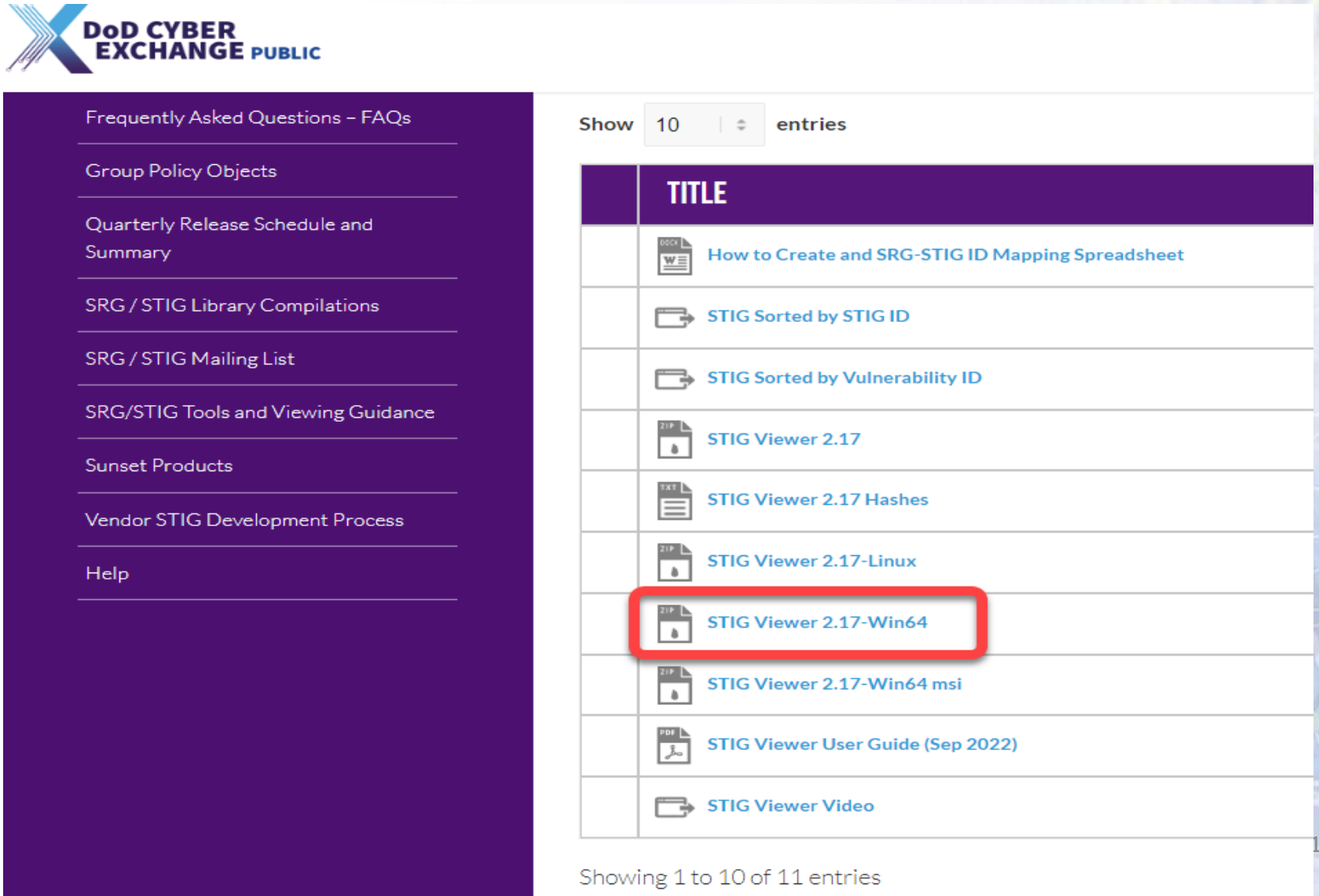
A Look At DISA STIGs (2)

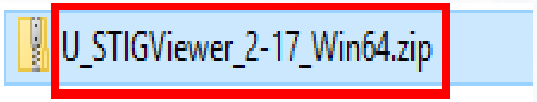
Vuln Information

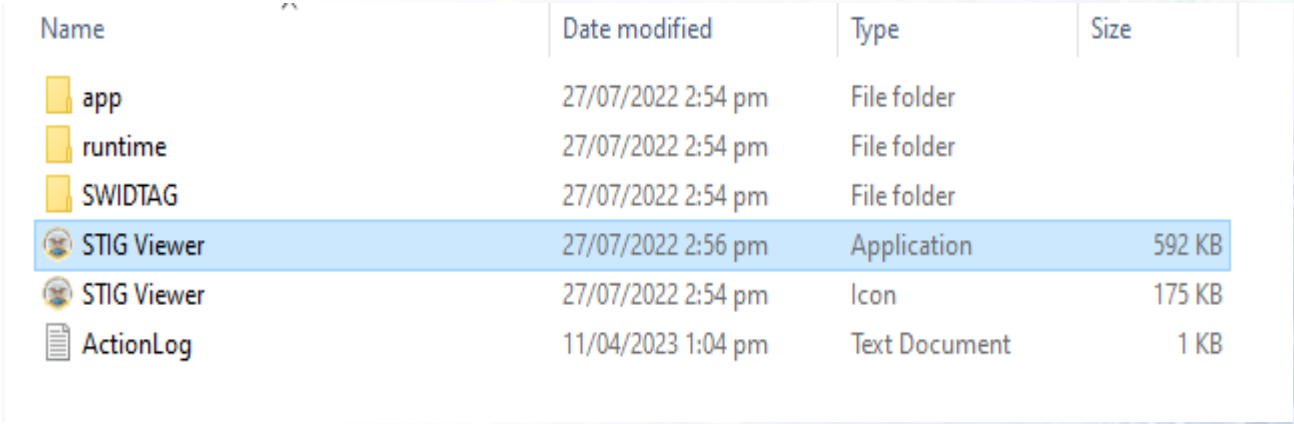


Week 8 Topic 9

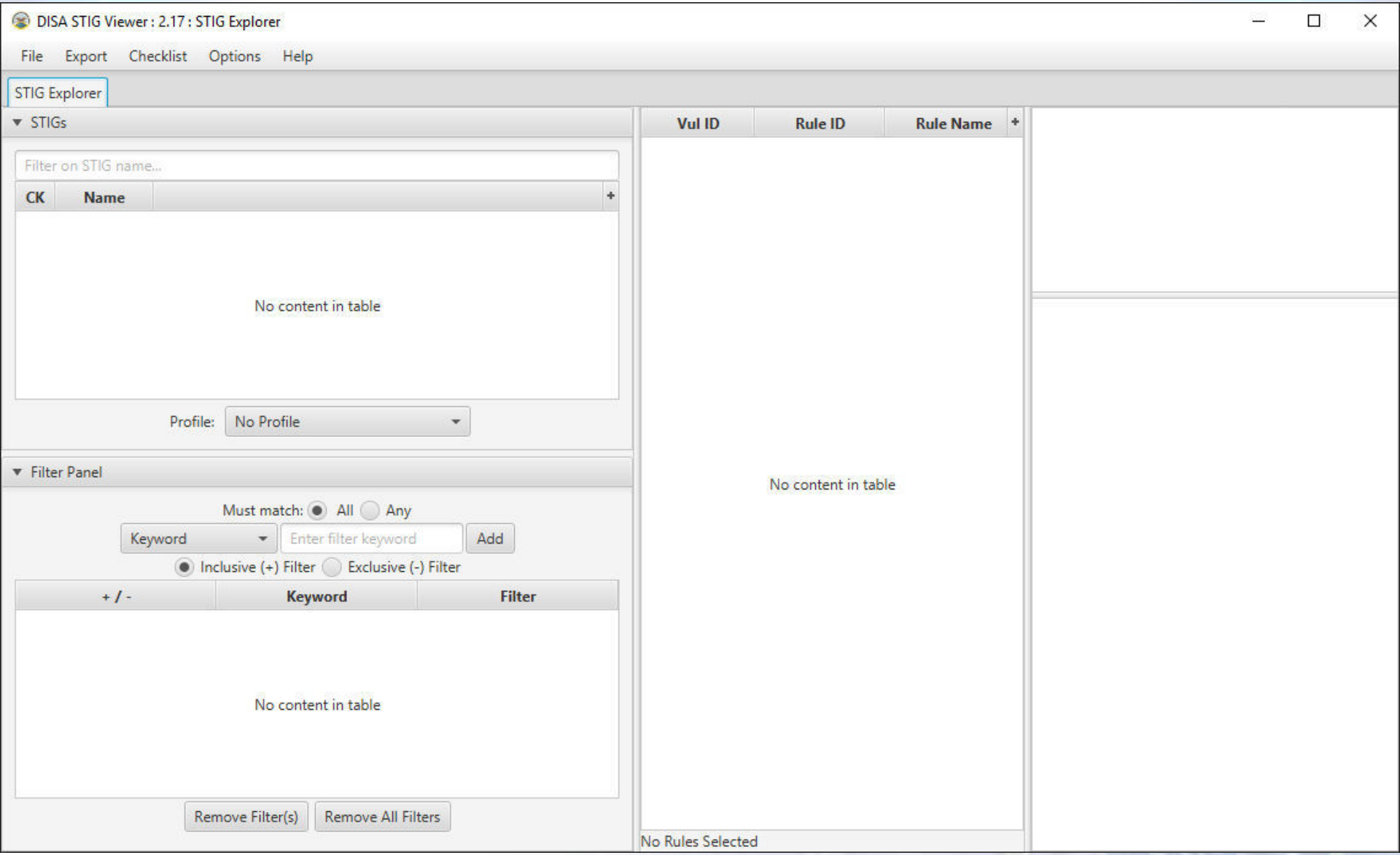
A Look At STIGs Viewer

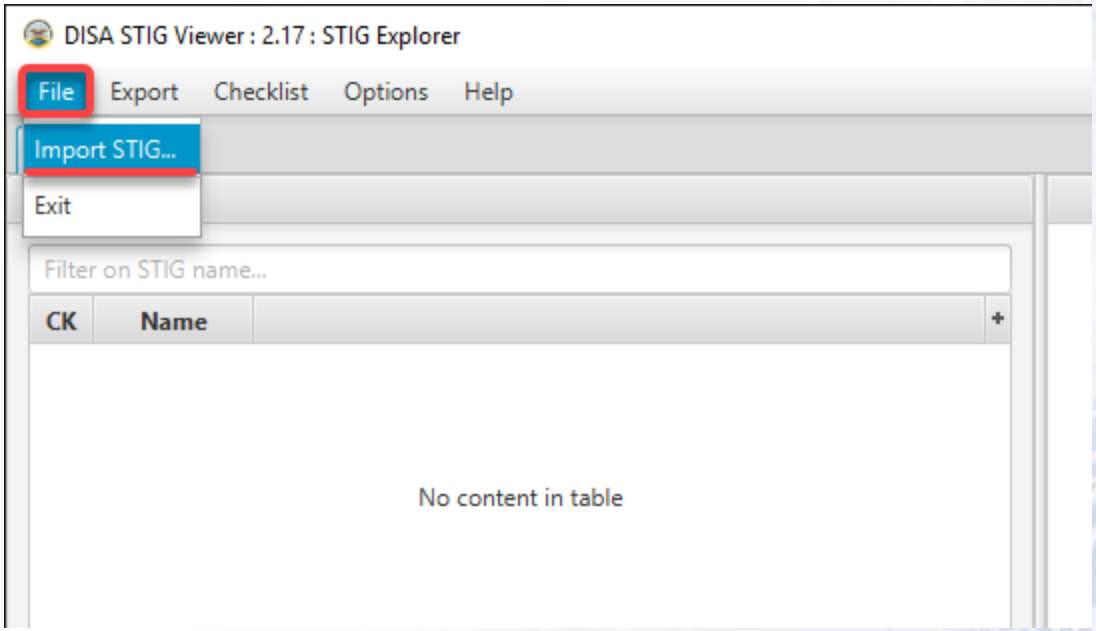




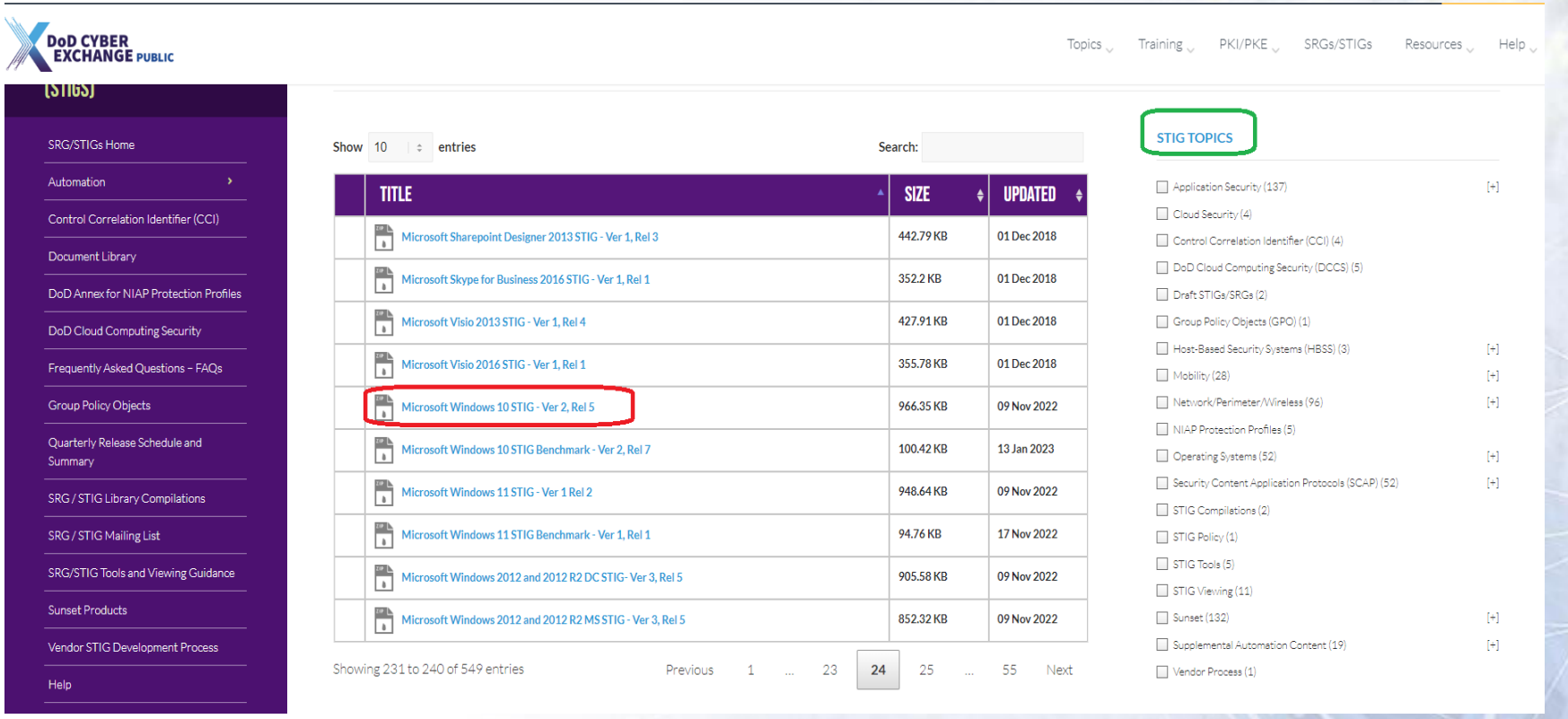


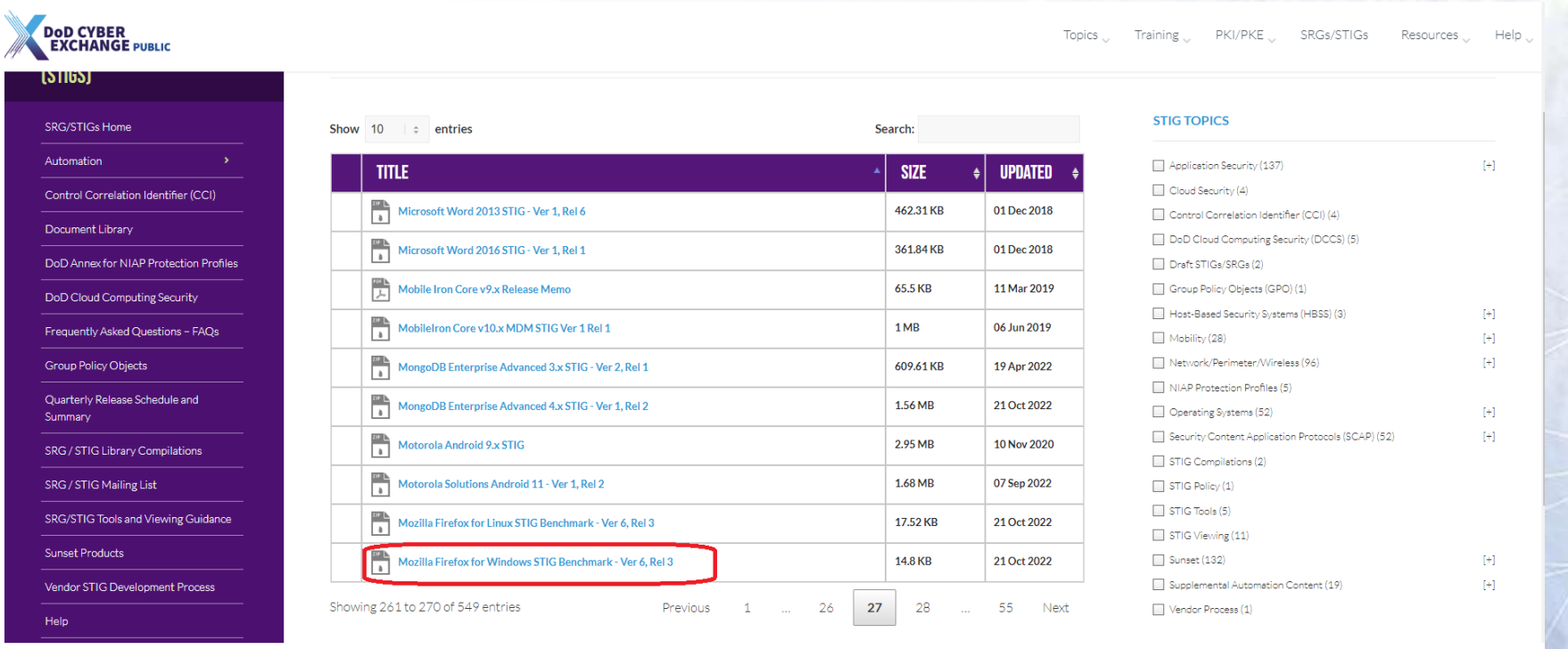
https://public.cyber.mil/stigs/srg-stig-tools/

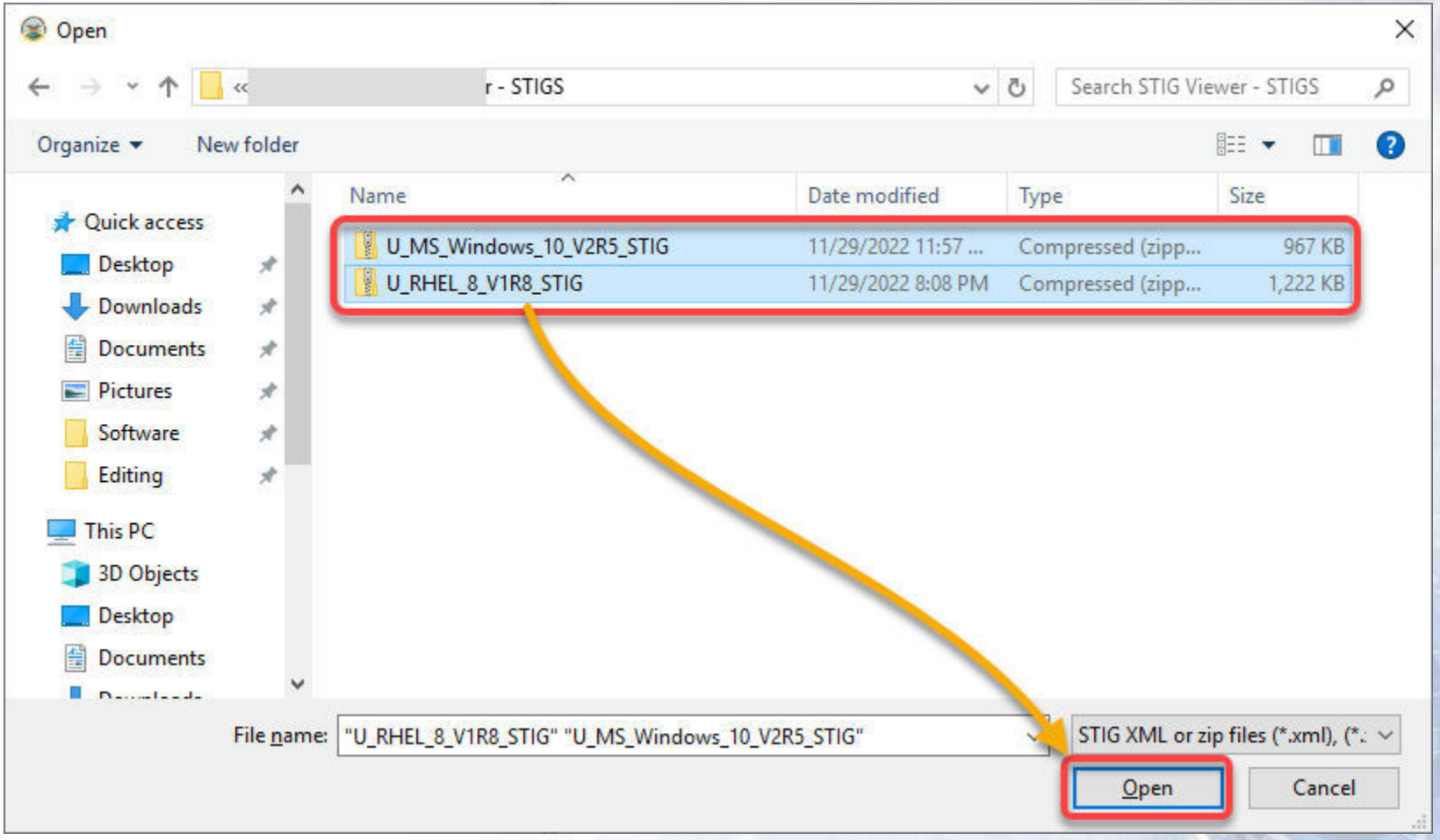

DISA STIG Viewer : 2.17 : STIG Explorer
File
Export
Checklist
Options
Help
STIG Explorer
STIGs
Filter on STIG name...
CK
Name
No content in table
Profile:
No Profile
Filter Panel
Must match:
All
Any
Keyword
Enter filter keyword
Add
Inclusive (+) Filter
Exclusive (-) Filter
+ / -
Keyword
Filter
No content in table
Remove Filter(s)
Remove All Filters
Vul ID
Rule ID
Rule Name
No content in table
No Rules Selected

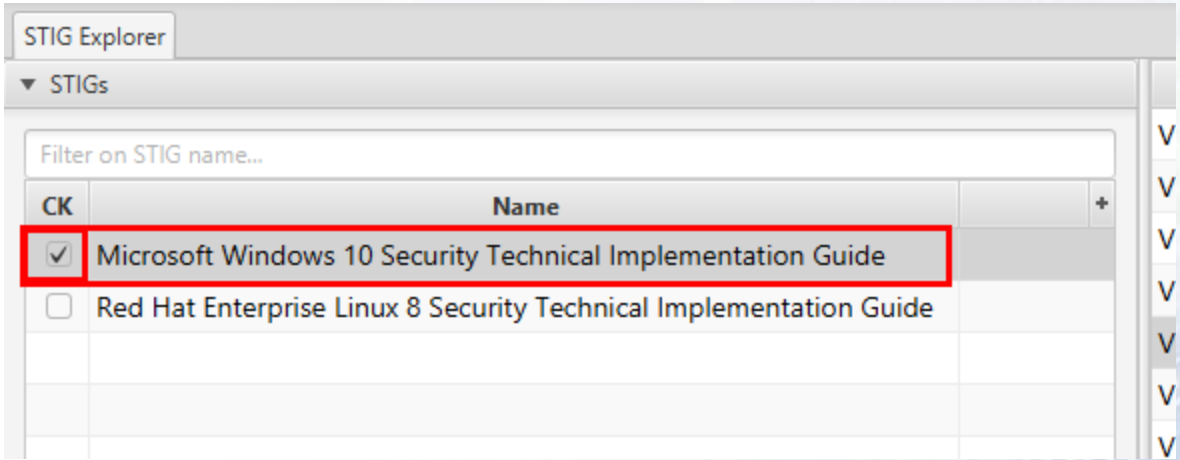

DISA STIG Viewer : 2.17 : STIG Explorer
File
Export
Checklist
Options
Help
Import STIG...
Exit
Filter on STIG name...
CK
Name
No content in table

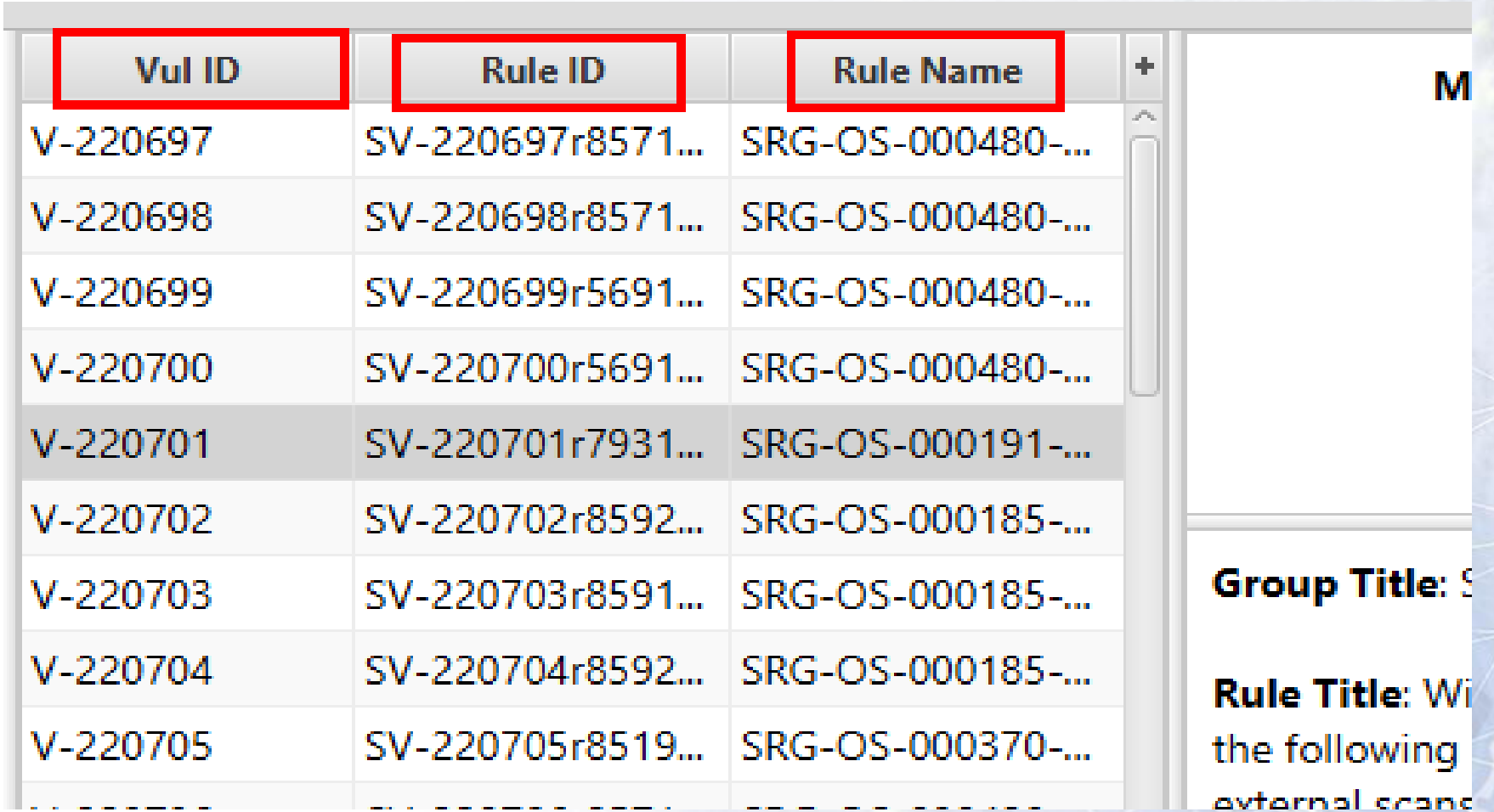
A Look At STIGs Viewer

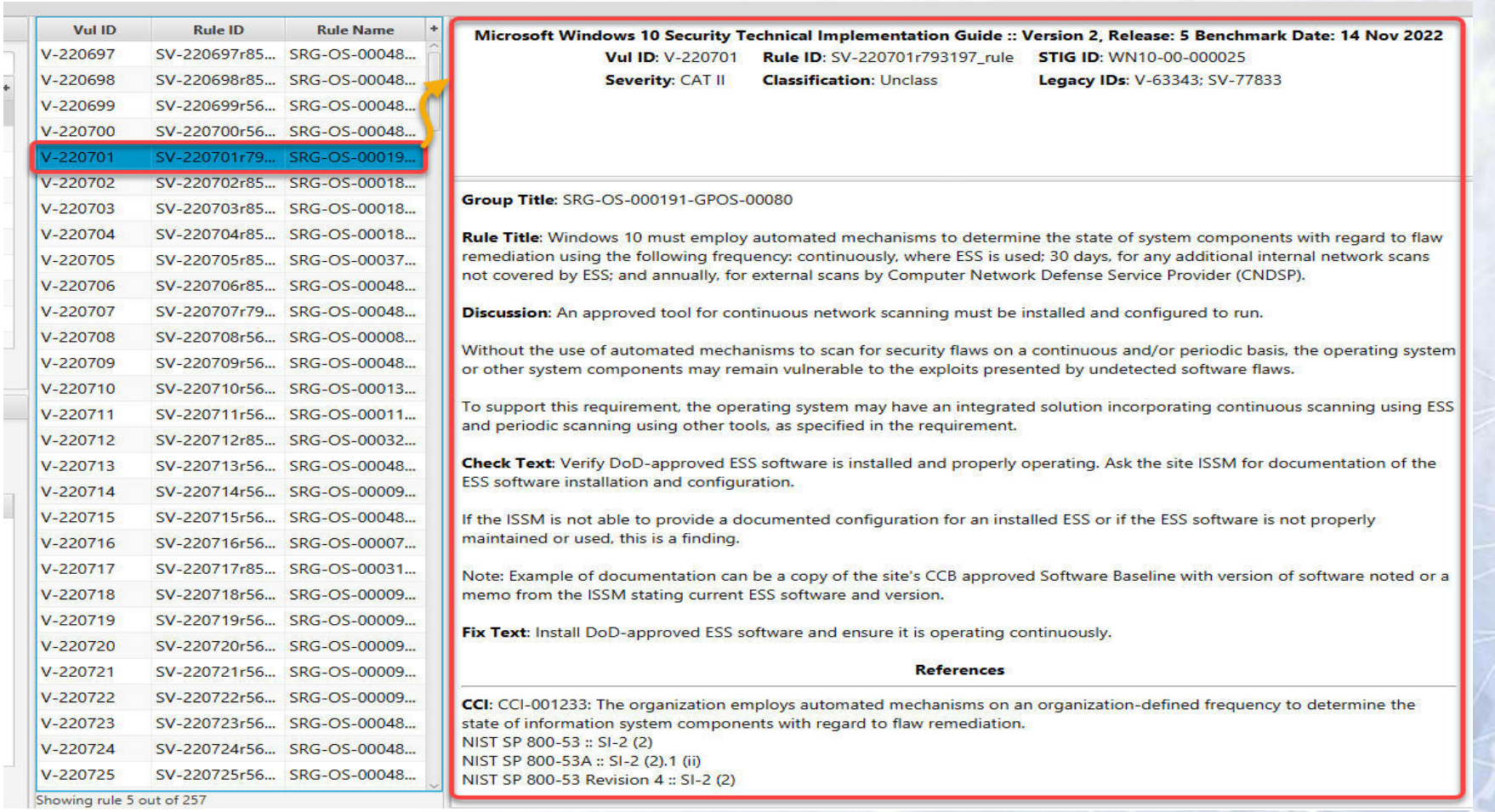



DoD CYBER EXCHANGE PUBLIC
Topics
Training
PKI/PKE
SRGs/STIGs
Resources
Help
(STIGS)
SRG/STIGs Home
Automation
Control Correlation Identifier (CCI)
Document Library
DoD Annex for NIAP Protection Profiles
DoD Cloud Computing Security
Frequently Asked Questions – FAQs
Group Policy Objects
Quarterly Release Schedule and Summary
SRG / STIG Library Compilations
SRG / STIG Mailing List
SRG/STIG Tools and Viewing Guidance
Sunset Products
Vendor STIG Development Process
Help
Show 10 entries
Search:
TITLE
SIZE
UPDATED
Microsoft Word 2013 STIG - Ver 1, Rel 6 462.31 KB 01 Dec 2018
Microsoft Word 2016 STIG - Ver 1, Rel 1 361.84 KB 01 Dec 2018
Mobile Iron Core v9.x Release Memo 65.5 KB 11 Mar 2019
MobileIron Core v10.x MDM STIG Ver 1 Rel 1 1 MB 06 Jun 2019
MongoDB Enterprise Advanced 3.x STIG - Ver 2, Rel 1 609.61 KB 19 Apr 2022
MongoDB Enterprise Advanced 4.x STIG - Ver 1, Rel 2 1.56 MB 21 Oct 2022
Motorola Android 9.x STIG 2.95 MB 10 Nov 2020
Motorola Solutions Android 11 - Ver 1, Rel 2 1.68 MB 07 Sep 2022
Mozilla Firefox for Linux STIG Benchmark - Ver 6, Rel 3 17.52 KB 21 Oct 2022
Mozilla Firefox for Windows STIG Benchmark - Ver 6, Rel 3 14.8 KB 21 Oct 2022
Showing 261 to 270 of 549 entries
Previous 1 ... 26 27 28 ... 55 Next
STIG TOPICS
Application Security (137) [+]
Cloud Security (4)
Control Correlation Identifier (CCI) (4)
DoD Cloud Computing Security (DCCS) (5)
Draft STIGs/SRGs (2)
Group Policy Objects (GPO) (1)
Host-Based Security Systems (HBSS) (3) [+]
Mobility (28) [+]
Network/Perimeter/Wireless (96) [+]
NIAP Protection Profiles (5)
Operating Systems (52) [+]
Security Content Application Protocols (SCAP) (52) [+]
STIG Compilations (2)
STIG Policy (1)
STIG Tools (5)
STIG Viewing (11)
Sunset (132) [+]
Supplemental Automation Content (19) [+]
Vendor Process (1)


Open
r - STIGS
Search STIG Viewer - STIGS
Organize
New folder
Quick access
Desktop
Downloads
Documents
Pictures
Software
Editing
This PC
3D Objects
Desktop
Documents
Name
Date modified
Type
Size
U_MS_Windows_10_V2R5_STIG
11/29/2022 11:57 ...
Compressed (zipp...
967 KB
U_RHEL_8_V1R8_STIG
11/29/2022 8:08 PM
Compressed (zipp...
1,222 KB
File name:
"U_RHEL_8_V1R8_STIG" "U_MS_Windows_10_V2R5_STIG"
STIG XML or zip files (*.xml), (*.
Open
Cancel


STIG Explorer
STIGs
Filter on STIG name...
CK
Name
Microsoft Windows 10 Security Technical Implementation Guide
Red Hat Enterprise Linux 8 Security Technical Implementation Guide


Vul ID
Rule ID
Rule Name
V-220697
SV-220697r8571...
SRG-OS-000480-...
V-220698
SV-220698r8571...
SRG-OS-000480-...
V-220699
SV-220699r5691...
SRG-OS-000480-...
V-220700
SV-220700r5691...
SRG-OS-000480-...
V-220701
SV-220701r7931...
SRG-OS-000191-...
V-220702
SV-220702r8592...
SRG-OS-000185-...
V-220703
SV-220703r8591...
SRG-OS-000185-...
V-220704
SV-220704r8592...
SRG-OS-000185-...
V-220705
SV-220705r8519...
SRG-OS-000370-...
Group Title:
Rule Title:
the following


Vul ID
Rule ID
Rule Name
V-220697 SV-220697r85... SRG-OS-00048...
V-220698 SV-220698r85... SRG-OS-00048...
V-220699 SV-220699r56... SRG-OS-00048...
V-220700 SV-220700r56... SRG-OS-00048...
V-220701 SV-220701r79... SRG-OS-00019...
V-220702 SV-220702r85... SRG-OS-00018...
V-220703 SV-220703r85... SRG-OS-00018...
V-220704 SV-220704r85... SRG-OS-00018...
V-220705 SV-220705r85... SRG-OS-00037...
V-220706 SV-220706r85... SRG-OS-00048...
V-220707 SV-220707r79... SRG-OS-00048...
V-220708 SV-220708r56... SRG-OS-00008...
V-220709 SV-220709r56... SRG-OS-00048...
V-220710 SV-220710r56... SRG-OS-00013...
V-220711 SV-220711r56... SRG-OS-00011...
V-220712 SV-220712r85... SRG-OS-00032...
V-220713 SV-220713r56... SRG-OS-00048...
V-220714 SV-220714r56... SRG-OS-00009...
V-220715 SV-220715r56... SRG-OS-00048...
V-220716 SV-220716r56... SRG-OS-00007...
V-220717 SV-220717r85... SRG-OS-00031...
V-220718 SV-220718r56... SRG-OS-00009...
V-220719 SV-220719r56... SRG-OS-00009...
V-220720 SV-220720r56... SRG-OS-00009...
V-220721 SV-220721r56... SRG-OS-00009...
V-220722 SV-220722r56... SRG-OS-00009...
V-220723 SV-220723r56... SRG-OS-00048...
V-220724 SV-220724r56... SRG-OS-00048...
V-220725 SV-220725r56... SRG-OS-00048...
Showing rule 5 out of 257
Microsoft Windows 10 Security Technical Implementation Guide :: Version 2, Release: 5 Benchmark Date: 14 Nov 2022
Vul ID: V-220701 Rule ID: SV-220701r793197_rule STIG ID: WN10-00-000025
Severity: CAT II Classification: Unclass Legacy IDs: V-63343; SV-77833
Group Title: SRG-OS-000191-GPOS-00080
Rule Title: Windows 10 must employ automated mechanisms to determine the state of system components with regard to flaw remediation using the following frequency: continuously, where ESS is used; 30 days, for any additional internal network scans not covered by ESS; and annually, for external scans by Computer Network Defense Service Provider (CNDSP).
Discussion: An approved tool for continuous network scanning must be installed and configured to run.
Without the use of automated mechanisms to scan for security flaws on a continuous and/or periodic basis, the operating system or other system components may remain vulnerable to the exploits presented by undetected software flaws.
To support this requirement, the operating system may have an integrated solution incorporating continuous scanning using ESS and periodic scanning using other tools, as specified in the requirement.
Check Text: Verify DoD-approved ESS software is installed and properly operating. Ask the site ISSM for documentation of the ESS software installation and configuration.
If the ISSM is not able to provide a documented configuration for an installed ESS or if the ESS software is not properly maintained or used, this is a finding.
Note: Example of documentation can be a copy of the site's CCB approved Software Baseline with version of software noted or a memo from the ISSM stating current ESS software and version.
Fix Text: Install DoD-approved ESS software and ensure it is operating continuously.
References
CCI: CCI-001233: The organization employs automated mechanisms on an organization-defined frequency to determine the state of information system components with regard to flaw remediation.
NIST SP 800-53 :: SI-2 (2)
NIST SP 800-53A :: SI-2 (2).1 (ii)
NIST SP 800-53 Revision 4 :: SI-2 (2)

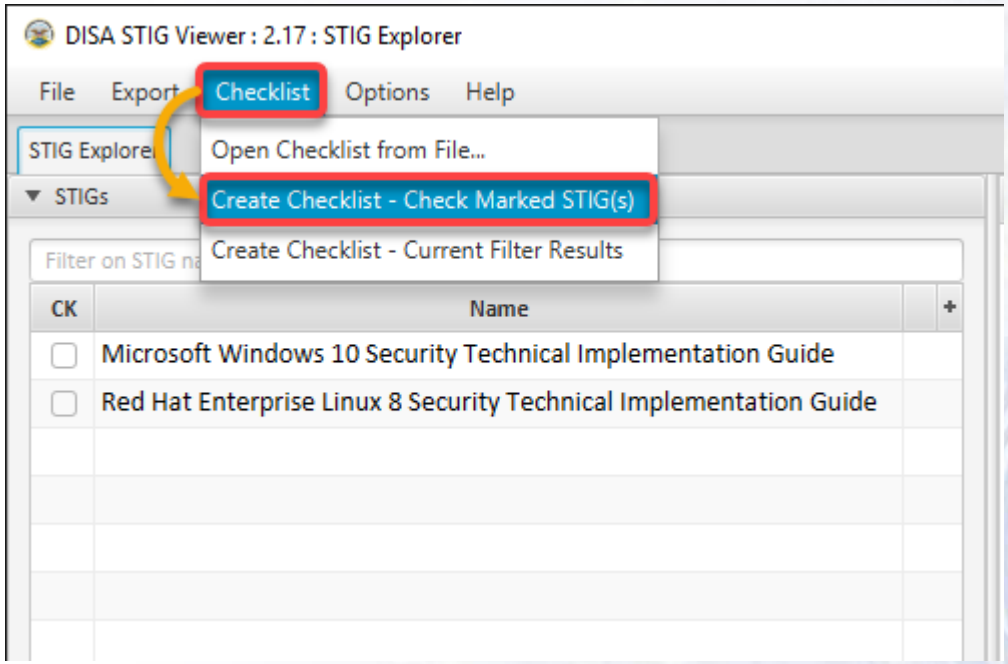
Microsoft Windows 10 Security Technical Implementation Guide :: Version 2, Release: 5 Benchmark Date: 14 Nov 2022

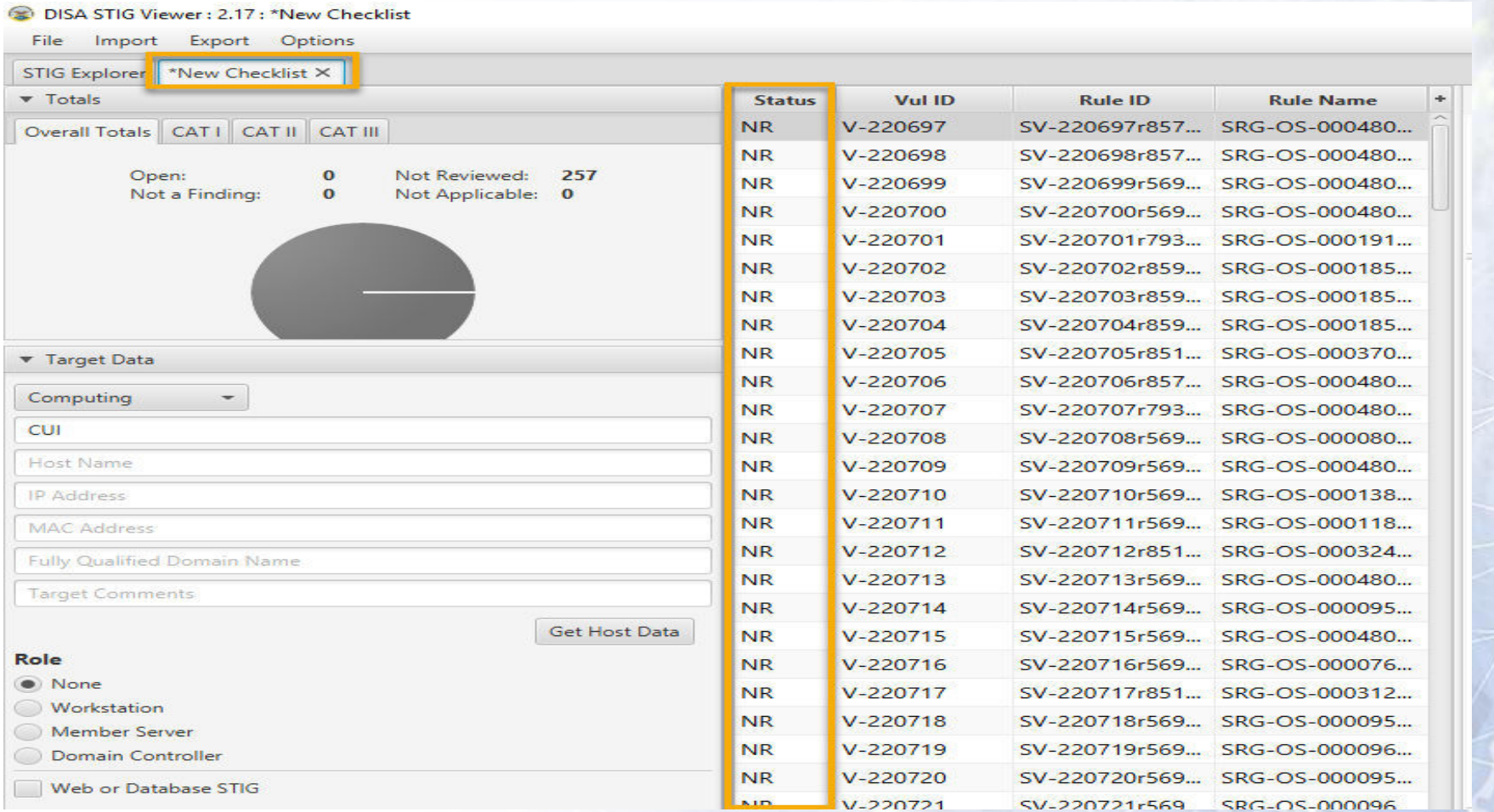
Vul ID: V-220701	**Rule ID**: SV-220701r793197_rule	**STIG ID**: WN10-00-000025
Severity: CAT II	**Classification**: Unclass	**Legacy IDs**: V-63343; SV-77833

Group Title: SRG-OS-000191-GPOS-00080

Rule Title: Windows 10 must employ automated mechanisms to determine the state of system components with regard to flaw remediation using the following frequency: continuously, where ESS is used; 30 days, for any additional internal network scans not covered by ESS; and annually, for external scans by Computer Network Defense Service Provider (CNDSP).

Discussion: An approved tool for continuous network scanning must be installed and configured to run.


DISA STIG Viewer : 2.17 : STIG Explorer
File
Export
Checklist
Options
Help
STIG Explorer
Open Checklist from File...
Create Checklist - Check Marked STIG(s)
Create Checklist - Current Filter Results
STIGs
Filter on STIG na
CK
Name
Microsoft Windows 10 Security Technical Implementation Guide
Red Hat Enterprise Linux 8 Security Technical Implementation Guide


DISA STIG Viewer : 2.17 : *New Checklist
File
Import
Export
Options
STIG Explorer
*New Checklist
Totals
Overall Totals
CAT I
CAT II
CAT III
Open: 0
Not a Finding: 0
Not Reviewed: 257
Not Applicable: 0
Target Data
Computing
CUI
Host Name
IP Address
MAC Address
Fully Qualified Domain Name
Target Comments
Get Host Data
Role
None
Workstation
Member Server
Domain Controller
Web or Database STIG
Status
Vul ID
Rule ID
Rule Name
NR V-220697 SV-220697r857... SRG-OS-000480...
NR V-220698 SV-220698r857... SRG-OS-000480...
NR V-220699 SV-220699r569... SRG-OS-000480...
NR V-220700 SV-220700r569... SRG-OS-000480...
NR V-220701 SV-220701r793... SRG-OS-000191...
NR V-220702 SV-220702r859... SRG-OS-000185...
NR V-220703 SV-220703r859... SRG-OS-000185...
NR V-220704 SV-220704r859... SRG-OS-000185...
NR V-220705 SV-220705r851... SRG-OS-000370...
NR V-220706 SV-220706r857... SRG-OS-000480...
NR V-220707 SV-220707r793... SRG-OS-000480...
NR V-220708 SV-220708r569... SRG-OS-000080...
NR V-220709 SV-220709r569... SRG-OS-000480...
NR V-220710 SV-220710r569... SRG-OS-000138...
NR V-220711 SV-220711r569... SRG-OS-000118...
NR V-220712 SV-220712r851... SRG-OS-000324...
NR V-220713 SV-220713r569... SRG-OS-000480...
NR V-220714 SV-220714r569... SRG-OS-000095...
NR V-220715 SV-220715r569... SRG-OS-000480...
NR V-220716 SV-220716r569... SRG-OS-000076...
NR V-220717 SV-220717r851... SRG-OS-000312...
NR V-220718 SV-220718r569... SRG-OS-000095...
NR V-220719 SV-220719r569... SRG-OS-000096...
NR V-220720 SV-220720r569... SRG-OS-000095...
NR V-220721 SV-220721r569 SRG-OS-000096

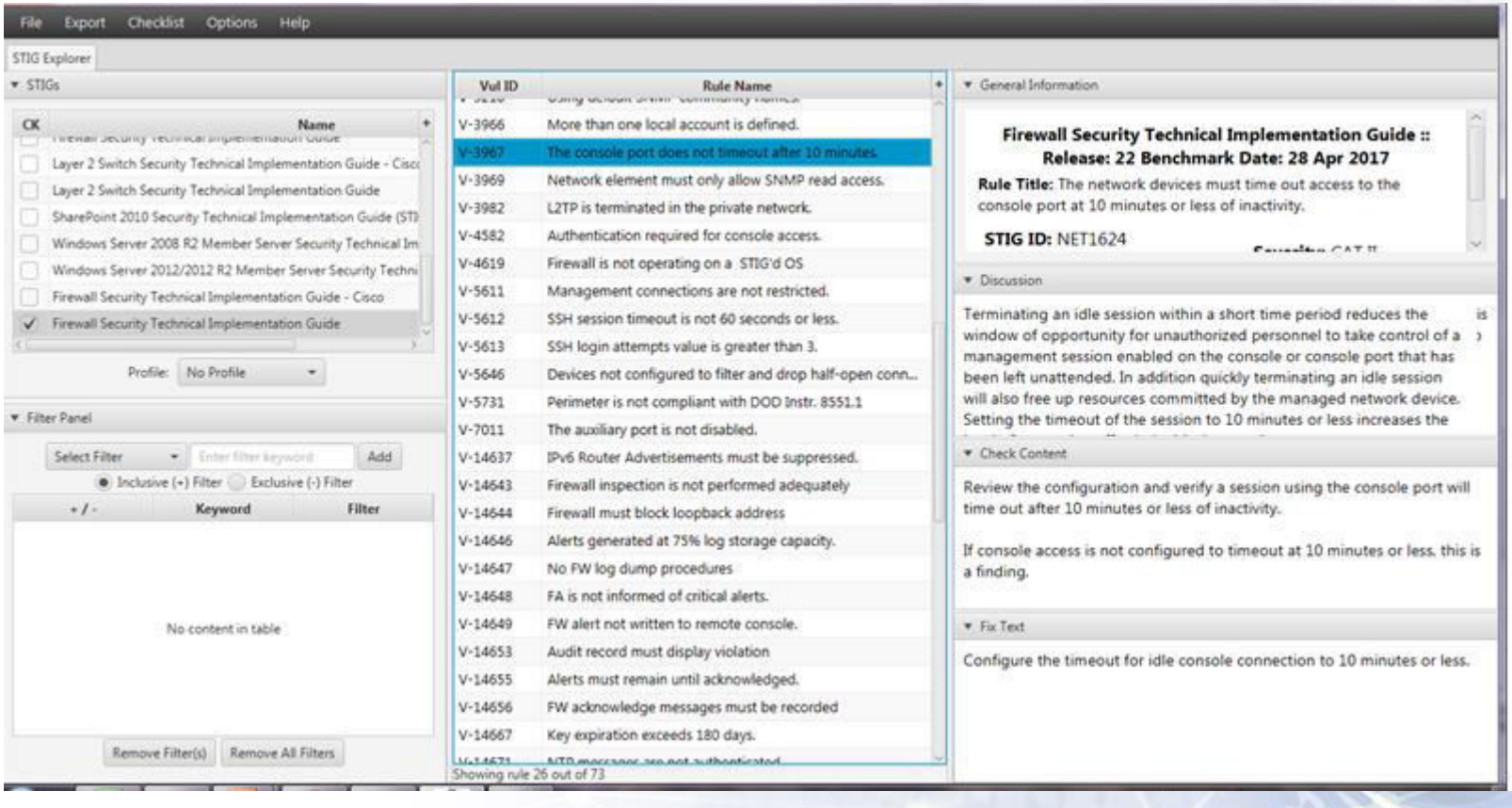
Week 8 Topic 10

A Look At DISA STIGs (4)

- Firewall Security Technical Implementation Guide.
- Vulnerability ID: V-3967
- Rule name: The console port does not timeout after 10 mins

A Look At DISA STIGs (4)

STIGVIEWER WINDOW



A Look At DISA STIGs (4)

- **General Information:**
  - **Rule Title:** The network devices must time out access to the console port at 10 minutes or less of inactivity
  - **STIG ID:** NET1624
  - **Severity:** CAT II

A Look At DISA STIGs (4)

- **Discussion:**
  - Terminating an idle session within a short time period reduces the window of opportunity for unauthorized personnel to take control of a management session enabled on the console or console.
  - port that has been left unattended. In addition quickly terminating an idle session will also free up resources committed by the managed network device. Setting the timeout of the session to 10 minutes
  - or less increases the level of protection afforded critical network components

A Look At DISA STIGs (4)

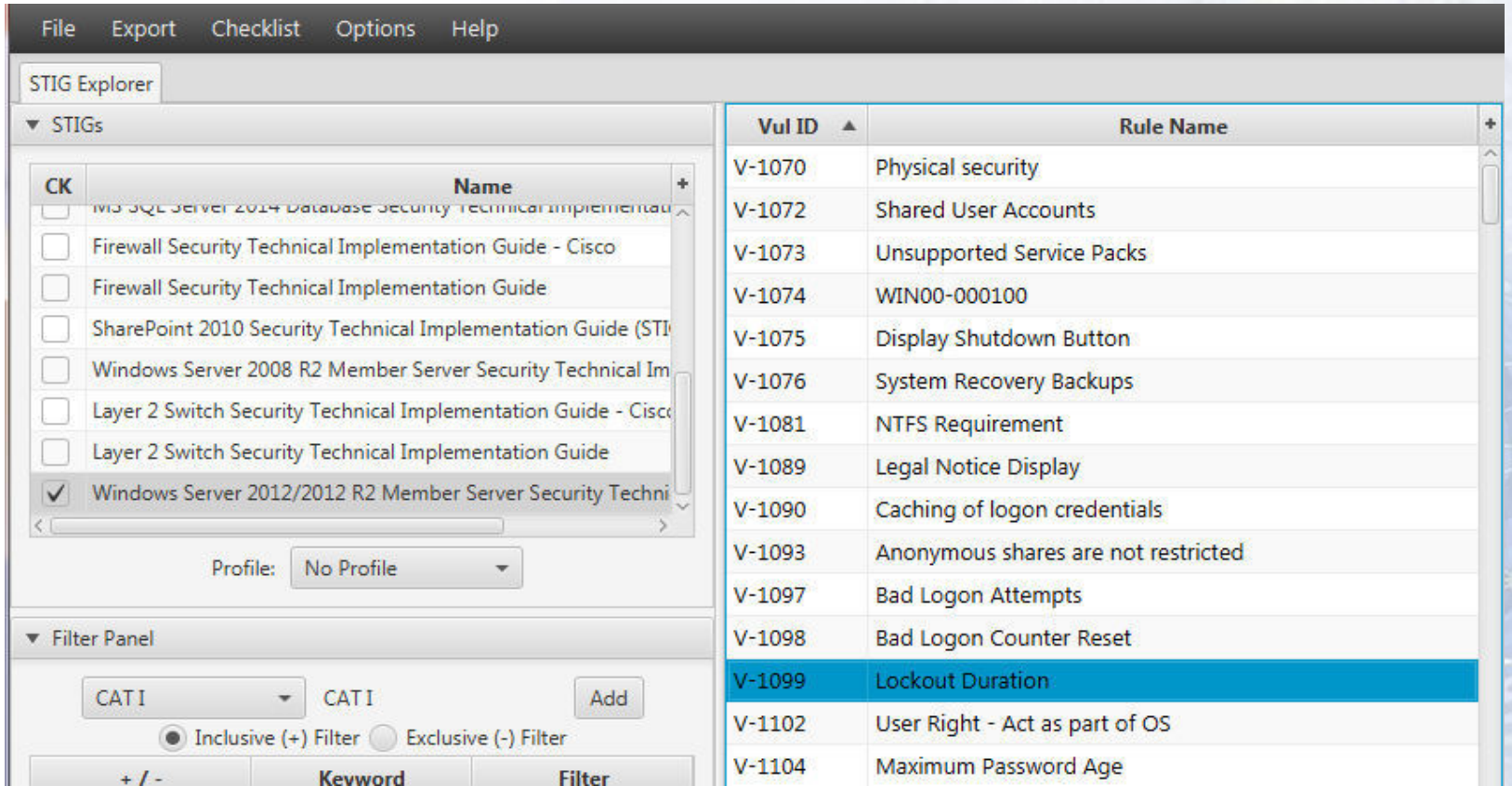
- **Check Content:**
  - Review the configuration and verify a session using the console port will time out after 10 mins or less of inactivity.
  - If console access is not configured to timeout at 10 minutes or less, this is a finding.

- **Fix Text:**
  - Configure the timeout for idle console connection to 10 minutes or less.

Week 8 Topic 12

A Look At DISA STIGs (3)

- Windows Server 2012 R2 Member Server
  - Import STIG
  - V1099 (Lockout duration)



A Look At DISA STIGs (2)

Vul ID	Rule Name
V-1070	Physical security
V-1072	Shared User Accounts
V-1073	Unsupported Service Packs
V-1074	WIN00-000100
V-1075	Display Shutdown Button
V-1076	System Recovery Backups
V-1081	NTFS Requirement
V-1089	Legal Notice Display
V-1090	Caching of logon credentials
V-1093	Anonymous shares are not restricted
V-1097	Bad Logon Attempts
V-1098	Bad Logon Counter Reset
V-1099	Lockout Duration
V-1102	User Right - Act as part of OS
V-1104	Maximum Password Age
V-1105	Minimum Password Age
V-1107	Password Uniqueness
V-1112	Dormant Accounts
V-1113	Disable Guest Account
V-1114	Rename Built-in Guest Account
V-1115	Rename Built-in Administrator Account

▼ General Information

Windows Server 2012/2012 R2 Member Server Security Technical Implementation Guide :: Release: 8 Benchmark Date: 28 Apr 2017

Rule Title: The lockout duration must be configured to require an

▼ Discussion

The account lockout feature, when enabled, prevents brute-force password attacks on the system. This parameter specifies the period of time that an account will remain locked after the specified number of failed logon attempts. A value of 0 will require an administrator to unlock the account.

▼ Check Content

Verify the effective setting in Local Group Policy Editor.
Run "gpedit.msc".

Navigate to Local Computer Policy -> Computer Configuration -> Windows Settings -> Security Settings -> Account Policies -> Account

▼ Fix Text

Configure the policy value for Computer Configuration -> Windows Settings -> Security Settings -> Account Policies -> Account Lockout Policy -> "Account lockout duration" to "0" minutes, "Account is locked out until administrator unlocks it".

▼ CCI

A Look At DISA STIGs (3)

- **Rule Title:**
  - The lockout duration must be configured to require an administrator to unlock an account
  - Severity: CAT II

- **Discussion:**
  - The account lockout feature, when enabled, prevents brute-force password attacks on the system. This parameter specifies the period of time that an account will remain locked after the specified number
  - of failed logon attempts. A value of 0 will require an administrator to unlock the account.

A Look At DISA STIGs (3)

- **Check Content:**
  - Verify the effective setting in Local Group Policy Editor.
    Run "gpedit.msc".
  - Navigate to Local Computer Policy -> Computer Configuration -> Windows Settings -> Security Settings -> Account Policies -> Account Lockout Policy.
  - If the "Account lockout duration" is not set to "0", requiring an administrator to unlock the account, this is a finding.

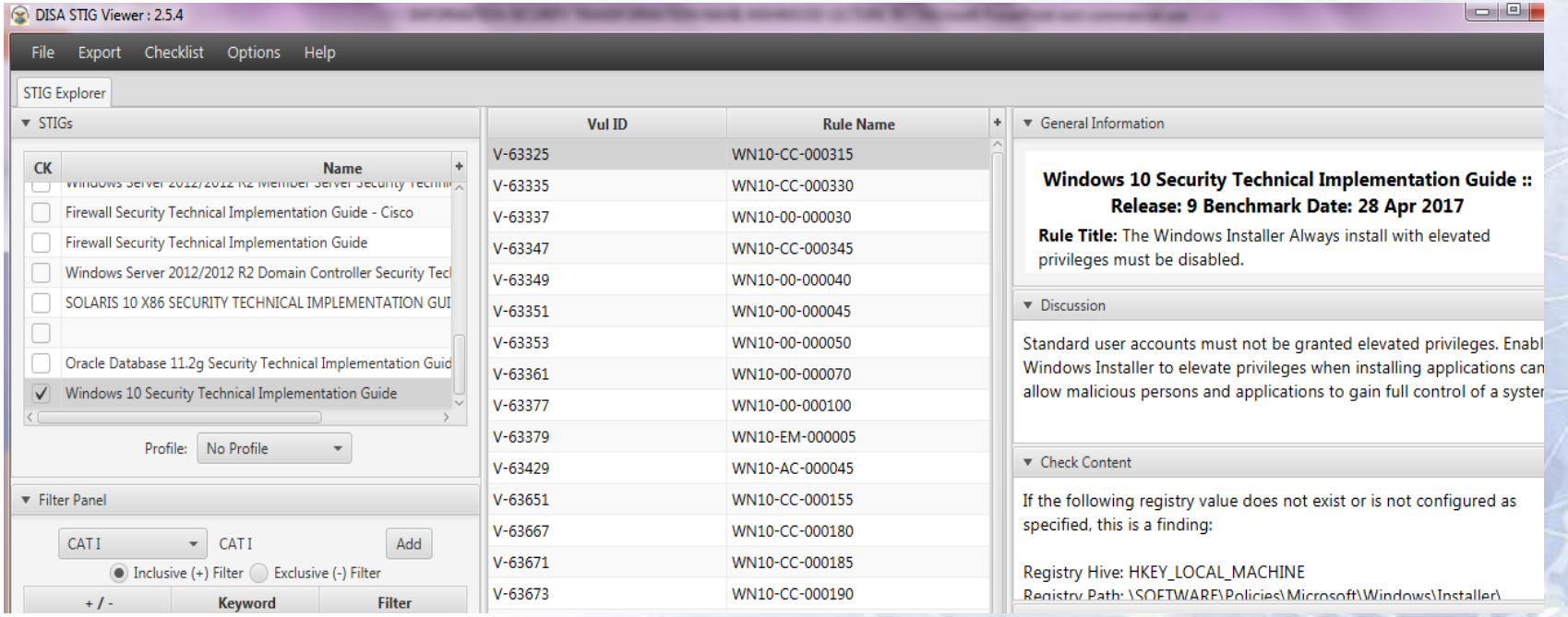
A Look At DISA STIGs (3)

- **Fix Text:**
  - Configure the policy value for Computer Configuration -> Windows Settings -> Security Settings -> Account Policies -> Account Lockout Policy -> "Account lockout duration" to "0" minutes,
  - "Account is locked out until administrator unlocks it".
- CCI: NIST SP 800-53 Revision 4 :: AC-7 b

Week 8 Topic 13

Security Hardening – Case Study – Win 10

- Windows 10
- DISA, Release 9
  - 28 April 2017



STIGVIEWER WINDOW

Security Hardening – Case Study – Win 10

- General Information:
  - **Rule Title:** The antivirus program must be configured to update signature files on a daily basis.
  - **STIG ID**: WN10-00-000046
  - **Severity**: CAT I

Security Hardening – Case Study – Win 10

- **Discussion:**
  - Virus scan programs are a primary line of defense against the introduction of viruses and malicious code that can destroy data and even render a computer inoperable. Using a virus scan program provides the ability to…
  - detect malicious code before extensive damage occurs. Updated virus scan data files help protect a system, as constantly changing malware is identified by the antivirus software vendors

Security Hardening – Case Study – Win 10

- **Check Content:**
  - This requirement is NA if McAfee VirusScan Enterprise (VSE) is used. It will be addressed with the corresponding McAfee VSE STIG.
  - Configurations will vary depending on the product.

- **Fix Text:**
  - Configure the antivirus program to update signature files at least daily. Ensure the updates are occurring on timely basis and are not more than a week old.

Security Hardening – Case Study – Win 10

- CCI (Control Correlation Identifier):
  - CCI: 000366
    The org implements the security config settings.
    NIST SP 800-53 :: CM-6 b
    NIST SP 800-53A :: CM-6.1 (iv)
    NIST SP 800-53 Revision 4 :: CM-6

END

Week 8 Topic 14

Comparison of CIS Vs DISA

- Many controls are common
- Approaches are different
- Organization styles are different

Comparison of CIS Vs DISA

FEATURE	CIS	DISA
CONTROL COVERAGE	GOOD	EXCELLENT
ORG SUITABILITY	SMALL AND MEDIUM ORGS	LARGE ORGS
USER FRIENDLINESS	GOOD	SATISFACTORY
UNUSABLE TERMINOLOGY	NO	YES
CONTROL DETAIL	GOOD	SATISFACTORY
TOOLS	CAT (COMMERCIAL)	SCAP (MILITARY USE)

Comparison of CIS Vs DISA

FEATURE	CIS	DISA
CONTROL PRIORITIZATION	LEVEL 1, LEVEL 2	CAT I - CAT III
TRACKING EASE	CAT TOOL (COMMERCIAL)	FREE STIG VIEWER (CHECKLIST)
FREQUENCY OF UPDATES	FAIR	QUARTERLY
INDUSTRY CREDIBILITY	HIGH	VERY HIGH
INDUSTRY ADOPTION	HIGH	MODERATE

Comparison of CIS Vs DISA

- **How to select CIS/DISA:**
  - Size of organization
  - IT infrastructure extent
  - Nature of business
  - Security program goals
  - Maturity of IT & security staff

- **Rule of thumb:**
  - Smaller orgs use CIS
  - Larger orgs use DISA
  - CIS is part of Homeland Security, DISA is part of US Military
  - DISA more frequently updated and maintained with wider coverage

Week 8 Topic 15

ISMS:Leading InfoSec Governance Framework

- ISO27001:2013 (ISMS)
  - Specifies the requirements for establishing, implementing, maintaining and continually improving an information security management system
  - Ten short clauses
  - Long annex

ISMS:Leading InfoSec Governance Framework

ISO27001:2013 MANDATORY CLAUSES

Reference		Description
Mandatory	Clause 4	Context of the organization
	Clause 5	Leadership
	Clause 6	Planning
	Clause 7	Support
	Clause 8	Operation
	Clause 9	Performance evaluation
	Clause 10	Improvement

https://chapters.theiia.org/bermuda/Events/ChapterDocuments/Information%20Security%20Management%20System%20%28ISMS%29%20Overview.pdf

ISMS:Leading InfoSec Governance Framework

ISO27001:2013 DISCRETIONARY CONTROLS

Reference		Description	Control Total
Discretionary	A5	Information security policies	2
	A6	Organization of information security	7
	A7	Human resource security	6
	A8	Asset management	10
	A9	Access control	13
	A10	Cryptography	2
	A11	Physical and environmental security	15
	A12	Operations security	14
	A13	Communications security	7
	A14	System acquisition, development and maintenance	13
	A15	Supplier relationships	5
	A16	Information security incident management	7
	A17	Information security aspects of business continuity management	4
	A18	Compliance	8

TOTAL: 113

https://chapters.theiia.org/bermuda/Events/ChapterDocuments/Information%20Security%20Management%20System%20%28ISMS%29%20Overview.pdf

ISMS:Leading InfoSec Governance Framework

- **Merits of ISO27001:2013 (ISMS):**
  - Exceptional framework with comprehensive coverage of mandatory requirements (clauses 4-10) and discretionary controls (annex)
  - Highly beneficial as a framework for security program
  - Covers all domain areas
  - Provides a structure and organized sequence for security controls
  - Complements security transformation model as serves as a reference and guideline for activities and controls

ISMS:Leading InfoSec Governance Framework

- **De-merits** of ISO27001:2013 (ISMS):
  - Very broad
  - Generic framework – leaves it to organization how to implement the measures and controls
  - Not suited for orgs that are new to security program

- **How to best use advantages of ISO27001:2013 (ISMS):**
  - **Implement security transformational model**
  - **Cap off security transformation project with ISO27001:2013 (ISMS) certification**
  - **ISMS as a complementary reference and checklist rather than main framework**

Week 8 Topic 16

How to Use ISO27002:2013

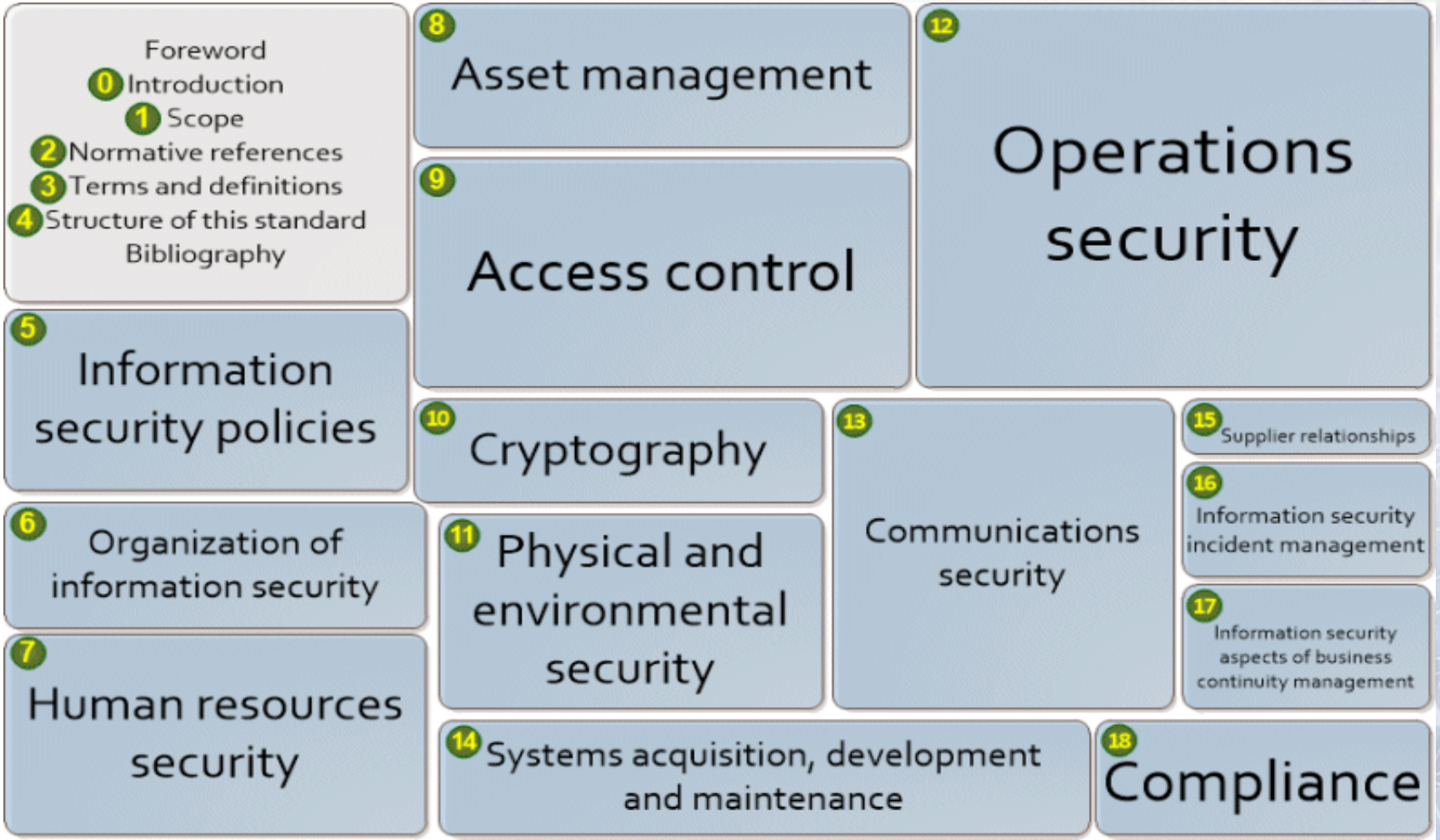
- What is ISO27002:2013 ?
  - Information technology -- Security techniques -- Code of practice for information security controls
  - Renamed from ISO 17799

0.1 Background & Context

- This Int'l Standard is designed for orgs to use as a **reference** for selecting controls within the process of implementing an Information Security Management System (ISMS) based on ISO/IEC 27001;
- or as a **guidance document** for organizations implementing commonly accepted information security controls.

How to Use ISO27002:2013

STRUCTURE OF ISO27002:2013



http://www.iso27001security.com/html/27002.html

How to Use ISO27002:2013

- Lets have a look at control A.5.1.2 (**Review Of The Policies Of Information Security**)

A.5.1.2	REVIEW OF THE POLICIES FOR INFORMATION SECURITY	***Control:*** The policies for information security shall be reviewed at planned intervals or if significant changes occur to ensure their continuing suitability, adequacy and effectiveness.

ISO27001:2013

How to Use ISO27002:2013

ISO27002:2013

TITLE

5.1.2 Review of the policies for information security

Control

The policies for information security should be reviewed at planned intervals or if significant changes occur to ensure their continuing suitability, adequacy and effectiveness.

Implementation guidance

Each policy should have an owner who has approved management responsibility for the development, review and evaluation of the policies. The review should include assessing opportunities for improvement of the organization's policies and approach to managing information security in response to changes to the organizational environment, business circumstances, legal conditions or technical environment.

How to Use ISO27002:2013

ISO27002:2013

CONTROL

5.1.2 Review of the policies for information security

Control

The policies for information security should be reviewed at planned intervals or if significant changes occur to ensure their continuing suitability, adequacy and effectiveness.

Implementation guidance

Each policy should have an owner who has approved management responsibility for the development, review and evaluation of the policies. The review should include assessing opportunities for improvement of the organization's policies and approach to managing information security in response to changes to the organizational environment, business circumstances, legal conditions or technical environment.

How to Use ISO27002:2013

ISO27002:2013

5.1.2 Review of the policies for information security

Control

The policies for information security should be reviewed at planned intervals or if significant changes occur to ensure their continuing suitability, adequacy and effectiveness.

Implementation guidance

Each policy should have an owner who has approved management responsibility for the development, review and evaluation of the policies. The review should include assessing opportunities for improvement of the organization's policies and approach to managing information security in response to changes to the organizational environment, business circumstances, legal conditions or technical environment.

27002 IMPLEMENTATION GUIDANCE

How to Use ISO27002:2013

ISO27002:2013

A.5.1.2	REVIEW OF THE POLICIES FOR INFOSEC	***27002: Implementation Guidance:*** Each policy should have an owner who has approved mngmt responsibility for the development, review and evaluation of the policies. The review should include assessing opportunities for improvement of the org's policies and approach to managing Infosec in response to changes to the org environment, business circumstances, legal conditions or tech environment.

How to Use ISO27002:2013

- Practically:
  - ISO27001:2013 controls are brief and generic
  - **27002 clarifies further what is being referred to, gives further context & very useful implementation guidance**

END

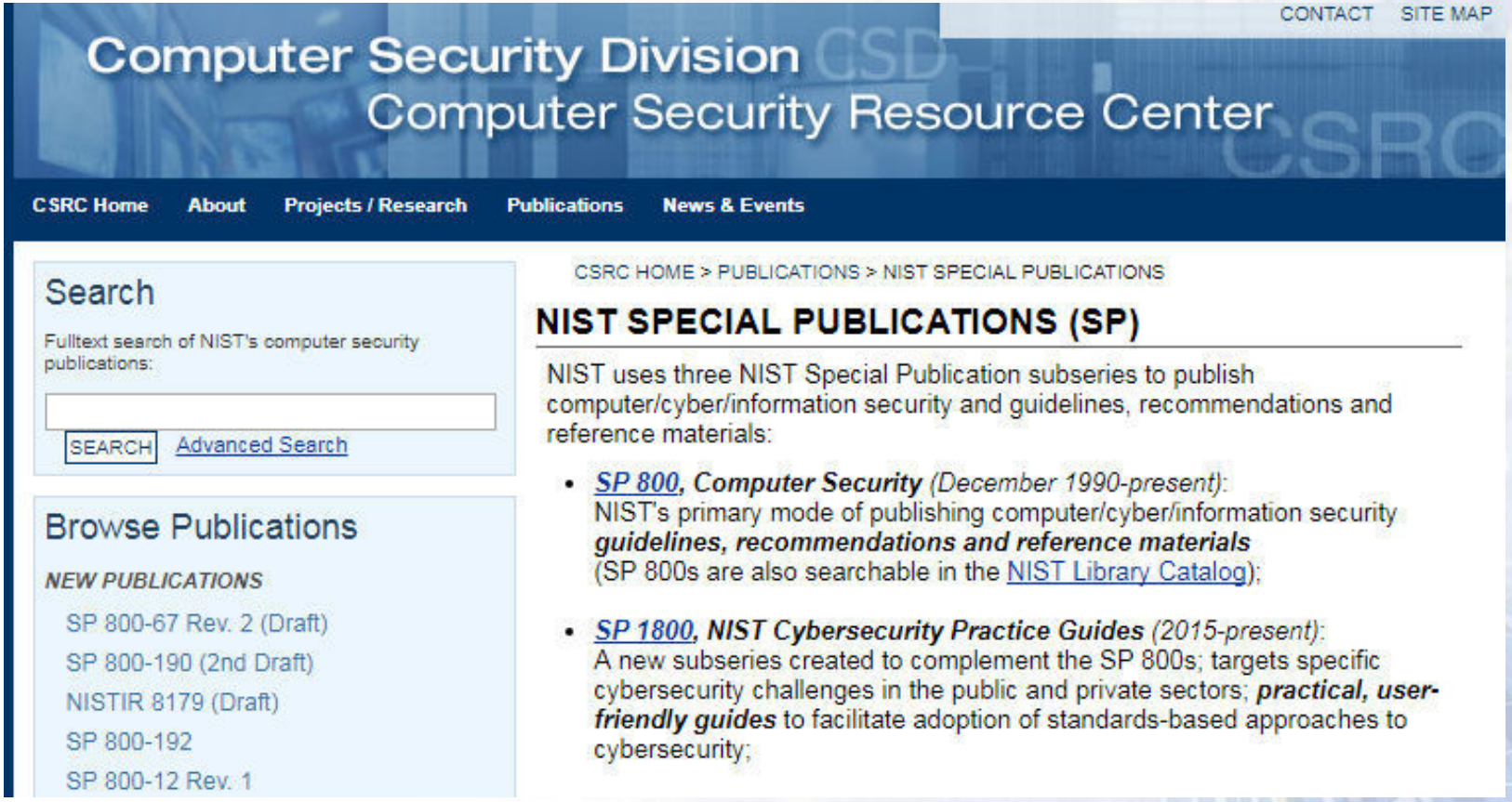
Week 8 Topic 17

NIST FRAMEWORK

- The Computer Security Resource Center (CSRC) website guides users to NIST resources on **computer, cyber, and information security and privacy.**

- Its content includes **publications, projects, research, news and events** from the NIST Information Technology Laboratory's (ITL) two security divisions

NIST FRAMEWORK



- ***SP 800*, Computer Security** (*December 1990-present*):
  NIST's primary mode of publishing computer/cyber/information security ***guidelines, recommendations and reference materials***
  (SP 800s are also searchable in the NIST Library Catalog);

NIST FRAMEWORK

PUBLICATIONS BY TOPIC/PROJECT

Browse FIPS, Special Publications, NISTIRs and ITL Bulletins ***by topic or project***:

Security Concepts
General IT Security
Audit & Accountability
Authentication
Awareness & Training
Certification & Accreditation
Contingency Planning
Cryptography
Digital Signatures
Incident Response
Maintenance
Personal Identity Verification (PIV)
Planning
Privacy
Public Key Infrastructure (PKI)
Research
Risk Assessment
Security Automation
Services & Acquisition
Threats & Vulnerability Management
Usability

Technologies
Biometrics
Cloud Computing & Virtualization
Communications & Wireless
Mobile
Smart Cards

Applications
Cyber-Physical Systems / Smart Grid
Forensics
Healthcare
Internet of Things (IoT)
Public Safety
Supply Chain
Voting

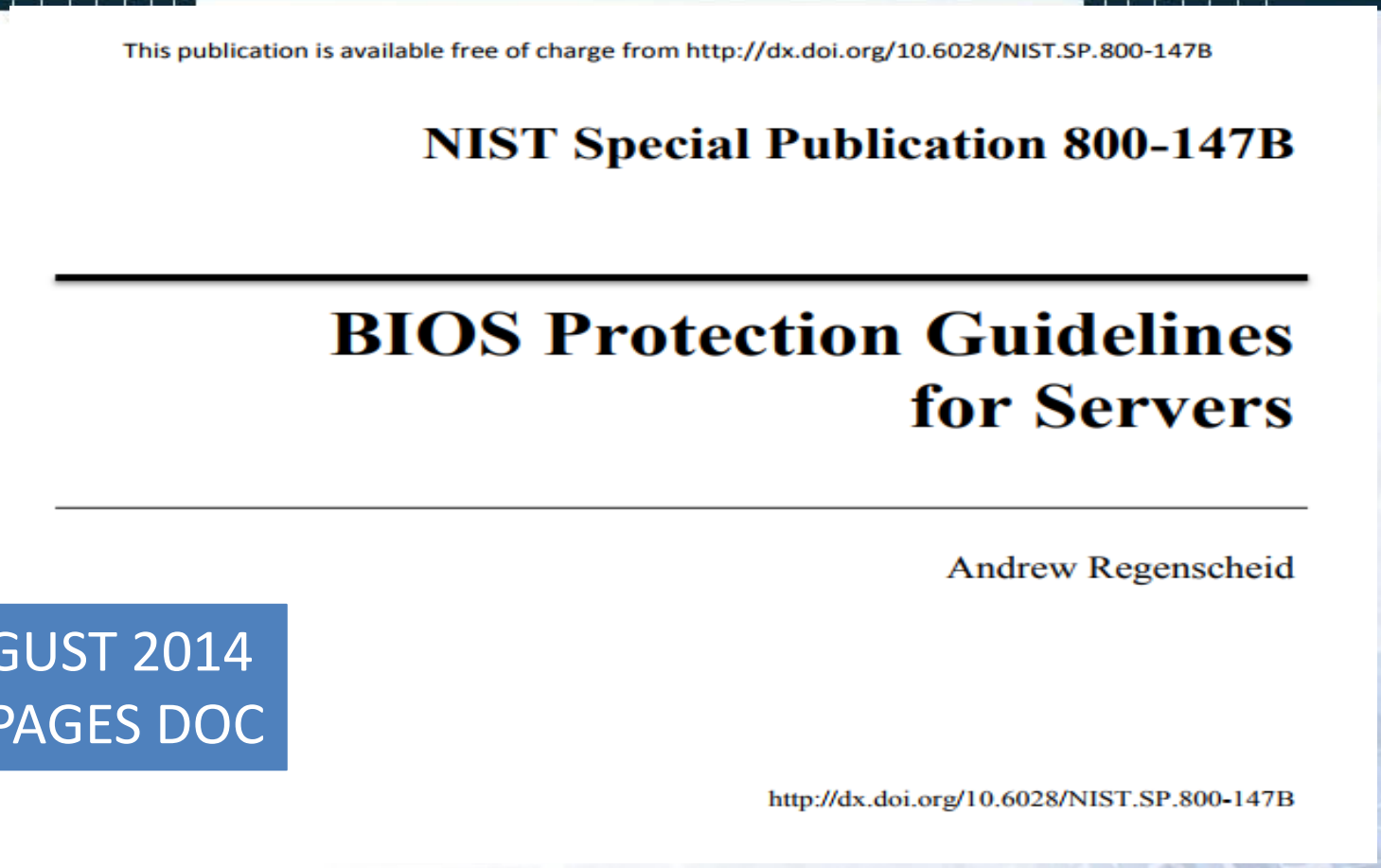
Activities
Annual Reports
Conferences & Workshops

NIST FRAMEWORK

SP 800s - Computer Security

Number	Date	Title
SP 800-193 (Draft)	May 2017	**DRAFT Platform Firmware Resiliency Guidelines** Announcement and Draft Publication
SP 800-192	June 2017	**Verification and Test Methods for Access Control Policies/Models** SP 800-192 FAQ doi:10.6028/NIST.SP.800-192 [Direct Link]
SP 800-190 (Draft)	July 2017	**DRAFT Application Container Security Guide (2nd Draft)** Announcement and Draft Publication
SP 800-188 (Draft)	December 2016	**DRAFT De-Identifying Government Datasets (2nd Draft)** Announcement and Draft Publication
SP 800-187 (Draft)	November 2016	**DRAFT Guide to LTE Security** Announcement and Draft Publication
SP 800-185	December 2016	**SHA-3 Derived Functions: cSHAKE, KMAC, TupleHash, and ParallelHash** SP 800-185 FAQ doi:10.6028/NIST.SP.800-185 [Direct Link] Comments Received on Draft SP 800-185
SP 800-184	December	**Guide for Cybersecurity Event Recovery**

NIST FRAMEWORK



AUGUST 2014
32 PAGES DOC

- NIST has a tremendous library of free documentation on a diverse range of topics
- **Relevance is often average, however, depth and detail of material is extra-ordinary**

Week 8 Topic 18

PCI DSS V3

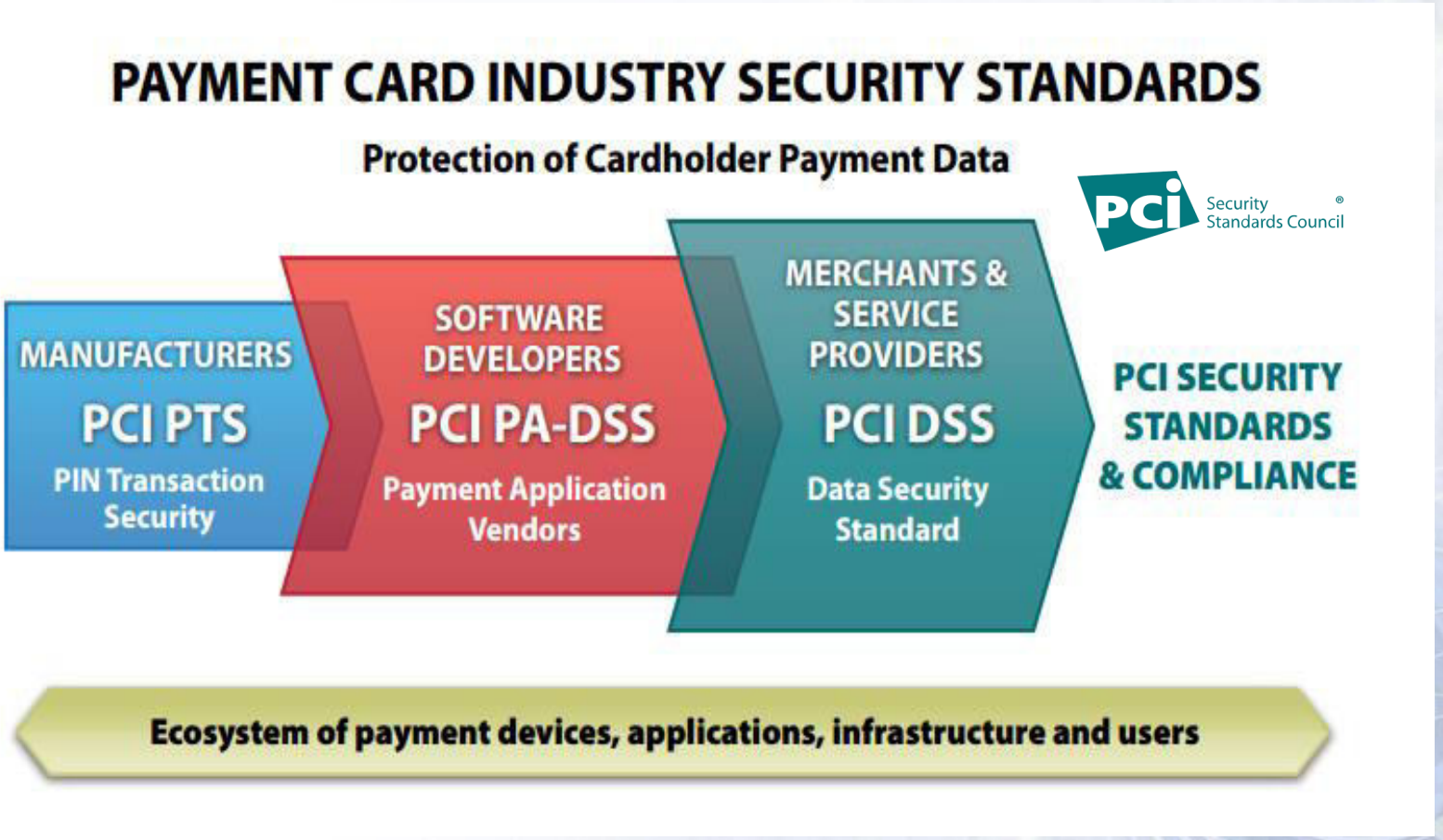
- PCI Data Security Standard (DSS):
  - Designed to ensure that ALL companies that accept, process, store or transmit credit card info maintain a secure environment
  - Managed by Security Standards Council

- PCI DSS:
  - SSC is an independent body that was created by the major payment card brands (Visa, MasterCard, American Express, Discover and JCB
  - 6 Broad goals and 12 requirements

https://www.pcicomplianceguide.org/pci-faqs-2/

REF: PCI Best Practices For Implementing Security Awareness; https://www.pcisecuritystandards.org/documents/

PCI DSS V3



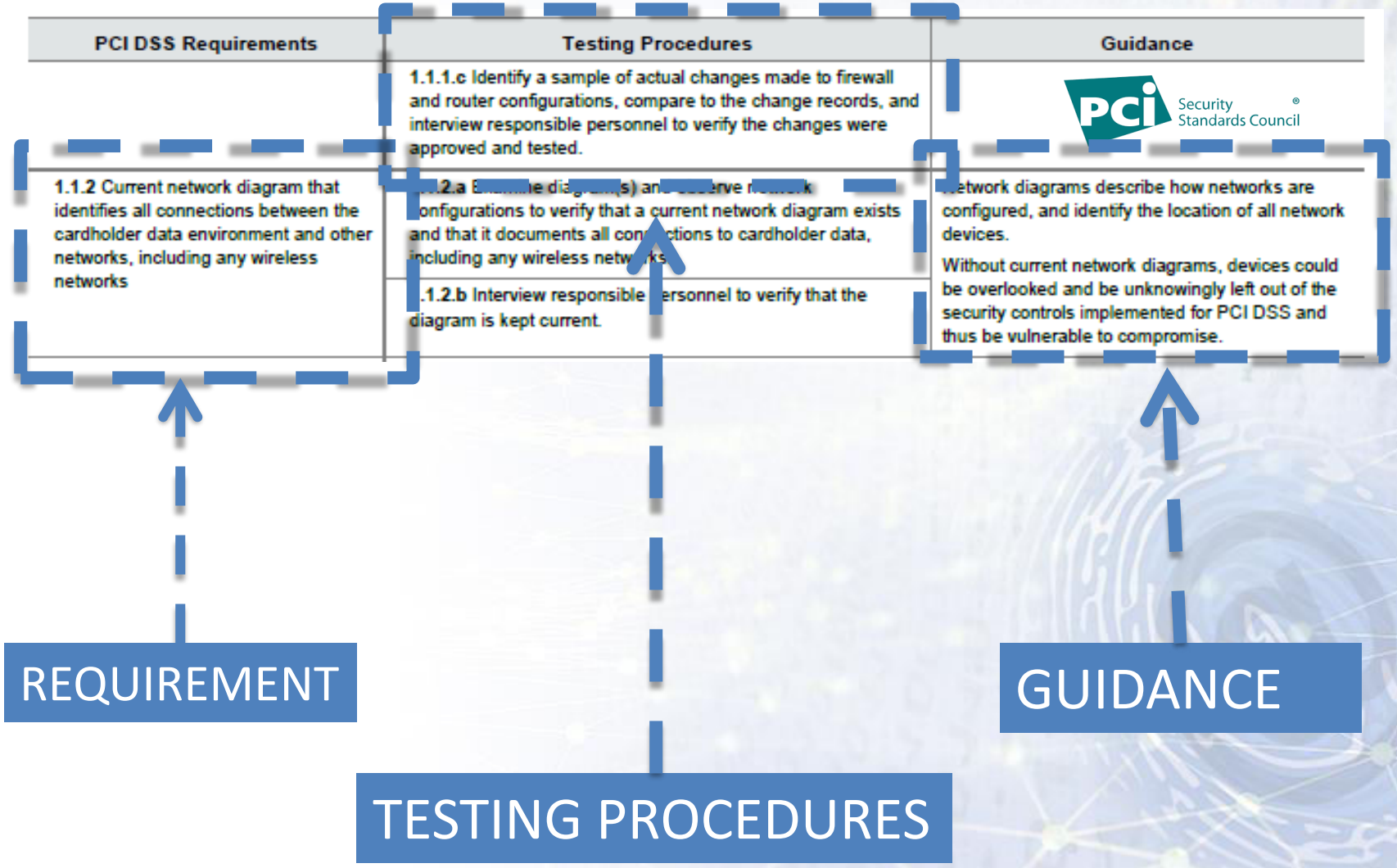
https://www.pcisecuritystandards.org/documents/PCI%20SSC%20Quick%20Reference%20Guide.pdf

PCI DSS V3

Goals	PCI DSS Requirements
Build and Maintain a Secure Network and Systems	1. Install and maintain a firewall configuration to protect cardholder data 2. Do not use vendor-supplied defaults for system passwords and other security parameters
Protect Cardholder Data	3. Protect stored cardholder data 4. Encrypt transmission of cardholder data across open, public networks
Maintain a Vulnerability Management Program	5. Protect all systems against malware and regularly update anti-virus software or programs 6. Develop and maintain secure systems and applications
Implement Strong Access Control Measures	7. Restrict access to cardholder data by business need to know 8. Identify and authenticate access to system components 9. Restrict physical access to cardholder data
Regularly Monitor and Test Networks	10. Track and monitor all access to network resources and cardholder data 11. Regularly test security systems and processes
Maintain an Information Security Policy	12. Maintain a policy that addresses information security for all personnel

https://www.pcisecuritystandards.org/documents/PCI%20SSC%20Quick%20Reference%20Guide.pdf

PCI DSS V3


PCI DSS Requirements
Testing Procedures
Guidance
1.1.1.c Identify a sample of actual changes made to firewall and router configurations, compare to the change records, and interview responsible personnel to verify the changes were approved and tested.
PCI Security Standards Council
1.1.2 Current network diagram that identifies all connections between the cardholder data environment and other networks, including any wireless networks
onfigurations to verify that a current network diagram exists and that it documents all con ctions to cardholder data, ncluding any wireless net ks
etwork diagrams describe how networks are configured, and identify the location of all network devices.
Without current network diagrams, devices could be overlooked and be unknowingly left out of the security controls implemented for PCI DSS and thus be vulnerable to compromise.
.1.2.b Interview responsible ersonnel to verify that the diagram is kept current.
REQUIREMENT
TESTING PROCEDURES
GUIDANCE

PCI DSS V3

REQMT	TEST PROCEDURES	GUIDANCE
7.1.4 Require documented approval by authorized parties specifying required privileges.	**7.1.4** Select a sample of user IDs & compare with documented approvals to verify that: -Documented approval exists for the assigned privileges -The approval was by authorized parties -That specified privileges match the roles assigned to the individual.	Documented approval (for example, in writing or electronically) assures that those with access and privileges are known and authorized by management, and that their access is necessary for their job function.

PCI DSS V3

REQMT	TEST PROCEDURES	GUIDANCE
8.1.4 Remove/ disable inactive user accounts within 90 days.	**8.1.4** Observe user accounts to verify that any inactive accounts over 90 days old are either removed or disabled.	Accounts that are not used regularly are often targets of attack since it is less likely that any changes (such as a changed password) will be noticed. As such, these accounts may be more easily exploited and used to access cardholder data.

PCI DSS V3

- PCI is specific to the card environment to protect cardholder data
- PCI controls are very specific and in-depth compared to generic and high-level controls of ISO27001

END

Week 8 Topic 19

COBIT

- ISACA framework for IT Governance

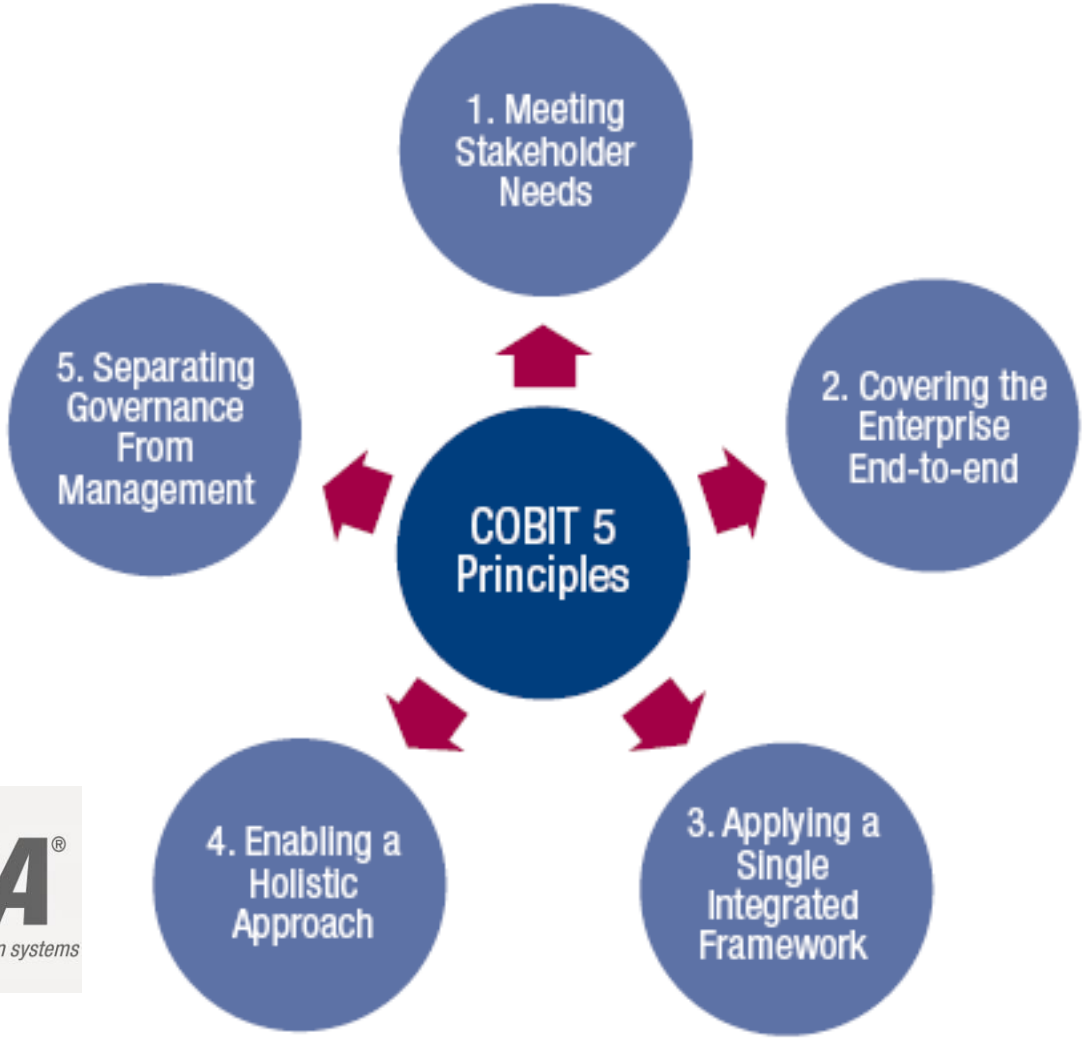
- COBIT 5 helps enterprises to create optimal value from IT by maintaining a balance between realising benefits and optimising risk levels and resource use (ISACA)

COBIT

- COBIT 5 brings together **five principles** that allow the enterprise to build an effective governance and management framework (ISACA)
- Based on a holistic set of **seven enablers** that optimises IT investment and use for the benefit of stakeholders (ISACA)

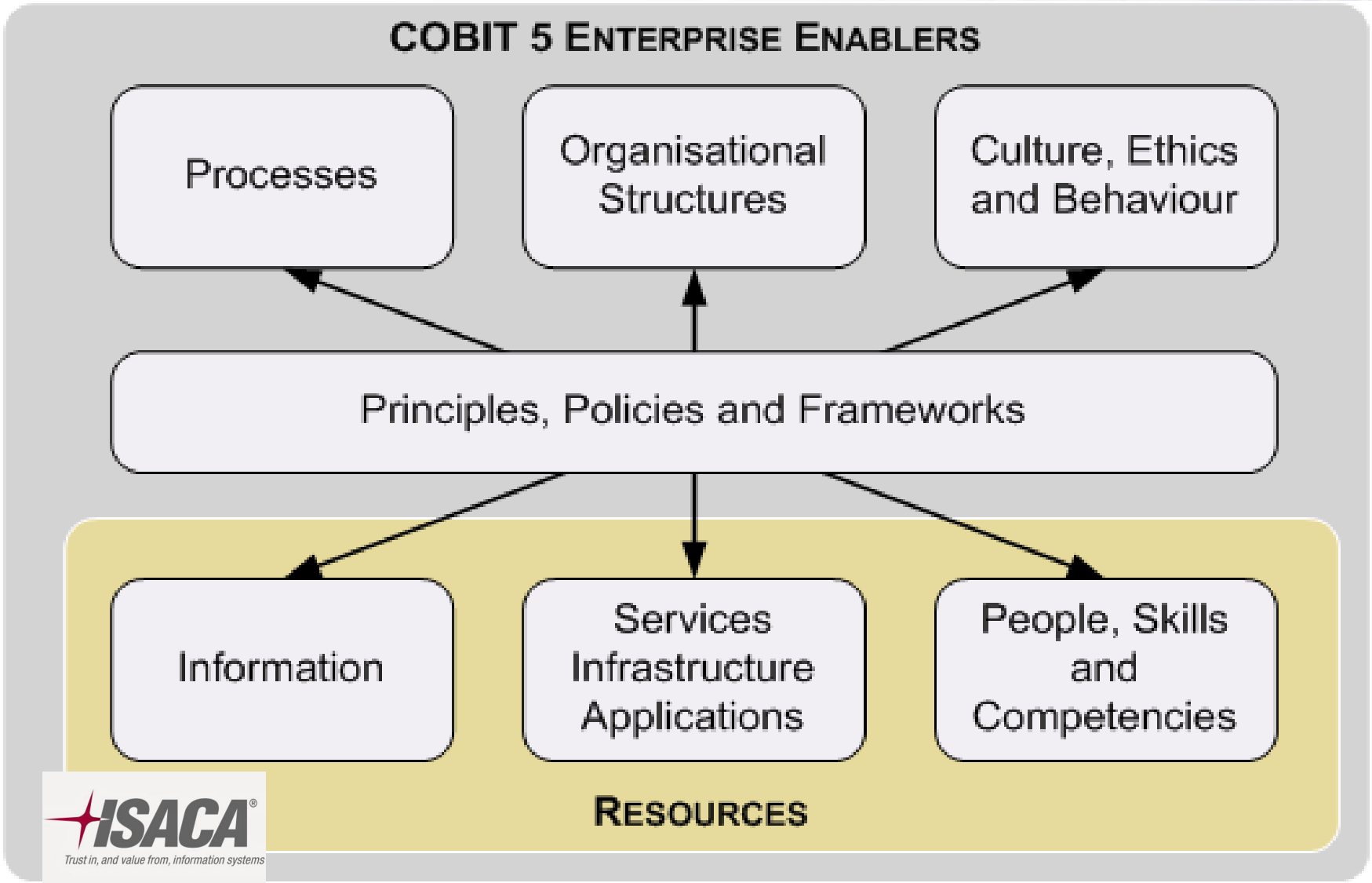
ISACA®
Trust in, and value from, information systems

COBIT



ISACA®
Trust in, and value from, information systems

COBIT


COBIT 5 ENTERPRISE ENABLERS
Processes
Organisational Structures
Culture, Ethics and Behaviour
Principles, Policies and Frameworks
Information
Services Infrastructure Applications
People, Skills and Competencies
RESOURCES

COBIT

- **Governance** ensures that enterprise objectives are achieved by **evaluating** stakeholder needs, conditions & options; setting **direction** through prioritisation & decision making;

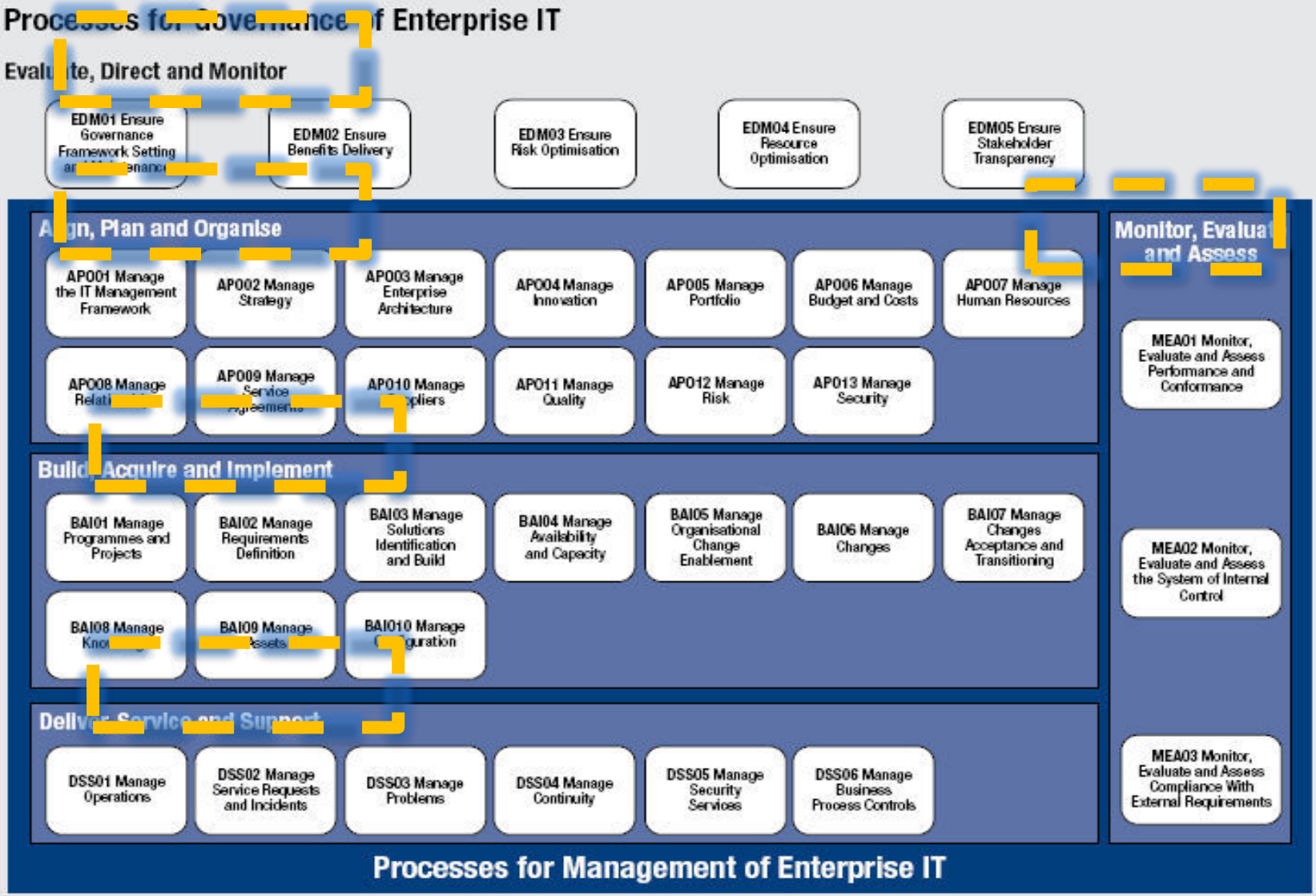
- …& **monitoring** performance, compliance and progress against agreed direction and objectives **(EDM)**

- **Management plans, builds, runs** and **monitors** activities in alignment with the direction set by the governance body to achieve the enterprise objectives **(PBRM)**

COBIT 5
AN ISACA® FRAMEWORK

COBIT

COBIT 5
AN ISACA® FRAMEWORK



COBIT

- COBIT 5 is a detailed framework for IT governance developed by ISACA which has principles, enablers, and processes
- These tools assist implementers and customer organizations to successfully deploy the framework
- Certifiable framework

END

Week No. 9

Malware

Week 9 Topic 1

Malware Introduction

Malware

- Malicious software

- Rogue software

- Rogue
  - Large, destructive, anomalous or unpredictable
  - Operating outside normal or desirable control

Week 9 Topic 2

Malware Description

- Malware (a portmanteau for malicious software) is any software intentionally designed to cause disruption to a computer, server, client, or computer network, leak private information, gain unauthorized access to information or systems, deprive access to information, or which unknowingly interferes with the user's computer security and privacy.
- Researchers tend to classify malware into one or more sub-types (i.e. computer viruses, worms, Trojan horses, ransomware, spyware, adware, rogue software, wiper and keyloggers).

- Malware poses serious problems to individuals and businesses on the Internet.
- According to Symantec's 2018 Internet Security Threat Report (ISTR), malware variants number has increased to 669,947,865 in 2017, which is twice as many malware variants as in 2016.

Malware Description

- Cybercrime, which includes malware attacks as well as other crimes committed by computer, was predicted to cost the world economy $6 trillion USD in 2021, and is increasing at a rate of 15% per year.
- Since 2021, malware has been designed to target computer systems that run critical infrastructure such as the electricity distribution network.
- The defense strategies against malware differ according to the type of malware but most can be thwarted by installing antivirus software, firewalls, applying regular patches to reduce zero-day attacks, securing networks from intrusion, having regular backups and isolating infected systems.
- Malware is now being designed to evade antivirus software detection algorithms

Week 9 Topic 3

Malware Types

- Virus
- Worm
- Trojan
- Logic bomb
- Time bomb
- Drive-by download
- Backdoor
- PUPs(Potentially Unwanted Program)

Spyware

- Adware
- Scareware
- Ransomware
- FakeAV/ Rogue security software
- Malvertising
- APT
- Fileless malwarez

Week 9 Topic 4

Writing Malware

Software versus Malware

- Writing a typical software
  - Software application (word processor, spreadsheet, graphics)
  - Higher-level programming language
  - Usually independent of OS
- Writing a virus
  - Detailed knowledge of OS

Week 9 Topic 5

Computer Virus

- Biological virus
  - Latin → Poison
  - Shell filled with genetic material that it injects into a living cell, thereby infecting it
  - The cell then starts manufacturing copies of virus

- Computer virus behaves similarly
  - Injects its contents, a short computer program /instructions, into a host computer and infecting it
  - When the computer executes the virus code, it replicates the code, performs some task(s), usually of a malicious nature

Computer Virus

- A computer virus is a type of computer program that, when executed, replicates itself by modifying other computer programs and inserting its own code into those programs.
- Computer viruses generally require a host program.
- The virus writes its own code into the host program.
- When the program runs, the written virus program is executed first, causing infection and damage.
- By contrast, a computer worm does not need a host program, as it is an independent program or code chunk.
- Therefore, it is not restricted by the host program, but can run independently and actively carry out attacks
- Any virus will by definition make unauthorised changes to a computer, which is undesirable even if no damage is done or intended.
- The first page of Dr Solomon's Virus Encyclopaedia explains the undesirability of viruses, even those that do nothing but reproduce.

Week 9 Topic 6

Computer Virus History-I

- The first academic work on the theory of self-replicating computer programs was done in 1949 by John von Neumann who gave lectures at the University of Illinois about the "Theory and Organization of Complicated Automata".
- The work of von Neumann was later published as the "Theory of self reproducing automata".
- In his essay von Neumann described how a computer program could be designed to reproduce itself.
- Von Neumann's design for a self-reproducing computer program is considered the world's first computer virus, and he is considered to be the theoretical "father" of computer virology.

Computer Virus History-I

- In 1972, Veith Risak directly building on von Neumann's work on self replication, published his article "Selbstreproduzierende Automaten mit minimaler Informationsübertragung" (Self-reproducing automata with minimal information exchange).
- The article describes a fully functional virus written in assembler programming language for a SIEMENS 4004/35 computer system.
- In 1980 Jürgen Kraus wrote his diplom thesis "Selbstreproduktion bei Programmen" (Self-reproduction of programs) at the University of Dortmund.
- In his work Kraus postulated that computer programs can behave in a way similar to biological viruses.

Week 9 Topic 7

Computer Virus History-II

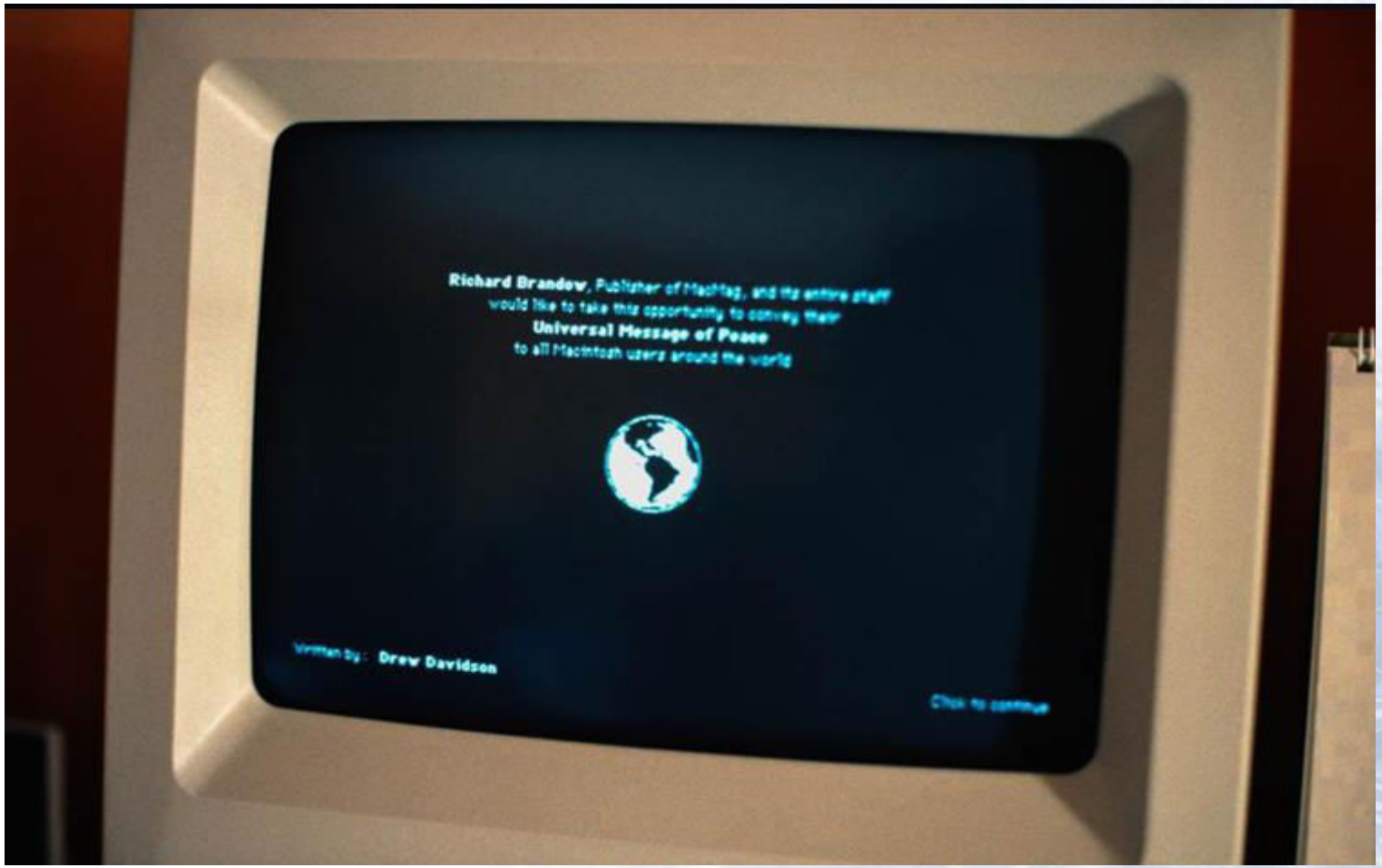
Creeper / Reaper

- The Creeper virus was first detected on ARPANET, the forerunner of the Internet, in the early 1970s.
- Creeper was an experimental self-replicating program written by
- Bob Thomas at BBN Technologies in 1971.
- Creeper used the ARPANET to infect DEC PDP-10 computers running the TENEX operating system.
- Creeper gained access via the ARPANET and copied itself to the
- remote system where the message, "I'M THE CREEPER. CATCH ME
- IF YOU CAN!" was displayed.
- The Reaper program was created to delete Creeper.

- In 1984 Fred Cohen from the University of Southern California wrote his paper "Computer Viruses – Theory and Experiments".

- It was the first paper to explicitly call a self-reproducing program a "virus", a term introduced by Cohen's mentor Leonard Adleman

This is the MacMag virus 'Universal Peace' message, as displayed on my Mac in March of 1988.



Brain virus

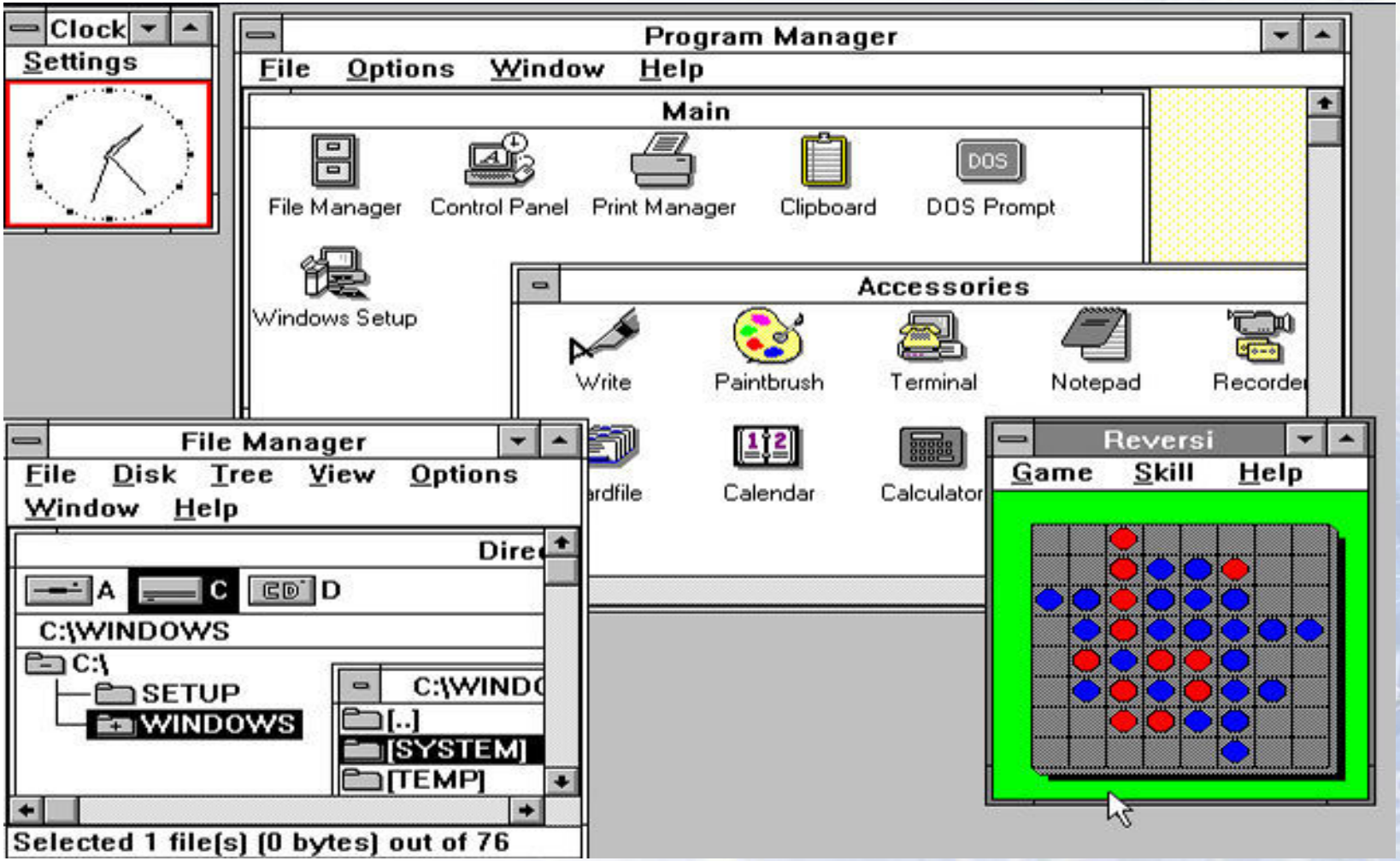
- 26 September 1988: front cover of Time magazine
- Title: “Invasion of the data snatchers”

- The first IBM PC compatible virus in the "wild" was a boot sector virus dubbed (c)Brain, created in 1986 and was released in 1987 by Amjad Farooq Alvi and Basit Farooq Alvi in Lahore, Pakistan, reportedly to deter unauthorized copying of the software they had written

Hex dump of the Brain virus, generally regarded as the first computer virus for the IBM Personal Computer (IBM PC) and compatibles

Displacement	Hex codes															ASCII value
0000(0000)	FA	E9	4A	01	34	12	00	07	14	00	01	00	00	00	00	20	-0J04↕●╥0
0016(0010)	20	20	20	20	20	20	57	65	6C	63	6F	6D	65	20	74	6F	Welcome to
0032(0020)	20	74	68	65	20	44	75	6E	67	65	6F	6E	20	20	20	20	the Dungeon
0048(0030)	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
0064(0040)	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
0080(0050)	20	28	63	29	20	31	39	38	36	20	42	61	73	69	74	20	(c) 1986 Basit
0096(0060)	26	20	41	6D	6A	61	64	20	28	70	76	74	29	20	4C	74	& Amjad (put) Lt
0112(0070)	64	2E	20	20	20	20	20	20	20	20	20	20	20	20	20	20	d.
0128(0080)	20	42	52	41	49	4E	20	43	4F	4D	50	55	54	45	52	20	BRAIN COMPUTER
0144(0090)	53	45	52	56	49	43	45	53	2E	2E	37	33	30	20	4E	49	SERVICES.. 730 NI
0160(00A8)	5A	41	4D	20	42	4C	4F	43	4B	20	41	4C	4C	41	4D	41	ZAM BLOCK ALLAMA
0176(00B0)	20	49	51	42	41	4C	20	54	4F	57	4E	20	20	20	20	20	.IQBAL TOWN
0192(00C0)	20	20	20	20	20	20	20	20	20	20	20	4C	41	48	4F	52	LAHDR
0208(0BD0)	45	2D	50	41	4B	49	53	54	41	4E	2E	2E	50	48	4F	4E	E-PAKISTAN..PHJN
0224(00E0)	45	20	3A	34	33	30	37	39	31	2C	34	34	33	32	34	3B	E :430791,443248
0240(00F0)	2C	32	38	30	35	33	30	2E	20	20	20	20	20	20	20	20	,280530.

The first virus to specifically target Microsoft Windows, WinVir was discovered in April 1992, two years after the release of Windows 3.0



Week 9 Topic 8

Parts of a Virus

A computer virus generally contains three parts:

- the infection mechanism, which finds and infects new files
- the payload, which is the malicious code to execute.
- the trigger, which determines when to activate the payload.

Week 9 Topic 9

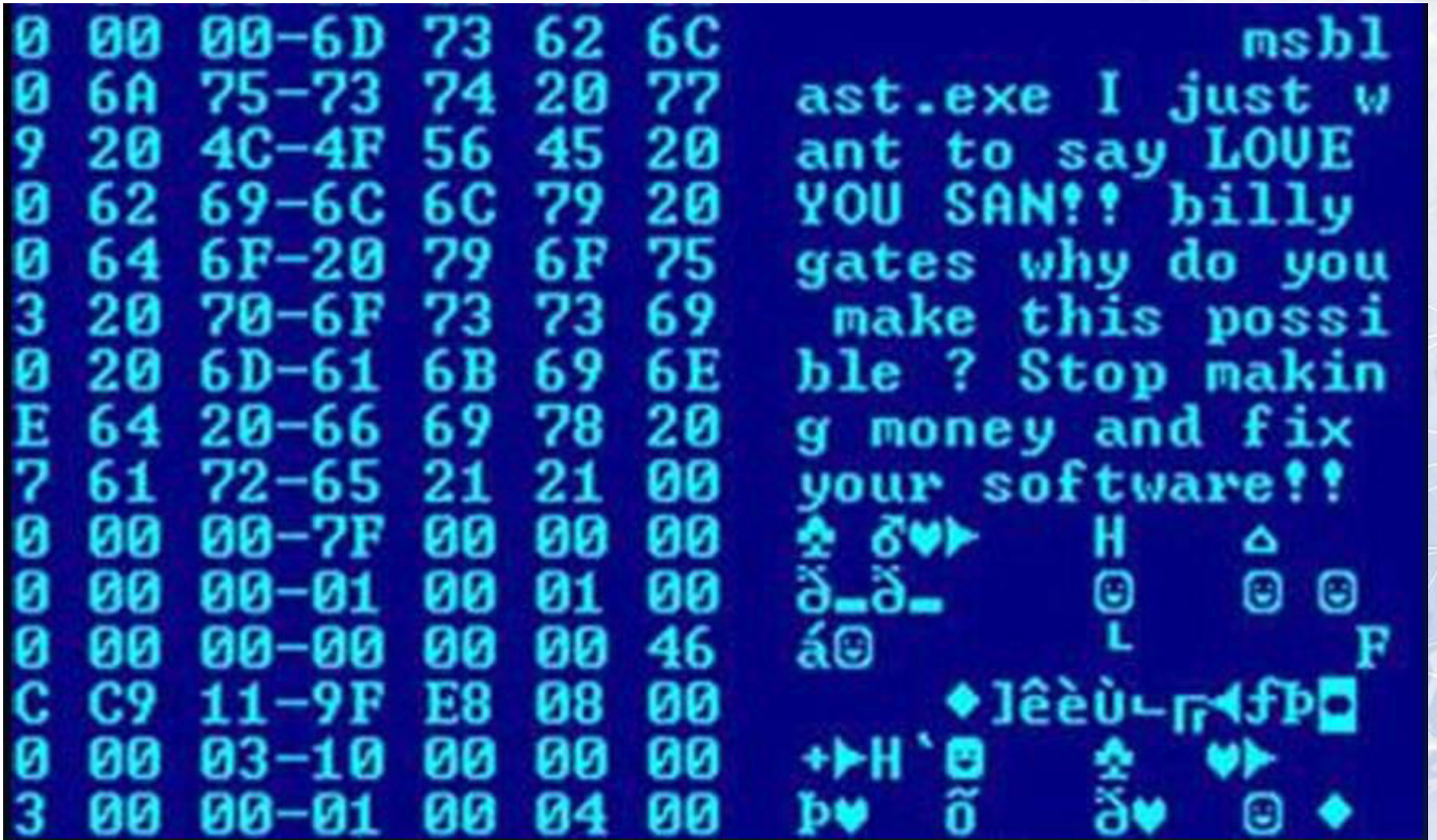
Computer Worm

- A computer worm is a standalone malware computer program that replicates itself in order to spread to other computers.
- It often uses a computer network to spread itself, relying on security failures on the target computer to access it.
- It will use this machine as a host to scan and infect other computers.
- When these new worm-invaded computers are controlled, the worm will continue to scan and infect other computers using these computers as hosts, and this behaviour will continue.

Computer Worm

- Computer worms use recursive methods to copy themselves without host programs and distribute themselves based on exploiting the advantages of exponential growth, thus controlling and infecting more and more computers in a short time.
- Worms almost always cause at least some harm to the network, even if only by consuming bandwidth, whereas viruses almost always corrupt or modify files on a targeted computer.
- Many worms are designed only to spread, and do not attempt to change the systems they pass through.
- However, as the Morris worm and Mydoom showed, even these "payload-free" worms can cause major disruption by increasing network traffic and other unintended effects.

Hex dump of the Blaster worm, showing a message left for Microsoft CEO Bill Gates by the worm's creator



Week 9 Topic 10

Ransomware

• Ransomware is a type of malware from cryptovirology that threatens to publish the victim's personal data or permanently block access to it unless a ransom is paid off.

• While some simple ransomware may lock the system without damaging any files, more advanced malware uses a technique called cryptoviral extortion.

• It encrypts the victim's files, making them inaccessible, and demands a ransom payment to decrypt them

CryptoLocker
Payment for private key
Choose a convenient payment method and click «Next»:
Bitcoin (most cheap option)
bitcoin
Bitcoin is a cryptocurrency where the creation and transfer of bitcoins is based on an open-source cryptographic protocol that is independent of any central authority. Bitcoins can be transferred through a computer or smartphone without an intermediate financial institution.
You have to send 2 BTC to Bitcoin address
and specify the Transaction ID on the next page, which will be verified and confirmed.
Home Page
Getting started with Bitcoin
Private key will be destroyed on
10/13/2013
1:21 PM
Time left
71 : 33 : 17
<< Back
Next >>

Wana Decrypt0r 2.0
Ooops, your files have been encrypted!
English
What Happened to My Computer?
Your important files are encrypted.
Many of your documents, photos, videos, databases and other files are no longer accessible because they have been encrypted. Maybe you are busy looking for a way to recover your files, but do not waste your time. Nobody can recover your files without our decryption service.
Can I Recover My Files?
Sure. We guarantee that you can recover all your files safely and easily. But you have not so enough time.
You can decrypt some of your files for free. Try now by clicking <Decrypt>.
But if you want to decrypt all your files, you need to pay.
You only have 3 days to submit the payment. After that the price will be doubled.
Also, if you don't pay in 7 days, you won't be able to recover your files forever.
We will have free events for users who are so poor that they couldn't pay in 6 months.
How Do I Pay?
Payment is accepted in Bitcoin only. For more information, click <About bitcoin>.
Please check the current price of Bitcoin and buy some bitcoins. For more information, click <How to buy bitcoins>.
And send the correct amount to the address specified in this window.
After your payment, click <Check Payment>. Best time to check: 9:00am - 11:00am
Payment will be raised on
5/16/2017 00:47:55
Time Left
02:23:57:37
Your files will be lost on
5/20/2017 00:47:55
Time Left
06:23:57:37
About bitcoin
How to buy bitcoins?
Contact Us
bitcoin
ACCEPTED HERE
Send $300 worth of bitcoin to this address:
12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw
Copy
Check Payment
Decrypt

Week 9 Topic 11

Trojan Horse

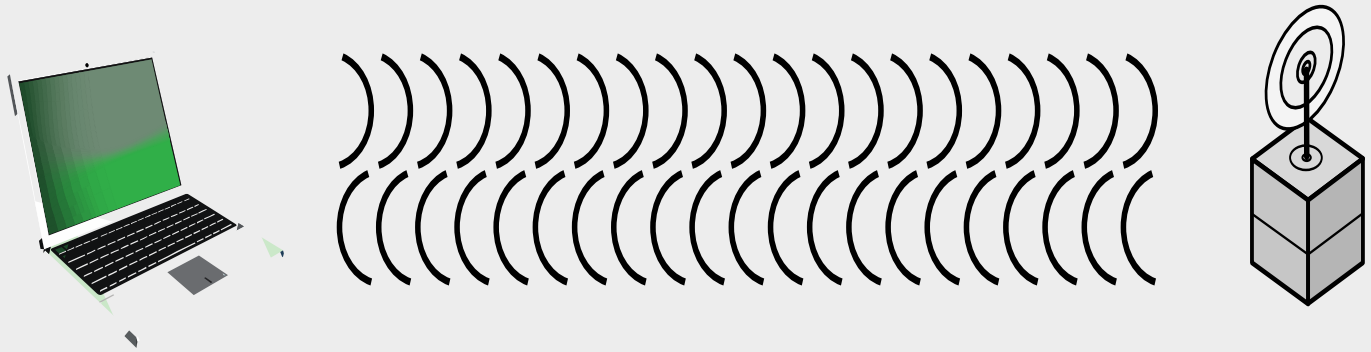
- In computing, a Trojan horse is any malware that misleads users of its true intent by disguising itself as a standard program.
- The term is derived from the ancient Greek story of the deceptive Trojan Horse that led to the fall of the city of Troy.
- Once installed, Trojans may perform a range of malicious actions.
- Many tend to contact one or more Command and Control (C2) servers across the Internet and await instruction.
- Since individual Trojans typically use a specific set of ports for this communication, it can be relatively simple to detect them

Week No. 10

Wireless Network Security

Week 10 Topic 1

Wireless Networking Components



Endpoint

Access point

Wireless Security

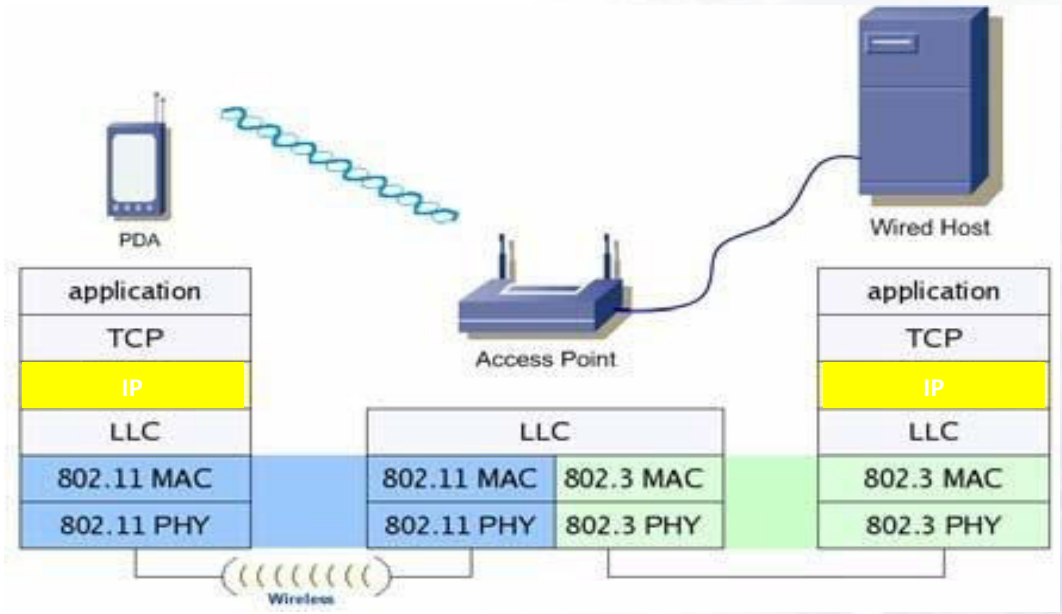
- Key factors contributing to higher security risk of wireless networks compared to wired networks include:
- **Channel**
  - Wireless networking typically involves broadcast communications, which is far more susceptible to eavesdropping and jamming than wired networks
  - Wireless networks are also more vulnerable to active attacks that exploit vulnerabilities in communications protocols
- **Mobility**
  - Wireless devices are far more portable and mobile, thus resulting in a number of risks
- **Limited Resources**
  - Some wireless devices, such as smartphones and tablets, have sophisticated operating systems but limited memory and processing resources with which to counter threats, including denial of service and malware
- **Accessibility**
  - Some wireless devices, such as sensors and robots, may be left unattended in remote and/or hostile locations, thus greatly increasing their vulnerability to physical attacks

Week 10 Topic 2

IEEE 802.11 Standard

Wired and Wireless LANs

- LANs are designed for use in a **small area** (offices, campuses)
- IEEE has defined the specifications for a
  - wired LAN, called IEEE 802.3
  - wireless LAN, called IEEE 802.11,
- These specifications are concerning the physical and data link layers.



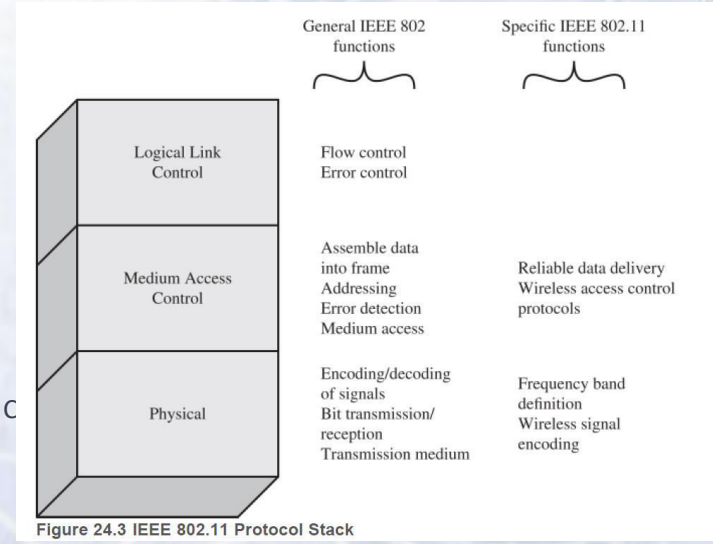


Figure 24.3 IEEE 802.11 Protocol Stack

Brief History of IEEE 802

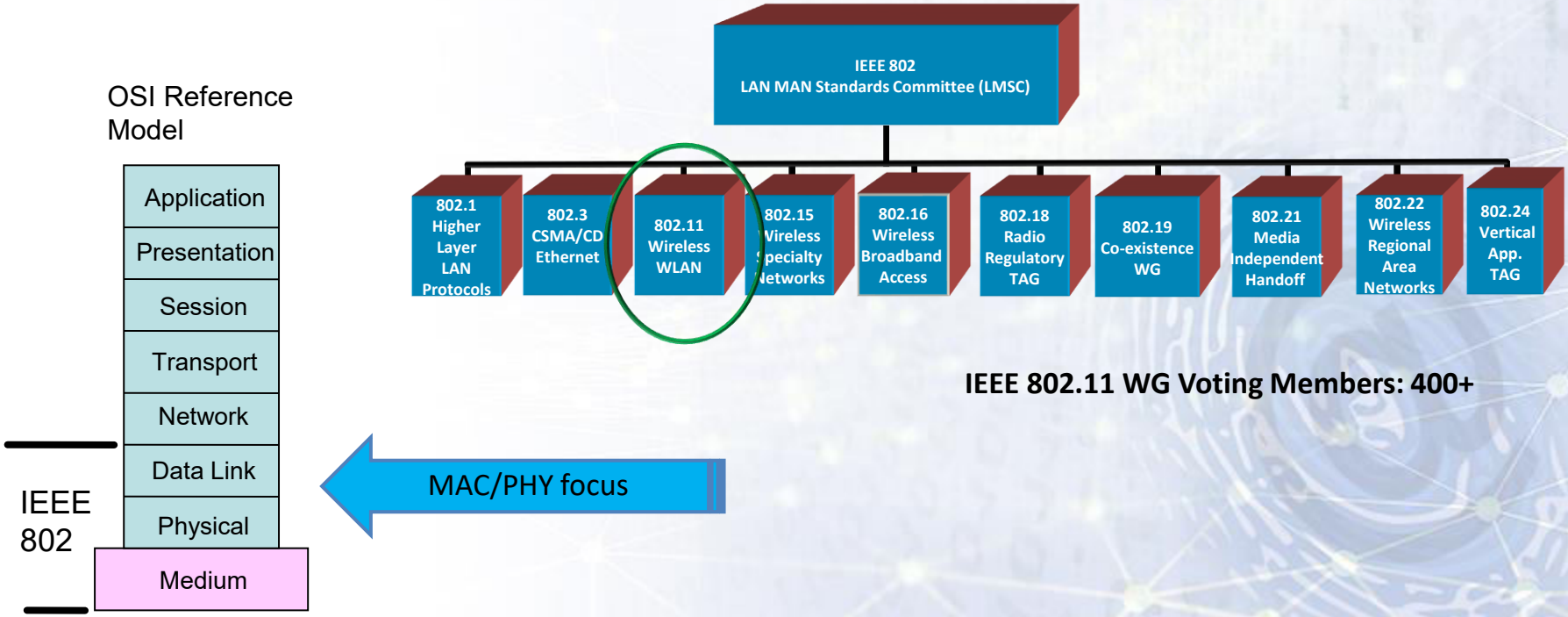
- Since there are various types of computer system manufacturers, the IEEE's computer society started a project in 1985 called **Project 802** to enable standard communication between various devices.

- The **IEEE 802 LAN/MAN Standards Committee (LMSC)** develops and maintains networking standards and recommended practices for **local, metropolitan, and other area networks**, using an open and accredited process, and advocates them on a global basis.

https://www.ieee802.org/

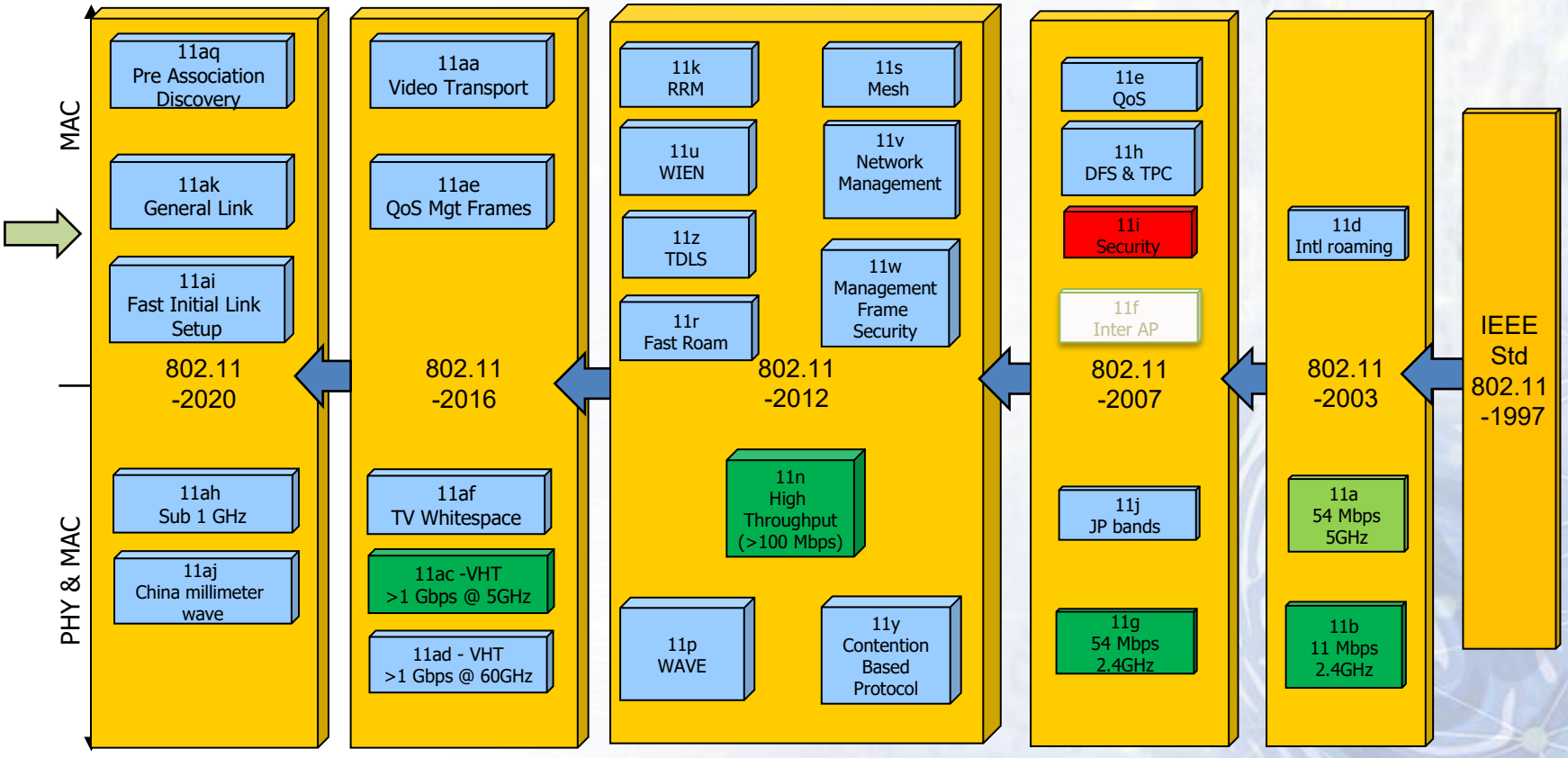
The IEEE 802.11 Working Group is one of the most active WGs in 802

- The 802.11 Working Group was established in 1990.
- Focus on **link and physical layers** of the network stack
- Leverage IETF protocols for upper layers



IEEE 802.11 WG Voting Members: 400+

Development of the IEEE 802.11 Standard is ongoing since 1997



IEEE802.11 Wi-Fi standard evolution: 802.11be is under development

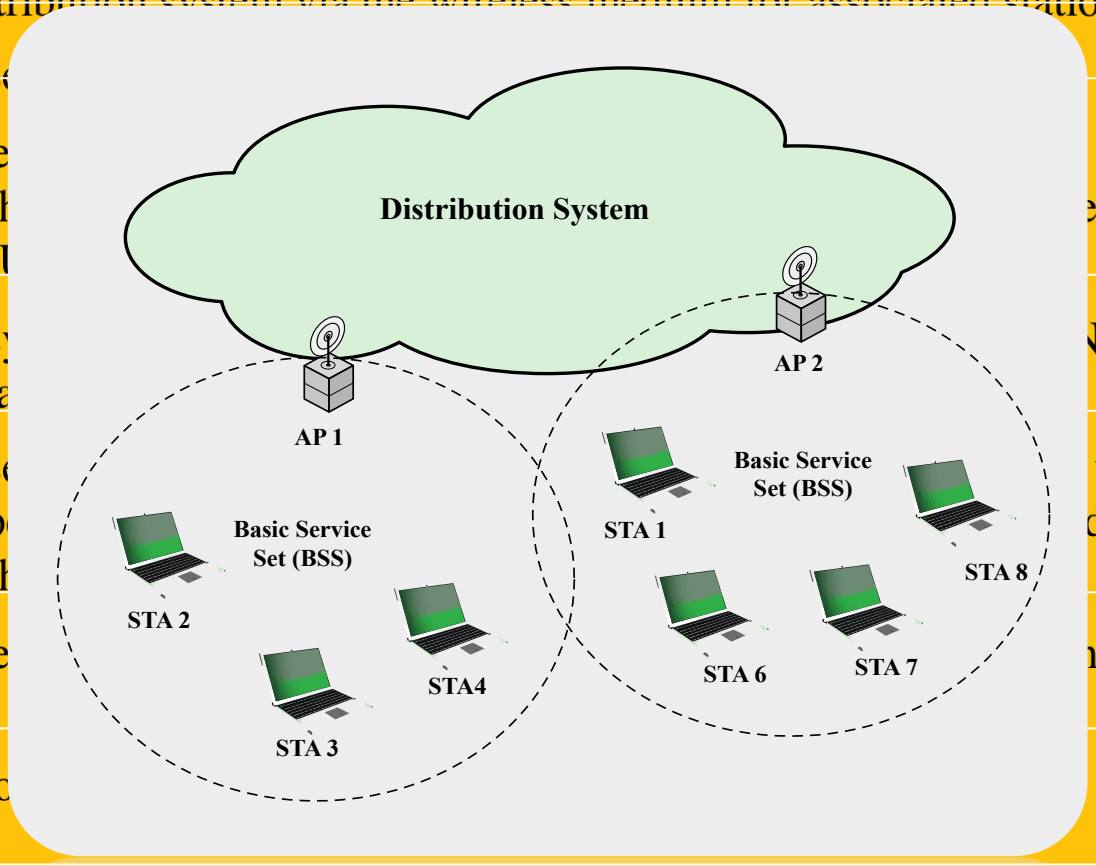
802.11n (2009) Wi-Fi 4	802.11ac (2013) Wi-Fi 5	802.11ax (2021) Wi-Fi 6 6E	802.11be (est. 2024) Wi-Fi 7
• 2.4GHz and 5GHz supported • Wider channels (40MHz) • Better modulation (64-QAM) • Additional streams (Up to 4) • Backward compatibility with 11a/b/g • Standard supports up to 600Mbps	• 5GHz only • Wider channels (80, 160MHz) • Better modulation (256-QAM) • Additional streams (Up to 8, implemented up to 4) • Backward compatibility with 11a/b/g/n • Standard supports up to 7Gbps	• 2.4GHz, 5GHz and **6GHz** supported • Wider channels (80, 160MHz) • Better modulation (1024-QAM) • Additional streams (Up to 8, implemented) • Backward compatibility with 11a/b/g/n/ac • Standard supports up to 9.6Gbps	• 2.4GHz, 5GHz and **6GHz** supported • Wider channels (40, 80, 160, 240, **320MHz**) • Better modulation (4096-QAM) • Backward compatibility with 11a/b/g/n/ac/ax • Throughput minimum of 30Gbps, expect 40Gbps+ • Low latency features • Draft 2.0 now in Working Group Ballot

(Ratification date) Products available in the market typically ~2 years prior

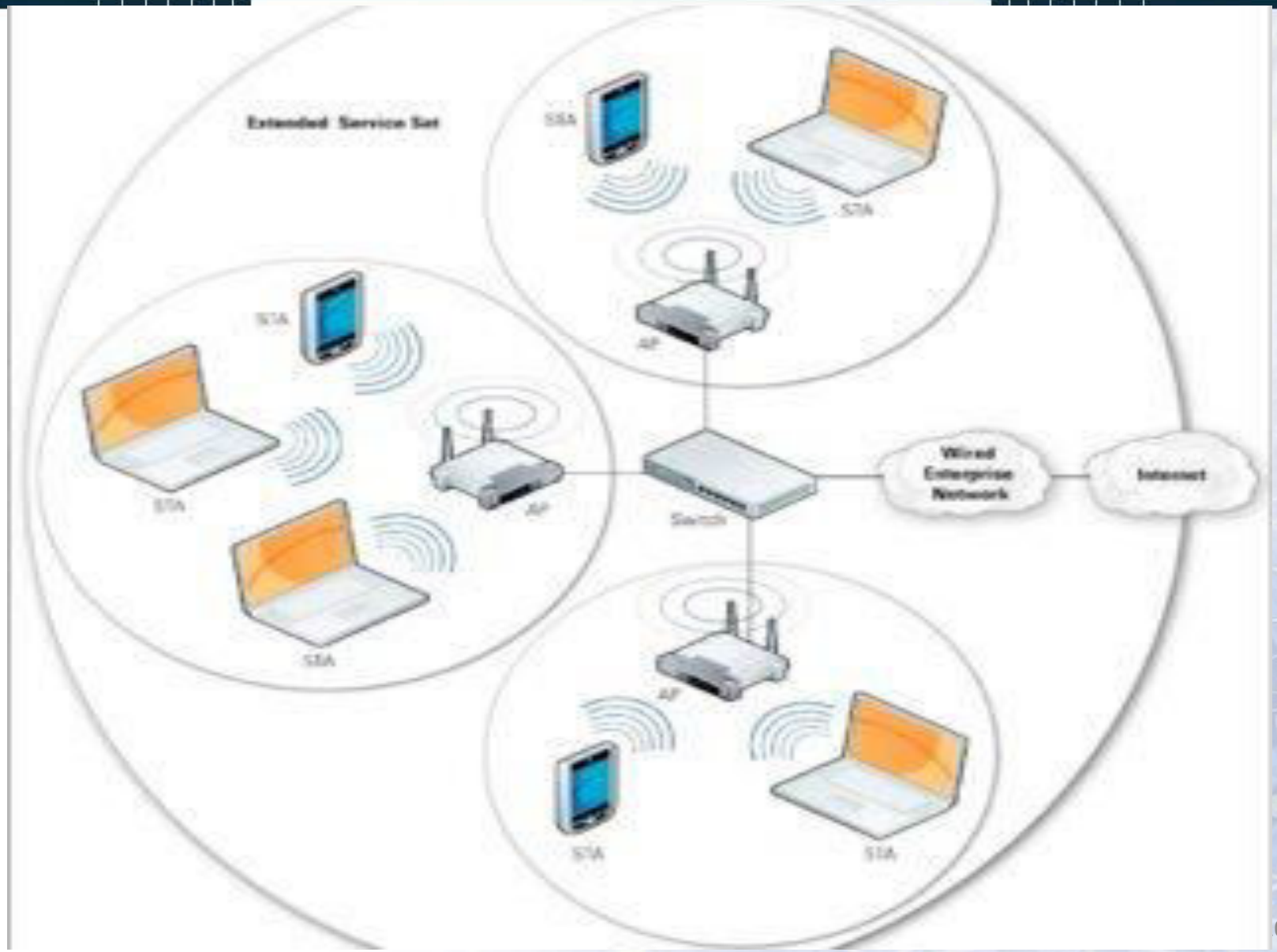
Week 10 Topic 3

IEEE 802.11 Terminology

Access point (AP)	Any entity that has station functionality and provides access to the distribution system via the wireless medium for associated stations.
Basic service set (BSS)	A s
Coordination function	The with PDU
Distribution system (DS)	A sy crea Ns to
Extended service set (ESS)	A s app with that d
MAC protocol data unit (MPDU)	The the ng
MAC service data unit (MSDU)	Info
Station	Any device that contains an IEEE 802.11 conformant MAC and physical layer.



IEEE 802.11 Extended Service Set Example

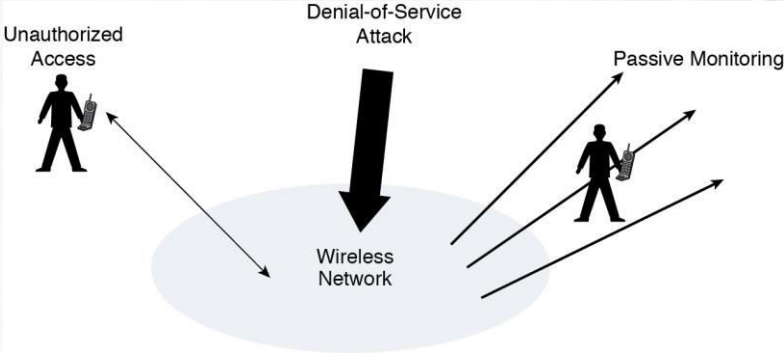


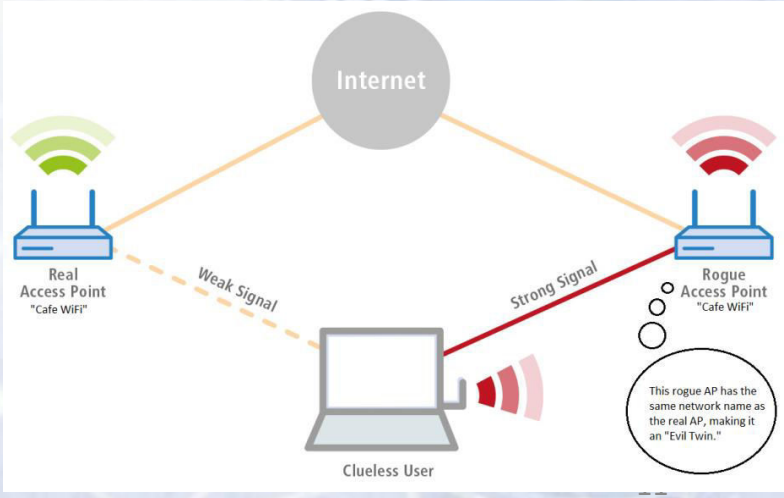
Week 10 Topic 4

IEEE 802.1x Standard

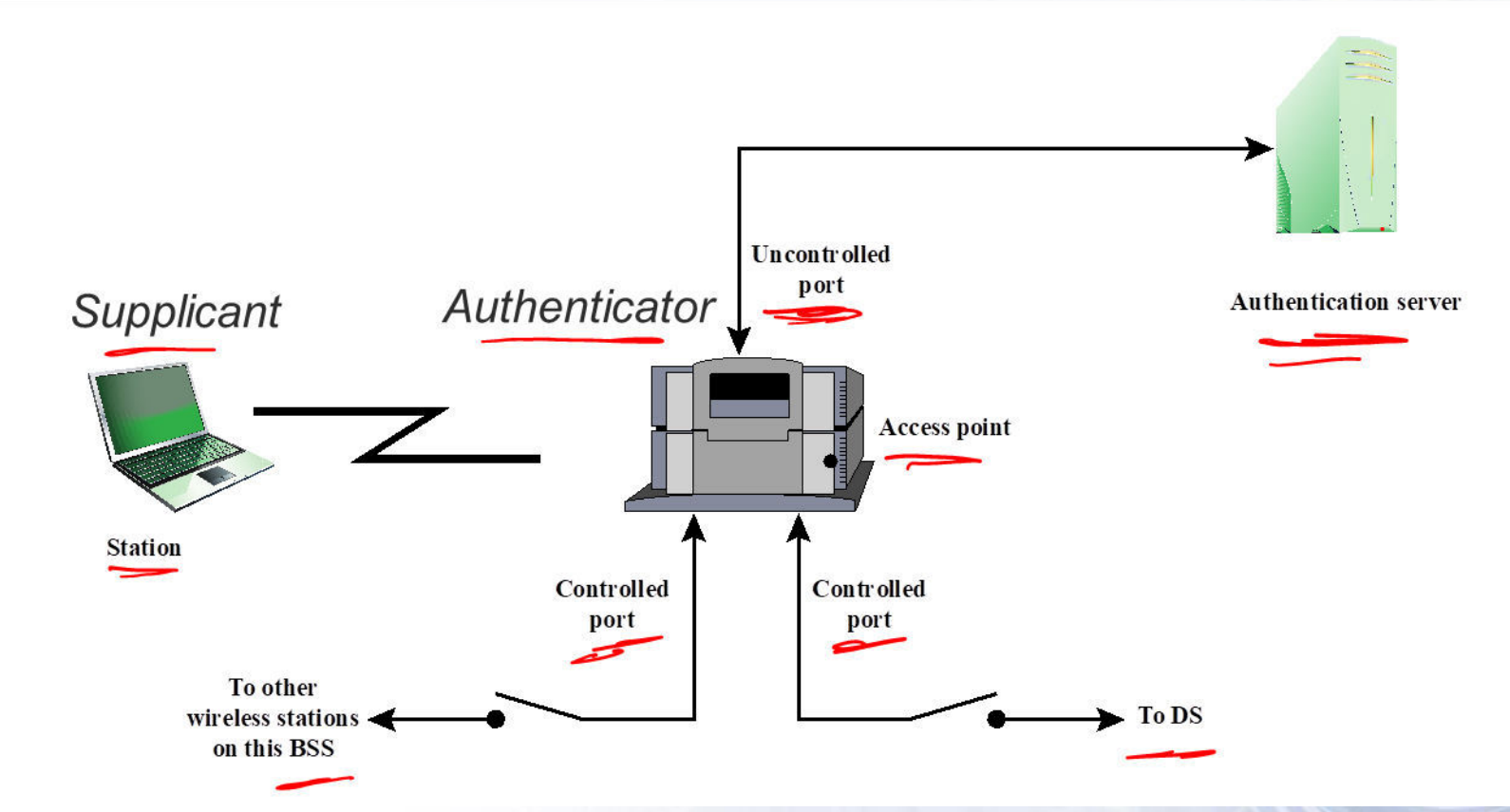
Securing Wireless Networks

- The main threat involving wireless access points is unauthorized access to the network
- Principal approach for preventing such access is the IEEE 802.1X standard for port-based network access control
  - The standard provides an authentication mechanism for devices wishing to attach to a LAN or wireless network
- Use of 802.1X can prevent rogue access points and other unauthorized devices from becoming insecure backdoors

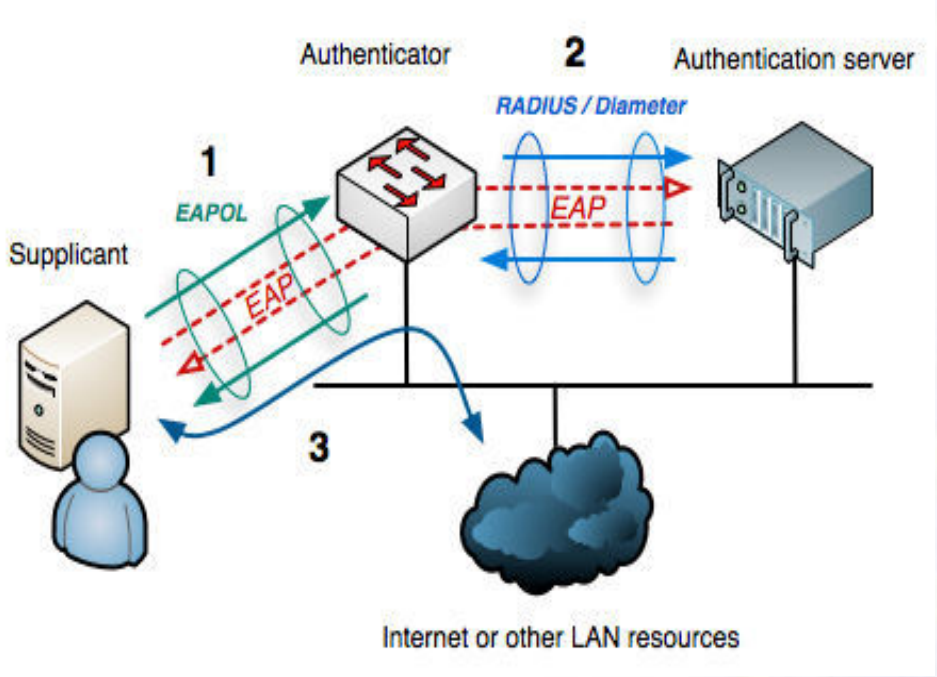




802.1X Access Control



IEEE 802.1X standard for Securing Wired/Wireless Networks



Extensible Authentication Protocol (EAP) originally designed for Point-to-Point Protocol (PPP)

EAPoL (EAP over LAN)

EAPoW (EAP over Wireless)

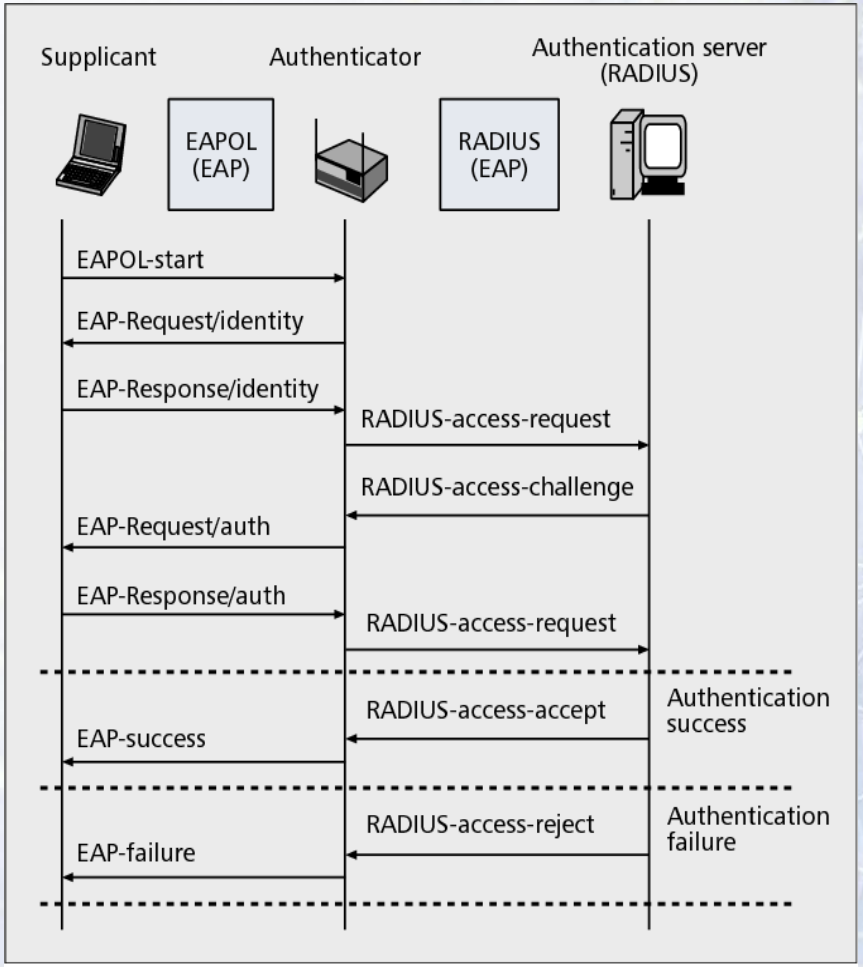


FIGURE 4. *A typical 802.1x message flow.*

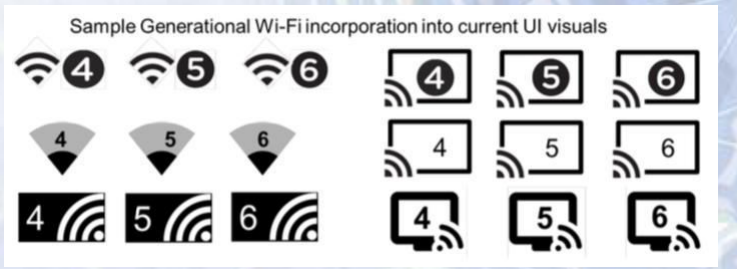
Week 10 Topic 5

Wireless Fidelity (Wi-Fi) Alliance

- The Wi-Fi Alliance is a non-profit organization that owns the Wi-Fi trademark.
  - Manufacturers may use the trademark to brand products certified for Wi-Fi interoperability.
- 802.11b
  - First 802.11 standard to gain broad industry acceptance
- Wireless Ethernet Compatibility Alliance (WECA)
  - Industry consortium formed in 1999 to address the concern of products from different vendors successfully interoperating
  - Later renamed the Wi-Fi Alliance in 2002
- Term used for certified 802.11b products is *Wi-Fi* in 1999
- In 2018, the Wi-Fi Alliance introduced simplified Wi-Fi generational numbering to indicate supported 802.11 standard by the equipment e.g.
- Other programs
  - Wi-Fi Direct: In October 2010, the Alliance began to certify Wi-Fi Direct
  - Wi-Fi Easy Connect protocol via QR Code
  - Wi-Fi Passpoint : Wi-Fi Passpoint, alternatively known as Hotspot 2.0
  - and may others…

https://www.wi-fi.org/

Year	WiFi Versions/Protocols	Old name	New name
1999	First Generation	WiFi 802.11b	WiFI 1
1999	Second Generation	WiFi 802.11a	WiFi 2
2003	Third Generation	WiFI 802.11g	WiFi 3
2009	Four Generation	WiFi 802.11n	WiFi 4
2014	Fifth Generation	WiFi 802.11ac	WiFi 5
2019	Sixth Generation	WiFi 802.11ax	WiFi 6

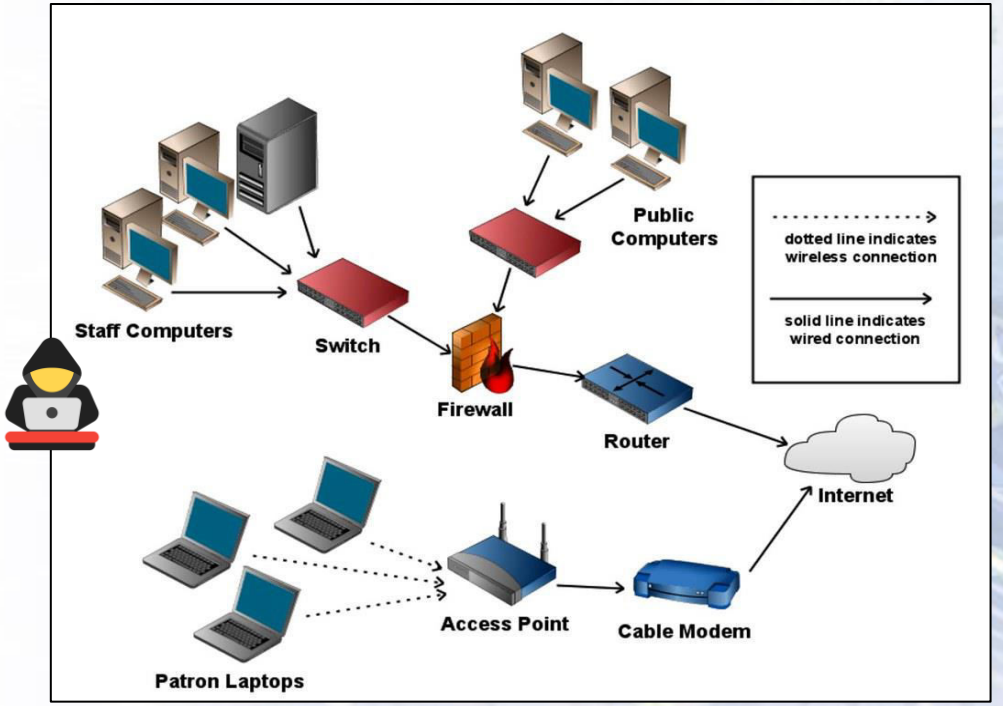


Demo Wi-Fi Info…

Week 10 Topic 6

Wireless LAN Security

- There are two characteristics of a wired LAN that are not inherent in a wireless LAN.
  - In order to transmit over a wired LAN, a station must be physically connected to the LAN, which is not the case in Wireless…
  - Similarly, in order to receive a transmission from a station that is part of a wired LAN, the receiving station also must be attached to the wired LAN, which is not the case in Wireless...



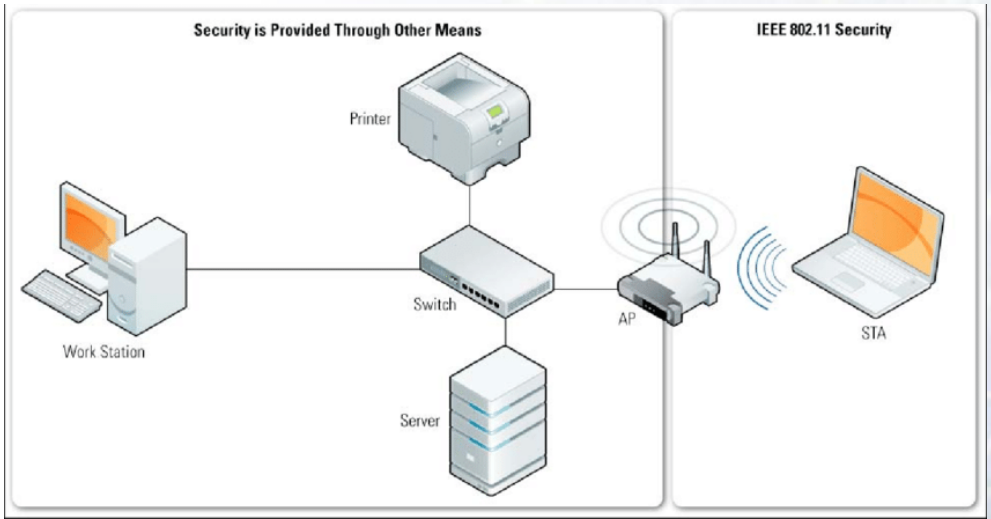
Wireless LAN Security

- The following security measures are often considered as trivial solutions for risk mitigation:
  - Disable the SSID broadcast
  - Maintain a list of authorized MAC addresses and block unauthorized ones from the network
  - Select AP locations in the middle of the building (not close to windows), use directional antennas and line walls and windows with metal foil to minimize the signal leakage to the outside of the building
- How much security do these measures bring?
- Need something more…real security solutions

Demo AP Setting…

Wireless LAN Security

- The original 802.11 specification included a set of security features for privacy and authentication that were quite weak.
  - For privacy, 802.11 defined the Wired Equivalent Privacy (WEP) algorithm.
  - The privacy portion of the 802.11 standard contained major weaknesses.
- Subsequently, the 802.11i task group has developed a set of capabilities to address the WLAN security issues. In order to accelerate the introduction of strong security into WLANs, the Wi-Fi Alliance promulgated Wi-Fi Protected Access (WPA) as a Wi-Fi standard.
  - WPA is a set of security mechanisms that eliminates most 802.11 security issues and was based on the trimmed down version of 802.11i standard.
- The final form of the 802.11i standard is referred to as Robust Security Network (RSN).
  - The Wi-Fi Alliance certifies vendors in compliance with the full 802.11i specification under the WPA2 program.



Demo Wi-Fi Info...

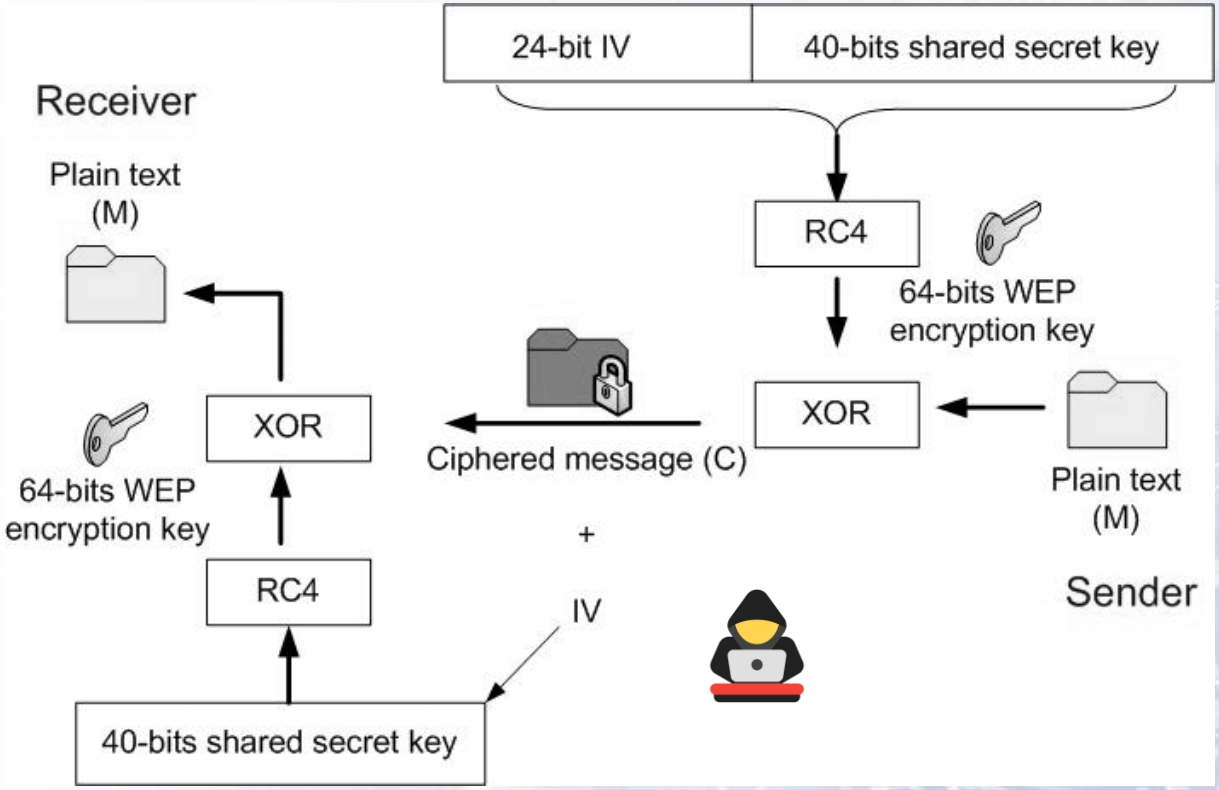
Week 10 Topic 7

Wired Equivalent Privacy (WEP) Algorithm

WEP was part of the IEEE 802.11 standard ratified in 1997.

Provide Support for

- Authentication
- Confidentiality
- Integrity



WEP is notoriously weak…

Week 10 Topic 8

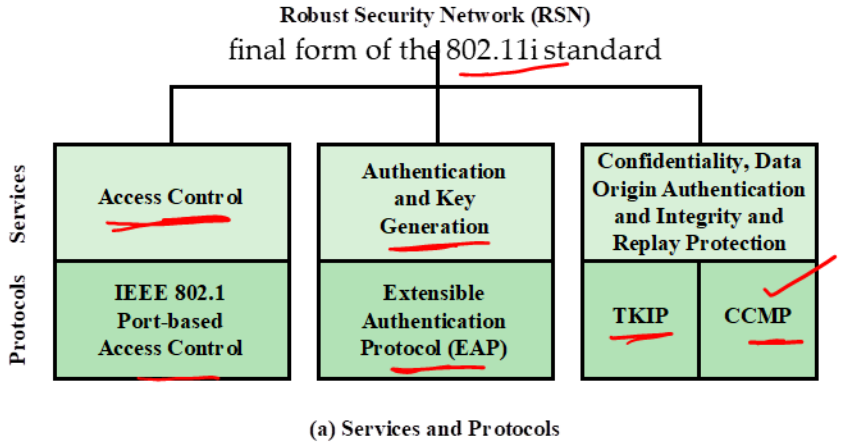
Wi-Fi Protected Access (WPA)

Brief History of Wi-Fi Protected Access (WPA)

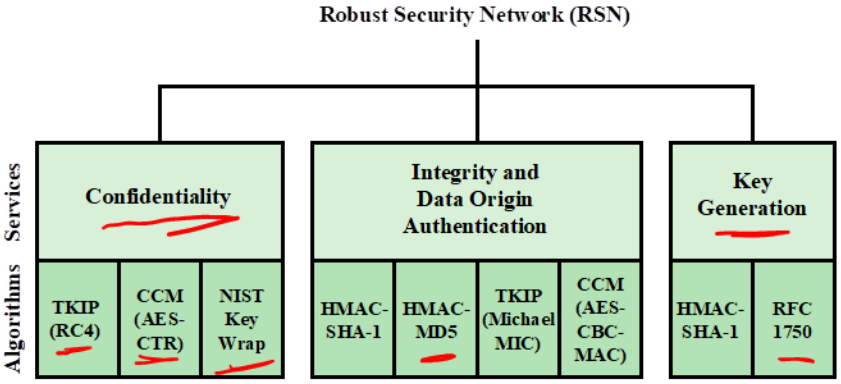
- As WEP was not secure…
- Subsequently, the 802.11i task group has developed a set of capabilities to address the WLAN security issues.
  - In order to accelerate the introduction of strong security into WLANs, the Wi-Fi Alliance promulgated Wi-Fi Protected Access (WPA) as a Wi-Fi standard.
- WPA is a set of security mechanisms that eliminates most 802.11 security issues and was based on trimmed down version of 802.11i standard…initially released in 2003.
  - No new/specialized equipment was required…just firmware update
  - Based on TKIP / CCMP for encryption
    - TKIP - Temporal Key Integrity Protocol (TKIP)
    - CCMP - Counter mode CBC-MAC Protocol
- WPA2 is improved version of original WPA released in 2004.
  - Based on CCMP for encryption
- WPA3 comes as a successor to WPA2 - introduced in 2018
  - after July 2020, all Wi-Fi devices that are certified must support it.

WPA2

The Wi-Fi Alliance certifies vendors in compliance with the full 802.11i specification under the WPA2 program.



(a) Services and Protocols



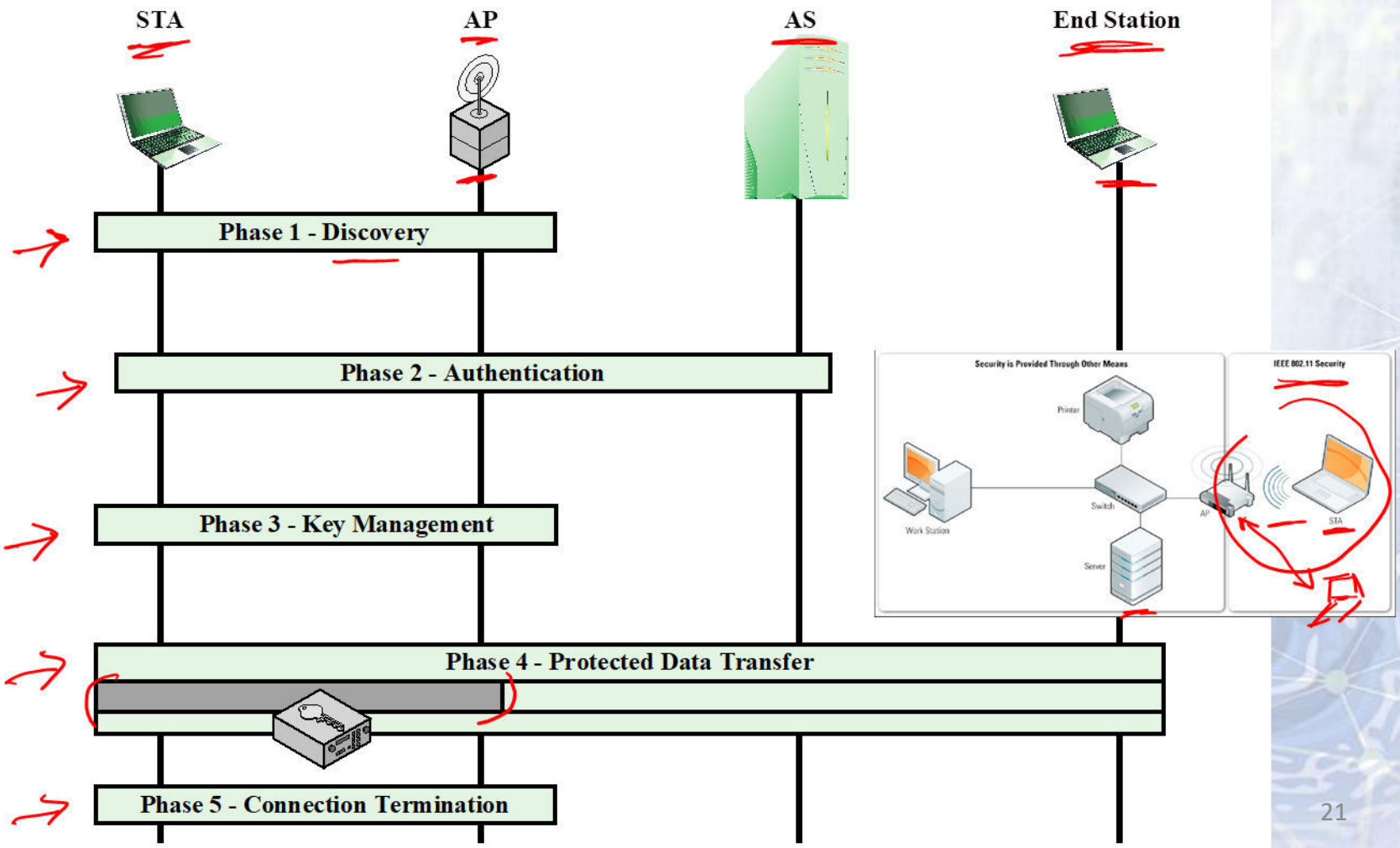
(b) Cryptographic Algorithms

CBC-MAC = Cipher Block Block Chaining Message Authentication Code (MAC)
CCM = Counter Mode with Cipher Block Chaining Message Authentication Code
CCMP = Counter Mode with Cipher Block Chaining MAC Protocol
TKIP = Temporal Key Integrity Protocol

Figure 24.6 Elements of IEEE 802.11i

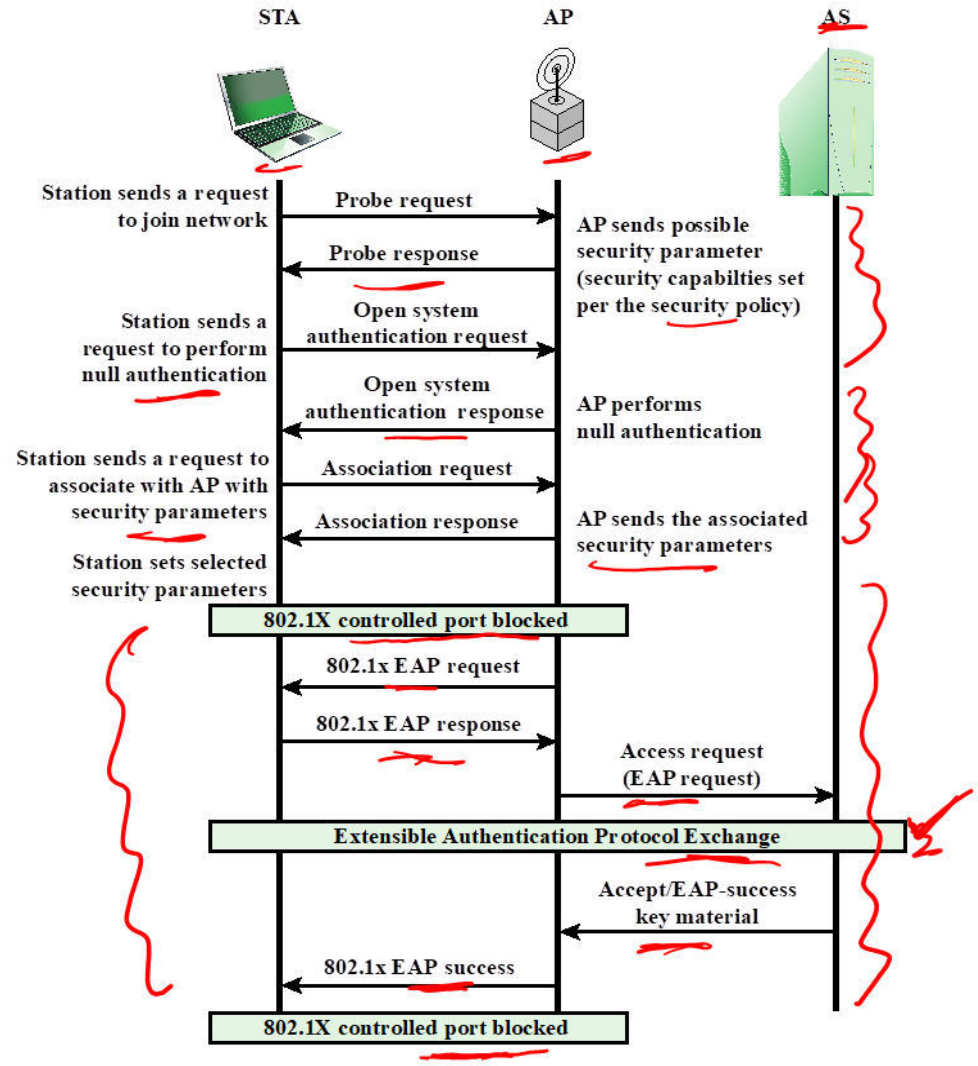
Week 10 Topic 9

IEEE 802.11i Phases of Operation

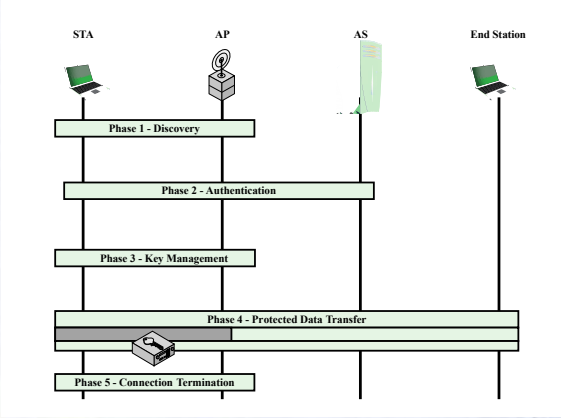


Week 10 Topic 10

IEEE 802.11i Standard



Capability Discovery, Association. Authentication

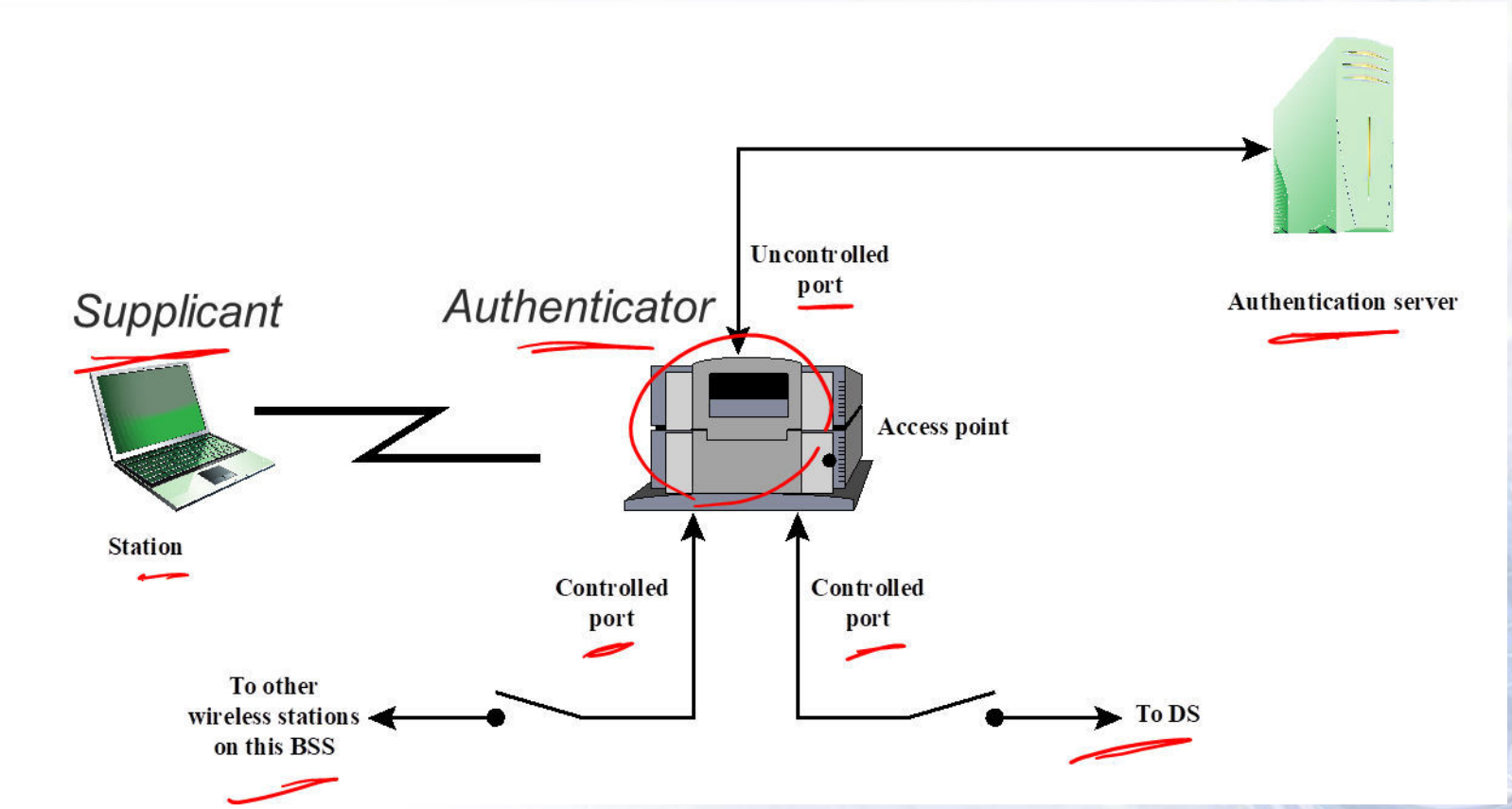


Capability Discovery, Association, & Authentication

Authentication phase consists of three phases:

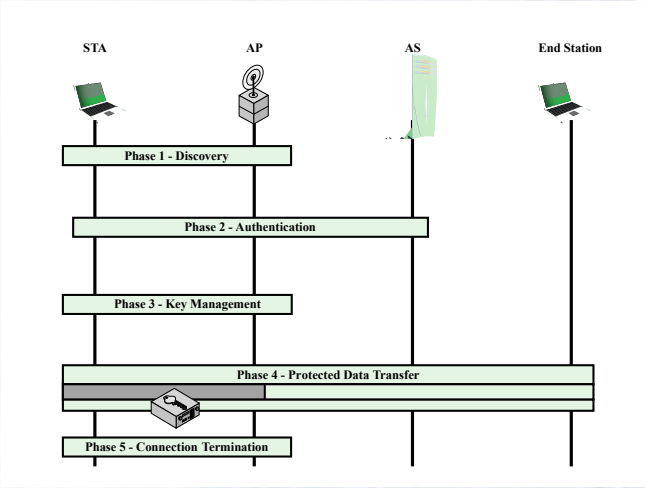
- **Connect to AS**
  - The STA sends a request to its AS through AP
- **EAP exchange**
  - Authenticates the STA and AS to each other
- **Secure key delivery**
  - On success, the AS generates a Master Session Key and sends it to the STA

802.1X Access Control

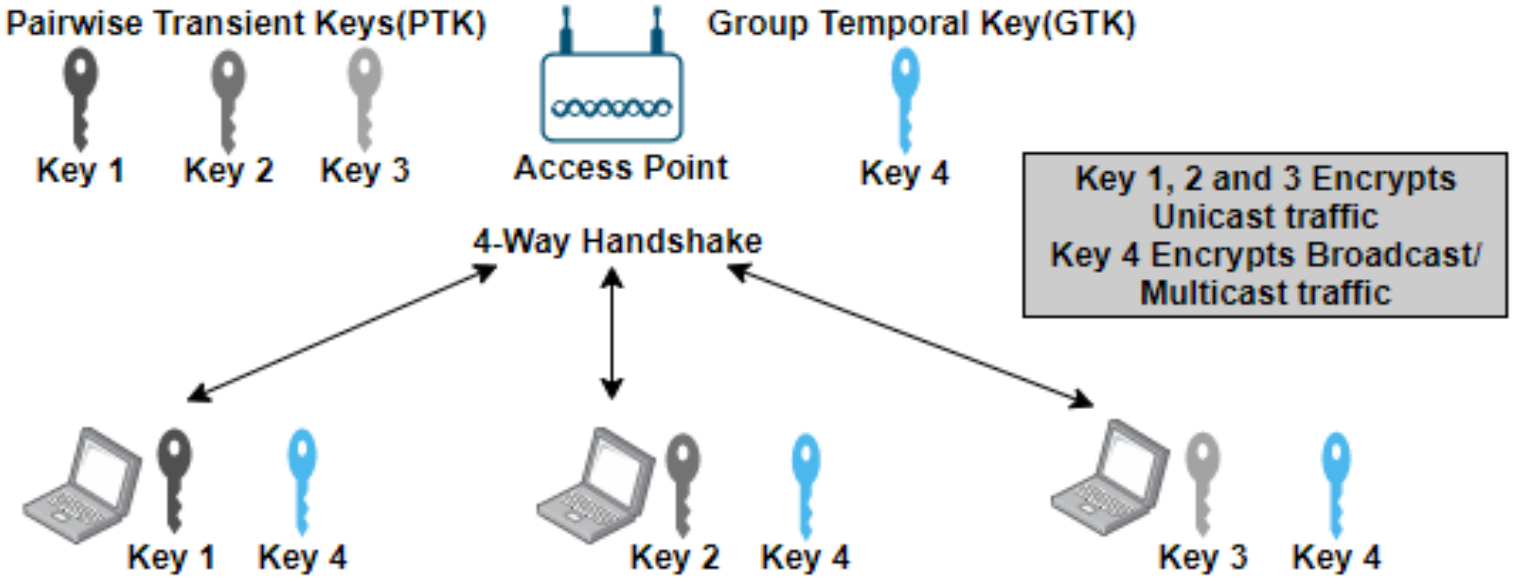


Week 10 Topic 11

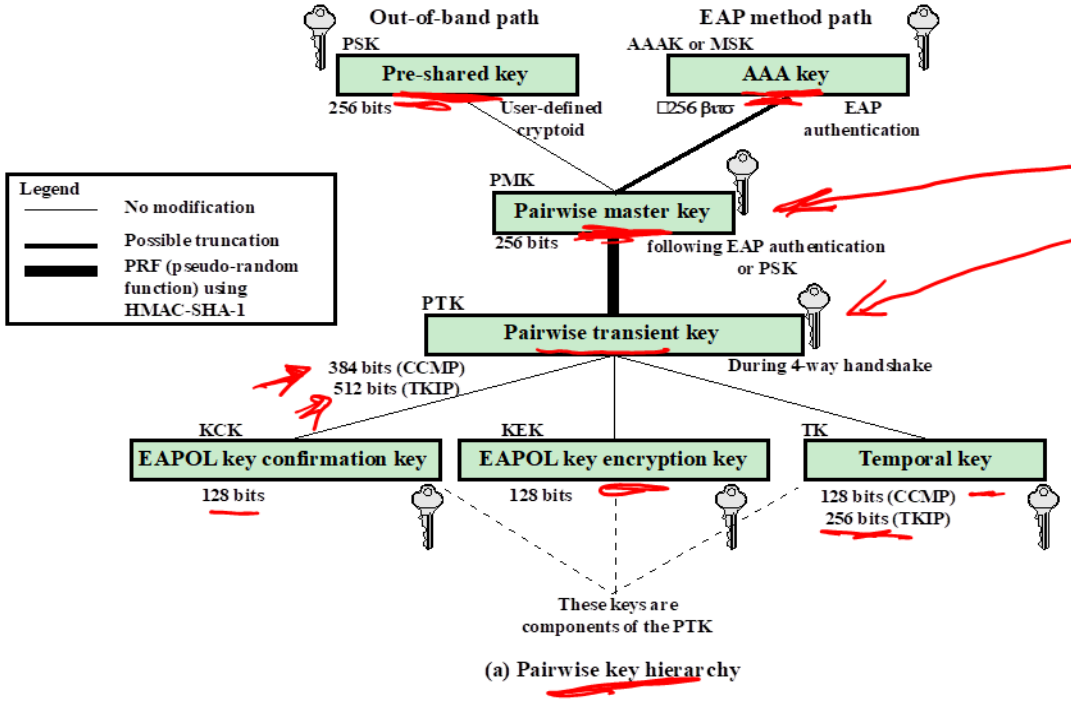
IEEE 802.11i Key Hierarchies



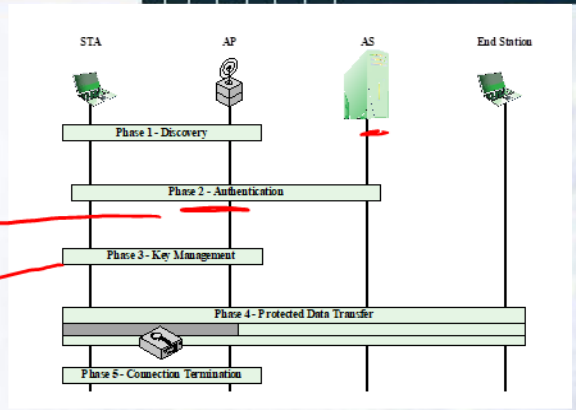
Pairwise keys and Group keys

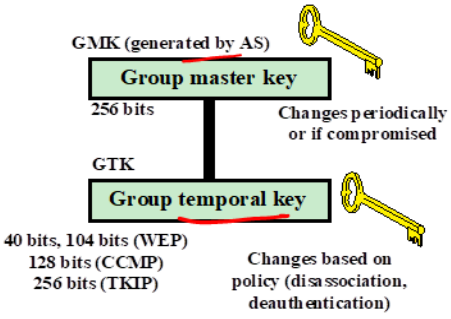


IEEE 802.11i Key Hierarchies



(a) Pairwise key hierarchy





(b) Group key hierarchy

Abbrev-iation	Name	Description / Purpose	Size (bits)	Type
AAA Key	Authentication, Accounting, and Authorization Key	Used to derive the PMK. Used with the IEEE 802.1X authentication and key management approach. Same as MMSK.	≥ 256	Key generation key, root key
PSK	Pre-Shared Key	Becomes the PMK in pre-shared key environments.	256	Key generation key, root key
PMK	Pairwise Master Key	Used with other inputs to derive the PTK.	256	Key generation key
GMK	Group Master Key	Used with other inputs to derive the GTK.	128	Key generation key
PTK	Pair-wise Transient Key	Derived from the PMK. Comprises the EAPOL-KCK, EAPOL-KEK, and TK and (for TKIP) the MIC key.	512 (TKIP) 384 (CCMP)	Composite key
TK	Temporal Key	Used with TKIP or CCMP to provide confidentiality and integrity protection for unicast user traffic.	256 (TKIP) 128 (CCMP)	Traffic key
GTK	Group Temporal Key	Derived from the GMK. Used to provide confidentiality and integrity protection for multicast/broadcast user traffic.	256 (TKIP) 128 (CCMP) 40, 104 (WEP)	Traffic key
MIC Key	Message Integrity Code Key	Used by TKIP's Michael MIC to provide integrity protection of messages.	64	Message integrity key
EAPOL-KCK	EAPOL-Key Confirmation Key	Used to provide integrity protection for key material distributed during the 4-Way Handshake.	128	Message integrity key
EAPOL-KEK	EAPOL-Key Encryption Key	Used to ensure the confidentiality of the GTK and other key material in the 4-Way Handshake.	128	Traffic key / key encryption key
WEP Key	Wired Equivalent Privacy Key	Used with WEP.	40, 104	Traffic key

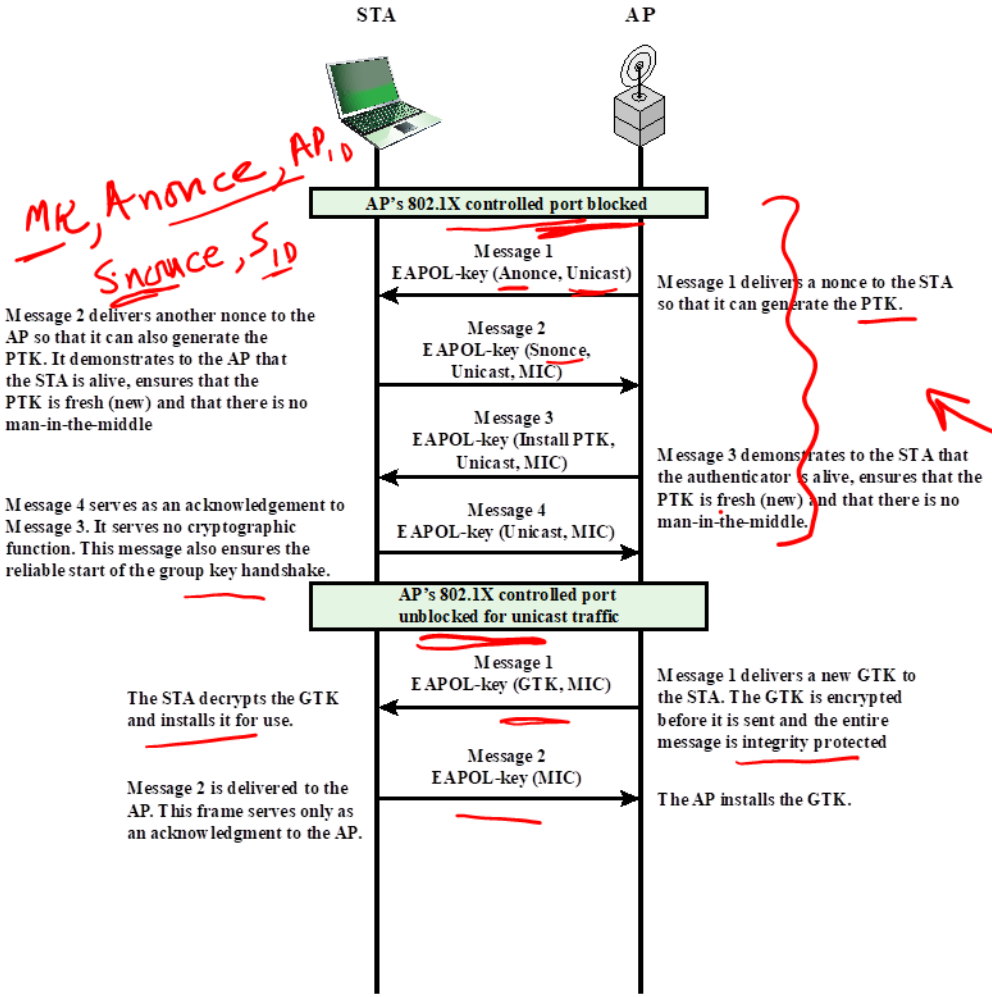
IEEE 802.11i

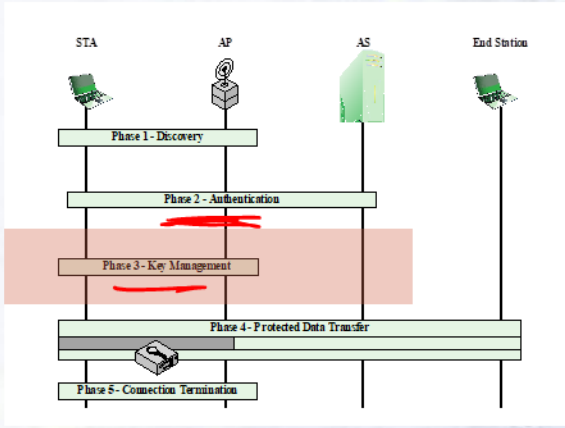
Keys for Data Confidentiality and Integrity Protocols

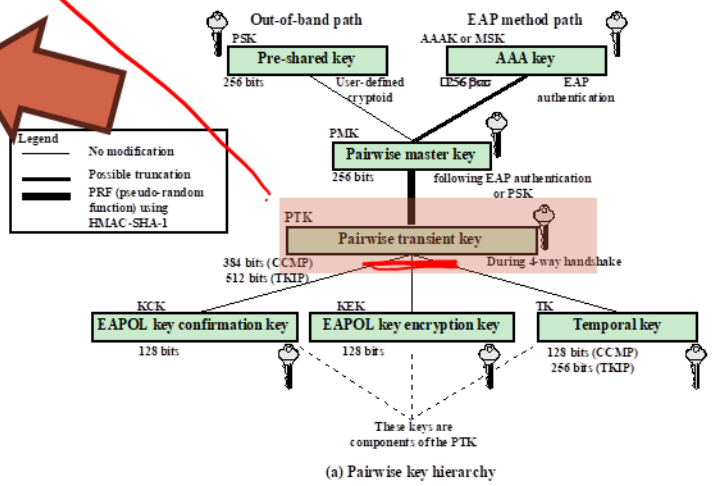
Week 10 Topic 12

IEEE 802.11i Phases of Operation

4-Way Handshake and Group Key Handshake







(a) Pairwise key hierarchy

About Strength of WPA2

- WPA2 ensures that data sent or received over your wireless network is encrypted,
  - only people with your network password have access to it.
- WPA2 use AES to replace the more vulnerable TKIP algorithm used in the original WPA protocol.
- WPA2 uses dynamic key encryption,
  - Regularly changes the key PTK and makes it more difficult to crack.
- However, Pre-Shared Key is derived from passphrase (Wi-Fi password)
  - This is common in Home networks… Enterprise networks use AS and are more safe…
- Weak Wif-Fi passwords can easily be cracked
  - All you need is to capture 4-way handshake exchange messages **“airodump-ng”**
  - Then use offline dictionary attacks…to crack the password **“aircrack-ng”**
- That’s why we have WPA3 coming as a successor to WPA2…

Week No. 11

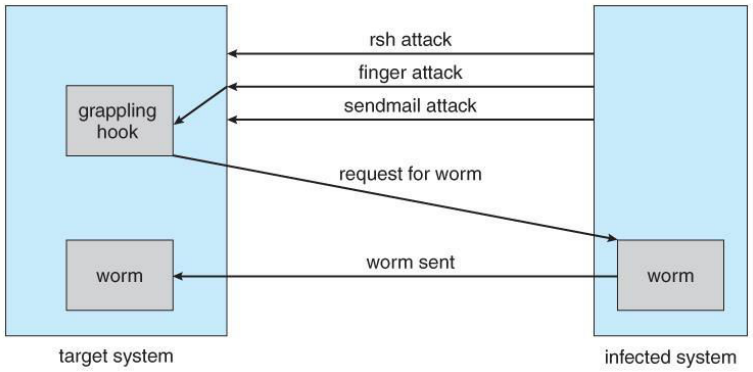
Buffer Overflow Attacks

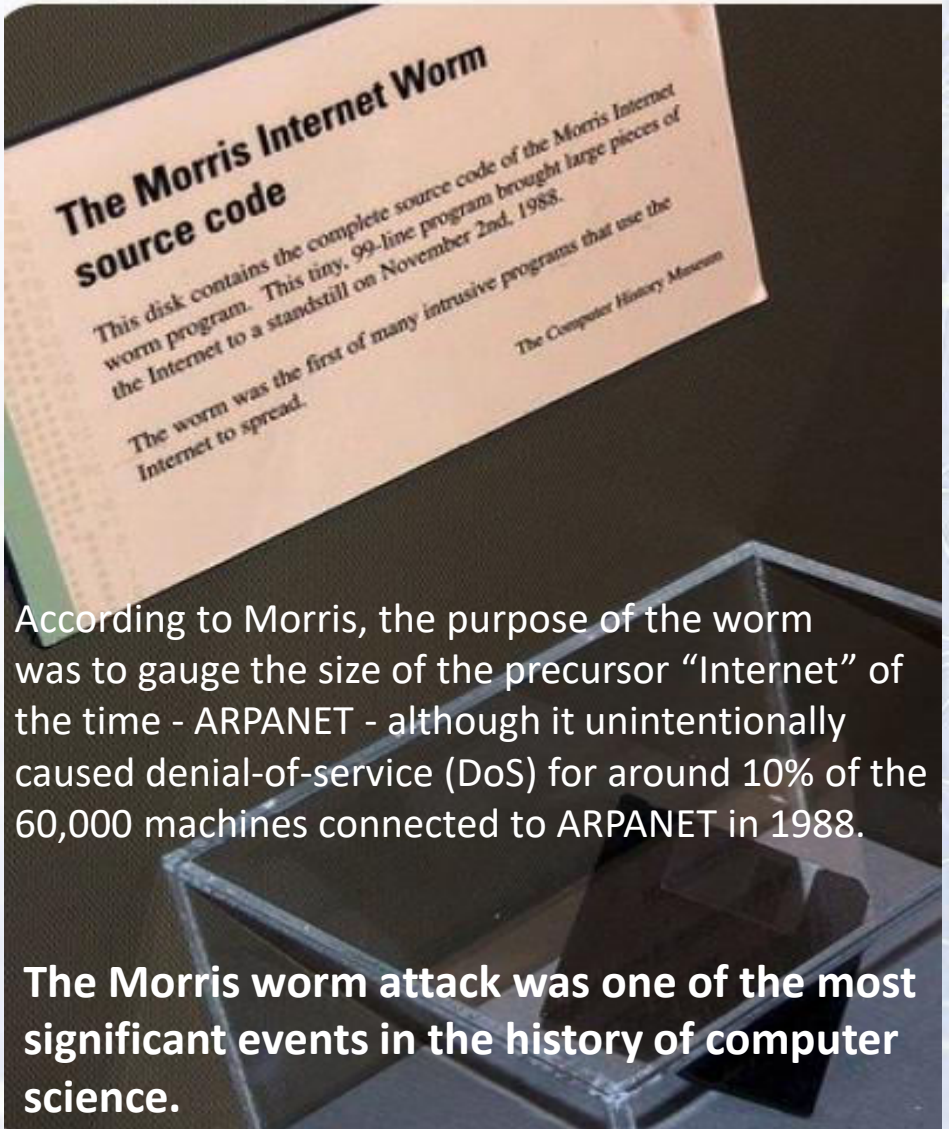
Week 11 Topic 1

A Brief History of Some Buffer Overflow Attacks

Robert Tappan Morris

The Morris worm functioned as a denial-of-service (DoS) attack.
The Morris worm spread by exploiting vulnerabilities in UNIX sendmail (using a backdoor), finger (through buffer overflow), and rsh/rexec (to explore whether it could operate).





According to Morris, the purpose of the worm was to gauge the size of the precursor “Internet” of the time - ARPANET - although it unintentionally caused denial-of-service (DoS) for around 10% of the 60,000 machines connected to ARPANET in 1988.

The Morris worm attack was one of the most significant events in the history of computer science.

The front page of an infected server might have been changed by the worm following:

Code Red was a computer worm observed on the Internet on July 15, 2001.

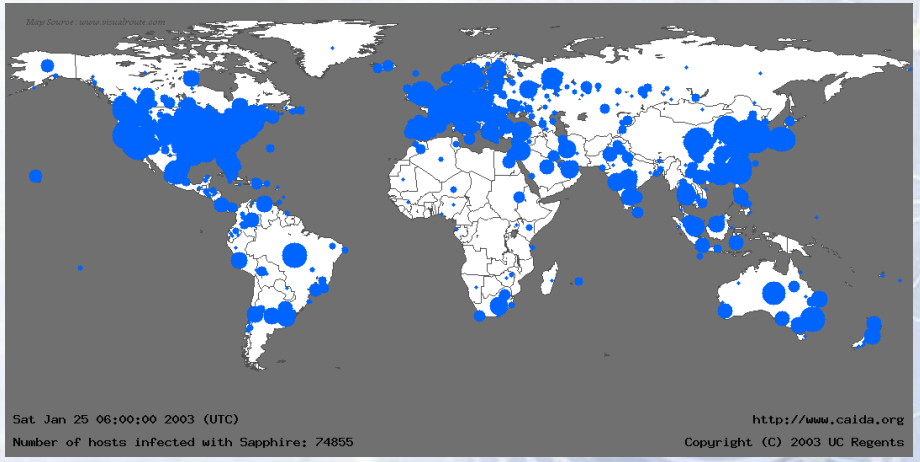
This worm targeted Windows-based systems with Microsoft IIS (Internet Information Services for Windows Server) installed exploiting **buffer overflow**.

The Code Red worm has three phases

- Discovery and propagation… between day 1 and day 19 of any month
- Attack …between day 20 and day 27 of any month…DDoS attack on White House Web site
- Dormancy…From day 28 to the end of the month

SQL Slammer Virus Attack

SQL Slammer is a 2003 computer worm that caused a denial of service on some Internet hosts and dramatically slowed general Internet traffic and crashing routers all around the world. It spread rapidly, infecting most of its 75,000 victims within 10 minutes.



Slammer exploited a buffer overflow vulnerability in Microsoft SQL server 2000 in order to propagate.

Buffer overflow vulnerability in WhatsApp

On May 14, 2019, a buffer overflow vulnerability was identified in WhatsApp.

The vulnerability allows a remote attacker to execute arbitrary code on the target system.

The vulnerability exists due to a boundary error within the WhatsApp VOIP stack when processing SRTCP packets.

A remote attacker can send a series of specially crafted SRTCP packets sent to a target phone number, trigger buffer overflow and execute arbitrary code on the target device.

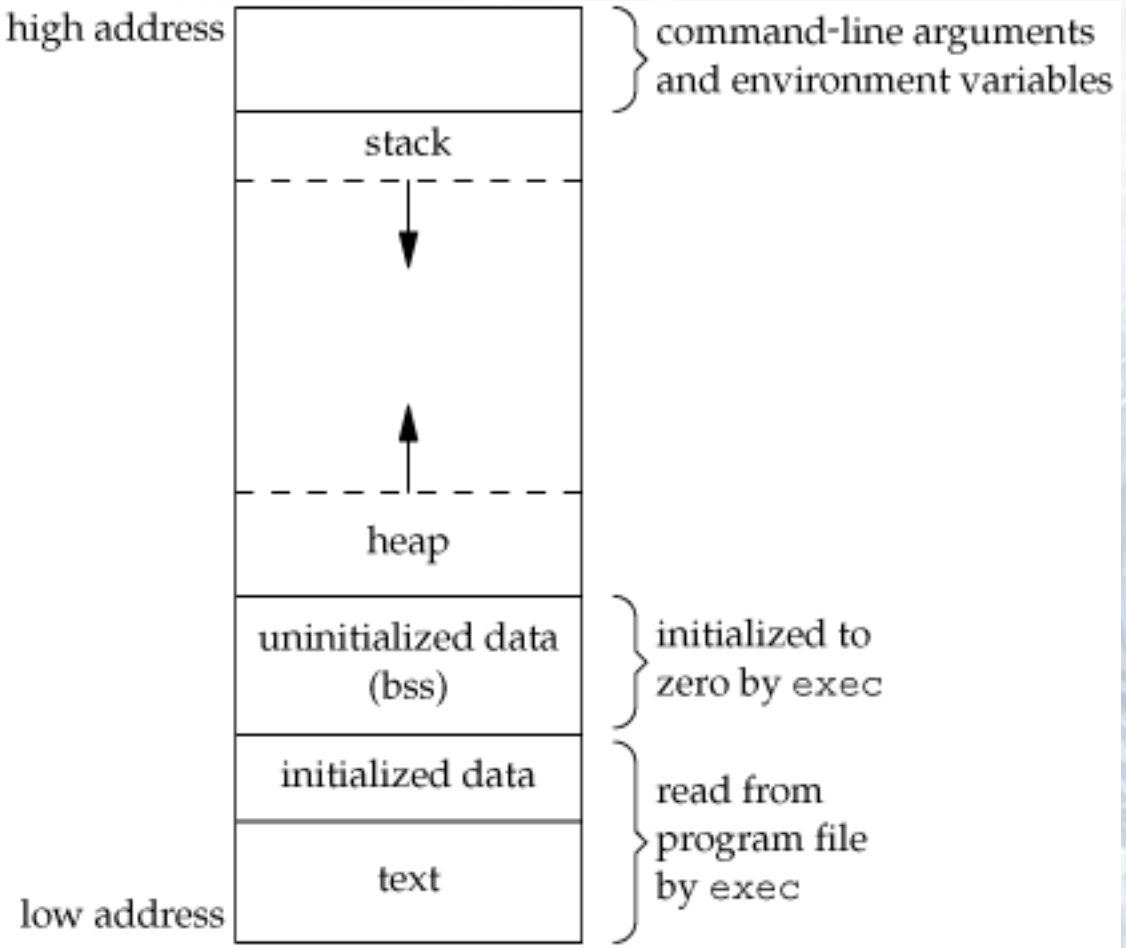
Successful exploitation of this vulnerability may result in complete compromise of vulnerable system.

https://www.whatsapp.com/security/advisories/archive?lang=en_US

Week 11 Topic 2

Program Memory Layout and Stack Buffer

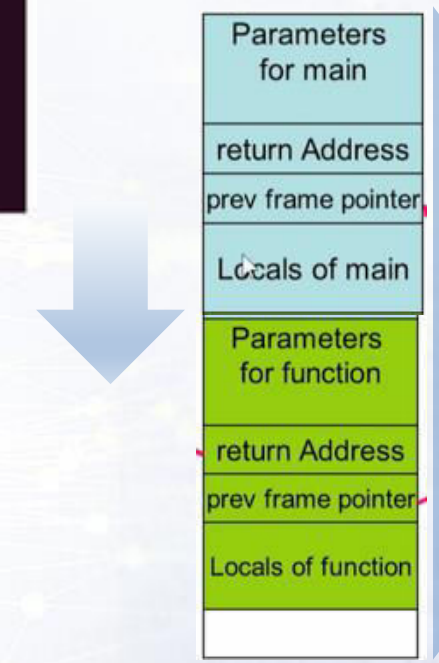
Memory Layout of a C Program

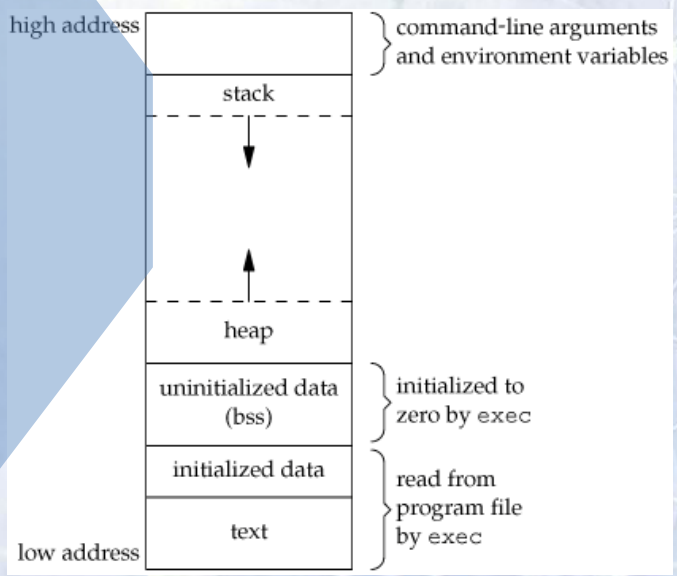


Stack in Program

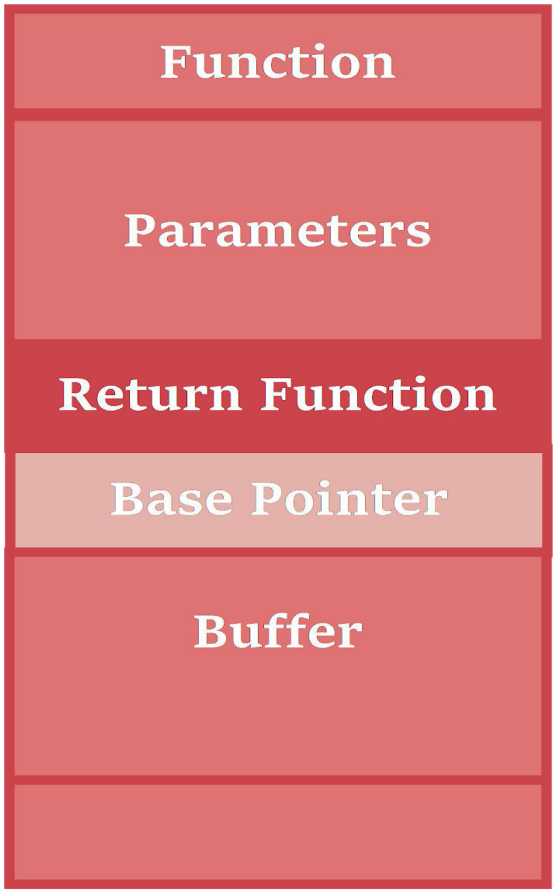
(when functions are executing)

```
void function(int a, int b, int c){
  char buffer1[5];
  char buffer2[10];
}

int main(int argc, char **argv){
  function(1,2,3);
}
```

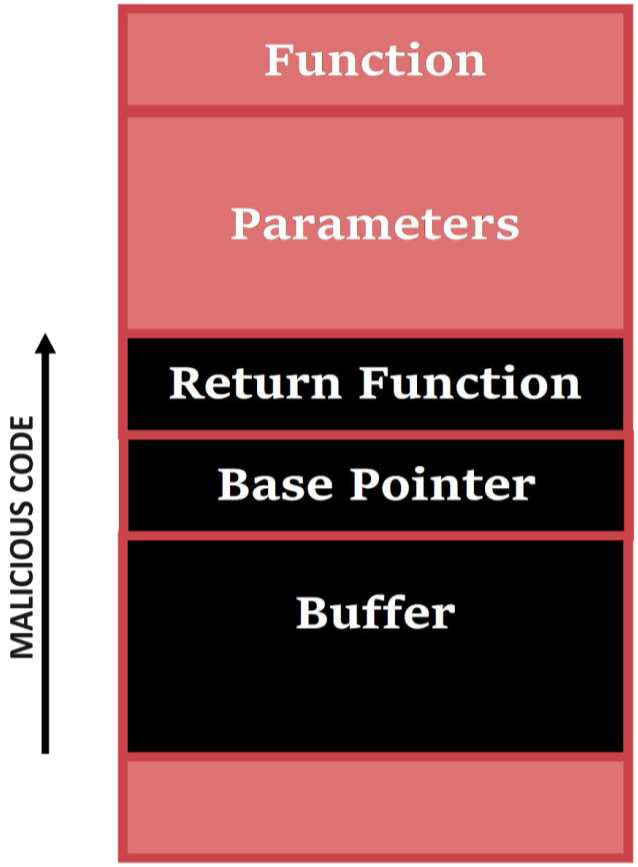




Buffer Overflow Attack



BEFORE ATTACK



AFTER ATTACK

Week 11 Topic 3

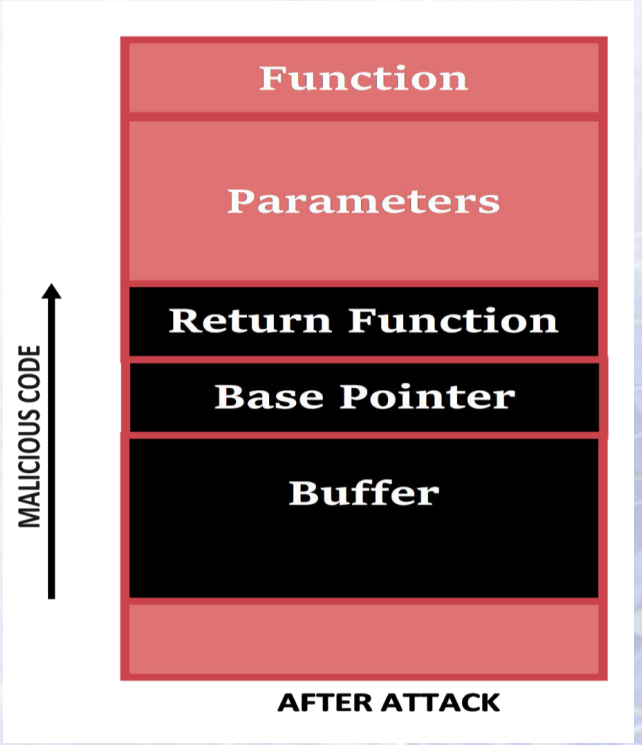
Buffer Overflow Attack

- A very common attack mechanism
  - First widely used by the Morris Worm in 1988
- Prevention techniques known
- Still of major concern
  - Legacy of buggy code in widely deployed operating systems and applications
  - Continued careless programming practices by programmers

Buffer Overflow

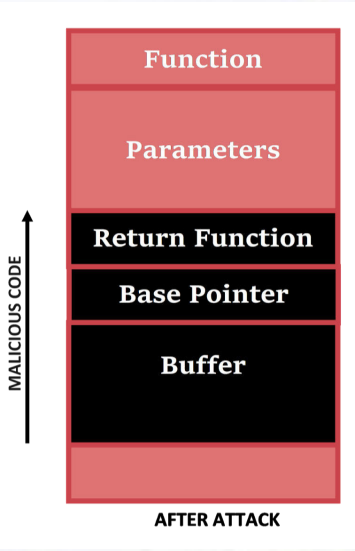
A buffer overflow, also known as a buffer overrun, is defined in the NIST *Glossary of Key Information Security Terms* as follows:

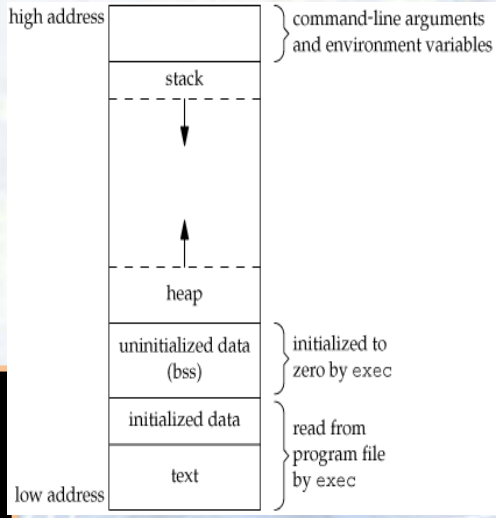
"A condition at an interface under which more input can be placed into a buffer or data holding area than the capacity allocated, overwriting other information. Attackers exploit such a condition to crash a system or to insert specially crafted code that allows them to gain control of the system."



Buffer Overflow Basics

- Programming error when a process attempts to store data beyond the limits of a fixed-sized buffer
- Overwrites adjacent memory locations
  - Locations could hold other program variables, parameters, or program control flow data
- Buffer could be located on the stack, in the heap, or in the data section of the process





Consequences:

- Corruption of program data
- Unexpected transfer of control
- Memory access violations
- Execution of code chosen by attacker

Buffer Overflow Attacks

- To exploit a buffer overflow an attacker needs:
  - To identify a buffer overflow vulnerability in some program that can be triggered using externally sourced data under the attacker's control
  - To understand how that buffer is stored in memory and determine potential for corruption
- Identifying vulnerable programs can be done by:
  - Inspection of program source
  - Tracing the execution of programs as they process oversized input
  - Using tools such as fuzzing to automatically identify potentially vulnerable programs

Buffer Overflows Made Easy - YouTube

By The Cyber Mentor

Week 11 Topic 4

Role of Programming Language in Handling Buffer Overflow Attacks

- At the machine level data manipulated by machine instructions executed by the computer processor are stored in either the processor's registers or in memory
- Assembly language programmer is responsible for the correct interpretation of any saved data value

Modern high-level languages have a strong notion of type and valid operations

- **Not vulnerable to buffer overflows**
- **Does incur overhead, some limits on use**

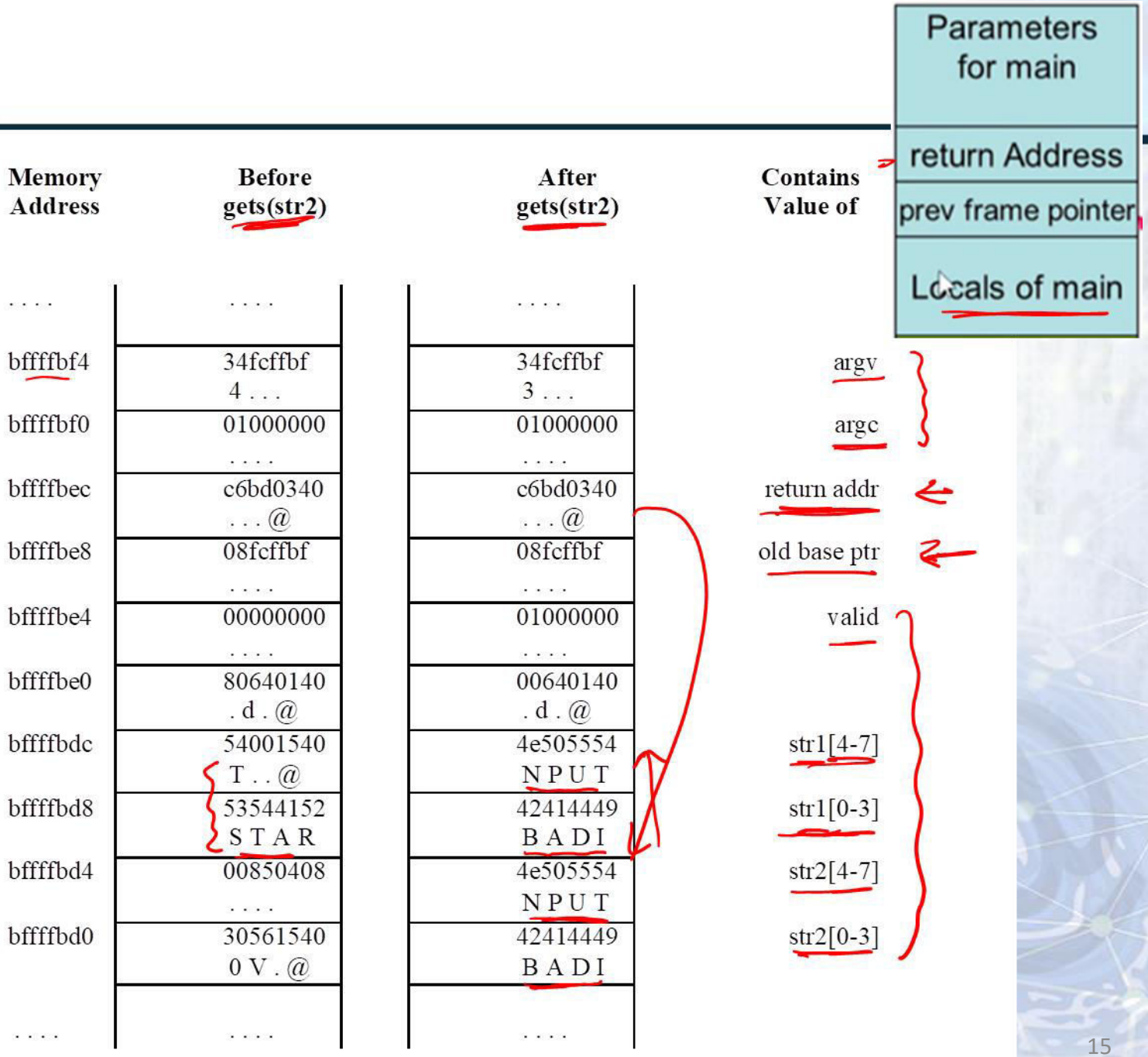
C and related languages have high-level control structures, but allow direct access to memory

- **Hence are vulnerable to buffer overflow**
- **Have a large legacy of widely used, unsafe, and hence vulnerable code**

Basic Buffer Overflow Example

```
int main(int argc, char *argv[]) {
    int valid = FALSE;
    char str1[8];
    char str2[8];

    next tag(str1);
    gets(str2);
    if (strncmp(str1, str2, 8) == 0)
        valid = TRUE;
    printf("buffer1: str1(%s), str2(%s), valid(%d)\n", str1, str2, valid);
}
```

```
$ cc -g -o buffer1 buffer1.c
$ ./buffer1
START
buffer1: str1(START), str2(START), valid(1)
$ ./buffer1
EVILINPUTVALUE
buffer1: str1(TVALUE), str2(EVILINPUTVALUE), valid(0)
$ ./buffer1
BADINPUTBADINPUT
buffer1: str1(BADINPUT), str2(BADINPUTBADINPUT), valid(1)
```

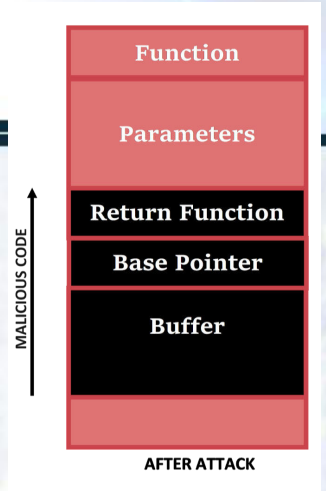

Parameters for main
return Address
prev frame pointer
Locals of main
Memory Address
Before gets(str2)
After gets(str2)
Contains Value of
bffffbf4
34fcffbf 4 . . .
34fcffbf 3 . . .
argv
bffffbf0
01000000 . . . .
01000000 . . . .
argc
bffffbec
c6bd0340 . . . @
c6bd0340 . . . @
return addr
bffffbe8
08fcffbf . . . .
08fcffbf . . . .
old base ptr
bffffbe4
00000000 . . . .
01000000 . . . .
valid
bffffbe0
80640140 . d . @
00640140 . d . @
bffffbdc
54001540 T . . @
4e505554 N P U T
str1[4-7]
bffffbd8
53544152 S T A R
42414449 B A D I
str1[0-3]
bffffbd4
00850408 . . . .
4e505554 N P U T
str2[4-7]
bffffbd0
30561540 0 V . @
42414449 B A D I
str2[0-3]

Some Common Unsafe C Standard Library Routines

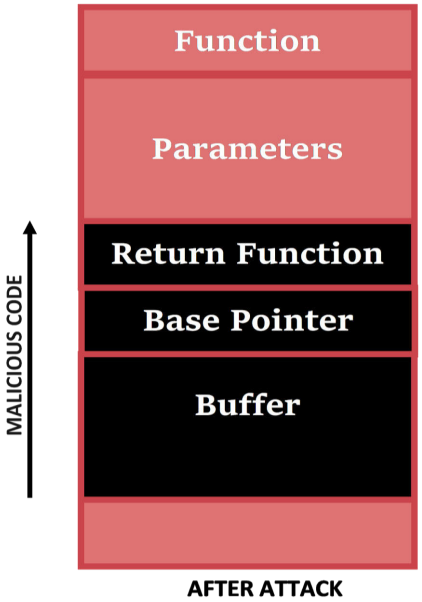
`gets(char *str)`	read line from standard input into str
`sprintf(char *str, char *format, ...)`	create str according to supplied format and variables
`strcat(char *dest, char *src)`	append contents of string src to string dest
`strcpy(char *dest, char *src)`	copy contents of string src to string dest
`vsprintf(char *str, char *fmt, va_list ap)`	create str according to supplied format and variables

Week 11 Topic 5

Shellcode



- Code supplied by attacker
  - Often saved in buffer being overflowed
  - Traditionally transferred control to a user command-line interpreter (shell)
- Machine code
  - Specific to processor and operating system
  - Traditionally needed good assembly language skills to create
  - More recently a number of sites and tools have been developed that automate this process
- Metasploit Project
  - Provides useful information to people who perform penetration, IDS signature development, and exploit research



Example UNIX Shellcode

```
int main(int argc, char *argv[])
{
   char *sh;
   char *args[2];

   sh = "/bin/sh";
   args[0] = sh;
   args[1] = NULL;
   execve(sh, args, NULL);
}
```

```
      nop
      nop                  // end of nop sled
      jmp    find          // jump to end of code
cont:  pop    %esi          // pop address of sh off stack into %esi
      xor    %eax,%eax     // zero contents of EAX
      mov    %al,0x7(%esi)  // copy zero byte to end of string sh (%esi)
      lea    (%esi),%ebx    // load address of sh (%esi) into %ebx
      mov    %ebx,0x8(%esi)  // save address of sh in args[0] (%esi+8)
      mov    %eax,0xc(%esi)  // copy zero to args[1] (%esi+c)
      mov    $0xb,%al       // copy execve syscall number (11) to AL
      mov    %esi,%ebx      // copy address of sh (%esi) t0 %ebx
      lea    0x8(%esi),%ecx  // copy address of args (%esi+8) to %ecx
      lea    0xc(%esi),%edx  // copy address of args[1] (%esi+c) to %edx
      int    $0x80          // software interrupt to execute syscall
find:  call   cont           // call cont which saves next address on stack
sh:    .string "/bin/sh "     // string constant
args:  .long 0               // space used for args array
      .long 0               // args[1] and also NULL for env array
```

```
90 90 eb 1a 5e 31 c0 88 46 07 8d 1e 89 5e 08 89
46 0c b0 0b 89 f3 8d 4e 08 8d 56 0c cd 80 e8 e1
ff ff ff 2f 62 69 6e 2f 73 68 20 20 20 20 20 20
```

Stack Overflow Variants

Target program can be:

- A trusted system utility
- Network service daemon
- Commonly used library code

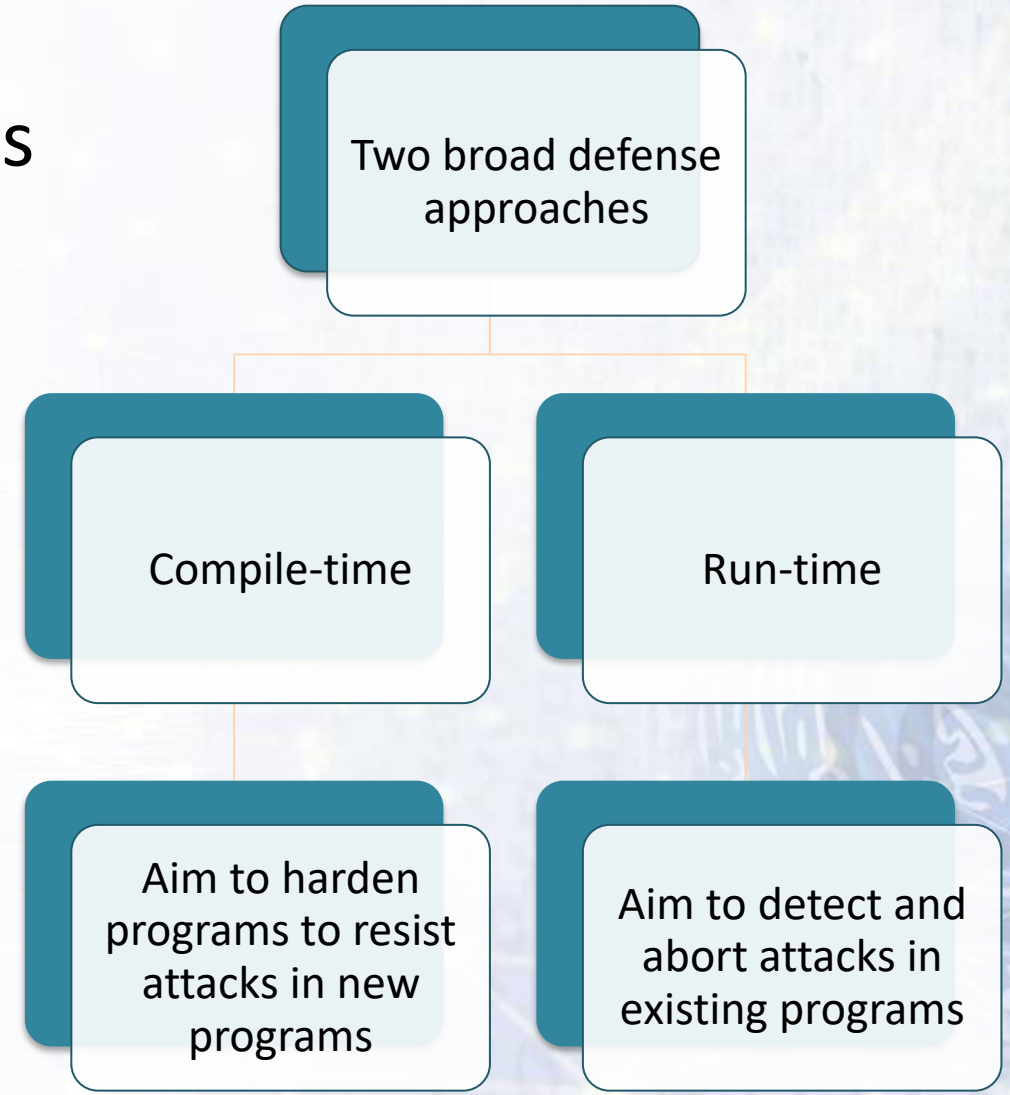
Shellcode functions

- **Launch a remote shell when connected to**
- **Create a reverse shell that connects back to the hacker**
- **Use local exploits that establish a shell**
- **Flush firewall rules that currently block other attacks**
- **Break out of a chroot (restricted execution) environment, giving full access to the system**

Week 11 Topic 6

Buffer Overflow Defenses

- Buffer overflows are widely exploited



Compile-Time Defenses: Programming Language

- Use a modern high-level language
  - Not vulnerable to buffer overflow attacks
  - Compiler enforces range checks and permissible operations on variables

Disadvantages

- Additional code must be executed at run time to impose checks
- Flexibility and safety comes at a cost in resource use
- Distance from the underlying machine language and architecture means that access to some instructions and hardware resources is lost
- Limits their usefulness in writing code, such as device drivers, that must interact with such resources

Compile-Time Defenses: Safe Coding Techniques

- C designers placed much more emphasis on space efficiency and performance considerations than on type safety
  - Assumed programmers would exercise due care in writing code
- Programmers need to inspect the code and rewrite any unsafe coding
  - An example of this is the OpenBSD project
- Programmers have audited the existing code base, including the operating system, standard libraries, and common utilities
  - This has resulted in what is widely regarded as one of the safest operating systems in widespread use

Run-Time Defenses:

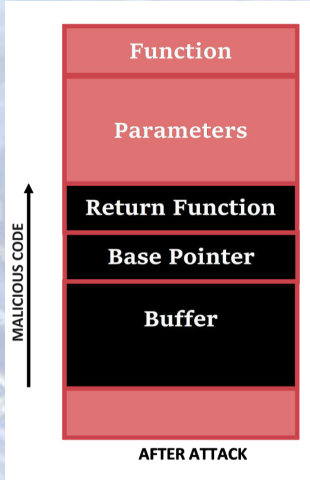
Executable Address Space Protection

Use virtual memory support to make some regions of memory non-executable

- Requires support from memory management unit (MMU)
- Long existed on SPARC / Solaris systems
- Recent on x86 Linux/Unix/Windows systems

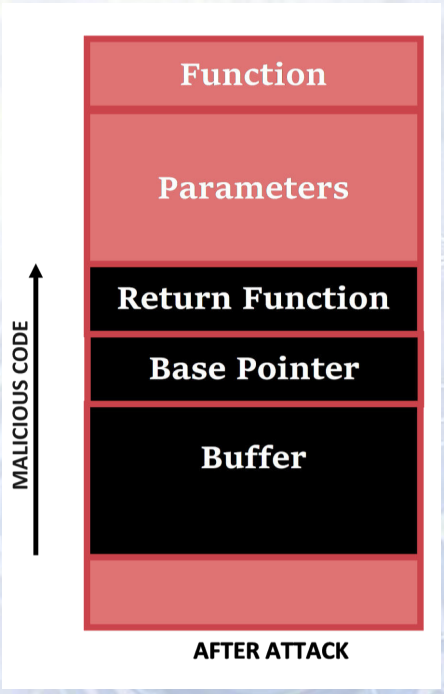
Issues

- Support for executable stack code
- Special provisions are needed



Run-Time Defenses: Address Space Randomization

- Manipulate location of key data structures
  - Stack, heap, global data
  - Using random shift for each process
  - Large address range on modern systems means wasting some has negligible impact
- Randomize location of heap buffers
- Random location of standard library functions



Week 11 Topic 7

CWE/SANS TOP 25 Most Dangerous Software Errors (2011)

Software Error Category: Insecure Interaction Between Components
Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')
Improper Neutralization of Special Elements used in an OS Command ('OS Command Injection')
Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')
Unrestricted Upload of File with Dangerous Type
Cross-Site Request Forgery (CSRF)
URL Redirection to Untrusted Site ('Open Redirect')

Software Error Category: Risky Resource Management
Buffer Copy without Checking Size of Input ('Classic Buffer Overflow')
Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')
Download of Code Without Integrity Check
Inclusion of Functionality from Untrusted Control Sphere
Use of Potentially Dangerous Function
Incorrect Calculation of Buffer Size
Uncontrolled Format String
Integer Overflow or Wraparound

Software Error Category: Porous Defenses
Missing Authentication for Critical Function
Missing Authorization
Use of Hard-coded Credentials
Missing Encryption of Sensitive Data
Reliance on Untrusted Inputs in a Security Decision
Execution with Unnecessary Privileges
Incorrect Authorization
Incorrect Permission Assignment for Critical Resource
Use of a Broken or Risky Cryptographic Algorithm
Improper Restriction of Excessive Authentication Attempts
Use of a One-Way Hash without a Salt

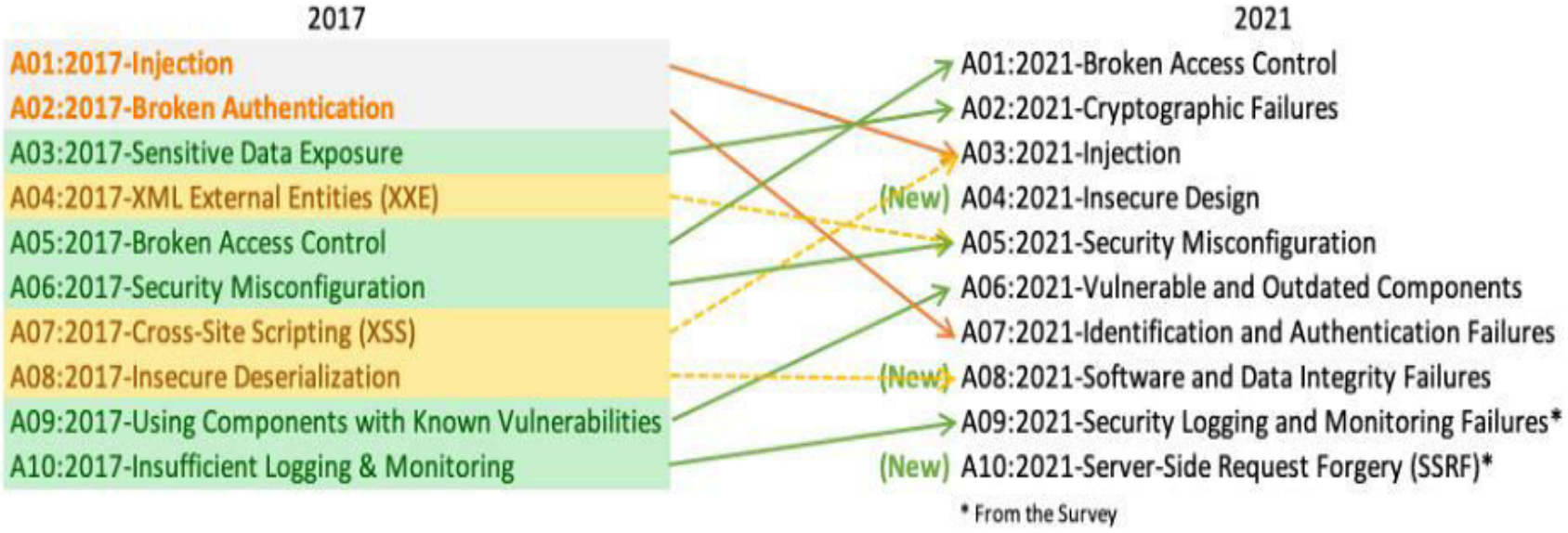
https://cwe.mitre.org/top25/archive/2023/2023_top25_list.html

OWASP Top 10

Web Application Security Risks

Open Web Application Security Project (OWASP)

There are three new categories, four categories with naming and scoping changes, and some consolidation in the Top 10 for 2021.



https://owasp.org/www-project-top-ten/

Security Flaws

- Critical Web application security flaws include five related to insecure software code
  - Unvalidate input
  - Cross-site scripting
  - Buffer overflow
  - Injection flaws
  - Improper error handling
- These flaws occur as a consequence of insufficient checking and validation of data and error codes in programs
- Awareness of these issues is a critical initial step in writing more secure program code
- Emphasis should be placed on the need for software developers to address these known areas of concern

Week 11 Topic 8

Reducing Software Vulnerabilities

- The NIST report NISTIR 8151 presents a range of approaches to reduce the number of software vulnerabilities
- It recommends:
  - Stopping vulnerabilities before they occur by using improved methods for specifying and building software
  - Finding vulnerabilities before they can be exploited by using better and more efficient testing techniques
  - Reducing the impact of vulnerabilities by building more resilient architectures

Week 11 Topic 9

Software Security, Quality and Reliability

- Software quality and reliability

- Software Security

Software Security, Quality and Reliability

- Software quality and reliability:
  - Concerned with the accidental failure of program as a result of some theoretically random, unanticipated input, system interaction, or use of incorrect code
  - Improve using structured design and testing to identify and eliminate as many bugs as possible from a program
  - Concern is not how many bugs, but how often they are triggered
- Software security:
  - Attacker chooses probability distribution, specifically targeting bugs that result in a failure that can be exploited by the attacker
  - Triggered by inputs that differ dramatically from what is usually expected
  - Unlikely to be identified by common testing approaches

Week 11 Topic 10

Defensive Programming

- Designing and implementing software so that it continues to function even when under attack
- Requires attention to all aspects of program execution, environment, and type of data it processes
- Software is able to detect erroneous conditions resulting from some attack
- Also referred to as secure programming
- Key rule is to never assume anything, check all assumptions and handle any possible error states

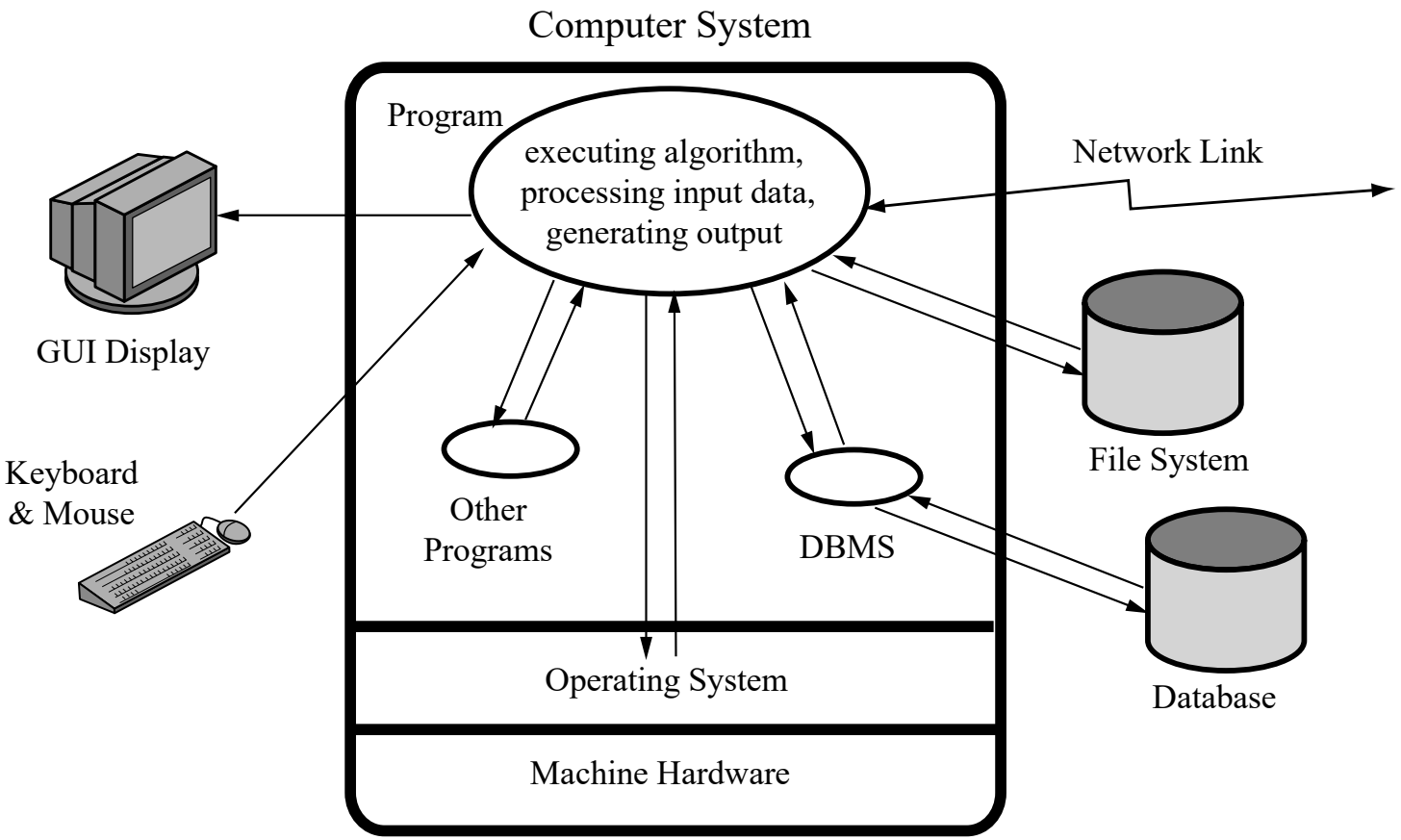


Figure 11.1 Abstract View of Program

Defensive Programming

- Programmers often make assumptions about the type of inputs a program will receive and the environment it executes in
  - Assumptions need to be validated by the program and all potential failures handled gracefully and safely
- Requires a changed mindset to traditional programming practices
  - Programmers have to understand how failures can occur and the steps needed to reduce the chance of them occurring in their programs

Conflicts with business pressures to keep development times as short as possible to maximize market advantage

Week 11 Topic 11

Handling Program Input

Incorrect handling of input data is a very common failing

Input is any source of data from outside and whose value is not explicitly known by the programmer when the code was written

Must identify all data sources

Explicitly validate assumptions on size and type of values before use

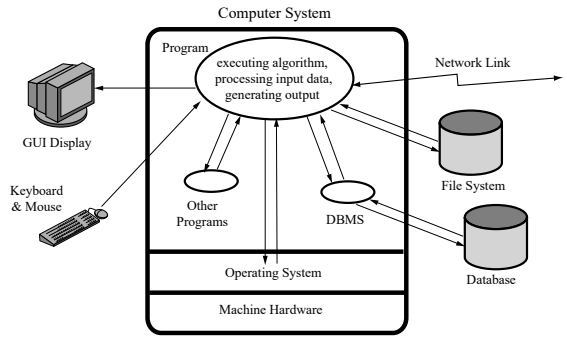


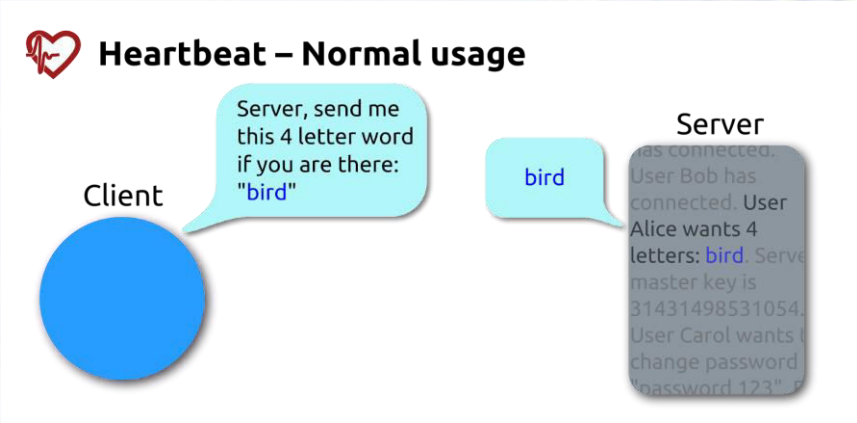
Figure 11.1 Abstract View of Program

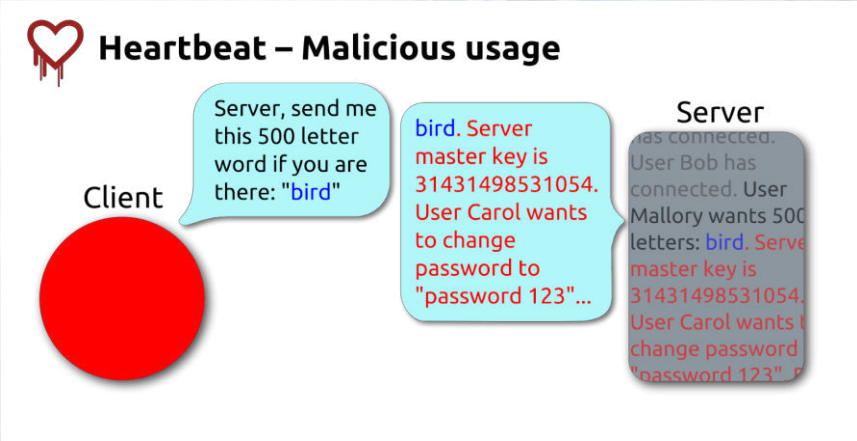
Input Size & Buffer Overflow

- Programmers often make assumptions about the maximum expected size of input
  - Allocated buffer size is not confirmed
  - Resulting in buffer overflow
- Testing may not identify vulnerability
  - Test inputs are unlikely to include large enough inputs to trigger the overflow
- Safe coding treats all input as dangerous

Interpretation of Program Input

- Program input may be binary or text
  - Binary interpretation depends on encoding and is usually application specific
- There is an increasing variety of character sets being used
  - Care is needed to identify just which set is being used and what characters are being read
- Failure to validate may result in an exploitable vulnerability
- 2014 Heartbleed OpenSSL bug is a recent example of a failure to check the validity of a binary input value





Week 11 Topic 12

Injection Attacks

- Flaws relating to invalid handling of input data, specifically when program input data can accidentally or deliberately influence the flow of execution of the program

Most often occur in scripting languages
• Encourage reuse of other programs and system utilities where possible to save coding effort

SQL Injection Attack Demo

```
$name = $_REQUEST['name'];
$query = "SELECT * FROM suppliers WHERE name = '" . $name . "';"
$result = mysql_query($query);
```

(a) Vulnerable PHP code

```
$name = $_REQUEST['name'];
$query = "SELECT * FROM suppliers WHERE name = '" .
    mysql_real_escape_string($name) . "';"
$result = mysql_query($query);
```

(b) Safer PHP code

Figure 11.3 SQL Injection Example

https://www.codingame.com/playgrounds/154/sql-injection-demo/sql-injection

```
$name = $_REQUEST['name'];
$query = "SELECT * FROM suppliers WHERE name = '" . $name . "';"
$result = mysql_query($query);
```

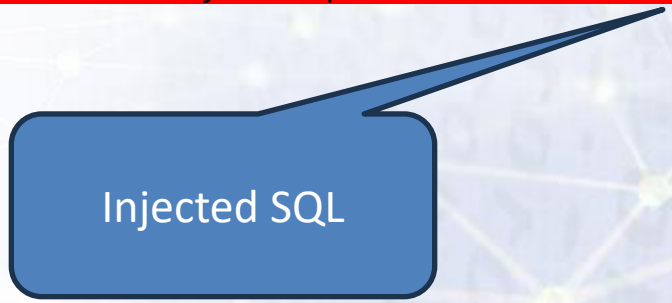
(a) Vulnerable PHP code

```
$name = $_REQUEST['name'];
$query = "SELECT * FROM suppliers WHERE name = '" .
    mysql_real_escape_string($name) . "';"
$result = mysql_query($query);
```

(b) Safer PHP code

Figure 11.3 SQL Injection Example

```
query = "SELECT name FROM user where username = '" + username + "' and password = '" + password  + "'";

query = "SELECT name FROM user where username = 'admin' and password = 'admin123' ";

query = "SELECT name FROM user where username = 'any' and password = '123' or '1'='1'";
```



Week No. 12

Digital Forensic

Week 12 Topic 1

Forensic Science

- Forensic science, also known as criminalistics, is the application of science to criminal and civil laws.

- During criminal investigation in particular, it is governed by the legal standards of admissible evidence and criminal procedure.

- It is a broad field utilizing numerous practices such as the analysis of DNA, fingerprints, bloodstain patterns, firearms, ballistics, and toxicology.

Forensic Science

- Forensic scientists collect, preserve, and analyze scientific evidence during the course of an investigation.
- While some forensic scientists travel to the scene of the crime to collect the evidence themselves, others occupy a laboratory role, performing analysis on objects brought to them by other individuals.
- Still others are involved in analysis of financial, banking, or other numerical data for use in financial crime investigation, and can be employed as consultants from private firms, academia, or as government employees.
- In addition to their laboratory role, forensic scientists testify as expert witnesses in both criminal and civil cases and can work for either the prosecution or the defense.

Origin of Forensic Science

Origins of forensic science and early methods:

- The ancient world lacked standardized forensic practices, which enabled criminals to escape punishment.
- Criminal investigations and trials relied heavily on forced confessions and witness testimony.
- However, ancient sources do contain several accounts of techniques that foreshadow concepts in forensic science developed centuries later.

Week 12 Topic 2

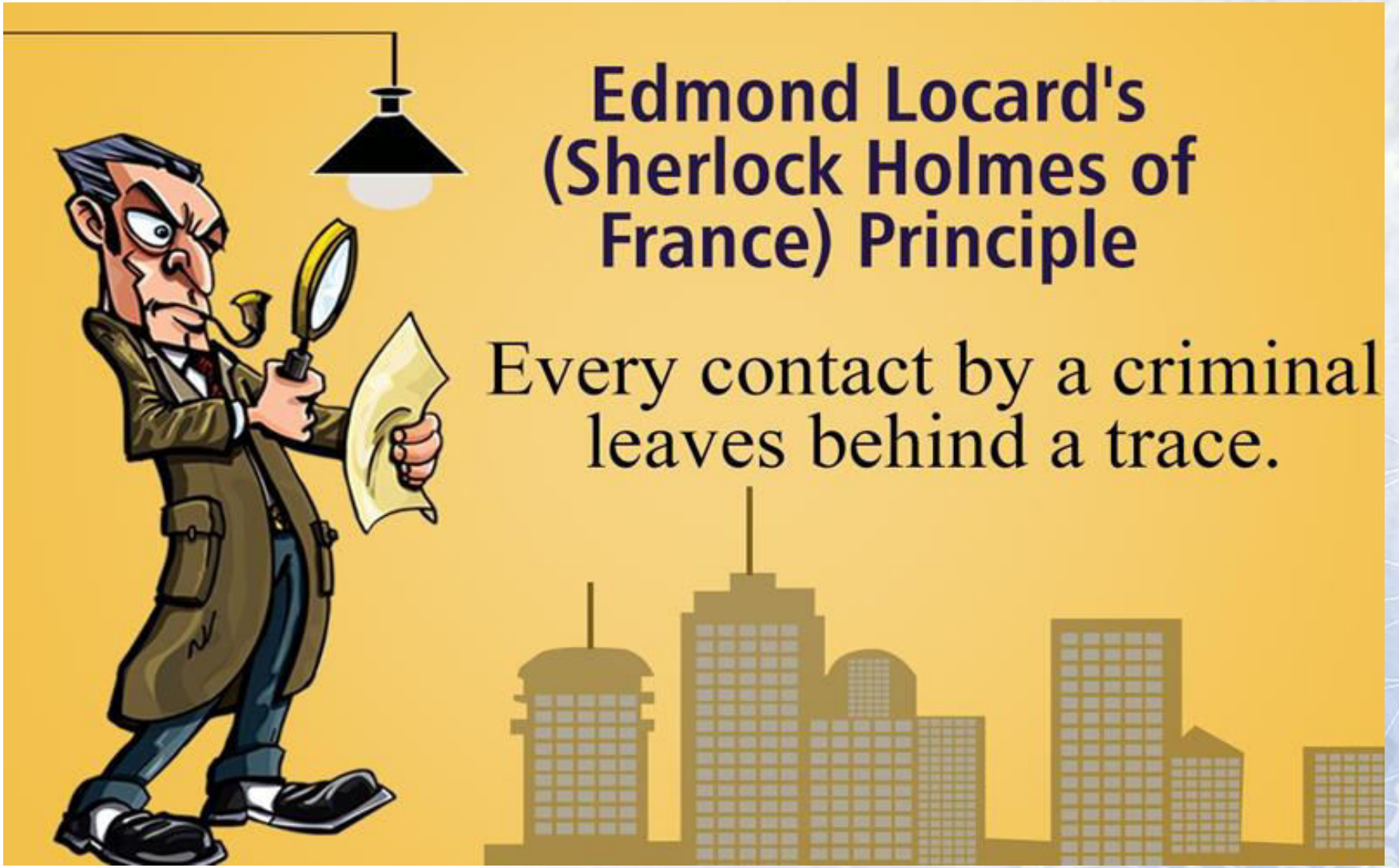
Locard's Exchange Principle

- In forensic science, Locard's principle holds that the perpetrator of a crime will bring something into the crime scene and leave with something from it, and that both can be used as forensic evidence.
- Dr. Edmond Locard (1877–1966) was a pioneer in forensic science who became known as the Sherlock Holmes of Lyon, France.
- He formulated the basic principle of forensic science as: "Every contact leaves a trace".
- It is generally understood as "with contact between two items, there will be an exchange."

Locard's Exchange Principle

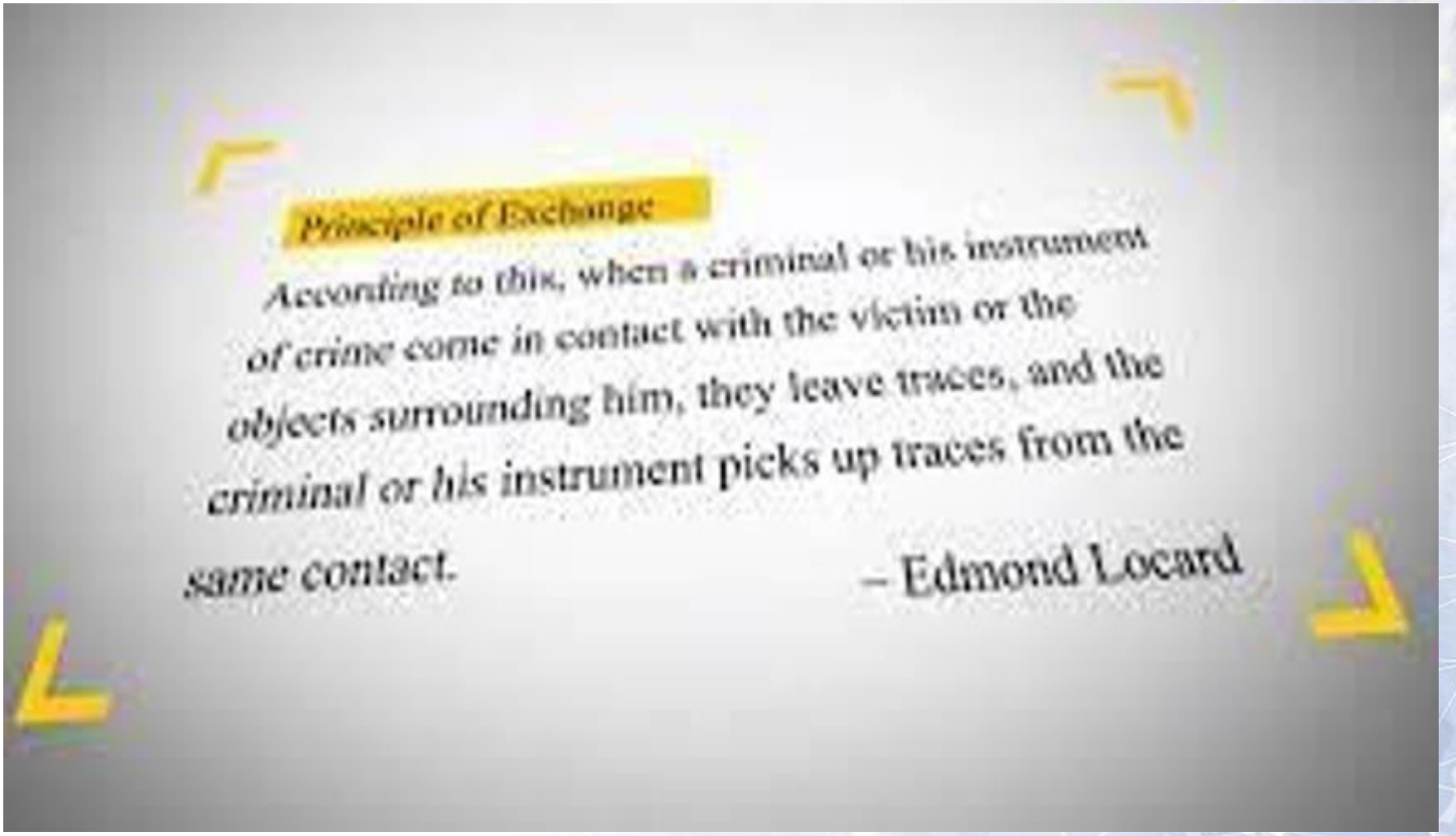
Paul L. Kirk expressed the principle as follows:

- Wherever he steps, whatever he touches, whatever he leaves, even unconsciously, will serve as a silent witness against him.
- Not only his fingerprints or his footprints, but his hair, the fibres from his clothes, the glass he breaks, the tool mark he leaves, the paint he scratches, the blood or semen he deposits or collects.
- All of these and more, bear mute witness against him. This is evidence that does not forget. It is not confused by the excitement of the moment. It is not absent because human witnesses are. It is factual evidence. Physical evidence cannot be wrong, it cannot perjure itself, it cannot be wholly absent. Only human failure to find it, study and understand it, can diminish its value.



https://sciencestruck.com/locards-exchange-principle-in-forensic-science

- Fragmentary or trace evidence is any type of material left at (or taken from) a crime scene, or the result of contact between two surfaces, such as shoes and the floor covering or soil, or fibres from where someone sat on an upholstered chair.
- When a crime is committed, fragmentary (or trace) evidence needs to be collected from the scene.
- A team of specialised police technicians goes to the scene of the crime and seals it off. • They record video and take photographs of the crime scene, victim/s (if there are any) and items of evidence.
- If necessary, they undertake ballistics examinations.
- They check for foot, shoe, and tire mark impressions, plus hair as well as examine any vehicles and check for fingerprints – whole or partial



https://www.facebook.com/evidigators/videos/principle-of-exchange-locards-law/1061852050665075/

Week 12 Topic 3

Digital Forensics & Digital Evidence

What is computer/digital/cyber forensics?

- Cyber forensics is the discovery, analysis, and reconstruction of evidence extracted from any element of Computer systems, Computer networks, Computer media, and Computer peripherals that allow investigators to solve a crime.
- Digital forensics (sometimes known as digital forensic science) is a branch of forensic science encompassing the recovery, investigation, examination, and analysis of material found in digital devices, often in relation to mobile devices and computer crime.
- The term "digital forensics" was originally used as a synonym for computer forensics but has expanded to cover investigation of all devices capable of storing digital data.

Media types used for computer forensic analysis: a Fujifilm FinePix digital camera, two flash memory cards, a USB flash drive, a 5GB iPod, a CD-R or DVD recordable, and a Mini CD.

https://en.wikipedia.org/wiki/File:PersonalStorageDevices.agr.jpg

https://dxhub.calpoly.edu/challenges/digital-evidence/

11:59
13:45

Week 12 Topic 4

Nature of Evidence

Nature of Digital Evidence:

- Physical Form
- Virtual Locations / Paths
- Volatile vs Non-volatile

Week 12 Topic 5

Applications & Uses

- Digital forensics investigations have a variety of applications.
- The most common is to support or refute a hypothesis before criminal or civil courts.
- Criminal cases involve the alleged breaking of laws that are defined by legislation and enforced by the police and prosecuted by the state, such as murder, theft, and assault against the person.
- Civil cases, on the other hand, deal with protecting the rights and property of individuals (often associated with family disputes), but may also be concerned with contractual disputes between commercial entities where a form of digital forensics referred to as electronic discovery (ediscovery) may be involved.
- Forensics may also feature in the private sector, such as during internal corporate investigations or intrusion investigations (a special probe into the nature and extent of an unauthorized network intrusion).
- As well as identifying direct evidence of a crime, digital forensics can be used to attribute evidence to specific suspects, confirm alibis or statements, determine intent, identify sources (for example, in copyright cases), or authenticate documents.
- Investigations are much broader in scope than other areas of forensic analysis (where the usual aim is to provide answers to a series of simpler questions), often involving complex time-lines or hypotheses.

- Digital forensics is commonly used in both criminal law and private investigation.
- Traditionally it has been associated with criminal law, where evidence is collected to support or oppose a hypothesis before the courts.
- As with other areas of forensics this is often a part of a wider investigation spanning a number of disciplines.
- In some cases, the collected evidence is used as a form of intelligence gathering, used for other purposes than court proceedings (for example to locate, identify or halt other crimes).
- In civil litigation or corporate matters digital forensics forms part of the electronic discovery (or eDiscovery) process.
- Forensic procedures are similar to those used in criminal investigations, often with different legal requirements and limitations.
- Outside of the courts digital forensics can form a part of internal corporate investigations.
- A common example might be following unauthorized network intrusion.
- A specialist forensic examination into the nature and extent of the attack is performed as a damage limitation exercise, both to establish the extent of any intrusion and in an attempt to identify the attacker.

Week 12 Topic 6

Branches & Divisions

- Digital forensics is the application of proven scientific methods and techniques in order to recover data from electronic / digital media. Digital Forensic specialists work in the field as well as in the lab.
- Digital forensics investigation is not restricted to retrieve data merely from the computer, as laws are breached by the criminals and small digital devices (e.g. tablets, smartphones, flash drives) are now extensively used.
- Some of these devices have volatile memory while some have non-volatile memory.
- Sufficient methodologies are available to retrieve data from volatile memory, however, there is lack of detailed methodology or a framework for data retrieval from non-volatile memory sources.
- Also, the technical aspect of an investigation is divided into several sub-branches related to the type of digital devices involved:
  - computer forensics, network forensics, forensic data analysis, and mobile device forensics, router forensics.
- Depending on the type of devices, media or artifacts, digital forensics investigation is branched into various types.

Digital Forensic Categories / Subdivisions / Branches

- Computer Forensics
- Mobile Device Forensics
- Network Forensics [Logs]
- Forensic Data Analysis
- Digital Image Forensics
- Database Forensics
- Cloud Forensics
- IoT Forensics
- Dark Web Forensics
- Audio Analysis/Forensics
- Video Analysis/Forensics
- E-Mail Forensics

Computer forensics

- The goal of computer forensics is to explain the current state of a digital artifact; such as a computer system, storage medium or electronic document.
- The discipline usually covers computers, embedded systems (digital devices with rudimentary computing power and onboard memory) and static memory (such as USB pen drives).
- Computer forensics can deal with a broad range of information; from logs (such as internet history) through to the actual files on the drive.
- In 2007 prosecutors used a spreadsheet recovered from the computer of Joseph Edward Duncan to show premeditation and secure the death penalty.
- Sharon Lopatka's killer was identified in 2006 after email messages from him detailing torture and death fantasies were found on her computer.

Mobile device forensics

- Mobile device forensics is a sub-branch of digital forensics relating to recovery of digital evidence or data from a mobile device.
- Mobile device forensics is the scientific examination and evaluation of evidence found in mobile phones, e.g. Call History and Deleted SMS, and includes SIM Card Forensics.
- It differs from Computer forensics in that a mobile device will have an inbuilt communication system (e.g. GSM) and, usually, proprietary storage mechanisms.
- Investigations usually focus on simple data such as call data and communications (SMS/Email) rather than in-depth recovery of deleted data.
- SMS data from a mobile device investigation helped to exonerate Patrick Lumumba in the murder of Meredith Kercher.
- Mobile devices are also useful for providing location information; either from inbuilt gps/location tracking or via cell site logs, which track the devices within their range.
- Such information used to track down kidnappers of Thomas Onofri in 2006.

Network forensics

- Network forensics is concerned with the monitoring and analysis of computer network traffic, both local and WAN/internet, for the purposes of information gathering, evidence collection, or intrusion detection.
- Traffic is usually intercepted at the packet level, and either stored for later analysis or filtered in real-time.
- Unlike other areas of digital forensics network data is often volatile and rarely logged, making the discipline often reactionary.

- **Forensic data analysis**
  - Forensic Data Analysis is a branch of digital forensics. It examines structured data with the aim to discover and analyze patterns of fraudulent activities resulting from financial crime.

- **Digital image forensics**
  - Digital image forensics (or forensic image analysis) is a branch of digital forensics that deals with examination and verification of an image's authenticity and content.
  - These can range from Stalin-era airbrushed photos to elaborate deepfake videos.
  - This has broad implications for a wide variety of crimes, for determining the validity of information presented in civil and criminal trials, and for verifying images and information that are circulated through news and social media.

- **Forensic video analysis** is the scientific examination, comparison and evaluation of video in legal matters

- **Database forensics**
  - Database forensics is a branch of digital forensics relating to the forensic study of databases and their metadata.
  - Investigations use database contents, log files and in-RAM data to build a timeline or recover relevant information.

- **IoT Forensics**
  - IoT forensics is a branch of Digital forensics that has the goal of identifying and extracting digital information from devices belonging to the Internet of things field, to be used for forensics investigations as potential source of evidence

Week 12 Topic 7

Digital Forensic Standards & Frameworks

- Since 2000, in response to the need for standardization, various bodies and agencies have published guidelines for digital forensics.

1. The Scientific Working Group on Digital Evidence (SWGDE) produced a 2002 paper, Best practices for Computer Forensics.
2. This was followed, in 2005, by the publication of an ISO standard (ISO 17025, General requirements for the competence of testing and calibration laboratories).
3. NIST

Week 12 Topic 8

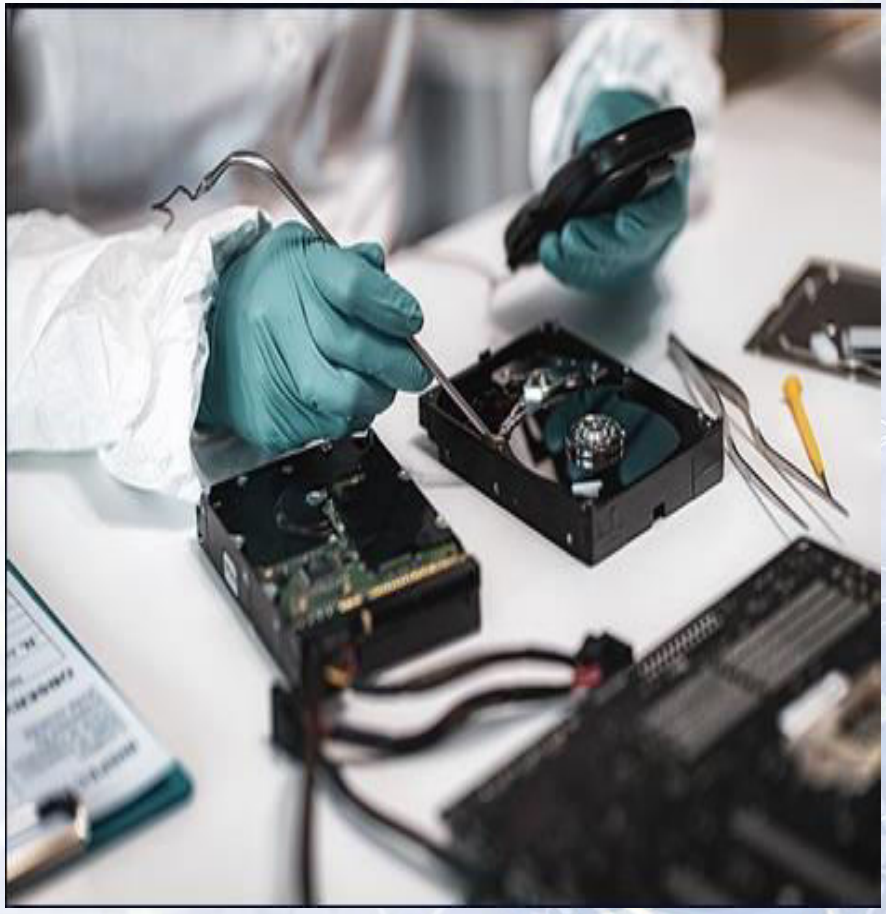
Forensic Lab Setup

- Laws & Regulations
- Types of Incidents
- Important Numbers /Emergency Contacts
- Physical Location
- Team Members &Personnel
- Forensic Expertise
- HVAC & Lighting
- Building Structure
- **Tools**
  - Hardware
    - Forensic Machines
    - Cables & Connectors
  - Software
  - Write Blocker
  - Duplicator
  - Storage
  - Data & Equipment

Week 12 Topic 9

Digital Forensic Process

- The typical forensic process encompasses the seizure, forensic imaging (acquisition), and analysis of digital media, followed with the production of a report of the collected evidence.



https://www.uco.edu/programs/microcredentials/digital-evidence

Digital forensic investigation

- A digital forensic investigation commonly consists of 3 stages:
    - acquisition or imaging of exhibits, analysis, and reporting.

1. Ideally acquisition involves capturing an image of the computer's volatile memory (RAM) and creating an exact sector level duplicate (or "forensic duplicate") of the media, often using a write blocking device to prevent modification of the original.
    - However, the growth in size of storage media and developments such as cloud computing have led to more use of 'live' acquisitions whereby a 'logical' copy of the data is acquired rather than a complete image of the physical storage device.
    - Both acquired image (or logical copy) and original media/data are hashed (using an algorithm such as SHA-1 or MD5) and the values compared to verify the copy is accurate.

Digital forensic investigation

2. During the analysis phase an investigator recovers evidence material using a number of different methodologies and tools.

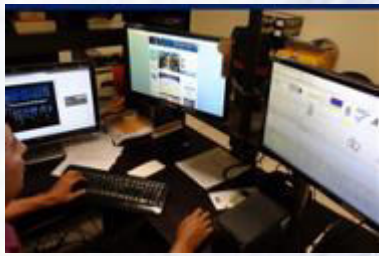
- In 2002, an article in the International Journal of Digital Evidence referred to this step as "an in-depth systematic search of evidence related to the suspected crime."
- In 2006, forensics researcher Brian Carrier described an "intuitive procedure" in which obvious evidence is first identified and then "exhaustive searches are conducted to start filling in the holes."
- The actual process of analysis can vary between investigations, but common methodologies include conducting keyword searches across the digital media (within files as well as unallocated and slack space), recovering deleted files and extraction of registry information (for example to list user accounts, or attached USB devices).
- The evidence recovered is analysed to reconstruct events or actions and to reach conclusions, work that can mostly be performed by specialised staff.

3. When an investigation is complete the data is presented, usually in the form of a written report, in lay persons' terms.

Digital forensic

Identification

Where is the evidence? In which country?
Which device(s) does it need to be gathered from?
Is the data “at rest” or “in transit”?

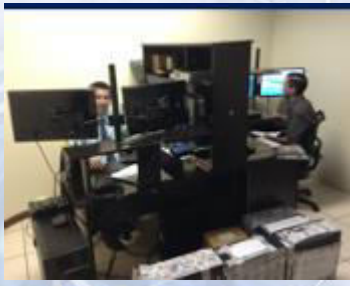


Preservation

How will the data be accessed? Which legislation applies? Is the action proportionate, lawful, accountable and necessary?

Analysis

Are the required technical and people skills available to acquire, analyze and exploit the evidence in a timely manner with full chain-of-custody?



Documentation

Ensure EVERYTHING is documented to prevent losing evidence due to legal or procedural mistakes

Presentation

Evidence should be presented accurately, concisely & clearly. Can the judge and/or jury understand it? Can it be tested and challenged by the defense?

An example of an image's Exif metadata that might be used to prove its origin

https://en.wikipedia.org/wiki/File:En_exif_data.png

Camera manufacturer	Canon
Camera model	Canon EOS 400D DIGITAL
Exposure time	1/60 sec (0.016666666666667)
F-number	f/4.5
ISO speed rating	400
Date and time of data generation	11:06, August 27, 2010
Lens focal length	31 mm
Show extended details	

Week 12 Topic 10

Forensic Reporting

- Title Page [Case number/title, reporting date, number of pages, examiner name]
- Case Summary
- Evidence Details or Particulars
- Tools Used
- Methodology / Steps
- Forensic Analysis & Findings
- Forensic Investigator/Examiner Signature
- Images of Evidentiary Items
- Chain of Custody

Week 9 Topic 11

Development of Digital Forensic Tools

Development of Digital Forensic Tools:

- During the 1980s, very few specialized digital forensic tools existed.
- Consequently, investigators often performed live analysis on media, examining computers from within the operating system using existing sysadmin tools to extract evidence.
- This practice carried the risk of modifying data on the disk, either inadvertently or otherwise, which led to claim of evidence tampering.
- A number of tools were created during the early 1990s to address the problem.
- The need for such software was first recognized in 1989 at the Federal Law Enforcement Training Center, resulting in the creation of IMDUMP (by Michael White) and in 1990, SafeBack (developed by Sydex).

Development of Digital Forensic Tools

- Similar software was developed in other countries; DIBS (a hardware and software solution) was released commercially in the UK in 1991, and Rob McKemmish released Fixed Disk Image free to Australian law enforcement.
- These tools allowed examiners to create an exact copy of a piece of digital media to work on, leaving the original disk intact for verification.
- By the end of the 1990s, as demand for digital evidence grew, more advanced commercial tools such as EnCase and FTK were developed, allowing analysts to examine copies of media without using any live forensics.
- More recently, a trend towards "live memory forensics" has grown, resulting in the availability of tools such as WindowsSCOPE.
- The same progression of tool development has occurred for mobile devices; initially investigators accessed data directly on the device, but soon specialist tools such as Cellebrite, XRY or Radio Tactics Aceso appeared.

https://interworks.com/blog/bstephens/2016/02/05/what-digital-forensics/

Digital Forensic Tools

- Building Structure
- **Tools**
  - Hardware
    - Forensic Machines
    - Cables & Connectors
  - Software
  - Write Blocker
  - Duplicator
  - Storage
  - Data & Equipment

Mobile phones in an Evidence bag

https://en.wikipedia.org/wiki/File:Mobiles.JPG

A portable Tableau write-blocker attached to a hard drive

https://en.wikipedia.org/wiki/File:Portable_forensic_tableau.JPG

Week 12 Topic 12

Limitations

- **Encryption**
  - One major limitation to a forensic investigation is the use of encryption; this disrupts initial examination where pertinent evidence might be located using keywords.
  - Laws to compel individuals to disclose encryption keys are still relatively new and controversial but always more frequently there are solutions to brute force passwords or bypass encryption, such as in smartphones or PCs where by means of bootloader techniques the content of the device can be first acquired and later forced in order to find the password or encryption key.
- **Steganography**
- **Cloud Computing/Forensics**
  - Jurisdiction
  - Virtualization
  - Elasticity

Anti-Forensic Techniques

- Passwords
- Encryption
- Jurisdiction
- Proxy Chaining
- Steganography
- NTFS Data Streams

Data hiding

- Data hiding is the process of making data difficult to find while also keeping it accessible for future use.
- "Obfuscation and encryption of data give an adversary the ability to limit identification and collection of evidence by investigators while allowing access and use to themselves."
- Some of the more common forms of data hiding include encryption, steganography and other various forms of hardware/software based data concealment.
- Each of the different data hiding methods makes digital forensic examinations difficult.
- When the different data hiding methods are combined, they can make a successful forensic investigation nearly impossible.
- Encryption
- Steganography
- Other forms of data hiding

Data hiding: Encryption

- One of the more commonly used techniques to defeat computer forensics is data encryption.
- In a presentation given on encryption and anti-forensic methodologies, the Vice President of Secure Computing, Paul Henry, referred to encryption as a "forensic expert's nightmare".
- The majority of publicly available encryption programs allow the user to create virtual encrypted disks which can only be opened with a designated key.
  - Through the use of modern encryption algorithms and various encryption techniques these programs make the data virtually impossible to read without the designated key.
- File level encryption encrypts only the file contents.
  - This leaves important information such as file name, size and timestamps unencrypted.
  - Parts of the content of the file can be reconstructed from other locations, such as temporary files, swap file and deleted, unencrypted copies.
- Most encryption programs have the ability to perform a number of additional functions that make digital forensic efforts increasingly difficult.
  - Some of these functions include the use of a keyfile, full-volume encryption, and plausible deniability.
  - The widespread availability of software containing these functions has put the field of digital forensics at a great disadvantage.

Data hiding: Steganography

- Steganography is a technique where information or files are hidden within another file in an attempt to hide data by leaving it in plain sight.
- "Steganography produces dark data that is typically buried within light data (e.g., a non-perceptible digital watermark buried within a digital photograph).“
- While some experts have argued that the use of steganography techniques is not very widespread and therefore the subject shouldn't be given a lot of thought, most experts agree that steganography has the capability of disrupting the forensic process when used correctly.
- According to Jeffrey Carr, a 2007 edition of Technical Mujahid (a bi-monthly terrorist publication) outlined the importance of using a steganography program called Secrets of the Mujahideen. According to Carr, the program was touted as giving the user the capability to avoid detection by current steganalysis programs. It did this through the use of steganography in conjunction with file compression

Data hiding: Other forms of data hiding

- Other forms of data hiding involve the use of tools and techniques to hide data throughout various locations in a computer system.
    - Some of these places can include "memory, slack space, hidden directories, bad blocks, alternate data streams, (and) hidden partitions."
- One of the more well known tools that is often used for data hiding is called Slacker (part of the Metasploit framework).
    - Slacker breaks up a file and places each piece of that file into the slack space of other files, thereby hiding it from the forensic examination software.
- Another data hiding technique involves the use of bad sectors.
    - To perform this technique, the user changes a particular sector from good to bad and then data is placed onto that particular cluster.
    - The belief is that forensic examination tools will see these clusters as bad and continue on without any examination of their contents

Artifact wiping

- The methods used in artifact wiping are tasked with permanently eliminating particular files or entire file systems.

- This can be accomplished through the use of a variety of methods that include disk cleaning utilities, file wiping utilities and disk degaussing/destruction techniques.

- Disk cleaning utilities
- File wiping utilities
- Disk degaussing / destruction techniques

Artifact wiping: Disk cleaning utilities

- Disk cleaning utilities use a variety of methods to overwrite the existing data on disks .
- The effectiveness of disk cleaning utilities as anti-forensic tools is often challenged as some believe they are not completely effective.
- Experts who don't believe that disk cleaning utilities are acceptable for disk sanitization base their opinions of current DOD policy, which states that the only acceptable form of sanitization is degaussing.
- Disk cleaning utilities are also criticized because they leave signatures that the file system was wiped, which in some cases is unacceptable.
- Some of the widely used disk cleaning utilities include DBAN, srm, BCWipe Total WipeOut, KillDisk, PC Inspector and CyberScrubs cyberCide.
- Another option which is approved by the NIST and the NSA is CMRR Secure Erase, which uses the Secure Erase command built into the ATA specification.

Artifact wiping: File wiping utilities

- File wiping utilities are used to delete individual files from an operating system.
- The advantage of file wiping utilities is that they can accomplish their task in a relatively short amount of time as opposed to disk cleaning utilities which take much longer.
- Another advantage of file wiping utilities is that they generally leave a much smaller signature than disk cleaning utilities.
- There are two primary disadvantages of file wiping utilities, first they require user involvement in the process and second some experts believe that file wiping programs don't always correctly and completely wipe file information.
- Some of the widely used file wiping utilities include BCWipe, R-Wipe & Clean, Eraser, Aevita Wipe & Delete and CyberScrubs PrivacySuite.
- On Linux tools like shred and srm can be also used to wipe single files.
- SSDs are by design more difficult to wipe, since the firmware can write to other cells therefore allowing data recovery.
- In these instances ATA Secure Erase should be used on the whole drive, with tools like hdparm that support it.

Artifact wiping: Disk degaussing / destruction techniques

- Disk degaussing is a process by which a magnetic field is applied to a digital media device.
- The result is a device that is entirely clean of any previously stored data.
- Degaussing is rarely used as an anti-forensic method despite the fact that it is an effective means to ensure data has been wiped.
- This is attributed to the high cost of degaussing machines, which are difficult for the average consumer to afford.
- A more commonly used technique to ensure data wiping is the physical destruction of the device.
- The NIST recommends that "physical destruction can be accomplished using a variety of methods, including disintegration, incineration, pulverizing, shredding and melting

Trail obfuscation

- The purpose of trail obfuscation is to confuse, disorient, and divert the forensic examination process.
- Trail obfuscation covers a variety of techniques and tools that include "log cleaners, spoofing, misinformation, backbone hopping, zombied accounts, trojan commands."
- One of the more widely known trail obfuscation tools is Timestomp (part of the Metasploit Framework).
    - Timestompgives the user the ability to modify file metadata pertaining to access, creation and modification times/dates.
    - By using programs such as Timestomp, a user can render any number of files useless in a legal setting by directly calling into question the files' credibility.
- • Another well known trail-obfuscation program is Transmogrify (also part of the Metasploit Framework).
    - In most file types the header of the file contains identifying information.
    - A (.jpg) would have header information that identifies it as a (.jpg), a (.doc) would have information that identifies it as (.doc) and so on.
    - Transmogrify allows the user to change the header information of a file, so a (.jpg) header could be changed to a (.doc) header.
    - If a forensic examination program or operating system were to conduct a search for images on a machine, it would simply see a (.doc) file and skip over it.

Attacks against computer forensics

- In the past anti-forensic tools have focused on attacking the forensic process by destroying data, hiding data, or altering data usage information.
- Anti-forensics has recently moved into a new realm where tools and techniques are focused on attacking forensic tools that perform the examinations.
- These new anti-forensic methods have benefited from a number of factors to include well documented forensic examination procedures, widely known forensic tool vulnerabilities, and digital forensic examiners' heavy reliance on their tools.
- During a typical forensic examination, the examiner would create an image of the computer's disks.
    - This keeps the original computer (evidence) from being tainted by forensic tools.
    - Hashes are created by the forensic examination software to verify the integrity of the image.
    - One of the recent anti-tool techniques targets the integrity of the hash that is created to verify the image.
    - By affecting the integrity of the hash, any evidence that is collected during the subsequent investigation can be challenged

Week 12 Topic 13

Digital Forensic Jobs

- Government, Military, Law Enforcement Agencies
- Private sector companies
- Designations:
  - Forensic Examiner / Specialist / Scientist / Manager
- Relevant Degree, Certifications and Experienc

Week 12 Topic 14

Digital Forensic Certifications

- There are several computer forensics certifications available, such as the ISFCE Certified Computer Examiner, Digital Forensics Investigation Professional (DFIP) and IACRB Certified Computer Forensics Examiner.
- The top vendor independent certification (especially within EU) is considered the CCFP - Certified Cyber Forensics Professional.
- Others, worth to mention for USA or APAC are: The International Association of Computer Investigative Specialists offers the Certified Computer Examiner program. The International Society of Forensic Computer Examiners offers the Certified Computer Examiner program.
- Many commercial based forensic software companies are now also offering proprietary certifications on their products.
  - For example, Guidance Software offering the (EnCE) certification on their tool EnCase, AccessData offering (ACE) certification on their tool FTK, PassMark Software offering certification on their tool OSForensics, and X-Ways Software Technology offering (X-PERT) certification for their software, X-Ways Forensics.
- Tool Specific: EnCase, AccessData FTK, Cellebrite
- General: CHFI (Certified Hacking Forensic Investigator by EC-Council), Mile2, eLearnSecurity, SANS GIAC and many more

Week No. 13

Internet Security Protocols

Week 13 Topic 1

HTTPS (Hypertext Transfer Protocol Secure)

- HTTPS (Hypertext Transfer Protocol Secure) is a secure version of the HTTP protocol used for communication between web browsers and web servers.
- HTTPS adds an extra layer of security by encrypting data exchanged between the client and server, ensuring confidentiality and integrity.

How HTTPS Works

HTTPS utilizes the SSL/TLS (Secure Sockets Layer/Transport Layer Security) protocol to establish a secure connection:

Client Request: The client (web browser) initiates a request to the server using the HTTPS protocol.

Server Response: The server responds by presenting its SSL/TLS certificate to the client.

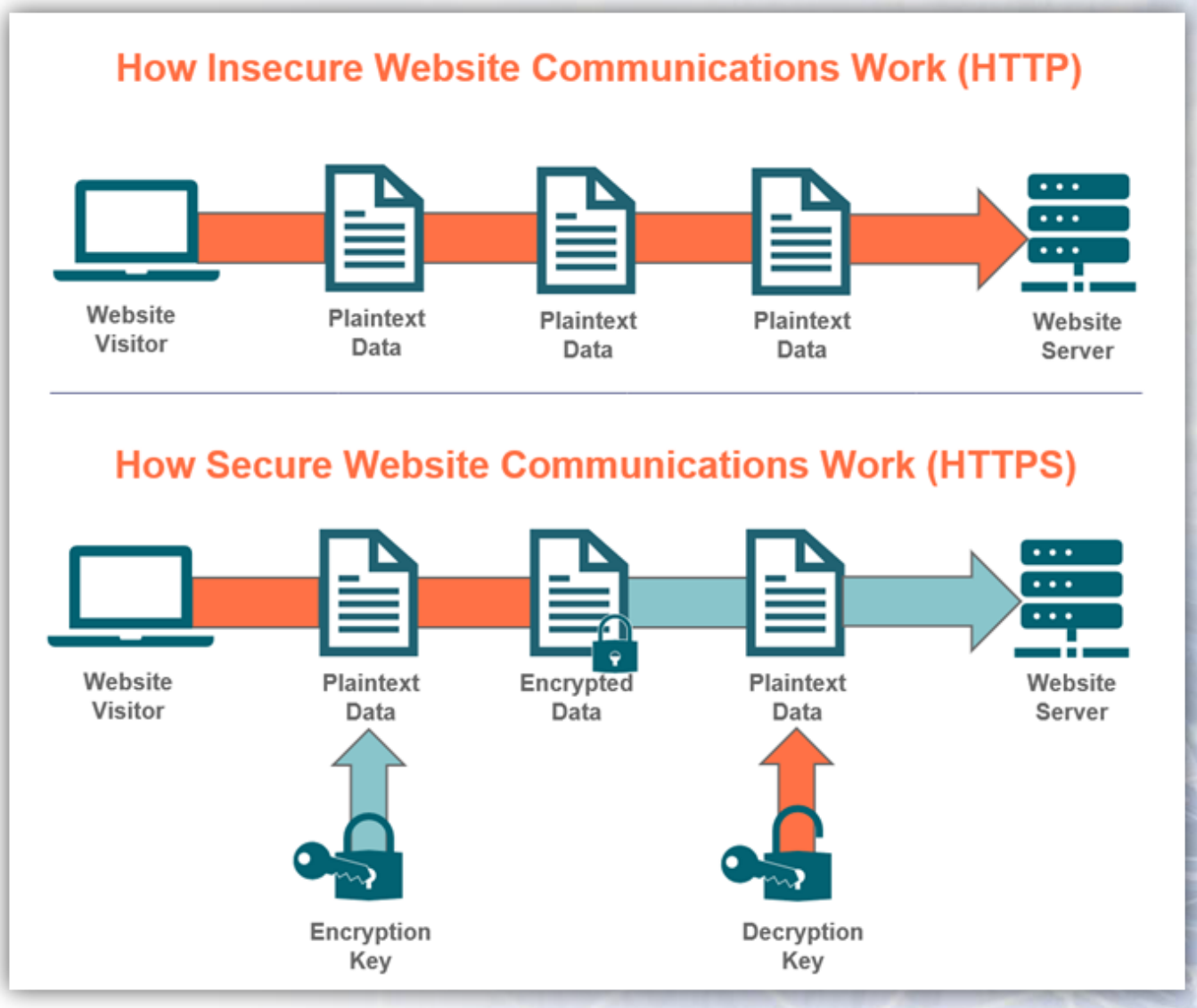
Certificate Verification: The client verifies the authenticity of the server's certificate using trusted Certificate Authorities (CAs).

Secure Connection: Once the certificate is validated, a secure connection is established, and data exchange occurs over an encrypted channel.

Encryption and Decryption: The client and server encrypt and decrypt data using symmetric encryption algorithms and a shared session key.

Secure Data Transfer: All subsequent data exchanged between the client and server remains encrypted, protecting it from eavesdropping and tampering.

HTTPS Working



Benefits of HTTPS

HTTPS offers numerous benefits for secure web communication:

- **Data Confidentiality**: HTTPS encrypts sensitive information, such as passwords, credit card details, and personal data, preventing unauthorized access.
- **Data Integrity**: HTTPS ensures that data remains intact during transmission, detecting any modifications or tampering attempts.
- **Authentication**: HTTPS verifies the identity of the server through SSL/TLS certificates, preventing impersonation or man-in-the-middle attacks.
- **Trust and User Confidence**: Displaying the padlock symbol and "https" in the URL instills trust in users and reassures them about the security of the website.
- **SEO Benefits**: Search engines consider HTTPS as a ranking signal, incentivizing website owners to implement HTTPS for improved search engine visibility.

Implementing HTTPS

Steps to implement HTTPS on a website:

1. Obtain an SSL/TLS certificate from a trusted Certificate Authority (CA).
2. Install the SSL/TLS certificate on the web server.
3. Configure the server to redirect HTTP traffic to HTTPS.
4. Ensure all web content (images, scripts, stylesheets, etc.) is loaded securely using HTTPS.
5. Test the HTTPS implementation to verify proper functioning and resolve any issues.

HTTPS Best Practices

Best practices for implementing and maintaining HTTPS:

- Use the latest version of the SSL/TLS protocol supported by your infrastructure.
- Enable strong cipher suites and prioritize secure encryption algorithms.
- Implement HTTP Strict Transport Security (HSTS) to enforce HTTPS usage and prevent downgrade attacks.
- Regularly renew and update SSL/TLS certificates to prevent expiration or revocation.
- Perform security audits and vulnerability scans to identify and address any weaknesses in the HTTPS configuration.

Conclusion

- HTTPS is a crucial technology for securing web communication and protecting sensitive information.
- It ensures data confidentiality, integrity, and authenticity, building trust with users and search engines.
- By implementing HTTPS correctly and following best practices, organizations can enhance the security of their websites and safeguard user data.

Week 13 Topic 2

SSH (Secure Shell) - Secure Remote Access and File Transfer

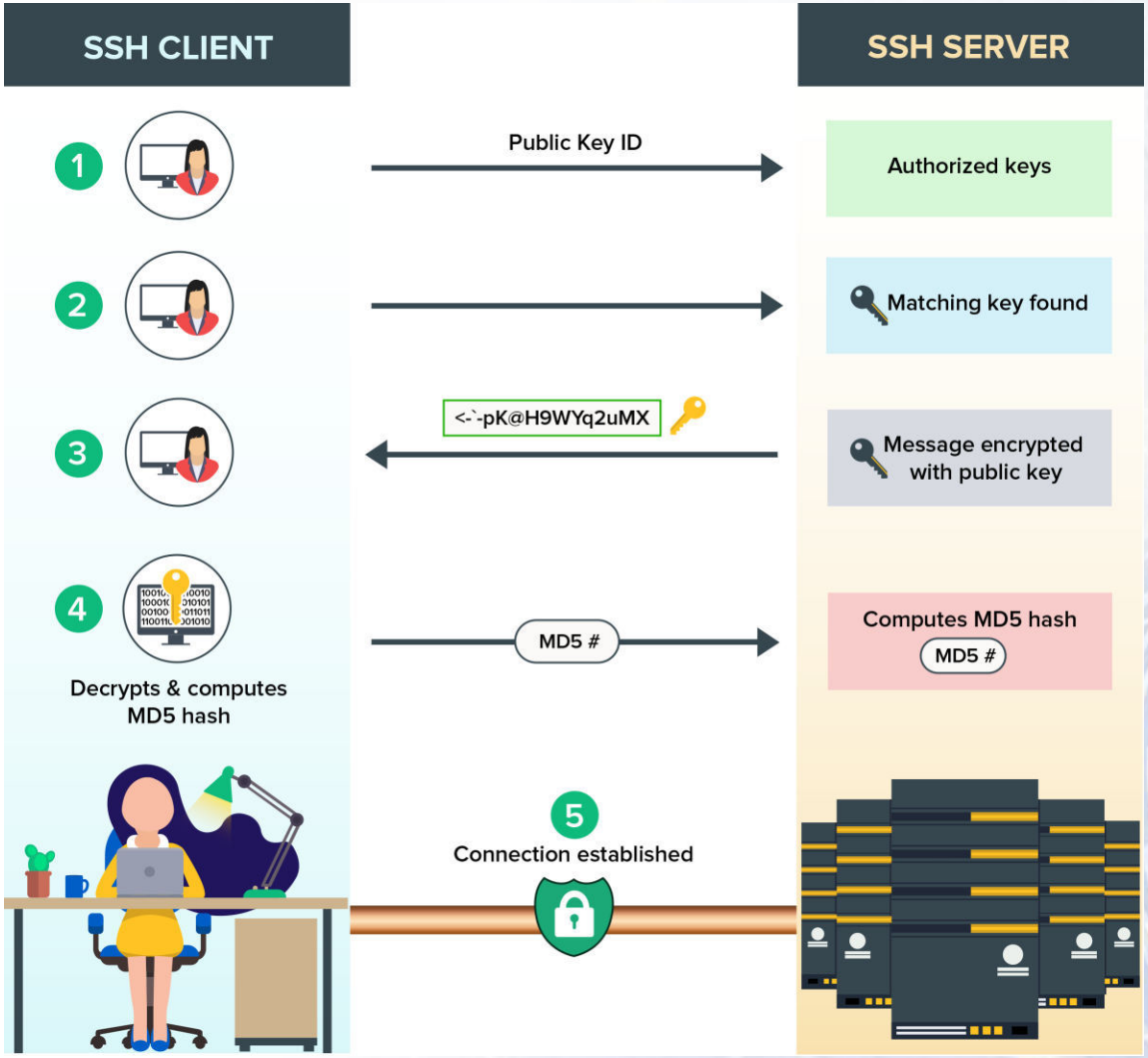
- SSH (Secure Shell) is a cryptographic network protocol that provides secure remote access and file transfer between two systems.
- It offers a secure alternative to traditional remote access protocols like Telnet and FTP.
- SSH ensures the confidentiality, integrity, and authenticity of data exchanged between client and server.

How SSH Works

SSH uses public-key cryptography and encryption to establish a secure connection:

1. **Client Authentication:** The client initiates a connection to the server using SSH and provides its public key.
2. **Server Authentication:** The server presents its public key to the client, which the client verifies to ensure the authenticity of the server.
3. **Session Key Exchange:** Both client and server agree on a session key, which is used for symmetric encryption and decryption of data during the session.
4. **Secure Data Exchange:** All subsequent data exchanged between the client and server is encrypted using the session key.
5. **Command Execution and File Transfer:** The client can securely execute remote commands or transfer files to and from server.

HTTPS Working



Benefits of SSH

SSH offers several advantages for secure remote access and file transfer:

- **Encryption:** All data transmitted over SSH is encrypted, protecting it from unauthorized access or interception.
- **Authentication:** SSH uses public-key cryptography for user authentication, ensuring secure access to remote systems.
- **Data Integrity:** SSH verifies the integrity of data during transmission, preventing tampering or unauthorized modification.
- **Port Forwarding:** SSH allows secure tunneling of other protocols, enabling remote access to services running on a server's local network.
- **Cross-Platform Compatibility:** SSH is available on various operating systems, making it widely accessible for remote administration.

SSH Implementation

Steps to implement SSH for remote access and file transfer:

1. Install an SSH server on the remote system you want to access.
2. Generate an SSH key pair on the client machine.
3. Configure the SSH server to allow remote access using the generated SSH public key.
4. Use an SSH client (e.g., OpenSSH, PuTTY) to establish a secure connection to the remote system using the SSH key pair.

SSH Best Practices

Best practices for implementing and using SSH:

- Use strong and unique SSH key pairs for authentication.
- Disable password-based authentication to enhance security.
- Regularly update SSH software and apply security patches.
- Implement firewall rules to allow SSH access only from trusted IP addresses.
- Monitor SSH logs for suspicious activity and enable two-factor authentication for added security.

Conclusion

- SSH is a critical protocol for secure remote access and file transfer.
- It provides encryption, authentication, and data integrity, ensuring secure communication between client and server.
- By implementing SSH correctly and following best practices, organizations can protect sensitive information and maintain secure remote administration.

Week 13 Topic 3

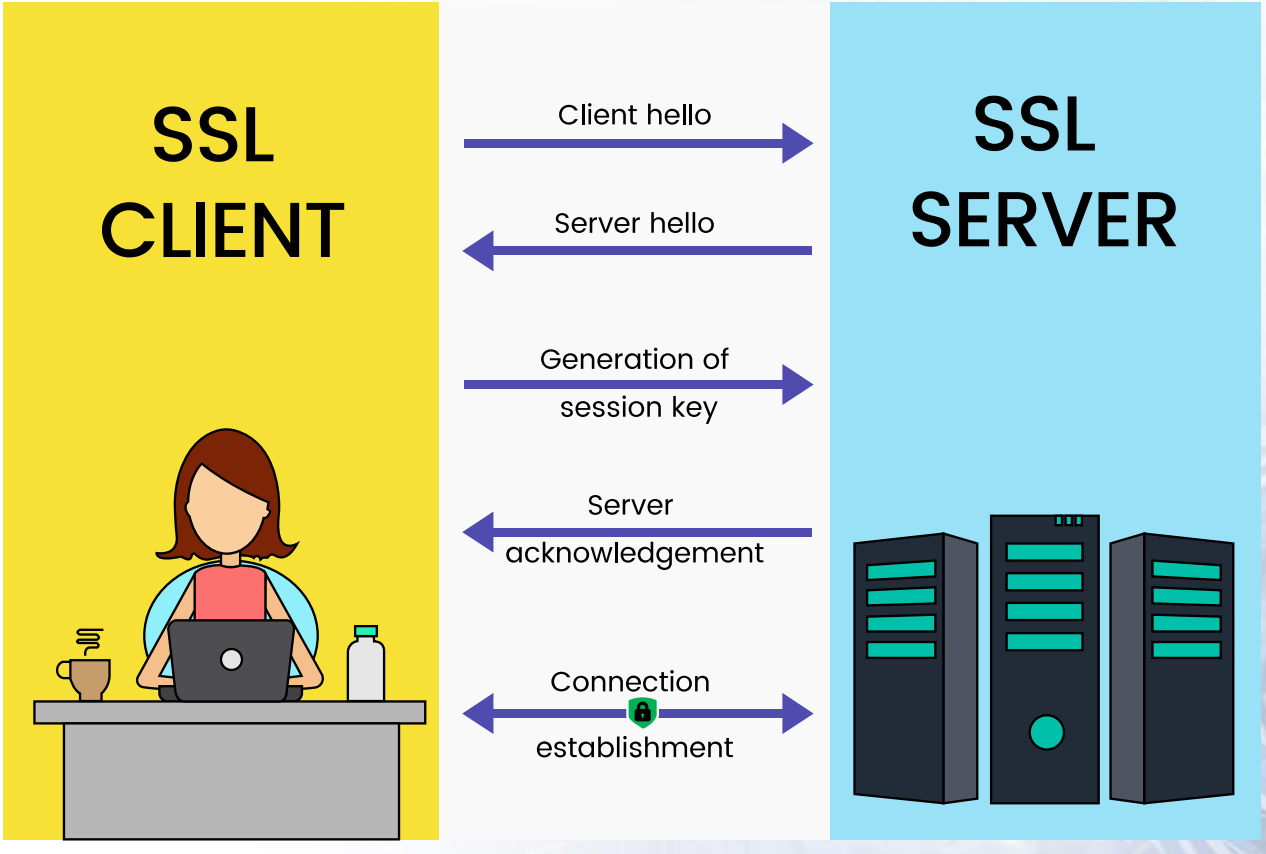
SSL (Secure Sockets Layer) – Ensuring Secure Communication

- SSL (Secure Sockets Layer) is a cryptographic protocol that provides secure communication over the internet.
- It ensures the confidentiality, integrity, and authenticity of data transmitted between clients and servers.
- SSL has been succeeded by Transport Layer Security (TLS), but the term "SSL" is still commonly used.

How SSL Works

- SSL handshake process:
- **Client Hello**: The client initiates a connection request to the server, presenting its supported SSL/TLS versions and cipher suites.
- **Server Hello**: The server responds by selecting the appropriate SSL/TLS version and cipher suite from the client's options.
- **Certificate Exchange**: The server sends its SSL certificate to the client, containing the server's public key and digitally signed by a trusted Certificate Authority (CA).
- **Client Authentication**: If required, the client presents its own SSL certificate to the server for authentication.
- **Session Key Exchange**: The client and server establish a session key for symmetric encryption and decryption of data during the session.
- **Secure Data Exchange**: All subsequent data exchanged between the client and server is encrypted using the session key.
- **Session Closure**: When the session ends, the client and server exchange closure messages to properly terminate the connection.

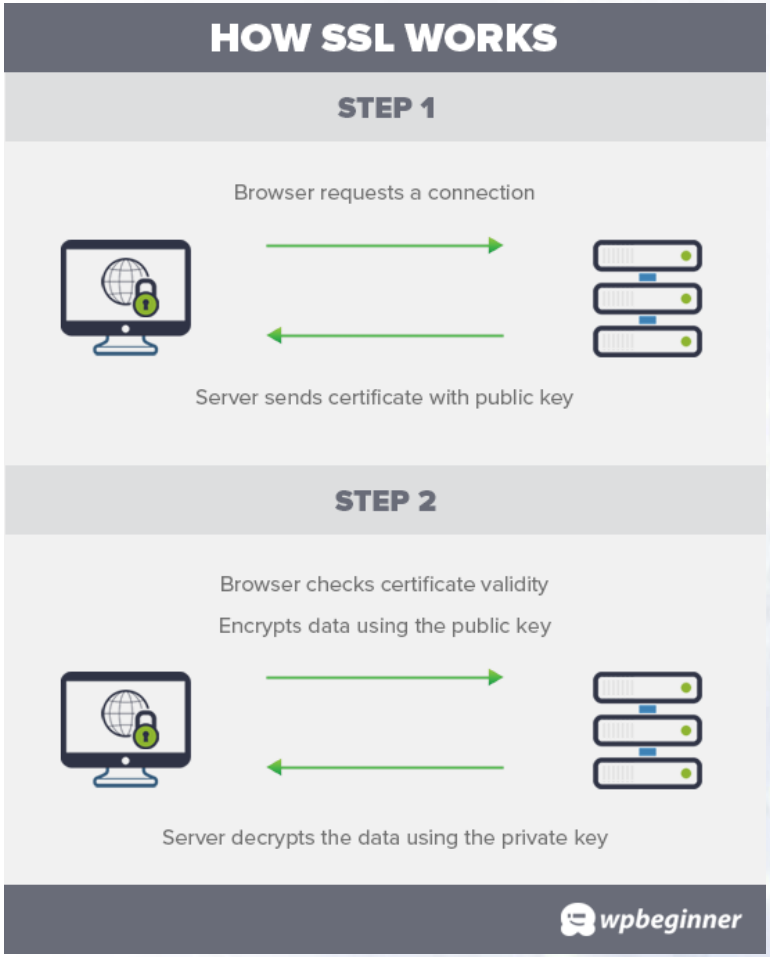
SSL Working


SSL
CLIENT
Client hello
Server hello
Generation of
session key
Server
acknowledgement
Connection
establishment
SSL
SERVER

SSL Certificate

- SSL certificates play a crucial role in SSL/TLS communication:
- They are digital files that bind an organization's identity with a public key.
- Issued by trusted Certificate Authorities (CAs), SSL certificates provide authenticity and trust to the SSL/TLS connection.
- SSL certificates contain information such as the domain name, organization name, CA's digital signature, and the public key.

SSL Working with certificate



Benefits of SSL

- SSL provides several advantages for secure communication:

- **Data Encryption**: SSL encrypts sensitive information, such as login credentials and financial details, ensuring their confidentiality.
- **Data Integrity**: SSL ensures that data remains intact during transmission, preventing unauthorized modification.
- **Authentication**: SSL certificates verify the identity of the server, assuring users that they are connecting to the intended website and not an imposter.
- **Trust and User Confidence**: Displaying the padlock symbol and "https" in the URL enhances trust and reassures users about the security of the website.
- **Compliance**: SSL is often required to meet regulatory and compliance standards, such as PCI DSS for secure payment processing.

SSL Implementation

Steps to implement SSL on a website:

- Obtain an SSL certificate from a trusted CA.
- Install the SSL certificate on the web server.
- Configure the server to use SSL/TLS protocols and enable HTTPS.
- Test the SSL implementation to ensure it is functioning correctly.

SSL Best Practices

Best practices for SSL implementation and management:

- Keep SSL certificates up to date and renew them before expiration.
- Use strong encryption algorithms and secure cipher suites.
- Implement HTTP Strict Transport Security (HSTS) to enforce HTTPS usage.
- Regularly monitor and audit SSL configurations for vulnerabilities.
- Educate users on how to identify SSL indicators, such as padlock symbols and "https" in URLs.

Conclusion

- SSL is a vital technology for ensuring secure communication over the internet.
- It provides data encryption, authentication, and integrity, fostering trust and user confidence.
- By implementing SSL correctly and following best practices, organizations can enhance the security of their online services and protect sensitive information.

Week 13 Topic 4

TLS (Transport Layer Security) - Ensuring Secure Communication

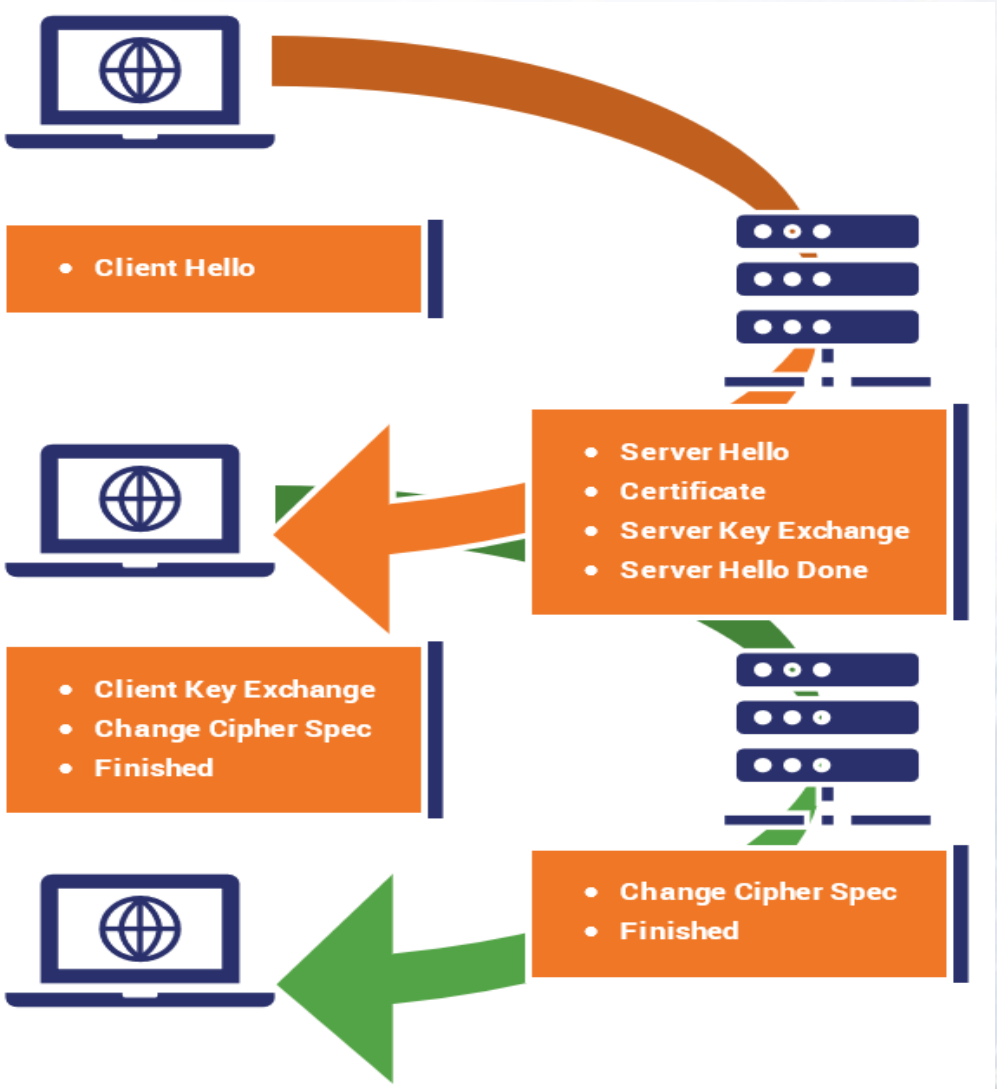
- TLS (Transport Layer Security) is a cryptographic protocol that provides secure communication over the internet.
- It is the successor to SSL (Secure Sockets Layer) and offers enhanced security features.
- TLS ensures the confidentiality, integrity, and authenticity of data transmitted between clients and servers.

How TLS Works

TLS handshake process:

1. **Client Hello**: The client initiates a connection request to the server, presenting its supported TLS versions and cipher suites.
2. **Server Hello**: The server responds by selecting the appropriate TLS version and cipher suite from the client's options.
3. **Certificate Exchange**: The server sends its TLS certificate to the client, containing the server's public key and digitally signed by a trusted Certificate Authority (CA).
4. **Client Authentication**: If required, the client presents its own TLS certificate to the server for authentication.
5. **Key Exchange**: The client and server negotiate a shared secret key for symmetric encryption and decryption of data during the session.
6. **Secure Data Exchange**: All subsequent data exchanged between the client and server is encrypted using the shared secret key.
7. **Session Closure**: When the session ends, the client and server exchange closure messages to properly terminate the connection.

TSL Working


Client Hello
Server Hello
Certificate
Server Key Exchange
Server Hello Done
Client Key Exchange
Change Cipher Spec
Finished
Change Cipher Spec
Finished

TLS Versions and Cipher Suites

TLS supports various versions and cipher suites to ensure secure communication:

- **TLS versions**: TLS 1.0, TLS 1.1, TLS 1.2, TLS 1.3 (the latest version).
- **Cipher suites**: Combinations of encryption algorithms, authentication methods, and key exchange protocols.

Benefits of TLS

TLS provides several advantages for secure communication:

- **Strong Encryption**: TLS uses robust encryption algorithms to protect data from unauthorized access.
- **Data Integrity**: TLS ensures that data remains intact during transmission and detects any tampering attempts.
- **Authentication**: TLS certificates verify the identity of the server, assuring users they are connecting to the correct website.
- **Forward Secrecy**: TLS supports forward secrecy, ensuring that even if a private key is compromised, past communications remain secure.
- **Compatibility**: TLS is widely supported by modern web browsers and applications, enabling secure communication across platforms.

TLS Implementation

Steps to implement TLS on a website:

1. Obtain a TLS certificate from a trusted CA.
2. Install the TLS certificate on the web server.
3. Configure the server to use TLS protocols and enable HTTPS.
4. Regularly update TLS software and libraries to stay protected against vulnerabilities.

TLS Best Practices

Best practices for TLS implementation and management:

- Use the latest TLS version supported by your infrastructure.
- Select strong cipher suites that prioritize security and performance.
- Enable Perfect Forward Secrecy (PFS) to enhance security.
- Implement strict certificate validation and revocation checking.
- Regularly monitor and audit TLS configurations for vulnerabilities.

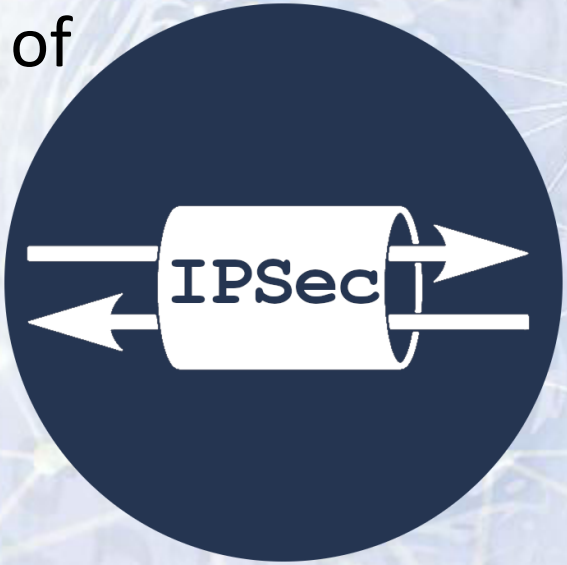
Conclusion

- TLS is a critical technology for ensuring secure communication over the internet.
- It offers strong encryption, data integrity, and authentication, enhancing the overall security of online services.
- By implementing TLS correctly and following best practices, organizations can protect sensitive information and establish trust with their users.

Week 13 Topic 5

IPsec (Internet Protocol Security) – Securing Network Communications

- IPsec (Internet Protocol Security) is a suite of protocols and cryptographic techniques used to secure network communications at the IP layer.
- It provides a robust framework for ensuring the confidentiality, integrity, and authenticity of IP packets transmitted over a network.

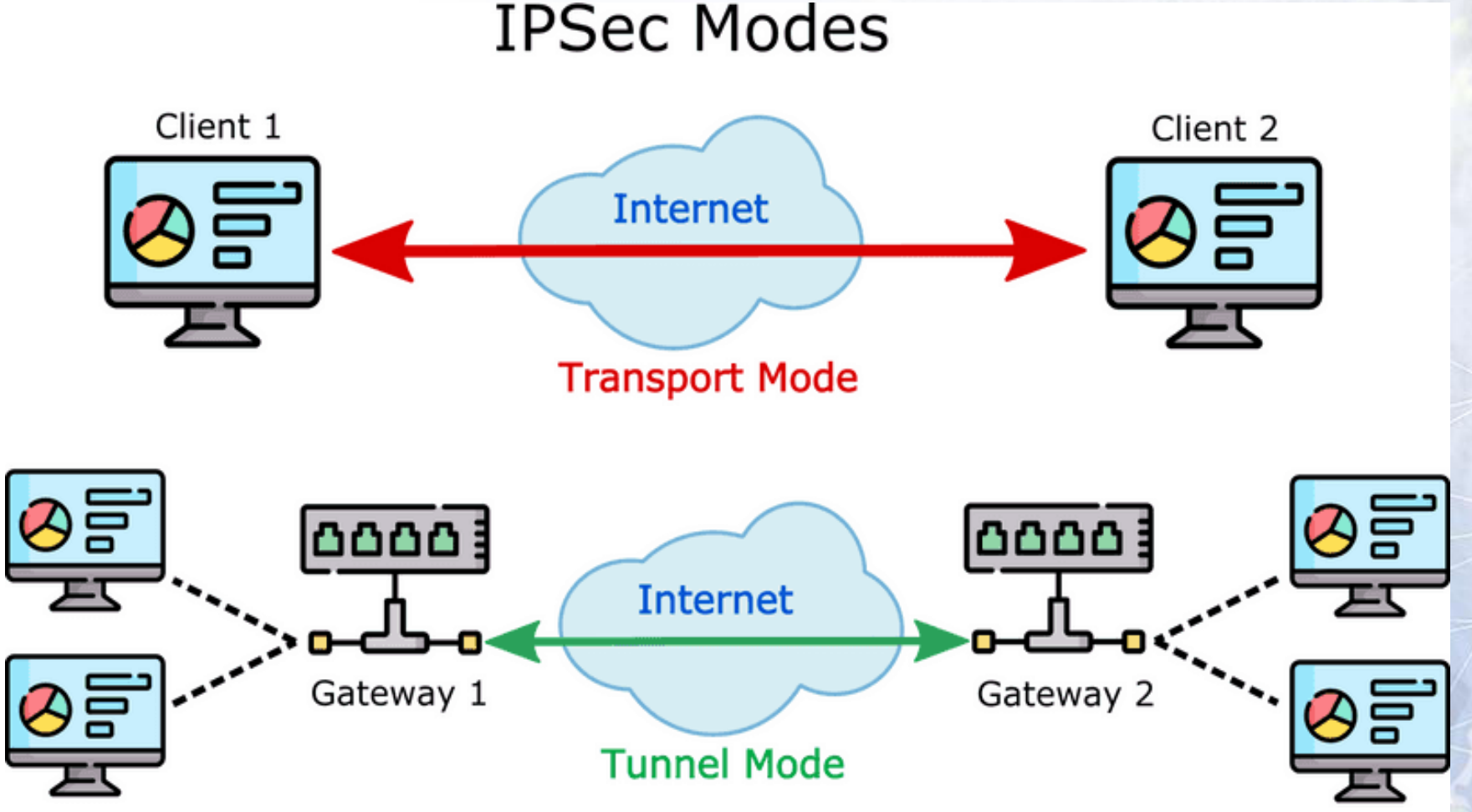


How IPsec Works?

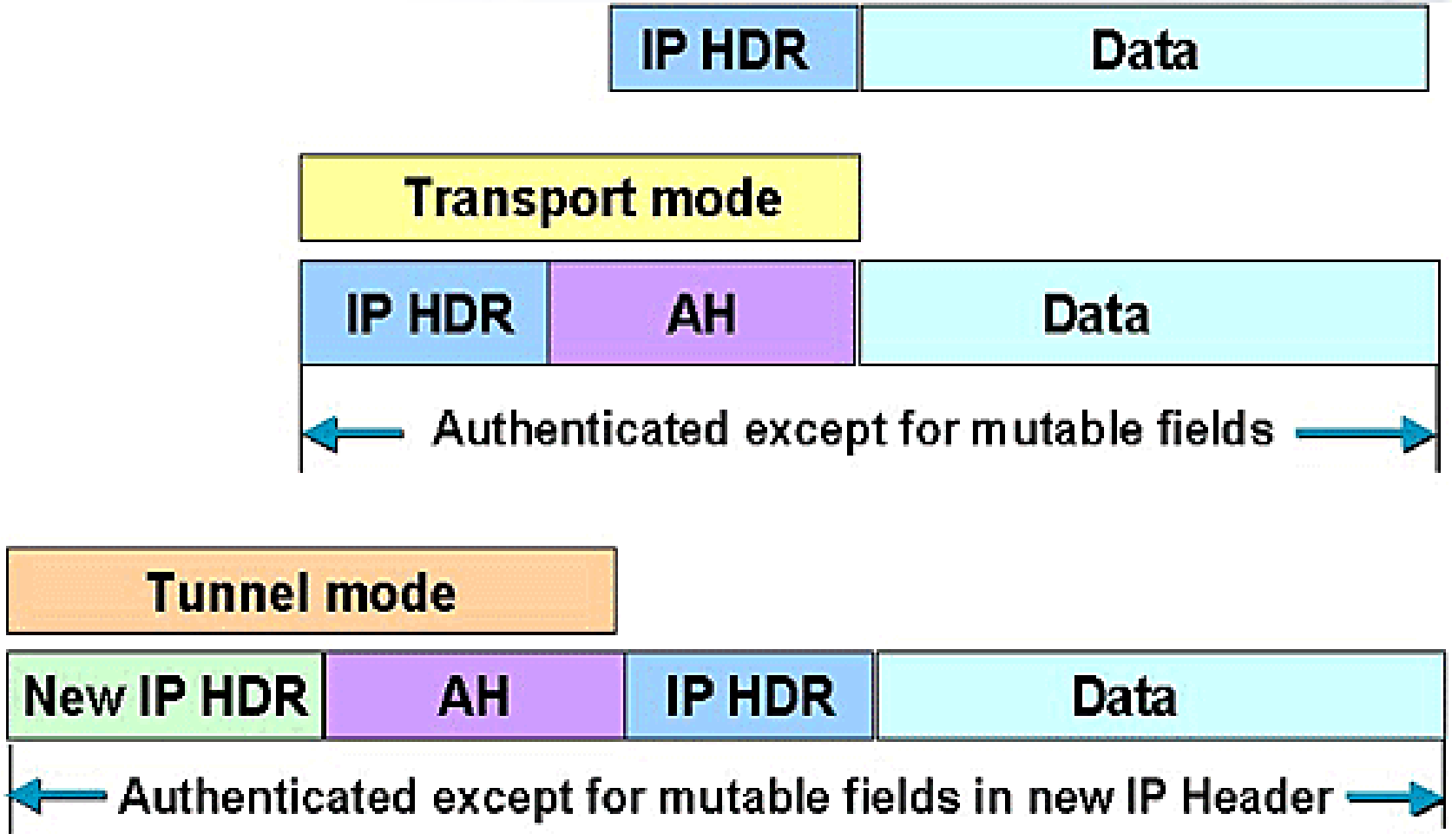
IPsec operates in two main modes: Transport mode and Tunnel mode:

1. **Transport Mode**: Protects the data payload of IP packets, leaving the original IP header intact. It is typically used for end-to-end communication within a network.
2. **Tunnel Mode**: Encrypts the entire IP packet, including the original IP header, and encapsulates it within a new IP packet. It is often used for secure communication between networks.

IPsec Working



IPsec Modes

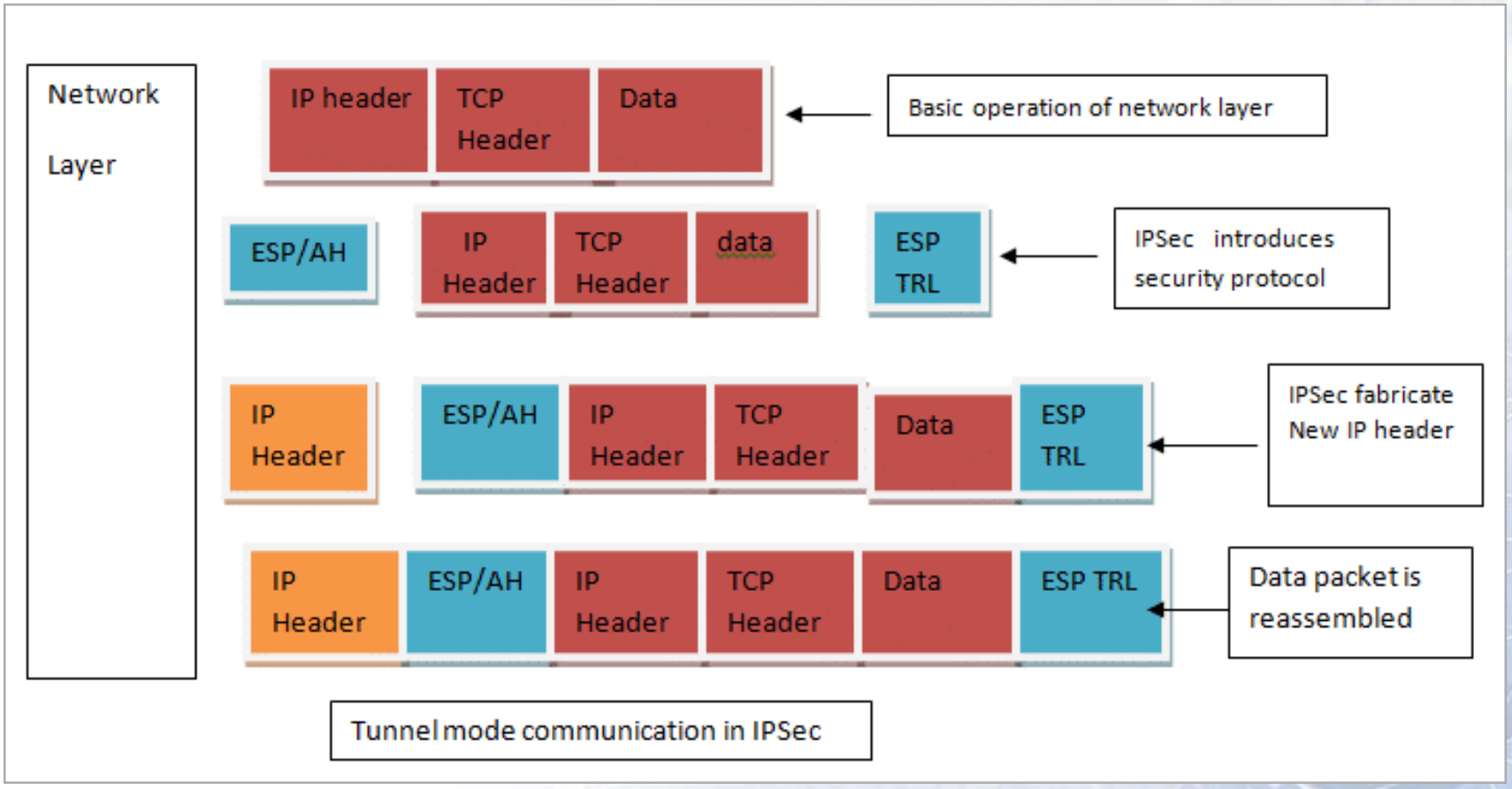

IP HDR
Data
Transport mode
IP HDR
AH
Data
Authenticated except for mutable fields
Tunnel mode
New IP HDR
AH
IP HDR
Data
Authenticated except for mutable fields in new IP Header

Components of IPsec

IPsec consists of several key components:

1. **Authentication Header (AH)**: Provides data integrity and authentication of IP packets.
2. **Encapsulating Security Payload (ESP)**: Offers data confidentiality, integrity, and authentication through encryption and encapsulation.
3. **Security Associations (SA)**: Establishes the security parameters, including encryption algorithms, authentication methods, and key management, between two communicating entities.
4. **Key Management**: Involves the secure exchange, generation, and management of cryptographic keys used by IPsec for encryption and authentication.

Tunnel Mode Detail



Benefits of IPsec

IPsec offers numerous advantages for securing network communications:

1. **Confidentiality**: Encrypts IP packets, preventing unauthorized access and eavesdropping of sensitive data.
2. **Integrity**: Verifies the integrity of IP packets, ensuring they have not been tampered with during transit.
3. **Authentication**: Authenticates the source of IP packets, preventing spoofing and unauthorized access.
4. **Seamless Integration**: IPsec can be implemented at the network layer, providing security for all applications and protocols that use IP.
5. **Interoperability**: IPsec is a widely supported standard, allowing secure communication between different vendors and platforms.

IPsec Implementation

Steps to implement IPsec in a network:

1. **Assess Network Requirements**: Determine the specific security needs and objectives of the network.
2. **Plan IPsec Policy**: Define security policies, including encryption algorithms, authentication methods, and key management approaches.
3. **Configure IPsec Devices**: Configure IPsec parameters on network devices, such as routers, firewalls, or VPN concentrators, based on the defined policies.
4. **Test and Monitor**: Perform thorough testing to ensure IPsec is functioning correctly and continuously monitor the network for any potential issues or vulnerabilities.

Conclusion

- IPsec is a powerful security protocol suite that protects network communications at the IP layer, ensuring confidentiality, integrity, and authenticity.
- By implementing IPsec correctly and following best practices, organizations can secure their network infrastructure and protect sensitive data from unauthorized access.

Week 13 Topic 6

SHTTPS (Secure Hypertext Transfer Protocol Secure) - Strengthening Web Security

- SHTTPS (Secure Hypertext Transfer Protocol Secure) is an enhanced version of HTTPS that integrates additional security measures.
- It combines the secure communication features of HTTPS with additional mechanisms to protect against various web security threats.

Why SHTTPS?

- Traditional HTTPS provides secure communication between clients and servers, but SHTTPS further strengthens web security by addressing specific vulnerabilities and attacks.
- SHTTPS aims to mitigate risks related to website tampering, data integrity, and man-in-the-middle attacks.

Key Features of SHTTPS

SHTTPS incorporates the following key features to enhance web security:

- **Subresource Integrity (SRI):**
  - SRI ensures that external resources, such as scripts and stylesheets, loaded by a webpage are not tampered with during transit.
  - It uses cryptographic hashes to verify the integrity of the resources, allowing the browser to block or warn if any modifications are detected.
- **Certificate Transparency (CT):**
  - CT is a framework that provides public auditing and monitoring of SSL/TLS certificates issued by Certificate Authorities (CAs).
  - It aims to detect and mitigate misissued or fraudulent certificates, enhancing trust and accountability in the certificate ecosystem.
- **HTTP Public Key Pinning (HPKP):**
  - HPKP allows website operators to specify a list of cryptographic hashes of their SSL/TLS public keys in the HTTP response header.
  - This helps protect against rogue or compromised CAs by instructing the browser to only accept specific public keys for future connections to website.

Benefits of SHTTPS

SHTTPS offers several advantages for web security:

- **Enhanced Protection:** SHTTPS strengthens the overall security of websites by addressing specific vulnerabilities and threats.
- **Tamper Detection:** SRI helps ensure the integrity of external resources and prevents unauthorized modifications.
- **Certificate Accountability:** CT provides transparency and oversight of SSL/TLS certificates, reducing the risk of misissued or fraudulent certificates.
- **Key Validation:** HPKP allows website operators to enforce the use of specific public keys, protecting against compromised or unauthorized certificates.
- **User Trust:** Implementing SHTTPS demonstrates a commitment to web security, enhancing user confidence and trust in the website.

Implementing SHTTPS

Steps to implement SHTTPS on a website:

- **Enable HTTPS:** Ensure that your website is already using HTTPS for secure communication.
- **Subresource Integrity (SRI):** Implement SRI by adding integrity attributes to external resource references (e.g., script and stylesheet tags) in your web pages.
- **Certificate Transparency (CT):** Utilize CT logs and monitoring services to track and validate SSL/TLS certificates issued for your website.
- **HTTP Public Key Pinning (HPKP):** Carefully consider the implications of HPKP, as it requires careful management of public keys and potential risks associated with pinning.

Conclusion

- SHTTPS is an advanced approach to web security, integrating additional measures to strengthen HTTPS and protect against specific vulnerabilities.
- By implementing SHTTPS and leveraging its features, website owners can enhance the integrity, trustworthiness, and overall security of their web presence.

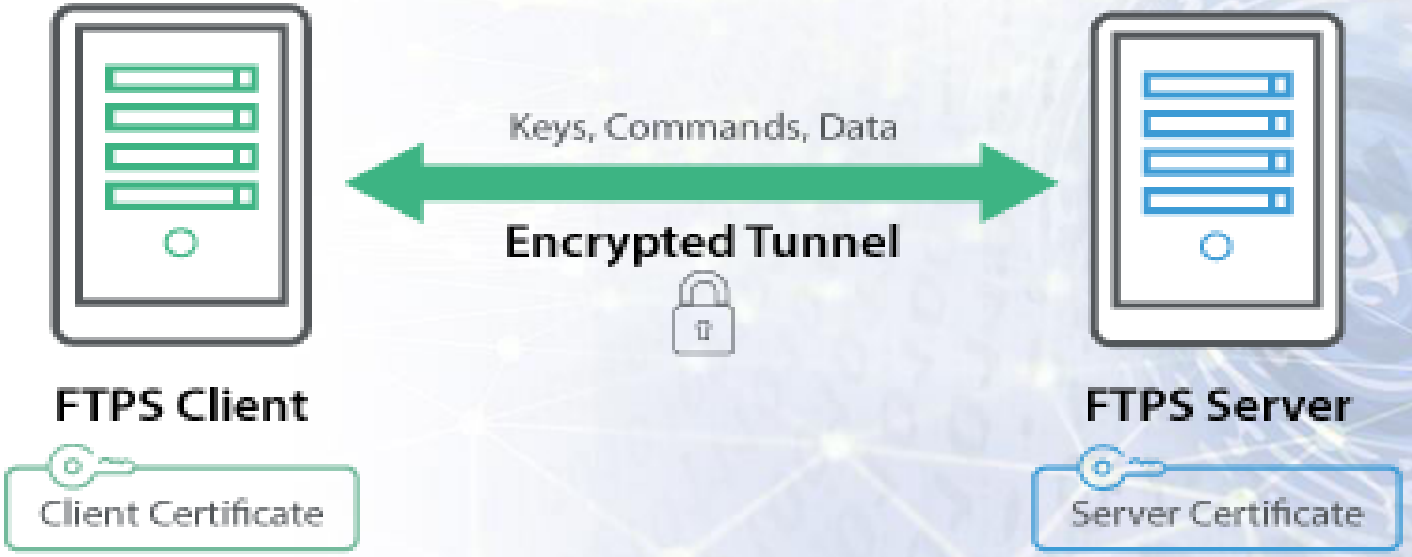
Week 13 Topic 7

FTPS (File Transfer Protocol Secure)

- FTPS is a secure file transfer protocol that combines the traditional FTP protocol with the security features of SSL/TLS encryption.
- It provides a secure and reliable method for transferring files over a network, ensuring data integrity and confidentiality.
- FTPS is widely used in various industries, including banking, healthcare, and e-commerce, where secure file transfers are crucial.

How FTPS Works?

- FTPS utilizes SSL/TLS encryption to secure the communication between the client and the server.
- SSL/TLS establishes a secure and encrypted connection, preventing unauthorized access and eavesdropping.
- FTPS supports both implicit and explicit modes, offering flexibility in choosing the level of security required.



Key Features of FTPS

- **Encryption**: FTPS uses SSL/TLS encryption to protect data during transit, safeguarding it from interception and tampering.
- **Authentication**: FTPS provides strong authentication mechanisms, ensuring that only authorized users can access and transfer files.
- **Data Integrity**: Through SSL/TLS, FTPS guarantees the integrity of transferred files, detecting any unauthorized modifications.
- **Portability**: FTPS is supported by various operating systems and FTP client software, making it widely compatible and accessible.

Benefits of Using FTPS

Enhanced Security: FTPS offers a higher level of security compared to standard FTP, making it suitable for sensitive data transfers.

Compliance with Regulations: FTPS helps organizations meet regulatory requirements for data protection, such as HIPAA and GDPR.

Reliability: FTPS ensures the reliable delivery of files, minimizing errors and interruptions during transfers.

Flexibility: With implicit and explicit modes, FTPS allows organizations to choose the level of security that best suits their needs.

Use Cases for FTPS

- **Banking and Finance**: FTPS enables secure transfer of financial data, such as transactions, statements, and sensitive customer information.
- **Healthcare**: FTPS is used for transferring confidential patient records, medical images, and other sensitive healthcare data.
- **E-commerce**: FTPS ensures secure exchange of product information, inventory updates, and customer data between online retailers and suppliers.

Best Practices for FTPS Implementation

1. **Use Strong Authentication**: Implement strong username/password combinations or consider additional authentication methods like certificates.
2. **Regularly Update SSL/TLS Certificates**: Keep SSL/TLS certificates up to date to maintain the security of the FTPS server.
3. **Employ Access Controls**: Restrict FTPS access to authorized users and employ granular permissions to limit access to specific files and directories.
4. **Monitor and Audit**: Implement logging and monitoring mechanisms to track FTPS activities and detect any suspicious or unauthorized actions.

Conclusion

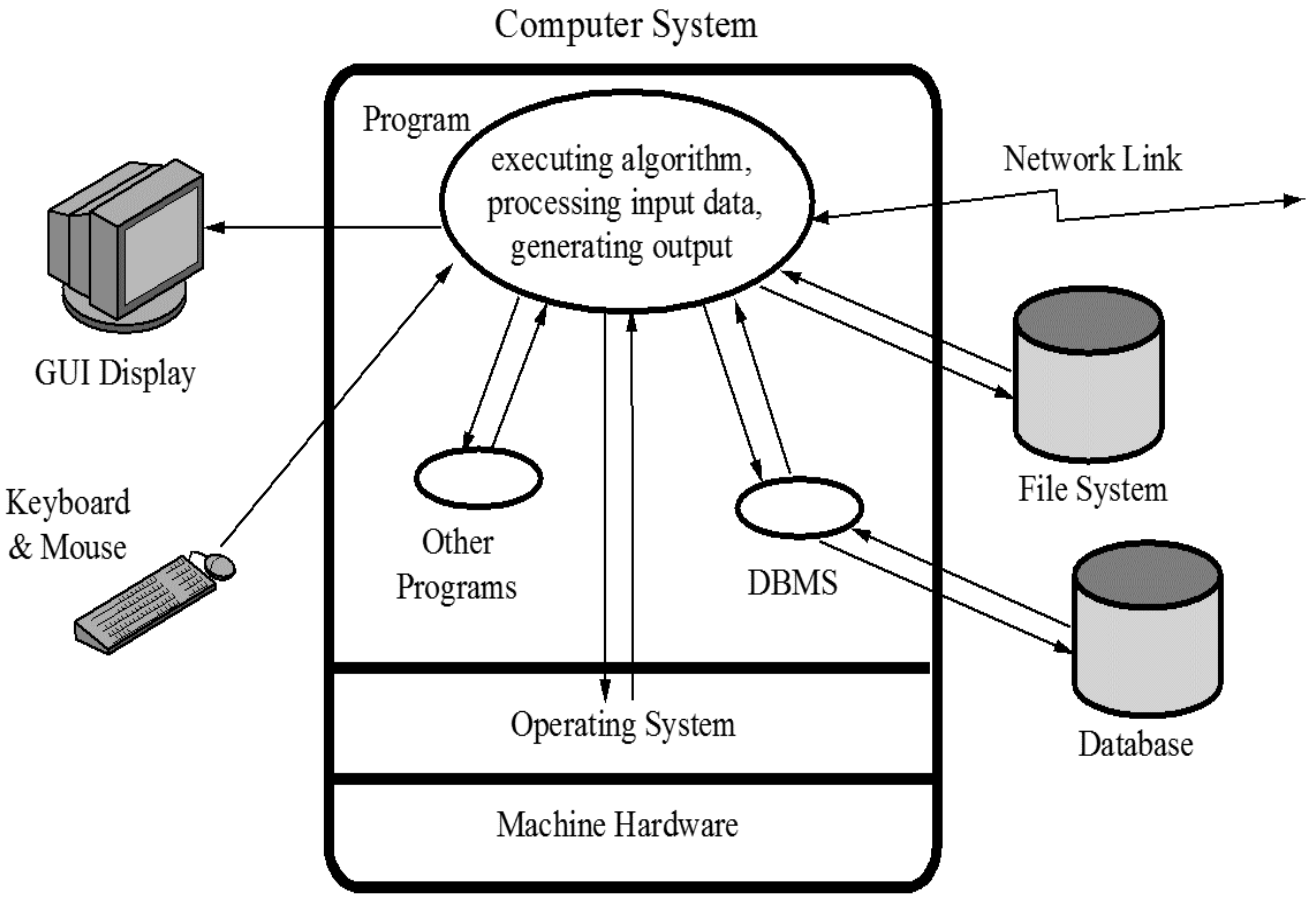
- FTPS is a secure and reliable file transfer protocol that combines the benefits of FTP with SSL/TLS encryption.
- By implementing FTPS, organizations can ensure the confidentiality, integrity, and secure transfer of sensitive data.
- With its wide compatibility and industry adoption, FTPS remains a trusted solution for secure file transfers in various sectors.

Week No. 14

Operating System Security

Week 14 Topic 1

Recall… Abstract View of Program



Operating System Security

Computer are rarely used in isolation

- Usually connected to some kind of network, Internet etc.

Networked computers

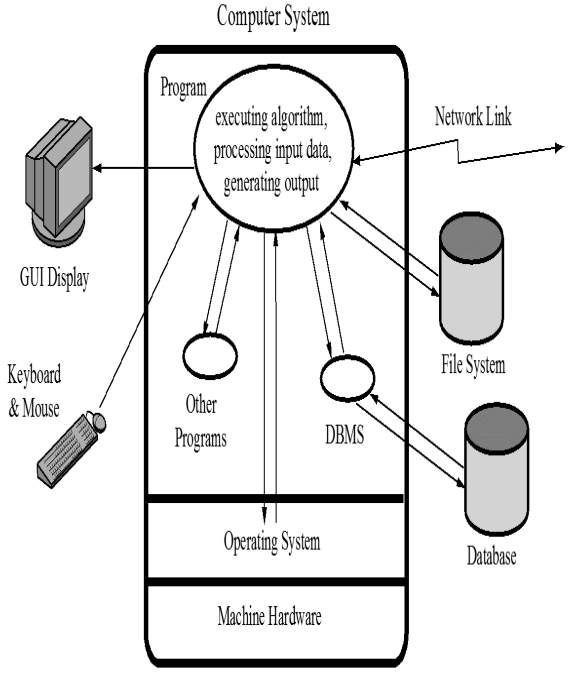
- ensure secure operation in networked environment

New threat?

- Attackers coming from the network.
- Network-facing programs on computers may be buggy.
- Users may be hurt via online communication.

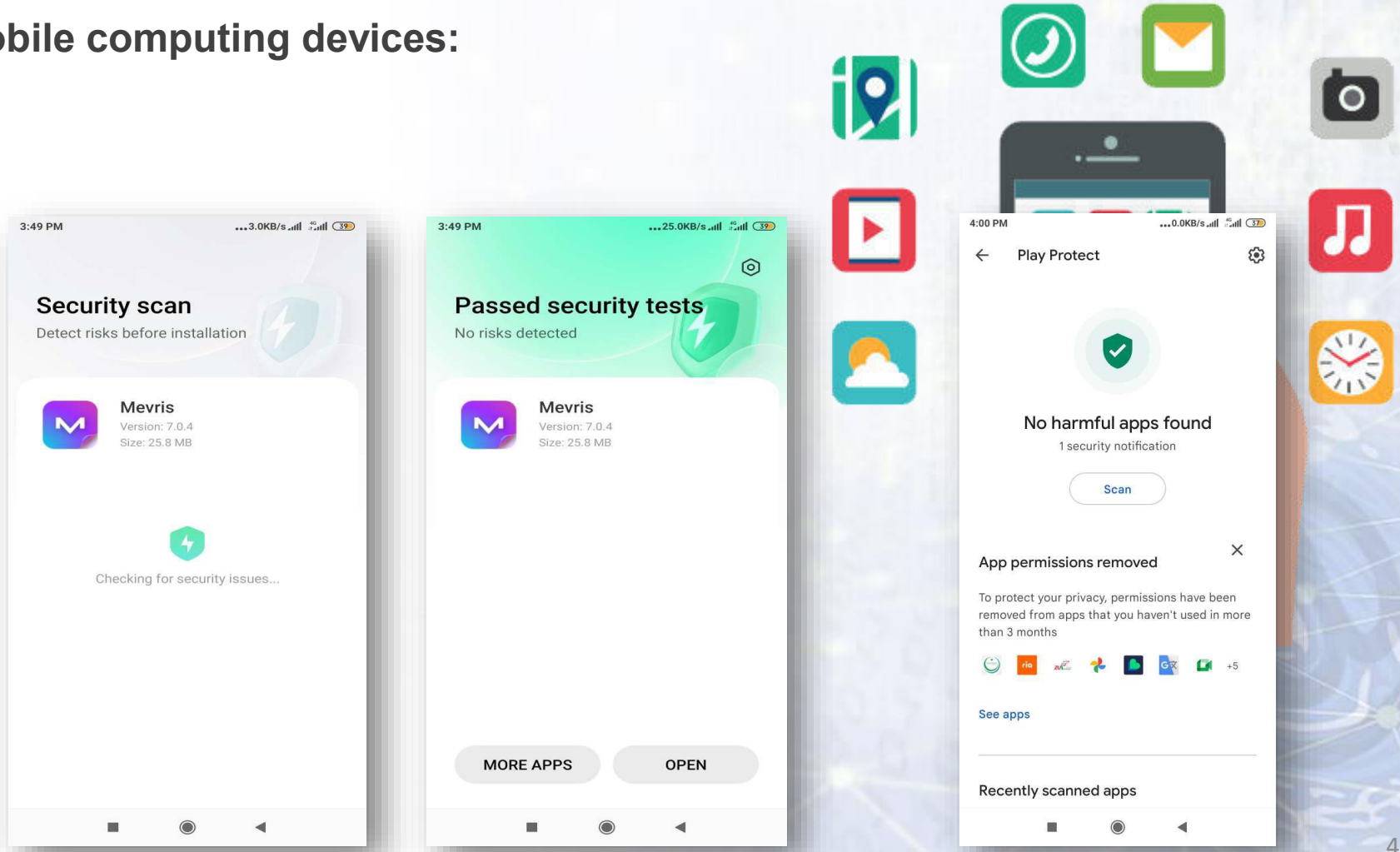
Security mechanisms

- Authentication; Access Control
- Secure Communication (using cryptography)
- Logging & Auditing
- Intrusion Prevention and Detection



Mobile Operating System Security

Mobile computing devices:



Mobile Operating System Security

Mobile computing devices:

New threat?

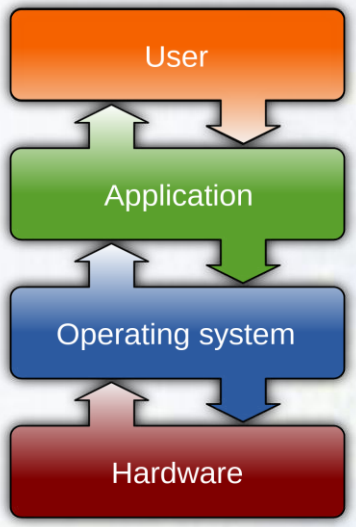
- Apps (programs) may be malicious.
- More tightly connected with personal life of the user e.g. personal health and banking apps.

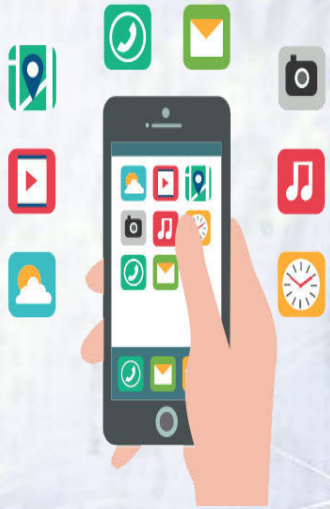
Security mechanisms?

- Isolation of each app.
- Users assess risks of apps.
- Risk communication.

Google Play Protect

https://support.google.com/googleplay/answer/2812853?hl=en





Use Google Play Protect to help keep your apps safe and your data private

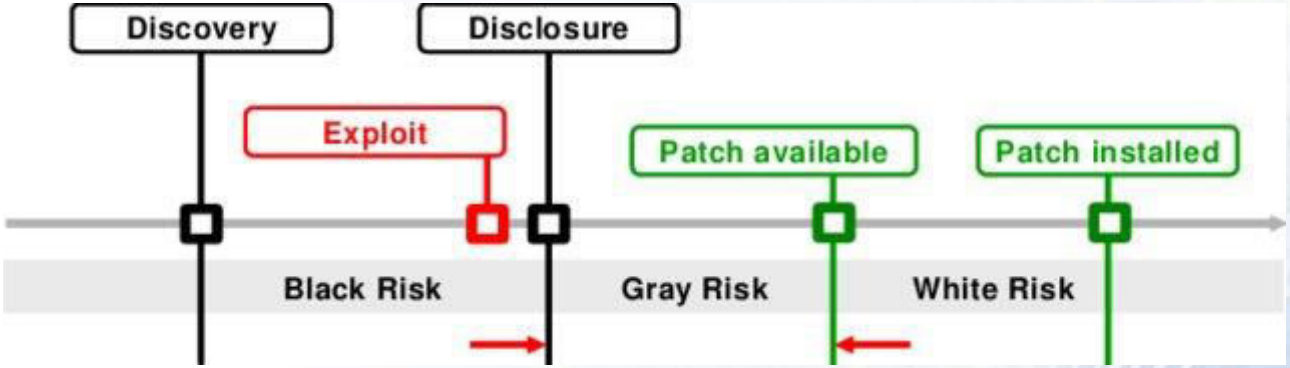
Google Play Protect checks your apps and devices for harmful behavior.

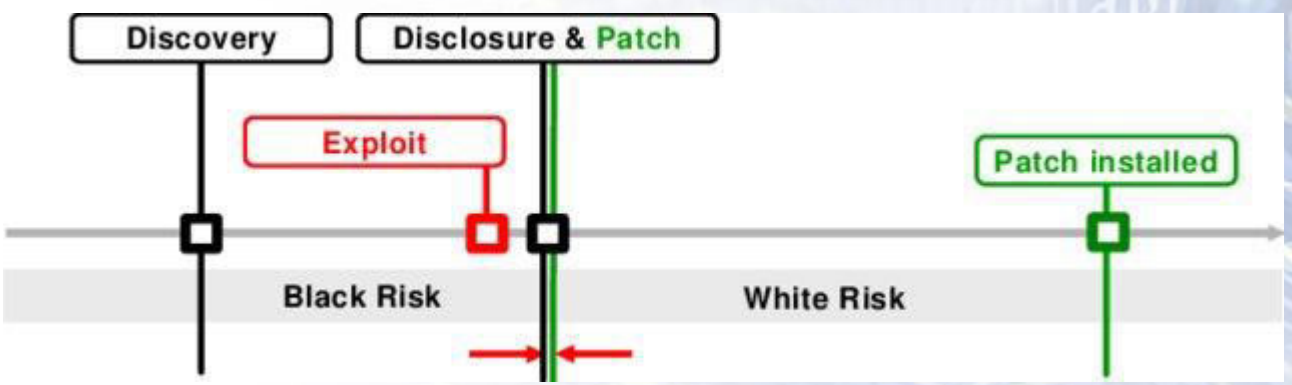
- It runs a safety check on apps from the Google Play Store before you download them.
- It checks your device for potentially harmful apps from other sources. These harmful apps are sometimes called malware.
- It warns you about potentially harmful apps.
- It may deactivate or remove harmful apps from your device.
- It warns you about detected apps that violate our Unwanted Software Policy by hiding or misrepresenting important information.
- It sends you privacy alerts about apps that can get user permissions to access your personal information, violating our Developer Policy.
- It may reset app permissions to protect your privacy on certain Android versions.

Week 14 Topic 2

Operating System Hardening

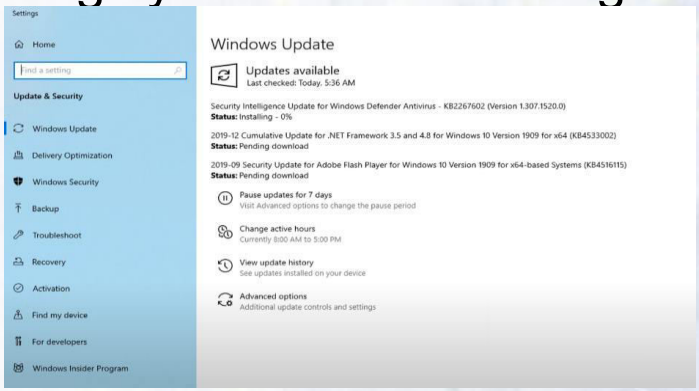
Operating Systems' Risks
Vulnerability, Exploit, Patch…





Operating System Hardening

- The idea of OS hardening is to minimize a computer's exposure to current and future threats by fully configuring the operating system and removing unnecessary applications.
  - Updates in Windows
  - Patch Management System in Linux/Unix



- Basic steps
  - Install and patch the operating system
  - Harden and configure the operating system to adequately address the identified security needs of the system by:
    - Removing unnecessary services, applications, and protocols
    - Configuring users, groups, and permissions
    - Configuring resource controls
  - Install and configure additional security controls, such as anti-virus, host-based firewalls, and intrusion detection system (IDS)
  - Test the security of the basic operating system to ensure that the steps taken adequately address its security needs

- When performing the initial installation the supplied defaults should not be used
  - Default configuration is set to maximize ease of use and functionality rather than security
  - If additional packages are needed later they can be installed when they are required

If fewer software packages are available to run the risk is reduced

System planning process should identify what is actually required for a given system

Not all users with access to a system will have the same access to all data and resources on that system

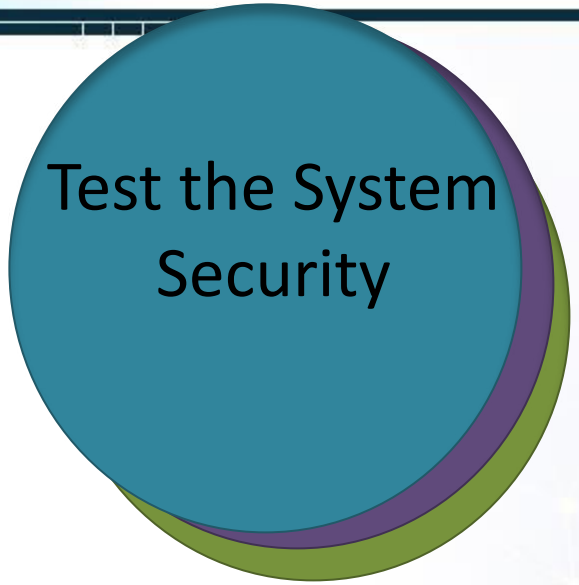
Elevated privileges should be restricted to only those users that require them, and then only when they are needed to perform a task

- System planning process should consider:
  - Categories of users on the system
  - Privileges they have
  - Types of information they can access
  - How and where they are defined and authenticated
- Default accounts included as part of the system installation should be secured
  - Those that are not required should be either removed or disabled
  - Policies that apply to authentication credentials configured

Once the users and groups are defined, appropriate permissions can be set on data and resources

Many of the security hardening guides provide lists of recommended changes to the default access configuration

- Further security possible by installing and configuring additional security tools:
  - Anti-virus software
  - Host-based firewalls
  - IDS or IPS software
  - Application white-listing



- Final step in the process of initially securing the base operating system is security testing
- Goal:
  - Ensure the previous security configuration steps are correctly implemented
  - Identify any possible vulnerabilities

- There are programs specifically designed to:
  - Review a system to ensure that a system meets the basic security requirements
  - Scan for known vulnerabilities and poor configuration practices
- Should be done following the initial hardening of the system
- Repeated periodically as part of the security maintenance process

Week 14 Topic 3

Security Maintenance

Process of maintaining security is continuous

Security maintenance includes:

- Monitoring and analyzing logging information
- Performing regular backups
- Recovering from security compromises
- Regularly testing system security
- Using appropriate software maintenance processes to patch and update all critical software

Week 14 Topic 4

Logging

Can only inform you about bad things that have already happened

In the event of a system breach or failure, system administrators can more quickly identify what happened

Key is to ensure you capture the correct data and then appropriately monitor and analyze this data

Information can be generated by the system, network and applications

Range of data acquired should be determined during the system planning stage

Generates significant volumes of information and it is important that sufficient space is allocated for them

- automatic log rotation

Automated analysis is preferred

Week 14 Topic 5

Data Backup and Archive

Performing regular backups of data is a critical control that assists with maintaining the integrity of the system and user data

- May be legal or operational requirements for the retention of data

Backup

- The process of making copies of data at regular intervals

Archive

- The process of retaining copies of data over extended periods of time in order to meet legal and operational requirements to access past data

Needs and policy relating to backup and archive should be determined during the system planning stage

- Kept online or offline
- Stored locally or transported to a remote site
  - Trade-offs include ease of implementation and cost versus greater security and robustness against different threats

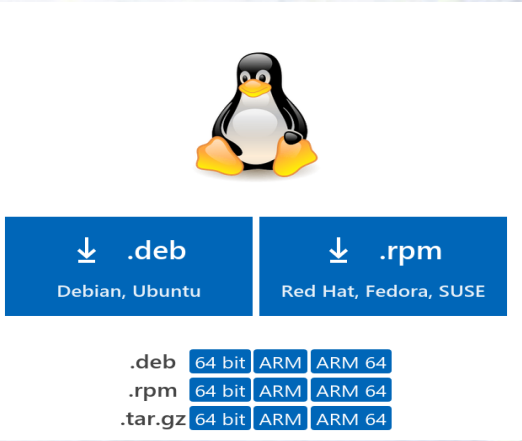
Example: Attack on an Australian hosting provider in early 2011

Week 14 Topic 6

Linux/Unix Security

Patch Management

- Keeping security patches up to date is a widely recognized and critical control for maintaining security
  - Debian based distribution e.g. Ubuntu…include apt-get
  - RPM… Redhat based distribution e.g. Fedora…include up2date or yum

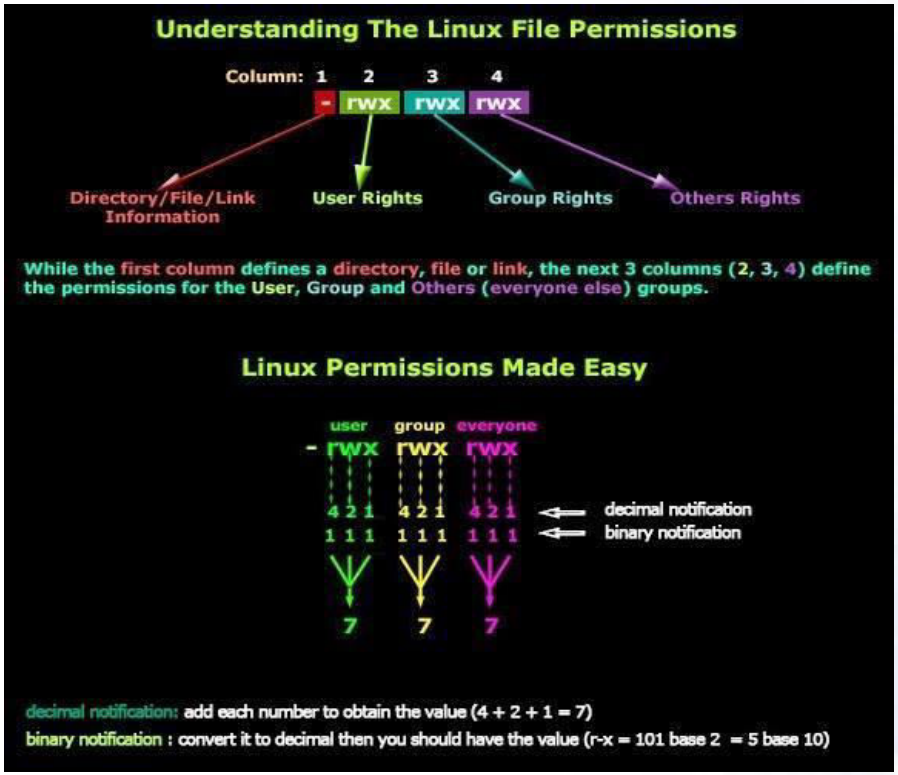


- Application and service configuration
  - Most commonly implemented using separate text files for each application and service
  - Generally located either in the /etc directory or in the installation tree for a specific application
- Individual user configurations that can override the system defaults are located in hidden “dot” files in each user’s home directory
- Most important changes needed to improve system security are to disable services and applications that are not required

Linux/Unix Security

Users, groups, and permissions

- Access is specified as granting read, write, and execute permissions to each of owner, group, and others for each resource
  - These are set using the **chmod** command.




```
israr@israr-VirtualBox: ~/Desktop/CS627/Assignment2
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$ ls
AES  aes-brute-force-master  AES.c
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$ ls -l
total 32
-rwxrwxr-x 1 israr israr 17784 2021  31 شمبر AES
drwxrwxr-x 4 israr israr  4096 2021  27 شمبر aes-brute-force-master
-rw-rw-r-- 1 israr israr  5661 2021  27 شمبر AES.c
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$
```

```
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$ chmod 777 AES.c
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$ ls -l
total 32
-rwxrwxr-x 1 israr israr 17784 2021  31 شمبر AES
drwxrwxr-x 4 israr israr  4096 2021  27 شمبر aes-brute-force-master
-rwxrwxrwx 1 israr israr  5661 2021  27 شمبر AES.c
```

```
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$ chmod 700 AES.c
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$ ls -l
total 32
-rwxrwxr-x 1 israr israr 17784 2021  31 شمبر AES
drwxrwxr-x 4 israr israr  4096 2021  27 شمبر aes-brute-force-master
-rwx------ 1 israr israr  5661 2021  27 شمبر AES.c
israr@israr-VirtualBox:~/Desktop/CS627/Assignment2$
```

Linux/Unix Security

- Guides recommend changing the access permissions for critical directories and files

- Local exploit
  - Software vulnerability that can be exploited by an attacker to gain elevated privileges
- Remote exploit
  - Software vulnerability in a network server that could be triggered by a remote attacker

Week 14 Topic 7

Windows Security

Patch management

- "Windows Update" and "Windows Server Update Service" assist with regular maintenance and should be used
- Third party applications also provide automatic update support

Users administration and access controls

- Systems implement discretionary access controls resources
- Vista and later systems also include mandatory integrity controls (MIC)
- Objects are labeled as being of low, medium, high, or system integrity level
- System ensures the subject's integrity is equal or higher than the object's level

Mandatory Integrity Control (MIC) provides a mechanism for controlling access to securable objects. This mechanism is in addition to discretionary access control and evaluates access before access checks against an object's discretionary access control list (DACL) are evaluated

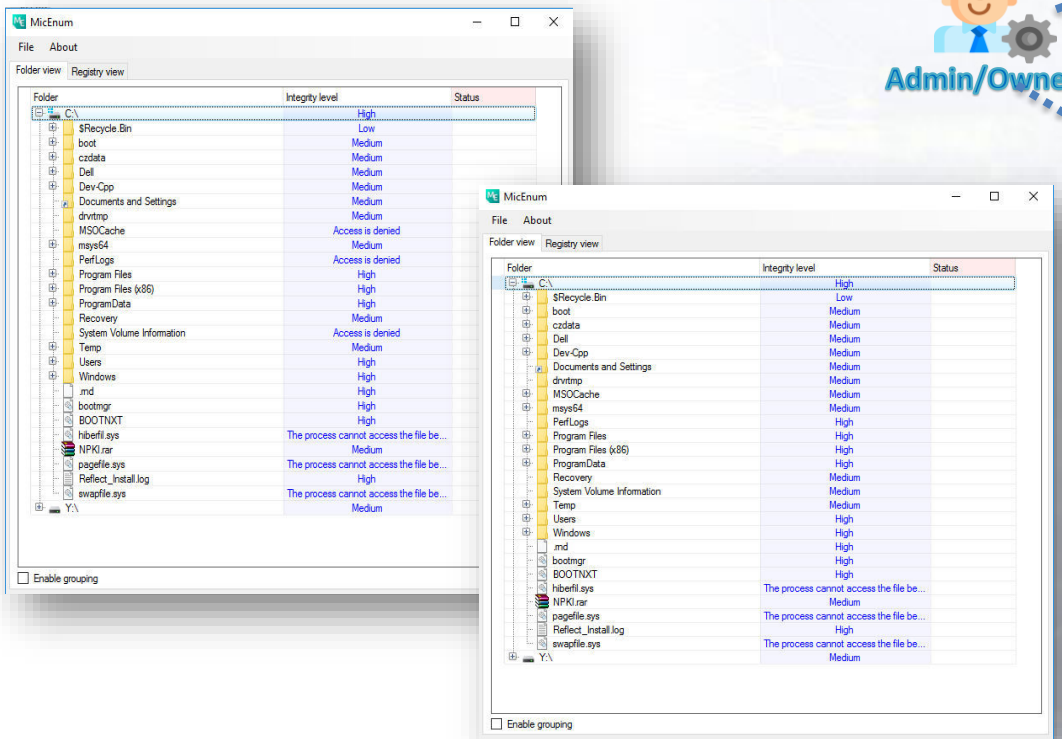
Windows Security

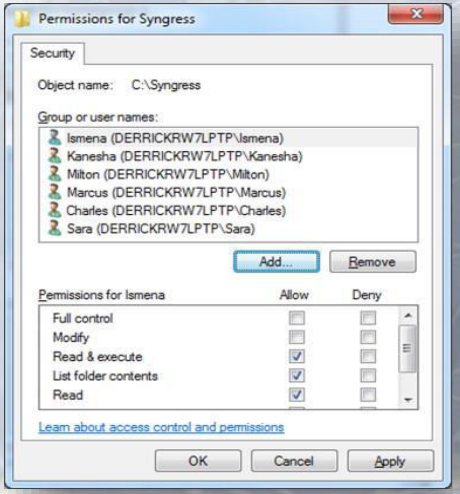
Mandatory Integrity Control (MIC) provides a mechanism for controlling access to securable objects. This mechanism is in addition to discretionary access control and evaluates access before access checks against an object's discretionary access control list (DACL) are evaluated

MANDATORY INTEGRITY CONTROL (MIC)

MicEnum Tool, Mandatory Integrity Control Enumerator

DISCRETIONARY ACCESS CONTROL LIST (DACL)

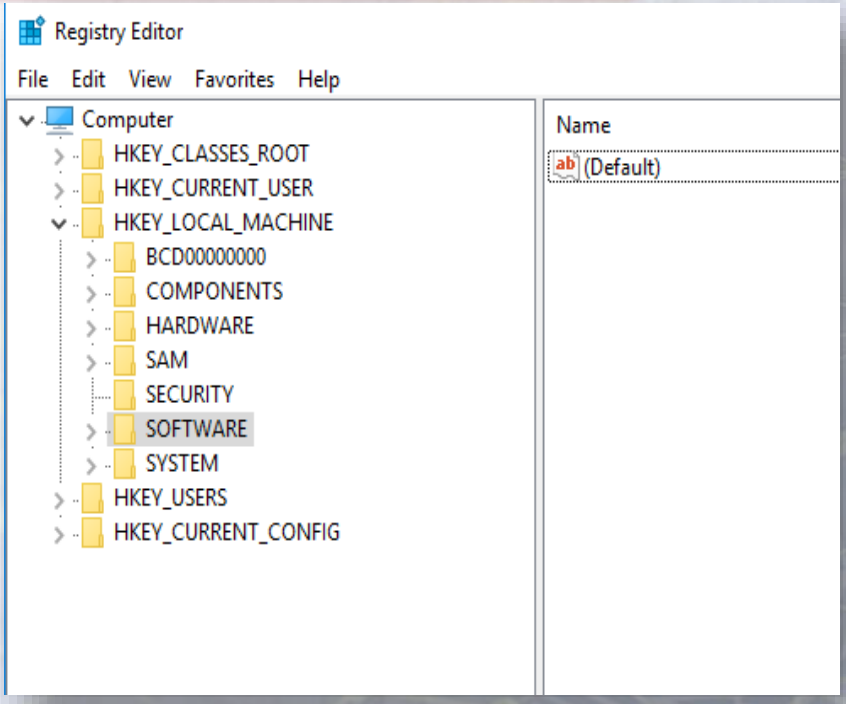




Windows Security

Application and service configuration

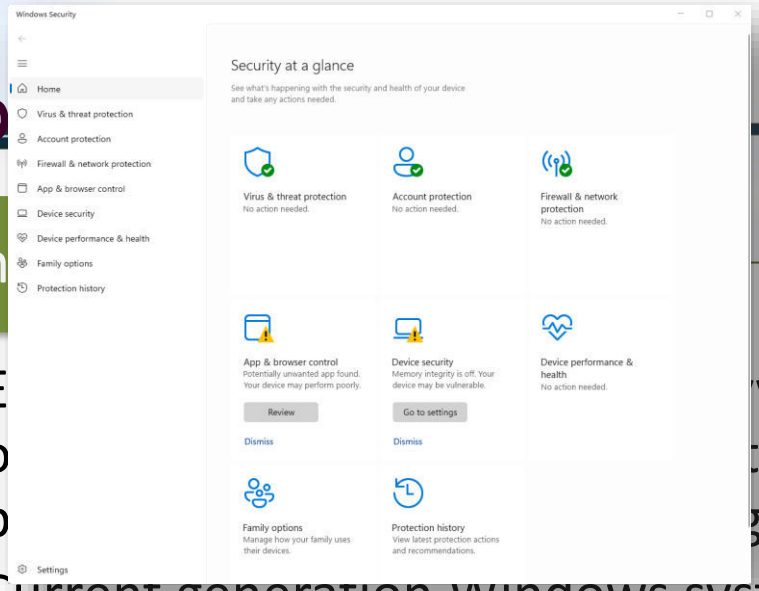
- Much of the configuration information is centralized in the Registry
  - Forms a database of keys and values that may be queried and interpreted by applications
- Registry keys can be directly modified using the “Registry Editor”
  - More useful for making bulk changes

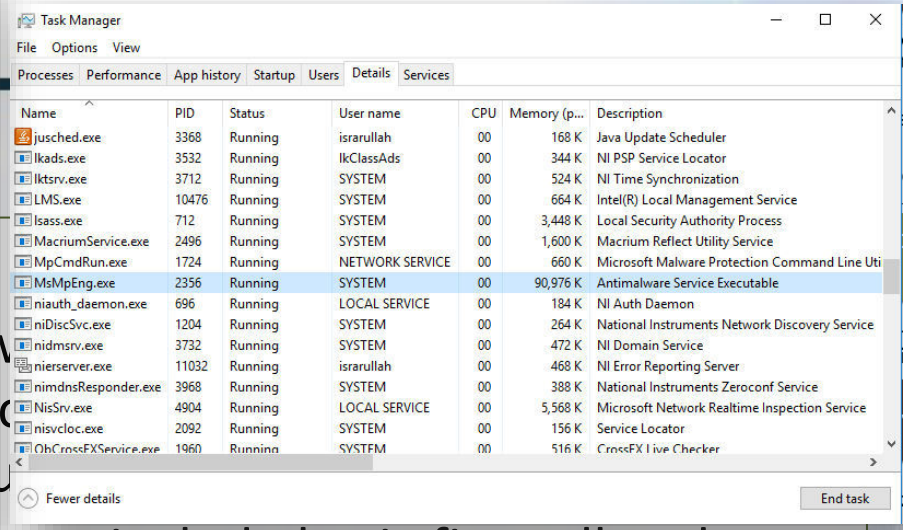


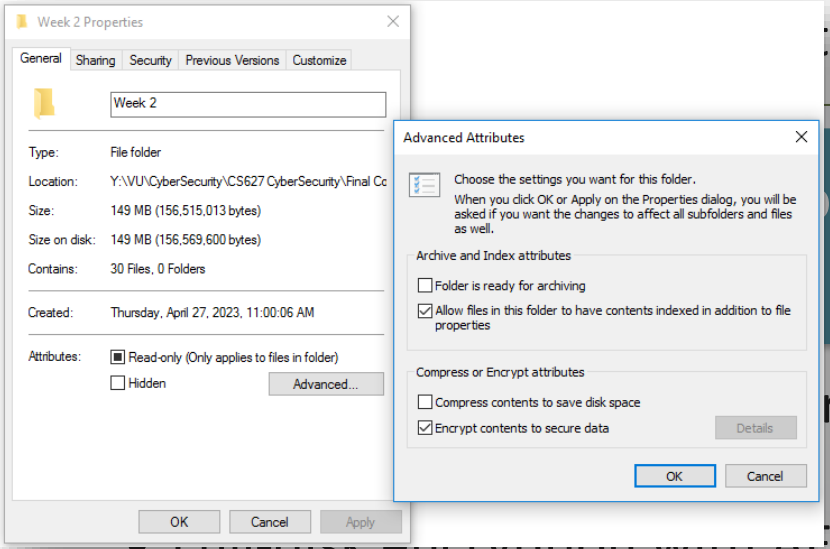
Windo

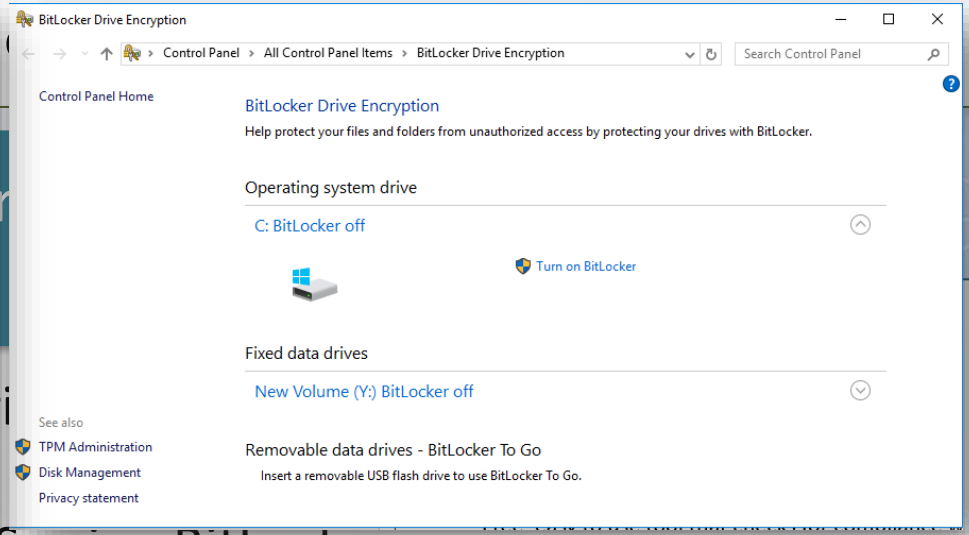
Oth

- E
  o
  p
- Current generation Windows systems include basic firewall and malware countermeasure capabilities
- Full-disk encryption with AES using BitLocker









Week No. 15

Cyber Crimes

Week 15 Topic 1

Cyber Crime

- Cybercrime is a type of crime involving a computer or a computer network.
- The computer may have been used in committing the crime, or it may be the target.
- Cybercrime may harm someone, someone's security or finances

Actors and Stakeholders

- Internationally, both state and non-state actors engage in cybercrimes, including espionage, financial theft, and other cross border crimes.
- Cybercrimes crossing international borders and involving the actions of at least one nation-state are sometimes referred to as cyberwarfare.
- Warren Buffett describes cybercrime as the "number one problem with mankind" and said that it "poses real risks to humanity".

- A 2014 report sponsored by McAfee estimated that cybercrime had resulted in $445 billion USD in annual damage to the global economy.
- Approximately $1.5 billion was lost in 2012 to online credit and debit card fraud in the US.
- In 2018, a study by the Center for Strategic and International Studies (CSIS), in partnership with McAfee, concluded that nearly 1 percent of global gross domestic product (GDP), close to $600 billion, is lost to cybercrime each year.
- The World Economic Forum 2020 Global Risk Report confirmed that organized cybercrime groups are joining forces to commit criminal activities online, while estimating the likelihood of their detection and prosecution to be less than 1 percent in the US.

Privacy

- There are also many privacy concerns surrounding cybercrime when confidential information is intercepted or disclosed, legally or otherwise

Week 15 Topic 2

Cyber Crime Categories

- Computer Fraud
- Cyberterrorism
  - Ransomware
- Cybersex Trafficking
- Cyberwarfare
- Obscene/Offensive Content
- Ad-Fraud
- Online Harassment
  - Cyberbullying
  - Online predator
  - Cyberstalking
  - Cyber Racism
  - Internet troll
- Drug Trafficking
- Identity Theft
- Phishing

Week 15 Topic 3

Computer Fraud

- Computer fraud is the act of using a computer to take or alter electronic data, or to gain unlawful use of a computer or system.
- If computer fraud involves the use of the internet, it can be considered Internet fraud.
- The legal definition of computer fraud varies by jurisdiction, but typically involves accessing a computer without permission or authorization.
- Forms of computer fraud include hacking into computers to alter information, distributing malicious code such as computer worms or viruses, installing malware or spyware to steal data, phishing, and advance-fee scams.
- Other forms of fraud may be committed using computer systems, including bank fraud, carding, identity theft, extortion, and theft of classified information.
- These types of crimes often result in loss of personal or financial information

Week 15 Topic 4

Cyberterrorism

- Cyberterrorism are acts of terrorism committed through the use of cyberspace or computer resources.
- Acts of disruption of computer networks and personal computers through viruses, worms, phishing, malicious software, hardware, or programming scripts can all be forms of cyberterrorism.
- Government officials and information technology (IT) security specialists have documented a significant increase in network problems and server scams since early 2001.
- Within the United States, there is an increasing concern from agencies such as the Federal Bureau of Investigation (FBI) and the Central Intelligence Agency (CIA).
- Such incidents are part of an organized effort by cyberterrorist foreign intelligence services or other groups to map potential security flaws in critical systems

Week 15 Topic 5

Cyberextortion (Ransomware)

- Biological virus
  - Latin → Poison
  - Shell filled with genetic material that it injects into a living cell, thereby infecting it
  - The cell then starts manufacturing copies of virus

- Computer virus behaves similarly
  - Injects its contents, a short computer program /instructions, into a host computer and infecting it
  - When the computer executes the virus code, it replicates the code, performs some task(s), usually of a malicious nature

Week 15 Topic 6

Cybersex Trafficking

- Cybersex trafficking is the transportation of victims and then the live streaming of coerced sexual acts or rape on webcam.
- Victims are abducted, threatened, or deceived and transferred to "cybersex dens".
- The dens can be in any location where the cybersex traffickers have a computer, tablet, or phone with an internet connection.
- Perpetrators use social media networks, video conferences, dating pages, online chat rooms, apps, dark web sites, and other platforms.
- They use online payment systems and cryptocurrencies to hide their identities.
- Millions of reports of its occurrence are sent to authorities annually.
- New legislation and police procedures are needed to combat this type of cybercrime.
- There are an estimated 6.3 million victims of cybersex trafficking, according to a recent report by the International Labour Organization and IOM.
- This number includes about 1.7 million child victims
- An example of cybersex trafficking is the 2018-2020 Nth room case in South Korea.

Week 15 Topic 7

Cyberwarfare

- The U.S. Department of Defense notes that cyberspace has emerged as a national-level concern through several recent events of geostrategic importance, including the attack on Estonia's infrastructure in 2007, allegedly by Russian hackers.
- In August 2008, Russia again allegedly conducted cyberattacks, this time in a coordinated and synchronized kinetic and non-kinetic campaign against the country of Georgia.
- Fearing that such attacks may become normalized in future warfare among nation-states, the military commanders intend to adapt the concept of cyberspace operations impact in the future.

Week 15 Topic 8

Obscene or offensive content

- The content of websites and other electronic communications may be distasteful, obscene, or offensive for a variety of reasons. In some instances, these communications may be illegal.
- The extent to which these communications are unlawful varies greatly between countries, and even within nations. It is a sensitive area in which the courts can become involved in arbitrating between groups with strong beliefs.
- One area of internet pornography that has been the target of the strongest efforts at curtailment is child pornography, which is illegal in most jurisdictions in the world

Week 15 Topic 9

Computer as a tool

- When the individual is the main target of cybercrime, the computer can be considered as the tool rather than the target.
- These crimes generally involve less technical expertise.
- Human weaknesses are generally exploited.
- The damage dealt is largely psychological and intangible, making legal action against the variants more difficult.
- These are the crimes which have existed for centuries in the offline world.
- Scams, theft, and the like existed before the development of computers and the internet.
- The same criminal has simply been given a tool which increases their potential pool of victims and makes them all the harder to trace and apprehend.

Computer as a tool

- Crimes that use computer networks or devices to advance other ends include:
  - Fraud and identity theft (although this increasingly uses malware, hacking or phishing, making it an example of both "computer as target" and "computer as tool" crime)
  - Information warfare
  - Phishing scams
  - Spam
  - Propagation of illegal obscene or offensive content, including harassment and threats
- The unsolicited sending of bulk email for commercial purposes (spam) is unlawful in some jurisdictions.
- Phishing is mostly propagated via email.
- Phishing emails may contain links to other websites that are affected by malware.
- Or they may contain links to fake online banking or other websites used to steal private account information

Week 15 Topic 10

Ad-fraud

- Ad-frauds are particularly popular among cybercriminals, as such frauds are lucrative and less likely to be prosecuted.
- Jean-Loup Richet, a professor at the Sorbonne Business School, classified the large variety of ad-fraud committed by cybercriminals into three categories:
    - identity fraud, attribution fraud, and ad-fraud services.

Ad-fraud

- Identity fraud aims to impersonate real users and inflate audience numbers.
- Several ad-fraud techniques relate to this category and include
  - traffic from bots (coming from a hosting company or a data center, or from compromised devices);
  - cookie stuffing;
  - falsifying user characteristics, such as location and browser type;
  - fake social traffic (misleading users on social networks into visiting the advertised website);
  - the creation of fake social media accounts to make the bot appear legitimate.

Ad-fraud

- Attribution fraud impersonates the activities of real users, such as clicks and conversations.
- Multiple ad-fraud techniques belong to this category:
  - hijacked devices and the use of infected users (through malware) as part of a botnet to participate in ad fraud campaigns;
  - click farms (companies where low-wage employees are paid to click or engage in conversations and affiliates' offers);
  - incentivized browsing;
  - video placement abuse (delivered in display banner slots);
  - hidden ads (which will never be viewed by real users);
  - domain spoofing (ads served on a website other than the advertised real-time bidding website);
  - clickjacking, in which the user is forced to click on an ad.

Ad-fraud

- Ad fraud services are related to all online infrastructure and hosting services that might be needed to undertake identity or attribution fraud.
- Services can involve
    - the creation of spam websites (fake networks of websites created to provide artificial backlinks);
    - link building services;
    - hosting services;
    - creation of fake and scam pages impersonating a famous brand and used as part of an ad fraud campaign

Week 15 Topic 11

Online harassment

- Cyberbullying
- Online predator
- Cyberstalking
- Cyber Racism
- Internet troll

- Whereas content may be offensive in a non-specific way, harassment directs obscenities and derogatory comments at specific individuals focusing, for example, on gender, race, religion, nationality, or sexual orientation.

- There are instances when committing a crime using a computer can lead to an enhanced sentence.
  - For example, in the case of United States v. Neil Scott Kramer, the defendant was given an enhanced sentence according to the U.S. Sentencing Guidelines Manual §2G1.3(b)(3) for his use of a cell phone to "persuade, induce, entice, coerce, or facilitate the travel of, the minor to engage in prohibited sexual conduct."
  - Kramer appealed the sentence on the grounds that there was insufficient evidence to convict him under this statute because his charge included persuading through a computer device and his cellular phone technically is not a computer.
  - Although Kramer tried to argue this point, the U.S. Sentencing Guidelines Manual states that the term "computer" means "an electronic, magnetic, optical, electrochemical, or other high-speed data processing device performing logical, arithmetic, or storage functions, and includes any data storage facility or communications facility directly related to or operating in conjunction with such device.

- In the United States, over 41 states have passed laws and regulations that regard extreme online harassment as a criminal act.
- These acts can be punished on a federal scale, such as US Code 18 Section 2261A, which states that using computers to threaten or harass can lead to a sentence of up to 20 years, depending on the action taken.
- Several countries outside the US have also created laws to combat online harassment.
- In China, a country that supports over 20 percent of the world's internet users, the Legislative Affairs Office of the State Council passed a strict law against the bullying of young people through a bill in response to the Human Flesh Search Engine.
- The United Kingdom passed the Malicious Communications Act, among other acts from 1997 to 2013, which stated that sending messages or letters electronically that the government deemed "indecent or grossly offensive" and/or language intended to cause "distress and anxiety" can lead to a prison sentence of six months and a potentially large fine.
- Australia, while not directly addressing the issue of harassment, has grouped the majority of online harassment under the Criminal Code Act of 1995.
- Using telecommunication to send threats or harass and cause offense was a direct violation of this act

Freedom of Speech

- Although freedom of speech is protected by law in most democratic societies (in the US this is done by the First Amendment), it does not include all types of speech.
- In fact, spoken or written "true threat" speech or text is criminalized because of "intent to harm or intimidate".
- That also applies to online or network-related threats in written text or speech.

Cyberbullying

- Cyberbullying has increased drastically with the growing popularity of online social networking.
- As of January 2020, 44 percent of adult internet users in the United States have "personally experienced online harassment".
- Children who experience online harassment deal with negative and sometimes life-threatening side effects.
- In 2021, reports displayed 41 percent of children developing social anxiety, 37 percent of children developing depression, and 26 percent of children having suicidal thoughts.

Spying

- The United Arab Emirates was named in a spying scandal where the Gulf nation along with other repressive governments purchased NSO Group's mobile spyware Pegasus for mass surveillance.
- Prominent activists and journalists were targeted as part of the campaign, including Ahmed Mansoor, Princess Latifa, Princess Haya, and more.
- GhadaOueiss was one of the many high-profile female journalists and activists who became the target of online harassment.
- Oueiss filed a lawsuit against UAE ruler Mohamed bin Zayed Al Nahyan along with other defendants, accusing them of sharing her photos online.
- The defendants, including the UAE ruler, filed motions to dismiss the case of the hack-and-leak attack.

Week 15 Topic 12

Drug trafficking

- Darknet markets are used to buy and sell recreational drugs online.
- Some drug traffickers use encrypted messaging tools to communicate with drug mules or potential customers.
- The dark web site Silk Road was the first major online marketplace for drugs, starting operation in 2011.
- It was permanently shut down in 2014 by the FBI and Europol.
- After Silk Road 2.0 went down, Silk Road 3 Reloaded emerged.
- However, it was just an older marketplace named Diabolus Market, that used the name for more exposure from the brand's previous success.
- Darknet markets have had a rise in traffic in recent years for many reasons, one of the biggest contributors being the anonymity offered in purchases, and often a seller-review system.

Drug trafficking

- There are many ways in which darknet markets can financially drain individuals.
- Vendors and customers alike go to great lengths to keep their identities a secret while online.
- Commonly used tools are virtual private networks (VPN), Tails, and the Tor Browser to help hide their online presence.
- Darknet markets entice customers by making them feel comfortable.
- People can easily gain access to a Tor browser with DuckDuckGo browser that allows a user to explore much deeper than other browsers such as Google Chrome.
- However, actually gaining access to an illicit market is not as simple as typing it in on a search engine like one would with Google.
- Darknet markets have special links that change frequently, ending in .onion as opposed to the typical .com, .net, and .org domain extensions

Digital/Virtual Currency:

- To add to privacy, the most prevalent currency on these markets is Bitcoin.
- Bitcoin allows transactions to be anonymous, with the only information available to the public being the record that a transaction occurred between two parties.

- An issue marketplace users face is when vendors or the market itself are exit scamming.
- This is when a vendor with a high rating will act as if they are selling on the market and have users pay for products they will not receive.
- The vendor will then close off their account after receiving money from multiple buyers and never send what was purchased.
- The vendors all being involved in illegal activities have a low chance of not exiting scamming when they no longer want to be a vendor.
- In 2019, an entire market known as Wall Street Market had allegedly exit scammed, stealing $30 million dollars from the vendors' and buyers' wallets in bitcoin

- FBI has cracked down on these markets.
- In July 2017, FBI seized one of the biggest markets, commonly called Alphabay, which later re-opened in August 2021 under the control of DeSnake, one of the original administrators.
- Investigators will pose as a buyer and order products from darknet vendors in the hopes that vendors leave a trail the investigators can follow.
- One investigation had an investigator pose as a firearms seller and for six months people purchased from them and provided home addresses.
- The FBI was able to make over a dozen arrests during this six-month investigation.

- Another one of law enforcement's crackdowns was on vendors selling fentanyl and opiates.
- With thousands of people dying each year due to drug overdose, investigators have made it a priority.
- Many vendors do not realize the extra criminal charges that go along with selling drugs online.
- Commonly they get charged with money laundering and charges for when the drugs are shipped in the mail on top of being a drug distributor.
- In 2019, a vendor was sentenced to 10 years in prison after selling cocaine and methamphetamine under the name JetSetLife.
- Although investigators spend large amounts of time tracking down people in 2018, only 65 suspects who bought and sold illegal goods on some of the biggest markets were identified.
- This is compared to the thousands of transactions taking place daily on these markets

Week 15 Topic 13

Famous Incidents:

- Data Leaks
- Celebrity Leaks
- Blackmailing
- Web Defacements
- Hacking / Spying
- Data Stealing
- Impersonation
- Superimposing / Fake Media
- ATM Fraud/Hacking
- Car Hacking
- Cryptocurrency / Forex

Week 15 Topic 14

Identity Theft

- Identity theft occurs when someone uses another person's personal identifying information, like their name, identifying number, or credit card number, without their permission, to commit fraud or other crimes.
- The term identity theft was coined in 1964.
- Since that time, the definition of identity theft has been statutorily defined throughout both the U.K. and the U.S. as the theft of personally identifiable information [PII].
- Identity theft deliberately uses someone else's identity as a method to gain financial advantages or obtain credit and other benefits, and perhaps to cause other person's disadvantages or loss.
- The person whose identity has been stolen may suffer adverse consequences, especially if they are falsely held responsible for the perpetrator's actions.
- Personally identifiable information generally includes a person's name, date of birth, social security number, driver's license number, bank account or credit card numbers, PINs, electronic signatures, fingerprints, passwords, or any other information that can be used to access a person's financial resources

Week 15 Topic 15

Phishing and its Types

Phishing

- Phishing is a form of social engineering where attackers deceive people into revealing sensitive information or installing malware such as ransomware.
- Phishing attacks have become increasingly sophisticated and often transparently mirror the site being targeted, allowing the attacker to observe everything while the victim is navigating the site, and transverse any additional security boundaries with the victim.
- As of 2020, it is the most common type of cybercrime, with the FBI's Internet Crime Complaint Centre reporting more incidents of phishing than any other type of computer crime.

Phishing

- The term "phishing" was first recorded in 1995 in the cracking toolkit AOHell, but may have been used earlier in the hacker magazine 2600.
- It is a variation of fishing and refers to the use of lures to "fish" for sensitive information.
- Measures to prevent or reduce the impact of phishing attacks include legislation, user education, public awareness, and technical security measures.
- The importance of phishing awareness has increased in both personal and professional settings, with phishing attacks among businesses rising from 72% to 86% from 2017 to 2020.

Phishing Types

- Email phishing
- Spear phishing
- Whaling and CEO fraud
- Clone phishing
- Voice phishing
- SMS phishing
- Page hijacking
- Calendar phishing

Phishing Type: Email phishing

- Phishing attacks, often delivered via email spam, attempt to trick individuals into giving away sensitive information or login credentials.
- Most attacks are "bulk attacks" that are not targeted and are instead sent in bulk to a wide audience.
- The goal of the attacker can vary, with common targets including financial institutions, email and cloud productivity providers, and streaming services.
- The stolen information or access may be used to steal money, install malware, or spear phish others within the target organization.
- Compromised streaming service accounts may also be sold on darknet markets.
- This method of social engineering attack involve sending fraud email or messages that appear to be from a trusted source, such as bank, amazon, or government agency.
- These messages will typically contain a link or attachment that, when you click, will install malware automatically on the targeted device or redirect them to fake login page of any trusted website where they will be promoted to enter their login credential.

Phishing Type: Spear phishing

- Spear phishing is a targeted phishing attack that uses personalized emails to trick a specific individual or organization into believing they are legitimate.
- It often utilizes personal information about the target to increase the chances of success.
- These attacks often target executives or those in financial departments with access to sensitive financial data and services.
- Accountancy and audit firms are particularly vulnerable to spear phishing due to the value of the information their employees have access to.
- Threat Group-4127 (Fancy Bear) targeted Hillary Clinton's 2016 presidential campaign with spear phishing attacks on over 1,800 Google accounts, using the accounts-google.com domain to threaten targeted users.
- A study on spear phishing susceptibility among different age groups found that 43% of 100 young and 58 older users clicked on simulated phishing links in daily emails over 21 days.
- Older women had the highest susceptibility, while susceptibility in young users declined over the study, but remained stable in older users.

Phishing Type: Whaling and CEO fraud

- Whaling attacks use spear phishing techniques to target senior executives and other high-profile individuals[with customized content, often related to a subpoena or customer complaint.
- CEO fraud involves sending fake emails from senior executives to trick employees into sending money to an offshore account.
- It has a low success rate, but can result in organizations losing large sums of money.

Phishing Type: Clone phishing

- Clone phishing is a type of attack where a legitimate email with an attachment or link is copied and modified to contain malicious content.
- The modified email is then sent from a fake address made to look like it's from the original sender.
- The attack may appear to be a resend or update of the original email.
- It often relies on the sender or recipient being previously hacked so the attacker can access the legitimate email.

Phishing Type: Voice phishing

- Voice over IP (VoIP) is used in vishing or voice phishing attacks, where attackers make automated phone calls to large numbers of people, often using text-to-speech synthesizers, claiming fraudulent activity on their accounts.
- The attackers spoof the calling phone number to appear as if it is coming from a legitimate bank or institution.
- The victim is then prompted to enter sensitive information or connected to a live person who uses social engineering tactics to obtain information.
- Vishing takes advantage of the public's lower awareness and trust in voice telephony compared to email phishing.

Phishing Type: SMS phishing (Smishing)

- SMS phishing or smishing is a type of phishing attack that uses text messages from a cell phone or smartphone to deliver a bait message.
- The victim is usually asked to click a link, call a phone number, or contact an email address provided by the attacker.
- They may then be asked to provide private information, such as login credentials for other websites.
- The difficulty in identifying illegitimate links can be compounded on mobile devices due to the limited display of URLs in mobile browsers.
- Smishing can be just as effective as email phishing, as many smartphones have fast internet connectivity. Smishing messages may also come from unusual phone numbers.

Phishing Type: Page hijacking

- Page hijacking involves redirecting users to malicious websites or exploit kits through the compromise of legitimate web pages, often using cross site scripting.
- Hackers may insert exploit kits such as MPack into compromised websites to exploit legitimate users visiting the server.
- Page hijacking can also involve the insertion of malicious inline frames, allowing exploit kits to load.
- This tactic is often used in conjunction with watering hole attacks on corporate targets

Phishing Type: Calendar phishing

- Calendar phishing involves sending fake calendar invitations with phishing links.
- These invitations often mimic common event requests and can easily be added to calendars automatically.
- To protect against this form of fraud, former Google click fraud czar Shuman Ghosemajumder recommends changing calendar settings to not automatically add new invitations

Week 15 Topic 16

Phishing Techniques

- Link manipulation
- Social engineering
- Filter evasion

Filter evasion

- Phishers have sometimes used images instead of text to make it harder for anti-phishing filters to detect the text commonly used in phishing emails.
- In response, more sophisticated anti-phishing filters are able to recover hidden text in images using optical character recognition (OCR).

Link manipulation

- Phishing attacks often involve creating fake links that appear to be from a legitimate organization.
- These links may use misspelled URLs or subdomains to deceive the user.
- Another tactic is to make the displayed text for a link appear trustworthy, while the actual link goes to the phisher's site.
- To check the destination of a link, many email clients and web browsers will show the URL in the status bar when the mouse is hovering over it.
- However, some phishers may be able to bypass this security measure.

Social engineering

- Phishing often uses social engineering techniques to trick users into performing actions such as clicking a link or opening an attachment, or revealing sensitive information.
- It often involves pretending to be a trusted entity and creating a sense of urgency, like threatening to close or seize a victim's bank or insurance account.
- An alternative technique to impersonation-based phishing is the use of fake news articles to trick victims into clicking on a malicious link.
- These links often lead to fake websites that appear legitimate, but are actually run by attackers who may try to install malware or present fake "virus" notifications to the victim.

Week 15 Topic 17

Combating computer crime

- It is difficult to find and combat cybercrime perpetrators due to their use of the internet in support of cross-border attacks.
- Not only does the internet allow people to be targeted from various locations, but the scale of the harm done can be magnified.
- Cybercriminals can target more than one person at a time.
- The availability of virtual spaces to public and private sectors has allowed cybercrime to become an everyday occurrence.
- In 2018, The Internet Crime Complaint Center received 351,937 complaints of cybercrime, which lead to $2.7 billion lost.

Combating computer crime

- Investigation
- Prevention
- Legislation
- Penalties
- Awareness
- Intelligence

Week 15 Topic 18

Cyber Crime Jobs

- Government, Military, Law Enforcement Agencies
- Private sector companies
- Designations:
  - Cyber Crime Analyst / Forensic Examiner / Scientist / Specialist / Investigator
- Relevant Degree,Certifications and Experienc

Week 15 Topic 19

Cyber Crime Agencies and Organizations

- FIA – CCW
- Military
- Government Sector
- Private Companies
- CERTs

Week 15 Topic 20

Cyber Crime Law in Pakistan

- PECA (2016)
  - Prevention of Electronic Crimes Act
- PPC (Pakistan Penal Code)
- CrPC (Criminal Procedure Code)
- FERA (Foreign Exchange Regulation Act, 1947)
- AMLA (Anti-Money Laundering Act, 2010)
- The Anti-Terrorism Act,1997

Week 15 Topic 21

Cyber Crime Reporting in Pakistan

- **Modes:**
  - Phone
  - Online
  - In-Person
- **Process:**
  - Application submission
  - Verification
  - Enquiry
  - FIR

Required Documents

- Application / Matter
- CNIC
- Proofs